Basic Concepts of String Theory: Reading Companion

Consolidated questions, derivations, source map, and book roadmap

Study notebook edition

August 2026

Contents

Chapter 0

Consolidated Reading Summary

This chapter collects the short conclusions from the reading conversations. Detailed derivations and textbook locations appear in the chapters that follow. Exact duplicate files were consolidated once; the empty `Superstrings.md` file contributes no Q&A material.

0.1 Bosonic One-Loop Amplitudes

- **Question 1:** For the closed bosonic string torus amplitude, why does the Hamiltonian trace over a fixed torus have to be integrated over τ, and why is the measure $d^2\tau/\tau_2^2$? The key point is that τ is a genuine complex structure modulus left after conformal gauge fixing, and $d^2\tau/\tau_2^2$ is the natural Poincare/Faddeev-Popov volume form on $\mathbb{H}/SL(2,\mathbb{Z})$, not an extra physical Boltzmann weight.
- **Question 2:** For the open bosonic string annulus/cylinder amplitude, where does the factor $dt/(2t)$ come from? The key point is that the annulus has one real modulus t with logarithmic moduli-space measure dt/t, and the standard extra factor $1/2$ comes from the residual vacuum-cylinder automorphism/conformal-Killing normalization.
- **Question 3:** A static flat D-brane with $F = 0$ has no source linear in the antisymmetric B-field because the trace of a symmetric inverse metric with $B_{\alpha\beta}$ vanishes. Hence there is no one-B exchange, whereas the linear graviton and dilaton sources generate the NS–NS attraction.

0.2 Torus Partition Functions for RCFTs

- **Question 1:** The modular transformation S exchanges the two torus cycles. Applying it twice reverses both cycles and acts on a chiral sector by conjugating its representation. Therefore $S^2 = C$, where $C_{ij} = \delta_{i,j^*}$ pairs each primary with its conjugate.

0.3 The Classical Fermionic String and Spin Structures

- **Question 1:** On a compact Riemann surface, the chiral Dirac operator splits into first-order operators on the two spin bundles; in a conformal coordinate the zero-mode equations reduce, after the standard Weyl rescaling, to holomorphic or anti-holomorphic half-differential equations. For a spin structure L with $L^2 = K$, the parity $h^0(\Sigma, L) \bmod 2$ is topological: it equals the Arf invariant of the quadratic refinement associated to the spin structure. Under gluing by connected sum, the quadratic refinements add orthogonally, so the mod-two number of chiral zero modes is additive.
- **Question 2:** The 2^{2g} spin structures are equivalently quadratic refinements q of the mod-two intersection form on $H_1(\Sigma_g, \mathbb{Z}_2)$. In a symplectic basis, $\mathrm{Arf}(q) = \sum_i q(a_i)q(b_i)$, so each handle has three even local choices and one odd local choice. Hence the odd spin structures are counted by choosing an odd number of odd handles, giving $2^{g-1}(2^g - 1)$, while the even spin structures are counted by choosing an even number, giving $2^{g-1}(2^g + 1)$.

- **Question 3:** The apparent $\vartheta\begin{bmatrix}\alpha\\-\beta\end{bmatrix}$ versus $\vartheta\begin{bmatrix}\alpha\\\beta\end{bmatrix}$ mismatch is only a characteristic-convention phase, and for $\beta = 0, \frac{1}{2}$ the spin structure is the same modulo integers; any remaining sign can be absorbed into the spin-structure phase $\eta_{\alpha\beta}$. Also $\vartheta_1(\tau) = 0$ means $\vartheta_1(0|\tau) = 0$, while $\vartheta_1'(\tau) = 2\pi\eta^3(\tau)$ means $\partial_\nu\vartheta_1(\nu|\tau)|_{\nu=0} = 2\pi\eta^3(\tau)$, so there is no contradiction.
- **Question 4:** The $\eta = \pm 1$ in the closed-string boundary state is a boundary gluing label, not by itself the open-string R/NS sector. Under the open-closed channel exchange, the two closed-string boundaries have opposite induced orientations. Bosons are insensitive to the square root of this orientation reversal, but fermions are half-differentials, so the orientation reversal contributes a relative phase whose square is -1; with the usual gluing convention this gives $s_0 s_l = -\eta_1\eta_2$. Hence same closed labels give the open NS trace, while opposite closed labels give the open R trace. In the NS-NS sector, $(-1)^F$ flips the sign of the left-moving fermion oscillators and the odd NS vacuum gives $(-1)^F|\mathrm{D}p, \eta\rangle_{\mathrm{NS}} = -|\mathrm{D}p, -\eta\rangle_{\mathrm{NS}}$, and similarly for $(-1)^{\bar{F}}$.
- **Question 5:** The sign of a static exchange potential comes from the propagator numerator contracted with static source components. In mostly-plus signature, scalar and graviton exchange give $V \sim -q_1 q_2 G(r)$ for positive/equal charges, hence attraction, while vector exchange gives $V \sim +q_1 q_2 G(r)$, hence repulsion. A R-R $(p+1)$-form coupled to a Dp-brane behaves like the vector case for parallel branes: reversing orientation reverses the R-R charge, and equal R-R charges repel. For BPS parallel D-branes this R-R repulsion exactly cancels the NS-NS attraction from graviton, dilaton, and related fields.
- **Question 6:** The type II torus vacuum amplitude vanishes because the GSO-summed worldsheet path integral computes a spacetime supertrace, not an ordinary thermal trace. In unbroken spacetime supersymmetry, each bosonic physical state has a fermionic partner of the same mass and degeneracy, so $\mathrm{Str}\, e^{-2\pi t M^2} = 0$ level by level. On the worldsheet this cancellation is the Jacobi abstruse identity $\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4 = 0$.
- **Question 7:** In tree channel, the tadpole factorization has the schematic form $\frac{1}{2}\langle B+C|\Delta|B+C\rangle = \frac{1}{2}\langle B|\Delta|B\rangle + \langle B|\Delta|C\rangle + \frac{1}{2}\langle C|\Delta|C\rangle$. The annulus and Klein bottle are the two half-square terms, while the single Möbius strip amplitude is already the full boundary-crosscap cross term. Adding two copies of the Möbius amplitude would double-count the same unoriented worldsheet/moduli region.
- **Question 8:** The orientifold projection Ω acts on a BPS Dp boundary state by leaving the NS-NS GSO-projected part invariant but multiplying the R-R GSO-projected part by $(-1)^{(p+3)/2}$ for odd p. Therefore the R-R charged BPS branes invariant under type I Ω are $p = 1, 5, 9$. Equivalently, Ω keeps the R-R potentials C_2, its magnetic dual C_6, and the top-form C_{10}, while projecting out C_0, C_4, C_8.

0.4 The Quantized Fermionic String

- **Question 1:** In old-covariant quantization the oscillator Fock space has indefinite metric because of timelike oscillators. The no-ghost theorem says that, at the critical values, the physical state space modulo null/spurious states is positive definite and is isomorphic to the transverse light-cone/DDF Fock space: for the bosonic string $D = 26$, $a = 1$; for the RNS superstring $D = 10$, with $a_{\mathrm{NS}} = \frac{1}{2}$ and $a_{\mathrm{R}} = 0$. Thus negative-norm states are gauge artifacts or null states and decouple from amplitudes.
- **Question 2:** The NS oscillator $b^i_{-1/2}$ is fermionic as a worldsheet oscillator, but it carries an $SO(8)$ vector index, so the states $b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}$ form a vector basis of the massless one-string Hilbert space. The polarization ζ_i is not more physical than the basis state; rather $|\zeta;k\rangle = \zeta_i b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}$ is the physical one-particle state. In second quantization, the momentum-space gauge field $A_i(k)$ is the coefficient multiplying this string state in the string field. Spacetime fermions instead come from the R sector, whose zero modes form a little-group Clifford algebra and produce spinor representations.

0.5 Kac–Moody Algebras

- **Question 1:** A representation weight pairs integrally with every coroot; for simply-laced roots normalized by $\vec{\alpha}^2 = 2$, this gives $\vec{\alpha} \cdot \vec{m} \in \mathbb{Z}$. Affine unitarity then requires $\vec{\psi} \cdot \vec{m}_0 \leq k$, where $\vec{\psi}$ and

$\vec{m}_0$ are the highest root and highest weight. At level one, the ADE primaries are the vacuum and the minuscule fundamental representations, one minimal representative for each class in P/Q.

0.6 Covariant Vertex Operators, BRST, and Covariant Lattices

- **Question 1:** For the bosonic string, an integrated open-string matter vertex must have $h = 1$, while an integrated closed-string matter vertex must have $(h, \bar{h}) = (1, 1)$, so that its insertion is invariant under the residual conformal reparametrizations. Equivalently, these are the L_0 physical-state condition with intercept one and the condition for the ghost-dressed vertex cV_{m} or $c\bar{c}V_{\mathrm{m}}$ to be BRST closed. Weight one is the on-shell condition, not by itself a masslessness condition: tachyon vertices also have weight one.
- **Question 2:** In the RNS superstring, the complete integrated operator, including its $\beta\gamma$ superghost dressing but excluding the c ghost, has $h = 1$ for an open string and $(h, \bar{h}) = (1, 1)$ for a closed string. The matter factor alone is picture-dependent: for example, the massless NS matter factor has $h = 1/2$ in picture -1 but the superghost $e^{-\phi}$ supplies the missing $1/2$.
- **Question 3:** The exponential map from the open-string strip to the upper half-plane sends $\sigma = 0$ to the positive real axis and $\sigma = l$ to the negative real axis. Because the worldsheet fermions have weight $1/2$, their square-root Jacobians have relative sign -1 on the negative axis. Thus the general gluing is $D(0)$ for $x > 0$ and $-\eta_{\mathrm{strip}}D(l)$ for $x < 0$; when both ends lie on the same D-brane, this becomes $+D$ everywhere in the NS sector and $\mathrm{sign}(x)D$ in the R sector, as in Eq. (13.99).
- **Question 4:** In BCFT, $\bar{T}(\bar{z})$ denotes the antiholomorphic stress-tensor component, not automatically the Hermitian adjoint of $T(z)$. The conditions $T(x) = \bar{T}(x)$ and $\epsilon(x) = \bar{\epsilon}(x)$ express vanishing boundary flux and boundary-preserving conformal transformations, leaving one diagonal Virasoro algebra rather than two independent copies. Applying the same contour argument to D-brane supersymmetry currents preserves $Q_L + PQ_R$; GSO chirality then requires even p in type IIA and odd p in type IIB.
- **Question 5:** The NS/R sector of a bosonized vertex V_λ is determined by the monodromy of a fundamental fermion around its insertion, not by rotating V_λ itself. The OPE $\Psi^{\pm j}(z)V_\lambda(0) \sim z^{\pm\lambda_j}(\cdots)$ gives monodromy $e^{\pm 2\pi i\lambda_j}$: integer λ_j give the plane NS sector and comprise $(0) \cup (V)$, while half-integer λ_j give the plane R sector and comprise $(S) \cup (C)$.
- **Question 6:** The one-loop trace is not a trace over already on-shell covariant states with $L_0 = 1$. It is either a trace over the gauge-fixed worldsheet CFT state space or, in light-cone gauge, over the unconstrained transverse oscillator Fock space together with off-shell Euclidean loop momentum. The cylinder Hamiltonian is $L_0 - c/24$; for the 24 transverse bosons this is $L_0 - 1$. In Eq. (13.114), $L_0 - 1$ is likewise the chiral worldsheet propagation operator before the c-ghost dressing, while $(-1)^{F_R}$ makes the trace a spacetime supertrace.

0.7 CFTs for Type II and Heterotic String Vacua

- **Question 1:** For a toroidal orbifold $T^6/\mathbb{Z}_N$, the untwisted Hodge classes are the $\mathbb{Z}_N$-invariant constant forms on T^6. The twisted classes are computed from the cohomology of the fixed loci of each θ^k, shifted in bidegree by the age of the twist and projected by the full orbifold group. For case 1, $T^6/\mathbb{Z}_3$ on A_2^3, this gives $(h^{1,1}, h^{2,1}) = (9, 0) + (27, 0) = (36, 0)$. For case 2, $T^6/\mathbb{Z}_4$ on $A_1^2 \times B_2^2$, the θ fixed points and the θ^2 fixed tori give $(h^{1,1}, h^{2,1}) = (5, 1) + (16, 0) + (10, 6) = (31, 7)$.
- **Question 2:** The extra orbifold gauge bosons do not arise from twisted sectors. They are untwisted ten-dimensional E_8 current-algebra states whose generators commute with the discrete gauge shift $\gamma_{\mathbf{V}}$ used to embed the orbifold point group. A smooth standard embedding has full $SU(3)$ gauge-bundle holonomy and leaves only its commutant E_6, whereas the orbifold background has only a discrete subgroup generated by $\gamma_{\mathbf{V}}$ and hence the larger centralizer $E_6 \times G_{\mathrm{enh}}$. For equal, two equal, or three distinct eigenvalues of $\gamma_{\mathbf{V}}$ in $SU(3)$, G_{enh} is respectively $SU(3)$, $SU(2) \times U(1)$, or $U(1)^2$. Charged twisted blow-up modes Higgs this extra factor when they acquire expectation values.
- **Question 3:** T-duality along n coordinates conjugates the type-IIB orientifold generator to $\mathcal{T}_n\Omega\mathcal{T}_n^{-1} = \Omega I_n[(-1)^{F_L}]^{n(n-1)/2}$. The reflection I_n comes from the bosonic asymmetric reflection, while the $(-1)^{F_L}$ factor for $n = 2, 3 \pmod 4$ comes from the projective action of reflections on Ramond spinors. For $n = 0, 1, 2, 3, 4$ this reproduces respectively the O9, O8, O7, O6, and O5

rows of Table 10.3, including their type-II theory, orientifold action, fixed-plane multiplicity, and local Chan–Paton gauge group.

- **Question 4:** Tensoring the chiral states in Table 15.3, imposing equal I_4 parity, and then projecting with $\Omega R(-1)^{F_L}$ gives the untwisted spectrum $(3,3)+11(1,1)$ in NS–NS, $(3,1)+(1,3)+6(1,1)$ in R–R, and $2(2,3)+10(2,1)$ in the mixed sectors. The chosen extra minus sign on each twisted sector leaves, at each of the 16 fixed points, $3(1,1)$ in NS–NS, $(1,1)$ in R–R, and $2(2,1)$ in the mixed sectors. These states form one gravity, one tensor, and four hypermultiplets in the untwisted sector, plus sixteen twisted hypermultiplets.
- **Question 5:** Because the theory is already orbifolded by I_4, its orientifold group contains two orientation-reversing elements, $\Omega R(-1)^{F_L}$ and $\Omega R I_4(-1)^{F_L}$. The corresponding terms in the Klein-bottle trace identify two crosscaps whose geometric parts fix respectively four two-tori parallel to (x^1,x^2) and four two-tori parallel to (y^1,y^2) in the covering T^4. Equivalently, a fixed point of the induced reflection on $T^4/\mathbb{Z}_2$ may obey either $Rp=p$ or $Rp=I_4p$. The insertion and its zero-mode lattice sum therefore determine the O7-plane locus, but not as the locus of open-string endpoints: the transverse Klein bottle is closed-string propagation between crosscaps and has no worldsheet boundary. Open-string endpoints attach only to D-branes; annulus and Möbius amplitudes contain boundary states.
- **Question 6:** For an I_4-invariant D7-brane wrapping (x^1,x^2), the I_4 projection retains the six-dimensional gauge vector while removing transverse-position and Wilson-line scalars. After the annulus modular transformation, the inserted R contribution is I_4-twisted R–R propagation, proving that the fractional boundary state carries twisted charge under $C_6^{(i)}=\int_{E_i}C_8$. The two fractional branes differ by the sign of this twisted boundary-state component. Because this model chooses $\Omega R(-1)^{F_L}$ to act with an extra minus sign on every I_4-twisted state, it leaves the untwisted component fixed but reverses the twisted component, giving $|D7^{\pm}\rangle_{\text{frac}} \mapsto |D7^{\mp}\rangle_{\text{frac}}$. An invariant configuration therefore contains the pair, whose twisted charges and tadpoles cancel.
- **Question 7:** Equation (15.52) follows by requiring the vertex in (15.51), before the bc ghosts are attached, to have conformal weights $(h,\bar{h})=(1,1)$. The oscillator factors contribute N_L, N_{1R}, N_{2R}, the fermion-lattice exponential contributes $\lambda_R^2/2$, the picture factor contributes $-q(q+2)/2$, the internal operator contributes $(h_{\text{int}},\bar{h}_{\text{int}})$, and the plane wave contributes $-\alpha' m^2/4$ to each chirality. The book packages the two chiral weight conditions as $m^2=m_L^2+m_R^2$, while level matching gives $m_L^2=m_R^2=m^2/2$ for a physical closed-string state.
- **Question 8:** The significant momentum factors in (15.84) are the extra $\epsilon_\mu k_\nu$ and k_μ, not the universal plane wave $e^{ik\cdot X}$. Antisymmetry of $\sigma^{\mu\nu}$ turns $\epsilon_\mu k_\nu$ into the graviphoton field strength $F_{\mu\nu}$, while $k_\mu C(k)$ is the Fourier transform of $\partial_\mu C$. This is the four-dimensional form of the general R–R fact that BRST-invariant spin-field polarizations are field-strength bispinors. Hence perturbative local bulk couplings contain R–R potentials through their field strengths and do not produce a minimal coupling $A_\mu J^\mu$ to perturbative charged matter. Wrapped D-branes can nevertheless carry R–R charge, and topological or Chern–Simons terms require the usual qualification.
- **Question 9:** In the right-moving NS sector, BRST invariance of an integrated picture-zero weight-one operator requires its supercurrent variation to be a total derivative. If $G(z)W(w)\sim A/(z-w)^2+B/(z-w)$, the γG part of the BRST charge gives $(\partial\gamma)A+\gamma B$, which is a total derivative precisely when $B=\partial A$. For a weight-1/2 superprimary $\hat{J}^a$, the $N=1$ algebra gives $J^a=G_{-1/2}\hat{J}^a$ and $G(z)J^a(w)\sim \hat{J}^a/(z-w)^2+\partial\hat{J}^a/(z-w)$. This derives the superfield condition and the canonical vertex (15.91). A full finite-momentum picture change gives $J^a+(k\cdot\psi)\hat{J}^a$; therefore (15.92), as printed, is exact at $k=0$ or shorthand for its internal-current term, but omits the universal momentum-dependent completion required at $k\neq 0$.

0.8 String Scattering Amplitudes and Low-Energy Effective Field Theory

- **Question 1:** A conventional first-quantized string calculation determines only the on-shell spacetime S-matrix. An “off-shell amplitude” in the low-energy field theory is more precisely an amputated Green function or an n-point vertex obtained by differentiating an effective action at momenta not obeying the free equations of motion. Such quantities are needed to define propagators, internal-line exchange, equations of motion, background dependence, and loop calculations, but they are not separately observable and are not uniquely fixed by the string S-matrix: local field redefinitions, gauge choices, and operators proportional to the equations of motion change them while leaving every on-shell amplitude unchanged.

- **Question 2:** The three-tachyon matter factor $\prod_{i<j}|z_{ij}|^{-2}$ exactly cancels the sphere ghost factor $|z_{12}z_{13}z_{23}|^2$, leaving $\mathcal{A}_3 = (2\pi)^d\delta^{(d)}(\sum_i k_i)C_{S^2}g_c^3$, or $C_{S^2}g_c^3$ with the delta function suppressed. The line below (16.24) has a factor-of-two typo: the correct condition is $\frac{\alpha'}{2}k_i\cdot k_j = -1$, not -2.
- **Question 3:** On the upper half-plane, the Dp-brane gluing matrix $D^\mu{}_\nu = +1$ in Neumann directions and -1 in Dirichlet directions reflects a right mover into a holomorphic image field in the lower half-plane. Applying the ordinary plane propagators to the doubled fields gives $\langle X^\mu(z)\bar{X}^\nu(\bar{w})\rangle = -\frac{\alpha'}{2}D^{\mu\nu}\log(z-\bar{w})$ and $\langle\psi^\mu(z)\bar{\psi}^\nu(\bar{w})\rangle = D^{\mu\nu}/(z-\bar{w})$, with the Dirichlet sign entirely encoded by D.
- **Question 4:** The sigma-model couplings $G_{\mu\nu}(X)$, $B_{\mu\nu}(X)$, and $\Phi(X)$ are spacetime fields, and their Weyl-anomaly coefficients measure whether the corresponding background defines a consistent worldsheet CFT. The spacetime string action is constructed from the same worldsheet theory, and its variational derivatives are invertible field-space combinations of these coefficients. Explicitly at leading order, varying (14.8) reproduces (14.5), so $\beta^G = \beta^B = \beta^\Phi = 0$ is equivalent to the spacetime equations of motion, modulo gauge transformations, field redefinitions, and matched orders in α' and g_s.

0.9 Type-II Compactifications with D-Branes and Fluxes

- **Question 1:** An infinitesimal displacement of an embedded D6-brane cycle is kinematically a section of its normal bundle, hence an element of $H^0(\Sigma, N_\Sigma) = \Gamma(N_\Sigma)$. For a special Lagrangian cycle, however, a displacement is a massless supersymmetric modulus only if it preserves both $\omega|_\Sigma = 0$ and $\mathrm{Im}(e^{-i\theta}\Omega)|_\Sigma = 0$. The map $v \mapsto -\iota_v\omega|_\Sigma$ converts the linearized conditions into $d\alpha = d^\dagger\alpha = 0$, so McLean's theorem identifies the deformation space with $H^1(\Sigma,\mathbb{R})$ and gives $b_1(\Sigma)$ real geometric moduli. A continuous Wilson-line fluctuation is likewise a closed one-form modulo exact gauge transformations, and is therefore counted by $H^1(\Sigma,\mathbb{R})$. The two sets of $b_1(\Sigma)$ real modes combine into $b_1(\Sigma)$ complex adjoint scalars.
- **Question 2:** At tree level, the type-IIB Kähler-modulus Kähler potential obeys the no-scale identity $K^{\alpha\bar{\beta}}K_\alpha K_{\bar{\beta}} = 3$ in Planck units. Since the flux superpotential is independent of T_α, one has $D_\alpha W = K_\alpha W$, and its contribution cancels the universal $-3|W|^2$ term in the $\mathcal{N}=1$ F-term potential. The remaining potential is a positive sum of the complex-structure and axiodilaton F-terms, so it vanishes only when those F-terms vanish. If also $W=0$, all F-terms vanish and the vacuum is supersymmetric Minkowski; if $W\neq 0$, the vacuum can still have zero tree-level energy but has $D_\alpha W \neq 0$ and broken supersymmetry. The Kähler moduli remain flat, and α' corrections, string loops, nonperturbative T_α dependence, or nonzero D-terms generally spoil the zero-energy conclusion.

0.10 String Dualities and M-Theory

- **Question 1:** Writing the full dilaton as $\Phi = \Phi_0 + \phi$ with $g_s = e^{\Phi_0}$, the three metrics are related by $G^{(E)}_{MN} = e^{-\Phi/2}G^{(S)}_{MN}$ and $g^{(mE)}_{MN} = e^{-\phi/2}G^{(S)}_{MN} = \sqrt{g_s}\,G^{(E)}_{MN}$. The string frame is natural for the world-sheet, loop counting, and T-duality; the ordinary Einstein frame gives a canonical Einstein–Hilbert term and makes type-IIB $SL(2)$ duality manifest; the modified Einstein frame retains the canonical bulk action while normalizing the asymptotic metric to η_{MN}, which is why it is used for the brane solutions in Section 18.5 and in equation (18.26).
- **Question 2:** For the warped p-brane ansatz, a single brane projector removes half of the spinors. After imposing it, the world-volume and transverse gravitino equations reduce to first-order BPS equations $A' = -2\tilde{n}\,u/[\Delta(d-2)]$ and $B' = 2n\,u/[\Delta(d-2)]$ for one common flux function $u(r)$. Their weighted sum vanishes, so $nA + \tilde{n}B$ is constant; asymptotic Minkowski normalization sets it to zero. For a Dp-brane this follows explicitly from $B'_S = -A'_S$ and $\phi' = (p-3)A'_S$ in string frame, followed by $A = A_S - \phi/4$ and $B = B_S - \phi/4$.
- **Question 3:** In type-IIB spinor-doublet notation the D5 and NS5 conditions are the commuting,

independent projectors $K_D = \Gamma^{012345}\sigma_1$ and $K_N = \Gamma^{012346}\sigma_3$. Each halves the 32 supercharges, so their intersection preserves $32/4 = 8$. For a (p,q) five-brane, the $SL(2)/SO(2)$ scalar vielbein dresses the integer charge vector into the local central-charge matrix $\mathcal{Z}_5 = T_{\mathrm{D5}}[\mathrm{Re}(p+\tau q)\sigma_1 + \mathrm{Im}(p+\tau q)\sigma_3]$; dividing by the BPS tension gives $\widehat{\mathcal{Z}}_5 = \cos\alpha\,\sigma_1 + \sin\alpha\,\sigma_3$. Saturation of the BPS bound makes the supercharge anticommutator vanish on preserved spinors and therefore forces $K_{(p,q)} = \Gamma_{(5)}\Gamma_\theta\widehat{\mathcal{Z}}_5$. It preserves the same eight supercharges precisely when $\theta = \alpha = \arg(p+\tau q)$. Normalized coordinates in the real basis $(1,\tau)$ convert every leg to $\delta\tilde{x}^5 + i\delta\tilde{x}^6 \propto p + iq$; this is the web convention $\tau = i$, not generally an $SL(2,\mathbb{Z})$ equivalence of the full string background.

- **Question 4:** An oriented type-II Dp-brane carries the dimensional reduction of ten-dimensional $\mathcal{N}=1$ super-Yang–Mills: a world-volume gauge field A_α, $9-p$ transverse scalars Φ^i, and a GSO-projected gaugino. For a single type-I D1, Ω removes $A_{0,1}$ and one two-dimensional chirality of the D1–D1 Ramond ground state, leaving eight transverse bosons X^i and eight right-moving Green–Schwarz fermions ψ^A. The eight mixed directions make the D1–D9 NS zero-point energy positive, but its Ramond ground state is massless; the GSO and Ω projections leave 32 real left-moving fermions λ^a. These are exactly the Green–Schwarz world-sheet fields of the Spin(32)/$\mathbb{Z}_2$ heterotic string. Summing over the four flat $O(1) \cong \mathbb{Z}_2$ bundles on a toroidal D1 world-sheet supplies the heterotic GSO projection.
- **Question 5:** In the eleven-to-ten-dimensional Kaluza–Klein reduction, the metric exponents are fixed by demanding the type-IIA string-frame couplings: for $ds^2_{11} = e^{2a\Phi}ds^2_{10,S} + e^{2b\Phi}(dx^{11}+C_1)^2$, the NS–NS Ricci term requires $8a+b=-2$, while the R–R $F_2 = dC_1$ term must have no $e^{-2\Phi}$ prefactor and requires $2a+b=0$. Hence $a=-1/3$ and $b=2/3$. Explicitly, $\sqrt{-G} = e^{-8\Phi/3}\sqrt{-g}$ and $R_{11} = e^{2\Phi/3}[R_{10} + \frac{14}{3}\Box\Phi - \frac{16}{3}(\partial\Phi)^2 - \frac{1}{4}e^{2\Phi}F_2^2]$. After integration by parts this gives $\sqrt{-g}\{e^{-2\Phi}[R_{10} + 4(\partial\Phi)^2] - \frac{1}{4}F_2^2\}$, so the gravitational prefactor relation does not imply equality of the local curvature densities.
- **Question 6:** For the Green–Schwarz M2-brane, the supersymmetric pullback makes the kinetic term strictly invariant, but the Wess–Zumino potential is invariant only up to an exact form. The resulting Noether-current improvement contributes a term proportional to $T_{\mathrm{M2}}\epsilon^{rs}\partial_r X^M\partial_s X^N\Gamma_{MN}\Theta$ to the supercharge. Its Dirac bracket with the fermionic momentum gives the two-form extension $Z^{MN} = T_{\mathrm{M2}}\int_{\Sigma_2} dX^M \wedge dX^N$, while the ordinary term gives P_M; in the book's conventions this is exactly the displayed M-theory algebra. The M5 Wess–Zumino term analogously supplies the five-form charge.
- **Question 7:** On the punctured base $B^\circ = \mathbb{P}^1 \setminus \{z_i\}$, the loops around all 24 seven-branes obey $\gamma_1\cdots\gamma_{24} = 1$, so their ordered monodromies must satisfy $K_{24}\cdots K_1 = \mathbf{1}$. This is the non-Abelian $SL(2,\mathbb{Z})$ form of the integrated axion Bianchi identity and hence of seven-brane tadpole cancellation. Since a D7 has $K_A = T$, 24 D7-branes would give $T^{24} \neq \mathbf{1}$, so some mutually nonlocal (p,q) branes are necessary. The sharper number 18 does not follow from this product equation alone: it counts the 18 independent complex deformations/vector multiplets of elliptic K3, obtained from $9+13-3-1=18$, and therefore bounds the independent brane gauge rank or independently movable mutually local branes. Read literally as a bound on D7 constituents, the statement is too strong; an I_{19} fiber has monodromy T^{19} but only rank-18 algebra A_{18}.
- **Question 8:** At the $SO(8)^4$ orientifold point, the open D7 sector has rank $4\times4=16$, while four closed-string vectors survive from $B_{\mu a}$ and $C_{\mu a}$, $a=1,2$. Their survival follows from the combined projection: $B_{\mu a}$ is odd under both Ω and I_2, while $C_{\mu a}$ is even under Ω and odd under both $(-1)^{F_L}$ and I_2, so both products are even. Hence the full gauge group is $SO(8)^4 \times U(1)^4$ of rank 20. For the heterotic string on T^2, generic Wilson lines break the ten-dimensional rank-16 group to $U(1)^{16}$; reduction of $G_{\mu a}$ and $B_{\mu a}$ supplies two Kaluza–Klein and two winding vectors, giving $U(1)^{20}$. Equivalently, the Narain lattice $\Gamma^{18,2}$ gives $U(1)^{18}_L \times U(1)^2_R$, with extra ADE roots becoming massless only at special moduli.
- **Question 9:** An ab open string has one endpoint on each D-brane, so its mixed boundary conditions force its internal center-of-mass zero mode to lie on the common intersection; it can propagate only along the directions shared by both branes. At an intersection its classical stretching length vanishes, the Ramond ground state is massless, and supersymmetry supplies the corresponding NS boson. For stacks of N_a and N_b branes these states transform in $(\mathbf{N}_a, \overline{\mathbf{N}}_b)$, with multiplicity and chirality fixed by the oriented intersection number. In F-theory this becomes codimension-two fiber enhancement: additional fibral curves collapse only over the brane-intersection locus, producing localized matter states, while G_4 flux determines the net four-dimensional chirality.

- **Question 10:** The inequalities in the question are reversed relative to page 734: both descriptions require $g_s \ll 1$; the flat-space perturbative D-brane regime has $g_s N \ll 1$, whereas classical supergravity requires $g_s N \gg 1$. Since $\kappa_{10}^2 N \tau_p \sim g_s N \alpha'^{(7-p)/2}$, the metric perturbation at the string scale is controlled by $g_s N$. For D3-branes, $L^4 = 4\pi g_s N \alpha'^2$, so $\alpha'/L^2 = (4\pi g_s N)^{-1/2}$ and α' corrections are small precisely when $g_s N \gg 1$. Combining this with $g_s \ll 1$ requires large N and produces a strongly warped but gently curved geometry.
- **Question 11:** The ambient null cone in $\mathbb{R}^{d,2}$ has one redundant scale. On the patch $v \neq 0$, fixing that scale by $v = 1$ gives $u = x^2$ and the embedding $Y^M = (x^\mu, (x^2-1)/2, (x^2+1)/2)$, whose induced metric is exactly the Minkowski metric. Fixing the scale globally with the Euclidean sphere gives $S^1 \times S^{d-1}$; the projective compactification is more precisely $(S^1 \times S^{d-1})/\mathbb{Z}_2$, while the product is its double cover. Ordinary Minkowski space is the open conformal diamond $v \neq 0$, with $v = 0$ becoming null, timelike, and spatial infinity at finite locations on the Einstein cylinder.
- **Question 12:** Finite-temperature $\mathcal{N} = 4$ $SU(N)$ SYM is defined by the thermal density matrix $e^{-\beta H}$; antiperiodic fermions break supersymmetry, but flat-space conformal invariance still implies $\epsilon = 3p$. Homogeneity and isotropy give the perfect-fluid form $\langle T_{\mu\nu} \rangle = \mathrm{diag}(\epsilon, p, p, p)$. At large N and strong 't Hooft coupling, the dual AdS$_5$ black brane has $T = U_0/\pi$ and entropy density $s = \pi^2 N^2 T^3/2$, so $p = \pi^2 N^2 T^4/8$ and $\epsilon = 3\pi^2 N^2 T^4/8$. This is 3/4 of the large-N free-theory energy density.

Chapter 1

Bosonic One-Loop Amplitudes

1.1 Q1: Integrating over the torus modulus

Consider the closed bosonic string one-loop partition function $A_0^{(g=1)}$. We denote the torus moduli as $\tau = \tau_1 + i\tau_2$ and the fundamental region as $\mathcal{F}$. From the path integral formalism we can find the torus partition function up to numerical constants

$$A_0^{(g=1)} \sim \frac{V_{26}}{\alpha'^{13}} \int_{\mathcal{F}} \frac{d^2\tau}{\tau_2^2} \tau_2^{12} \int [\mathcal{D}' X \mathcal{D}' c \mathcal{D}' b] e^{-S[X,\tau]-S[b,c,\tau]}$$

Where $\mathcal{D}'$ means that we integrate only over the non-zero modes. After evaluating these integrals we can obtain the answer

$$A_0^{(g=1)} = \frac{V_{26}}{l_s^{26}} \int_{\mathcal{F}} \frac{d^2\tau}{4\tau_2^2} Z(\tau,\bar{\tau}) \ , \ Z(\tau,\bar{\tau}) = \tau_2^{-12} |\eta(\tau)|^{-48}$$

While in the Hamiltonian formalism this partition function can be written as

$$A_0^{(g=1)} = \int_{\mathcal{F}} \frac{d^2\tau}{4\tau_2^2} \mathrm{Tr}_{\mathcal{H}} (q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24}) \ , \ q = e^{2\pi i \tau}$$

I can understand the factor in the trace as the partition function of the closed bosonic string on a fixed torus, but how to understand the integral over τ? Why we have the factor $\frac{d^2\tau}{4\tau_2^2}$? I can understand that we should integrate over $d^2\tau$, but why it will appear as the modular invariant measure $\frac{d^2\tau}{\tau_2^2}$? I may regard this as for different τ the weight for the corresponding partition function is different and they are weighted by $\frac{1}{\tau_2^2}$, why?

> **Textbook location and related material.** Chapter 6, especially Sections 6.2–6.3 (pp. 126–152): gauge fixing of the genus-one Polyakov integral, Teichmüller modulus τ, modular identifications, and the torus partition function.

1.1.1 Answer

1.1.1.1 Core point

The trace

$$\mathrm{Tr}_{\mathcal{H}} \left(q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24} \right)$$

is the CFT partition function on one fixed complex torus of modulus τ. In string perturbation theory, however, the worldsheet metric is dynamical. After fixing diffeomorphisms and Weyl transformations, most metric fluctuations are gauge, but the complex structure modulus τ is not gauge. Therefore the one-loop vacuum amplitude must still integrate over inequivalent complex structures of the torus.

The quotient by large diffeomorphisms identifies

$$\tau \sim \frac{a\tau + b}{c\tau + d}, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z}),$$

so the integral is over the modular fundamental region $\mathcal{F}$, not over the whole upper half-plane.

The factor

$$\frac{d^2\tau}{\tau_2^2}$$

is not an extra physical Boltzmann weight saying that some tori are preferred. It is the natural volume form on the moduli space of flat tori, induced by the metric on the space of worldsheet metrics after quotienting by diffeomorphisms and Weyl transformations. Equivalently, it is the Faddeev-Popov measure left over after conformal gauge fixing. The numerical factor 1/4 in your convention is an overall normalization convention coming from the normalization of zero modes, residual conformal Killing symmetries, and the chosen definition of $d^2\tau$. The important universal dependence is the modular invariant Poincare measure $d^2\tau/\tau_2^2$.

1.1.1.2 Derivation

Start from the Polyakov path integral for the genus-one vacuum amplitude,

$$A_0^{(g=1)} \sim \int \frac{\mathcal{D}g\,\mathcal{D}X}{\mathrm{Vol}(\mathrm{Diff}\times\mathrm{Weyl})} e^{-S_P[X,g]}.$$

For the critical bosonic string, the matter plus ghost central charge vanishes,

$$c_X + c_{bc} = 26 - 26 = 0,$$

so Weyl transformations are genuine gauge transformations at the quantum level. Thus one can decompose a genus-one metric into

$$g_{ab} = e^{2\omega} f^* \hat{g}_{ab}(\tau),$$

where f is a diffeomorphism, ω is a Weyl factor, and $\hat{g}_{ab}(\tau)$ is a representative flat metric with complex structure τ.

The crucial point is that τ is not removable by small diffeomorphisms or Weyl transformations. It labels a genuine modulus. For a torus, there is one complex modulus, hence two real moduli τ_1, τ_2. Therefore after gauge fixing we schematically get

$$\frac{\mathcal{D}g}{\mathrm{Vol}(\mathrm{Diff}\times\mathrm{Weyl})} \longrightarrow d\mu_{\mathcal{M}_1}(\tau),$$

where $d\mu_{\mathcal{M}_1}$ is the induced measure on the moduli space of complex tori.

Now let us derive the τ_2^{-2} dependence.

Use coordinates $\sigma^1, \sigma^2 \in [0,1]$ on the torus and choose the unit-area flat metric

$$ds^2 = \hat{g}_{ab}(\tau) d\sigma^a d\sigma^b = \frac{1}{\tau_2} \left| d\sigma^1 + \tau d\sigma^2 \right|^2.$$

In components,

$$\hat{g}_{ab}(\tau) = \frac{1}{\tau_2} \begin{pmatrix} 1 & \tau_1 \\ \tau_1 & |\tau|^2 \end{pmatrix}.$$

This metric has determinant

$$\det \hat{g} = \frac{1}{\tau_2^2} \left(|\tau|^2 - \tau_1^2 \right) = \frac{1}{\tau_2^2} \tau_2^2 = 1,$$

so the coordinate area is normalized to one. The overall area is not a physical modulus in the critical string because it can be removed by a Weyl transformation. The remaining data is the shape of the torus, namely τ.

The space of metrics has a natural ultralocal metric, often called the DeWitt metric. For variations δg_{ab}, take for definiteness

$$\|\delta g\|^2 = \int d^2\sigma \sqrt{g}\, g^{ac} g^{bd} \delta g_{ab} \delta g_{cd}.$$

Different authors insert harmless numerical factors such as 1/2 in this definition; these change only the overall normalization, not the τ_2 dependence.

Restrict this metric on the full space of metrics to the slice of flat unit-area representatives $\hat{g}_{ab}(\tau)$. A variation of the modulus gives

$$\delta\hat{g}_{ab} = \frac{\partial \hat{g}_{ab}}{\partial \tau_1}\delta\tau_1 + \frac{\partial \hat{g}_{ab}}{\partial \tau_2}\delta\tau_2.$$

Because the upper half-plane metric we obtain will be $SL(2,\mathbb{R})$ invariant, it is enough to evaluate the local form at $\tau_1 = 0$, and then rewrite the answer covariantly. At $\tau_1 = 0$, the metric is

$$\hat{g}_{ab} = \begin{pmatrix} 1/\tau_2 & 0 \\ 0 & \tau_2 \end{pmatrix}, \qquad \hat{g}^{ab} = \begin{pmatrix} \tau_2 & 0 \\ 0 & 1/\tau_2 \end{pmatrix}.$$

The two independent variations are

$$\frac{\partial \hat{g}}{\partial \tau_1} = \begin{pmatrix} 0 & 1/\tau_2 \\ 1/\tau_2 & 0 \end{pmatrix},$$

and

$$\frac{\partial \hat{g}}{\partial \tau_2} = \begin{pmatrix} -1/\tau_2^2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Now compute their norms using

$$G_{ij} = \int d^2\sigma \sqrt{\hat{g}}\, \hat{g}^{ac}\hat{g}^{bd} \frac{\partial \hat{g}_{ab}}{\partial m^i}\frac{\partial \hat{g}_{cd}}{\partial m^j}, \qquad m^i = (\tau_1, \tau_2).$$

Since $\sqrt{\hat{g}} = 1$ and the coordinate volume is one, the integral only contributes a factor 1. For the τ_1 variation,

$$G_{\tau_1\tau_1} = 2\hat{g}^{11}\hat{g}^{22}\left(\frac{1}{\tau_2}\right)^2 = 2\,(\tau_2)\left(\frac{1}{\tau_2}\right)\frac{1}{\tau_2^2} = \frac{2}{\tau_2^2}.$$

For the τ_2 variation,

$$G_{\tau_2\tau_2} = (\hat{g}^{11})^2\left(-\frac{1}{\tau_2^2}\right)^2 + (\hat{g}^{22})^2(1)^2 = \tau_2^2\frac{1}{\tau_2^4} + \frac{1}{\tau_2^2} = \frac{2}{\tau_2^2}.$$

The mixed term vanishes:

$$G_{\tau_1\tau_2} = 0.$$

Therefore the induced line element on the torus moduli is

$$ds^2_{\text{moduli}} = \frac{2}{\tau_2^2}\left(d\tau_1^2 + d\tau_2^2\right) = \frac{2\, d\tau d\bar{\tau}}{\tau_2^2}.$$

Up to the conventional factor 2, this is the standard Poincare metric on the upper half-plane. Its volume element is

$$d\mu_{\mathcal{T}_1} = \sqrt{\det G}\, d\tau_1 d\tau_2 = \frac{2\, d\tau_1 d\tau_2}{\tau_2^2} \propto \frac{d^2\tau}{\tau_2^2}.$$

Thus the factor τ_2^{-2} is simply the Jacobian obtained by using τ_1, τ_2 as coordinates on the space of inequivalent flat metrics.

This is the geometric reason. The BRST/Faddeev-Popov derivation gives the same result in a slightly different language. In conformal gauge, metric variations split as

$$\delta g_{ab} = 2\delta\omega\, g_{ab} + \nabla_a v_b + \nabla_b v_a + \delta\tau\, \frac{\partial \hat{g}_{ab}}{\partial \tau} + \delta\bar{\tau}\, \frac{\partial \hat{g}_{ab}}{\partial \bar{\tau}}.$$

The first term is Weyl, the second is diffeomorphism, and the final two terms are true moduli. Gauge fixing produces the bc ghost determinant. Since the torus has one complex modulus, the ghost path integral contains one holomorphic and one antiholomorphic antighost insertion schematically of the form

$$(b,\mu_\tau)(\bar{b},\bar{\mu}_{\bar{\tau}}),$$

where μ_τ is the Beltrami differential representing the change of complex structure. Thus the genus-one amplitude has the schematic form

$$A_0^{(g=1)} \sim \int_{\mathcal{M}_1} d^2\tau \, \left\langle (b,\mu_\tau)(\bar{b},\bar{\mu}_{\bar{\tau}}) \right\rangle_\tau Z_X(\tau,\bar{\tau}).$$

When one evaluates the normalization of the modulus zero modes and ghost zero modes with the standard torus metric, this ghost/moduli measure gives precisely the same Poincare density

$$d\mu_{\mathcal{M}_1} \propto \frac{d^2\tau}{\tau_2^2}.$$

Now let us connect this to the Hamiltonian trace. The fixed-torus partition function can be written as

$$Z(\tau,\bar{\tau}) = \mathrm{Tr}_{\mathcal{H}}\left(q^{L_0-c/24}\bar{q}^{\bar{L}_0-\bar{c}/24}\right), \qquad q = e^{2\pi i\tau}.$$

Equivalently,

$$q^{L_0-c/24}\bar{q}^{\bar{L}_0-\bar{c}/24} = e^{-2\pi\tau_2\left(L_0+\bar{L}_0-\frac{c+\bar{c}}{24}\right)}e^{2\pi i\tau_1\left(L_0-\bar{L}_0-\frac{c-\bar{c}}{24}\right)}.$$

For the bosonic string, $c=\bar{c}$, so

$$q^{L_0-c/24}\bar{q}^{\bar{L}_0-\bar{c}/24} = e^{-2\pi\tau_2\left(L_0+\bar{L}_0-\frac{c}{12}\right)}e^{2\pi i\tau_1\left(L_0-\bar{L}_0\right)}.$$

This makes the roles of τ_2 and τ_1 transparent:

- τ_2 is like a Euclidean proper time for propagation around one cycle of the torus.
- τ_1 is a twist inserted before gluing the closed string back to itself.

If one integrates τ_1 over a full unit interval in the strip representation, the factor

$$e^{2\pi i\tau_1(L_0-\bar{L}_0)}$$

projects onto level-matched states:

$$L_0 - \bar{L}_0 = 0.$$

This is the Hamiltonian version of imposing invariance under translations around the closed string. However, the full string amplitude is not merely a trace at one value of τ and not merely an integral over a strip. A single torus can be cut into a spatial circle and Euclidean time circle in many equivalent ways. These different choices are related by large diffeomorphisms,

$$\tau \mapsto \frac{a\tau+b}{c\tau+d}.$$

Therefore we integrate once over the quotient

$$\mathcal{M}_1 = \mathbb{H}/SL(2,\mathbb{Z}),$$

which is represented by the fundamental region $\mathcal{F}$.

The modular invariance of the measure follows directly. Under

$$\tau' = \frac{a\tau+b}{c\tau+d},$$

one has

$$d^2\tau' = \frac{d^2\tau}{|c\tau+d|^4}, \qquad \tau_2' = \frac{\tau_2}{|c\tau+d|^2}.$$

Therefore

$$\frac{d^2\tau'}{(\tau_2')^2} = \frac{d^2\tau/|c\tau+d|^4}{\tau_2^2/|c\tau+d|^4} = \frac{d^2\tau}{\tau_2^2}.$$

So the measure is invariant under large diffeomorphisms. This is exactly what we need, because τ and τ' describe the same worldsheet geometry.

The factor should therefore not be interpreted as follows:

"The string statistically prefers small or large τ_2 with weight $1/\tau_2^2$."

Rather, it is like the $r\,dr\,d\theta$ factor in polar coordinates. The factor r does not mean that points at larger radius are physically preferred; it is the coordinate volume element. Here $1/\tau_2^2$ is the coordinate volume element on the moduli space of torus complex structures.

Putting this together, the one-loop vacuum amplitude has the form

$$A_0^{(g=1)} = \frac{V_{26}}{l_s^{26}} \int_{\mathcal{F}} \frac{d^2\tau}{4\tau_2^2} Z(\tau,\bar{\tau}),$$

with

$$Z(\tau,\bar{\tau}) = \tau_2^{-12}|\eta(\tau)|^{-48}.$$

Here $Z(\tau,\bar{\tau})$ is the fixed-complex-structure CFT partition function after the nonzero oscillator modes and zero-mode Gaussian factors have been evaluated, while

$$\frac{d^2\tau}{4\tau_2^2}$$

is the gauge-fixed measure over inequivalent torus complex structures. The factor $1/4$ is convention-dependent, while the τ_2^{-2} factor is fixed by the geometry of moduli space and by modular invariance.

1.1.1.3 Clarifications

First, the integral over τ is not an ordinary thermal average over different temperatures. In the trace expression, τ_2 does look like an inverse temperature or Schwinger proper time, and τ_1 looks like a twist. But in string theory these are worldsheet geometric moduli, not external thermodynamic parameters. The string path integral sums over worldsheets, so it must integrate over their inequivalent complex structures.

Second, $d^2\tau/\tau_2^2$ is not chosen merely because it is modular invariant, although modular invariance is a very efficient way to recognize it. More fundamentally, it is the measure induced by the metric on the space of worldsheet metrics after quotienting by gauge transformations. Modular invariance then follows because large diffeomorphisms are gauge redundancies.

Third, the fundamental domain $\mathcal{F}$ is not an arbitrary restriction. If we integrated over all $\mathbb{H}$, we would count the same torus infinitely many times, because τ, $\tau+1$, $-1/\tau$, and more generally $(a\tau+b)/(c\tau+d)$ represent the same torus with different choices of homology basis.

Fourth, the numerical coefficient $1/4$ should not be over-interpreted. Different books distribute constants differently between the normalization of the path integral, the definition of l_s, the ghost zero-mode measure, and the volume of residual conformal Killing symmetries. What is structurally important is

$$d\mu_{\mathcal{M}_1} \propto \frac{d^2\tau}{\tau_2^2}.$$

1.1.1.4 Result

The fixed-torus Hamiltonian trace is

$$Z(\tau,\bar{\tau}) = \mathrm{Tr}_{\mathcal{H}}\left(q^{L_0-c/24}\bar{q}^{\bar{L}_0-\bar{c}/24}\right).$$

The string one-loop vacuum amplitude integrates this fixed-torus partition function over inequivalent complex structures:

$$A_0^{(g=1)} = \frac{V_{26}}{l_s^{26}} \int_{\mathcal{F}} \frac{d^2\tau}{4\tau_2^2} Z(\tau,\bar{\tau}).$$

The quotient region is

$$\mathcal{M}_1 = \mathbb{H}/SL(2,\mathbb{Z}),$$

because large diffeomorphisms identify

$$\tau \sim \frac{a\tau+b}{c\tau+d}.$$

The measure comes from the induced metric on the moduli space of flat tori:

$$ds^2_{\text{moduli}} \propto \frac{d\tau d\bar{\tau}}{\tau_2^2}, \qquad d\mu_{\mathcal{M}_1} \propto \frac{d^2\tau}{\tau_2^2}.$$

Equivalently, it is the Faddeev-Popov measure left over after conformal gauge fixing. Thus the τ_2^{-2} factor is a moduli-space volume element, not an additional physical weighting assigned by hand to different fixed-torus partition functions.

1.2 Q2: The annulus modulus measure

Now consider open bosonic string one-loop partition function. In the cylinder case there is one real modular t with $0 \le t < \infty$; the inequivalent cylinders are described by $\{(\tau,\sigma) : 0 \le \sigma \le \pi, 0 \le \tau \le 2\pi t\}$. The cylinder one-loop partition function can be written as

$$A_0^{(g=1,b=2)} = \int_0^\infty \frac{dt}{2t} \mathrm{Tr}_{\mathcal{H}}(q^{L_0-c/24}) \ , \ q = e^{-2\pi t}$$

Derive the factor $\frac{dt}{2t}$.

> **Textbook location and related material.** Chapter 6, Section 6.5 (pp. 157–163): the cylinder/annulus modulus, its open-string trace, and the open–closed channel transformation. Section 6.1 supplies the general vacuum-amplitude normalization.

1.2.1 Answer

1.2.1.1 Core point

For the annulus/cylinder, the trace

$$\mathrm{Tr}_{\mathcal{H}}\left(q^{L_0-c/24}\right), \qquad q = e^{-2\pi t},$$

is the open-string CFT partition function on a fixed cylinder of modulus t. The parameter t measures the shape of the cylinder: roughly, the ratio between its periodic Euclidean-time direction and its open-string spatial width.

After conformal gauge fixing, there is one genuine real modulus t. The natural metric on this one-dimensional moduli space is logarithmic:

$$ds^2_{\text{moduli}} \propto \frac{dt^2}{t^2}.$$

Therefore the corresponding volume element is

$$d\mu_{\text{annulus}} \propto \frac{dt}{t}.$$

The conventional open-string one-loop vacuum amplitude has

$$\frac{dt}{2t}$$

rather than just dt/t because the vacuum cylinder also has a residual conformal Killing symmetry: translations around the periodic direction. In the standard Faddeev-Popov normalization, dividing by this residual symmetry gives the factor $1/(2t)$ multiplying dt. Equivalently, in the Hamiltonian sewing picture, the trace describes a cylinder with a marked cut; dividing by the freedom to move this cut around the loop gives the $1/t$ factor, with the remaining $1/2$ being the standard vacuum-diagram/automorphism normalization.

So the factor should be thought of as

$$\frac{dt}{2t} = \text{gauge-fixed measure over inequivalent annuli,}$$

not as an extra physical probability weight assigned by hand to different cylinders.

1.2.1.2 Derivation

Let us describe the annulus as a Euclidean open-string worldsheet with coordinates

$$0 \leq \sigma \leq \pi, \qquad \tau \sim \tau + 2\pi t.$$

The boundaries are at

$$\sigma = 0, \qquad \sigma = \pi.$$

The coordinate τ is periodic, so the annulus can be viewed as an open string propagating in Euclidean worldsheet time and then being traced over:

$$Z_{\text{open}}(t) = \mathrm{Tr}_{\mathcal{H}}\left(e^{-2\pi t(L_0 - c/24)}\right) = \mathrm{Tr}_{\mathcal{H}}\left(q^{L_0 - c/24}\right), \qquad q = e^{-2\pi t}.$$

This is the fixed-modulus partition function. The remaining question is the measure over t.

As in the torus case, the starting point is the Polyakov path integral

$$A_0^{(g=1,b=2)} \sim \int \frac{\mathcal{D}g\,\mathcal{D}X}{\mathrm{Vol}(\mathrm{Diff} \times \mathrm{Weyl})} e^{-S_P[X,g]}.$$

Here the topology is genus zero with two boundaries, equivalently an annulus. After fixing conformal gauge, the metric can be written schematically as

$$g_{ab} = e^{2\omega} f^* \hat{g}_{ab}(t),$$

where f is a diffeomorphism, ω is a Weyl factor, and $\hat{g}_{ab}(t)$ is a representative metric for the annulus with modulus t.

The annulus has one real modulus. A convenient fixed-area representative is obtained by using coordinates

$$0 \leq \sigma \leq \pi, \qquad 0 \leq \theta < 2\pi,$$

and writing

$$ds^2 = \hat{g}_{ab}(t) dx^a dx^b = \frac{1}{t} d\sigma^2 + t\, d\theta^2.$$

This is conformally equivalent to the flat cylinder

$$ds^2 = d\sigma^2 + d\tau^2, \qquad \tau = t\theta, \qquad 0 \leq \tau < 2\pi t.$$

The fixed-area form is useful because the Weyl factor has already been separated off. The determinant is

$$\det \hat{g} = \frac{1}{t} \cdot t = 1,$$

so the area is independent of t:

$$\int_0^\pi d\sigma \int_0^{2\pi} d\theta\, \sqrt{\hat{g}} = 2\pi^2.$$

Thus t is not the area; it is the shape modulus. More explicitly, the circumference of the periodic direction is proportional to $\sqrt{t}$, while the distance between the two boundaries is proportional to $1/\sqrt{t}$, so their ratio is proportional to t.

Now restrict the natural metric on the space of worldsheet metrics to this one-parameter family. Use the same ultralocal metric as in the torus discussion:

$$\|\delta g\|^2 = \int d^2x\,\sqrt{g}\,g^{ac}g^{bd}\delta g_{ab}\delta g_{cd}.$$

For

$$\hat{g}_{\sigma\sigma} = \frac{1}{t}, \qquad \hat{g}_{\theta\theta} = t,$$

we have

$$\hat{g}^{\sigma\sigma} = t, \qquad \hat{g}^{\theta\theta} = \frac{1}{t}.$$

The variation with respect to t is

$$\delta\hat{g}_{\sigma\sigma} = -\frac{\delta t}{t^2}, \qquad \delta\hat{g}_{\theta\theta} = \delta t.$$

Therefore

$$\hat{g}^{ac}\hat{g}^{bd}\delta\hat{g}_{ab}\delta\hat{g}_{cd} = (\hat{g}^{\sigma\sigma})^2(\delta\hat{g}_{\sigma\sigma})^2 + (\hat{g}^{\theta\theta})^2(\delta\hat{g}_{\theta\theta})^2.$$

Substituting the explicit expressions gives

$$(\hat{g}^{\sigma\sigma})^2(\delta\hat{g}_{\sigma\sigma})^2 = t^2\left(\frac{\delta t}{t^2}\right)^2 = \frac{(\delta t)^2}{t^2},$$

and

$$(\hat{g}^{\theta\theta})^2(\delta\hat{g}_{\theta\theta})^2 = \frac{1}{t^2}(\delta t)^2 = \frac{(\delta t)^2}{t^2}.$$

Thus

$$\|\delta\hat{g}\|^2 = \int_0^{\pi} d\sigma \int_0^{2\pi} d\theta\,\left[\frac{(\delta t)^2}{t^2} + \frac{(\delta t)^2}{t^2}\right].$$

Since $\sqrt{\hat{g}} = 1$, this becomes

$$\|\delta\hat{g}\|^2 = 2(2\pi^2)\frac{(\delta t)^2}{t^2}.$$

The overall constant depends on the normalization chosen for the metric on the space of metrics, but the t-dependence is fixed:

$$ds^2_{\text{moduli}} \propto \frac{dt^2}{t^2}.$$

Since the moduli space is one-dimensional, the induced volume element is

$$d\mu_{\text{Teich}} \propto \sqrt{G_{tt}}\,dt \propto \frac{dt}{t}.$$

This is the annulus analogue of the torus result

$$d\mu_{\mathcal{M}_1} \propto \frac{d^2\tau}{\tau_2^2}.$$

For the torus the modulus is complex, so the Poincare metric gives a two-dimensional volume form. For the annulus the modulus is real, so the logarithmic metric gives the one-dimensional volume form dt/t.

Now let us explain the extra factor of 1/2 in

$$\frac{dt}{2t}.$$

The annulus has a residual conformal Killing symmetry generated by translations around the periodic direction:

$$\theta \mapsto \theta + \epsilon.$$

Equivalently, in the original cylinder coordinate,

$$\tau \mapsto \tau + \epsilon.$$

This symmetry remains after choosing the conformal representative of the annulus. In a vacuum amplitude there are no vertex operators fixing a point on the worldsheet, so this residual symmetry must still be divided out.

In the Hamiltonian trace picture, this has a very concrete meaning. The expression

$$\mathrm{Tr}_{\mathcal{H}}\left(e^{-2\pi t(L_0-c/24)}\right)$$

is naturally obtained by cutting the annulus open at some value of τ, evolving the open-string state for Euclidean time $2\pi t$, and gluing the final state back to the initial state. But the uncut annulus does not contain a distinguished cut. Moving the cut around the periodic direction gives the same geometric annulus. Therefore the trace computation overcounts by the volume of translations around the loop.

Schematically, if

$$s = 2\pi t$$

is the open-string proper time around the loop, then the Schwinger representation gives an integration over s, while the unmarked cylinder requires division by the translation volume proportional to s:

$$\frac{ds}{s} = \frac{d(2\pi t)}{2\pi t} = \frac{dt}{t}.$$

With the standard normalization of the annulus vacuum diagram, or equivalently the standard normalization of the residual conformal Killing group in the Faddeev-Popov procedure, this becomes

$$\frac{dt}{2t}.$$

One often phrases this as

$$d\mu_{\text{annulus}} = \frac{dt}{2t}.$$

The factor $1/2$ is a conventional but standard normalization associated with the automorphism/symmetry factor of the vacuum cylinder. The important modulus dependence is the logarithmic measure dt/t.

Putting the measure and fixed-cylinder partition function together gives

$$A_0^{(g=1,b=2)} = \int_0^\infty \frac{dt}{2t}\mathrm{Tr}_{\mathcal{H}}\left(q^{L_0-c/24}\right), \qquad q = e^{-2\pi t}.$$

There is a useful parallel with ordinary one-loop field theory. A one-loop vacuum amplitude written in Schwinger proper time has the schematic form

$$\int_0^\infty \frac{dT}{T}\,\mathrm{Tr}\left(e^{-TH}\right).$$

The factor dT/T appears because a closed loop has no preferred starting point: translating the starting point around the loop is a redundancy. The open-string annulus is the stringy version of this statement, with

$$T = 2\pi t, \qquad H = L_0 - \frac{c}{24}.$$

1.2.1.3 Clarifications

First, t is not an external inverse temperature chosen by hand. It plays a role similar to Euclidean proper time in the Hamiltonian trace, but in the string path integral it is a modulus of the worldsheet geometry. We integrate over it because the worldsheet metric is dynamical.

Second, the factor $1/t$ is not a physical preference for smaller values of t. It is the coordinate volume element on the moduli space of inequivalent annuli. A better coordinate is

$$u = \log t.$$

Then

$$\frac{dt}{t} = du.$$

So the measure is simply flat in the logarithmic modulus u.

Third, unlike the torus, the annulus modulus is real rather than complex. Therefore the measure is not

$$\frac{d^2\tau}{\tau_2^2},$$

but its one-dimensional analogue

$$\frac{dt}{t}.$$

Fourth, the open-channel and closed-channel descriptions are related by a change of variables, often of the form

$$t \sim \frac{1}{\ell}.$$

This does not mean that t and $1/t$ are being identified inside the same integration domain in the same way that τ and $-1/\tau$ are identified for the torus. Rather, it is a dual description of the same annulus amplitude as either an open-string loop or a closed-string tree-level exchange between two boundaries.

1.2.1.4 Result

Choose a fixed-area representative metric

$$ds^2 = \frac{1}{t}d\sigma^2 + t\, d\theta^2, \qquad 0 \le \sigma \le \pi, \qquad \theta \sim \theta + 2\pi.$$

Then

$$\delta g_{\sigma\sigma} = -\frac{\delta t}{t^2}, \qquad \delta g_{\theta\theta} = \delta t,$$

and the induced metric on the modulus is

$$ds^2_{\text{moduli}} \propto \frac{dt^2}{t^2}.$$

Therefore

$$d\mu_{\text{Teich}} \propto \frac{dt}{t}.$$

Dividing by the residual translation symmetry of the vacuum cylinder, with the standard annulus normalization, gives

$$d\mu_{\text{annulus}} = \frac{dt}{2t}.$$

Thus

$$A_0^{(g=1,b=2)} = \int_0^\infty \frac{dt}{2t} \mathrm{Tr}_{\mathcal{H}}\left(q^{L_0 - c/24}\right), \qquad q = e^{-2\pi t}.$$

1.3 Q3: Massless exchange between parallel D-branes

Consider two parallel Dp-branes in a d-dimensional space-time. In order to compare the string computation with a field theory computation, one needs to study the exchange of massless modes between the two branes. D-branes are dynamical objects which are expected to have a mass-density (tension) and they therefore couple to gravity and possibly other massless excitations of the closed string, such as the dilaton and the anti-symmetric tensor. It turns out that the latter does not contribute to lowest order because there is no field-theoretical one-particle exchange involving this field (I do not fully understand this).

To derive the one-particle exchange amplitude between two Dp-branes, for example the graviton, we consider the following effective space-time action (neglected fields that do not contribute)

$$S = S_{\text{brane}} + S_{\text{bulk}}$$

Where

$$S_{\text{bulk}} = \frac{1}{2\kappa_d^2}\int d^dx\sqrt{-g}\left[R - \frac{4}{d-2}\partial_\mu\phi\partial^\mu\phi\right] \ , \ \Phi = \Phi_0 + \phi$$

$$S_{\text{brane}} = -\tau_p\int d^{p+1}\xi \exp\left(\frac{4-d+2p}{d-2}\phi\right)\sqrt{-\Gamma}$$

We choose the normalization and conventions

$$\phi = \frac{1}{2}\kappa_d\sqrt{d-2}D \ , \ g_{\mu\nu} = \eta_{\mu\nu} + 2\kappa_d h_{\mu\nu}$$

And the static gauge

$$X^\alpha = \xi^\alpha \ , \ X^i = \text{constant} \ , \ \alpha = 0, \cdots, p \ , \ i = p+1, \cdots, d-1$$

We ignored the fluctuation of the Dp-brane and the gauge field on the Dp-brane since they do not contribute. And we ignore the B-field for now.

Then to the first order

$$S_{\text{brane}} = \tau_p\kappa_d\int d^{p+1}\xi\left(D^{\mu\nu}h_{\mu\nu} - \left(\frac{4-d+2p}{2\sqrt{d-2}}\right)D\right) \ , \ D^{\mu\nu} = \begin{pmatrix}\eta^{\alpha\beta} & 0\\ 0 & 0\end{pmatrix}$$

Textbook location and related material. Chapter 6, Section 6.8 (pp. 171–174): comparison of the cylinder with massless closed-string exchange and extraction of the D-brane tension. Related spacetime couplings appear in Section 16.5 (pp. 630–636), especially the DBI action.

1.3.1 Answer

1.3.1.1 Core point

For a static, flat Dp-brane with vanishing world-volume gauge field, there is no source term linear in the NS–NS two-form B_{muv}. A one-particle exchange diagram needs one linear source on each brane, so the B-field exchange vanishes at this order. The graviton and dilaton do have linear sources and therefore contribute.

1.3.1.2 Why the B-field has no linear source

The DBI action is

$$S_{\text{DBI}} = -\tau_p\int d^{p+1}\xi\, e^{-\Phi}\sqrt{-\det\big(P[G+B]_{\alpha\beta} + 2\pi\alpha' F_{\alpha\beta}\big)}.$$

Set $F=0$ and expand around a flat induced metric, writing the fluctuation matrix as $M_{\alpha\beta} = h_{\alpha\beta} + B_{\alpha\beta}$. Since

$$\sqrt{\det(1+M)} = 1 + \frac{1}{2}\operatorname{tr} M + O(M^2),$$

the linear term contains $\eta^{\alpha\beta}h_{\alpha\beta}$ but

$$\eta^{\alpha\beta}B_{\alpha\beta} = 0$$

because the inverse metric is symmetric and $B_{\alpha\beta}$ is antisymmetric. Thus $J_B^{\mu\nu} = 0$ for the background used in the comparison. A nonzero background flux F or a more general brane configuration can generate B couplings through the gauge-invariant combination $B + 2\pi\alpha' F$.

1.3.1.3 Graviton and dilaton exchange

With the canonically normalized fields in the question, the source terms are

$$S_{\text{source}} = \kappa_d \tau_p \int d^{p+1}\xi \left(D^{\mu\nu} h_{\mu\nu} - a_p D\right), \qquad a_p = \frac{4-d+2p}{2\sqrt{d-2}}.$$

For two static parallel branes separated by r in the $d-p-1$ transverse directions, the exchange amplitude is obtained by contracting the two sources with the massless propagators and integrating only over transverse momentum:

$$\mathcal{A}_{\text{NSNS}}(r) \propto \kappa_d^2 \tau_p^2 V_{p+1} \int \frac{d^{d-p-1}k}{(2\pi)^{d-p-1}} \frac{e^{ik\cdot r}}{k^2} \left(\mathcal{C}_h + a_p^2\right).$$

Here $\mathcal{C}_h$ is the contraction of $D^{\mu\nu}$ with the de Donder-gauge graviton projector. Both contributions are attractive for equal positive-tension branes. In a supersymmetric theory, R–R form exchange supplies an equal repulsive term, yielding the no-force condition for parallel BPS branes.

1.3.1.4 Result

At leading order about the static $F = 0$ background,

$$J_h \neq 0, \qquad J_D \neq 0, \qquad J_B = 0.$$

The absence of B exchange is therefore not a statement that D-branes never interact with B; it is the statement that this particular background has no one-B linear vertex.

Chapter 2

Torus Partition Functions for RCFTs

2.1 Q1: Why $S^2 = C$ in RCFT

In RCFT, the characters transform as

$$\chi_i\left(-\frac{1}{\tau}\right) = \sum_j S_{ij}\chi_j(\tau).$$

It is said that

$$S^2 = C,$$

Where C is the charge conjugation matrix.

How can we derive $S^2 = C$? What is C in the context of RCFT?

> **Textbook location and related material.** Chapter 6, Section 6.4 (pp. 152–157): RCFT characters, modular S and T matrices, modular invariants, and charge conjugation. The geometric action of S^2 is the twice-applied torus cycle exchange.

2.1.1 Answer

2.1.1.1 Core point

In an RCFT, the modular matrix S is not just a matrix describing the scalar map $\tau \mapsto -1/\tau$. More precisely, it is the representation of the mapping class group of the torus on the finite-dimensional vector space spanned by the chiral characters, or equivalently by genus-one conformal blocks. The modular transformation

$$s = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

satisfies

$$s^2 = -I$$

inside $SL(2,\mathbb{Z})$. Although $-I$ acts trivially on the complex modulus τ, it does not act trivially on the labelled chiral block: geometrically it reverses the orientation of the torus cycles, so a primary label i is replaced by its conjugate, or dual, label i^*. Therefore the representation of s^2 is the charge conjugation matrix:

$$S^2 = C, \qquad C_{ij} = \delta_{i,j^*}.$$

Here C is the matrix which sends each primary label, or equivalently each basis vector in the character space, to its contragredient label:

$$Ce_i = e_{i^*}.$$

If every primary is self-conjugate, then $i^* = i$ for all i, and $C = I$. In that special case $S^2 = I$. But in a general RCFT, C can be a nontrivial permutation matrix.

2.1.1.2 Derivation

Let the RCFT have finitely many irreducible chiral representations, or primary labels,

$$i, j, k \in \mathcal{I}.$$

For each primary label i, let $\mathcal{H}_i$ be the corresponding irreducible module of the chiral algebra. Its character is

$$\chi_i(\tau) = \mathrm{Tr}_{\mathcal{H}_i}\, q^{L_0 - c/24}, \qquad q = e^{2\pi i \tau}.$$

The defining RCFT property is that the span of these finitely many characters is closed under modular transformations. In particular, under

$$s : \tau \mapsto -\frac{1}{\tau},$$

we have

$$\chi_i\left(-\frac{1}{\tau}\right) = \sum_j S_{ij}\chi_j(\tau).$$

Equivalently, if we package the characters into a column vector

$$\chi(\tau) = \begin{pmatrix} \chi_0(\tau) \\ \chi_1(\tau) \\ \vdots \end{pmatrix},$$

then, up to the chosen row-versus-column convention, the modular transformation is represented by a finite matrix S.

The key point is that S represents the mapping class group action, not merely the induced action on the upper half-plane coordinate τ. The torus mapping class group is

$$SL(2, \mathbb{Z}),$$

with standard generators

$$s = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad t = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The generator s acts on the modulus by

$$\tau \mapsto -\frac{1}{\tau},$$

while t acts by

$$\tau \mapsto \tau + 1.$$

Now compute s^2 as a matrix:

$$s^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I.$$

As a fractional linear transformation,

$$-I : \tau \mapsto \frac{-\tau}{-1} = \tau.$$

So if we only look at the numerical point τ in the upper half-plane, s^2 appears to do nothing. This is why one might naively expect $S^2 = I$. But this misses the labelled conformal block data.

To see what is really happening, remember that a character can be viewed geometrically as a genus-one chiral conformal block with a label i. Equivalently, one can think of it as a torus amplitude with an i-labelled chiral sector propagating along one cycle of the torus.

Let a and b denote a basis of one-cycles of the torus. The modular transformation s exchanges the two cycles, up to orientation:

$$s : (a, b) \mapsto (b, -a),$$

with convention-dependent signs. Applying s twice gives

$$s^2 : (a, b) \mapsto (-a, -b).$$

Thus s^2 reverses the orientation of both fundamental cycles.

Now the important RCFT fact is:

> Reversing the orientation of a labelled chiral line replaces the label by its dual, or charge-conjugate, label.

If a primary representation is $\mathcal{H}_i$, its dual or contragredient representation is usually denoted

$$\mathcal{H}_{i^*}.$$

Therefore, under s^2, the block labelled by i is mapped to the block labelled by i^*:

$$s^2 : i \mapsto i^*.$$

Passing to the matrix representation on characters, this says

$$\rho(s)^2 = \rho(s^2) = \rho(-I) = C,$$

where $\rho(s) = S$. Hence

$$S^2 = C.$$

This is the clean geometric derivation. The central element $-I \in SL(2, \mathbb{Z})$ acts trivially on the complex number τ, but nontrivially on the primary label: it sends a representation to its conjugate representation.

There is also a useful algebraic derivation using standard properties of the modular S-matrix. In a unitary RCFT, the S-matrix is unitary and symmetric:

$$S^\dagger S = I, \qquad S_{ij} = S_{ji}.$$

Moreover, charge conjugation of a primary corresponds to complex conjugation of the S-matrix row:

$$S_{i^*j} = S_{ij}^*.$$

This relation is natural because the S-matrix can be interpreted as the invariant associated to a Hopf link whose components are labelled by primaries. Reversing the orientation of a labelled component replaces the label by the dual representation, and in a unitary theory this complex-conjugates the amplitude.

Now compute the matrix square:

$$(S^2)_{ij} = \sum_k S_{ik} S_{kj}.$$

Using symmetry, $S_{kj} = S_{jk}$, so

$$(S^2)_{ij} = \sum_k S_{ik} S_{jk}.$$

Using

$$S_{j^*k} = S_{jk}^*,$$

we can rewrite

$$S_{jk} = S_{j^*k}^*.$$

Therefore

$$(S^2)_{ij} = \sum_k S_{ik} S_{j^*k}^*.$$

But unitarity says that the rows of S are orthonormal:

$$\sum_k S_{ik} S_{\ell k}^* = \delta_{i\ell}.$$

Taking $\ell = j^*$ gives

$$(S^2)_{ij} = \delta_{i,j^*}.$$

Thus

$$(S^2)_{ij} = C_{ij},$$

where

$$C_{ij} = \delta_{i,j^*}.$$

So again,

$$S^2 = C.$$

This algebraic derivation is often the most efficient way to check the statement once the standard RCFT properties of S are known. The geometric derivation explains why the answer is charge conjugation rather than the identity.

Let us also connect this to the torus partition function. A full nonchiral CFT has left-moving and right-moving sectors. A common modular invariant, often called the charge-conjugation modular invariant, is

$$Z_C(\tau, \bar{\tau}) = \sum_{i,j} C_{ij} \chi_i(\tau) \overline{\chi_j(\tau)} = \sum_i \chi_i(\tau) \overline{\chi_{i^*}(\tau)}.$$

This pairs the left-moving representation i with the right-moving representation i^*. In many familiar diagonal theories the characters of i and i^* are identical as q-series, so this is often written schematically as

$$Z(\tau, \bar{\tau}) = \sum_i |\chi_i(\tau)|^2.$$

But conceptually the pairing is really between a representation and its conjugate. This is why the same matrix C appears both in the modular relation

$$S^2 = C$$

and in the canonical charge-conjugation modular invariant.

2.1.1.3 Explanation of notation

- $\mathcal{I}$ is the finite set of primary labels of the RCFT.
- $\mathcal{H}_i$ is the irreducible chiral module associated with the primary label i.
- $\chi_i(\tau)$ is the character of $\mathcal{H}_i$:

$$\chi_i(\tau) = \mathrm{Tr}_{\mathcal{H}_i} q^{L_0 - c/24}.$$

- S is the modular matrix representing

$$\tau \mapsto -\frac{1}{\tau}$$

on the finite-dimensional space spanned by the RCFT characters.

- i^* is the conjugate, dual, or contragredient primary label. It is defined by

$$\mathcal{H}_{i^*} \cong \mathcal{H}_i^\vee.$$

Physically, i^* is the representation carried by the oppositely oriented chiral line. In WZW models, it is the conjugate finite-dimensional Lie algebra representation. For example, in an $SU(3)$ WZW model, the fundamental representation $\mathbf{3}$ and antifundamental representation $\overline{\mathbf{3}}$ are conjugate.

- C is the charge conjugation matrix:

$$C_{ij} = \delta_{i,j^*}.$$

Equivalently,

$$Ce_j = e_{j^*},$$

where e_j is the basis vector corresponding to the primary label j. Since conjugation is an involution,

$$(i^*)^* = i,$$

we have

$$C^2 = I.$$

Also, because C is a permutation matrix,

$$C^{-1} = C.$$

Depending on whether one writes characters as row vectors or column vectors, one may write the action as either

$$(C\chi)_i = \chi_{i^*}$$

or

$$(\chi C)_i = \chi_{i^*}.$$

The invariant content is the same: C permutes primary labels by charge conjugation.

2.1.1.4 Clarifications

Confusion 1: Since applying $\tau \mapsto -1/\tau$ twice gives τ, shouldn't $S^2 = I$?

Only if one forgets that the modular transformation acts on labelled conformal blocks, not just on the complex number τ. In $PSL(2, \mathbb{Z})$, where matrices differing by $\pm I$ are identified, the element s has order two. But the mapping class group relevant for chiral conformal blocks is more naturally represented before forgetting the central element $-I$. That central element acts as charge conjugation:

$$\rho(-I) = C.$$

So the correct RCFT statement is

$$S^2 = C,$$

not necessarily $S^2 = I$.

Confusion 2: Does charge conjugation mean electric charge conjugation?

Not necessarily. In RCFT, charge conjugation means taking the dual representation of the chiral algebra. If the chiral algebra contains a current algebra, this often coincides with ordinary representation conjugation, such as

$$\mathbf{3} \leftrightarrow \overline{\mathbf{3}}$$

for $SU(3)$. But the abstract definition is

$$i \mapsto i^*,$$

where i^* labels the contragredient module.

Confusion 3: If $\chi_i(\tau) = \chi_{i^*}(\tau)$ as functions, is C still meaningful?

Yes. The equality of the q-series can hide the distinction between labels. The modular representation is defined on the labelled vector space of conformal blocks. Even if two conjugate characters happen to have the same numerical expansion, the labels i and i^* can still be conceptually distinct.

This is why one should not derive $S^2 = I$ by treating the characters as unlabeled scalar functions. The statement

$$S^2 = C$$

remembers the primary-label data.

Confusion 4: Is C always nontrivial?

No. In many simple RCFTs all primaries are self-conjugate. Then

$$i^* = i$$

for every primary label i, so

$$C = I.$$

For example, in the Ising model all three primaries are self-conjugate, so its modular matrix satisfies $S^2 = I$. But in theories with genuinely complex representations, C is nontrivial.

Confusion 5: Is $S^2 = C$ an assumption or a theorem?

It is a theorem under the standard RCFT axioms. Geometrically, it follows from how the torus mapping class group acts on labelled conformal blocks. Algebraically, in a unitary RCFT it follows from symmetry, unitarity, and the conjugation property of the modular S-matrix:

$$S_{ij} = S_{ji}, \qquad S^\dagger S = I, \qquad S_{i^*j} = S_{ij}^*.$$

These imply

$$(S^2)_{ij} = \delta_{i,j^*}.$$

2.1.1.5 Result

The modular generator

$$s = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

satisfies

$$s^2 = -I$$

in $SL(2,\mathbb{Z})$. Although $-I$ fixes the modulus τ, it reverses the orientation of the torus cycles and hence sends a primary label to its conjugate:

$$i \mapsto i^*.$$

Therefore, on the vector space of RCFT characters,

$$\rho(s)^2 = \rho(-I) = C.$$

Since $\rho(s) = S$, one obtains

$$S^2 = C.$$

The charge conjugation matrix is

$$C_{ij} = \delta_{i,j^*}.$$

Equivalently,

$$Ce_j = e_{j^*}.$$

Using symmetry and unitarity of S, the same statement is

$$(S^2)_{ij} = \sum_k S_{ik} S_{kj} = \sum_k S_{ik} S_{jk} = \sum_k S_{ik} S^*_{j^*k} = \delta_{i,j^*} = C_{ij}.$$

Thus C is the matrix implementing charge conjugation on primary labels, and $S^2 = C$ means that applying the modular S-transformation twice sends every primary sector to its conjugate sector.

Chapter 3

The Classical Fermionic String and Spin Structures

3.1 Q1: Dirac operators on Riemann surfaces and their zero modes

Consider the Dirac operator on compact Riemann surfaces. Denote the Riemann surface with genus g as Σ_g. In 2d, a Dirac spinor has two complex dimensions i.e. it can be written as

$$\psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}$$

And we choose the flat Dirac matrices to be

$$\gamma^0 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \ , \ \gamma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \ , \ \gamma^2 = i\gamma^0\gamma^1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Derive the explicit form of the Dirac operator for the two chiralities.

On Σ_g there are 2^{2g} spin structures. Prove the following statements: 1. For a given spin structure, the number of chiral Dirac zero modes modulo two is a topological invariant; 2. The number of chiral Dirac zero modes modulo two is additive when we glue together two Riemann surfaces.

> **Textbook location and related material.** Chapter 7, Sections 7.3–7.5 (pp. 180–194) for the two-dimensional Dirac operator and spinors; Chapter 9, Section 9.1 (pp. 223–232) for spin structures on the torus. The Arf-invariant formulation used in the answer is additional mathematical background beyond the book.

3.1.1 Answer

3.1.1.1 Core point

The question has two connected parts. First, one wants the explicit chiral form of the Dirac operator on a compact Riemann surface. Second, one wants to understand why, for a fixed spin structure, the number of chiral zero modes modulo two is topological and additive under gluing.

The key geometric fact is that a spin structure on a Riemann surface is a square root of the canonical bundle:

$$L^2 \simeq K.$$

Chiral Dirac zero modes are holomorphic sections of this spin bundle, up to the convention that the opposite chirality gives the complex conjugate anti-holomorphic equation. Thus the number of positive or negative chiral zero modes is controlled by

$$h^0(\Sigma, L) = \dim_{\mathbb{C}} H^0(\Sigma, L).$$

The ordinary integer index of the chiral Dirac operator vanishes in two dimensions for an honest spin bundle, because

$$\deg L = g - 1, \qquad \deg L + 1 - g = 0.$$

So the integer index does not distinguish spin structures. The subtle invariant is instead the mod-two index

$$h^0(\Sigma, L) \bmod 2.$$

This parity is the Arf invariant of the quadratic refinement determined by the spin structure:

$$h^0(\Sigma, L) \equiv \mathrm{Arf}(q_L) \pmod 2.$$

Since q_L is topological, the parity of chiral zero modes is topological. Under connected-sum gluing, the quadratic refinements combine by orthogonal direct sum, and the Arf invariant is additive. Therefore the parity of chiral zero modes is additive.

3.1.1.2 A small correction to the chirality matrix

Using the gamma matrices in the question,

$$\gamma^0 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \gamma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

one computes

$$\gamma^0\gamma^1 = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix},$$

and therefore

$$\gamma^2 = i\gamma^0\gamma^1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

So I will use

$$\gamma^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

With this convention,

$$\psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}, \qquad \gamma^2\psi_\pm = \pm\psi_\pm.$$

The matrix written in the question,

$$\begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix},$$

is not equal to $i\gamma^0\gamma^1$ for the displayed γ^0, γ^1, and it is not the usual chirality matrix because it is not diagonal with eigenvalues ±1.

3.1.1.3 Local form of the Dirac operator

Let us work locally in conformal coordinates

$$z = x^0 + ix^1, \qquad \bar{z} = x^0 - ix^1,$$

with

$$ds^2 = e^{2\omega}\left((dx^0)^2 + (dx^1)^2\right) = e^{2\omega} dz\, d\bar{z}.$$

Choose the orthonormal coframe

$$e^0 = e^{\omega} dx^0, \qquad e^1 = e^{\omega} dx^1.$$

The curved gamma matrices are

$$\gamma^\alpha = e^\alpha{}_a \gamma^a,$$

so in these coordinates the Dirac operator is

$$\not{D} = \gamma^\alpha \nabla_\alpha = e^{-\omega}\gamma^a \nabla_a.$$

The torsion-free spin connection satisfies

$$de^a + \omega^a{}_b \wedge e^b = 0.$$

Writing $\omega^{01} = \omega_\mu{}^{01} dx^\mu$, one finds

$$\omega^{01} = (\partial_1 \omega)\, dx^0 - (\partial_0 \omega)\, dx^1.$$

The spinor covariant derivative is

$$\nabla_\mu = \partial_\mu + \frac{1}{4}\omega_\mu{}^{ab}\gamma_a\gamma_b.$$

Since only the 01 component is present, this becomes

$$\nabla_\mu = \partial_\mu + \frac{1}{2}\omega_\mu{}^{01}\gamma_0\gamma_1.$$

Using

$$\gamma_0\gamma_1 = -i\gamma^2,$$

we get

$$\nabla_0 = \partial_0 - \frac{i}{2}(\partial_1\omega)\gamma^2, \qquad \nabla_1 = \partial_1 + \frac{i}{2}(\partial_0\omega)\gamma^2.$$

Therefore

$$\not{D} = e^{-\omega}\left[\gamma^0\left(\partial_0 - \frac{i}{2}(\partial_1\omega)\gamma^2\right) + \gamma^1\left(\partial_1 + \frac{i}{2}(\partial_0\omega)\gamma^2\right)\right].$$

Acting on

$$\psi = \begin{pmatrix}\psi_+ \\ \psi_-\end{pmatrix},$$

this gives

$$\not{D}\begin{pmatrix}\psi_+ \\ \psi_-\end{pmatrix} = e^{-\omega}\begin{pmatrix}-2i\left(\partial_{\bar z} + \frac{1}{2}\partial_{\bar z}\omega\right)\psi_- \\ 2i\left(\partial_z + \frac{1}{2}\partial_z\omega\right)\psi_+\end{pmatrix}.$$

Equivalently,

$$\not{D} = 2ie^{-\omega}\begin{pmatrix}0 & -\left(\partial_{\bar z} + \frac{1}{2}\partial_{\bar z}\omega\right) \\ \partial_z + \frac{1}{2}\partial_z\omega & 0\end{pmatrix}.$$

Thus the chiral pieces are

$$D_+\psi_+ = 2ie^{-\omega}\left(\partial_z + \frac{1}{2}\partial_z\omega\right)\psi_+,$$

and

$$D_-\psi_- = -2ie^{-\omega}\left(\partial_{\bar z} + \frac{1}{2}\partial_{\bar z}\omega\right)\psi_-.$$

The full Dirac equation $\not{D}\psi = 0$ is therefore equivalent to

$$\left(\partial_z + \frac{1}{2}\partial_z\omega\right)\psi_+ = 0, \qquad \left(\partial_{\bar z} + \frac{1}{2}\partial_{\bar z}\omega\right)\psi_- = 0.$$

After the Weyl rescaling

$$\widehat{\psi}_\pm = e^{\omega/2}\psi_\pm,$$

these become simply

$$\partial_z\widehat{\psi}_+ = 0, \qquad \partial_{\bar z}\widehat{\psi}_- = 0.$$

So, with the coordinate and gamma-matrix convention used here, $\widehat{\psi}_+$ is anti-holomorphic and $\widehat{\psi}_-$ is holomorphic. If one reverses the orientation convention or swaps the labels ψ_+ and ψ_-, the words "holomorphic" and "anti-holomorphic" are interchanged. The invariant statement is that the two chiral Dirac operators are the Dolbeault operator and its conjugate, acting on the two spin bundles.

3.1.1.4 Spin bundles and chiral zero modes

Globally, one should not think of ψ_+ and ψ_- as ordinary functions. They are spinors, so they are sections of spin bundles.

On a Riemann surface, a spin structure is a holomorphic line bundle L such that

$$L^2 \simeq K,$$

where $K = T^{*(1,0)}\Sigma$ is the canonical bundle. Such an L is also called a theta characteristic.

Depending on the chirality convention, one chirality is represented by sections of L, while the opposite chirality is represented by sections of the conjugate bundle. With the convention above, the holomorphic zero-mode equation is naturally

$$\bar{\partial}_L \widehat{\psi}_- = 0,$$

so negative-chirality zero modes are holomorphic sections of L:

$$\widehat{\psi}_- \in H^0(\Sigma, L).$$

The opposite chirality satisfies the complex conjugate equation. Thus its zero modes are anti-holomorphic spinors, equivalently conjugates of holomorphic sections:

$$\widehat{\psi}_+ \in \overline{H^0(\Sigma, L)}.$$

In particular, for the untwisted Dirac operator on a compact Riemann surface, the two chiral zero-mode spaces have the same complex dimension:

$$\dim_{\mathbb{C}} \ker D_+ = \dim_{\mathbb{C}} \ker D_- = h^0(\Sigma, L),$$

up to the assignment of which sign of chirality is called plus or minus.

This is also consistent with the ordinary index theorem. Since $L^2 = K$,

$$\deg L = \frac{1}{2} \deg K = \frac{1}{2}(2g-2) = g-1.$$

Riemann-Roch gives

$$h^0(\Sigma, L) - h^0(\Sigma, K \otimes L^{-1}) = \deg L + 1 - g.$$

But

$$K \otimes L^{-1} \simeq L,$$

because $L^2 \simeq K$. Hence

$$h^0(\Sigma, L) - h^0(\Sigma, L) = 0.$$

So the integer index of the chiral Dirac operator is zero:

$$\operatorname{ind} D = 0.$$

The interesting invariant is not this integer index, but the parity

$$h^0(\Sigma, L) \bmod 2.$$

3.1.1.5 The 2^{2g} spin structures

The set of spin structures on Σ_g is not a vector space canonically, but it is a torsor for

$$H^1(\Sigma_g, \mathbb{Z}_2).$$

Since

$$\dim_{\mathbb{Z}_2} H^1(\Sigma_g, \mathbb{Z}_2) = 2g,$$

there are

$$\left|H^1(\Sigma_g, \mathbb{Z}_2)\right| = 2^{2g}$$

spin structures.

Equivalently, once one spin bundle L is chosen, all others are of the form

$$L \otimes M,$$

where M is a flat line bundle of order two:

$$M^2 \simeq \mathcal{O}.$$

The group of such M is

$$\mathrm{Jac}(\Sigma_g)[2] \simeq H^1(\Sigma_g, \mathbb{Z}_2),$$

which again has 2^{2g} elements.

3.1.1.6 Why the mod-two number of chiral zero modes is topological

The statement is

$$h^0(\Sigma, L) \bmod 2$$

depends only on the topological spin structure, not on the chosen metric or complex structure.

This is not obvious from the formula $h^0(\Sigma, L)$ itself. The actual number $h^0(\Sigma, L)$ can jump as the complex structure varies. What is invariant is its parity.

The clean topological explanation uses the quadratic refinement associated to a spin structure. Let

$$V = H_1(\Sigma_g, \mathbb{Z}_2).$$

The mod-two intersection pairing

$$\langle\ ,\ \rangle : V \times V \to \mathbb{Z}_2$$

is symplectic. A spin structure determines a function

$$q_L : V \to \mathbb{Z}_2$$

satisfying

$$q_L(a+b) = q_L(a) + q_L(b) + \langle a, b\rangle.$$

Such a function is called a quadratic refinement of the intersection form.

Geometrically, $q_L(\gamma)$ measures whether the restriction of the spin structure to the loop γ is periodic or antiperiodic, with a convention-dependent shift chosen so that the quadratic-refinement identity holds. The important point is that q_L depends only on the topological spin structure.

Given a symplectic basis

$$a_1, b_1, \dots, a_g, b_g$$

of $H_1(\Sigma_g, \mathbb{Z}_2)$, the Arf invariant is

$$\mathrm{Arf}(q_L) = \sum_{i=1}^{g} q_L(a_i) q_L(b_i) \pmod 2.$$

Although this formula uses a symplectic basis, the resulting number is independent of the chosen symplectic basis. Thus

$$\mathrm{Arf}(q_L) \in \mathbb{Z}_2$$

is a topological invariant of the spin structure.

The theorem relating this to zero modes is the Riemann-Mumford parity theorem:

$$h^0(\Sigma, L) \equiv \mathrm{Arf}(q_L) \pmod 2.$$

Therefore

$$h^0(\Sigma, L) \bmod 2$$

is topological.

This proves the first requested statement:

$$\boxed{\text{For a fixed spin structure } L,\ \dim_{\mathbb{C}} \ker D_{\text{chiral}} \bmod 2 \text{ is a topological invariant.}}$$

The word "fixed" is important. If one changes the spin structure, the parity may change. Spin structures are divided into even and odd spin structures:

$$L \text{ even} \quad \Longleftrightarrow \quad h^0(\Sigma, L) \equiv 0 \pmod 2,$$

and

$$L \text{ odd} \quad \Longleftrightarrow \quad h^0(\Sigma, L) \equiv 1 \pmod 2.$$

For genus g, the numbers of even and odd spin structures are

$$N_{\text{even}} = 2^{g-1}(2^g + 1), \qquad N_{\text{odd}} = 2^{g-1}(2^g - 1).$$

This is another way to see that the parity is not the same for all spin structures; it is an invariant attached to a chosen spin structure.

3.1.1.7 Additivity under gluing

Now consider gluing two compact Riemann surfaces by connected sum:

$$\Sigma_{g_1} \# \Sigma_{g_2} \simeq \Sigma_{g_1+g_2}.$$

Assume the spin structures L_1 and L_2 are glued compatibly to produce a spin structure

$$L = L_1 \# L_2$$

on $\Sigma_{g_1+g_2}$.

Topologically,

$$H_1(\Sigma_{g_1+g_2}, \mathbb{Z}_2) \simeq H_1(\Sigma_{g_1}, \mathbb{Z}_2) \oplus H_1(\Sigma_{g_2}, \mathbb{Z}_2),$$

and the mod-two intersection pairing is the orthogonal direct sum of the two intersection pairings:

$$\langle\ ,\ \rangle = \langle\ ,\ \rangle_1 \oplus \langle\ ,\ \rangle_2.$$

The glued spin structure gives the quadratic refinement

$$q_L = q_{L_1} \oplus q_{L_2},$$

meaning that for

$$x \in H_1(\Sigma_{g_1}, \mathbb{Z}_2), \qquad y \in H_1(\Sigma_{g_2}, \mathbb{Z}_2),$$

one has

$$q_L(x, y) = q_{L_1}(x) + q_{L_2}(y).$$

The Arf invariant is additive under orthogonal direct sums:

$$\operatorname{Arf}(q_{L_1} \oplus q_{L_2}) = \operatorname{Arf}(q_{L_1}) + \operatorname{Arf}(q_{L_2}) \pmod 2.$$

Using the Riemann-Mumford parity theorem on each surface gives

$$h^0(\Sigma_{g_1+g_2}, L_1 \# L_2) \equiv \operatorname{Arf}(q_L) \pmod 2,$$

and therefore

$$h^0(\Sigma_{g_1+g_2}, L_1 \# L_2) \equiv \operatorname{Arf}(q_{L_1}) + \operatorname{Arf}(q_{L_2}) \pmod 2.$$

Applying the same theorem again,

$$\operatorname{Arf}(q_{L_i}) \equiv h^0(\Sigma_{g_i}, L_i) \pmod 2,$$

so

$$h^0(\Sigma_{g_1+g_2}, L_1 \# L_2) \equiv h^0(\Sigma_{g_1}, L_1) + h^0(\Sigma_{g_2}, L_2) \pmod 2.$$

This proves the second statement:

$$\boxed{\dim \ker D_{\text{chiral},\, L_1 \# L_2} \equiv \dim \ker D_{\text{chiral},\, L_1} + \dim \ker D_{\text{chiral},\, L_2} \pmod 2.}$$

The proof shows exactly why the result is true: gluing corresponds to direct sum on $H_1(-, \mathbb{Z}_2)$, and the mod-two zero-mode count is the Arf invariant of the associated quadratic refinement.

3.1.1.8 Checks

For genus $g = 0$, there is only one spin structure. Since

$$K_{\mathbb{P}^1} \simeq \mathcal{O}(-2),$$

the spin bundle is

$$L \simeq \mathcal{O}(-1).$$

It has no holomorphic sections:

$$h^0(\mathbb{P}^1, \mathcal{O}(-1)) = 0.$$

Thus the spin structure is even.

For genus $g = 1$, the canonical bundle is trivial:

$$K \simeq \mathcal{O}.$$

There are four spin structures, corresponding to the four choices of periodic or antiperiodic boundary conditions around the two fundamental cycles. The trivial spin bundle has

$$h^0(T^2, \mathcal{O}) = 1,$$

so it is odd. The other three flat square roots have no holomorphic sections, so they are even. This matches

$$N_{\text{odd}} = 2^0(2^1 - 1) = 1, \qquad N_{\text{even}} = 2^0(2^1 + 1) = 3.$$

As a gluing check, if one connected-sums two tori and chooses the odd spin structure on both, the parity on the genus-two surface is

$$1 + 1 \equiv 0 \pmod 2.$$

So the glued spin structure is even. If one glues an odd torus spin structure with an even torus spin structure, the resulting genus-two spin structure is odd.

3.1.1.9 Result

With

$$z = x^0 + ix^1, \qquad ds^2 = e^{2\omega} dz\, d\bar{z},$$

and with

$$\gamma^2 = i\gamma^0\gamma^1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the Dirac operator is

$$\not{D} = 2ie^{-\omega} \begin{pmatrix} 0 & -\left(\partial_{\bar{z}} + \frac{1}{2}\partial_{\bar{z}}\omega\right) \\ \partial_z + \frac{1}{2}\partial_z\omega & 0 \end{pmatrix}.$$

Thus

$$\not{D}\psi = 0 \quad \Longleftrightarrow \quad \left(\partial_z + \frac{1}{2}\partial_z\omega\right)\psi_+ = 0, \qquad \left(\partial_{\bar{z}} + \frac{1}{2}\partial_{\bar{z}}\omega\right)\psi_- = 0.$$

After $\widehat{\psi}_\pm = e^{\omega/2}\psi_\pm$,

$$\partial_z \widehat{\psi}_+ = 0, \qquad \partial_{\bar{z}} \widehat{\psi}_- = 0.$$

A spin structure is

$$L^2 \simeq K,$$

and chiral zero modes are counted by

$$h^0(\Sigma, L) = \dim_{\mathbb{C}} H^0(\Sigma, L),$$

up to the convention of which chirality is called plus or minus.

The parity theorem says

$$h^0(\Sigma, L) \equiv \operatorname{Arf}(q_L) \pmod 2.$$

Therefore

$$\boxed{h^0(\Sigma, L) \bmod 2 \text{ is a topological invariant of the spin structure.}}$$

For gluing by connected sum,

$$q_{L_1 \# L_2} = q_{L_1} \oplus q_{L_2},$$

so

$$\mathrm{Arf}(q_{L_1\#L_2}) = \mathrm{Arf}(q_{L_1}) + \mathrm{Arf}(q_{L_2}) \pmod 2.$$

Equivalently,

$$\boxed{h^0(\Sigma_{g_1+g_2}, L_1\#L_2) \equiv h^0(\Sigma_{g_1}, L_1) + h^0(\Sigma_{g_2}, L_2) \pmod 2.}$$

3.2 Q2: Counting even and odd spin structures

We call a spin structure even if the number of zero modes of the chiral Dirac operator is even and we call it odd otherwise. Prove: there are $\sum_{m \text{ odd}} \binom{g}{m} 3^{g-m} = 2^{g-1}(2^g - 1)$ odd and $\sum_{m \text{ even}} \binom{g}{m} 3^{g-m} = 2^{g-1}(2^g + 1)$ even spin structures.

> **Textbook location and related material.** Chapter 9, Section 9.1 (pp. 223–232), together with the theta-function appendix in Section 9.6 (pp. 254–262). The genus-g counting by quadratic refinements is a generalization of the book's genus-one discussion.

3.2.1 Answer

3.2.1.1 Core point

The question asks for the number of even and odd spin structures on a genus-g compact Riemann surface Σ_g. By definition here,

$$L \text{ is even} \quad \Longleftrightarrow \quad h^0(\Sigma_g, L) \equiv 0 \pmod 2,$$

and

$$L \text{ is odd} \quad \Longleftrightarrow \quad h^0(\Sigma_g, L) \equiv 1 \pmod 2.$$

The key point from Question 1 is the Riemann-Mumford parity theorem:

$$h^0(\Sigma_g, L) \equiv \mathrm{Arf}(q_L) \pmod 2.$$

So the problem is equivalent to counting quadratic refinements

$$q : H_1(\Sigma_g, \mathbb{Z}_2) \to \mathbb{Z}_2$$

with Arf invariant 0 or 1.

In a symplectic basis

$$a_1, b_1, \dots, a_g, b_g$$

of $H_1(\Sigma_g, \mathbb{Z}_2)$, the Arf invariant is

$$\mathrm{Arf}(q) = \sum_{i=1}^{g} q(a_i) q(b_i) \pmod 2.$$

For each handle, there are four possible pairs

$$(q(a_i), q(b_i)) \in \mathbb{Z}_2^2.$$

Among these four choices, three contribute 0 to the Arf sum and one contributes 1. Therefore an odd spin structure is obtained by choosing an odd number of handles of the fourth type; an even spin structure is obtained by choosing an even number.

3.2.1.2 Spin structures as quadratic refinements

Let

$$V = H_1(\Sigma_g, \mathbb{Z}_2).$$

The mod-two intersection pairing

$$\langle\ ,\ \rangle : V \times V \to \mathbb{Z}_2$$

is a nondegenerate symplectic pairing. A spin structure determines a quadratic refinement

$$q : V \to \mathbb{Z}_2$$

satisfying

$$q(x + y) = q(x) + q(y) + \langle x, y\rangle.$$

Conversely, every such quadratic refinement comes from a spin structure. Thus the set of spin structures can be counted by counting quadratic refinements of the intersection form.

Choose a symplectic basis

$$a_1, b_1, \dots, a_g, b_g$$

with

$$\langle a_i, b_j\rangle = \delta_{ij}, \qquad \langle a_i, a_j\rangle = 0, \qquad \langle b_i, b_j\rangle = 0.$$

The values

$$q(a_i), \qquad q(b_i), \qquad i = 1, \dots, g,$$

determine q on all of V. Indeed, if

$$x = \sum_{i=1}^{g} (\alpha_i a_i + \beta_i b_i)\,, \qquad \alpha_i, \beta_i \in \mathbb{Z}_2,$$

then the quadratic-refinement identity gives

$$q(x) = \sum_{i=1}^{g} \alpha_i q(a_i) + \sum_{i=1}^{g} \beta_i q(b_i) + \sum_{i=1}^{g} \alpha_i \beta_i \pmod 2.$$

The final term is the correction coming from the nonzero intersections

$$\langle a_i, b_i\rangle = 1.$$

Therefore, once the $2g$ numbers

$$q(a_1), q(b_1), \dots, q(a_g), q(b_g)$$

are chosen, the quadratic refinement is fixed.

Since each of these $2g$ numbers can be 0 or 1, the total number of quadratic refinements is

$$2^{2g}.$$

This agrees with the standard count of spin structures on Σ_g.

3.2.1.3 Arf invariant handle by handle

The Arf invariant is

$$\mathrm{Arf}(q) = \sum_{i=1}^{g} q(a_i) q(b_i) \pmod 2.$$

For a fixed handle i, there are four choices:

$$(q(a_i), q(b_i)) = (0,0), (1,0), (0,1), (1,1).$$

Their contributions to

$$q(a_i)q(b_i)$$

are

$(q(a_i), q(b_i))$	$q(a_i)q(b_i)$
$(0,0)$	0
$(1,0)$	0
$(0,1)$	0
$(1,1)$	1

So, for each handle:

$$\text{three choices contribute } 0, \qquad \text{one choice contributes } 1.$$

Now let m be the number of handles for which

$$(q(a_i), q(b_i)) = (1,1).$$

Then

$$\operatorname{Arf}(q) \equiv m \pmod 2.$$

Thus:

$$q \text{ is odd} \iff m \text{ is odd},$$

and

$$q \text{ is even} \iff m \text{ is even}.$$

This already gives the desired combinatorial structure.

3.2.1.4 Counting odd spin structures

To construct an odd spin structure, choose an odd number m of handles to have the locally odd choice

$$(q(a_i), q(b_i)) = (1,1).$$

There are

$$\binom{g}{m}$$

ways to choose those handles.

On each of these m handles, the choice is forced to be $(1,1)$. On each of the remaining $g-m$ handles, the contribution must be 0, so there are three possible choices:

$$(0,0), \qquad (1,0), \qquad (0,1).$$

Therefore, for fixed odd m, the number of spin structures with exactly m contributing handles is

$$\binom{g}{m} 3^{g-m}.$$

Summing over odd m gives

$$N_{\text{odd}} = \sum_{m \text{ odd}} \binom{g}{m} 3^{g-m}.$$

To evaluate this sum, use the binomial identities

$$\sum_{m=0}^{g} \binom{g}{m} 3^{g-m} = (3+1)^g = 4^g,$$

and

$$\sum_{m=0}^{g} (-1)^m \binom{g}{m} 3^{g-m} = (3-1)^g = 2^g.$$

The first sum is

$$N_{\text{even}} + N_{\text{odd}},$$

while the second is

$$N_{\text{even}} - N_{\text{odd}},$$

because even m terms have sign $+1$ and odd m terms have sign -1.

Therefore

$$N_{\text{odd}} = \frac{1}{2}\left(4^g - 2^g\right) = \frac{1}{2}\left(2^{2g} - 2^g\right) = 2^{g-1}(2^g - 1).$$

So

$$\boxed{N_{\text{odd}} = \sum_{m \text{ odd}} \binom{g}{m} 3^{g-m} = 2^{g-1}(2^g - 1).}$$

3.2.1.5 Counting even spin structures

The same reasoning gives the even count. An even spin structure has

$$\mathrm{Arf}(q) = 0,$$

so the number m of handles with

$$(q(a_i), q(b_i)) = (1, 1)$$

must be even.

For fixed even m, there are

$$\binom{g}{m}$$

ways to choose the contributing handles, and

$$3^{g-m}$$

ways to choose the remaining noncontributing handles. Thus

$$N_{\text{even}} = \sum_{m \text{ even}} \binom{g}{m} 3^{g-m}.$$

Using the same binomial identities,

$$N_{\text{even}} = \frac{1}{2}\left(4^g + 2^g\right) = \frac{1}{2}\left(2^{2g} + 2^g\right) = 2^{g-1}(2^g + 1).$$

Therefore

$$\boxed{N_{\text{even}} = \sum_{m \text{ even}} \binom{g}{m} 3^{g-m} = 2^{g-1}(2^g + 1).}$$

As a check,

$$N_{\text{even}} + N_{\text{odd}} = 2^{g-1}(2^g + 1) + 2^{g-1}(2^g - 1) = 2^{2g},$$

which is the total number of spin structures.

3.2.1.6 Checks

For $g = 1$, there are

$$N_{\text{odd}} = 2^0(2^1 - 1) = 1$$

odd spin structure and

$$N_{\text{even}} = 2^0(2^1 + 1) = 3$$

even spin structures. This is the familiar torus result: the fully periodic spin structure is odd, while the other three are even.

For $g = 2$,

$$N_{\text{odd}} = 2^1(2^2 - 1) = 6,$$

and

$$N_{\text{even}} = 2^1(2^2 + 1) = 10.$$

Together they give

$$6 + 10 = 16 = 2^4,$$

as required.

3.2.1.7 Result

A spin structure L determines a quadratic refinement

$$q_L : H_1(\Sigma_g, \mathbb{Z}_2) \to \mathbb{Z}_2, \qquad q_L(x+y) = q_L(x) + q_L(y) + \langle x, y \rangle.$$

In a symplectic basis,

$$\mathrm{Arf}(q_L) = \sum_{i=1}^{g} q_L(a_i) q_L(b_i) \pmod 2.$$

The parity theorem says

$$h^0(\Sigma_g, L) \equiv \mathrm{Arf}(q_L) \pmod 2.$$

For each handle, the four possible pairs are

$$(0,0), (1,0), (0,1), (1,1).$$

The first three contribute 0 to the Arf invariant, while $(1,1)$ contributes 1. Hence if m handles are of type $(1,1)$, then

$$h^0(\Sigma_g, L) \equiv m \pmod 2.$$

Therefore

$$\boxed{N_{\text{odd}} = \sum_{m \text{ odd}} \binom{g}{m} 3^{g-m} = 2^{g-1}(2^g - 1)}$$

and

$$\boxed{N_{\text{even}} = \sum_{m \text{ even}} \binom{g}{m} 3^{g-m} = 2^{g-1}(2^g + 1).}$$

3.3 Q3: Fermion spin structures and theta-function conventions

Consider the closed superstring partition function on torus. The contribution to the partition function of the right-moving fermions with fixed spin structure are

$$\begin{aligned}
\chi^{(++)}_{\text{fermion}}(\tau) &= \eta_{++} \mathrm{Tr} e^{2\pi i \tau H_R} (-1)^F \\
\chi^{(+-)}_{\text{fermion}}(\tau) &= \eta_{+-} \mathrm{Tr} e^{2\pi i \tau H_R} \\
\chi^{(--)}_{\text{fermion}}(\tau) &= \eta_{--} \mathrm{Tr} e^{2\pi i \tau H_{NS}} \\
\chi^{(-+)}_{\text{fermion}}(\tau) &= \eta_{-+} \mathrm{Tr} e^{2\pi i \tau H_{NS}} (-1)^F
\end{aligned}$$

If we choose the complex coordinates on the torus to be

$$z = \sigma^1 + i\sigma^0 = 2\pi(\xi^1 + \tau \xi^2)$$

Then for a general twisted boundary condition

$$\psi(\xi^1 + 1, \xi^2) = -e^{2\pi i \alpha} \psi(\xi^1, \xi^2) \ , \ \psi(\xi^1, \xi^2 + 1) = -e^{2\pi i \beta} \psi(\xi^1, \xi^2)$$

The partition function can be computed to be

$$\begin{aligned}
Z \begin{bmatrix} \alpha \\ \beta \end{bmatrix} (\tau) &= \mathrm{Tr}(e^{2\pi i \beta (N - \bar{N})} q^{L_0 - 1/24}) \\
&= q^{\alpha^2/2 - 1/24} \prod_{n=1}^{\infty} (1 + q^{n+\alpha-1/2} e^{-2\pi i \beta}) \qquad \times (1 + q^{n-\alpha-1/2} e^{2\pi i \beta}) \\
&= \frac{e^{2\pi i \alpha \beta} \vartheta \begin{bmatrix} \alpha \\ -\beta \end{bmatrix} (\tau)}{\eta(\tau)}.
\end{aligned}$$

In the textbook, the different spin structures correspond to

$$\begin{array}{lll} (++) & \alpha=\frac{1}{2},\beta=\frac{1}{2} & \vartheta\begin{bmatrix}1/2\\1/2\end{bmatrix}=\vartheta_1 \\ (+-) & \alpha=\frac{1}{2},\beta=0 & \vartheta\begin{bmatrix}1/2\\0\end{bmatrix}=\vartheta_2 \\ (--) & \alpha=0,\beta=0 & \vartheta\begin{bmatrix}0\\0\end{bmatrix}=\vartheta_3 \\ (-+) & \alpha=0,\beta=\frac{1}{2} & \vartheta\begin{bmatrix}0\\1/2\end{bmatrix}=\vartheta_4 \end{array}$$

Q1: the expression computed for twisted boundary condition is $\vartheta\begin{bmatrix}\alpha\\-\beta\end{bmatrix}$, while the above are $\vartheta\begin{bmatrix}\alpha\\\beta\end{bmatrix}$, are these equivalent?

There are many identities among the Jacobi ϑ functions. Some of them are

$$\begin{aligned} &\vartheta_2^4(\tau)-\vartheta_3^4(\tau)+\vartheta_4^4(\tau)=0 \\ &\vartheta_2(\tau)\vartheta_3(\tau)\vartheta_4(\tau)=2\eta^3(\tau) \\ &\vartheta_1'(\tau)=2\pi\eta^3(\tau) \end{aligned}$$

It is said that $\vartheta_1(\tau)=0$. Q2: I am confused with this: if $\vartheta_1(\tau)=0$ then why $\vartheta_1'(\tau)=2\pi\eta^3(\tau)$ holds? How to compute $\vartheta_1(\tau)=0$?

> **Textbook location and related material.** Chapter 9, Sections 9.1 and 9.6 (pp. 223–232, 254–262): fermionic boundary conditions, spin-structure traces, theta functions with characteristics, and their identities.

3.3.1 Answer

3.3.1.1 Core point

There are two notation issues here.

First, the expression

$$Z\begin{bmatrix}\alpha\\\beta\end{bmatrix}(\tau)\propto e^{2\pi i\alpha\beta}\frac{\vartheta\begin{bmatrix}\alpha\\-\beta\end{bmatrix}(0|\tau)}{\eta(\tau)}$$

does not really conflict with the table written using

$$\vartheta\begin{bmatrix}\alpha\\\beta\end{bmatrix}(0|\tau).$$

For the four spin structures one only has

$$\alpha,\beta\in\left\{0,\frac{1}{2}\right\}.$$

Since characteristics are defined modulo integers, $-\beta$ and β describe the same spin structure. The theta function itself can pick up a phase under this integer shift, but that phase is precisely the sort of convention-dependent sign absorbed into the spin-structure coefficient $\eta_{\alpha\beta}$.

Second,

$$\vartheta_1(\tau)=0$$

is shorthand for

$$\vartheta_1(0|\tau)=0.$$

But

$$\vartheta_1'(\tau)=2\pi\eta^3(\tau)$$

means

$$\frac{\partial}{\partial\nu}\vartheta_1(\nu|\tau)\Big|_{\nu=0} = 2\pi\eta^3(\tau).$$

So there is no contradiction: a function can vanish at a point and still have a nonzero first derivative there.

3.3.1.2 Theta functions with characteristics

I will use the convention

$$\vartheta\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\nu|\tau) = \sum_{n\in\mathbb{Z}} \exp\left[\pi i(n+\alpha)^2\tau + 2\pi i(n+\alpha)(\nu+\beta)\right].$$

At $\nu = 0$ this becomes

$$\vartheta\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(0|\tau) = \sum_{n\in\mathbb{Z}} q^{\frac{1}{2}(n+\alpha)^2} e^{2\pi i(n+\alpha)\beta}, \qquad q = e^{2\pi i\tau}.$$

The four standard Jacobi theta constants are then

$$\vartheta_1(\tau) = \vartheta\begin{bmatrix}\frac{1}{2}\\ \frac{1}{2}\end{bmatrix}(0|\tau),$$

$$\vartheta_2(\tau) = \vartheta\begin{bmatrix}\frac{1}{2}\\ 0\end{bmatrix}(0|\tau),$$

$$\vartheta_3(\tau) = \vartheta\begin{bmatrix}0\\ 0\end{bmatrix}(0|\tau),$$

and

$$\vartheta_4(\tau) = \vartheta\begin{bmatrix}0\\ \frac{1}{2}\end{bmatrix}(0|\tau).$$

Different textbooks sometimes differ by signs in the lower characteristic or by phases in the definition of the trace. The physical spin structure is the periodicity data; the corresponding theta constant can differ by an overall phase depending on convention.

3.3.1.3 Why β and $-\beta$ describe the same spin structure

From the definition,

$$\vartheta\begin{bmatrix}\alpha\\ \beta+1\end{bmatrix}(\nu|\tau) = e^{2\pi i\alpha}\vartheta\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\nu|\tau).$$

This is because shifting $\beta \mapsto \beta+1$ multiplies each term in the sum by

$$e^{2\pi i(n+\alpha)} = e^{2\pi i\alpha}.$$

Similarly,

$$\vartheta\begin{bmatrix}\alpha\\ \beta-1\end{bmatrix}(\nu|\tau) = e^{-2\pi i\alpha}\vartheta\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\nu|\tau).$$

Now for the spin structures on the torus,

$$\beta = 0 \qquad \text{or} \qquad \beta = \frac{1}{2}.$$

If $\beta = 0$, then

$$-\beta = 0,$$

so there is no change.

If $\beta = \frac{1}{2}$, then

$$-\frac{1}{2} = \frac{1}{2} - 1.$$

Therefore

$$\vartheta\begin{bmatrix}\alpha\\-\frac{1}{2}\end{bmatrix}(\nu|\tau) = e^{-2\pi i\alpha}\vartheta\begin{bmatrix}\alpha\\\frac{1}{2}\end{bmatrix}(\nu|\tau).$$

For $\alpha = 0$, this phase is 1. For $\alpha = \frac{1}{2}$, this phase is -1.

Thus

$$\vartheta\begin{bmatrix}0\\-\frac{1}{2}\end{bmatrix} = \vartheta\begin{bmatrix}0\\\frac{1}{2}\end{bmatrix},$$

while

$$\vartheta\begin{bmatrix}\frac{1}{2}\\-\frac{1}{2}\end{bmatrix} = -\vartheta\begin{bmatrix}\frac{1}{2}\\\frac{1}{2}\end{bmatrix}.$$

At $\nu = 0$, the latter is just

$$\vartheta_1(\tau) \mapsto -\vartheta_1(\tau),$$

but

$$\vartheta_1(\tau) = 0,$$

so even this sign is invisible for the theta constant itself.

More generally, even when the theta function is not zero, this sign is not a physical change of spin structure. It is an overall phase convention. In the fermion partition function, such phases are included in the coefficients

$$\eta_{++}, \quad \eta_{+-}, \quad \eta_{--}, \quad \eta_{-+}.$$

These are the GSO or spin-structure phases. Thus the replacement

$$\beta \longleftrightarrow -\beta$$

is harmless after one is consistent about the phase convention.

3.3.1.4 Matching the four spin structures

The boundary conditions are

$$\psi(\xi^1 + 1, \xi^2) = -e^{2\pi i\alpha}\psi(\xi^1, \xi^2),$$

and

$$\psi(\xi^1, \xi^2 + 1) = -e^{2\pi i\beta}\psi(\xi^1, \xi^2).$$

For

$$\alpha = 0,$$

the first boundary condition is antiperiodic:

$$\psi(\xi^1 + 1, \xi^2) = -\psi(\xi^1, \xi^2).$$

For

$$\alpha = \frac{1}{2},$$

it is periodic:

$$\psi(\xi^1 + 1, \xi^2) = +\psi(\xi^1, \xi^2).$$

Thus the sign convention is

$$- \quad \text{means NS}, \qquad + \quad \text{means R}.$$

The same interpretation applies to β around the other cycle.

Therefore

$$(++) \quad \Longleftrightarrow \quad \alpha = \frac{1}{2}, \quad \beta = \frac{1}{2},$$

$$(+-) \quad \Longleftrightarrow \quad \alpha = \frac{1}{2}, \quad \beta = 0,$$

$$(--) \quad \Longleftrightarrow \quad \alpha = 0, \quad \beta = 0,$$

and

$$(-+) \quad \Longleftrightarrow \quad \alpha = 0, \quad \beta = \frac{1}{2}.$$

This matches the table:

$$\begin{aligned}
&(++) \quad \alpha = \tfrac{1}{2}, \beta = \tfrac{1}{2} \quad \vartheta \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \vartheta_1, \\
&(+-) \quad \alpha = \tfrac{1}{2}, \beta = 0 \quad \vartheta \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} = \vartheta_2, \\
&(--) \quad \alpha = 0, \beta = 0 \quad \vartheta \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \vartheta_3, \\
&(-+) \quad \alpha = 0, \beta = \tfrac{1}{2} \quad \vartheta \begin{bmatrix} 0 \\ 1/2 \end{bmatrix} = \vartheta_4.
\end{aligned}$$

If the determinant computation gives $-\beta$ in the lower characteristic, then the same table is recovered after using the periodicity of the characteristic and adjusting the η phases.

3.3.1.5 Why $\vartheta_1(0|\tau) = 0$

Now let us compute the vanishing of ϑ_1 directly from the series.

By definition,

$$\vartheta_1(\nu|\tau) = \vartheta \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} (\nu|\tau).$$

Therefore

$$\vartheta_1(\nu|\tau) = \sum_{n\in\mathbb{Z}} \exp\left[\pi i \left(n + \frac{1}{2}\right)^2 \tau + 2\pi i \left(n + \frac{1}{2}\right)\left(\nu + \frac{1}{2}\right)\right].$$

At $\nu = 0$,

$$\vartheta_1(0|\tau) = \sum_{n\in\mathbb{Z}} \exp\left[\pi i \left(n + \frac{1}{2}\right)^2 \tau + \pi i \left(n + \frac{1}{2}\right)\right].$$

Equivalently,

$$\vartheta_1(0|\tau) = \sum_{n\in\mathbb{Z}} i(-1)^n q^{\frac{1}{2}(n+\frac{1}{2})^2}.$$

Now pair the term with index n with the term with index

$$n' = -n - 1.$$

Then

$$n' + \frac{1}{2} = -n - \frac{1}{2} = -\left(n + \frac{1}{2}\right),$$

so the q-power is the same:

$$\left(n' + \frac{1}{2}\right)^2 = \left(n + \frac{1}{2}\right)^2.$$

But the phase changes sign:

$$i(-1)^{n'} = i(-1)^{-n-1} = -i(-1)^n.$$

Therefore the n and $-n-1$ terms cancel pairwise, so

$$\boxed{\vartheta_1(0|\tau) = 0.}$$

Equivalently, $\vartheta_1(\nu|\tau)$ is an odd function of ν:

$$\vartheta_1(-\nu|\tau) = -\vartheta_1(\nu|\tau).$$

Every odd function vanishes at the origin, so

$$\vartheta_1(0|\tau) = 0.$$

3.3.1.6 Why the derivative is nonzero

The identity

$$\vartheta_1'(\tau) = 2\pi\eta^3(\tau)$$

is shorthand for

$$\vartheta_1'(0|\tau) \equiv \frac{\partial}{\partial\nu}\vartheta_1(\nu|\tau)\Big|_{\nu=0} = 2\pi\eta^3(\tau).$$

The derivative is with respect to the elliptic variable ν, not with respect to the modular parameter τ.

There is no contradiction with

$$\vartheta_1(0|\tau) = 0.$$

For example, the elementary function $f(\nu) = \nu$ satisfies

$$f(0) = 0, \qquad f'(0) = 1.$$

The same thing happens here: $\vartheta_1(\nu|\tau)$ has a simple zero at $\nu = 0$.

The cleanest derivation uses the Jacobi triple product:

$$\vartheta_1(\nu|\tau) = 2q^{1/8}\sin(\pi\nu)\prod_{n=1}^{\infty}(1-q^n)(1-2q^n\cos(2\pi\nu)+q^{2n}),$$

with

$$q = e^{2\pi i\tau}.$$

At $\nu = 0$,

$$\sin(\pi\nu)\big|_{\nu=0} = 0,$$

so immediately

$$\vartheta_1(0|\tau) = 0.$$

Now differentiate with respect to ν and then set $\nu = 0$. Since the zero comes from the factor $\sin(\pi\nu)$, only the derivative of that factor contributes at $\nu = 0$. Thus

$$\vartheta_1'(0|\tau) = 2q^{1/8}\pi\prod_{n=1}^{\infty}(1-q^n)(1-2q^n+q^{2n}).$$

But

$$1-2q^n+q^{2n} = (1-q^n)^2,$$

so

$$\vartheta_1'(0|\tau) = 2\pi q^{1/8}\prod_{n=1}^{\infty}(1-q^n)^3.$$

Using the Dedekind eta function

$$\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n),$$

we get

$$\eta^3(\tau) = q^{1/8}\prod_{n=1}^{\infty}(1-q^n)^3.$$

Therefore

$$\boxed{\vartheta_1'(0|\tau) = 2\pi\eta^3(\tau).}$$

3.3.1.7 Checks

For the $(-+)$ spin structure,

$$\alpha = 0, \qquad \beta = \frac{1}{2}.$$

Replacing β by $-\beta$ gives

$$\vartheta \begin{bmatrix} 0 \\ -1/2 \end{bmatrix} = \vartheta \begin{bmatrix} 0 \\ 1/2 \end{bmatrix},$$

because $e^{-2\pi i\alpha} = 1$. So ϑ_4 is completely unchanged.

For the $(++)$ spin structure,

$$\alpha = \frac{1}{2}, \qquad \beta = \frac{1}{2}.$$

Replacing β by $-\beta$ gives

$$\vartheta \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix} = -\vartheta \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = -\vartheta_1.$$

At zero elliptic argument,

$$\vartheta_1(0|\tau) = 0,$$

so the sign is irrelevant for the theta constant. Away from $\nu = 0$, this sign is an overall convention that can be absorbed into η_{++}.

3.3.1.8 Result

With

$$\vartheta \begin{bmatrix} \alpha \\ \beta \end{bmatrix} (\nu|\tau) = \sum_{n\in\mathbb{Z}} \exp\left[\pi i(n+\alpha)^2\tau + 2\pi i(n+\alpha)(\nu+\beta)\right],$$

one has

$$\vartheta \begin{bmatrix} \alpha \\ \beta - 1 \end{bmatrix} = e^{-2\pi i\alpha} \vartheta \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

Therefore, for $\beta = 0, \frac{1}{2}$,

$$\vartheta \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$$

is the same spin structure as

$$\vartheta \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

possibly up to a convention-dependent phase. That phase is absorbed into the spin-structure coefficient $\eta_{\alpha\beta}$.

The notation

$$\vartheta_1(\tau) = 0$$

means

$$\vartheta_1(0|\tau) = 0.$$

This follows either because $\vartheta_1(\nu|\tau)$ is odd in ν, or by pairwise cancellation in the series. The notation

$$\vartheta_1'(\tau) = 2\pi\eta^3(\tau)$$

means

$$\partial_\nu \vartheta_1(\nu|\tau)\big|_{\nu=0} = 2\pi\eta^3(\tau).$$

Using the product formula,

$$\vartheta_1(\nu|\tau) = 2q^{1/8}\sin(\pi\nu)\prod_{n=1}^{\infty}(1-q^n)(1-2q^n\cos(2\pi\nu)+q^{2n}),$$

one obtains

$$\vartheta_1'(0|\tau) = 2\pi q^{1/8}\prod_{n=1}^{\infty}(1-q^n)^3 = 2\pi\eta^3(\tau).$$

3.4 Q4: Boundary-state gluing signs and NS-NS GSO action

Consider the open superstring. Use $\sigma^{\pm} = \tau \pm \sigma$ as the world-sheet coordinates, the fermion needs to satisfy the boundary condition

$$\psi_+^\mu(0) = \pm\psi_-^\mu(0)\ ,\ \psi_+^\mu(l) = \pm\psi_-^\mu(l)$$

Only the relative sign is relevant We separate the spacetime indices $\mu = (\alpha, i), \alpha = 0, \cdots, p, i = p+1, \cdots, 9$, and denote $D^\mu{}_\nu = (\delta^\alpha{}_\beta, -\delta^i{}_j)$, then the boundary condition can be summarized as

$$\psi_+^\mu(0) = D^\mu{}_\nu(0)\psi^\nu(0)\ ,\ \psi_+^\mu(l) = \eta D^\mu{}_\nu(l)\psi^\nu(l)$$

Where $\eta = +1$ is called R sector and $\eta = -1$ is the NS sector. Only the relative sign is relevant and we can always move the sign η to the $\sigma = l$ boundary.

Now we want to construct boundary states in closed string theory via open-closed duality. The transformation from the closed to the open string channel via the exchange $\tau \leftrightarrow \sigma$ or $\sigma^{\pm} \to \sigma'^{\pm} = \pm\sigma^{\pm}$; and $\psi'_+(\sigma'^{\pm}) = \left(\frac{\partial\sigma'^+}{\partial\sigma^+}\right)^{-1.2}\psi_+(\sigma^+)$. Then the open string boundary condition will become a gluing condition on a boundary state:

$$(\psi_+'^\mu(\sigma'^{\pm}) + \eta D^\mu{}_\nu \psi_-'^\nu(\sigma'^-))|B, \eta\rangle = 0\ ,\ \text{at } \tau = 0$$

Where $\eta = \pm 1$. Q: Here in my understanding this η is not the η for R/NS sector in open string. Then what is it? In the following paragraph there may be some hints.

Through mode expansion of this gluing condition

$$(b_r^\mu + i\eta D^\mu{}_\nu \bar{b}_{-r}^\nu)|B, \eta\rangle = 0$$

We can solve this condition. For simplicity we only consider one complex fermion $\psi = \psi^1 + i\psi^2$. In the NS sector

$$|B_{NS}, \eta\rangle = \exp\left(\mp i\eta \sum_{i=1}^{2}\sum_{r>0} b_{-r}^i \bar{b}_{-r}^i\right)|0\rangle_{NS}$$

Where $-$ sign is for N and the $+$ sign for D boundary condition. By computing the two NN or two DD overlaps

$$\langle B_{NS}, \eta|e^{-2\pi l H_{\text{closed}}}|B_{NS}, \eta\rangle = \frac{\vartheta_3(2il)}{\eta(2il)}\ ,\ \langle B_{NS}, \eta|e^{-2\pi l H_{\text{closed}}}|B_{NS}, -\eta\rangle = \frac{\vartheta_4(2il)}{\eta(2il)}$$

We find that different relative signs of η correspond to different spin structures. We see that $(\eta, \eta) = (+,+), (-,-)$ overlaps yield the $\mathrm{Tr}_{NS} q^{L_0 - c/24}$ sector while the $(-,+), (+,-)$ overlaps yield the $\mathrm{Tr}_R q^{L_0 - c/24}$ sector. Here the relative sign -1 will corresponds to the trace over R sector. Also in the R sector

$$|B_R, \eta\rangle = \exp\left(\mp i\eta \sum_{i=1}^{2}\sum_{r>0} b_{-r}^i \bar{b}_{-r}^i\right)|R\rangle\ ,\ |R\rangle = (b_0^+ + i\eta\bar{b}_0^+)|0\rangle_R$$

Where $b_0^{\pm} = \frac{1}{\sqrt{2}}(b_0^1 \pm i b_0^2)$. Again we can compute the NN and DD overlaps.

Now consider the boundary state corresponding to a Dp-brane. It requires the gluing conditions

$$(\alpha_n^i + D^i{}_j \bar{\alpha}_{-n}^i)|\mathrm{Dp}, \eta\rangle = 0\ ,\ (b_r^i + i\eta D^i{}_j \bar{b}_{-r}^j)|\mathrm{Dp}, \eta\rangle = 0$$

Here we are in the light-cone gauge, the (0,1)-directions are light-cone directions with N boundary conditions; and $i = 2, \cdots, 9$ are transverse coordinates i.e. we have N for $i = 0, \cdots, p$ and D for $i = p+1, \cdots, 9$. The solution to these conditions are

$$|\mathrm{Dp}, \eta\rangle = \int dk_\perp \exp\left[-\sum_{n>0}\frac{1}{n}\alpha_{-n}^i D_{ij}\bar{\alpha}_{-n}^j - i\eta\sum_{r>0} b_{-r}^i D_{ij}\bar{b}_{-r}^j\right] e^{ik_\perp\cdot x_0^\perp}|k_\perp\rangle,$$

where $x_0^\perp$ denotes the position of the D-brane in the Dirichlet directions, i.e. transverse to the brane's world-volume. In the NS-NS sector the ground state $|k_\perp\rangle$ is simply the unique ground state with transverse momentum $k_\perp$. In the R-R sector the degenerate ground state also has to satisfy the gluing condition for the zero modes.

In the NS-NS sector, acting with the GSO projections on the $|\mathrm{Dp}, \eta\rangle_{NS}$ boundary states we get

$$(-1)^F|\mathrm{Dp}, \eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp}, -\eta\rangle_{\mathrm{NS}}, \qquad (-1)^{\overline{F}}|\mathrm{Dp}, \eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp}, -\eta\rangle_{\mathrm{NS}},$$

Where the minus sign is due to the fact that the NS ground state carries odd fermion number. Q: how to derive this equations?

> **Textbook location and related material.** Chapter 9, Section 9.2 (pp. 232–235): fermionic boundary states, gluing signs, overlaps, and GSO projection. Section 7.4 (pp. 189–192) contains the open-string NS/R boundary conditions used in the channel exchange.

3.4.1 Answer

3.4.1.1 Core point

There are two different uses of signs here, and separating them resolves the confusion.

In the open-string channel, the relative sign between the two ends of the strip determines whether the open-string fermion is periodic or antiperiodic around the doubled interval. That is what gives the open-string R or NS sector.

In the closed-string boundary-state channel, the sign

$$\eta = \pm 1$$

in

$$\left(b^\mu_r + i\eta D^\mu{}_\nu \bar{b}^\nu_{-r}\right)|B, \eta\rangle = 0$$

is a boundary gluing label. It says which linear combination of left- and right-moving fermions vanishes at that boundary. A single boundary label η is not itself the open-string R/NS sector. On an annulus with two boundary states,

$$\langle B, \eta_1|e^{-2\pi l H_{\text{closed}}}|B, \eta_2\rangle,$$

the relative sign

$$\eta_1\eta_2$$

does determine the spin structure, but not by the naive identification

$$s_0 s_l = \eta_1\eta_2.$$

With the conventions in the question, the open-channel endpoint sign is instead

$$\boxed{s_0 s_l = -\eta_1\eta_2.}$$

The extra minus sign comes from the open-closed channel exchange: the two boundaries of the closed-string cylinder induce opposite orientations on the open-string strip, and a worldsheet fermion is a half-differential. Equivalently, the square-root Jacobian in the map

$$\sigma^\pm \mapsto \sigma'^\pm = \pm\sigma^\pm$$

contributes a relative phase between the two ends. This is not a new sign in the bosonic gluing condition; it is specifically a sign/phase in the spin lift of the coordinate transformation. This is why

$$\eta_1\eta_2 = +1 \quad\Longrightarrow\quad s_0 s_l = -1 \quad\Longrightarrow\quad \text{open NS},$$

whereas

$$\eta_1\eta_2 = -1 \quad\Longrightarrow\quad s_0 s_l = +1 \quad\Longrightarrow\quad \text{open R}.$$

The GSO equations

$$(-1)^F|\mathrm{Dp}, \eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp}, -\eta\rangle_{\mathrm{NS}}, \qquad (-1)^{\bar{F}}|\mathrm{Dp}, \eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp}, -\eta\rangle_{\mathrm{NS}}$$

come from two facts:

1. $(-1)^F$ flips the sign of every left-moving fermion oscillator b_r^i, while $(-1)^{\bar{F}}$ flips the sign of every right-moving fermion oscillator $\bar{b}_r^i$.
2. In the NS sector, the ground state is conventionally assigned odd fermion number:

$$(-1)^F|0\rangle_{\rm NS} = -|0\rangle_{\rm NS}, \qquad (-1)^{\bar{F}}|0\rangle_{\rm NS} = -|0\rangle_{\rm NS}.$$

3.4.1.2 Open-channel relative sign versus boundary-state label

For an open string on a strip, the fermion boundary conditions may be written schematically as

$$\psi_+^\mu(0) = s_0 D^\mu{}_\nu \psi_-^\nu(0), \qquad \psi_+^\mu(l) = s_l D^\mu{}_\nu \psi_-^\nu(l),$$

where

$$s_0, s_l = \pm 1.$$

A simultaneous flip

$$s_0 \mapsto -s_0, \qquad s_l \mapsto -s_l$$

can be absorbed by redefining one chiral fermion, so only the relative sign

$$s_0 s_l$$

is invariant. This relative sign determines whether the open-string fermion is periodic or antiperiodic on the doubled interval. With the convention in the question,

$$s_0 s_l = +1 \quad \text{is called R}, \qquad s_0 s_l = -1 \quad \text{is called NS}.$$

Equivalently, one can put all the relative sign at $\sigma = l$ and write it as the open-channel η.

In the closed-string channel, the cylinder is viewed as a closed string propagating between two boundary states. Each boundary state has its own fermionic gluing sign:

$$(\psi_+^\mu + i\eta_a D^\mu{}_\nu \psi_-^\nu)\,|B, \eta_a\rangle = 0, \qquad a = 1, 2.$$

Here η_a is attached to a boundary, not to the open string as a whole. Therefore η_a is not the open R/NS label. The open-channel spin structure is recovered only after comparing the two boundaries.

There is one further sign that matters. When we rotate the cylinder and reinterpret the closed-string propagation as an open-string trace, the two closed-string boundaries become the two ends of the open string with opposite induced boundary orientations. In the convention used in the question, the result is

$$s_0 s_l = -\eta_1 \eta_2.$$

Let us unpack this statement carefully.

3.4.1.3 Boundary orientation and why fermions see it

Take the closed-channel cylinder coordinates to be

$$0 \le \tau_c \le T, \qquad \sigma_c \sim \sigma_c + 2\pi,$$

with worldsheet orientation

$$d\tau_c \wedge d\sigma_c.$$

The two boundary circles are

$$\tau_c = 0, \qquad \tau_c = T.$$

The induced boundary orientation is defined by the convention that

$$\text{outward normal} \wedge \text{boundary orientation} = \text{surface orientation}.$$

At the upper boundary $\tau_c = T$, the outward normal is $+\partial_{\tau_c}$. Therefore the induced boundary tangent is $+\partial_{\sigma_c}$. At the lower boundary $\tau_c = 0$, the outward normal is $-\partial_{\tau_c}$. Therefore the induced boundary tangent is $-\partial_{\sigma_c}$. Thus the two boundary components have opposite induced orientations:

$$\tau_c = T: \quad +\partial_{\sigma_c}, \qquad \tau_c = 0: \quad -\partial_{\sigma_c}.$$

This does not mean that one cannot use the same coordinate symbol σ_c on both boundaries. One can. The point is that σ_c increasing agrees with the induced orientation on one boundary and disagrees with it on the other.

A bosonic coordinate X^μ is a scalar on the worldsheet:

$$X'^\mu(\sigma') = X^\mu(\sigma).$$

So if we reverse a boundary coordinate,

$$\sigma' = -\sigma,$$

then X^μ itself does not acquire a square-root phase. Its derivatives transform as ordinary vectors or one-forms. Any sign from reversing the normal derivative is already part of the usual distinction between Neumann and Dirichlet gluing. There is no extra spin-lift ambiguity.

By contrast, the chiral fermions are spin fields on the worldsheet. Locally, in light-cone coordinates, one may view them as half-differentials:

$$\psi_+ \, (d\sigma^+)^{1/2}, \qquad \psi_- \, (d\sigma^-)^{1/2}.$$

Under a change of chiral coordinate,

$$\sigma^\pm \mapsto \sigma'^\pm(\sigma^\pm),$$

the components transform as

$$\psi'_\pm(\sigma'^\pm) = \left(\frac{\partial \sigma'^\pm}{\partial \sigma^\pm}\right)^{-1/2} \psi_\pm(\sigma^\pm).$$

If the coordinate transformation reverses the relevant chiral coordinate,

$$\sigma'^\pm = -\sigma^\pm,$$

then

$$\frac{\partial \sigma'^\pm}{\partial \sigma^\pm} = -1,$$

and therefore the fermion picks a square-root phase:

$$(-1)^{-1/2} = \pm i.$$

The choice of sign is a choice of spin lift. Once the spin lift is fixed consistently, the two boundaries of the cylinder get opposite induced orientation factors. In the standard boundary-state convention this is exactly the origin of the i in

$$(\psi_+^\mu + i\eta D^\mu{}_\nu \psi_-^\nu)\,|B,\eta\rangle = 0,$$

and of the extra minus in

$$s_0 s_l = -\eta_1\eta_2.$$

Said differently: each closed boundary gluing condition contains a factor $i\eta_a$ because rotating an open boundary into a closed boundary involves a square root of -1. When we compare the two boundaries to reconstruct the open-string relative sign, the two i-type factors multiply to

$$i^2 = -1,$$

up to the convention-dependent placement of signs. This is the extra minus between $s_0 s_l$ and $\eta_1\eta_2$.

So the closed-channel overlaps distinguish

$$\eta_1\eta_2 = +1 \qquad \text{from} \qquad \eta_1\eta_2 = -1.$$

In the NS-NS oscillator overlap, one finds factors of the schematic form

$$\prod_{r>0} \left(1 + \eta_1\eta_2\, q^{2r}\right).$$

Thus the same-sign overlap and opposite-sign overlap give different theta functions:

$$\eta_1\eta_2 = +1 \quad \Longrightarrow \quad \vartheta_3,$$

whereas

$$\eta_1\eta_2 = -1 \quad \Longrightarrow \quad \vartheta_4.$$

After the modular transformation to the open-string channel, the same-sign closed overlap gives the open NS character, while the opposite-sign closed overlap gives the open R character. In theta-function language, this is the familiar exchange in which the closed-channel ϑ_4 contribution turns into the open-channel ϑ_2 contribution. So the slogan is:

$$\boxed{\eta_1\eta_2 = +1 \text{ gives open NS, while } \eta_1\eta_2 = -1 \text{ gives open R.}}$$

This is consistent with the open-string rule

$$s_0 s_l = +1 \text{ for R}, \qquad s_0 s_l = -1 \text{ for NS},$$

because the open and closed signs are related by

$$s_0 s_l = -\eta_1\eta_2.$$

3.4.1.4 The NS-NS boundary state

Ignoring normalization and ghosts, the NS-NS fermionic part of the Dp-brane boundary state is

$$|\mathrm{D}p,\eta\rangle_{\mathrm{NS}} = \exp\left[-i\eta\sum_{r>0} b^i_{-r} D_{ij}\bar{b}^j_{-r}\right]|0\rangle_{\mathrm{NS}}.$$

Here

$$r\in\mathbb{Z}+\frac{1}{2}, \qquad r>0,$$

and i,j run over the physical transverse directions in light-cone gauge. The matrix D_{ij} is

$$D_{ij} = \begin{cases} \delta_{ij}, & \text{Neumann direction,} \\ -\delta_{ij}, & \text{Dirichlet direction.} \end{cases}$$

The precise sign convention in front of $i\eta$ can differ between books. What matters for the following derivation is that η appears linearly in the exponent:

$$\eta \mapsto -\eta$$

is equivalent to flipping the sign of the fermion bilinear in the exponential.

The state satisfies

$$\left(b^i_r + i\eta D^i{}_j \bar{b}^j_{-r}\right)|\mathrm{D}p,\eta\rangle_{\mathrm{NS}} = 0.$$

The sign η therefore labels the two possible fermionic gluing conditions.

3.4.1.5 Action of $(-1)^F$ on the oscillators

The left-moving fermion number operator F is defined so that

$$(-1)^F b^i_r (-1)^F = -b^i_r,$$

while it leaves right-moving oscillators invariant:

$$(-1)^F \bar{b}^i_r (-1)^F = \bar{b}^i_r.$$

Equivalently,

$$(-1)^F b^i_{-r} = -b^i_{-r}(-1)^F, \qquad (-1)^F \bar{b}^i_{-r} = \bar{b}^i_{-r}(-1)^F.$$

Therefore, acting on the bilinear,

$$(-1)^F\left(b^i_{-r} D_{ij}\bar{b}^j_{-r}\right)(-1)^F = -b^i_{-r} D_{ij}\bar{b}^j_{-r}.$$

Thus $(-1)^F$ changes the sign of the exponent:

$$(-1)^F \exp\left[-i\eta\sum_{r>0} b^i_{-r}D_{ij}\bar{b}^j_{-r}\right](-1)^F = \exp\left[+i\eta\sum_{r>0} b^i_{-r}D_{ij}\bar{b}^j_{-r}\right].$$

But this is the same exponential that appears in

$$|\mathrm{Dp}, -\eta\rangle_{\mathrm{NS}},$$

because replacing η by $-\eta$ gives

$$-i(-\eta) = +i\eta.$$

Now we must also act on the NS ground state. With the standard superstring convention,

$$(-1)^F|0\rangle_{\mathrm{NS}} = -|0\rangle_{\mathrm{NS}}.$$

Therefore

$$\begin{aligned}(-1)^F|\mathrm{Dp},\eta\rangle_{\mathrm{NS}} &= (-1)^F \exp\left[-i\eta\sum_{r>0} b^i_{-r}D_{ij}\bar{b}^j_{-r}\right]|0\rangle_{\mathrm{NS}} \\ &= \exp\left[+i\eta\sum_{r>0} b^i_{-r}D_{ij}\bar{b}^j_{-r}\right](-1)^F|0\rangle_{\mathrm{NS}} \\ &= -\exp\left[+i\eta\sum_{r>0} b^i_{-r}D_{ij}\bar{b}^j_{-r}\right]|0\rangle_{\mathrm{NS}} \\ &= -|\mathrm{Dp},-\eta\rangle_{\mathrm{NS}}.\end{aligned}$$

This proves

$$\boxed{(-1)^F|\mathrm{Dp},\eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp},-\eta\rangle_{\mathrm{NS}}.}$$

3.4.1.6 Action of $(-1)^{\bar{F}}$

The right-moving fermion number operator acts oppositely:

$$(-1)^{\bar{F}}\bar{b}^i_r(-1)^{\bar{F}} = -\bar{b}^i_r,$$

while leaving left-moving oscillators invariant:

$$(-1)^{\bar{F}}b^i_r(-1)^{\bar{F}} = b^i_r.$$

Therefore

$$(-1)^{\bar{F}}\left(b^i_{-r}D_{ij}\bar{b}^j_{-r}\right)(-1)^{\bar{F}} = -b^i_{-r}D_{ij}\bar{b}^j_{-r}.$$

So $(-1)^{\bar{F}}$ also sends

$$\eta \mapsto -\eta$$

in the oscillator exponential.

The right-moving NS ground state is also odd:

$$(-1)^{\bar{F}}|0\rangle_{\mathrm{NS}} = -|0\rangle_{\mathrm{NS}}.$$

Repeating the same calculation gives

$$\begin{aligned}(-1)^{\bar{F}}|\mathrm{Dp},\eta\rangle_{\mathrm{NS}} &= \exp\left[+i\eta\sum_{r>0} b^i_{-r}D_{ij}\bar{b}^j_{-r}\right](-1)^{\bar{F}}|0\rangle_{\mathrm{NS}} \\ &= -\exp\left[+i\eta\sum_{r>0} b^i_{-r}D_{ij}\bar{b}^j_{-r}\right]|0\rangle_{\mathrm{NS}} \\ &= -|\mathrm{Dp},-\eta\rangle_{\mathrm{NS}}.\end{aligned}$$

Thus

$$\boxed{(-1)^{\bar{F}}|\mathrm{Dp},\eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp},-\eta\rangle_{\mathrm{NS}}.}$$

3.4.1.7 Why the NS ground state is assigned odd fermion number

The minus sign in the final result is sometimes the most confusing part. It is not produced by the oscillator exponential alone. The oscillator exponential only gives

$$\eta \mapsto -\eta.$$

The extra overall minus sign comes from the convention

$$(-1)^F|0\rangle_{\text{NS}} = -|0\rangle_{\text{NS}}.$$

This convention is the standard one in the NS sector of the superstring. It makes the NS tachyon odd under $(-1)^F$, so the usual GSO projection removes it and keeps states with one fermionic oscillator, such as

$$b^i_{-1/2}|0\rangle_{\text{NS}}.$$

Indeed,

$$(-1)^F b^i_{-1/2}|0\rangle_{\text{NS}} = \left((-1)^F b^i_{-1/2}(-1)^F\right)(-1)^F|0\rangle_{\text{NS}} = (-b^i_{-1/2})(-|0\rangle_{\text{NS}}) = b^i_{-1/2}|0\rangle_{\text{NS}}.$$

So the massless NS vector is even and survives the usual projection, while the NS ground state is odd.

3.4.1.8 GSO-invariant NS-NS boundary-state combination

Because

$$(-1)^F|\text{Dp},\eta\rangle_{\text{NS}} = -|\text{Dp},-\eta\rangle_{\text{NS}},$$

neither

$$|\text{Dp},+\rangle_{\text{NS}}$$

nor

$$|\text{Dp},-\rangle_{\text{NS}}$$

is separately GSO invariant.

The invariant NS-NS combination is

$$|\text{Dp}\rangle_{\text{NSNS}} \propto |\text{Dp},+\rangle_{\text{NS}} - |\text{Dp},-\rangle_{\text{NS}}.$$

Check:

$$\begin{aligned}(-1)^F\left(|\text{Dp},+\rangle_{\text{NS}} - |\text{Dp},-\rangle_{\text{NS}}\right) &= -|\text{Dp},-\rangle_{\text{NS}} + |\text{Dp},+\rangle_{\text{NS}} \\ &= |\text{Dp},+\rangle_{\text{NS}} - |\text{Dp},-\rangle_{\text{NS}}.\end{aligned}$$

The same check works for $(-1)^{\bar{F}}$. This is why the physical NS-NS part of the D-brane boundary state is built from a difference of the two η gluing states.

3.4.1.9 Checks

If the NS ground state were assigned even fermion number instead, the derivation would give

$$(-1)^F|\text{Dp},\eta\rangle_{\text{NS}} = |\text{Dp},-\eta\rangle_{\text{NS}}.$$

So the overall minus sign is convention-dependent in the precise sense that it follows from the convention for the vacuum fermion number. The standard superstring convention is the odd NS vacuum convention used above.

Also, applying both left and right fermion parities gives

$$(-1)^{F+\bar{F}}|\text{Dp},\eta\rangle_{\text{NS}} = |\text{Dp},\eta\rangle_{\text{NS}}.$$

Indeed, the first action sends

$$|\eta\rangle \mapsto -|-\eta\rangle,$$

and the second sends

$$|-\eta\rangle \mapsto -|\eta\rangle.$$

The two minus signs cancel. This is consistent with the fact that the NS-NS boundary state lies in a closed-string sector with matched left and right GSO parity after projection.

3.4.1.10 Result

The open-channel R/NS sign is a relative sign between the two strip boundaries. The closed-channel boundary-state sign

$$\eta = \pm 1$$

is instead the local fermionic gluing label:

$$\left(b^i_r + i\eta D^i{}_j \bar{b}^j_{-r}\right)|\mathrm{Dp}, \eta\rangle = 0.$$

On an annulus, the open spin structure is determined by

$$\eta_1 \eta_2,$$

not by either η_1 or η_2 alone. More precisely, in the conventions of the question,

$$\boxed{s_0 s_l = -\eta_1 \eta_2.}$$

Therefore

$$\eta_1\eta_2 = +1 \quad \Longleftrightarrow \quad s_0 s_l = -1 \quad \Longleftrightarrow \quad \text{open NS},$$

while

$$\eta_1\eta_2 = -1 \quad \Longleftrightarrow \quad s_0 s_l = +1 \quad \Longleftrightarrow \quad \text{open R}.$$

The extra minus sign is the spinor orientation factor from the open-closed channel exchange. Bosonic coordinates do not acquire this factor because they are worldsheet scalars, while fermions do because their components transform with

$$\left(\frac{\partial \sigma'}{\partial \sigma}\right)^{-1/2}.$$

In the NS-NS sector,

$$|\mathrm{Dp}, \eta\rangle_{\mathrm{NS}} = \exp\left[-i\eta \sum_{r>0} b^i_{-r} D_{ij} \bar{b}^j_{-r}\right]|0\rangle_{\mathrm{NS}}.$$

The fermion parities act as

$$(-1)^F b^i_r (-1)^F = -b^i_r, \qquad (-1)^{\bar{F}} \bar{b}^i_r (-1)^{\bar{F}} = -\bar{b}^i_r,$$

and the NS vacuum convention is

$$(-1)^F |0\rangle_{\mathrm{NS}} = (-1)^{\bar{F}} |0\rangle_{\mathrm{NS}} = -|0\rangle_{\mathrm{NS}}.$$

Therefore

$$\boxed{(-1)^F |\mathrm{Dp}, \eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp}, -\eta\rangle_{\mathrm{NS}}, \qquad (-1)^{\bar{F}} |\mathrm{Dp}, \eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp}, -\eta\rangle_{\mathrm{NS}}.}$$

Consequently the GSO-invariant NS-NS boundary state is proportional to

$$\boxed{|\mathrm{Dp}\rangle_{\mathrm{NSNS}} \propto |\mathrm{Dp}, +\rangle_{\mathrm{NS}} - |\mathrm{Dp}, -\rangle_{\mathrm{NS}}.}$$

3.5 Q5: Force signs from even-spin, odd-spin, and R-R exchange

Consider two parallel Dp-branes.There are competing forces between these two D-branes, namely attraction due to exchange of gravitons and dilatons and repulsion due to the exchange of the R-R form. These two forces precisely cancel, independent of the distance between the two branes. Q: it is said that if the exchange particle has even spin, equal sign charges attract and opposite charges repel. If the spin is odd, the situation is reversed. Prove this in details. The exchange of an antisymmetric tensor particle leads to repulsion between equal charges.

Textbook location and related material. Chapter 9, Section 9.3 (pp. 235–241): D-brane boundary states, NS–NS and R–R exchange, and the BPS force cancellation. Section 6.8 gives the corresponding long-distance field-theory comparison.

3.5.1 Answer

3.5.1.1 Core point

The question is why the sign of a force depends on the spin of the exchanged field, and in particular why R-R form exchange between two identical parallel D-branes is repulsive.

The clean rule for static sources is:

$$\boxed{\text{even-spin exchange: equal charges attract, opposite charges repel,}}$$

whereas

$$\boxed{\text{odd-spin exchange: equal charges repel, opposite charges attract.}}$$

For ordinary fields this means:

$$\text{scalar exchange} \quad \Rightarrow \quad \text{attraction for equal scalar charges,}$$
$$\text{vector exchange} \quad \Rightarrow \quad \text{repulsion for equal electric charges,}$$

and

$$\text{graviton exchange} \quad \Rightarrow \quad \text{attraction for positive masses.}$$

For Dp-branes, the NS-NS fields couple to the tension T_p and produce attraction between two parallel branes. The R-R potential C_{p+1} couples to the oriented R-R charge μ_p and produces repulsion between two branes with equal R-R charge. For BPS branes, these two effects exactly cancel:

$$V_{\mathrm{NSNS}}(r) + V_{\mathrm{RR}}(r) = 0.$$

3.5.1.2 Static potential from integrating out a field

Let a field Φ couple linearly to a source J:

$$S[\Phi, J] = \frac{1}{2}\Phi K \Phi + J\Phi,$$

where K is the kinetic operator. Solving the classical equation

$$K\Phi + J = 0$$

gives

$$\Phi = -K^{-1}J.$$

Substituting back into the action gives the source-source effective action

$$S_{\mathrm{eff}}[J] = -\frac{1}{2}JK^{-1}J.$$

For two separated static sources,

$$J = J_1 + J_2,$$

the interaction term is

$$S_{\mathrm{int}} = -J_1 K^{-1} J_2.$$

For a static potential, the effective action is related to the potential by

$$S_{\mathrm{int}} = -\int dt\, V(r).$$

Therefore the sign of the potential is determined by the sign of the propagator contraction between the two source currents.

This sounds abstract, so let us do the standard cases explicitly.

3.5.1.3 Scalar exchange: equal charges attract

Take a real scalar ϕ with mostly-plus signature,

$$S_\phi = \int d^D x \left[-\frac{1}{2}\partial_\mu \phi \, \partial^\mu \phi + J\phi \right].$$

For static sources, the energy functional is

$$E[\phi] = \int d^{D-1} x \left[\frac{1}{2}(\nabla\phi)^2 - J\phi \right].$$

The static equation is

$$-\nabla^2 \phi = J.$$

Let

$$J(x) = q_1 \delta(x - x_1) + q_2 \delta(x - x_2).$$

The Green's function satisfies

$$-\nabla^2 G(r) = \delta^{(D-1)}(r), \qquad G(r) > 0.$$

The solution is

$$\phi(x) = q_1 G(x - x_1) + q_2 G(x - x_2).$$

Substituting the solution back into the energy gives

$$E_{\text{on-shell}} = -\frac{1}{2} \int d^{D-1} x \, J\phi.$$

The cross term is therefore

$$V_\phi(r) = -q_1 q_2 G(r).$$

Thus for equal signs,

$$q_1 q_2 > 0 \quad \Longrightarrow \quad V_\phi(r) < 0.$$

A negative potential that becomes more negative as the sources approach means attraction. So scalar exchange attracts equal scalar charges and repels opposite scalar charges.

3.5.1.4 Vector exchange: equal charges repel

Now take Maxwell theory:

$$S_A = \int d^D x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J_\mu A^\mu \right].$$

For static electric sources, only

$$J^0 = \rho$$

is nonzero. The interaction is mediated by A_0. The electrostatic equation is

$$-\nabla^2 A_0 = \rho.$$

Thus a point charge produces

$$A_0(x) = qG(x - x_a).$$

The potential energy of a second charge q_2 in the potential produced by the first charge is

$$V_A(r) = q_2 A_0^{(1)}(x_2) = q_1 q_2 G(r).$$

Equivalently, the electric field is

$$E_i = -\partial_i A_0,$$

and the field energy

$$E_{\text{field}} = \int d^{D-1} x \frac{1}{2} E_i E_i$$

has a cross term

$$\int d^{D-1}x\, \nabla A_0^{(1)} \cdot \nabla A_0^{(2)} = q_1 q_2 G(r).$$

So

$$V_A(r) = +q_1 q_2 G(r).$$

In covariant language, the same sign comes from the contraction of the vector propagator with the time components of the current. In mostly-plus signature,

$$J_\mu A^\mu = J_0 A^0 + \cdots = -J^0 A^0 + \cdots.$$

The propagator contraction for static sources contains

$$\eta_{00} = -1.$$

Together with the standard relation between exchange amplitude and potential, this reverses the scalar sign. Thus

$$q_1 q_2 > 0 \quad \Longrightarrow \quad V_A(r) > 0,$$

which means repulsion. Opposite charges have

$$q_1 q_2 < 0,$$

so

$$V_A(r) < 0,$$

which means attraction.

The vector case is the prototype of odd-spin exchange.

3.5.1.5 Graviton exchange: positive masses attract

The linearized graviton couples to the stress tensor as

$$S_{\text{int}} = \frac{\kappa}{2} \int d^D x\, h_{\mu\nu} T^{\mu\nu}.$$

For a nonrelativistic static source,

$$T^{00} \simeq m\, \delta^{(D-1)}(x - x_a),$$

and the spatial components T^{ij} are negligible. The graviton propagator numerator in de Donder gauge is

$$P_{\mu\nu,\rho\sigma} = \frac{1}{2}\left(\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho} - \frac{2}{D-2}\eta_{\mu\nu}\eta_{\rho\sigma}\right).$$

For static sources, the relevant component is

$$P_{00,00} = \frac{1}{2}\left(2\eta_{00}^2 - \frac{2}{D-2}\eta_{00}^2\right) = 1 - \frac{1}{D-2} = \frac{D-3}{D-2}.$$

This is positive. Including the universal sign relating the exchange amplitude to the potential, one finds

$$V_h(r) = -\frac{\kappa^2}{4}\frac{D-3}{D-2} m_1 m_2 G(r).$$

Therefore

$$m_1 m_2 > 0 \quad \Longrightarrow \quad V_h(r) < 0.$$

Positive masses attract.

This is the spin-2 analogue of scalar attraction. The two time indices in T^{00} effectively produce

$$\eta_{00}^2 = +1,$$

whereas the vector current had one time index and produced

$$\eta_{00} = -1.$$

This is the intuitive origin of the even-spin versus odd-spin rule.

3.5.1.6 General spin sign rule for static sources

For a symmetric spin-s field coupled schematically as

$$h_{\mu_1\cdots\mu_s}J^{\mu_1\cdots\mu_s},$$

a static source mainly has the component

$$J^{00\cdots 0}.$$

The propagator contraction contains one metric factor for each time index, schematically

$$\eta_{00}^s = (-1)^s.$$

After including the standard relation between exchange amplitude and potential, the static potential behaves as

$$V_s(r) \sim (-1)^{s+1} q_1 q_2 G(r).$$

Thus:

$$s \text{ even} \quad \Longrightarrow \quad V_s(r) \sim -q_1 q_2 G(r),$$

so equal charges attract and opposite charges repel.

For

$$s \text{ odd},$$

one gets

$$V_s(r) \sim +q_1 q_2 G(r),$$

so equal charges repel and opposite charges attract.

This is a useful rule of thumb. For gauge fields with traces, constraints, or mixed symmetry, the detailed numerator matters, but the static sign is still controlled by how the source indices contract with the metric and by the sign of the kinetic term.

3.5.1.7 R-R form exchange between D-branes

A Dp-brane couples electrically to the R-R $(p+1)$-form potential:

$$S_{\text{WZ}} = \mu_p \int_{\text{Dp}} C_{p+1}.$$

For a static flat Dp-brane extended along

$$x^0, x^1, \dots, x^p,$$

this coupling is

$$S_{\text{WZ}} = \mu_p \int d^{p+1}\xi\, C_{01\cdots p}.$$

So the brane is a source for the component

$$C_{01\cdots p}.$$

The R-R kinetic term has the conventional sign

$$S_{\text{RR}} = -\frac{1}{2}\int |F_{p+2}|^2$$

up to normalization. The propagator contraction between two static parallel brane currents contains the metric factor

$$\eta_{00}\eta_{11}\cdots\eta_{pp} = -1.$$

There is exactly one time index and p spatial indices. Therefore the static sign is the same vector-like sign as electromagnetic exchange, not the scalar-like sign.

Integrating out C_{p+1} gives a source-source interaction. For two parallel branes with the same orientation, the R-R currents have the same sign:

$$\mu_1\mu_2 > 0.$$

The resulting potential has the vector-like sign

$$V_{\mathrm{RR}}(r) = +\mu_1\mu_2 G_{9-p}(r)$$

up to a positive normalization. Thus equal R-R charges repel.

Why is this vector-like rather than scalar-like? The important physical point is that a p-form gauge field couples to an oriented conserved brane current, just as a vector field couples to an oriented particle current. Reversing the brane orientation reverses the R-R charge:

$$\mu_p \mapsto -\mu_p.$$

So a brane and an anti-brane have opposite R-R charges. Consequently:

$$\text{brane-brane:} \quad \mu_1\mu_2 > 0 \quad \Longrightarrow \quad V_{\mathrm{RR}} > 0 \quad \Longrightarrow \quad \text{repulsion},$$

whereas

$$\text{brane-antibrane:} \quad \mu_1\mu_2 < 0 \quad \Longrightarrow \quad V_{\mathrm{RR}} < 0 \quad \Longrightarrow \quad \text{attraction.}$$

One sometimes loosely says that antisymmetric tensor exchange is “odd-spin-like” in its force sign. More precisely, a R-R form is a gauge potential coupling to an oriented conserved current. The sign of its static exchange between equal charges is the same as for electromagnetic vector exchange: equal charges repel.

3.5.1.8 NS-NS exchange between D-branes

The NS-NS sector contains the graviton $g_{\mu\nu}$, dilaton Φ, and Kalb-Ramond field $B_{\mu\nu}$. For two identical static parallel Dp-branes with no worldvolume flux, the relevant universal attractive contribution comes from the tension coupling in the DBI action:

$$S_{\mathrm{DBI}} = -T_p \int d^{p+1}\xi \, e^{-\Phi} \sqrt{-\det P[g+B]}.$$

Expanding around flat space gives couplings to the graviton and dilaton proportional to T_p. Since

$$T_p > 0$$

for both a brane and an anti-brane, the NS-NS force is attractive for both brane-brane and brane-antibrane pairs:

$$V_{\mathrm{NSNS}}(r) = -T_1 T_2 G_{9-p}(r)$$

up to a positive normalization and the correct tensor coefficient.

For two identical BPS Dp-branes,

$$T_1 = T_2 = T_p, \qquad \mu_1 = \mu_2 = \mu_p,$$

and supersymmetry fixes

$$|\mu_p| = T_p$$

in matching units. The total static potential is therefore

$$V_{\mathrm{total}}(r) = V_{\mathrm{NSNS}}(r) + V_{\mathrm{RR}}(r) = -T_p^2 G_{9-p}(r) + \mu_p^2 G_{9-p}(r) = 0.$$

This is the no-force condition for parallel BPS D-branes.

For a brane-antibrane pair,

$$\mu_1\mu_2 < 0,$$

so the R-R term becomes attractive:

$$V_{\mathrm{RR}}(r) < 0.$$

The NS-NS term is also attractive, so the forces add rather than cancel:

$$V_{\text{brane-antibrane}}(r) < 0.$$

3.5.1.9 Checks

For ordinary electromagnetism in four dimensions,

$$G_3(r) = \frac{1}{4\pi r},$$

so

$$V_A(r) = \frac{q_1 q_2}{4\pi r}.$$

Equal charges repel because $V_A > 0$ and decreases as r increases.

For Newtonian gravity,

$$V_h(r) = -\frac{G_N m_1 m_2}{r}.$$

Positive masses attract because $V_h < 0$ and becomes more negative as r decreases.

For D-branes, the schematic long-distance potentials are

$$V_{\mathrm{NSNS}}(r) \sim -T_p^2 G_{9-p}(r), \qquad V_{\mathrm{RR}}(r) \sim +\mu_p^2 G_{9-p}(r).$$

For BPS branes,

$$T_p^2 = \mu_p^2,$$

so they cancel.

3.5.1.10 Result

For static exchange of a spin-s field between sources with charges q_1, q_2,

$$V_s(r) \sim (-1)^{s+1} q_1 q_2 G(r).$$

Therefore:

$$s \text{ even} \quad \Longrightarrow \quad V_s(r) \sim -q_1 q_2 G(r),$$

so equal charges attract and opposite charges repel.

For

$$s \text{ odd},$$

one has

$$V_s(r) \sim +q_1 q_2 G(r),$$

so equal charges repel and opposite charges attract.

For Dp-branes,

$$S_{\mathrm{DBI}} = -T_p \int e^{-\Phi} \sqrt{-\det P[g+B]}, \qquad S_{\mathrm{WZ}} = \mu_p \int C_{p+1}.$$

The NS-NS exchange is attractive:

$$V_{\mathrm{NSNS}}(r) \sim -T_1 T_2 G_{9-p}(r).$$

The R-R form exchange is repulsive for equal R-R charges:

$$V_{\mathrm{RR}}(r) \sim +\mu_1 \mu_2 G_{9-p}(r).$$

For two identical BPS Dp-branes,

$$T_p = |\mu_p|,$$

so

$$\boxed{V_{\mathrm{NSNS}}(r) + V_{\mathrm{RR}}(r) = 0.}$$

3.6 Q6: Type II torus partition function and supersymmetric cancellation

Consider the type II closed superstring one-loop partition function $Z(\tau,\bar{\tau})$ on a torus with moduli τ. The explicit computation shows that

$$Z(\tau,\bar{\tau}) = \frac{1}{4}\frac{1}{(\mathrm{Im}\tau)^4}\frac{1}{|\eta|^{24}}|\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4|^2 = 0$$

It is said that this is a consequence of a supersymmetric spectrum: the contributions from space-time bosons and fermions cancel. Why? Suppose we have a supersymmetric space-time theory, write down its one-loop partition function and show this result.

> **Textbook location and related material.** Chapter 9, Section 9.1 (pp. 223–232): the type-II torus amplitude, GSO sum, and Jacobi identity. Section 8.3 supplies the supersymmetric spectrum whose level-by-level degeneracy explains the cancellation.

3.6.1 Answer

3.6.1.1 Core point

The question is why the type II torus vacuum amplitude

$$Z(\tau,\bar{\tau}) = \frac{1}{4}\frac{1}{(\mathrm{Im}\,\tau)^4}\frac{1}{|\eta|^{24}}\left|\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4\right|^2$$

vanishes, and how this is related to spacetime supersymmetry.

The key point is that the torus vacuum amplitude is not an ordinary thermal partition function

$$\mathrm{Tr}\, e^{-\beta H}.$$

It is a one-loop vacuum functional. In spacetime field theory, a one-loop vacuum diagram receives a plus sign from a bosonic loop and a minus sign from a fermionic loop. Therefore it is a supertrace:

$$\mathrm{Str}\, e^{-tH} = \mathrm{Tr}_{\mathcal{H}_B}\, e^{-tH} - \mathrm{Tr}_{\mathcal{H}_F}\, e^{-tH}.$$

If spacetime supersymmetry is unbroken, then bosonic and fermionic physical states pair with the same mass. Thus the supertrace cancels level by level:

$$\mathrm{Str}\, e^{-tH} = 0.$$

In type II string theory, the same statement is encoded worldsheet-wise by the Jacobi abstruse identity:

$$\vartheta_3^4(\tau) - \vartheta_4^4(\tau) - \vartheta_2^4(\tau) = 0.$$

3.6.1.2 Ordinary partition function versus vacuum amplitude

First separate two objects that are often both called "partition functions."

An ordinary thermal partition function is

$$Z_{\mathrm{thermal}}(\beta) = \mathrm{Tr}_{\mathcal{H}}\, e^{-\beta H}.$$

This does not vanish in a supersymmetric theory. Bosonic and fermionic states both contribute positively:

$$Z_{\mathrm{thermal}}(\beta) = \sum_B e^{-\beta E_B} + \sum_F e^{-\beta E_F}.$$

Supersymmetry makes the bosonic and fermionic energies equal, but it does not put a minus sign between them in the ordinary thermal trace.

By contrast, the one-loop vacuum amplitude is the logarithm of the vacuum functional. In a spacetime QFT it has the schematic form

$$\Gamma^{(1)} = \frac{1}{2}\,\mathrm{Tr}_B \log \Delta_B - \frac{1}{2}\,\mathrm{Tr}_F \log \Delta_F.$$

The minus sign is the usual minus sign from a closed fermion loop, equivalently from the Grassmann functional integral:

$$\int D\phi\, e^{-\frac{1}{2}\phi\Delta_B\phi} \propto (\det \Delta_B)^{-1/2},$$

whereas

$$\int D\psi\, e^{-\psi\Delta_F\psi} \propto \det \Delta_F.$$

Taking logarithms gives opposite signs for bosons and fermions.

Using the Schwinger representation

$$\log A = -\int_0^\infty \frac{dt}{t} e^{-tA}$$

up to an irrelevant constant, the one-loop vacuum amplitude becomes schematically

$$\Gamma^{(1)} \sim -\frac{1}{2}\int_0^\infty \frac{dt}{t}\,\mathrm{Str}\, e^{-t\Delta}.$$

This is why the relevant object is a supertrace, not an ordinary trace:

$$\mathrm{Str}_{\mathcal{H}}(\cdots) = \mathrm{Tr}_{\mathcal{H}}\left((-1)^{F_{\mathrm{st}}}(\cdots)\right).$$

3.6.1.3 Cancellation in a supersymmetric spectrum

Let the Hilbert space decompose into bosonic and fermionic states:

$$\mathcal{H} = \mathcal{H}_B \oplus \mathcal{H}_F.$$

Suppose supersymmetry is unbroken. Then for every massive bosonic state $|B\rangle$ there is a fermionic state

$$|F\rangle \sim Q|B\rangle$$

with the same energy or mass. This follows because the supercharges commute with the Hamiltonian:

$$[H, Q] = 0.$$

Thus if

$$H|B\rangle = E|B\rangle,$$

then

$$H(Q|B\rangle) = QH|B\rangle = E(Q|B\rangle).$$

So supersymmetry pairs bosons and fermions at equal energy.

Now compute the supertrace:

$$\mathrm{Str}\, e^{-tH} = \sum_B e^{-tE_B} - \sum_F e^{-tE_F}.$$

If the spectrum is paired with equal degeneracies at every energy,

$$d_B(E) = d_F(E),$$

then

$$\mathrm{Str}\, e^{-tH} = \sum_E \left(d_B(E) - d_F(E)\right) e^{-tE} = 0.$$

This is the field-theory version of the type II torus cancellation.

For a relativistic field theory in D noncompact dimensions, the one-loop vacuum energy can be written more explicitly as

$$\Lambda^{(1)} = \frac{1}{2}\sum_i (-1)^{F_i} d_i \int \frac{d^D k_E}{(2\pi)^D} \log(k_E^2 + m_i^2).$$

Using the Schwinger parameter,

$$\log(k_E^2 + m_i^2) = -\int_0^\infty \frac{dt}{t} e^{-t(k_E^2+m_i^2)},$$

and integrating over k_E,

$$\int \frac{d^D k_E}{(2\pi)^D} e^{-tk_E^2} = \frac{1}{(4\pi t)^{D/2}},$$

one obtains

$$\Lambda^{(1)} = -\frac{1}{2}\int_0^\infty \frac{dt}{t}\frac{1}{(4\pi t)^{D/2}} \sum_i (-1)^{F_i} d_i e^{-tm_i^2}.$$

The sum

$$\sum_i (-1)^{F_i} d_i e^{-tm_i^2}$$

is exactly a mass-weighted supertrace. If supersymmetry pairs all bosonic and fermionic states at the same mass, then

$$\sum_i (-1)^{F_i} d_i e^{-tm_i^2} = 0$$

for every t. Hence

$$\Lambda^{(1)} = 0.$$

3.6.1.4 String theory version

In closed string theory, the torus amplitude is the one-loop vacuum amplitude:

$$\Lambda^{(1)}_{\text{string}} = \int_{\mathcal{F}} \frac{d^2\tau}{(\operatorname{Im}\tau)^2} Z(\tau,\bar\tau),$$

where $\mathcal{F}$ is the fundamental domain of $SL(2,\mathbb{Z})$.

The integrand can be written as a trace over the closed-string Hilbert space:

$$Z(\tau,\bar\tau) = \mathrm{Tr}_{\mathcal{H}_{\text{closed}}}\left[(-1)^{F_{\text{st}}} q^{L_0 - c/24}\bar q^{\bar L_0 - \bar c/24}\right],$$

with

$$q = e^{2\pi i\tau},$$

after the GSO sum is included. This expression is schematic because in the worldsheet derivation one sums over spin structures rather than inserting a single explicit spacetime $(-1)^{F_{\text{st}}}$ from the start. But after GSO projection, the result is exactly the spacetime supertrace over physical bosonic and fermionic states.

The type II answer is

$$Z(\tau,\bar\tau) = \frac{1}{4}\frac{1}{(\operatorname{Im}\tau)^4}\frac{1}{|\eta(\tau)|^{24}}\left|\vartheta_3^4(\tau) - \vartheta_4^4(\tau) - \vartheta_2^4(\tau)\right|^2.$$

The three terms have a spacetime interpretation:

$$\vartheta_3^4 \quad \text{comes from NS states without } (-1)^F,$$
$$\vartheta_4^4 \quad \text{comes from NS states with } (-1)^F,$$

and

$$\vartheta_2^4 \quad \text{comes from R states.}$$

The relative signs are the GSO phases. They are chosen so that the physical spectrum is spacetime supersymmetric.

The Jacobi abstruse identity says

$$\vartheta_3^4(\tau) - \vartheta_4^4(\tau) - \vartheta_2^4(\tau) = 0.$$

Therefore

$$Z(\tau, \bar{\tau}) = 0$$

pointwise in τ, before even integrating over the modular fundamental domain.

3.6.1.5 Level-by-level interpretation

The identity

$$\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4 = 0$$

is stronger than saying only the total integrated amplitude vanishes. It says that the generating function for boson-minus-fermion degeneracies vanishes.

Schematically, one may expand the left-moving contribution as

$$\frac{1}{\eta^{12}} \left(\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4\right) = \sum_N \left(d_B(N) - d_F(N)\right) q^N.$$

The Jacobi identity says

$$d_B(N) - d_F(N) = 0$$

at every oscillator level N. The right-moving sector satisfies the same identity. For type II closed strings, the full closed-string spectrum is built from left and right movers with the level-matching constraint

$$L_0 = \bar{L}_0.$$

Spacetime supersymmetry then pairs every physical bosonic state with a physical fermionic state of the same mass.

Thus the worldsheet expression

$$\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4$$

is the character-level implementation of

$$\mathrm{Str}\, e^{-tM^2} = 0.$$

3.6.1.6 Why this is not a contradiction with nonzero thermal free energy

At finite temperature, spacetime fermions are antiperiodic around the Euclidean time circle. This explicitly breaks spacetime supersymmetry. Then the thermal partition function is

$$Z_{\text{thermal}}(\beta) = \mathrm{Tr}\, e^{-\beta H},$$

not

$$\mathrm{Str}\, e^{-\beta H}.$$

Therefore bosonic and fermionic contributions add instead of canceling. This is why a supersymmetric theory can have

$$\Lambda^{(1)} = 0$$

for the zero-temperature vacuum amplitude but a nonzero finite-temperature free energy.

3.6.1.7 Checks

For a simple supersymmetric harmonic spectrum with one bosonic state and one fermionic state at the same energy E,

$$Z_{\text{thermal}} = e^{-\beta E} + e^{-\beta E} = 2e^{-\beta E},$$

which is nonzero. But the one-loop vacuum supertrace gives

$$\operatorname{Str} e^{-\beta H} = e^{-\beta E} - e^{-\beta E} = 0.$$

This is the same distinction appearing in the type II torus amplitude.

For type II strings, the pointwise vanishing

$$Z(\tau, \bar{\tau}) = 0$$

comes from

$$\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4 = 0.$$

If supersymmetry is broken, for example by thermal boundary conditions or by a non-supersymmetric GSO projection, the spin-structure sum is modified and this identity no longer forces the integrand to vanish.

3.6.1.8 Result

The ordinary thermal partition function is

$$Z_{\text{thermal}}(\beta) = \operatorname{Tr} e^{-\beta H},$$

so bosons and fermions both contribute with positive sign.

The one-loop vacuum amplitude is instead a supertrace:

$$\Gamma^{(1)} \sim \int_0^\infty \frac{dt}{t} \operatorname{Str} e^{-tH}, \qquad \operatorname{Str}(\cdots) = \operatorname{Tr}\left((-1)^{F_{\text{st}}}(\cdots)\right).$$

In an unbroken supersymmetric theory,

$$\mathcal{H}_B \simeq \mathcal{H}_F$$

at equal energy or mass, hence

$$\operatorname{Str} e^{-tH} = 0.$$

For the type II closed superstring,

$$Z(\tau, \bar{\tau}) = \frac{1}{4} \frac{1}{(\operatorname{Im} \tau)^4} \frac{1}{|\eta|^{24}} \left|\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4\right|^2.$$

The Jacobi identity

$$\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4 = 0$$

is the worldsheet character form of spacetime Bose-Fermi cancellation. Therefore

$$\boxed{Z(\tau, \bar{\tau}) = 0}$$

for type II strings with unbroken spacetime supersymmetry.

3.7 Q7: Möbius amplitude and the boundary-crosscap cross term

Consider the type I string. The sum of the cylinder, the two Mobius and the Klein bottle diagrams must be a perfect square such that the sum of the disc and the crosscap tadpoles cancel. This is the requirement of no spacetime R-R 10-form charge.

These amplitudes can be computed as follows: 1. The loop-channel annulus amplitude for open strings stretched between a stack of N D9-branes:

$$\mathcal{A} = N^2 \frac{V_{10}}{(4\pi^2\alpha')^5} \int_0^\infty \frac{dt}{2t} \frac{1}{(2t)^5} \left(\frac{\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4}{2\eta^{12}} \right) (it)$$

The transformation to tree-channel via $t = \frac{1}{2l}$ gives (here $q^0 = 1$ mean 0th square)

$$\tilde{\mathcal{A}} = N^2 \frac{V_{10}}{(4\pi^2\alpha')^5} \int_0^\infty dl 2^{-6} (1_{\text{NS}} - 1_{\text{R}})[16q^0 + \mathcal{O}(e^{-2\pi l})]$$

2. The O9-O9 amplitude can be viewed as the type IIB superstring amplitude on a Klein bottle

$$\mathcal{K} = \frac{V_{10}}{(4\pi^2\alpha')^5} \int_0^\infty \frac{dt}{2t} \frac{1}{t^5} \left(\frac{\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4}{2\eta^{12}} \right) (2it)$$

Transform to tree-channel via $t = \frac{1}{4l}$ we have

$$\tilde{\mathcal{K}} = \frac{V_{10}}{(4\pi^2\alpha')^5} \int_0^\infty dl 2^4 \left(\frac{\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4}{2\eta^{12}} \right) (2il) = \frac{V_{10}}{(4\pi^2\alpha')^5} \int_0^\infty dl 2^4 (1_{\text{NS}} - 1_{\text{R}})[16q^0 + \mathcal{O}(e^{-2\pi l})]$$

3. The N D9-O9 amplitude can be computed in open string sector (Chan-Paton factors supposed to be SO (N)) as Mobius strip one-loop amplitude

$$\mathcal{M} = -N \frac{V_{10}}{(4\pi^2\alpha')^5} \int_0^\infty \frac{dt}{2t} \frac{1}{(2t)^5} \left(\frac{\vartheta_3^4 - \vartheta_4^4 - \vartheta_2^4}{2\eta^{12}} \right) \left(it + \frac{1}{2} \right)$$

Transform to tree-channel via $t = \frac{1}{8l}$ we have

$$\tilde{\mathcal{M}} = -N \frac{V_{10}}{(4\pi^2\alpha')^5} \int_0^\infty dl (1_{\text{NS}} - 1_{\text{R}})[16q^0 + \mathcal{O}(e^{-2\pi l})]$$

Combine these we find the factor

$$2^{-6}(N^2 - 2^6 N + 2^{10}) = 2^{-6}(N - 2^5)^2$$

The cancellation requires that $N = 32$ i.e. type I $SO(32)$ superstring is consistent.

Question: the square of the sum of the disc and the crosscap tadpoles should be

$$(\text{disc} + \text{crosscap})^2 = (\text{disc-disc}) + 2(\text{disc-crosscap}) + (\text{crosscap-crosscap})$$

Why in the above computation we only sum one copy of the Mobius amplitude, not two?

> **Textbook location and related material.** Chapter 9, Section 9.4 (pp. 241–251): type-I annulus, Klein-bottle, and Möbius amplitudes and R–R tadpole factorization. Section 6.7 (pp. 166–171) gives the general crosscap and Möbius formalism.

3.7.1 Answer

3.7.1.1 Core point

The question is why the square

$$(\text{disc} + \text{crosscap})^2 = (\text{disc-disc}) + 2(\text{disc-crosscap}) + (\text{crosscap-crosscap})$$

seems to contain two disc-crosscap terms, but the one-loop type I computation contains only one Möbius amplitude.

The answer is that the tree-channel vacuum amplitude has an overall factor $\frac{1}{2}$ in the closed-string exchange between tadpole sources. Schematically,

$$\Gamma_{\text{tree}} = \frac{1}{2} \langle B + C | \Delta | B + C \rangle.$$

Expanding this gives

$$\Gamma_{\text{tree}} = \frac{1}{2}\langle B|\Delta|B\rangle + \langle B|\Delta|C\rangle + \frac{1}{2}\langle C|\Delta|C\rangle.$$

Thus:

$$\tilde{\mathcal{A}} = \frac{1}{2}\langle B|\Delta|B\rangle, \qquad \tilde{\mathcal{M}} = \langle B|\Delta|C\rangle, \qquad \tilde{\mathcal{K}} = \frac{1}{2}\langle C|\Delta|C\rangle.$$

The single Möbius amplitude is already the full cross term. If one added two copies of the Möbius amplitude, one would double-count the boundary-crosscap exchange.

3.7.1.2 Factorization of tadpoles in tree channel

Let $|B\rangle$ be the boundary state sourced by the D9-branes and $|C\rangle$ the crosscap state sourced by the O9-plane. The tree-channel diagrams represent closed-string propagation between these sources:

$$\tilde{\mathcal{A}} \sim \text{D9-D9 exchange},$$

$$\tilde{\mathcal{M}} \sim \text{D9-O9 exchange},$$

and

$$\tilde{\mathcal{K}} \sim \text{O9-O9 exchange}.$$

The total source for the massless R-R field is the sum of the boundary and crosscap sources:

$$|J\rangle = |B\rangle + |C\rangle.$$

The closed-string exchange energy is quadratic in the source, but with the standard symmetry factor:

$$\Gamma_{\text{tree}} = \frac{1}{2}\langle J|\Delta|J\rangle.$$

Therefore

$$\begin{aligned}\Gamma_{\text{tree}} &= \frac{1}{2}\left(\langle B|\Delta|B\rangle + \langle B|\Delta|C\rangle + \langle C|\Delta|B\rangle + \langle C|\Delta|C\rangle\right)\\ &= \frac{1}{2}\langle B|\Delta|B\rangle + \langle B|\Delta|C\rangle + \frac{1}{2}\langle C|\Delta|C\rangle.\end{aligned}$$

In the second line we used

$$\langle B|\Delta|C\rangle = \langle C|\Delta|B\rangle.$$

So the algebraic factor of 2 in the cross term is precisely canceled by the overall $\frac{1}{2}$ in the quadratic exchange functional.

This is the same basic reason that, for ordinary fields, integrating out a field sourced by $J = J_1 + J_2$ gives

$$\frac{1}{2}(J_1 + J_2)\Delta(J_1 + J_2) = \frac{1}{2}J_1\Delta J_1 + J_1\Delta J_2 + \frac{1}{2}J_2\Delta J_2.$$

There is only one interaction term $J_1\Delta J_2$, not two separately counted ones.

3.7.1.3 Relation to annulus, Möbius strip, and Klein bottle

The three unoriented one-loop worldsheets are:

$$\mathcal{A}: \quad \text{annulus},$$

$$\mathcal{M}: \quad \text{Möbius strip},$$

and

$$\mathcal{K}: \quad \text{Klein bottle}.$$

After modular transformation to tree channel, they are interpreted as closed-string propagation:

$$\tilde{\mathcal{A}} = \frac{1}{2}\langle B|\Delta|B\rangle,$$

$$\tilde{\mathcal{M}} = \langle B|\Delta|C\rangle,$$
$$\tilde{\mathcal{K}} = \frac{1}{2}\langle C|\Delta|C\rangle.$$

The factors of $\frac{1}{2}$ for the annulus and Klein bottle are not arbitrary. They are symmetry factors for diagrams with two identical ends:

annulus: two boundary ends,

Klein bottle: two crosscap ends.

By contrast, the Möbius strip has one boundary and one crosscap. These two ends are different types of objects, so there is no additional identical-end symmetry factor. The Möbius strip is precisely the mixed boundary-crosscap exchange.

Equivalently, in the loop-channel unoriented open-string description, the Möbius strip is already the worldsheet with one boundary and one crosscap. There are not two different Möbius worldsheets corresponding to

$$\langle B|\Delta|C\rangle \qquad \text{and} \qquad \langle C|\Delta|B\rangle.$$

They are the same unoriented surface seen from opposite ends of the closed-string propagator. Counting both would double-count the same moduli region.

3.7.1.4 Seeing the square in your coefficients

Your tree-channel massless coefficients are

$$\tilde{\mathcal{A}} \sim 2^{-6}N^2,$$
$$\tilde{\mathcal{M}} \sim -N,$$

and

$$\tilde{\mathcal{K}} \sim 2^4.$$

Their sum is

$$2^{-6}N^2 - N + 2^4.$$

Rewriting everything with the same overall factor,

$$2^{-6}N^2 - N + 2^4 = 2^{-6}\left(N^2 - 2^6 N + 2^{10}\right).$$

But

$$N^2 - 2^6 N + 2^{10} = (N - 2^5)^2.$$

Thus

$$\tilde{\mathcal{A}} + \tilde{\mathcal{M}} + \tilde{\mathcal{K}} \sim 2^{-6}(N-32)^2.$$

To see the cross term explicitly, write

$$2^{-6}(N-32)^2 = \left(\frac{N}{8} - 4\right)^2.$$

Then

$$\left(\frac{N}{8} - 4\right)^2 = \left(\frac{N}{8}\right)^2 - 2\left(\frac{N}{8}\right)(4) + 4^2.$$

These are

$$\left(\frac{N}{8}\right)^2 = 2^{-6}N^2 \quad \text{from } \tilde{\mathcal{A}},$$
$$-2\left(\frac{N}{8}\right)(4) = -N \quad \text{from } \tilde{\mathcal{M}},$$

and

$$4^2 = 2^4 \quad \text{from } \tilde{\mathcal{K}}.$$

So the single Möbius amplitude already carries the full coefficient

$$2(\text{disc})(\text{crosscap})$$

inside the square. That is exactly why adding a second Möbius amplitude would spoil the square.

3.7.1.5 Tadpole cancellation and $N = 32$

The R-R tadpole is proportional to the total R-R source:

$$Q_{\text{total}} = Q_{\text{D9}} + Q_{\text{O9}}.$$

The tree-channel massless R-R exchange is proportional to

$$Q_{\text{total}}^2.$$

Your coefficient

$$2^{-6}(N-32)^2$$

therefore means

$$Q_{\text{D9}} \propto N, \qquad Q_{\text{O9}} \propto -32.$$

Tadpole cancellation requires

$$Q_{\text{total}} = 0,$$

so

$$N - 32 = 0.$$

Hence

$$\boxed{N = 32.}$$

This is the usual type I result: the consistent supersymmetric ten-dimensional type I string has gauge group $SO(32)$.

3.7.1.6 Checks

If one incorrectly added two copies of the Möbius amplitude, the massless coefficient would become

$$2^{-6}N^2 - 2N + 2^4.$$

This is

$$2^{-6}\left(N^2 - 2^7 N + 2^{10}\right),$$

which is not

$$2^{-6}(N-32)^2.$$

It would instead correspond to the wrong cross term and would not factorize into the square of the total R-R tadpole with the correct O9-plane charge.

The correct counting is:

$$\boxed{\tilde{\mathcal{A}} + \tilde{\mathcal{M}} + \tilde{\mathcal{K}} = \frac{1}{2}\langle B + C|\Delta|B + C\rangle.}$$

The Möbius strip appears once because it is the one mixed boundary-crosscap surface.

3.7.1.7 Result

The tree-channel factorization is

$$\Gamma_{\text{tree}} = \frac{1}{2}\langle B + C|\Delta|B + C\rangle.$$

Expanding,

$$\Gamma_{\text{tree}} = \frac{1}{2}\langle B|\Delta|B\rangle + \langle B|\Delta|C\rangle + \frac{1}{2}\langle C|\Delta|C\rangle.$$

Thus

$$\tilde{\mathcal{A}} = \frac{1}{2}\langle B|\Delta|B\rangle, \qquad \tilde{\mathcal{M}} = \langle B|\Delta|C\rangle, \qquad \tilde{\mathcal{K}} = \frac{1}{2}\langle C|\Delta|C\rangle.$$

So one sums only one Möbius amplitude because that one amplitude is already the full cross term.

For the massless tadpoles,

$$\tilde{\mathcal{A}} + \tilde{\mathcal{M}} + \tilde{\mathcal{K}} \sim 2^{-6}N^2 - N + 2^4 = 2^{-6}(N-32)^2.$$

The tadpole vanishes only for

$$\boxed{N = 32.}$$

3.8 Q8: Orientifold action on D-brane boundary states

Previously we have talked about the Dp-brane state

$$|\mathrm{Dp},\eta\rangle = \int dk_\perp \exp\left[-\sum_{n>0}\frac{1}{n}\alpha^i_{-n}D_{ij}\overline{\alpha}^j_{-n} - i\eta\sum_{r>0} b^i_{-r}D_{ij}\overline{b}^j_{-r}\right] e^{ik_\perp\cdot x_0^\perp}|k_\perp\rangle,$$

A complete Dp-brane states is

$$|\mathrm{Dp}\rangle = \frac{1}{\mathcal{N}_{\mathrm{Dp}}}(|\mathrm{Dp},\eta=1\rangle_{\mathrm{NS}} - |\mathrm{Dp},\eta=-1\rangle_{\mathrm{NS}} + i|\mathrm{Dp},\eta=1\rangle_{\mathrm{R}} + i|\mathrm{Dp},\eta=-1\rangle_{\mathrm{R}})$$

Under the orientifold projection Ω, in the resulting type I string theory only R-R 2-form C_2 left. Which means that only D9, D5, and D1 branes survive. Can we see this from the action of ω on the Dp-brane state?

> **Textbook location and related material.** Chapter 9, Sections 9.3–9.4 (pp. 235–251): GSO-projected D-brane boundary states and the Ω projection in type I. Table 10.3 and Section 10.6 relate the surviving branes to toroidal orientifolds.

3.8.1 Answer

3.8.1.1 Core point

Yes. One can see the type I restriction

$$p = 1, 5, 9$$

directly from the action of worldsheet orientation reversal Ω on the Dp boundary state.

The BPS Dp boundary state has the schematic form

$$|\mathrm{Dp}\rangle = |\mathrm{Dp}\rangle_{\mathrm{NSNS}} + |\mathrm{Dp}\rangle_{\mathrm{RR}},$$

where, with your convention,

$$|\mathrm{Dp}\rangle_{\mathrm{NSNS}} \propto |\mathrm{Dp},+\rangle_{\mathrm{NS}} - |\mathrm{Dp},-\rangle_{\mathrm{NS}},$$

and

$$|\mathrm{Dp}\rangle_{\mathrm{RR}} \propto i|\mathrm{Dp},+\rangle_{\mathrm{R}} + i|\mathrm{Dp},-\rangle_{\mathrm{R}}.$$

The orientifold projection keeps only Ω-even closed-string states and Ω-invariant boundary-state sources. The NS-NS part is Ω-even:

$$\Omega|\mathrm{Dp}\rangle_{\mathrm{NSNS}} = |\mathrm{Dp}\rangle_{\mathrm{NSNS}}.$$

The R-R part transforms with a p-dependent sign:

$$\Omega|\mathrm{Dp}\rangle_{\mathrm{RR}} = (-1)^{(p+3)/2}|\mathrm{Dp}\rangle_{\mathrm{RR}}, \qquad p \text{ odd}.$$

Therefore the R-R charged BPS Dp boundary state survives the type I orientifold projection only if

$$(-1)^{(p+3)/2} = +1.$$

For type IIB odd p,

$$p = -1, 1, 3, 5, 7, 9,$$

this gives

$$p = 1, 5, 9.$$

These are precisely the BPS D-branes of type I string theory.

3.8.1.2 How Ω acts on the oscillator part

Worldsheet parity exchanges left and right movers:

$$\Omega\,\alpha_n^\mu\,\Omega^{-1} = \bar\alpha_n^\mu, \qquad \Omega\,\bar\alpha_n^\mu\,\Omega^{-1} = \alpha_n^\mu,$$

and similarly for the closed-string fermion oscillators:

$$\Omega\,b_r^\mu\,\Omega^{-1} = \bar b_r^\mu, \qquad \Omega\,\bar b_r^\mu\,\Omega^{-1} = b_r^\mu.$$

The bosonic oscillator exponential in the Dp boundary state is

$$\exp\left[-\sum_{n>0}\frac{1}{n}\alpha_{-n}^i D_{ij}\bar\alpha_{-n}^j\right].$$

Since the bosonic oscillators commute and D_{ij} is symmetric, exchanging left and right movers leaves this part invariant:

$$\alpha_{-n}^i D_{ij}\bar\alpha_{-n}^j \longmapsto \bar\alpha_{-n}^i D_{ij}\alpha_{-n}^j = \alpha_{-n}^i D_{ij}\bar\alpha_{-n}^j.$$

The fermionic nonzero-mode exponential is

$$\exp\left[-i\eta\sum_{r>0} b_{-r}^i D_{ij}\bar b_{-r}^j\right].$$

Under Ω,

$$b_{-r}^i D_{ij}\bar b_{-r}^j \longmapsto \bar b_{-r}^i D_{ij} b_{-r}^j.$$

Because fermionic oscillators anticommute,

$$\bar b_{-r}^i D_{ij} b_{-r}^j = -b_{-r}^j D_{ji}\bar b_{-r}^i = -b_{-r}^i D_{ij}\bar b_{-r}^j.$$

Therefore

$$-i\eta\sum_{r>0} b_{-r}D\bar b_{-r} \quad\longmapsto\quad +i\eta\sum_{r>0} b_{-r}D\bar b_{-r}.$$

This is the same as replacing

$$\eta \mapsto -\eta.$$

So the oscillator part of the boundary state naturally satisfies

$$\Omega : |\mathrm{D}p,\eta\rangle \longmapsto \text{phase}\times|\mathrm{D}p,-\eta\rangle.$$

The remaining question is the phase. In the NS-NS sector the phase is fixed so that the GSO-projected NS-NS boundary state is invariant. In the R-R sector the phase depends on p because of the R-R zero modes.

3.8.1.3 NS-NS part

The GSO-projected NS-NS boundary state is

$$|\mathrm{Dp}\rangle_{\mathrm{NSNS}} \propto |\mathrm{Dp},+\rangle_{\mathrm{NS}} - |\mathrm{Dp},-\rangle_{\mathrm{NS}}.$$

With the standard ghost and superghost conventions, the action on the unprojected NS-NS components is

$$\Omega|\mathrm{Dp},\eta\rangle_{\mathrm{NS}} = -|\mathrm{Dp},-\eta\rangle_{\mathrm{NS}}.$$

Therefore

$$\begin{aligned}\Omega|\mathrm{Dp}\rangle_{\mathrm{NSNS}} &\propto \Omega\left(|\mathrm{Dp},+\rangle_{\mathrm{NS}} - |\mathrm{Dp},-\rangle_{\mathrm{NS}}\right)\\ &= -|\mathrm{Dp},-\rangle_{\mathrm{NS}} + |\mathrm{Dp},+\rangle_{\mathrm{NS}}\\ &= |\mathrm{Dp},+\rangle_{\mathrm{NS}} - |\mathrm{Dp},-\rangle_{\mathrm{NS}}.\end{aligned}$$

Thus

$$\boxed{\Omega|\mathrm{Dp}\rangle_{\mathrm{NSNS}} = |\mathrm{Dp}\rangle_{\mathrm{NSNS}}.}$$

This is why the tension coupling to the Ω-even NS-NS fields, such as $g_{\mu\nu}$ and Φ, is not the obstruction. The obstruction is the R-R charge.

3.8.1.4 R-R zero modes and the p-dependent sign

The R-R ground states are spacetime spinors. The zero-mode part of the R-R boundary state is usually written as a bispinor matrix involving the gamma-matrix product along the brane worldvolume:

$$M_p(\eta) \sim C\Gamma^0\Gamma^1 \cdots \Gamma^p \,(1 + i\eta\Gamma_{11})\,,$$

up to normalization and convention-dependent phases.

Worldsheet parity Ω exchanges the left and right R-R ground-state spinors. In bispinor language, this transposes the matrix $M_p(\eta)$. The transpose of a product of $p+1$ gamma matrices gives a p-dependent sign. Equivalently, this is the same sign that appears in the Ω transformation of R-R potentials:

$$\Omega : C_n \longmapsto (-1)^{n/2+1} C_n \qquad \text{for even } n \text{ in type IIB.}$$

Since a Dp-brane couples to

$$C_{p+1},$$

we set

$$n = p + 1.$$

Hence

$$\Omega : C_{p+1} \longmapsto (-1)^{(p+1)/2+1} C_{p+1} = (-1)^{(p+3)/2} C_{p+1}.$$

The R-R part of the Dp boundary state has the same eigenvalue:

$$\boxed{\Omega|\mathrm{Dp}\rangle_{\mathrm{RR}} = (-1)^{(p+3)/2}|\mathrm{Dp}\rangle_{\mathrm{RR}}.}$$

This is the boundary-state way of saying that the R-R potential sourced by the brane is either kept or projected out by Ω.

3.8.1.5 Which D-branes survive

Type IIB has BPS Dp-branes with odd p:

$$p = -1, 1, 3, 5, 7, 9.$$

The R-R eigenvalue is

$$\lambda_p = (-1)^{(p+3)/2}.$$

Let us list it:

p	C_{p+1}	λ_p	Ω-even?
-1	C_0	-1	no
1	C_2	$+1$	yes
3	C_4	-1	no
5	C_6	$+1$	yes
7	C_8	-1	no
9	C_{10}	$+1$	yes

Therefore the Ω-invariant R-R charged BPS branes are

$$\boxed{\mathrm{D1}, \quad \mathrm{D5}, \quad \mathrm{D9}.}$$

This agrees with the spacetime field content. In type I, the propagating R-R potential is often described as

$$C_2.$$

Its magnetic dual is

$$C_6,$$

so D5-branes are also allowed. The top-form

$$C_{10}$$

has no propagating field strength degree of freedom, but it is crucial for the D9/O9 R-R tadpole. D9-branes couple electrically to this top-form.

Thus the statement "only C_2 is left" should be understood together with its dual and the top-form tadpole sector:

$$C_2, \quad C_6, \quad C_{10}$$

are the R-R potentials relevant to

$$\mathrm{D1}, \quad \mathrm{D5}, \quad \mathrm{D9}.$$

3.8.1.6 Branes projected out versus branes mapped to non-BPS objects

For

$$p = -1, 3, 7,$$

the R-R part is Ω-odd:

$$\Omega|\mathrm{Dp}\rangle_{\mathrm{RR}} = -|\mathrm{Dp}\rangle_{\mathrm{RR}}.$$

So the R-R charge carried by such a BPS Dp-brane is projected out in type I. The corresponding BPS R-R charged brane of type IIB therefore does not survive as a BPS R-R charged object in type I.

This does not mean there are no possible non-BPS D-brane-like objects in type I. There are indeed non-BPS branes in type I with more subtle stability properties. But the simple R-R charged BPS boundary-state argument selects precisely

$$p = 1, 5, 9.$$

3.8.1.7 Checks

For a D1-brane,

$$p = 1, \qquad (-1)^{(p+3)/2} = (-1)^2 = +1.$$

It couples to the Ω-even R-R two-form C_2, so it survives.

For a D3-brane,

$$p = 3, \qquad (-1)^{(p+3)/2} = (-1)^3 = -1.$$

It couples to C_4, which is Ω-odd in type I, so the BPS D3-brane does not survive the orientifold projection.

For a D9-brane,

$$p = 9, \qquad (-1)^{(p+3)/2} = (-1)^6 = +1.$$

It couples to C_{10} and is needed to cancel the O9-plane R-R charge. This is the same D9-brane that led to the $SO(32)$ tadpole cancellation condition in Q7.

3.8.1.8 Result

Worldsheet parity exchanges left and right movers:

$$\Omega b_r^\mu \Omega^{-1} = \bar{b}_r^\mu, \qquad \Omega \bar{b}_r^\mu \Omega^{-1} = b_r^\mu.$$

Because fermion oscillators anticommute, the fermionic gluing label flips:

$$\Omega : \eta \mapsto -\eta.$$

For the GSO-projected NS-NS state,

$$\Omega|\mathrm{Dp}\rangle_{\mathrm{NSNS}} = |\mathrm{Dp}\rangle_{\mathrm{NSNS}}.$$

For the R-R part,

$$\Omega|\mathrm{Dp}\rangle_{\mathrm{RR}} = (-1)^{(p+3)/2}|\mathrm{Dp}\rangle_{\mathrm{RR}}.$$

Thus the R-R charged BPS Dp-brane survives the type I orientifold projection only when

$$(-1)^{(p+3)/2} = +1.$$

For odd type IIB p, this gives

$$\boxed{p = 1, 5, 9.}$$

Equivalently, type I keeps

$$C_2, \quad C_6, \quad C_{10},$$

corresponding to

$$\boxed{\mathrm{D1}, \quad \mathrm{D5}, \quad \mathrm{D9}.}$$

Chapter 4

The Quantized Fermionic String

4.1 Q1: The no-ghost theorem

In the canonical quantization of the bosonic string and superstring, due to the negative eigenvalue of $\eta^{\mu\nu}$ there are negative norm states (ghosts). One can prove a no-ghost theorem which states that the negative norm states decouple in the critical dimension d for a particular value of the normal ordering constant a. Explain the no-ghost theorem for me in details.

> **Textbook location and related material.** Chapter 8, Sections 8.1–8.4 (pp. 195–210), read together with Chapter 3, Sections 3.1–3.3 (pp. 35–53): covariant constraints, light-cone quantization, critical dimension, and removal of negative-norm states. The DDF formulation is supplementary to the book's presentation.

4.1.1 Answer

4.1.1.1 Core point

The issue is that old-covariant quantization keeps manifest Lorentz covariance. Therefore one quantizes all target-space components

$$X^\mu(\tau,\sigma), \qquad \mu = 0, 1, \dots, D-1,$$

including the timelike component X^0. Since

$$\eta^{00} < 0,$$

the naive oscillator Fock space has an indefinite inner product. For example, in the bosonic string,

$$[\alpha_m^\mu, \alpha_n^\nu] = m\delta_{m+n,0}\eta^{\mu\nu},$$

so

$$\|\alpha_{-1}^0|0;p\rangle\|^2 = \langle 0;p|\alpha_1^0\alpha_{-1}^0|0;p\rangle = \eta^{00} < 0.$$

These negative-norm states are the covariant-quantization ghosts.

The no-ghost theorem says that these ghosts do not survive as physical states, provided the string is quantized at the critical values of the dimension and intercept. For the open bosonic string:

$$D = 26, \qquad a = 1.$$

For the open RNS superstring:

$$D = 10, \qquad a_{\rm NS} = \frac{1}{2}, \qquad a_{\rm R} = 0.$$

More precisely, the theorem says that the physical state space modulo null states is positive definite and is isomorphic to the transverse light-cone Fock space. Symbolically,

$$\frac{\mathcal{H}_{\rm phys}}{\mathcal{H}_{\rm null}} \cong \mathcal{H}_{\rm transverse}.$$

Thus negative-norm excitations are either absent from the physical quotient or belong to null/spurious gauge sectors that decouple from physical amplitudes.

4.1.1.2 Why ghosts appear before imposing constraints

For the open bosonic string, the oscillator expansion can be written schematically as

$$X^\mu(\tau,\sigma) = x^\mu + 2\alpha' p^\mu \tau + i\sqrt{2\alpha'} \sum_{n\neq 0} \frac{1}{n} \alpha_n^\mu e^{-in\tau} \cos n\sigma.$$

Canonical quantization gives

$$[x^\mu, p^\nu] = i\eta^{\mu\nu}, \qquad [\alpha_m^\mu, \alpha_n^\nu] = m\delta_{m+n,0}\eta^{\mu\nu}.$$

The Fock vacuum obeys

$$\alpha_n^\mu |0;p\rangle = 0, \qquad n > 0,$$

and excited states are produced by α_{-n}^μ. Since the target metric is indefinite, the Fock space inner product is also indefinite.

At level one, a general state is

$$|\zeta;p\rangle = \zeta_\mu \alpha_{-1}^\mu |0;p\rangle.$$

Its norm is

$$\langle \zeta;p|\zeta;p\rangle = \zeta_\mu^* \zeta_\nu \eta^{\mu\nu}.$$

If ζ^μ is timelike, this norm is negative. So before constraints, the covariant string Fock space definitely contains ghosts.

The purpose of the Virasoro constraints is to remove precisely the unphysical longitudinal and timelike excitations.

4.1.1.3 Physical-state conditions in the bosonic string

Classically the Polyakov string has worldsheet diffeomorphism and Weyl gauge symmetry. In conformal gauge, the residual constraints are

$$T_{\alpha\beta} = 0.$$

In holomorphic language this becomes

$$T(z) = 0,$$

or equivalently

$$L_n = 0$$

on physical configurations.

Quantum mechanically, the Virasoro generators are

$$L_m = \frac{1}{2} \sum_{n\in\mathbb{Z}} :\alpha_{m-n} \cdot \alpha_n:,$$

and the zero mode is

$$L_0 = \alpha' p^2 + N$$

for the open string. Because of normal ordering, the mass-shell condition is shifted:

$$(L_0 - a)|\psi\rangle = 0.$$

The old-covariant physical-state conditions are therefore

$$(L_0 - a)|\psi\rangle = 0, \qquad L_n|\psi\rangle = 0 \quad n > 0.$$

The first condition gives the mass formula

$$\alpha' M^2 = N - a,$$

where $M^2 = -p^2$.

The positive-mode constraints $L_n|\psi\rangle = 0$ with $n > 0$ are the quantum remnants of $T(z) = 0$. They remove many timelike and longitudinal states, but it is not obvious a priori that they remove all negative-norm physical states. The no-ghost theorem is precisely the nontrivial statement that they do, at the critical values.

4.1.1.4 Spurious states and null states

A key idea is the distinction between physical states, spurious states, and null states.

A physical state $|\phi\rangle$ obeys

$$L_n|\phi\rangle = 0, \qquad n > 0,$$

and the mass-shell condition

$$(L_0 - a)|\phi\rangle = 0.$$

A spurious state is a state of the form

$$|\chi\rangle_{\text{sp}} = \sum_{n>0} L_{-n}|\chi_n\rangle,$$

possibly with products of several negative Virasoro generators. Such states are generated by gauge transformations on the worldsheet.

If $|\phi\rangle$ is physical, then

$$\langle\phi|L_{-n}|\chi_n\rangle = \langle L_n\phi|\chi_n\rangle = 0, \qquad n > 0.$$

Therefore every spurious state has zero inner product with every physical state. If a spurious state is itself physical, then it is a null physical state:

$$|\chi\rangle_{\text{sp}} \in \mathcal{H}_{\text{phys}} \quad \Longrightarrow \quad \langle\phi|\chi\rangle_{\text{sp}} = 0 \quad \text{for all } |\phi\rangle \in \mathcal{H}_{\text{phys}}.$$

In particular,

$$\||\chi\rangle_{\text{sp}}\|^2 = 0.$$

Such states decouple from amplitudes because amplitudes depend only on inner products with physical external states, or equivalently on BRST cohomology classes.

However, this argument alone only says that physical spurious states are null. It does not yet prove that every negative-norm physical state is spurious or null. That stronger statement is the content of the no-ghost theorem.

4.1.1.5 The first level as a warm-up

At level one in the open bosonic string,

$$|\zeta; p\rangle = \zeta_\mu \alpha^\mu_{-1}|0; p\rangle.$$

The mass-shell condition gives

$$\alpha' M^2 = 1 - a.$$

At the critical intercept $a = 1$, the level-one state is massless:

$$p^2 = 0.$$

The L_1 constraint gives

$$0 = L_1|\zeta; p\rangle = \sqrt{2\alpha'}(\zeta \cdot p)|0; p\rangle,$$

so

$$\zeta \cdot p = 0.$$

Thus the polarization must be transverse to the null momentum p^μ.

But for a null momentum, the condition $\zeta \cdot p = 0$ still permits a polarization proportional to p^μ:

$$\zeta^\mu \sim p^\mu.$$

The corresponding state is

$$p_\mu \alpha^\mu_{-1}|0; p\rangle.$$

Using $\alpha^\mu_0 = \sqrt{2\alpha'}p^\mu$, one has

$$L_{-1}|0; p\rangle = \alpha_{-1} \cdot \alpha_0|0; p\rangle = \sqrt{2\alpha'}p_\mu \alpha^\mu_{-1}|0; p\rangle.$$

So the longitudinal polarization is a spurious state. Since $p^2 = 0$, it also has zero norm. The quotient removes it.

The remaining polarizations are transverse to both light-cone directions. They form a vector of $SO(D-2)$, so the level-one physical quotient has

$$D - 2$$

positive-norm states. This is exactly the light-cone answer.

This example already shows the mechanism:

1. the Virasoro constraints impose transversality;
2. longitudinal polarizations are spurious;
3. null/spurious states are quotiented out;
4. only transverse positive-norm polarizations remain.

The no-ghost theorem says this continues to hold at every oscillator level.

4.1.1.6 Why the critical values matter

The Virasoro algebra of the bosonic matter oscillators is

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{D}{12}(m^3 - m)\delta_{m+n,0}.$$

The central term is proportional to D because each target-space coordinate X^μ contributes $c = 1$ to the worldsheet central charge.

In a purely classical constraint algebra, the constraints are first class:

$$\{L_m, L_n\} \sim (m-n)L_{m+n}.$$

Quantum mechanically, the central term is an anomaly. It obstructs treating all Virasoro constraints as exact gauge constraints unless it is canceled by the ghost contribution in BRST quantization, or equivalently unless the old-covariant theory has the special intercept and dimension for which Lorentz covariance and positivity are compatible.

For the bosonic string, the reparametrization ghosts (b, c) have central charge

$$c_{bc} = -26.$$

The matter central charge is

$$c_X = D.$$

Thus BRST nilpotency requires

$$c_X + c_{bc} = 0,$$

or

$$D - 26 = 0.$$

So

$$D = 26.$$

The same computation also fixes the open-string intercept to

$$a = 1.$$

In old-covariant language, $D = 26$ and $a = 1$ are the values for which the Lorentz algebra closes without anomalous terms and the spurious physical states have the right null structure to decouple consistently.

If D and a are not at the allowed values, the constraints no longer remove the indefinite directions in the correct way. Equivalently, BRST nilpotency fails or Lorentz invariance fails. Then one cannot consistently identify physical states with a positive transverse Hilbert space.

There is a slightly more general mathematical form of the bosonic no-ghost theorem: for the open bosonic string, the physical state space has no negative-norm states for $D \leq 26$ and $a \leq 1$ under suitable assumptions. But the fully consistent critical bosonic string, with Lorentz invariance and BRST nilpotency, is the special case

$$D = 26, \qquad a = 1.$$

This is the case usually meant in string theory.

4.1.1.7 DDF operators and the precise structure of the theorem

The clean constructive proof uses DDF operators. The idea is to build physical excitations that are manifestly transverse, while still working in covariant quantization.

Choose a null vector k^μ satisfying

$$k^2 = 0.$$

For a momentum p^μ, one chooses the normalization of k^μ so that the vertex operators below have the correct conformal weight. A standard open-string choice is

$$2\alpha' p \cdot k = 1.$$

Let $i = 1, \dots, D-2$ label directions transverse to both p^μ and k^μ. The DDF oscillator is defined by a contour integral of a dimension-one current:

$$A_n^i = \sqrt{\frac{2}{\alpha'}} \oint \frac{dz}{2\pi i} \, \partial X^i(z) e^{ink \cdot X(z)}.$$

The current has conformal weight one because $k^2 = 0$. Therefore the contour integral commutes with the positive Virasoro constraints:

$$[L_m, A_n^i] = 0, \qquad m > 0.$$

More precisely, A_n^i maps physical states to physical states, shifting the spacetime momentum by nk^μ while increasing the oscillator level.

The DDF operators obey the ordinary transverse oscillator algebra

$$[A_m^i, A_n^j] = m\delta^{ij}\delta_{m+n,0}.$$

This algebra has positive definite norm because the indices i, j are transverse spacelike indices.

Starting from a suitable ground state $|p\rangle$, one constructs states

$$A_{-n_1}^{i_1} A_{-n_2}^{i_2} \cdots A_{-n_s}^{i_s} |p\rangle, \qquad n_r > 0.$$

These states are physical and have the same counting and inner products as light-cone transverse oscillator states.

The DDF form of the no-ghost theorem is:

$$\boxed{\text{Every physical state is equivalent, modulo null states, to a DDF state.}}$$

Equivalently,

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_{\text{DDF}} \oplus \mathcal{H}_{\text{null}},$$

with

$$\mathcal{H}_{\text{DDF}} \cong \mathcal{H}_{\text{light-cone}}.$$

Because $\mathcal{H}_{\text{DDF}}$ is generated only by transverse oscillators, its inner product is positive definite. Therefore any negative-norm contribution in $\mathcal{H}_{\text{phys}}$ must lie in the null/spurious sector and does not survive in

$$\mathcal{H}_{\text{phys}}/\mathcal{H}_{\text{null}}.$$

4.1.1.8 Relation to light-cone quantization

Light-cone quantization solves the constraints before quantization. Introduce light-cone coordinates

$$X^\pm = \frac{1}{\sqrt{2}}(X^0 \pm X^{D-1}).$$

One chooses light-cone gauge

$$X^+(\tau, \sigma) = x^+ + 2\alpha' p^+ \tau.$$

The Virasoro constraints then solve the longitudinal oscillators α_n^- in terms of the transverse oscillators α_n^i:

$$\alpha_n^- = \frac{1}{p^+} \times \text{quadratic expression in } \alpha_m^i.$$

Thus the only independent oscillators are

$$\alpha_{-n}^i, \qquad i = 1, \dots, D-2.$$

The light-cone Hilbert space is manifestly positive definite.

The price is that Lorentz invariance is no longer manifest. Quantum Lorentz invariance of the light-cone theory forces

$$D = 26, \qquad a = 1$$

for the bosonic string. Thus the covariant no-ghost theorem and the light-cone Lorentz anomaly calculation are two sides of the same fact:

$$\text{critical covariant physical quotient} \cong \text{positive light-cone Hilbert space}.$$

4.1.1.9 BRST formulation of the same result

In modern language, the cleanest conceptual statement uses BRST cohomology. After gauge fixing the worldsheet metric to conformal gauge, one introduces b, c ghosts. The BRST charge has the schematic form

$$Q_B = \sum_n c_{-n} \left(L_n^{\mathrm{m}} + \frac{1}{2} L_n^{\mathrm{gh}} - a\delta_{n,0} \right) - \frac{1}{2} \sum_{m,n} (m-n) : c_{-m} c_{-n} b_{m+n} : .$$

Here L_n^{m} are matter Virasoro generators and L_n^{gh} are ghost Virasoro generators.

Consistency requires

$$Q_B^2 = 0.$$

For the bosonic string this condition gives

$$D = 26, \qquad a = 1.$$

Physical states are BRST cohomology classes:

$$Q_B |\psi\rangle = 0, \qquad |\psi\rangle \sim |\psi\rangle + Q_B |\Lambda\rangle.$$

The BRST-exact states

$$Q_B |\Lambda\rangle$$

are gauge artifacts. They are the BRST version of spurious states. The no-ghost theorem becomes the statement that the BRST cohomology is represented by transverse excitations only:

$$H^\bullet(Q_B) \cong \mathcal{H}_{\text{transverse}}$$

at the appropriate ghost number.

This also explains the word "decouple." If a state is BRST exact, then inserting its vertex operator into a physical amplitude gives zero, assuming the usual absence of boundary contributions in moduli space:

$$\mathcal{A}(Q_B \Lambda, V_2, \dots, V_n) = 0.$$

So exact states, like null spurious states in old-covariant quantization, do not affect physical S-matrix elements.

4.1.1.10 Superstring: where the extra ghosts come from

In the RNS superstring, the matter fields are

$$X^\mu(\tau,\sigma), \qquad \psi^\mu(\tau,\sigma).$$

The bosonic oscillators still satisfy

$$[\alpha^\mu_m, \alpha^\nu_n] = m\delta_{m+n,0}\eta^{\mu\nu},$$

and the fermionic modes satisfy

$$\{\psi^\mu_r, \psi^\nu_s\} = \eta^{\mu\nu}\delta_{r+s,0}.$$

Because $\eta^{00} < 0$, fermionic timelike excitations also have indefinite norm. For example,

$$\|\psi^0_{-1/2}|0;p\rangle_{\rm NS}\|^2 = \eta^{00} < 0.$$

Thus the superstring has the same covariant ghost problem.

The new feature is local worldsheet supersymmetry. In conformal gauge, the constraints are not only

$$T(z) = 0,$$

but also

$$T_F(z) = 0,$$

where T_F is the worldsheet supercurrent. In modes, the physical-state conditions include both Virasoro and super-Virasoro constraints.

4.1.1.11 NS-sector physical-state conditions

In the NS sector, the fermions are half-integer moded:

$$r \in \mathbb{Z} + \frac{1}{2}.$$

The physical-state conditions are

$$(L_0 - a_{\rm NS})|\psi\rangle = 0,$$
$$L_n|\psi\rangle = 0, \qquad n > 0,$$

and

$$G_r|\psi\rangle = 0, \qquad r > 0.$$

At the critical dimension,

$$D = 10,$$

the normal-ordering constant is

$$a_{\rm NS} = \frac{1}{2}.$$

Thus the open-string NS mass formula is

$$\alpha' M^2 = N - \frac{1}{2}.$$

Before the GSO projection, the NS ground state is tachyonic. This tachyon is not a negative-norm ghost; it has positive norm but negative mass squared. The no-ghost theorem concerns the sign of the inner product, not the sign of M^2.

At the first excited NS level,

$$|\zeta;p\rangle = \zeta_\mu \psi^\mu_{-1/2}|0;p\rangle_{\rm NS},$$

and $N = \frac{1}{2}$. The mass-shell condition gives

$$p^2 = 0.$$

The $G_{1/2}$ constraint gives

$$p \cdot \zeta = 0.$$

The polarization $\zeta^\mu \sim p^\mu$ is a null gauge state. After quotienting it out, the remaining states are again the $D-2$ transverse polarizations. For $D = 10$, this gives

$$D - 2 = 8$$

positive physical polarizations in the open-string NS vector multiplet.

4.1.1.12 R-sector physical-state conditions

In the R sector, the fermions are integer moded:

$$r \in \mathbb{Z}.$$

There are fermion zero modes

$$\psi_0^\mu,$$

which satisfy a spacetime Clifford algebra:

$$\{\psi_0^\mu, \psi_0^\nu\} = \eta^{\mu\nu}.$$

Therefore the R-sector ground state is a spacetime spinor.

The physical-state conditions are

$$(L_0 - a_{\mathrm{R}})|\psi\rangle = 0,$$
$$L_n|\psi\rangle = 0, \qquad n > 0,$$

and

$$G_n|\psi\rangle = 0, \qquad n \geq 0.$$

The G_0 condition is special. Since

$$G_0^2 = L_0$$

up to the normal-ordering convention, the G_0 constraint is the spacetime Dirac equation on the R ground state. At the critical dimension,

$$D = 10, \qquad a_{\mathrm{R}} = 0.$$

The R ground state is massless and satisfies

$$\not{p}\,|u; p\rangle = 0.$$

After imposing this Dirac equation and quotienting null gauge components, the physical spinor states can be represented by transverse $SO(8)$ spinor degrees of freedom. Again, the indefinite timelike components do not survive in the physical quotient.

4.1.1.13 Super-Virasoro algebra and critical dimension

The matter super-Virasoro algebra has the form

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0},$$
$$[L_m, G_r] = \left(\frac{m}{2} - r\right) G_{m+r},$$
$$\{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}$$

in the NS sector. The matter central charge is

$$c_{\mathrm{m}} = c_X + c_\psi = D + \frac{D}{2} = \frac{3D}{2}.$$

Gauge fixing local worldsheet supersymmetry introduces not only the fermionic (b, c) ghosts, but also the bosonic superconformal ghosts (β, γ). Their total ghost central charge is

$$c_{\mathrm{gh}} = -26 + 11 = -15.$$

BRST nilpotency requires

$$c_{\mathrm{m}} + c_{\mathrm{gh}} = 0.$$

Therefore

$$\frac{3D}{2} - 15 = 0,$$

so

$$D = 10.$$

The same consistency condition gives

$$a_{\mathrm{NS}} = \frac{1}{2}, \qquad a_{\mathrm{R}} = 0.$$

Thus the superstring no-ghost theorem holds at the critical RNS values

$$D = 10, \qquad a_{\mathrm{NS}} = \frac{1}{2}, \qquad a_{\mathrm{R}} = 0.$$

4.1.1.14 Super-DDF form of the no-ghost theorem

Just as in the bosonic string, one can build physical transverse operators in the superstring. In addition to bosonic DDF oscillators A_n^i, there are fermionic DDF oscillators, usually denoted schematically by

$$B_r^i$$

in the NS sector, with half-integer r, and by integer-moded analogues in the R sector.

They obey transverse oscillator relations of the form

$$[A_m^i, A_n^j] = m\delta^{ij}\delta_{m+n,0},$$

$$\{B_r^i, B_s^j\} = \delta^{ij}\delta_{r+s,0}.$$

They commute with the positive super-Virasoro constraints in the appropriate sense:

$$[L_n, A_m^i] \sim 0, \qquad [L_n, B_r^i] \sim 0, \qquad [G_r, A_n^i] \sim 0, \qquad \{G_r, B_s^i\} \sim 0$$

on physical states, with the precise relations arranged so that DDF operators map physical states to physical states.

The superstring no-ghost theorem says:

$$\boxed{\frac{\mathcal{H}_{\mathrm{phys}}^{\mathrm{RNS}}}{\mathcal{H}_{\mathrm{null}}^{\mathrm{RNS}}} \cong \mathcal{H}_{\mathrm{transverse}}^{\mathrm{RNS}}}$$

where the right-hand side is generated only by transverse bosonic and fermionic oscillators:

$$A_{-n}^i, \qquad B_{-r}^i, \qquad i = 1, \dots, D-2.$$

For the critical superstring,

$$D - 2 = 8.$$

Thus the physical quotient is the positive-definite Fock space of eight transverse bosons and eight transverse worldsheet fermions, with the usual NS and R sector distinctions.

4.1.1.15 How this explains decoupling in amplitudes

Suppose $|\lambda\rangle$ is a physical null state. Then

$$\langle\phi|\lambda\rangle = 0$$

for every physical state $|\phi\rangle$. In operator language, the corresponding vertex operator is either spurious in old-covariant quantization or BRST exact in BRST quantization:

$$V_\lambda = \{Q_B, \Lambda\}$$

up to picture and sector-dependent details in the superstring.

If the other external vertex operators are physical, meaning BRST closed,

$$[Q_B, V_i] = 0,$$

then a physical amplitude with one null insertion is

$$\mathcal{A}(V_\lambda, V_2, \dots, V_n) = \mathcal{A}(\{Q_B, \Lambda\}, V_2, \dots, V_n).$$

By moving Q_B through the correlator, this becomes a total derivative on moduli space or vanishes by BRST invariance:

$$\mathcal{A}(V_\lambda, V_2, \dots, V_n) = 0,$$

assuming no anomalous boundary contribution. This is the precise sense in which null and spurious states decouple.

So the phrase "negative-norm states decouple" should be understood carefully. The theorem does not say that the full covariant Fock space has no negative-norm vectors. It certainly does. It says that after imposing the constraints and quotienting null gauge states, the physical Hilbert space is positive definite. Negative-norm directions are gauge artifacts of the covariant description.

4.1.1.16 Closed strings

For closed strings, there are left- and right-moving copies:

$$L_n, \qquad \widetilde{L}_n.$$

The physical-state conditions for the bosonic closed string are

$$(L_0 - a)|\psi\rangle = 0, \qquad (\widetilde{L}_0 - a)|\psi\rangle = 0,$$

$$L_n|\psi\rangle = 0, \qquad \widetilde{L}_n|\psi\rangle = 0, \qquad n > 0.$$

Equivalently, one imposes the mass-shell condition and level matching:

$$N - \widetilde{N} = 0.$$

The no-ghost theorem applies separately to the left- and right-moving sectors. The physical quotient is the tensor product of the two transverse Hilbert spaces, subject to level matching.

For the closed superstring, the same statement holds with left and right NS/R choices:

$$\text{NS-NS}, \qquad \text{NS-R}, \qquad \text{R-NS}, \qquad \text{R-R}.$$

Each sector has a positive physical quotient at

$$D = 10.$$

4.1.1.17 Checks

For the bosonic open string at level one, the critical values give

$$D = 26, \qquad a = 1.$$

The physical massless vector has

$$D - 2 = 24$$

positive transverse polarizations. The timelike and longitudinal polarizations are removed by the L_1 constraint and by quotienting the null state $L_{-1}|0;p\rangle$.

For the open superstring NS sector at the first massless level,

$$D = 10, \qquad a_{\text{NS}} = \frac{1}{2}.$$

The physical massless vector has

$$D - 2 = 8$$

positive transverse polarizations.

For the R sector, the ground state is a spacetime spinor. The G_0 constraint imposes the massless Dirac equation, and the physical quotient is represented by transverse $SO(8)$ spinor data, again with positive norm.

4.1.1.18 Result

The covariant oscillator Fock space is indefinite because

$$[\alpha_m^0, \alpha_n^0] = m\delta_{m+n,0}\eta^{00}$$

and, in the RNS string,

$$\{\psi_r^0, \psi_s^0\} = \eta^{00}\delta_{r+s,0}.$$

Physical states satisfy the old-covariant constraints. For the bosonic open string:

$$(L_0 - a)|\psi\rangle = 0, \qquad L_n|\psi\rangle = 0 \quad n > 0.$$

At the critical values

$$D = 26, \qquad a = 1,$$

the no-ghost theorem says

$$\frac{\mathcal{H}_{\text{phys}}}{\mathcal{H}_{\text{null}}} \cong \mathcal{H}_{\text{DDF}} \cong \mathcal{H}_{\text{light-cone}},$$

which is positive definite.

For the RNS superstring, the constraints are

$$(L_0 - a)|\psi\rangle = 0, \qquad L_n|\psi\rangle = 0, \qquad G_r|\psi\rangle = 0$$

with the appropriate NS or R moding. At the critical values

$$D = 10, \qquad a_{\text{NS}} = \frac{1}{2}, \qquad a_{\text{R}} = 0,$$

the superstring no-ghost theorem says

$$\frac{\mathcal{H}^{\text{RNS}}_{\text{phys}}}{\mathcal{H}^{\text{RNS}}_{\text{null}}} \cong \mathcal{H}^{\text{RNS}}_{\text{transverse}},$$

generated only by

$$A^i_{-n}, \qquad B^i_{-r}, \qquad i = 1, \dots, D-2.$$

Therefore the final physical Hilbert space has no negative-norm states. The ghosts of covariant quantization are removed by the Virasoro and super-Virasoro constraints, and the null/spurious remnants decouple from all physical amplitudes.

4.2 Q2: World-sheet fermionic oscillators and spacetime bosons

Consider the open superstring with NN boundary condition. In the light-cone quantization, the spectrum is given by

$$|0\rangle \ , \ b^i_{-\frac{1}{2}}|0\rangle \ , \ \alpha^i_{-1}|0\rangle + b^i_{-\frac{1}{2}} b^j_{-\frac{1}{2}}|0\rangle \ , \cdots$$

The first level is the tachyon. The second level which is massless has 8 states generated by fermionic creation operator $b^i_{-\frac{1}{2}}$ acting on the vacuum. It is a massless particle under the little group $SO(8)$ i.e. a massless vector. But I am confused that the operator b has its anti-commuting nature, why the states generated by them can be bosonic vector?

A general state at massless level is

$$|\zeta; k\rangle = \zeta_i b^i_{-1/2}|0; k\rangle_{\text{NS}}$$

If we regard ζ_i as the spacetime field, then it is an $SO(8)$ vector. Is that in the discussion of the string theory, it is fields such as ζ_i have physical importance while the state $b^i_{-1/2}|0; k\rangle_{\text{NS}}$ is not such important?

> **Textbook location and related material.** Chapter 8, Sections 8.2–8.3 (pp. 200–208): light-cone RNS spectrum, NS and R sectors, and the GSO projection. The spacetime-field interpretation is connected to the covariant vertices of Section 13.2.

4.2.1 Answer

4.2.1.1 Core point

The confusion comes from using the word "fermionic" in two different senses.

The oscillator

$$b^i_r$$

is fermionic as a worldsheet oscillator. This means it comes from the worldsheet fermion field $\psi^i(\tau,\sigma)$ and obeys anticommutation relations

$$\{b^i_r, b^j_s\} = \delta^{ij}\delta_{r+s,0}.$$

It does not mean that the one-string state created by $b^i_{-1/2}$ is automatically a spacetime fermion.

The massless NS state

$$|\zeta;k\rangle = \zeta_i b^i_{-1/2}|0;k\rangle_{\text{NS}}$$

has one transverse vector index i. Under the ten-dimensional massless little group

$$SO(8),$$

it transforms as the vector representation

$$8_v.$$

That is why it is a spacetime vector boson. Its spacetime statistics are determined by its target-space little-group representation and by second quantization of the string field, not by the anticommutation algebra of the internal worldsheet oscillator.

Spacetime fermions in the RNS string come from the Ramond sector, not from the NS oscillator $b^i_{-1/2}$. In the R sector the fermion zero modes form a spacetime Clifford algebra, and the ground states are spacetime spinors.

For your follow-up question: no, the basis state

$$b^i_{-1/2}|0;k\rangle_{\text{NS}}$$

is not unimportant. It is part of the physical one-string Hilbert space. The polarization ζ_i and the basis state play complementary roles. The physical state is the linear combination

$$|\zeta;k\rangle = \sum_{i=1}^{8} \zeta_i\, b^i_{-1/2}|0;k\rangle_{\text{NS}}.$$

The $b^i_{-1/2}|0;k\rangle_{\text{NS}}$ are the basis vectors of the $SO(8)$ vector representation, while the numbers ζ_i are the components of the polarization vector in that basis.

So the right statement is not

$$\text{only } \zeta_i \text{ is physical, while } b^i_{-1/2}|0;k\rangle_{\text{NS}} \text{ is not.}$$

Rather, the physical object is the state

$$|\zeta;k\rangle \in \mathcal{H}^{\text{NS}}_{k,\text{massless}},$$

or equivalently the corresponding vertex operator, or equivalently in second quantization the spacetime field mode multiplying this state.

4.2.1.2 Basis states, polarization components, and the physical vector

At fixed lightlike momentum k, the massless NS Hilbert space is eight-dimensional:

$$\mathcal{H}^{\mathrm{NS}}_{k,\mathrm{massless}} = \mathrm{span}\left\{b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}\right\}_{i=1}^{8}.$$

This is just like an ordinary vector space. If e_i is a basis of $\mathbb{R}^8$, a vector v is written as

$$v = \sum_{i=1}^{8} v_i e_i.$$

The basis vectors e_i are not "less physical" than the components v_i; rather, the vector is the combination of both:

$$\text{vector} = \text{components} \times \text{basis}.$$

In the string Hilbert space the analogous identification is

$$e_i \longleftrightarrow b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}, \qquad v_i \longleftrightarrow \zeta_i.$$

Thus

$$|\zeta;k\rangle = \zeta_i b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}$$

is the actual one-particle state. The polarization ζ_i is not by itself the state. It is the coordinate description of the state in the basis provided by the oscillator construction.

This is why it is meaningful to say both of the following:

$$b^i_{-1/2}|0;k\rangle_{\mathrm{NS}} \text{ transforms as } 8_v,$$

and

$$\zeta_i \text{ is a vector polarization.}$$

They are two sides of the same representation-theoretic statement.

4.2.1.3 Active and passive transformation viewpoints

Let $R \in SO(8)$. The basis states transform as

$$U(R)\, b^i_{-1/2}|0;k\rangle_{\mathrm{NS}} = R^i{}_j\, b^j_{-1/2}|0;Rk\rangle_{\mathrm{NS}}.$$

Therefore

$$U(R)|\zeta;k\rangle = \zeta_i R^i{}_j b^j_{-1/2}|0;Rk\rangle_{\mathrm{NS}}.$$

If we define the transformed polarization by

$$\zeta'_j = \zeta_i R^i{}_j,$$

then

$$U(R)|\zeta;k\rangle = |\zeta';Rk\rangle.$$

This is exactly what it means for the one-particle states to furnish the vector representation of $SO(8)$.

Depending on convention, one may instead say that the basis is fixed and the components transform, or that the basis transforms and the abstract vector is the object being acted on. These are the passive and active descriptions of the same mathematics. In neither description is the state unimportant.

4.2.1.4 First-quantized states versus spacetime fields

There are two levels of description.

In first-quantized string theory, the fundamental object is the single-string Hilbert space. A physical particle state is a vector in this Hilbert space:

$$|\zeta;k\rangle = \zeta_i b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}.$$

Equivalently, in the operator-state correspondence, it is represented by a vertex operator. In light-cone language the oscillator state is natural; in covariant language the corresponding massless NS vertex operator is

$$V_A^{(-1)} = g_o\, \zeta_\mu \psi^\mu e^{-\phi} e^{ik\cdot X}.$$

In second-quantized string theory, or in the low-energy effective field theory, one expands a string field in a basis of first-quantized string states. Schematically, for the massless open NS sector,

$$|\Psi\rangle = \int d^{10}k\, A_i(k)\, b^i_{-1/2}|0;k\rangle_{\mathrm{NS}} + \cdots.$$

Here

$$A_i(k)$$

is the momentum-space gauge field mode. Fourier transforming gives

$$A_i(x) = \int d^{10}k\, e^{ik\cdot x} A_i(k)$$

up to normalization conventions.

So if by ζ_i you mean the polarization of a single external string state, then ζ_i is a fixed vector labeling that one-particle state. If by $A_i(k)$ you mean a dynamical spacetime field mode, then it appears as the coefficient of the string basis state in the second-quantized string field:

$$\text{string field coefficient} \quad A_i(k) \quad \text{multiplies} \quad b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}.$$

This is the precise sense in which spacetime fields arise from string states.

4.2.1.5 Why the anticommuting oscillator does not make the coefficient Grassmann

One might still worry: if $b^i_{-1/2}$ is fermionic, should ζ_i or $A_i(k)$ be Grassmann-odd? For the NS gauge boson, the answer is no.

The oscillator $b^i_{-1/2}$ is odd with respect to the worldsheet fermion-number grading. Acting once gives a state with odd NS fermion number:

$$(-1)^F b^i_{-1/2}|0\rangle_{\mathrm{NS}} = -b^i_{-1/2}|0\rangle_{\mathrm{NS}},$$

up to the convention for the NS vacuum. But this $(-1)^F$ is a worldsheet grading used in the GSO projection. It is not identical to spacetime spin-statistics.

The spacetime statistics are determined by the spacetime little-group representation:

$$8_v \quad \text{is bosonic,}$$

while

$$8_s,\ 8_c \quad \text{are fermionic.}$$

Therefore the second-quantized coefficient $A_i(k)$ is an ordinary commuting bosonic field mode, even though the first-quantized oscillator $b^i_{-1/2}$ anticommutes with other b oscillators.

This is analogous to building a differential form with anticommuting basis elements dx^i. The components of an ordinary one-form,

$$A = A_i dx^i,$$

need not be Grassmann-odd just because the exterior basis satisfies

$$dx^i \wedge dx^j = -dx^j \wedge dx^i.$$

The anticommutation controls the internal algebra of the basis excitations, not the spacetime statistics of the coefficient field.

4.2.1.6 The light-cone NS spectrum

For the open RNS string with Neumann-Neumann boundary conditions in all target-space directions, the NS-sector worldsheet fermions are half-integer moded:

$$r \in \mathbb{Z} + \frac{1}{2}.$$

In light-cone gauge, only the transverse directions are independent:

$$i = 1, \dots, 8$$

in ten dimensions. The independent oscillators are

$$\alpha^i_{-n}, \qquad n = 1, 2, \dots,$$

and

$$b^i_{-r}, \qquad r = \frac{1}{2}, \frac{3}{2}, \dots .$$

They obey

$$[\alpha^i_m, \alpha^j_n] = m\delta^{ij}\delta_{m+n,0},$$

and

$$\{b^i_r, b^j_s\} = \delta^{ij}\delta_{r+s,0}.$$

The NS number operator is

$$N = \sum_{n=1}^{\infty} \alpha^i_{-n}\alpha^i_n + \sum_{r>0} r\, b^i_{-r} b^i_r.$$

The open-string NS mass formula is

$$\alpha' M^2 = N - \frac{1}{2}.$$

Therefore:

1. the NS vacuum has $N = 0$, so

   $$\alpha' M^2 = -\frac{1}{2},$$

   and is tachyonic before the GSO projection;

2. the state

   $$b^i_{-1/2}|0; k\rangle_{\mathrm{NS}}$$

   has

   $$N = \frac{1}{2},$$

   so

   $$M^2 = 0.$$

   This is the first massless NS state.

There are eight such states because $i = 1, \dots, 8$. In a covariant description this is the familiar massless gauge boson state

$$\zeta_\mu \psi^\mu_{-1/2}|0; k\rangle_{\mathrm{NS}},$$

with the physical constraints removing the timelike and longitudinal polarizations. In light-cone gauge only the eight transverse polarizations remain:

$$\zeta_i b^i_{-1/2}|0; k\rangle_{\mathrm{NS}}.$$

4.2.1.7 What anticommuting means here

The anticommutation relation says that the b oscillators generate an exterior algebra. For example,

$$b^i_{-1/2}b^j_{-1/2} = -b^j_{-1/2}b^i_{-1/2},$$

and in particular

$$(b^i_{-1/2})^2 = 0.$$

So each fermionic oscillator can be occupied at most once.

This affects the degeneracy and representation content of excited string states. For example,

$$b^i_{-1/2}|0\rangle_{\rm NS}$$

transforms as a vector of $SO(8)$, while

$$b^i_{-1/2}b^j_{-1/2}|0\rangle_{\rm NS}$$

is antisymmetric in i, j, so it transforms as an antisymmetric two-tensor of $SO(8)$:

$$\wedge^2 8_v.$$

The anticommuting nature of the b oscillators tells us that multiple b excitations are antisymmetrized. It does not tell us that the resulting target-space particle is a spacetime fermion.

A useful analogy is the exterior algebra of differential forms. The basis one-forms dx^i anticommute:

$$dx^i \wedge dx^j = -dx^j \wedge dx^i.$$

But a one-form is not a spacetime fermion. It is a tensorial object. Similarly, the $b^i_{-1/2}$ oscillators anticommute, but their index i is a vector index, not a spinor index.

4.2.1.8 Worldsheet spin versus spacetime spin

The RNS field ψ^μ has two kinds of transformation behavior:

1. it is a fermion on the worldsheet;
2. it is a vector in target spacetime.

The first statement means that ψ^μ has conformal weight

$$h = \frac{1}{2}$$

and its modes obey anticommutation relations.

The second statement means that under spacetime Lorentz transformations,

$$\psi^\mu$$

carries a vector index μ. In light-cone gauge the physical components are

$$\psi^i, \qquad i = 1, \dots, 8,$$

so their modes b^i_r carry the vector representation of the transverse rotation group $SO(8)$.

Thus

$$b^i_{-1/2}|0; k\rangle_{\rm NS}$$

is spacetime-vector-valued because of the index i. Its worldsheet origin as a fermionic oscillator only says that the oscillator is Grassmann-odd in the worldsheet operator algebra.

The distinction can be summarized as

$$\boxed{\text{worldsheet fermion} \neq \text{spacetime fermion.}}$$

In the RNS string, ψ^μ is a worldsheet fermion but a spacetime vector.

4.2.1.9 Why the NS massless state is a vector

Let $R \in SO(8)$ be a transverse rotation. The oscillators transform as

$$b^i_{-1/2} \longmapsto R^i{}_j b^j_{-1/2}.$$

Assume the NS ground state is a spacetime scalar:

$$|0;k\rangle_{\rm NS} \longmapsto |0;Rk\rangle_{\rm NS}.$$

Then the one-oscillator state transforms as

$$b^i_{-1/2}|0;k\rangle_{\rm NS} \longmapsto R^i{}_j b^j_{-1/2}|0;Rk\rangle_{\rm NS}.$$

That is exactly the transformation law of a vector.

Equivalently, a general polarization state

$$|\zeta;k\rangle = \zeta_i b^i_{-1/2}|0;k\rangle_{\rm NS}$$

transforms by

$$\zeta_i \longmapsto R_i{}^j \zeta_j.$$

So the physical polarization ζ_i is an $SO(8)$ vector. Since vectors are tensor representations, not spinor representations, this state is a spacetime boson.

Under a 2π spacetime rotation, a vector returns to itself:

$$8_v \longmapsto +8_v.$$

A spacetime spinor would pick up a minus sign under a 2π rotation. The NS massless state does not do that.

4.2.1.10 Where spacetime fermions come from: the R sector

Now compare this with the Ramond sector. In the R sector, the worldsheet fermions are integer moded:

$$r \in \mathbb{Z}.$$

In particular, there are zero modes

$$b^i_0.$$

They satisfy

$$\{b^i_0, b^j_0\} = \delta^{ij}.$$

This is a Clifford algebra. Therefore the R-sector ground states form a spinor representation of $SO(8)$:

$$|A\rangle_{\rm R}, \qquad A = 1, \dots, 8$$

after the usual GSO projection, with the chirality depending on the projection convention.

Schematically, the R ground state transforms as

$$|A\rangle_{\rm R} \longmapsto S(R)^A{}_B |B\rangle_{\rm R},$$

where $S(R)$ is a spinor representation matrix of $SO(8)$, not the vector matrix $R^i{}_j$.

Thus:

$$\text{NS sector} \quad \Rightarrow \quad \text{spacetime tensor/vector states,}$$

while

$$\text{R sector} \quad \Rightarrow \quad \text{spacetime spinor states.}$$

This is why the massless open superstring spectrum after the GSO projection contains a gauge boson from the NS sector and a gaugino from the R sector.

4.2.1.11 The role of the GSO projection

Before the GSO projection, the NS sector contains the tachyonic ground state

$$|0;k\rangle_{\mathrm{NS}},$$

with

$$\alpha' M^2 = -\frac{1}{2}.$$

The GSO projection removes this tachyon and keeps the states with the appropriate worldsheet fermion number. In the common open-superstring convention, it keeps the state

$$b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}$$

and removes the NS vacuum.

This sometimes adds to the confusion because the kept NS state has one b oscillator, hence odd worldsheet fermion number. But odd worldsheet fermion number is not the same as spacetime fermionic statistics. It is a quantum number of the worldsheet CFT Hilbert space.

The post-GSO massless open-string states are:

$$\text{NS:} \qquad b^i_{-1/2}|0;k\rangle_{\mathrm{NS}} \quad \in 8_v,$$

and

$$\text{R:} \qquad |A;k\rangle_{\mathrm{R}} \quad \in 8_s \text{ or } 8_c.$$

Together these form the field content of ten-dimensional $N = 1$ super Yang-Mills:

$$A_\mu \quad \text{and} \quad \lambda_A.$$

In light-cone gauge, the on-shell physical polarizations are

$$A_i, \qquad i = 1, \dots, 8,$$

and an $SO(8)$ spinor gaugino.

4.2.1.12 First-quantized oscillator algebra versus second-quantized statistics

There is one final conceptual point. The operators

$$\alpha^i_{-n}, \qquad b^i_{-r}$$

are not spacetime particle creation operators in a second-quantized quantum field theory. They are first-quantized string oscillators. They create internal excitations of a single string.

The spacetime statistics of the corresponding particle is determined after second quantization. A spacetime bosonic string state gives a spacetime field whose creation and annihilation operators commute. A spacetime fermionic string state gives a spacetime field whose creation and annihilation operators anticommute.

For the NS massless vector, the associated spacetime field is a gauge field:

$$A_\mu(x).$$

The one-particle state is

$$\zeta_i b^i_{-1/2}|0;k\rangle_{\mathrm{NS}},$$

but the second-quantized creation operator creating this gauge boson is not the same object as the worldsheet oscillator $b^i_{-1/2}$. The former is a spacetime particle creation operator; the latter is an internal oscillator acting on the single-string Hilbert space.

So there is no contradiction:

$$\boxed{b^i_{-1/2} \text{ anticommutes as a worldsheet oscillator, but } b^i_{-1/2}|0;k\rangle_{\mathrm{NS}} \text{ is a spacetime vector boson.}}$$

4.2.1.13 Vertex-operator viewpoint

The same conclusion appears in the covariant vertex-operator description. The massless NS vector vertex operator in the -1 picture is

$$V_A^{(-1)} = g_o\,\zeta_\mu \psi^\mu e^{-\phi} e^{ik\cdot X},$$

with

$$k^2 = 0, \qquad k\cdot\zeta = 0,$$

and the gauge equivalence

$$\zeta_\mu \sim \zeta_\mu + k_\mu.$$

The polarization is ζ_μ, a spacetime vector. The field ψ^μ is worldsheet-fermionic, but it carries a spacetime vector index. Thus this vertex operator creates a spacetime vector state.

By contrast, a massless R-sector fermion vertex involves a spin field:

$$V_\lambda^{(-1/2)} \sim u_A S^A e^{-\phi/2} e^{ik\cdot X}.$$

Here S^A is a spin field and u_A is a spacetime spinor polarization. This is the vertex operator that creates a spacetime fermion.

So the distinction is visible directly:

$$\psi^\mu \quad \text{carries a vector index,}$$

whereas

$$S^A \quad \text{carries a spinor index.}$$

4.2.1.14 Checks

For the NS massless state:

$$N = \frac{1}{2}, \qquad \alpha' M^2 = N - \frac{1}{2} = 0.$$

There are eight independent light-cone states:

$$b^i_{-1/2}|0;k\rangle_{\mathrm{NS}}, \qquad i = 1,\dots,8.$$

They transform as

$$8_v$$

of $SO(8)$, so they are the eight transverse polarizations of a ten-dimensional massless vector.

For the R ground state:

$$\{b_0^i, b_0^j\} = \delta^{ij}.$$

This Clifford algebra forces the ground state to transform as an $SO(8)$ spinor:

$$8_s \quad \text{or} \quad 8_c.$$

Those are the spacetime fermions.

4.2.1.15 Result

The NS fermionic oscillators obey

$$\{b_r^i, b_s^j\} = \delta^{ij}\delta_{r+s,0}.$$

This means they are worldsheet fermionic oscillators and generate antisymmetrized multi-oscillator states.

The massless NS state is

$$|\zeta;k\rangle = \zeta_i b^i_{-1/2}|0;k\rangle_{\mathrm{NS}},$$

with

$$N = \frac{1}{2}, \qquad M^2 = 0.$$

At fixed k,

$$\left\{b^i_{-1/2}|0;k\rangle_{\rm NS}\right\}_{i=1}^{8}$$

is a basis of the massless one-string Hilbert space, and ζ_i are the components of the polarization in that basis. The physical one-particle state is the full vector

$$|\zeta;k\rangle,$$

not ζ_i alone and not the basis state alone.

In second quantization, the corresponding string field contains

$$|\Psi\rangle \supset \int d^{10}k\, A_i(k) b^i_{-1/2}|0;k\rangle_{\rm NS},$$

so the spacetime gauge field mode $A_i(k)$ is the coefficient multiplying the first-quantized string state.

Because $b^i_{-1/2}$ carries a transverse vector index,

$$b^i_{-1/2}|0;k\rangle_{\rm NS} \in 8_v$$

under the massless little group $SO(8)$. Therefore it is a spacetime vector boson.

Spacetime fermions instead come from the R sector, where

$$\{b^i_0, b^j_0\} = \delta^{ij}$$

is a Clifford algebra, so the ground states transform as

$$8_s \quad \text{or} \quad 8_c.$$

Thus the correct distinction is

$$\boxed{\text{NS worldsheet fermion oscillator } b^i_{-1/2} \text{ creates a spacetime vector state, not a spacetime fermion.}}$$

Chapter 5

Kac–Moody Algebras

5.1 Q1: Integrable highest weights of simply-laced affine algebras

Consider the chiral algebra $\mathcal{A} = \hat{v} \ltimes \hat{g}$, where $\hat{v}$ is the Virasoro algebra and $\hat{g}$ is the level k Kac-Moody algebra of simply laced Lie algebra g whose roots are normalized to $(\text{length})^2 = 2$. Consider the $\mathfrak{su}(2)$ subalgebra of $\bar{g}$ generated by $E_1^{-\vec{\alpha}}, E_{-1}^{\vec{\alpha}}$ and $(k - \vec{\alpha} \cdot \vec{H}_0)$ where $\vec{\alpha}$ is a root, unitary requires that the eigenvalues of $(k - \vec{\alpha} \cdot \vec{H}_0)$ have to be integer. Acting on a primary state $|\vec{m}\rangle$ with weight $\vec{m}$. Since for simply-laced algebra we have $\vec{\alpha} \cdot \vec{m} \in \mathbb{Z}$, we find $k \in \mathbb{Z}$. Q1: why for simply-laced algebra we have $\vec{\alpha} \cdot \vec{m} \in \mathbb{Z}$?

Since $|\vec{m}\rangle$ is primary we have $E_1^{-\vec{\alpha}}|\vec{m}\rangle = 0$ and

$$||E_{-1}^{\vec{\alpha}}|\vec{m}\rangle||^2 = (k - \vec{\alpha} \cdot \vec{m})|||\vec{m}\rangle||^2 \geq 0$$

Gives that $k \geq \vec{\alpha} \cdot \vec{m}$. It is said that the right hand side of this inequality is maximized if we chose $\vec{\alpha}$ to be the highest root $\vec{\psi}$ and $\vec{m}$ to be the highest weight $\vec{m}_0$ of the given representations, we then have $k \geq \vec{\psi} \cdot \vec{m}_0$. Q2: what is a highest root and a highest weight, why the above statement is true? Explain with some examples.

Clearly for a given k only a finite number of highest weights satisfies this criterion. For $k = 1$ they are those belonging to the lowest dimensional representation in each conjugacy class. Q3: give some examples for the k=1 case.

> **Textbook location and related material.** Chapter 11, Sections 11.1, 11.5, and 11.6 (pp. 321–355): affine current algebras, unitary highest-weight representations, the level bound, characters, and the explicit $\widehat{\mathfrak{su}}(2)_k$ example. Sections 11.2–11.3 provide the root/weight-lattice background.

5.1.1 Answer

5.1.1.1 Core point

There are three related facts behind the statements in the question.

First, a representation weight is not an arbitrary vector in the real span of the roots. By definition it pairs integrally with every **coroot**:

$$\langle \vec{m}, \vec{\alpha}^\vee \rangle = \frac{2\vec{\alpha} \cdot \vec{m}}{\vec{\alpha}^{\,2}} \in \mathbb{Z}.$$

For a simply-laced algebra in the normalization $\vec{\alpha}^{\,2} = 2$, the coroot equals the root. Hence

$$\vec{\alpha} \cdot \vec{m} \in \mathbb{Z}.$$

Second, after choosing positive roots, a finite-dimensional irreducible g-representation has a dominant highest weight $\vec{m}_0 = \vec{\lambda}$. The root system itself has a highest root $\vec{\psi}$. The strongest affine positivity condition is

$$k - \vec{\psi} \cdot \vec{\lambda} \geq 0.$$

Together with integrality, this is the usual affine integrability condition

$$\vec{\psi} \cdot \vec{\lambda} \leq k.$$

Third, if

$$\vec{\psi} = \sum_{i=1}^{r} a_i \vec{\alpha}_i, \qquad \vec{\lambda} = \sum_{i=1}^{r} \lambda_i \vec{\omega}_i,$$

then, in the simply-laced normalization,

$$\vec{\psi} \cdot \vec{\lambda} = \sum_{i=1}^{r} a_i \lambda_i.$$

At level $k = 1$, this sum can be at most one. Thus the allowed nonzero highest weights are precisely the fundamental weights $\vec{\omega}_i$ attached to Dynkin nodes with $a_i = 1$. For ADE these are the minuscule fundamental weights. They give the smallest finite-dimensional representations in the different cosets of P/Q.

5.1.1.2 Weights pair integrally with coroots

Let $\vec{\alpha}_1, \dots, \vec{\alpha}_r$ be simple roots. The corresponding coroots are

$$\vec{\alpha}_i^\vee = \frac{2\vec{\alpha}_i}{\vec{\alpha}_i^2}.$$

For each simple root, the generators

$$E_0^{\vec{\alpha}_i}, \qquad E_0^{-\vec{\alpha}_i}, \qquad \vec{\alpha}_i^\vee \cdot \vec{H}_0$$

form an ordinary $\mathfrak{su}(2)$ subalgebra of the horizontal algebra g. If $|\vec{m}\rangle$ has weight $\vec{m}$, then

$$(\vec{\alpha}_i^\vee \cdot \vec{H}_0)|\vec{m}\rangle = \langle \vec{m}, \vec{\alpha}_i^\vee \rangle |\vec{m}\rangle.$$

Every finite-dimensional unitary $\mathfrak{su}(2)$ representation has integral eigenvalues of twice the third generator. Therefore

$$\langle \vec{m}, \vec{\alpha}_i^\vee \rangle \in \mathbb{Z}$$

for every simple coroot. This is exactly the statement that $\vec{m}$ lies in the weight lattice P.

Equivalently, introduce the fundamental weights $\vec{\omega}_j$ by

$$\langle \vec{\omega}_j, \vec{\alpha}_i^\vee \rangle = \delta_{ij}.$$

Every weight-lattice element can be written as

$$\vec{m} = \sum_{j=1}^{r} m_j \vec{\omega}_j, \qquad m_j \in \mathbb{Z}.$$

For a simply-laced algebra with $\vec{\alpha}_i^2 = 2$,

$$\vec{\alpha}_i^\vee = \vec{\alpha}_i,$$

and hence

$$\vec{\alpha}_i \cdot \vec{m} = m_i \in \mathbb{Z}.$$

Now let $\vec{\alpha}$ be any root, not necessarily simple. In a simply-laced root system it has an expansion

$$\vec{\alpha} = \sum_{i=1}^{r} n_i \vec{\alpha}_i, \qquad n_i \in \mathbb{Z}.$$

It follows that

$$\vec{\alpha} \cdot \vec{m} = \sum_{i=1}^{r} n_i (\vec{\alpha}_i \cdot \vec{m}) = \sum_{i=1}^{r} n_i m_i \in \mathbb{Z}.$$

This proves the statement used in Q1.

The simply-laced hypothesis is needed only to replace the coroot pairing by the ordinary inner product. For a general Lie algebra, the universal integrality statement is

$$\frac{2\vec{\alpha} \cdot \vec{m}}{\vec{\alpha}^2} = \langle \vec{m}, \vec{\alpha}^\vee \rangle \in \mathbb{Z},$$

not necessarily $\vec{\alpha} \cdot \vec{m} \in \mathbb{Z}$ for every root in one fixed normalization.

5.1.1.3 The affine $\mathfrak{su}(2)$ and level quantization

For each root $\vec{\alpha}$, define

$$\mathcal{J}_+ = E_1^{-\vec{\alpha}}, \qquad \mathcal{J}_- = E_{-1}^{\vec{\alpha}}, \qquad 2\mathcal{J}_3 = k - \vec{\alpha} \cdot \vec{H}_0.$$

With the normalization in the question, the affine commutation relations give

$$[\mathcal{J}_3, \mathcal{J}_\pm] = \pm \mathcal{J}_\pm, \qquad [\mathcal{J}_+, \mathcal{J}_-] = 2\mathcal{J}_3.$$

Thus these operators form an $\mathfrak{su}(2)$ algebra. On a weight state,

$$2\mathcal{J}_3 |\vec{m}\rangle = (k - \vec{\alpha} \cdot \vec{m}) |\vec{m}\rangle.$$

In a unitary representation, the eigenvalues of $2\mathcal{J}_3$ are integers, so

$$k - \vec{\alpha} \cdot \vec{m} \in \mathbb{Z}.$$

Since the previous section gives $\vec{\alpha} \cdot \vec{m} \in \mathbb{Z}$, one obtains

$$k \in \mathbb{Z}.$$

There is also a positivity condition. An affine primary multiplet is annihilated by all positive modes, so

$$E_1^{-\vec{\alpha}} |\vec{m}\rangle = 0.$$

Using $(E_{-1}^{\vec{\alpha}})^\dagger = E_1^{-\vec{\alpha}}$,

$$\begin{aligned} \left\| E_{-1}^{\vec{\alpha}} |\vec{m}\rangle \right\|^2 &= \langle \vec{m} | E_1^{-\vec{\alpha}} E_{-1}^{\vec{\alpha}} | \vec{m} \rangle \\ &= \langle \vec{m} | [E_1^{-\vec{\alpha}}, E_{-1}^{\vec{\alpha}}] | \vec{m} \rangle \\ &= (k - \vec{\alpha} \cdot \vec{m}) \langle \vec{m} | \vec{m} \rangle. \end{aligned}$$

Therefore

$$k - \vec{\alpha} \cdot \vec{m} \geq 0.$$

In fact, because $|\vec{m}\rangle$ is a highest-weight vector for this affine $\mathfrak{su}(2)$, unitarity gives the stronger combined statement

$$k - \vec{\alpha} \cdot \vec{m} \in \mathbb{Z}_{\geq 0}.$$

To see why the first norm inequality strengthens to an integer condition, set

$$\ell_{\vec{\alpha}} := k - \vec{\alpha} \cdot \vec{m}.$$

Repeated use of the $\mathfrak{su}(2)$ commutators gives

$$\left\| \mathcal{J}_-^N |\vec{m}\rangle \right\|^2 = N! \prod_{s=0}^{N-1} (\ell_{\vec{\alpha}} - s) \langle \vec{m} | \vec{m} \rangle.$$

If $\ell_{\vec{\alpha}}$ were positive but nonintegral, then for sufficiently large N one factor would become negative without any earlier factor vanishing. This would produce a negative-norm state. Hence the lowering chain must terminate when one factor vanishes, which requires

$$\ell_{\vec{\alpha}} \in \mathbb{Z}_{\geq 0}.$$

5.1.1.4 Highest weights of finite-dimensional representations

Choose a set of positive roots Δ_+. The simple roots are the positive roots that cannot be decomposed into sums of other positive roots. This choice determines raising operators $E_0^{\vec{\alpha}}$ for $\vec{\alpha} \in \Delta_+$.

An irreducible finite-dimensional representation has a highest-weight state $|\vec{\lambda}\rangle$ satisfying

$$E_0^{\vec{\alpha}_i}|\vec{\lambda}\rangle = 0 \qquad \text{for every simple root } \vec{\alpha}_i,$$

and

$$\vec{H}_0|\vec{\lambda}\rangle = \vec{\lambda}|\vec{\lambda}\rangle.$$

The weight $\vec{\lambda}$ is dominant integral:

$$\lambda_i := \langle \vec{\lambda}, \vec{\alpha}_i^\vee \rangle \in \mathbb{Z}_{\geq 0}.$$

It can therefore be expanded as

$$\vec{\lambda} = \sum_{i=1}^{r} \lambda_i \vec{\omega}_i.$$

Every other weight $\vec{m}$ in the representation is obtained by applying lowering operators. Consequently,

$$\vec{m} = \vec{\lambda} - \sum_{i=1}^{r} N_i \vec{\alpha}_i, \qquad N_i \in \mathbb{Z}_{\geq 0}.$$

This is the precise sense in which $\vec{\lambda}$ is the highest weight. “Highest” depends on the chosen positive-root system; it is not a statement about Euclidean length.

For example, for $\mathfrak{su}(2)$ there is one simple root $\vec{\alpha}$ and one fundamental weight

$$\vec{\omega} = \frac{\vec{\alpha}}{2}.$$

The spin-j representation has highest weight

$$\vec{\lambda} = 2j\,\vec{\omega},$$

and its weights are

$$\vec{\lambda}, \quad \vec{\lambda} - \vec{\alpha}, \quad \ldots, \quad \vec{\lambda} - 2j\vec{\alpha}.$$

5.1.1.5 The highest root

Every positive root has a unique expansion

$$\vec{\alpha} = \sum_{i=1}^{r} n_i \vec{\alpha}_i, \qquad n_i \in \mathbb{Z}_{\geq 0}.$$

The root partial order is defined by

$$\vec{\beta} \succeq \vec{\alpha} \quad \Longleftrightarrow \quad \vec{\beta} - \vec{\alpha} = \sum_i N_i \vec{\alpha}_i, \qquad N_i \in \mathbb{Z}_{\geq 0}.$$

For a simple Lie algebra there is a unique maximal positive root in this order. It is the highest root,

$$\vec{\psi} = \sum_{i=1}^{r} a_i \vec{\alpha}_i.$$

The coefficients a_i are called marks.

Another useful characterization is that the roots are the nonzero weights of the adjoint representation, and $\vec{\psi}$ is the highest weight of the adjoint representation. In particular, $\vec{\psi}$ is dominant:

$$\langle \vec{\psi}, \vec{\alpha}_i^\vee \rangle \geq 0$$

for every simple root.

For $A_1 = \mathfrak{su}(2)$,

$$\vec{\psi} = \vec{\alpha}.$$

For $A_2 = \mathfrak{su}(3)$,

$$\vec{\psi} = \vec{\alpha}_1 + \vec{\alpha}_2.$$

For $D_r = \mathfrak{so}(2r)$, with the standard numbering in which $\vec{\alpha}_{r-1}$ and $\vec{\alpha}_r$ are the two spinor-end nodes,

$$\vec{\psi} = \vec{\alpha}_1 + 2\vec{\alpha}_2 + \cdots + 2\vec{\alpha}_{r-2} + \vec{\alpha}_{r-1} + \vec{\alpha}_r.$$

5.1.1.6 Why $\vec{\psi} \cdot \vec{\lambda}$ is the maximal pairing

The statement in the question should be understood as follows: for a fixed irreducible representation $V_{\vec{\lambda}}$, maximize

$$\vec{\alpha} \cdot \vec{m}$$

over all roots $\vec{\alpha}$ and all weights $\vec{m}$ of $V_{\vec{\lambda}}$. For a simply-laced root system, the maximum is

$$\vec{\psi} \cdot \vec{\lambda}.$$

Here is a direct proof.

Because the root system is irreducible and simply laced, all roots have the same length and form a single orbit under the Weyl group W. Therefore, for any root $\vec{\alpha}$, there is a Weyl transformation w such that

$$w(\vec{\alpha}) = \vec{\psi}.$$

The Weyl group preserves the inner product, and the set of weights of a finite-dimensional representation is Weyl invariant. Thus

$$\vec{\alpha} \cdot \vec{m} = w(\vec{\alpha}) \cdot w(\vec{m}) = \vec{\psi} \cdot \vec{\mu},$$

where $\vec{\mu} = w(\vec{m})$ is another weight of the same representation.

Since $\vec{\lambda}$ is the highest weight,

$$\vec{\lambda} - \vec{\mu} = \sum_i N_i \vec{\alpha}_i, \qquad N_i \in \mathbb{Z}_{\geq 0}.$$

Since $\vec{\psi}$ is dominant and the algebra is simply laced,

$$\vec{\psi} \cdot \vec{\alpha}_i = \langle \vec{\psi}, \vec{\alpha}_i^\vee \rangle \geq 0.$$

It follows that

$$\begin{aligned} \vec{\psi} \cdot \vec{\lambda} - \vec{\psi} \cdot \vec{\mu} &= \vec{\psi} \cdot (\vec{\lambda} - \vec{\mu}) \\ &= \sum_i N_i \, \vec{\psi} \cdot \vec{\alpha}_i \\ &\geq 0. \end{aligned}$$

Therefore

$$\vec{\alpha} \cdot \vec{m} \leq \vec{\psi} \cdot \vec{\lambda}.$$

The upper bound is attained by choosing

$$\vec{\alpha} = \vec{\psi}, \qquad \vec{m} = \vec{\lambda}.$$

This proves the maximization statement.

Applying the norm inequality to this pair gives

$$k \geq \vec{\psi} \cdot \vec{\lambda}.$$

This is not merely one convenient necessary bound. Together with $\vec{\lambda}$ being dominant integral, it is precisely the integrability condition for an untwisted affine highest-weight representation of a simply-laced algebra.

5.1.1.7 Dynkin-label form of the integrability condition

Write

$$\vec{\psi} = \sum_i a_i \vec{\alpha}_i, \qquad \vec{\lambda} = \sum_j \lambda_j \vec{\omega}_j.$$

Since

$$\vec{\alpha}_i \cdot \vec{\omega}_j = \delta_{ij}$$

in the simply-laced normalization,

$$\vec{\psi} \cdot \vec{\lambda} = \sum_{i,j} a_i \lambda_j (\vec{\alpha}_i \cdot \vec{\omega}_j) = \sum_i a_i \lambda_i.$$

Thus the allowed affine primary highest weights at level k obey

$$\boxed{\lambda_i \in \mathbb{Z}_{\geq 0}, \qquad \sum_i a_i \lambda_i \leq k.}$$

Equivalently, define the affine Dynkin label

$$\lambda_0 := k - \vec{\psi} \cdot \vec{\lambda} = k - \sum_i a_i \lambda_i.$$

Then affine integrability says that all affine Dynkin labels are nonnegative integers:

$$\lambda_0, \lambda_1, \dots, \lambda_r \in \mathbb{Z}_{\geq 0}.$$

The $\mathfrak{su}(2)$ built from

$$E_1^{-\vec{\psi}}, \qquad E_{-1}^{\vec{\psi}}, \qquad k - \vec{\psi} \cdot \vec{H}_0$$

is precisely the $\mathfrak{su}(2)$ associated with the affine simple root

$$\vec{\alpha}_0 = \delta - \vec{\psi}.$$

5.1.1.8 Examples of the bound

For $A_1 = \mathfrak{su}(2)$, let

$$\vec{\lambda} = n\vec{\omega}, \qquad n = 2j \in \mathbb{Z}_{\geq 0}.$$

Because $\vec{\psi} = \vec{\alpha} = 2\vec{\omega}$ and $\vec{\alpha} \cdot \vec{\omega} = 1$,

$$\vec{\psi} \cdot \vec{\lambda} = n = 2j.$$

The affine integrability condition is

$$n \leq k, \qquad \text{or equivalently} \qquad j \leq \frac{k}{2}.$$

At level one, only $j = 0$ and $j = \frac{1}{2}$ occur as affine primary representations.

For $A_2 = \mathfrak{su}(3)$, write

$$\vec{\lambda} = p\vec{\omega}_1 + q\vec{\omega}_2, \qquad p, q \in \mathbb{Z}_{\geq 0}.$$

Since

$$\vec{\psi} = \vec{\alpha}_1 + \vec{\alpha}_2,$$

one has

$$\vec{\psi} \cdot \vec{\lambda} = p + q.$$

Therefore

$$p + q \leq k.$$

At level one the possibilities are

$$(p, q) = (0, 0), (1, 0), (0, 1),$$

corresponding to

$$\mathbf{1}, \qquad \mathbf{3}, \qquad \overline{\mathbf{3}}.$$

For comparison, the adjoint **8** has highest weight

$$\vec{\lambda} = \vec{\omega}_1 + \vec{\omega}_2 = \vec{\psi},$$

so

$$\vec{\psi} \cdot \vec{\lambda} = 2.$$

It first occurs as an affine primary at level $k = 2$. At level one the eight current states still exist, but they are level-one descendants in the vacuum module, not a separate affine-primary module.

For $D_r = \mathfrak{so}(2r)$, write

$$\vec{\lambda} = \sum_{i=1}^{r} \lambda_i \vec{\omega}_i.$$

The highest-root expansion above gives

$$\vec{\psi} \cdot \vec{\lambda} = \lambda_1 + 2\lambda_2 + \cdots + 2\lambda_{r-2} + \lambda_{r-1} + \lambda_r.$$

At level one, the only possibilities are

$$\vec{\lambda} = 0, \qquad \vec{\omega}_1, \qquad \vec{\omega}_{r-1}, \qquad \vec{\omega}_r.$$

These are the trivial, vector, and two chiral-spinor representations.

5.1.1.9 Why only finitely many highest weights occur at fixed level

The Dynkin labels λ_i are nonnegative integers, and every mark a_i is a positive integer. Therefore

$$\sum_i a_i \lambda_i \le k$$

has only finitely many solutions. This is the elementary reason that an affine WZW theory at positive integer level has only finitely many integrable affine highest-weight representations.

At level one,

$$\sum_i a_i \lambda_i \le 1.$$

Because all $a_i \ge 1$ and all $\lambda_i \ge 0$ are integers, either

$$\lambda_i = 0 \qquad \text{for every } i,$$

or there is exactly one node j such that

$$a_j = 1, \qquad \lambda_j = 1,$$

with all other Dynkin labels zero. Hence

$$\vec{\lambda} = 0 \qquad \text{or} \qquad \vec{\lambda} = \vec{\omega}_j \text{ with } a_j = 1.$$

For ADE root systems, these $\vec{\omega}_j$ are precisely the nonzero minuscule fundamental weights.

A dominant weight $\vec{\omega}$ is minuscule when the weights of its irreducible representation form a single Weyl orbit. Equivalently, for every positive root $\vec{\alpha}$,

$$\langle \vec{\omega}, \vec{\alpha}^\vee \rangle \in \{0, 1\}.$$

In particular,

$$\langle \vec{\omega}, \vec{\psi}^\vee \rangle = 1,$$

which is exactly the nontrivial level-one condition in the simply-laced case.

5.1.1.10 Level-one ADE examples

The complete list is:

Algebra	P/Q	Integrable horizontal highest weights at $k=1$	Finite-dimensional representations
$A_r = \mathfrak{su}(r+1)$	$\mathbb{Z}_{r+1}$	$0, \vec{\omega}_1, \ldots, \vec{\omega}_r$	$\mathbf{1}$ and $\bigwedge^i (\mathbf{r}+\mathbf{1})$, $i=1,\ldots,r$
$D_r = \mathfrak{so}(2r)$, r even	$\mathbb{Z}_2 \times \mathbb{Z}_2$	$0, \vec{\omega}_1, \vec{\omega}_{r-1}, \vec{\omega}_r$	$\mathbf{1}$, vector, and the two chiral spinors
$D_r = \mathfrak{so}(2r)$, r odd	$\mathbb{Z}_4$	$0, \vec{\omega}_1, \vec{\omega}_{r-1}, \vec{\omega}_r$	$\mathbf{1}$, vector, and the two chiral spinors
E_6	$\mathbb{Z}_3$	$0, \vec{\omega}_1, \vec{\omega}_6$	$\mathbf{1}, \mathbf{27}, \overline{\mathbf{27}}$
E_7	$\mathbb{Z}_2$	$0, \vec{\omega}_7$	$\mathbf{1}, \mathbf{56}$
E_8	0	0	only $\mathbf{1}$ as an affine primary

The E_6 and E_7 node labels in the table use the standard Bourbaki convention.

Some concrete special cases are:

- $SU(2)_1$: $\mathbf{1}$ and $\mathbf{2}$.
- $SU(3)_1$: $\mathbf{1}$, $\mathbf{3}$, and $\overline{\mathbf{3}}$.
- $SU(4)_1$: $\mathbf{1}$, $\mathbf{4}$, $\mathbf{6}$, and $\overline{\mathbf{4}}$.
- $SO(8)_1$: $\mathbf{1}$, $\mathbf{8}_v$, $\mathbf{8}_s$, and $\mathbf{8}_c$.
- $(E_6)_1$: $\mathbf{1}$, $\mathbf{27}$, and $\overline{\mathbf{27}}$.
- $(E_7)_1$: $\mathbf{1}$ and $\mathbf{56}$.
- $(E_8)_1$: only the vacuum affine-primary module.

5.1.1.11 Meaning of "one lowest-dimensional representation in each conjugacy class"

The phrase "conjugacy class" here refers to a coset of the weight lattice modulo the root lattice:

$$[\vec{\lambda}] \in P/Q.$$

Two highest weights are in the same class when

$$\vec{\lambda} - \vec{\lambda}' \in Q.$$

For the simply connected compact group G with Lie algebra g, P/Q is naturally related to the center $Z(G)$; a weight-lattice coset records the central character of a representation. It is not a conjugacy class of elements of G, and it is not merely the distinction between a representation and its complex conjugate.

For simply-laced ADE algebras, each coset in P/Q has a distinguished dominant representative of minimal size: the zero weight for the trivial coset and a minuscule fundamental weight for every nontrivial coset. These are exactly the level-one integrable highest weights listed above. This explains the statement in the question.

For example,

$$P/Q \cong \mathbb{Z}_3$$

for $SU(3)$. Its three cosets are represented at level one by

$$0, \qquad \vec{\omega}_1, \qquad \vec{\omega}_2,$$

corresponding to

$$\mathbf{1}, \qquad \mathbf{3}, \qquad \overline{\mathbf{3}}.$$

Representations such as the **6** or **10** lie in one of the same three lattice cosets, but their highest weights have larger $\vec{\psi}\cdot\vec{\lambda}$ and therefore do not satisfy the level-one bound.

For E_8, one has

$$P = Q, \qquad P/Q = 0.$$

There is only the trivial coset, whose minimal representative is 0. Therefore $(E_8)_1$ has only the vacuum integrable affine module. This does not mean that the theory has no E_8 currents: the 248 adjoint current states occur inside the vacuum module as descendants.

5.1.1.12 Checks

For $SU(2)$, the level-one inequality gives

$$2j \le 1,$$

so only $j = 0, \frac{1}{2}$ are allowed. This agrees with the two classes of

$$P/Q \cong \mathbb{Z}_2.$$

For $SU(3)$, the level-one inequality gives

$$p + q \le 1,$$

so only $(0,0), (1,0), (0,1)$ occur. This agrees with the three classes of

$$P/Q \cong \mathbb{Z}_3.$$

For $SO(8)$, the highest-root marks are

$$(a_1, a_2, a_3, a_4) = (1, 2, 1, 1)$$

in the standard D_4 numbering. Hence the level-one weights are

$$0, \vec{\omega}_1, \vec{\omega}_3, \vec{\omega}_4,$$

which give

$$\mathbf{1}, \mathbf{8}_v, \mathbf{8}_s, \mathbf{8}_c.$$

This matches the four elements of

$$P/Q \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

5.1.1.13 Result

The weight-lattice integrality condition is

$$\langle \vec{m}, \vec{\alpha}^\vee \rangle = \frac{2\vec{\alpha}\cdot\vec{m}}{\vec{\alpha}^{\,2}} \in \mathbb{Z}.$$

For simply-laced g with $\vec{\alpha}^{\,2} = 2$,

$$\vec{\alpha}^\vee = \vec{\alpha}, \qquad \vec{\alpha}\cdot\vec{m} \in \mathbb{Z}.$$

The affine-root $\mathfrak{su}(2)$ gives

$$k - \vec{\alpha}\cdot\vec{m} \in \mathbb{Z}_{\ge 0},$$

and therefore

$$k \in \mathbb{Z}_{\ge 0}.$$

For a horizontal irreducible representation with dominant highest weight $\vec{\lambda}$, the maximal root-weight pairing is

$$\max_{\vec{\alpha}\in\Delta,\, \vec{m}\in \mathrm{Wt}(V_{\vec{\lambda}})} (\vec{\alpha}\cdot\vec{m}) = \vec{\psi}\cdot\vec{\lambda}.$$

Thus affine integrability is

$$\boxed{\vec{\psi}\cdot\vec{\lambda} = \sum_i a_i \lambda_i \le k.}$$

At level one,

$$\vec{\lambda} = 0 \qquad \text{or} \qquad \vec{\lambda} = \vec{\omega}_j \text{ for a node with } a_j = 1.$$

For ADE, these are the vacuum and the minuscule fundamental representations, one minimal representative for each class in P/Q.

Consider M-theory on a local CY4 given by $\mathbb{P}^1$ with normal bundle $\mathcal{O} \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$, in the 3d field theory, what is the susy multiplet given by a single M2-brane wrapping this $\mathbb{P}^1$?

The moduli space of the $\mathbb{P}^1$ is a Riemann surface Σ extended in the direction of $\mathcal{O}$.

Chapter 6

Covariant Vertex Operators, BRST, and Covariant Lattices

6.1 Q1: Why physical vertex operators have matter weight one

Why the vertex operators of physical massless states must have $h = 1$?

> **Textbook location and related material.** Chapter 13, Section 13.2 (pp. 403–411): BRST-invariant covariant vertex operators. Chapter 5, Section 5.2 (pp. 110–120) supplies the BRST cohomology, and Chapter 4 explains primary weights and integrated insertions.

6.1.1 Answer

6.1.1.1 Core point

The statement must first be phrased carefully. For the critical **bosonic** string, it is the **matter part** of an on-shell open-string vertex operator that has

$$h = 1,$$

whereas the matter part of an on-shell closed-string vertex has

$$(h, \bar{h}) = (1, 1).$$

The corresponding unintegrated, ghost-dressed operators are

$$cV_{\mathrm{m}} \qquad \text{and} \qquad c\bar{c}V_{\mathrm{m}},$$

and these have total conformal weight zero because c and $\bar{c}$ have weights -1.

The number one arises in three equivalent descriptions of the same constraint:

1. an integrated vertex must be invariant under residual worldsheet conformal reparametrizations;
2. the state created by the vertex must obey the old covariant physical-state condition $(L_0 - 1)|\Psi\rangle = 0$;
3. the ghost-dressed vertex must be BRST closed.

However, $h = 1$ is not uniquely a masslessness condition. It is the complete worldsheet on-shell condition. The bosonic tachyon vertex also has $h = 1$. For a massless state, the oscillator or derivative part already contributes one unit of conformal weight, while the momentum exponential contributes zero because $k^2 = 0$.

6.1.1.2 Residual reparametrization invariance of an integrated vertex

After fixing the worldsheet metric to conformal gauge, one still has residual conformal coordinate changes. Consider first an open-string vertex inserted on a boundary coordinate x. If $V_{\mathrm{m}}(x)$ is a boundary primary of weight h, then under $x \mapsto x' = f(x)$ it transforms as

$$V'_{\mathrm{m}}(x') = \left(\frac{dx}{dx'}\right)^h V_{\mathrm{m}}(x).$$

An integrated vertex appears in an amplitude as

$$\int dx\, V_{\mathrm{m}}(x).$$

Since $dx' = (dx'/dx)dx$, the transformed insertion is

$$\int dx'\, V'_{\mathrm{m}}(x') = \int dx\, \left(\frac{dx'}{dx}\right)^{1-h} V_{\mathrm{m}}(x).$$

It is invariant for an arbitrary residual reparametrization precisely when

$$h = 1.$$

Thus the matter operator must be a marginal boundary operator.

For a closed-string insertion on a complex worldsheet coordinate $(z, \bar{z})$,

$$\int d^2z\, V_{\mathrm{m}}(z, \bar{z}),$$

the measure transforms with one holomorphic and one antiholomorphic Jacobian. If V_{m} has weights $(h, \bar{h})$, invariance under independent local maps $z \mapsto f(z)$ and $\bar{z} \mapsto \bar{f}(\bar{z})$ requires

$$h = 1, \qquad \bar{h} = 1.$$

Therefore an integrated closed-string vertex is a marginal bulk operator of weights $(1, 1)$.

This argument also explains the unintegrated form. Gauge fixing the positions of some vertex insertions introduces Faddeev–Popov ghosts. Since

$$h[c] = -1, \qquad \bar{h}[\bar{c}] = -1,$$

one has

$$h[cV_{\mathrm{m}}] = -1 + 1 = 0$$

for the open string, and

$$(h, \bar{h})[c\bar{c}V_{\mathrm{m}}] = (0, 0)$$

for the closed string. Hence the fixed, ghost-dressed insertion is a worldsheet scalar, as required.

6.1.1.3 The same condition in old covariant quantization

By the state–operator correspondence, a matter operator inserted at the origin creates a matter state:

$$|\Psi\rangle = V_{\mathrm{m}}(0)|0\rangle.$$

If V_{m} is a Virasoro primary of weight h, then

$$L_0|\Psi\rangle = h|\Psi\rangle, \qquad L_n|\Psi\rangle = 0 \quad (n > 0).$$

In old covariant quantization, the bosonic open-string physical-state conditions are

$$(L_n - a\delta_{n0})|\Psi\rangle = 0, \qquad n \geq 0,$$

with the critical intercept $a = 1$. The $n = 0$ condition is therefore

$$(L_0 - 1)|\Psi\rangle = 0,$$

so the corresponding matter operator has $h = 1$. The conditions with $n > 0$ impose primarity and remove negative-norm polarizations; they contain information such as transversality that is not captured by the value of h alone.

For an open-string state of momentum k and oscillator level N,

$$L_0 = \alpha' k^2 + N.$$

Thus

$$\alpha' k^2 + N = 1.$$

Using $k^2 = -m^2$ gives

$$m^2 = \frac{N-1}{\alpha'}.$$

The condition $h = 1$ therefore includes every on-shell level: $N = 0$ is the tachyon and $N = 1$ is the massless level.

For a closed-string state,

$$L_0 = \frac{\alpha' k^2}{4} + N, \qquad \bar{L}_0 = \frac{\alpha' k^2}{4} + \bar{N}.$$

The two physical-state conditions are

$$(L_0 - 1)|\Psi\rangle = 0, \qquad (\bar{L}_0 - 1)|\Psi\rangle = 0.$$

Equivalently,

$$\frac{\alpha' k^2}{4} + N = 1, \qquad \frac{\alpha' k^2}{4} + \bar{N} = 1.$$

Subtracting the two equations gives level matching $N = \bar{N}$, while either equation gives

$$m^2 = \frac{4(N-1)}{\alpha'} = \frac{4(\bar{N}-1)}{\alpha'}.$$

Again, $(h, \bar{h}) = (1, 1)$ is the full on-shell condition; masslessness is the particular level $N = \bar{N} = 1$.

6.1.1.4 BRST closure and the role of the ghosts

The BRST description packages the Virasoro constraints and their gauge equivalences into the cohomology of the BRST charge Q_B. Let V_m be a matter primary of weight h. The operator product with the matter stress tensor is

$$T_\mathrm{m}(z) V_\mathrm{m}(w) \sim \frac{h V_\mathrm{m}(w)}{(z-w)^2} + \frac{\partial V_\mathrm{m}(w)}{z-w}.$$

Evaluating the BRST contour integral on the ghost-number-one operator cV_m gives, up to an overall sign convention,

$$Q_B(cV_\mathrm{m}) \propto (1-h) c\partial c\, V_\mathrm{m}.$$

Consequently,

$$Q_B(cV_\mathrm{m}) = 0$$

requires $h = 1$, provided V_m is a matter primary. If the candidate operator is not primary, additional poles in its stress-tensor OPE generate additional BRST constraints. For the massless vector, those extra conditions give transversality.

For the closed string there are independent left- and right-moving BRST charges. Acting on

$$c\bar{c} V_\mathrm{m}$$

tests the two sectors separately, so BRST closure requires

$$(h, \bar{h}) = (1, 1)$$

together with left- and right-moving primarity. Two BRST-closed operators differing by a BRST-exact operator describe the same physical state:

$$\mathcal{V} \sim \mathcal{V} + Q_B\Lambda.$$

This quotient is the covariant origin of spacetime gauge equivalence.

Thus there is no contradiction between saying that the matter vertex has weight one and saying that a physical unintegrated vertex has weight zero. They refer to different operators:

$$V_{\mathrm{m}} : h = 1, \qquad cV_{\mathrm{m}} : h_{\mathrm{tot}} = 0.$$

6.1.1.5 Free-boson conformal weights and why weight one is not synonymous with massless

For a free boson in the stated normalization, the exponential has open-string boundary weight

$$h_{\mathrm{open}}[: e^{ik\cdot X} :] = \alpha' k^2,$$

whereas a closed-string exponential has

$$h_{\mathrm{closed}}[: e^{ik\cdot X} :] = \bar{h}_{\mathrm{closed}}[: e^{ik\cdot X} :] = \frac{\alpha' k^2}{4}.$$

Every holomorphic derivative ∂X^μ contributes one unit of h, and every antiholomorphic derivative $\bar{\partial} X^\nu$ contributes one unit of $\bar{h}$. More generally, oscillator excitations contribute their oscillator level. Hence

$$h_{\mathrm{open}} = N + \alpha' k^2,$$

and

$$h_{\mathrm{closed}} = N + \frac{\alpha' k^2}{4}, \qquad \bar{h}_{\mathrm{closed}} = \bar{N} + \frac{\alpha' k^2}{4}.$$

For the open-string tachyon,

$$V_T =: e^{ik\cdot X} :,$$

there is no derivative contribution. The condition $h = 1$ becomes

$$\alpha' k^2 = 1,$$

or

$$m^2 = -\frac{1}{\alpha'}.$$

Thus this tachyonic matter vertex has weight one when it is on shell.

Similarly, the closed-string tachyon vertex

$$V_T =: e^{ik\cdot X} :$$

has $(h, \bar{h}) = (1, 1)$ when

$$\frac{\alpha' k^2}{4} = 1,$$

which gives

$$m^2 = -\frac{4}{\alpha'}.$$

By contrast, a massless vertex contains a level-one oscillator or derivative in each required chiral sector. That part supplies the unit weight, while $k^2 = 0$ makes the exponential contribute zero. This is the precise sense in which the massless vertex has weight one.

6.1.1.6 Open-string massless vector

The matter vertex for an open-string vector can be written on the boundary as

$$V_A(x) = \epsilon_\mu :\partial_t X^\mu e^{ik\cdot X}: (x),$$

where ∂_t is tangent to the boundary. Its weight is

$$h[V_A] = 1 + \alpha' k^2.$$

Requiring $h = 1$ gives

$$k^2 = 0,$$

so the state is massless. But one must also require that V_A be primary. The potentially unwanted higher-order pole in $T(z)V_A(x)$ vanishes only if

$$k \cdot \epsilon = 0.$$

This is the same condition obtained from $L_1|\Psi\rangle = 0$ for

$$|\Psi\rangle = \epsilon_\mu \alpha^\mu_{-1}|0;k\rangle.$$

A longitudinal polarization is physically trivial. Indeed,

$$k_\mu \partial_t X^\mu e^{ik\cdot X} = \frac{1}{i}\partial_t e^{ik\cdot X},$$

so its integrated vertex is a boundary total derivative. In BRST language the same state is BRST exact. Therefore

$$\epsilon_\mu \sim \epsilon_\mu + \lambda k_\mu,$$

which is the momentum-space gauge equivalence of a massless vector.

6.1.1.7 Closed-string massless tensor states

At the first excited closed-string level the general matter vertex is

$$V_\epsilon(z,\bar{z}) = \epsilon_{\mu\nu} : \partial X^\mu \bar{\partial} X^\nu e^{ik\cdot X} :$$

and has weights

$$(h,\bar{h}) = \left(1 + \frac{\alpha' k^2}{4}, 1 + \frac{\alpha' k^2}{4}\right).$$

The condition $(h,\bar{h}) = (1,1)$ gives

$$k^2 = 0.$$

Left- and right-moving primarity further require

$$k^\mu \epsilon_{\mu\nu} = 0, \qquad k^\nu \epsilon_{\mu\nu} = 0.$$

BRST-exact states generate the polarization equivalence

$$\epsilon_{\mu\nu} \sim \epsilon_{\mu\nu} + k_\mu \xi_\nu + k_\nu \tilde{\xi}_\mu,$$

with the usual on-shell restrictions on the gauge parameters. The symmetric traceless, antisymmetric, and trace combinations give the graviton, Kalb–Ramond field, and dilaton. Their spacetime gauge redundancies are the symmetric and antisymmetric parts of the displayed BRST equivalence.

6.1.1.8 Checks

1. **Integrated versus unintegrated:** V_{m} has $h = 1$ so that $dx\, V_{\mathrm{m}}$ is invariant; cV_{m} has total weight zero so that a fixed insertion is invariant. These statements are complementary.

2. **Tachyon check:** setting $N = 0$ in $h = N + \alpha' k^2 = 1$ gives the open-string tachyon mass, not a massless state. Therefore weight one cannot by itself mean massless.

3. **Massless check:** setting $N = 1$ gives $k^2 = 0$. The oscillator supplies one unit of weight and the exponential supplies zero.

4. **Closed-string check:** both sectors must independently have weight one. Requiring only $h+\bar{h} = 2$ would be too weak because it would not enforce the two Virasoro constraints or level matching separately.

5. **Superstring caveat:** in the RNS string the matter and superghost contributions depend on the picture. For example, in the NS -1 picture, $\psi^\mu e^{ik\cdot X}$ has matter weight $1/2$ on shell and $e^{-\phi}$ supplies another $1/2$, while c supplies -1. Thus the invariant statement is that the complete integrated or ghost-dressed vertex has the required total weight; the simple phrase "the matter vertex has $h = 1$" is literally the bosonic-string statement, or a statement made after specifying the superghost picture.

6.1.1.9 Result

For the bosonic open string,

$$\boxed{h[V_{\mathrm{m}}] = 1, \qquad h[cV_{\mathrm{m}}] = 0}$$

because

$$\int dx\, V_{\mathrm{m}}$$

must be invariant, equivalently

$$(L_0 - 1)|\Psi\rangle = 0, \qquad Q_B(cV_{\mathrm{m}}) = 0.$$

With

$$h = N + \alpha' k^2,$$

the on-shell condition gives

$$m^2 = \frac{N-1}{\alpha'}.$$

For the bosonic closed string,

$$\boxed{(h,\bar{h})[V_{\mathrm{m}}] = (1,1), \qquad (h,\bar{h})[c\bar{c}V_{\mathrm{m}}] = (0,0)}$$

with

$$h = N + \frac{\alpha' k^2}{4}, \qquad \bar{h} = \bar{N} + \frac{\alpha' k^2}{4}.$$

The two conditions imply level matching and the closed-string mass shell.

For massless states specifically,

$$N = 1 \quad \text{(open)}, \qquad N = \bar{N} = 1 \quad \text{(closed)}, \qquad k^2 = 0.$$

Therefore the derivative or oscillator part contributes the required unit weight, while the momentum exponential contributes zero. The value $h = 1$ expresses the worldsheet physical-state condition; masslessness is the special level-one solution of that condition.

6.2 Q2: Conformal weights and pictures in the RNS superstring

I know the case of the bosonic string theory, but what about the superstring, do their vertex operators have h=1 and why?

Textbook location and related material. Chapter 13, Sections 13.1–13.2 (pp. 391–411): bosonized superghosts, picture number, BRST invariance, and picture changing. The NS/R intercepts come from Chapter 8.

6.2.1 Answer

6.2.1.1 Core point

Yes, but in the RNS superstring one must say **which part** of the vertex operator is being assigned weight one. The universal statement is:

- an integrated open-string insertion, including its matter and $\beta\gamma$ superghost factors but not a c ghost, must have total weight $h = 1$;
- an integrated closed-string insertion must have total weights $(h, \bar{h}) = (1, 1)$;
- after multiplying by c for an open-string fixed insertion, or by $c\bar{c}$ for a closed-string fixed insertion, the unintegrated operator has total weight zero.

What changes relative to the bosonic string is that the RNS vertex carries a **picture number**. The matter factor alone need not have $h = 1$, because the bosonized superghost exponential $e^{q\phi}$ also contributes conformal weight. For example, the massless NS matter factor $\epsilon \cdot \psi e^{ik\cdot X}$ has $h = 1/2$ on shell in picture -1, and $e^{-\phi}$ contributes the remaining $1/2$. In picture zero the superghost exponential is absent and the massless NS matter operator itself has $h = 1$.

Thus picture changing redistributes conformal weight between the matter and superghost factors without changing the conformal weight or BRST cohomology class of the complete physical insertion.

6.2.1.2 Why the complete integrated insertion still has weight one

The residual reparametrization argument is unchanged by worldsheet supersymmetry. On an open-string boundary, an integrated insertion is

$$\int dx\, \mathcal{V}(x),$$

where $\mathcal{V}$ now denotes the complete local RNS operator apart from the c ghost. If $\mathcal{V}$ has weight h, then under $x \mapsto x' = f(x)$,

$$dx'\, \mathcal{V}'(x') = dx \left(\frac{dx'}{dx}\right)^{1-h} \mathcal{V}(x).$$

Invariance for arbitrary residual conformal maps requires

$$h[\mathcal{V}] = 1.$$

For a closed-string bulk insertion,

$$\int d^2z\, \mathcal{V}(z, \bar{z}),$$

the two Jacobians require

$$(h, \bar{h})[\mathcal{V}] = (1, 1).$$

The reparametrization ghosts still obey

$$h[c] = -1, \qquad \bar{h}[\bar{c}] = -1.$$

Therefore the corresponding fixed insertions have

$$h[c\mathcal{V}] = 0$$

for an open string and

$$(h, \bar{h})[c\bar{c}\mathcal{V}] = (0, 0)$$

for a closed string.

Worldsheet supersymmetry introduces additional gauge fixing, namely the $\beta\gamma$ superghost system. Its factors belong to $\mathcal{V}$ in the above weight count. This is why "the matter vertex has weight one" is not a picture-independent superstring statement, whereas "the complete integrated vertex has weight one" is.

6.2.1.3 Bosonized superghost weights

The commuting $\beta\gamma$ system has weights

$$h[\beta] = \frac{3}{2}, \qquad h[\gamma] = -\frac{1}{2}.$$

Bosonize it as

$$\beta = e^{-\phi}\partial\xi, \qquad \gamma = \eta e^{\phi},$$

where

$$\phi(z)\phi(w) \sim -\ln(z-w).$$

The scalar ϕ has a background charge, so its stress tensor is not that of an ordinary free scalar:

$$T_\phi(z) = -\frac{1}{2} : (\partial\phi)^2 : (z) - \partial^2\phi(z).$$

To compute the weight of $e^{q\phi}$, first use

$$\partial\phi(z)e^{q\phi(w)} \sim -\frac{q}{z-w}e^{q\phi(w)}.$$

The double contraction with the first term in T_ϕ contributes

$$-\frac{q^2}{2(z-w)^2}e^{q\phi(w)},$$

while the background-charge term contributes

$$-\frac{q}{(z-w)^2}e^{q\phi(w)}.$$

Hence

$$T_\phi(z)e^{q\phi(w)} \sim \frac{-\frac{1}{2}q(q+2)}{(z-w)^2}e^{q\phi(w)} + \frac{1}{z-w}\partial e^{q\phi(w)},$$

and therefore

$$\boxed{h[e^{q\phi}] = -\frac{1}{2}q(q+2)}.$$

The cases needed below are

$$h[e^{-\phi}] = \frac{1}{2}, \qquad h[e^{-\phi/2}] = \frac{3}{8}, \qquad h[e^{0\phi}] = 0.$$

The background charge also produces a picture-number selection rule in amplitudes. For example, tree-level amplitudes are commonly arranged to have total picture number -2 in each relevant chiral superghost sector. This affects how pictures are distributed among the vertices, but it does not change the local condition that every complete integrated insertion have weight one.

6.2.1.4 Massless NS vector in picture minus one

The unintegrated open-string massless vector in the NS sector and picture -1 is

$$U_A^{(-1)}(x) = c\, e^{-\phi} \epsilon_\mu \psi^\mu e^{ik\cdot X}(x).$$

For a massless state, $k^2 = 0$, so the open-string exponential contributes

$$h[e^{ik\cdot X}] = \alpha' k^2 = 0.$$

The remaining weights are

$$h[c] = -1, \qquad h[e^{-\phi}] = \frac{1}{2}, \qquad h[\epsilon \cdot \psi] = \frac{1}{2}.$$

Thus

$$h[U_A^{(-1)}] = -1 + \frac{1}{2} + \frac{1}{2} + 0 = 0.$$

Removing c gives the complete integrated local operator

$$V_A^{(-1)} = e^{-\phi} \epsilon \cdot \psi e^{ik\cdot X},$$

whose total weight is

$$h[V_A^{(-1)}] = 1.$$

This example exhibits the main point particularly clearly:

$$\underbrace{h[\epsilon \cdot \psi e^{ik\cdot X}]}_{\text{matter}} = \frac{1}{2}, \qquad \underbrace{h[e^{-\phi}]}_{\text{superghost}} = \frac{1}{2}, \qquad \underbrace{h[V_A^{(-1)}]}_{\text{complete integrated operator}} = 1.$$

BRST closure imposes more than the weight condition. For this vector it gives

$$k^2 = 0, \qquad k \cdot \epsilon = 0.$$

BRST-exact vertices identify

$$\epsilon_\mu \sim \epsilon_\mu + \lambda k_\mu,$$

which is the spacetime gauge equivalence. Thus conformal weight, transversality, and gauge redundancy are different parts of the same BRST-cohomology problem.

6.2.1.5 The same NS vector in picture zero

In picture zero, the standard integrated massless vector has the schematic form

$$V_A^{(0)}(x) = \epsilon_\mu : \left(i\partial_t X^\mu + C_\psi \alpha' (k \cdot \psi)\psi^\mu\right) e^{ik\cdot X} : (x).$$

The numerical coefficient C_ψ, as well as factors of i, depends on the normalization of X, ψ, and the boundary coordinate; BRST closure fixes it once those conventions are chosen. The weight count does not depend on that coefficient. On the mass shell,

$$h[\partial_t X^\mu] = 1, \qquad h[(k \cdot \psi)\psi^\mu] = 1, \qquad h[e^{ik\cdot X}] = 0.$$

Therefore

$$h[V_A^{(0)}] = 1.$$

There is no $e^{q\phi}$ factor in picture zero, so in this picture the displayed operator is entirely a matter operator and its weight is one.

At noncanonical pictures, especially for unintegrated representatives, picture changing can also generate additional ghost terms needed for exact BRST closure. Therefore one should regard the simple expression above as the standard **integrated** picture-zero representative, rather than assume that every unintegrated representative is obtained merely by multiplying it by c and nothing else.

6.2.1.6 Picture changing preserves the physical conformal weight

The picture-changing operator is

$$\mathcal{X}(z) = \{Q_B, \xi(z)\}.$$

It has picture number $+1$ and conformal weight zero:

$$h[\mathcal{X}] = 0.$$

Acting with $\mathcal{X}$ on a BRST-closed picture-q vertex produces a picture-$(q+1)$ representative of the same BRST cohomology class, up to the usual qualifications about singular operator products and zero-momentum exceptional states. Since $\mathcal{X}$ has weight zero,

$$h[\mathcal{X}\mathcal{V}] = h[\mathcal{V}].$$

In particular, picture changing maps the picture--1 NS vertex to the picture-zero vertex, up to BRST-exact terms and total derivatives, without changing the total weight of the complete insertion.

Picture changing therefore does not turn a nonphysical operator into a physical one by altering its weight. It only changes how the fixed total weight is divided among matter and superghost factors.

6.2.1.7 Ramond vertex in picture minus one half

The unintegrated massless open-string Ramond vertex in the canonical picture $-1/2$ is

$$U_u^{(-1/2)}(x) = c\, e^{-\phi/2} u_A S^A e^{ik\cdot X}(x).$$

In ten dimensions the spin field has

$$h[S^A] = \frac{D}{16} = \frac{5}{8},$$

and the superghost factor has

$$h[e^{-\phi/2}] = \frac{3}{8}.$$

For $k^2 = 0$ the exponential again has zero weight. Hence

$$h[U_u^{(-1/2)}] = -1 + \frac{3}{8} + \frac{5}{8} + 0 = 0,$$

whereas the integrated operator without c has

$$h[e^{-\phi/2} u_A S^A e^{ik\cdot X}] = \frac{3}{8} + \frac{5}{8} = 1.$$

BRST closure imposes the massless spacetime Dirac equation

$$k_\mu (\Gamma^\mu)^A{}_B u_A = 0,$$

with index placement depending on the charge-conjugation and spinor conventions. The GSO projection further selects the appropriate spacetime chirality. Thus the equality $3/8 + 5/8 = 1$ verifies conformal covariance, while the Dirac equation is an additional BRST constraint.

6.2.1.8 Relation to the NS and R intercepts

In old covariant quantization, the open RNS mass-shell condition is

$$(L_0 - a)|\Psi\rangle = 0.$$

In the NS sector,

$$a_{\mathrm{NS}} = \frac{1}{2}, \qquad L_0 = \alpha' k^2 + N_{\mathrm{NS}},$$

so

$$\alpha' k^2 + N_{\mathrm{NS}} = \frac{1}{2}.$$

The GSO-projected massless vector has $N_{\mathrm{NS}} = 1/2$, hence $k^2 = 0$. Its plane matter operator $\epsilon \cdot \psi e^{ik\cdot X}$ correspondingly has weight $1/2$, not one. The picture--1 superghost dressing supplies the other $1/2$ needed by the complete integrated vertex.

In the Ramond sector,

$$a_{\mathrm{R}} = 0, \qquad \alpha' k^2 + N_{\mathrm{R}} = 0.$$

The Ramond ground state has $N_{\mathrm{R}} = 0$ and is massless. One must not naively identify the plane spin-field weight $h[S^A] = 5/8$ with the cylinder intercept $a_{\mathrm{R}} = 0$. The spin field is a twist operator that changes the boundary conditions of the worldsheet fermions, and the plane-to-cylinder map includes the corresponding sector-dependent vacuum-energy shift. The reliable local vertex-operator statement is instead

$$h[S^A] + h[e^{-\phi/2}] = \frac{5}{8} + \frac{3}{8} = 1.$$

The old-covariant intercept and the plane spin-field weight are therefore compatible, but they encode the Ramond vacuum relative to different reference vacua.

6.2.1.9 Closed NS–NS and heterotic examples

For a closed type-II NS–NS massless state in picture $(-1,-1)$, the unintegrated vertex is

$$U_\epsilon^{(-1,-1)} = c\bar{c}\, e^{-\phi} e^{-\bar{\phi}} \epsilon_{\mu\nu} \psi^\mu \bar{\psi}^\nu e^{ik\cdot X}.$$

At $k^2 = 0$, each chiral sector has the weight count

$$-1 + \frac{1}{2} + \frac{1}{2} = 0.$$

Thus the full unintegrated operator has weights $(0,0)$, while removing $c\bar{c}$ gives an integrated operator of weights

$$(h, \bar{h}) = (1,1).$$

BRST closure further yields

$$k^\mu \epsilon_{\mu\nu} = 0, \qquad k^\nu \epsilon_{\mu\nu} = 0,$$

and BRST-exact shifts generate

$$\epsilon_{\mu\nu} \sim \epsilon_{\mu\nu} + k_\mu \xi_\nu + k_\nu \tilde{\xi}_\mu.$$

The symmetric and antisymmetric parts are the linearized gauge equivalences of the graviton and Kalb–Ramond field.

The heterotic string simply combines two different chiral constructions. In the conventional assignment, one sector is bosonic and the other is RNS supersymmetric. A physical integrated heterotic vertex must still have total weights $(1,1)$: the bosonic side obtains its unit weight from a derivative, current, or oscillator factor, while the supersymmetric side may divide its unit weight between a matter operator and an $e^{q\phi}$ superghost dressing. Multiplication by $c\bar{c}$ again produces an unintegrated operator of weights $(0,0)$.

6.2.1.10 BRST closure is stronger than conformal weight alone

Schematically, if $\mathcal{V}$ is a complete open-string operator of weight h before inserting c, the cT part of the RNS BRST charge gives

$$Q_B(c\mathcal{V}) \supset (1-h) c\partial c\, \mathcal{V}.$$

Therefore BRST closure requires $h = 1$. But the RNS BRST charge also contains terms built from the supercurrent and the $\beta\gamma$ ghosts. Their operator products impose the super-Virasoro constraints that are not visible from L_0 alone. Depending on the state, these give transversality, the Dirac equation, tracelessness conditions, or analogous polarization constraints.

Finally, physical vertices are cohomology classes,

$$Q_B U = 0, \qquad U \sim U + Q_B \Lambda.$$

The second relation is what removes longitudinal or otherwise gauge-redundant polarizations. Thus having the correct conformal weight is necessary, but it is not sufficient without BRST closure modulo BRST-exact states.

6.2.1.11 Checks

1. **NS picture -1:**
$$h[c]+h[e^{-\phi}]+h[\psi]+h[e^{ik\cdot X}]=-1+\frac{1}{2}+\frac{1}{2}+0=0.$$
2. **NS picture zero:** both ∂X and $(k\cdot\psi)\psi$ have weight one, so the integrated massless vertex has $h=1$ without a superghost exponential.
3. **Ramond picture $-1/2$:**
$$h[e^{-\phi/2}]+h[S^A]=\frac{3}{8}+\frac{5}{8}=1.$$
4. **Closed NS–NS sector:** each chiral half independently has integrated weight one; it is not enough to require only $h+\bar{h}=2$.
5. **Picture changing:** $h[\mathcal{X}]=0$, so changing picture preserves the complete conformal weight even though it changes the separate matter and superghost contributions.
6. **Intercept check:** $a_{\mathrm{NS}}=1/2$ explains the NS matter weight $1/2$ at the massless level, but $a_{\mathrm{R}}=0$ must not be identified directly with the plane spin-field weight $5/8$.

6.2.1.12 Result

For an RNS open string, the picture-independent weight statement is

$$\boxed{h[\mathcal{V}_{\text{integrated}}]=1,\qquad h[c\mathcal{V}_{\text{integrated}}]=0}.$$

For an RNS closed string,

$$\boxed{(h,\bar{h})[\mathcal{V}_{\text{integrated}}]=(1,1),\qquad (h,\bar{h})[c\bar{c}\mathcal{V}_{\text{integrated}}]=(0,0)}.$$

The superghost exponential contributes

$$\boxed{h[e^{q\phi}]=-\frac{1}{2}q(q+2)}.$$

Therefore the canonical massless vertices satisfy

$$\begin{aligned}
\text{NS, picture }-1: &\qquad h[e^{-\phi}]+h[\psi]=\frac{1}{2}+\frac{1}{2}=1,\\
\text{NS, picture }0: &\qquad h[\partial X]=h[\psi\psi]=1,\\
\text{R, picture }-\frac{1}{2}: &\qquad h[e^{-\phi/2}]+h[S]=\frac{3}{8}+\frac{5}{8}=1.
\end{aligned}$$

Picture changing by
$$\mathcal{X}=\{Q_B,\xi\},\qquad h[\mathcal{X}]=0,$$
preserves the complete conformal weight and the BRST cohomology class. BRST closure then adds the state-specific equations such as

$$k^2=0,\qquad k\cdot\epsilon=0,\qquad \not{k}\,u=0,$$

while BRST-exact shifts implement spacetime gauge equivalence. Hence superstring vertex operators obey the same geometric weight-one requirement as bosonic vertices, but only the **complete picture-dressed insertion**, not the matter factor by itself, satisfies that requirement in every picture.

6.3 Q3: Deriving the upper-half-plane fermion gluing condition

In chapter 6.3 we have the boundary conditions for open superstring

$$\psi_+^\mu(0) = D^\mu{}_\nu(0)\psi_-^\nu(0)$$
$$\psi_+^\mu(l) = \eta D^\mu{}_\nu(l)\psi_-^\nu(l)$$

Where $D^\mu{}_\nu = (\delta^\alpha{}_\beta, -\delta^i{}_j)$ for α, β the NN directions and i, j the DD directions; $\eta = +1$ in R sector and $\eta = -1$ in NS sector. In chapter 13.3 they use this boundary condition as

$$\psi^\mu(z) = \eta D^\mu{}_\nu \bar{\psi}^\nu(\bar{z})|_{\mathrm{Im}(z)=0}$$

Where $\eta = +1$ in NS sector and $\eta = \mathrm{sign}(\mathrm{Re}(z))$ in R sector. Derive this equation from the original boundary condition.

> **Textbook location and related material.** Chapter 13, Section 13.3 (pp. 411–415), especially the upper-half-plane fermion gluing condition near Eq. (13.99). The original strip boundary conditions are in Section 7.4.

6.3.1 Answer

6.3.1.1 Core point

The two formulas are the same boundary condition written in two coordinate systems. There are two changes between them:

1. the two edges $\sigma = 0$ and $\sigma = l$ of the open-string strip become the positive and negative halves of the real axis of the upper half-plane;
2. ψ^μ is a conformal field of weight $1/2$, so its transformation law contains a square root of the Jacobian. That square root differs by a sign between the positive and negative real axes.

It is useful to distinguish the sign in the strip formula from the sign in the upper-half-plane formula. Write

$$\eta_{\text{strip}} = \begin{cases} +1, & \text{R}, \\ -1, & \text{NS}. \end{cases}$$

Then the strip boundary conditions are

$$\psi_+^\mu(0) = D_0{}^\mu{}_\nu \psi_-^\nu(0), \qquad \psi_+^\mu(l) = \eta_{\text{strip}} D_l{}^\mu{}_\nu \psi_-^\nu(l).$$

After mapping to the upper half-plane, the general result is

$$\psi^\mu(x) = \begin{cases} D_0{}^\mu{}_\nu \bar{\psi}^\nu(x), & x > 0, \\ -\eta_{\text{strip}} D_l{}^\mu{}_\nu \bar{\psi}^\nu(x), & x < 0. \end{cases}$$

For the single-D-brane situation used in Chapter 13, one has

$$D_0 = D_l = D.$$

It follows that

$$\psi^\mu(x) = \begin{cases} D^\mu{}_\nu \bar{\psi}^\nu(x), & \text{NS}, \\ \mathrm{sign}(x) D^\mu{}_\nu \bar{\psi}^\nu(x), & \text{R}, \end{cases}$$

which is precisely Eq. (13.99).

In the supplied PDF, the original strip formula is Eq. (7.62) in §7.3, and the book itself refers back to §7.3 immediately before Eq. (13.99). The relevant conformal map is Eq. (4.138). The chapter number in the question may refer to a different numbering scheme, but the displayed equations identify the passage unambiguously.

6.3.1.2 What the book states and what must be supplied

The book explicitly gives three ingredients:

- Eq. (7.62), the gluing conditions at $\sigma = 0$ and $\sigma = l$;
- Eq. (4.138), the exponential map from a strip to the upper half-plane;
- immediately before Eq. (13.99), the statement that the worldsheet fermions are half-differentials and therefore acquire a nontrivial Jacobian.

The omitted intermediate step is the evaluation of the square-root Jacobian on the two halves of the real axis. That evaluation is what changes

$$\eta_{\text{strip}} = \begin{cases} +1, & \text{R}, \\ -1, & \text{NS} \end{cases}$$

into

$$\eta_{\text{UHP}}(x) = \begin{cases} +1, & \text{NS}, \\ \operatorname{sign}(x), & \text{R}. \end{cases}$$

There is therefore no change in the definition of the R and NS sectors. The two symbols called η in the two formulas encode different quantities: the first is a relative endpoint sign on the strip, while the second already includes the half-differential Jacobian on the upper half-plane.

6.3.1.3 Mapping the strip to the upper half-plane

Use Euclidean strip coordinates

$$w = \tau + i\sigma, \qquad 0 \le \sigma \le l, \qquad -\infty < \tau < \infty.$$

The exponential map

$$z = e^{\pi w/l} = e^{\pi\tau/l} e^{i\pi\sigma/l}$$

sends the strip to the upper half-plane because

$$0 \le \arg z \le \pi.$$

Its action on the two boundaries is

$$\begin{aligned} \sigma = 0 \quad &\Longrightarrow \quad z = e^{\pi\tau/l} > 0, \\ \sigma = l \quad &\Longrightarrow \quad z = -e^{\pi\tau/l} < 0. \end{aligned}$$

Thus

$$\boxed{\sigma = 0 \longleftrightarrow x > 0, \qquad \sigma = l \longleftrightarrow x < 0,}$$

where $x = \operatorname{Re} z$ on the boundary.

The book's Eq. (4.138) parametrizes the strip as $-l \le \sigma \le 0$ and uses $w = \tau - i\sigma$. Replacing that σ by $-\sigma$ gives exactly the convention above, which is adapted to the endpoints 0 and l used in the question.

The inverse map and its derivatives are

$$w = \frac{l}{\pi} \log z, \qquad \frac{dw}{dz} = \frac{l}{\pi z},$$

and

$$\bar{w} = \frac{l}{\pi} \log \bar{z}, \qquad \frac{d\bar{w}}{d\bar{z}} = \frac{l}{\pi \bar{z}}.$$

6.3.1.4 Why a fermion produces an additional sign

A holomorphic primary field of weight h transforms under $w \mapsto z(w)$ as

$$\Phi(z) = \left(\frac{dw}{dz}\right)^h \Phi(w).$$

The chiral worldsheet fermions have weight $h = 1/2$. Therefore define

$$\psi^\mu(z) = \left(\frac{dw}{dz}\right)^{1/2} \psi^\mu_+(w),$$

and

$$\bar\psi^\mu(\bar z) = \left(\frac{d\bar w}{d\bar z}\right)^{1/2} \psi^\mu_-(\bar w).$$

Suppose that on one edge of the strip the gluing condition is

$$\psi^\mu_+(w) = s\, D^\mu{}_\nu \psi^\nu_-(\bar w),$$

where s is the sign assigned to that edge. Substitution into the transformation laws gives

$$\begin{aligned}\psi^\mu(z) &= \left(\frac{dw}{dz}\right)^{1/2} s\, D^\mu{}_\nu \psi^\nu_-(\bar w)\\ &= s\left[\frac{dw/dz}{d\bar w/d\bar z}\right]^{1/2} D^\mu{}_\nu \bar\psi^\nu(\bar z).\end{aligned}$$

Using the derivatives of the logarithm,

$$\left[\frac{dw/dz}{d\bar w/d\bar z}\right]^{1/2} = \left(\frac{\bar z}{z}\right)^{1/2}.$$

This factor is the entire origin of the extra sign.

6.3.1.5 Evaluating the square root on the two halves of the real axis

The upper-half-plane limit fixes the branches:

$$\arg z \in (0, \pi), \qquad \arg \bar z \in (-\pi, 0).$$

For $x > 0$, one approaches the positive real axis with

$$\arg z \longrightarrow 0, \qquad \arg \bar z \longrightarrow 0.$$

Therefore

$$\left(\frac{\bar z}{z}\right)^{1/2} = +1, \qquad x > 0.$$

For $x < 0$, the two boundary values approach the negative real axis from opposite sides:

$$z = |x| e^{i\pi}, \qquad \bar z = |x| e^{-i\pi}.$$

Consequently,

$$\left(\frac{\bar z}{z}\right)^{1/2} = \left(e^{-2\pi i}\right)^{1/2} = e^{-i\pi} = -1, \qquad x < 0.$$

Equivalently, the two square-root Jacobians are

$$\left(\frac{dw}{dz}\right)^{1/2} = \sqrt{\frac{l}{\pi |x|}} \begin{cases} 1, & x > 0, \\ -i, & x < 0, \end{cases}$$

and

$$\left(\frac{d\bar{w}}{d\bar{z}}\right)^{1/2} = \sqrt{\frac{l}{\pi|x|}} \begin{cases} 1, & x > 0, \\ +i, & x < 0. \end{cases}$$

Their ratio is $+1$ for $x > 0$ and -1 for $x < 0$.

It is useful to summarize this Jacobian as

$$J_{1/2}(x) := \left(\frac{\bar{z}}{z}\right)^{1/2}_{\mathrm{Im}\, z \to 0^+} = \mathrm{sign}(x).$$

This formula depends on the continuous branch inherited from the upper half-plane. A simultaneous overall sign change of both square roots is conventional, but the relative sign between the two halves of the real axis is fixed.

6.3.1.6 Applying the two original endpoint conditions

At $\sigma = 0$, which maps to $x > 0$, the original condition is

$$\psi_+^\mu(0) = D_0{}^\mu{}_\nu \psi_-^\nu(0).$$

Here the strip sign is $s = +1$ and the Jacobian is $J_{1/2}(x) = +1$. Therefore

$$\psi^\mu(x) = D_0{}^\mu{}_\nu \bar{\psi}^\nu(x), \qquad x > 0.$$

At $\sigma = l$, which maps to $x < 0$, the original condition is

$$\psi_+^\mu(l) = \eta_{\mathrm{strip}} D_l{}^\mu{}_\nu \psi_-^\nu(l).$$

Here the strip sign is $s = \eta_{\mathrm{strip}}$, while the Jacobian is $J_{1/2}(x) = -1$. Hence

$$\psi^\mu(x) = -\eta_{\mathrm{strip}} D_l{}^\mu{}_\nu \bar{\psi}^\nu(x), \qquad x < 0.$$

Combining the two halves gives the general upper-half-plane boundary condition

$$\boxed{\psi^\mu(x) = \begin{cases} D_0{}^\mu{}_\nu \bar{\psi}^\nu(x), & x > 0, \\ -\eta_{\mathrm{strip}} D_l{}^\mu{}_\nu \bar{\psi}^\nu(x), & x < 0. \end{cases}}$$

6.3.1.7 NS and R sectors for a single D-brane

Equation (13.99) uses one matrix D, so it assumes

$$D_0 = D_l = D.$$

This is the case in which the two ends of the open string obey the same NN/DD assignment, as for an open string ending on a single Dp-brane.

In the NS sector,

$$\eta_{\mathrm{strip}} = -1.$$

Therefore

$$-\eta_{\mathrm{strip}} = +1,$$

and the boundary condition is the same on both halves of the real axis:

$$\psi^\mu(x) = D^\mu{}_\nu \bar{\psi}^\nu(x), \qquad x > 0 \text{ or } x < 0.$$

Thus

$$\eta_{\mathrm{UHP}}(x) = +1 \qquad \text{in the NS sector.}$$

In the R sector,

$$\eta_{\text{strip}} = +1.$$

The positive real axis still has coefficient $+1$, but the negative real axis has coefficient -1:

$$\psi^\mu(x) = \begin{cases} D^\mu{}_\nu \bar{\psi}^\nu(x), & x > 0, \\ -D^\mu{}_\nu \bar{\psi}^\nu(x), & x < 0. \end{cases}$$

Since

$$\text{sign}(x) = \begin{cases} +1, & x > 0, \\ -1, & x < 0, \end{cases}$$

this becomes

$$\psi^\mu(x) = \text{sign}(x) D^\mu{}_\nu \bar{\psi}^\nu(x) \qquad \text{in the R sector.}$$

The two results can be written together as

$$\boxed{\psi^\mu(z) = \eta_{\text{UHP}}(\operatorname{Re} z) D^\mu{}_\nu \bar{\psi}^\nu(\bar{z})\big|_{\operatorname{Im} z = 0}\,, \qquad \eta_{\text{UHP}}(x) = \begin{cases} +1, & \text{NS}, \\ \text{sign}(x), & \text{R}. \end{cases}}$$

This is Eq. (13.99).

6.3.1.8 Why the R/NS signs appear to have exchanged roles

The strip sign and the upper-half-plane sign answer different questions.

On the strip, η_{strip} compares the gluing condition at $\sigma = l$ with the one at $\sigma = 0$:

$$\eta_{\text{strip}} = \begin{cases} +1, & \text{same endpoint sign, R}, \\ -1, & \text{opposite endpoint sign, NS}. \end{cases}$$

On the upper half-plane, the two endpoints have become two parts of one real boundary, and the conformal transformation contributes

$$J_{1/2}(x) = \begin{cases} +1, & x > 0, \\ -1, & x < 0. \end{cases}$$

The negative-axis coefficient is therefore

$$\eta_{\text{UHP}}(x < 0) = J_{1/2}(x < 0)\eta_{\text{strip}} = -\eta_{\text{strip}}.$$

This gives

sector	η_{strip}	$-\eta_{\text{strip}}$ on $x < 0$
NS	-1	$+1$
R	$+1$	-1

and explains the apparent reversal.

Geometrically, the exponential map converts translation along the doubled strip into rotation around the origin of the z-plane. A fermion is a spin-1/2 field, so a full 2π rotation contributes a minus sign. The square-root Jacobian above is the local manifestation of that spinorial sign. In the R sector the resulting real-axis gluing changes sign at $x = 0$; this is consistent with the Ramond spin-field branch point located at the origin, with the conjugate insertion effectively at infinity.

6.3.1.9 Different branes at the two ends

The starting equation in the question allows

$$D_0 \neq D_l.$$

For example, this occurs when the string stretches between different D-branes or when the two endpoints have different NN/DD assignments. In that case the correct upper-half-plane condition remains

$$\psi^\mu(x) = \begin{cases} D_0{}^\mu{}_\nu \bar{\psi}^\nu(x), & x > 0, \\ -\eta_{\rm strip} D_l{}^\mu{}_\nu \bar{\psi}^\nu(x), & x < 0. \end{cases}$$

It cannot in general be reduced to a single matrix D multiplied by either 1 or $\operatorname{sign}(x)$. The origin then marks a boundary-condition-changing point between the D_0 and D_l boundary conditions. Equation (13.99) is the special case relevant to the single Dp-brane discussion in §13.3.

6.3.1.10 Checks

1. **NS sector:** On the negative axis the original endpoint sign is -1 and the half-differential Jacobian is also -1, so their product is $+1$. The gluing is therefore D on the whole real axis.

2. **R sector:** On the negative axis the original endpoint sign is $+1$, while the Jacobian is -1. Hence the coefficient is $-D$ for $x < 0$ and $+D$ for $x > 0$, namely $\operatorname{sign}(x)D$.

3. **Why this is special to fermions:** For an integer-weight field with $h = 1$, the analogous phase on the negative axis is
$$\left(\frac{\bar{z}}{z}\right)^1 = e^{-2\pi i} = +1.$$
The additional minus sign arises specifically because $h = 1/2$.

4. **Same-brane assumption:** If $D_0 = D_l$, the book's compact formula follows. If $D_0 \neq D_l$, retaining both matrices is essential.

5. **Branch convention:** Changing the overall sign of a square root redefines one chiral fermion by an overall sign. It does not remove the relative sign between $x > 0$ and $x < 0$, which is the physical content of the derivation.

6.3.1.11 Result

The strip conditions are

$$\psi_+^\mu(0) = D_0{}^\mu{}_\nu \psi_-^\nu(0), \qquad \psi_+^\mu(l) = \eta_{\rm strip} D_l{}^\mu{}_\nu \psi_-^\nu(l),$$

with

$$\eta_{\rm strip} = \begin{cases} +1, & \text{R}, \\ -1, & \text{NS}. \end{cases}$$

Under

$$z = e^{\pi(\tau + i\sigma)/l},$$

the boundaries map as

$$\sigma = 0 \longleftrightarrow x > 0, \qquad \sigma = l \longleftrightarrow x < 0.$$

Because ψ has conformal weight $1/2$,

$$J_{1/2}(x) = \left(\frac{\bar{z}}{z}\right)^{1/2}_{\operatorname{Im} z \to 0^+} = \begin{cases} +1, & x > 0, \\ -1, & x < 0. \end{cases}$$

Therefore

$$\boxed{\psi^\mu(x) = \begin{cases} D_0{}^\mu{}_\nu \bar{\psi}^\nu(x), & x > 0, \\ -\eta_{\rm strip} D_l{}^\mu{}_\nu \bar{\psi}^\nu(x), & x < 0. \end{cases}}$$

For $D_0 = D_l = D$ this becomes

$$\boxed{\psi^\mu(z) = \eta_{\rm UHP}(\operatorname{Re} z) D^\mu{}_\nu \bar\psi^\nu(\bar z)\big|_{\operatorname{Im} z=0}\,, \qquad \eta_{\rm UHP}(x) = \begin{cases} +1, & \text{NS},\\ \operatorname{sign}(x), & \text{R}.\end{cases}}$$

The apparent change of the R/NS signs is exactly the extra factor -1 acquired by a half-differential on the negative real axis.

6.4 Q4: Why a D-brane preserves the diagonal supercharge

Why it follows from this discussion that Q_L and PQ_R must have the same chirality and that the conserved supercharges are given by eq.(13.104)?

> **Textbook location and related material.** Chapter 13, Section 13.3 (pp. 411–415), especially Eqs. (13.100)–(13.104): boundary conformal invariance, doubled currents, and the diagonal spacetime supercharge preserved by a D-brane.

6.4.1 Answer

6.4.1.1 Core point

There are two logically distinct statements behind Eq. (13.104):

1. **Conservation:** a worldsheet boundary does not preserve the left- and right-moving supersymmetry currents independently. The boundary condition identifies them through the spinor gluing matrix P. Therefore only their diagonal combination

$$Q_\alpha = Q_{L\alpha} + (PQ_R)_\alpha$$

has no current flux through the boundary and is conserved.

2. **Chirality:** after the GSO projection, Q_L and Q_R are ten-dimensional Majorana-Weyl supercharges. The two terms in the preserved diagonal combination must lie in the same Weyl representation. Therefore Q_L and PQ_R must have the same chirality. Since P flips chirality when the number $9-p$ of Dirichlet directions is odd, this condition selects even p in type IIA and odd p in type IIB.

The conservation statement is the supersymmetry-current version of the general BCFT construction in Eqs. (4.146) and (4.147). The chirality statement is an additional compatibility condition with the type-II GSO projection.

In that BCFT construction, $\bar T(\bar z)$ means the antiholomorphic component $T_{\bar z\bar z}$, not automatically the Hermitian adjoint of $T(z) = T_{zz}$. The boundary condition $T = \bar T$ sets the normal energy flux to zero, while the restriction $\varepsilon = \bar\varepsilon$ links the two conformal-transformation parameters. The result is one **diagonal Virasoro algebra**, with one conserved generator for every mode n, rather than one isolated numerical charge. For time translations its conserved density is proportional to $T + \bar T$.

6.4.1.2 The two supercharges before introducing a boundary

For a closed type-II string without a boundary, the holomorphic and antiholomorphic sectors are independent. The canonical $-1/2$-picture supersymmetry currents are, schematically,

$$\mathcal{J}^\alpha_R(z) = S^\alpha_R(z) e^{-\phi_R(z)/2},$$

and

$$\mathcal{J}^\alpha_L(\bar z) = S^\alpha_L(\bar z) e^{-\phi_L(\bar z)/2}.$$

The labels R and L follow the book's convention that the holomorphic sector is called right-moving and the antiholomorphic sector left-moving.

Their contour integrals define independent spacetime supercharges:

$$Q_R^\alpha = \oint \frac{dz}{2\pi i}\,\mathcal{J}_R^\alpha(z), \qquad Q_L^\alpha = \oint \frac{d\bar{z}}{2\pi i}\,\mathcal{J}_L^\alpha(\bar{z}).$$

Thus, before a boundary is introduced, type II theory has two independently conserved sets of supercharges:

$$Q_L \qquad \text{and} \qquad Q_R.$$

After the GSO projection, each is Majorana-Weyl. In type IIB they have the same chirality, while in type IIA they have opposite chiralities.

6.4.1.3 Does $\bar{T}(\bar{z})$ mean the complex conjugate of $T(z)$?

The short answer is: **the bar primarily labels the antiholomorphic component, not Hermitian conjugation**.

Start with the Euclidean stress tensor T_{ab} and introduce

$$z = \tau + i\sigma, \qquad \bar{z} = \tau - i\sigma.$$

Its two chiral components are denoted

$$T(z) = T_{zz}(z,\bar{z}), \qquad \bar{T}(\bar{z}) = T_{\bar{z}\bar{z}}(z,\bar{z}).$$

Conservation and tracelessness imply, away from operator insertions,

$$\partial_{\bar{z}} T(z) = 0, \qquad \partial_z \bar{T}(\bar{z}) = 0.$$

Thus T is holomorphic and $\bar{T}$ is antiholomorphic. They generate the two independent Virasoro algebras of a bulk CFT.

If one starts from a real classical Euclidean tensor T_{ab}, then complex conjugation of tensor components exchanges z and $\bar{z}$, so on a real field configuration one has a reality relation of the schematic form

$$\left(T_{zz}(z,\bar{z})\right)^* = T_{\bar{z}\bar{z}}(\bar{z},z).$$

This is a reality property inherited from the underlying real theory. It is not the definition of the notation $\bar{T}$, and it does not identify the two chiral operator algebras.

In radial quantization the distinction is particularly clear. Hermitian conjugation includes inversion of the radial coordinate:

$$T(z)^\dagger = \bar{z}^{-4} T\left(\frac{1}{\bar{z}}\right),$$

and similarly

$$\bar{T}(\bar{z})^\dagger = z^{-4} \bar{T}\left(\frac{1}{z}\right).$$

Thus Hermitian conjugation maps each chiral stress tensor to itself at the radially reflected point; it does not simply replace T by $\bar{T}$ at the same point. A parity transformation may exchange the two sectors, but parity is a separate operation.

Therefore the boundary equation

$$T(x) = \bar{T}(x), \qquad x \in \mathbb{R},$$

is an additional physical gluing condition. It is not a tautology following from the bar notation.

6.4.1.4 Why $T(x) = \bar{T}(x)$ is the no-flux condition

The relation to energy flow can be seen directly in real coordinates. With the above choice of z and $\bar{z}$,

$$T_{zz} = \frac{1}{4}\left(T_{\tau\tau} - 2iT_{\tau\sigma} - T_{\sigma\sigma}\right),$$

and

$$T_{\bar{z}\bar{z}} = \frac{1}{4}\left(T_{\tau\tau} + 2iT_{\tau\sigma} - T_{\sigma\sigma}\right).$$

Conformal invariance gives tracelessness,

$$T_{\tau\tau} + T_{\sigma\sigma} = 0.$$

Consequently,

$$T_{zz} = \frac{1}{2}\left(T_{\tau\tau} - iT_{\tau\sigma}\right), \qquad T_{\bar{z}\bar{z}} = \frac{1}{2}\left(T_{\tau\tau} + iT_{\tau\sigma}\right).$$

Hence, up to the overall normalization used when the CFT fields $T(z)$ and $\bar{T}(\bar{z})$ are defined,

$$T - \bar{T} = -iT_{\tau\sigma},$$

while

$$T + \bar{T} = T_{\tau\tau}.$$

For a boundary at fixed σ, the component $T_{\tau\sigma}$ is the flux of τ-momentum, namely worldsheet energy, through the boundary. Therefore

$$T_{\tau\sigma}\big|_{\partial\Sigma} = 0 \qquad \Longleftrightarrow \qquad T(x) = \bar{T}(x).$$

The **difference** $T - \bar{T}$ measures normal flux, whereas the **sum** $T + \bar{T}$ is the tangential energy density. This is why the conserved time-translation charge contains the sum even though the boundary condition sets the difference to zero.

6.4.1.5 Why the boundary leaves one Ward identity

On the full complex plane, infinitesimal conformal transformations are

$$\delta z = \varepsilon(z), \qquad \delta\bar{z} = \bar{\varepsilon}(\bar{z}).$$

There is no boundary, so ε and $\bar{\varepsilon}$ can be chosen independently. Their Ward identities are generated separately by T and $\bar{T}$, producing two commuting Virasoro algebras:

$$\mathrm{Vir}_L \oplus \mathrm{Vir}_R.$$

On the upper half-plane, the real axis must be mapped to itself. At a boundary point $z = \bar{z} = x$, the normal displacement is

$$\delta\sigma = \frac{\delta z - \delta\bar{z}}{2i} = \frac{\varepsilon(x) - \bar{\varepsilon}(x)}{2i}.$$

Preserving the boundary therefore requires

$$\boxed{\varepsilon(x) = \bar{\varepsilon}(x), \qquad x \in \mathbb{R}.}$$

The holomorphic and antiholomorphic transformations are no longer independent; they are the two boundary values of one real-analytic transformation.

This is what the book means when it says that there is only one Ward identity. Equation (4.141) still contains a T term and a $\bar{T}$ term,

$$\begin{aligned}\langle \delta X\rangle = &-\oint_C \frac{dz}{2\pi i}\,\varepsilon(z)\langle T(z)X\rangle \\ &+\oint_C \frac{d\bar{z}}{2\pi i}\,\bar{\varepsilon}(\bar{z})\langle \bar{T}(\bar{z})X\rangle,\end{aligned}$$

but the two terms have the **same** transformation parameter at the boundary. They therefore describe one diagonal transformation rather than two independent transformations.

"One Ward identity" does not mean one numerical conserved charge. Expanding the one allowed function in modes gives one infinite family of conserved generators:

$$\{L_n^{\mathrm{bdy}}\}_{n\in\mathbb{Z}}.$$

Thus the boundary reduces

$$\mathrm{Vir}_L \oplus \mathrm{Vir}_R \quad\longrightarrow\quad \mathrm{Vir}_{\mathrm{diag}},$$

one diagonal Virasoro algebra.

6.4.1.6 The doubling trick and the diagonal Virasoro charge

The book implements the preceding statement by defining a single holomorphic stress tensor on the doubled plane:

$$\mathcal{T}_{\mathrm{dbl}}(z) = \begin{cases} T(z), & \operatorname{Im} z > 0, \\ \bar{T}(z), & \operatorname{Im} z < 0. \end{cases}$$

In the lower half-plane, the argument z on the second line is the reflected holomorphic coordinate obtained from the upper-half-plane coordinate $\bar{z}$. The boundary condition

$$T(x) = \bar{T}(x)$$

makes $\mathcal{T}_{\mathrm{dbl}}$ continuous across the real axis.

Reflect the antiholomorphic contour in Eq. (4.141) into the lower half-plane. Reflection reverses its orientation, which cancels the relative sign between the two terms in the Ward identity. The two semicircles then form one closed contour:

$$L_n^{\mathrm{bdy}} = \oint_C \frac{dz}{2\pi i}\, z^{n+1} \mathcal{T}_{\mathrm{dbl}}(z).$$

Because this is a closed contour integral of a holomorphic current, it can be deformed without changing its value, provided it does not cross an insertion. This is the contour meaning of conservation.

Written before doubling, the same generator contains both chiral contributions:

$$\begin{aligned} L_n^{\mathrm{bdy}} = & \int_{C_+} \frac{dz}{2\pi i}\, z^{n+1} T(z) \\ & - \int_{C_+} \frac{d\bar{z}}{2\pi i}\, \bar{z}^{n+1} \bar{T}(\bar{z}). \end{aligned}$$

The minus sign is an orientation sign for the antiholomorphic semicircle. When each term is expressed using the standard orientation used to define its chiral charge, this is the diagonal **sum** of the two charge contributions. Thus the phrase "the conserved charge is $T + \bar{T}$" is shorthand: the local fields are integrated with their appropriate contour orientations.

For ordinary worldsheet time translations, the shorthand becomes literal at the level of the charge density. In Lorentzian strip coordinates, write the two chiral energy densities as T_{++} and T_{--}. The total Hamiltonian is

$$H = \int_0^l d\sigma\, \left(T_{++} + T_{--}\right).$$

Using chirality,

$$\partial_\tau T_{++} = \partial_\sigma T_{++}, \qquad \partial_\tau T_{--} = -\partial_\sigma T_{--},$$

one finds

$$\frac{dH}{d\tau} = \left[T_{++} - T_{--}\right]_{\sigma=0}^{\sigma=l}.$$

The no-flux gluing condition $T_{++} = T_{--}$ at each endpoint gives

$$\frac{dH}{d\tau} = 0.$$

The left- and right-moving energies are not separately conserved: an incoming left-moving excitation reflects from the boundary as a right-moving excitation. Their sum is conserved because the boundary only transfers energy between the two chiral sectors; it does not absorb it.

When the same boundary is described in the closed-string channel by a boundary state, the gluing condition is often written

$$\left(L_n - \bar{L}_{-n}\right) |B\rangle = 0.$$

The minus sign and the index reversal arise from the orientation change in passing to the closed-string channel. This is the same diagonal symmetry, not a contradiction with saying that the physical open-channel charge contains the sum of left- and right-moving contributions.

6.4.1.7 General BCFT reason that only a diagonal charge survives

The stress-tensor discussion generalizes directly. Suppose $W(z)$ is a holomorphic current of weight h and $\bar{W}(\bar{z})$ is an antiholomorphic current with the boundary gluing

$$W(z) = \bar{W}(\bar{z})\big|_{\mathrm{Im}\, z=0} .$$

Equation (4.147) then combines the two semicircle integrals into a single closed contour integral:

$$\begin{aligned} W_n &= \int_{C_+} \frac{dz}{2\pi i}\, z^{n+h-1} W(z) - \int_{C_+} \frac{d\bar{z}}{2\pi i}\, \bar{z}^{n+h-1} \bar{W}(\bar{z}) \\ &= \oint_C \frac{dz}{2\pi i}\, z^{n+h-1} W_{\mathrm{dbl}}(z). \end{aligned}$$

The boundary condition makes the doubled current continuous, so the closed contour defines a conserved diagonal charge.

If the two currents fail to match on the real axis, the two semicircles cannot be joined smoothly. The difference between the charge evaluated at two radii is then a boundary integral of the mismatch. Physically, the current has a nonzero normal component, so charge flows into the boundary and that symmetry is broken.

6.4.1.8 Applying the BCFT argument to the supersymmetry currents

The D-brane boundary condition for the spin fields is Eq. (13.100):

$$S_\alpha(z) = P_\alpha{}^\beta \bar{S}_\beta(\bar{z})\big|_{\mathrm{Im}\, z=0} .$$

The book's footnote 13 adds an important fact: the superconformal ghost is continuous across the real axis. Therefore the full supersymmetry currents, not only the matter spin fields, are glued by the same spinor matrix P.

In the spinor frame used for Eq. (13.104), this relation is

$$\mathcal{J}_{L\alpha}(\bar{z}) = (P\mathcal{J}_R)_\alpha(z)\big|_{\mathrm{Im}\, z=0} .$$

Whether one writes this equation with P, P^{-1}, or the charge-conjugate transpose of P depends on which spinor index is transported from the right-moving to the left-moving frame. Equation (13.104) fixes the book's convention: $(PQ_R)_\alpha$ is the right-moving charge expressed with a left-moving spinor index. This convention choice does not affect the chirality or conservation argument.

Now substitute

$$W = P\mathcal{J}_R, \qquad \bar{W} = \mathcal{J}_L$$

into the BCFT construction. The two semicircle charges combine as

$$\boxed{Q_\alpha = Q_{L\alpha} + (PQ_R)_\alpha.}$$

This is Eq. (13.104).

The orthogonal combination

$$Q_{L\alpha} - (PQ_R)_\alpha$$

has the wrong sign at the boundary. Its two semicircle currents do not join into one analytic doubled current, so that combination is not conserved. A D-brane therefore reduces the two independent type-II supersymmetries to a diagonal half.

6.4.1.9 Why the two terms must have the same chirality

Before the GSO projection, the spin fields in this part of §13.3 are organized as 32-component Majorana spinors. The GSO projection retains one Weyl chirality in each chiral sector. Write

$$\Gamma Q_L = \chi_L Q_L, \qquad \Gamma Q_R = \chi_R Q_R, \qquad \chi_L, \chi_R = \pm 1,$$

where

$$\Gamma = \Gamma^0 \Gamma^1 \cdots \Gamma^9$$

is the ten-dimensional chirality matrix.

The sum

$$Q_L + PQ_R$$

is a single GSO-allowed Majorana-Weyl supercharge only if its two terms have the same chirality. Equivalently, the boundary relation between supersymmetry parameters must map the GSO-allowed right-moving Weyl space into the GSO-allowed left-moving Weyl space. If the chiralities disagree, the equation relating the two parameters has only the trivial solution, so no supersymmetry of that form is preserved.

This is the precise meaning of the book's statement that Q_L and PQ_R must have the same chirality. It is not merely that one cannot algebraically add two Dirac spinors of opposite chirality; one can formally do so. Rather, the preserved generator must belong to the Weyl subspace selected by the GSO projection and must be parametrized by one nonzero Majorana-Weyl supersymmetry parameter.

6.4.1.10 How P acts on chirality

The vector gluing matrix for a Dp-brane is

$$D^\mu{}_\nu = \begin{cases} +\delta^\mu{}_\nu, & \mu \text{ NN}, \\ -\delta^\mu{}_\nu, & \mu \text{ DD}. \end{cases}$$

Its spinor representative P satisfies Eq. (13.102):

$$P \left(D^\mu{}_\nu \Gamma^\nu \right) = \Gamma^\mu P.$$

The book solves this as

$$P = \prod_{i \in \mathrm{DD}} \left(\Gamma_i \Gamma \right),$$

up to an overall sign and the phase needed to preserve the Majorana condition.

There are

$$n_D = 9 - p$$

Dirichlet directions. Each factor $\Gamma_i \Gamma$ anticommutes with the chirality matrix:

$$\Gamma(\Gamma_i \Gamma) = -(\Gamma_i \Gamma)\Gamma.$$

Passing Γ through all n_D factors therefore gives

$$\boxed{\Gamma P = (-1)^{n_D} P \Gamma = (-1)^{9-p} P \Gamma.}$$

If Q_R has chirality χ_R, then

$$\begin{aligned} \Gamma(PQ_R) &= (-1)^{9-p} P \Gamma Q_R \\ &= (-1)^{9-p} \chi_R \, PQ_R. \end{aligned}$$

Thus the chirality of PQ_R is

$$\chi_{PQ_R} = (-1)^{9-p} \chi_R.$$

The condition that Q_L and PQ_R have the same chirality is therefore

$$\boxed{\chi_L = (-1)^{9-p} \chi_R.}$$

6.4.1.11 Type IIA and type IIB D-branes

In type IIB,

$$\chi_L = \chi_R.$$

The chirality condition becomes

$$(-1)^{9-p} = +1.$$

Hence $9 - p$ is even, so

$$\boxed{p \text{ is odd in type IIB.}}$$

In type IIA,

$$\chi_L = -\chi_R.$$

The chirality condition becomes

$$(-1)^{9-p} = -1.$$

Hence $9 - p$ is odd, so

$$\boxed{p \text{ is even in type IIA.}}$$

This is the result stated immediately after Eq. (13.104).

6.4.1.12 Relation to the supersymmetry parameters

The book next writes a general type-II supersymmetry generator as

$$\bar{\epsilon}_L Q_L + \bar{\epsilon}_R Q_R.$$

Without a boundary, ϵ_L and ϵ_R are independent, giving $16 + 16$ real supercharges. A D-brane imposes

$$\boxed{\epsilon_L = \pm P \epsilon_R,}$$

which is Eq. (13.106). The plus and minus signs correspond to a brane and an anti-brane, with the assignment depending on the orientation convention.

This equation leaves only one independent 16-component Majorana-Weyl parameter. Using the invariance of the Majorana spinor pairing under the spin lift P, the generator can be written as the common parameter contracted with

$$Q_L \pm P Q_R.$$

For the sign convention chosen in Eq. (13.104), the brane preserves

$$Q_L + P Q_R.$$

The anti-brane preserves the opposite diagonal combination

$$Q_L - P Q_R.$$

Thus the current-gluing statement and the parameter constraint are two equivalent descriptions of the same halving of supersymmetry.

On chiral spinors, the matrix P can equivalently be written, up to phase and sign conventions, as the product of gamma matrices tangent to the Dp-brane:

$$\epsilon_L = \pm \Gamma^0 \Gamma^1 \cdots \Gamma^p \epsilon_R,$$

which is Eq. (13.107).

6.4.1.13 Checks

1. **D9-brane:** Here $n_D = 0$, so P preserves chirality. Type IIB has $\chi_L = \chi_R$, and therefore $Q_L + PQ_R$ is a valid chiral conserved charge. This is the D9-brane underlying type-I string theory.
2. **D0-brane:** Here $n_D = 9$, so P reverses chirality. Type IIA has opposite left- and right-moving chiralities, so PQ_R has the same chirality as Q_L. A BPS D0-brane is therefore allowed in type IIA.
3. **D3-brane:** Here $n_D = 6$ is even, so P preserves chirality. It is compatible with the equal chiralities of type IIB, as expected.
4. **Counting:** The boundary relation $\epsilon_L = \pm P\epsilon_R$ determines one 16-component parameter from the other. A single flat Dp-brane therefore preserves 16 of the original 32 real type-II supercharges.
5. **Broken combination:** The current associated with $Q_L - PQ_R$ has a boundary mismatch for the brane sign used in Eq. (13.104). It cannot be represented by the closed doubled contour of Eq. (4.147), so it is not conserved.

6.4.1.14 Result

The bar on the antiholomorphic stress tensor labels its chirality:

$$T(z) = T_{zz}, \qquad \bar{T}(\bar{z}) = T_{\bar{z}\bar{z}},$$

and does not by itself mean Hermitian conjugation. At the boundary,

$$T - \bar{T} \propto T_{\tau\sigma}, \qquad T + \bar{T} \propto T_{\tau\tau}.$$

Thus

$$T(x) = \bar{T}(x)$$

sets the normal energy flux to zero, while

$$\varepsilon(x) = \bar{\varepsilon}(x)$$

leaves one boundary-preserving conformal parameter. The doubled stress tensor has modes

$$L_n^{\mathrm{bdy}} = \oint \frac{dz}{2\pi i}\, z^{n+1} \mathcal{T}_{\mathrm{dbl}}(z),$$

so the surviving symmetry is one diagonal Virasoro algebra, not one single numerical charge.

The canonical supersymmetry currents are

$$\mathcal{J}_R = S_R e^{-\phi_R/2}, \qquad \mathcal{J}_L = S_L e^{-\phi_L/2}.$$

The spin-field boundary condition and continuity of the superghost identify them through P:

$$\mathcal{J}_L = P\mathcal{J}_R\big|_{\mathrm{Im}\, z=0}.$$

Therefore the BCFT contour construction of Eq. (4.147) preserves the diagonal charge

$$\boxed{Q = Q_L + PQ_R,}$$

while the orthogonal combination is broken.

For a Dp-brane,

$$n_D = 9 - p, \qquad \Gamma P = (-1)^{9-p} P\Gamma.$$

Hence

$$\Gamma(PQ_R) = (-1)^{9-p}\chi_R(PQ_R).$$

The two terms in the preserved charge have the same GSO chirality precisely when

$$\boxed{\chi_L = (-1)^{9-p}\chi_R.}$$

Consequently,

$$\boxed{\begin{array}{ll} \text{type IIA:} & p \text{ even,} \\ \text{type IIB:} & p \text{ odd.} \end{array}}$$

Equivalently, the unbroken supersymmetry parameters obey

$$\boxed{\epsilon_L = \pm P \epsilon_R,}$$

so one flat Dp-brane preserves one diagonal set of 16 real supercharges.

6.5 Q5: Reading NS and R sectors from bosonized weights

Around eq. (11.102) there is the discussion of bosonization. The vertex operators are

$$V_{\vec{\lambda}}(z) =: e^{i\vec{\lambda}\cdot\vec{X}(z)} : c_{\vec{\lambda}}$$

Where $c_{\vec{\lambda}}$ is the cocycle factor and $\vec{\lambda}$ is a weight vector in the weight lattice of D_n. It is said that the $\vec{\lambda}$ of all states in the NS sector are either in the (0) or (V) conjugacy class of D_n while that of R sector are in (S) or (C). In my understanding, NS/R sector has vertex operators of corresponding periodicity on the complex plane: NS is periodic and R is anti-periodic. How to see the periodicity of a vertex operator with given $\vec{\lambda}$ so we can derive the above statement?

> **Textbook location and related material.** Chapter 11, Section 11.4 (pp. 340–343) for bosonization and D_n conjugacy classes; Chapter 13, Sections 13.1 and 13.4 (pp. 391–403, 415–422) for bosonized covariant vertices and the covariant lattice.

6.5.1 Answer

6.5.1.1 Core point

Your statement about the fermions on the complex plane is correct:

sector	fermion on the cylinder	fermion on the complex plane
NS	anti-periodic	periodic
R	periodic	anti-periodic.

The important refinement is that one should not determine the sector from the periodicity of $V_\lambda(z)$ under rotating the vertex operator itself. Instead, $V_\lambda(0)$ creates a state at the origin, and one asks for the monodromy of a **fundamental fermion** transported around that insertion.

Using

$$\Psi^{\pm j}(z) =: e^{\pm i X^j(z)} : c_{\pm j},$$

one finds

$$\boxed{\Psi^{\pm j}(z) V_\lambda(0) \sim z^{\pm\lambda_j} (\text{single-valued operator and cocycle factors}).}$$

After taking z once counterclockwise around the origin,

$$z^{\pm\lambda_j} \longrightarrow e^{\pm 2\pi i \lambda_j} z^{\pm\lambda_j}.$$

Therefore

$$\lambda_j \in \mathbb{Z} \quad \Longrightarrow \quad \text{fermion monodromy } +1 \quad \Longrightarrow \quad \text{NS},$$

while

$$\lambda_j \in \mathbb{Z} + \frac{1}{2} \quad \Longrightarrow \quad \text{fermion monodromy } -1 \quad \Longrightarrow \quad \text{R}.$$

The D_n weight lattice has exactly two conjugacy classes with integer components,

$$(0) \cup (V),$$

and two with half-integer components,

$$(S) \cup (C).$$

This gives the statement around Eq. (11.102).

The monodromy distinguishes the two unions

$$(0) \cup (V) \qquad \text{versus} \qquad (S) \cup (C).$$

It does not by itself distinguish (0) from (V) or (S) from (C). Those further distinctions encode fermion-number parity and spinor chirality, respectively.

6.5.1.2 Cylinder and plane periodicities

The book begins with the complex-plane mode expansion

$$\psi^i(z) = \sum_r b_r^i z^{-r-1/2},$$

where

$$r \in \begin{cases} \mathbb{Z} + \frac{1}{2}, & \text{NS}, \\ \mathbb{Z}, & \text{R}. \end{cases}$$

Under one circuit around the origin,

$$z \longrightarrow e^{2\pi i} z,$$

each term transforms by

$$z^{-r-1/2} \longrightarrow e^{-2\pi i(r+1/2)} z^{-r-1/2}.$$

For the NS sector, $r \in \mathbb{Z} + \frac{1}{2}$, so $r + \frac{1}{2} \in \mathbb{Z}$ and

$$e^{-2\pi i(r+1/2)} = +1.$$

Hence the fermion is periodic on the plane.

For the R sector, $r \in \mathbb{Z}$, so $r + \frac{1}{2}$ is half-integral and

$$e^{-2\pi i(r+1/2)} = -1.$$

Hence the fermion is anti-periodic on the plane.

This is opposite to the familiar cylinder boundary conditions because the fermion is a half-differential. Under the cylinder-to-plane map, Eq. (4.10) gives

$$\psi_{\text{plane}}(z) = \left(\frac{l}{2\pi}\right)^{1/2} z^{-1/2} \psi_{\text{cyl}}(w).$$

A circuit around the origin changes

$$z^{-1/2} \longrightarrow -z^{-1/2}.$$

If

$$\psi_{\text{cyl}}(w + il) = s_{\text{cyl}} \psi_{\text{cyl}}(w),$$

then

$$s_{\text{plane}} = -s_{\text{cyl}}.$$

Thus

$$\begin{aligned} \text{NS}: \qquad s_{\text{cyl}} &= -1 \quad \Longrightarrow \quad s_{\text{plane}} = +1, \\ \text{R}: \qquad s_{\text{cyl}} &= +1 \quad \Longrightarrow \quad s_{\text{plane}} = -1. \end{aligned}$$

This is precisely the Jacobian effect mentioned immediately before Eq. (11.102).

6.5.1.3 What periodicity should be tested

Let

$$|\lambda\rangle = V_\lambda(0)|0\rangle.$$

The question "is this an NS or an R state?" means:

What boundary condition does the fundamental fermion obey when radially quantized around the state $|\lambda\rangle$?

Equivalently, one studies

$$\Psi^{\pm j}(z)|\lambda\rangle = \Psi^{\pm j}(z)V_\lambda(0)|0\rangle$$

and analytically continues z around the origin.

This is different from asking how $V_\lambda(z)$ itself transforms under a 2π rotation. The latter is governed by the conformal weight

$$h_\lambda = \frac{\lambda^2}{2}$$

and therefore by its conformal-spin phase. That phase does not classify the NS/R sector. For example, the identity operator and a fermion operator both belong to the NS Hilbert space, although their conformal weights are 0 and 1/2.

6.5.1.4 Deriving the monodromy from the bosonized OPE

The chiral bosons obey

$$X^i(z)X^j(w) \sim -\delta^{ij}\ln(z-w).$$

For normal-ordered exponentials, Wick contraction gives

$$:e^{i\mathbf{a}\cdot\mathbf{X}(z)}::e^{i\mathbf{b}\cdot\mathbf{X}(w)}:\sim (z-w)^{\mathbf{a}\cdot\mathbf{b}}\;:e^{i(\mathbf{a}+\mathbf{b})\cdot\mathbf{X}(w)}:.$$

This is the book's Eq. (11.92).

The bosonized complex fermions carry weights

$$\mathbf{a} = \pm\mathbf{e}_j,$$

where $\mathbf{e}_j$ is the jth Euclidean unit vector. Taking

$$\mathbf{b} = \lambda$$

therefore gives

$$\Psi^{+j}(z)V_\lambda(0) \sim z^{\lambda_j}V_{\lambda+\mathbf{e}_j}(0)\times(\text{cocycles}),$$
$$\Psi^{-j}(z)V_\lambda(0) \sim z^{-\lambda_j}V_{\lambda-\mathbf{e}_j}(0)\times(\text{cocycles}).$$

Taking z once around the origin yields

$$\Psi^{+j}(e^{2\pi i}z)V_\lambda(0) = e^{2\pi i\lambda_j}\Psi^{+j}(z)V_\lambda(0),$$
$$\Psi^{-j}(e^{2\pi i}z)V_\lambda(0) = e^{-2\pi i\lambda_j}\Psi^{-j}(z)V_\lambda(0),$$

where only the branch monodromy is displayed.

If all λ_j are integers, both phases equal $+1$ for every j. The fundamental fermions are single-valued around the insertion, so V_λ creates an NS state on the plane.

If all λ_j are half-integers, both phases equal -1 for every j. Every fundamental fermion changes sign around the insertion, so V_λ creates an R state on the plane.

The book assumes that all $2n$ real fermions have the same spin structure. If some λ_j were integral and others half-integral, different complex fermions would have different boundary conditions. Such mixed twist sectors can be considered in more general models, but they are not the common NS/R sectors assumed in §11.4.

6.5.1.5 The same conclusion from the fermion mode indices

The OPE also determines which fermion modes act on the state. In radial quantization,

$$\Psi^{\pm j}(z) = \sum_r \Psi_r^{\pm j} z^{-r-1/2}.$$

Around $V_\lambda(0)$, the OPE powers are shifted by $\pm\lambda_j$:

$$\Psi^{\pm j}(z) V_\lambda(0) \sim \sum_{m\in\mathbb{Z}} z^{m\pm\lambda_j}(\cdots).$$

Comparing

$$-r - \frac{1}{2} = m \pm \lambda_j$$

shows that:

- if $\lambda_j \in \mathbb{Z}$, then $r \in \mathbb{Z} + \frac{1}{2}$, which is the NS moding;
- if $\lambda_j \in \mathbb{Z} + \frac{1}{2}$, then $r \in \mathbb{Z}$, which is the R moding.

Thus the branch-monodromy argument and the oscillator-moding argument are exactly equivalent.

6.5.1.6 The four D_n conjugacy classes

The D_n root lattice is

$$(0) = \left\{ (k_1, \dots, k_n) \in \mathbb{Z}^n \,\middle|\, \sum_{i=1}^n k_i = 0 \pmod 2 \right\}.$$

The vector conjugacy class is

$$(V) = \left\{ (k_1, \dots, k_n) \in \mathbb{Z}^n \,\middle|\, \sum_{i=1}^n k_i = 1 \pmod 2 \right\}.$$

Therefore

$$\boxed{(0) \cup (V) = \mathbb{Z}^n.}$$

All components are integral, so all fermions have monodromy $+1$. These are precisely the NS weights.

The two spinor classes are

$$(S) = \left\{ \left(k_1 + \frac{1}{2}, \dots, k_n + \frac{1}{2}\right) \,\middle|\, k_i \in \mathbb{Z},\ \sum_i k_i = 0 \pmod 2 \right\},$$

and

$$(C) = \left\{ \left(k_1 + \frac{1}{2}, \dots, k_n + \frac{1}{2}\right) \,\middle|\, k_i \in \mathbb{Z},\ \sum_i k_i = 1 \pmod 2 \right\}.$$

Therefore

$$\boxed{(S) \cup (C) = \left(\mathbb{Z} + \frac{1}{2}\right)^n.}$$

All components are half-integral, so all fermions have monodromy -1. These are precisely the R weights.

Combining the two results,

$$\boxed{\begin{array}{ccc} \text{NS} & \Longleftrightarrow & \lambda \in (0) \cup (V), \\ \text{R} & \Longleftrightarrow & \lambda \in (S) \cup (C). \end{array}}$$

6.5.1.7 Why both (0) and (V) occur in the NS sector

The NS vacuum is a scalar of $SO(2n)$ and corresponds to

$$\lambda = \mathbf{0} \in (0).$$

Each fermionic creation operator transforms in the vector representation and carries a weight

$$\pm \mathbf{e}_j \in (V).$$

Consequently:

- an even number of fermionic excitations has total weight in (0);
- an odd number of fermionic excitations has total weight in (V).

This is also immediate from the conjugacy-class addition rule

$$(V) + (V) = (0).$$

For example, a bilinear current

$$\Psi^{\pm i}\Psi^{\pm j}$$

has weight

$$\pm \mathbf{e}_i \pm \mathbf{e}_j,$$

which is a D_n root and hence belongs to (0). A single fermion belongs to (V). Both operators are local with respect to the NS vacuum and both belong to the NS Hilbert space.

6.5.1.8 Why (S) and (C) occur in the R sector

The Ramond ground state is created by a spin field

$$S_{\mathbf{s}}(0) =: \exp\left[\frac{i}{2}\sum_{j=1}^{n} s_j X^j(0)\right] : c_{\mathbf{s}}, \qquad s_j = \pm 1.$$

Its weight vector is

$$\lambda_{\mathbf{s}} = \frac{1}{2}(s_1, \dots, s_n).$$

The OPE with a fermion is

$$\Psi^{\pm j}(z) S_{\mathbf{s}}(0) \sim z^{\pm s_j/2}(\cdots).$$

Every exponent is half-integral, so a circuit around the origin gives

$$e^{\pm \pi i s_j} = -1.$$

Thus the spin field creates the branch cut required for an R state on the complex plane.

The spinor weights with an even number of minus signs form (S), while those with an odd number form (C). Acting with a fermion adds a vector weight:

$$(V) + (S) = (C), \qquad (V) + (C) = (S).$$

Hence fermionic excitations interchange the two spinor classes, but the components remain half-integral and the state remains in the R sector.

Before the GSO projection, the R Hilbert space contains the two spinor chiralities. The GSO projection selects one of (S) or (C) according to the chosen convention. Similarly, in applications to the superstring, the NS GSO projection selects a definite fermion-number parity. The NS/R classification itself precedes this further projection.

The spin-field conformal weight is

$$h_S = \frac{\lambda_{\mathbf{s}}^2}{2} = \frac{1}{2}\left(\frac{n}{4}\right) = \frac{n}{8},$$

in agreement with the book.

6.5.1.9 Role of the cocycle factors

The cocycles c_λ are essential because exponentials built from different bosons would otherwise commute, whereas different fermions must anticommute. They supply momentum-dependent, position-independent phases that correct the exchange algebra.

The branch power in the OPE is still

$$(z-w)^{\lambda\cdot\mu}.$$

Therefore the full monodromy obtained by taking one operator completely around another contains

$$e^{2\pi i\lambda\cdot\mu}.$$

For a fundamental fermion, $\mu = \pm\mathbf{e}_j$, so this becomes

$$e^{\pm 2\pi i\lambda_j}.$$

The cocycle fixes exchange signs and operator ordering, but it does not change whether λ_j is integral or half-integral and hence does not change the NS/R assignment.

This also explains the book's locality statement. Within $(0)\cup(V)$, the relevant mutual inner products are integral, so the NS operator algebra contains no branch cuts. A vector weight has half-integral inner product with a spinor weight modulo integers,

$$(V)\cdot(S) = (V)\cdot(C) = \frac{1}{2} \pmod 1,$$

so transporting a fermion around a Ramond spin field produces the expected minus sign.

6.5.1.10 Checks

1. **NS vacuum:** For $\lambda = \mathbf{0}$,
$$\Psi^{\pm j}(z)V_{\mathbf{0}}(0) \sim z^0(\cdots),$$
so the fermions are periodic on the plane.

2. **Single NS fermion:** For $\lambda = \mathbf{e}_k \in (V)$, the powers $\pm\lambda_j = \pm\delta_{jk}$ are integers. The fermion remains single-valued around the insertion even though $V_{\mathbf{e}_k}$ itself has $h = 1/2$.

3. **NS current:** For $\lambda = \mathbf{e}_i + \mathbf{e}_j \in (0)$, all components are integers and the operator is local with respect to the fermions.

4. **Ramond spin field:** For $\lambda = \frac{1}{2}(s_1,\dots,s_n)$, every fermion OPE contains a square-root power $z^{\pm 1/2}$ and therefore has monodromy -1.

5. **What monodromy cannot distinguish:** Both (0) and (V) give monodromy $+1$, while both (S) and (C) give monodromy -1. The parity conditions separating the two classes within each pair contain representation-theoretic information beyond the NS/R spin structure.

6.5.1.11 Result

The bosonized fermions and a general weight vertex are

$$\Psi^{\pm j}(z) =\,:e^{\pm iX^j(z)}:\, c_{\pm j}, \qquad V_\lambda(0) =\,:e^{i\lambda\cdot\mathbf{X}(0)}:\, c_\lambda.$$

Their OPE is

$$\boxed{\Psi^{\pm j}(z)V_\lambda(0) \sim z^{\pm\lambda_j} V_{\lambda\pm\mathbf{e}_j}(0) \times (\text{cocycles}).}$$

Therefore the fermion monodromy around the state created by V_λ is

$$\boxed{M_j^{\pm}(\lambda) = e^{\pm 2\pi i\lambda_j}.}$$

It follows that

$$\lambda_j \in \mathbb{Z}\ \forall j \quad \Longrightarrow \quad M_j^{\pm} = +1 \quad \Longrightarrow \quad \text{NS on the plane,}$$
$$\lambda_j \in \mathbb{Z} + \frac{1}{2}\ \forall j \quad \Longrightarrow \quad M_j^{\pm} = -1 \quad \Longrightarrow \quad \text{R on the plane.}$$

The D_n weight lattice decomposes as

$$(0) \cup (V) = \mathbb{Z}^n,$$
$$(S) \cup (C) = \left(\mathbb{Z} + \frac{1}{2}\right)^n.$$

Hence

$$\boxed{\begin{array}{lcl} \text{NS} & \Longleftrightarrow & \lambda \in (0) \cup (V), \\ \text{R} & \Longleftrightarrow & \lambda \in (S) \cup (C). \end{array}}$$

The split (0) versus (V) records even versus odd tensor/fermion-number parity, while (S) versus (C) records the two spinor chiralities. The sector itself is determined by the monodromy of the fundamental fermions around the insertion, not by the rotation phase of V_λ alone.

6.6 Q6: Why the one-loop trace is not trivialized by the physical-state condition

In the discussion of one-loop string partition function, for example in the open bosonic string case we will compute

$$\mathrm{Tr}_{\mathcal{H}_{\mathrm{op}}} q^{H_{\mathrm{op}}} = \mathrm{Tr}_{\mathcal{H}_{\mathrm{op}}} q^{L_0 - c/24}\ ,\ q = e^{2\pi i \tau}$$

Or in the closed bosonic string

$$\mathrm{Tr}_{\mathcal{H}_{\mathrm{cl}}} (q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24})$$

But a physical state $|\phi\rangle$ should satisfies $(L_0 - 1)|\phi\rangle = 0$. Does this mean that every physical state in the Hilbert space will have L_0 eigenvalue $L_0 = 1$? Then it seems strange for the trace over $q^{L_0 - c/24}$.

In the fermionic string case, the holomorphic part of the one loop partition function of the fermionic string in Hamiltonian description has the form of eq. (13.114)

$$\chi(\tau) \sim \mathrm{Tr}[e^{2\pi i \tau (L_0 - 1)} (-1)^{F_R}]$$

What is the Hamiltonian and how to derive the trace over $q^{L_0 - 1}$ in this case?

> **Textbook location and related material.** Chapter 6, Sections 6.3 and 6.5 for torus/cylinder traces, together with Chapter 13, Section 13.4 near Eq. (13.114). The relevant trace is over a gauge-fixed CFT or transverse Fock space, not only BRST cohomology at fixed target momentum.

6.6.1 Answer

6.6.1.1 Core point

The apparent contradiction comes from using the phrase "string Hilbert space" for two related but different spaces.

The physical-state equation

$$(L_0^{\mathrm{m}} - 1)|\phi\rangle = 0$$

is an on-shell condition in the covariant bosonic string. It defines external asymptotic states, or equivalently the BRST cohomology at fixed on-shell target-space momentum. If one literally restricted a trace to such states and used $L_0^{\mathrm{m}} - 1$ as the traced operator, then this operator would indeed vanish.

Thus, in that precise sense, the answer to the first question is **yes**: every on-shell covariant bosonic-string state has matter eigenvalue $L_0^{\mathrm{m}} = 1$ in the open string, and $L_0^{\mathrm{m}} = \bar{L}_0^{\mathrm{m}} = 1$ in the closed string. The crucial point is that this is not the restriction imposed inside the one-loop CFT trace.

That is not the trace appearing in the one-loop worldsheet partition function. The latter has either of the following equivalent descriptions:

1. in covariant gauge, it is a trace over the gauge-fixed worldsheet CFT state space, including the appropriate ghost insertions or ghost cancellations;
2. in light-cone gauge, it is a trace over all transverse oscillator states and an integral over Euclidean loop momentum. The Virasoro constraints have already been solved, so they do not impose $L_0 = 1$ as an additional restriction on the oscillator trace.

The operator in the exponential is a **worldsheet cylinder Hamiltonian**, not the target-space energy P^0. For a chiral CFT it is

$$\boxed{H_{\rm cyl} = L_0 - \frac{c}{24}.}$$

For the 24 transverse bosons of the critical bosonic string, $c = 24$, and hence

$$H_{\rm cyl} = L_0 - 1.$$

Equation (13.114) uses the same worldsheet propagation principle. In the covariant RNS vertex-operator description, a physical integrated chiral vertex, including its superghost picture dressing but before multiplying by c, has weight one. Consequently its inverse propagator is $L_0 - 1$, and the holomorphic one-loop trace is schematically

$$\boxed{\chi(\tau) \sim \operatorname{Tr}\left[q_\tau^{L_0-1}(-1)^{F_R}\right], \qquad q_\tau = e^{2\pi i\tau}.}$$

Here $(-1)^{F_R}$ changes the ordinary trace into a spacetime supertrace, with opposite signs for spacetime bosons and fermions.

6.6.1.2 The three notions that must be separated

It is useful to keep the following three operators or spaces distinct.

Object	Meaning	Condition or generator	
Plane CFT state	State created by a local operator at the origin	L_0 measures conformal weight	
On-shell string state	BRST cohomology class or old-covariant physical state	$(L_0^{\rm m} - 1)$	$\|\phi\rangle = 0$ in the bosonic open string
State propagating around a one-loop worldsheet	Gauge-fixed CFT state, or a transverse light-cone state with Euclidean loop momentum	$H_{\rm cyl} = L_0 - c/24$	

The same oscillator state can appear in the second and third rows in different ways. In the physical spectrum, its target-space momentum is constrained to its mass shell. In a vacuum loop, the corresponding particle species runs with arbitrary Euclidean loop momentum. Thus the oscillator polarization is physical, while the momentum carried around the loop is not constrained to its mass shell at every point in the integral.

There is a second, equivalent way to state this. In covariant string field theory, the kinetic operator is proportional to

$$L_0 - 1.$$

The propagator is schematically

$$\frac{b_0}{L_0 - 1} = b_0 \int_0^\infty ds\, e^{-s(L_0-1)}.$$

The on-shell condition $L_0 - 1 = 0$ locates a pole of this propagator. One must not impose the pole equation before constructing the propagator; doing so would erase the propagation that the loop diagram is meant to describe.

6.6.1.3 Deriving the cylinder Hamiltonian $L_0 - c/24$

Start with a chiral CFT on the complex plane and map it to a cylinder. In dimensionless coordinates one may write

$$z = e^w.$$

The stress tensor is not a primary field. Under a holomorphic coordinate transformation it obeys

$$T_{\text{cyl}}(w) = \left(\frac{dz}{dw}\right)^2 T_{\text{plane}}(z) + \frac{c}{12}\{z, w\},$$

where

$$\{z, w\} = \frac{z'''(w)}{z'(w)} - \frac{3}{2}\left(\frac{z''(w)}{z'(w)}\right)^2$$

is the Schwarzian derivative. For $z = e^w$,

$$\{e^w, w\} = 1 - \frac{3}{2} = -\frac{1}{2}.$$

Therefore

$$T_{\text{cyl}}(w) = z^2 T_{\text{plane}}(z) - \frac{c}{24}.$$

The zero mode generating translations along the cylinder is consequently

$$\boxed{H_{\text{cyl}} = L_0 - \frac{c}{24}.}$$

The subtraction is the Casimir energy of the CFT vacuum on the spatial circle. It is not, by itself, an extra physical-state constraint.

One must not identify $c/24$ with the string intercept in every formulation. For example, the covariant bosonic matter CFT has

$$c_{\text{m}} = 26,$$

so its matter Casimir shift alone would be $26/24$, not one. The covariant calculation must also include the b, c ghosts and their zero-mode insertions. After the longitudinal, timelike, and ghost contributions cancel, the answer is equivalent to 24 transverse bosons, for which

$$\frac{c_\perp}{24} = \frac{24}{24} = 1.$$

It is in this reduced light-cone description that the CFT Casimir shift directly reproduces the bosonic intercept.

The RNS string provides an even clearer warning. The complete critical matter-plus-ghost system has total central charge zero, but the propagation operator before the c-ghost dressing is still written as $L_0 - 1$. Here the one is fixed by the weight-one physical-vertex condition, while the familiar NS and R zero-point energies emerge only after resolving the spin sector and canceling the unphysical fields.

For a non-chiral closed-string theory, the Euclidean-time and spatial-translation generators are

$$H_{\text{cyl}} = \left(L_0 - \frac{c}{24}\right) + \left(\bar{L}_0 - \frac{\bar{c}}{24}\right),$$
$$P_{\text{cyl}} = \left(L_0 - \frac{c}{24}\right) - \left(\bar{L}_0 - \frac{\bar{c}}{24}\right).$$

When $c = \bar{c}$, the second equation reduces to

$$P_{\text{cyl}} = L_0 - \bar{L}_0.$$

6.6.1.4 Cutting and gluing the torus

Write the torus modulus as

$$\tau = \tau_1 + i\tau_2.$$

Cutting the torus along a spatial cycle gives a cylinder of Euclidean length proportional to τ_2. Gluing its two ends without a twist gives an evolution operator

$$e^{-2\pi\tau_2 H_{\rm cyl}}.$$

Before gluing, one may also shift one end relative to the other by $2\pi\tau_1$. This inserts

$$e^{2\pi i\tau_1 P_{\rm cyl}}.$$

Taking the trace performs the gluing:

$$Z(\tau,\bar\tau) = \mathrm{Tr}_{\mathcal{H}_{\rm CFT}}\left[e^{-2\pi\tau_2 H_{\rm cyl}} e^{2\pi i\tau_1 P_{\rm cyl}}\right].$$

Substituting the expressions for $H_{\rm cyl}$ and $P_{\rm cyl}$ gives

$$\boxed{Z(\tau,\bar\tau) = \mathrm{Tr}_{\mathcal{H}_{\rm CFT}}\left[q_\tau^{L_0-c/24}\bar q_\tau^{\bar L_0-\bar c/24}\right],}$$

where

$$q_\tau = e^{2\pi i\tau}, \qquad \bar q_\tau = e^{-2\pi i\bar\tau}.$$

The bar on $\bar q_\tau$ is important: with $\tau_2 > 0$, both factors decay exponentially.

For the open string, the spatial worldsheet is an interval. Making Euclidean time periodic produces an annulus. There is one real modulus t, and no independent left-moving sector:

$$\boxed{\mathcal{A} = \int_0^\infty \frac{dt}{2t}\,\mathrm{Tr}_{\mathcal{H}_{\rm op}}\left[e^{-2\pi t H_{\rm op}}\right], \qquad H_{\rm op} = L_0 - \frac{c}{24}.}$$

Equivalently, with $\tau = it$,

$$q_\tau = e^{-2\pi t}, \qquad \mathcal{A} = \int_0^\infty \frac{dt}{2t}\,\mathrm{Tr}_{\mathcal{H}_{\rm op}}\, q_\tau^{L_0-c/24}.$$

This is the book's Eq. (6.114).

6.6.1.5 Why the bosonic physical-state condition does not make the trace constant

For the open bosonic string, the matter zero mode is

$$L_0^{\rm m} = \alpha' p^2 + N.$$

The old-covariant physical-state equation is

$$(L_0^{\rm m} - 1)|\phi\rangle = 0,$$

so, with mostly-plus target-space signature,

$$\alpha' p^2 + N - 1 = 0 \quad\Longleftrightarrow\quad m_N^2 = \frac{N-1}{\alpha'}.$$

This equation does not say that only $N = 1$ states exist. It says that every oscillator level N has its own allowed target-space mass:

N	$\alpha' m_N^2$	state type
0	-1	tachyon
1	0	massless vector
2	1	first massive level
$\vdots$	$\vdots$	$\vdots$

On shell, the momentum contribution changes with N so that the complete matter eigenvalue is one. In the annulus trace, however, the loop momentum is Wick-rotated and integrated:

$$\mathrm{Tr}_{\mathcal{H}_{\mathrm{op}}} q_\tau^{L_0-1} = \sum_N d_N \int \frac{d^D k_E}{(2\pi)^D} e^{-2\pi t(\alpha' k_E^2 + N - 1)}.$$

Here d_N is the number of physical transverse polarizations at level N. The quantity

$$\alpha' k_E^2 + N - 1 = \alpha'(k_E^2 + m_N^2)$$

is not zero for a generic Euclidean loop momentum k_E. It is the inverse propagator in Schwinger proper-time form.

For the closed bosonic string,

$$L_0^{\mathrm{m}} = \frac{\alpha' p^2}{4} + N, \qquad \bar{L}_0^{\mathrm{m}} = \frac{\alpha' p^2}{4} + \bar{N}.$$

The two physical-state equations imply

$$N = \bar{N}, \qquad m_N^2 = \frac{4}{\alpha'}(N-1).$$

Again, all levels occur. The τ_1 dependence of the torus trace is

$$e^{2\pi i \tau_1 (N - \bar{N})}.$$

Integrating τ_1 over one period gives

$$\int_{-1/2}^{1/2} d\tau_1\, e^{2\pi i \tau_1 (N-\bar{N})} = \delta_{N,\bar{N}},$$

so the modulus integral enforces level matching. The τ_2 integral supplies the closed-string proper time.

6.6.1.6 Proper-time derivation from the target-space one-loop vacuum energy

The same result follows without first using the cylinder map. A physical string oscillator state I behaves in target space like a particle species of mass m_I. Its one-loop vacuum contribution is schematically

$$\Gamma_{1-\mathrm{loop}} \sim \frac{1}{2} \sum_I (-1)^{F_I} \int \frac{d^D k_E}{(2\pi)^D} \log(k_E^2 + m_I^2).$$

Using

$$\log A = -\int_0^\infty \frac{ds}{s}\, e^{-sA} + \text{constant},$$

one obtains

$$\Gamma_{1-\mathrm{loop}} \sim -\frac{1}{2} \int_0^\infty \frac{ds}{s} \sum_I (-1)^{F_I} \int \frac{d^D k_E}{(2\pi)^D} e^{-s(k_E^2 + m_I^2)}.$$

For the open bosonic spectrum,

$$\alpha'(k_E^2 + m_I^2) = \alpha' k_E^2 + N_I - 1 = L_0 - 1$$

on the off-shell loop state. After setting $s = 2\pi\alpha' t$, the exponential becomes

$$e^{-2\pi t (L_0 - 1)} = q_\tau^{L_0 - 1}, \qquad q_\tau = e^{-2\pi t}.$$

Thus the annulus modulus is precisely Schwinger proper time. The trace sums the physical particle species, while the momentum integral supplies their off-shell propagation.

This is the logic used around Eqs. (6.89)–(6.93) of the book. It also explains why imposing $L_0 - 1 = 0$ inside the trace would be incorrect: that condition identifies the poles of the propagators, whereas the proper-time integral constructs the propagators.

6.6.1.7 Covariant versus light-cone traces

There are two standard ways to obtain the same answer.

In light-cone gauge, only the $D-2$ transverse oscillators are present. For the critical bosonic string,

$$c_\perp = 24,$$

and hence

$$H_{\text{op}} = L_0^\perp - \frac{c_\perp}{24} = L_0^\perp - 1.$$

All transverse oscillator Fock states are allowed in the trace. As the book emphasizes after Eq. (6.82), the constraints leading to the physical-state conditions have already been solved and do not impose a further restriction on this trace.

In covariant gauge, one may include all 26 coordinate oscillators, but one must also trace over the b, c ghosts with the correct zero-mode insertions. The longitudinal and timelike coordinate contributions are canceled by the ghosts. The result is again the contribution of 24 transverse oscillators:

$$Z_{\text{covariant}} = Z_{26\,X}\, Z_{bc} = Z_{24\,X^\perp}.$$

This is why the covariant and light-cone calculations have the same level-generating function.

One should therefore not interpret

$$\mathcal{H}_{\text{op}}$$

in Eq. (6.114) as "the set of covariant states after imposing $L_0 = 1$ at fixed momentum." In that equation it means the Hilbert space appropriate to the chosen gauge-fixed formulation.

6.6.1.8 The Hamiltonian in Eq. (13.114)

Equation (13.114) is a **holomorphic** or chiral partition function. Its Hamiltonian is

$$\boxed{H_{\text{chiral}} = L_0 - 1.}$$

More precisely:

- L_0 is the plane Virasoro zero mode of the chiral RNS system used in the covariant vertex-operator construction;
- the subtraction of one is the intercept appropriate to an integrated, picture-dressed chiral string vertex before insertion of the reparametrization ghost c;
- for $\tau = it$, $e^{2\pi i\tau(L_0-1)} = e^{-2\pi t(L_0-1)}$ is Euclidean worldsheet evolution;
- for nonzero τ_1, the phase implements the twist used when gluing the cylinder into a torus;
- $(-1)^{F_R}$ supplies the minus sign for right-moving spacetime fermions.

It is not the target-space Hamiltonian P^0. It is the dimensionless generator of translations along the chiral worldsheet cylinder, including the intercept appropriate to the string propagation problem.

The reason the exponent is written uniformly as $L_0 - 1$ in both the NS and R sectors is that Chapter 13 includes the superghost picture dressing. Before multiplication by c, every physical chiral vertex has total conformal weight

$$h_{\text{matter+superghost}} = 1.$$

Thus the uniform physical equation is

$$(L_0 - 1)|V\rangle = 0.$$

After removing the longitudinal, timelike, and ghost degrees of freedom, this uniform covariant expression reduces to the familiar sector-dependent light-cone Hamiltonians

$$\boxed{\begin{aligned} H_{\text{NS}}^{\text{lc}} &= \frac{\alpha' k^2}{4} + N_{\text{NS}} - \frac{1}{2}, \\ H_{\text{R}}^{\text{lc}} &= \frac{\alpha' k^2}{4} + N_{\text{R}}, \end{aligned}}$$

for one chiral half of a closed RNS string. The corresponding intercepts are therefore

$$a_{\mathrm{NS}} = \frac{1}{2}, \qquad a_{\mathrm{R}} = 0.$$

There is no contradiction between these formulas and Eq. (13.114): the two descriptions distribute the normal-ordering and ghost contributions differently.

6.6.1.9 Deriving the Chapter 13 lattice trace

Chapter 13 bosonizes five complex matter fermions and the β, γ superghost system. A state is labeled by

$$\mathbf{w} = (\lambda, q_{\mathrm{gh}}) \in D_{5,1},$$

where λ is a D_5 weight and q_{gh} is the superghost picture charge. According to Eq. (13.65), the contribution to L_0 is

$$L_0 = \frac{1}{2}\lambda^2 - \frac{1}{2}q_{\mathrm{gh}}^2 - q_{\mathrm{gh}} + N + L_0^X,$$

where N denotes oscillator excitations omitted from the simple exponential vertex and L_0^X contains the spacetime-boson contribution.

Substitution into Eq. (13.114) gives

$$\chi(\tau) \sim \sum_{\lambda, q_{\mathrm{gh}}, N} e^{2\pi i\tau\left(\frac{1}{2}\lambda^2 - \frac{1}{2}q_{\mathrm{gh}}^2 - q_{\mathrm{gh}} + N + L_0^X - 1\right)} (-1)^{F_R}.$$

The book then first isolates the nontrivial lattice zero-mode part. Since q_{gh} is integral on spacetime bosons and half-integral on spacetime fermions,

$$\boxed{(-1)^{F_R} = e^{-2\pi i q_{\mathrm{gh}}}.}$$

Indeed,

$$\begin{aligned} q_{\mathrm{gh}} \in \mathbb{Z} \quad &\Longrightarrow \quad e^{-2\pi i q_{\mathrm{gh}}} = +1, \\ q_{\mathrm{gh}} \in \mathbb{Z} + \frac{1}{2} \quad &\Longrightarrow \quad e^{-2\pi i q_{\mathrm{gh}}} = -1. \end{aligned}$$

This yields the lattice factor displayed in Eq. (13.115):

$$\tilde{\chi}(\tau) = \sum_{\mathbf{w}=(\lambda, q_{\mathrm{gh}}) \in D_{5,1}} e^{2\pi i\tau\left(\frac{1}{2}\lambda^2 - \frac{1}{2}q_{\mathrm{gh}}^2 - q_{\mathrm{gh}}\right)} e^{-2\pi i q_{\mathrm{gh}}},$$

up to the universal oscillator and intercept factors temporarily suppressed by the proportionality sign in Eq. (13.114). They are restored in Eq. (13.122).

One can see the restoration of the full -1 explicitly. Let

$$\mathbf{e}_6 = (0, \dots, 0; 1).$$

Because the last direction of $D_{5,1}$ is timelike,

$$\mathbf{e}_6^2 = -1, \qquad \mathbf{w} \cdot \mathbf{e}_6 = -q_{\mathrm{gh}}.$$

The shifted lattice exponent in Eq. (13.122) is therefore

$$\begin{aligned} \frac{1}{2}(\mathbf{w} + \mathbf{e}_6)^2 &= \frac{1}{2}\mathbf{w}^2 + \mathbf{w} \cdot \mathbf{e}_6 + \frac{1}{2}\mathbf{e}_6^2 \\ &= \frac{1}{2}\lambda^2 - \frac{1}{2}q_{\mathrm{gh}}^2 - q_{\mathrm{gh}} - \frac{1}{2}. \end{aligned}$$

Moreover,

$$\frac{1}{\eta(\tau)^{12}} = q_\tau^{-1/2} \prod_{n=1}^{\infty} (1 - q_\tau^n)^{-12}.$$

The $-\frac{1}{2}$ in the shifted lattice exponent and the $-\frac{1}{2}$ from η^{-12} add to

$$-\frac{1}{2}-\frac{1}{2}=-1.$$

Consequently the complete expression has precisely the factor

$$q_\tau^{L_0-1}$$

announced in Eq. (13.114); Eq. (13.115) was displaying only the selected zero-mode part.

The book immediately warns that the unrestricted $D_{5,1}$ lattice sum is not yet the physical partition function. It sums over infinitely many equivalent ghost pictures and includes longitudinal, timelike, and ghost degrees of freedom. The subsequent decomposition

$$D_{5,1}=D_4\oplus D_{1,1}$$

separates the transverse $SO(8)$ degrees of freedom from the unphysical sector. Dividing by the $D_{1,1}$ lattice factor and fixing the canonical-picture representative leaves the physical light-cone partition function.

Thus Eq. (13.114) is a schematic covariant starting trace. It should not be read as a literal trace over BRST cohomology in which L_0-1 has already been set to zero.

6.6.1.10 Checks

1. **Open bosonic tachyon:** Omitting momentum, the $N=0$ state contributes

   $$q_\tau^{-1},$$

   which is the leading term of $\eta(\tau)^{-24}$.

2. **Open bosonic massless level:** The $N=1$ oscillator states contribute at order

   $$q_\tau^0.$$

   They are massless, but they do not remove the other oscillator levels from the trace.

3. **On-shell pole:** When a momentum is analytically continued so that

   $$L_0-1=0,$$

   the propagator $(L_0-1)^{-1}$ develops a pole. This confirms that the physical-state equation identifies a pole rather than an operator to be set to zero throughout the loop calculation.

4. **Closed-string level matching:** The τ_1 integral removes terms with

   $$L_0-\bar{L}_0\neq 0.$$

   This is why terms such as q_τ^{-1} without the matching $\bar{q}_\tau^{-1}$ partner occur in the unintegrated partition function but do not survive the full modulus integral.

5. **RNS intercepts:** The uniform covariant expression L_0-1 becomes $N_{\mathrm{NS}}-1/2$ in the light-cone NS sector and N_{R} in the light-cone R sector after the superghost and unphysical degrees of freedom are eliminated.

6. **Spacetime supersymmetry:** After the GSO projection, the supertrace weighted by $(-1)^{F_R}$ cancels equal-mass bosonic and fermionic states. This is the Hamiltonian interpretation of the vanishing Jacobi combination in the ten-dimensional supersymmetric partition function.

6.6.1.11 Result

The plane-to-cylinder map gives

$$\boxed{H_{\text{cyl}} = L_0 - \frac{c}{24}.}$$

Therefore

$$\boxed{\begin{aligned} Z_{\text{open}}(t) &= \mathrm{Tr}_{\mathcal{H}_{\text{CFT}}}\, e^{-2\pi t(L_0 - c/24)}, \\ Z_{\text{closed}}(\tau,\bar\tau) &= \mathrm{Tr}_{\mathcal{H}_{\text{CFT}}} \left(q_\tau^{L_0 - c/24} \bar q_\tau^{\bar L_0 - \bar c/24} \right). \end{aligned}}$$

For the 24 transverse bosons,

$$\frac{c_\perp}{24} = 1, \qquad H_{\text{cyl}} = L_0^\perp - 1.$$

The bosonic physical-state equation

$$(L_0^{\text{m}} - 1)|\phi\rangle = 0$$

means

$$\boxed{\begin{array}{lll} \text{open:} & m_N^2 = \dfrac{N-1}{\alpha'}, & \\ \text{closed:} & m_N^2 = \dfrac{4(N-1)}{\alpha'}, & N = \bar N. \end{array}}$$

It fixes the mass of each oscillator level; it does not restrict the spectrum to $N = 1$ and does not put the Euclidean loop momentum on shell.

The proper-time identity gives

$$\boxed{\frac{1}{L_0 - 1} = \int_0^\infty ds\, e^{-s(L_0 - 1)},}$$

so the on-shell equation $L_0 - 1 = 0$ determines the propagator poles, while the one-loop trace constructs the propagator.

Finally, Eq. (13.114) is

$$\boxed{\chi(\tau) \sim \mathrm{Tr}\left[q_\tau^{L_0 - 1} (-1)^{F_R} \right].}$$

Its Hamiltonian is the chiral worldsheet generator

$$\boxed{H_{\text{chiral}} = L_0 - 1,}$$

and not the target-space energy P^0. The trace is a spacetime supertrace over the gauge-fixed chiral string state space; after ghost-picture reduction and cancellation of unphysical degrees of freedom, it becomes the physical light-cone partition function.

$$A_\perp = \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{>0} \cup \{\perp\}$$

$$(a,b) \cdot (c,d) = \begin{cases} (c, d-a) & d > a \\ \perp & d \leq a \end{cases}$$

$$(a,b) \cdot \perp = \perp\ , \ \perp \cdot (a,b) = (a,b)$$

Chapter 7

CFTs for Type II and Heterotic String Vacua

7.1 Q1: Untwisted and twisted cohomology of toroidal orbifolds

In page 523-524, it is said that: >It is straightforward to work out which elements of the Dolbeault cohomology of the torus survive the orbifold projection. This is referred to as the untwisted cohomology of the orbifold. There will be additional contributions from the twisted sectors which are related to moduli controlling the repair of the orbifold singularities.

The results are listed in Table 15.2.

I don't know how to obtain both the untwisted cohomology and the twisted cohomology. Consider some examples in table 15.2, give the explicit computation.

> **Textbook location and related material.** Chapter 15, Section 15.2 (pp. 522–534), especially pp. 523–524 and Table 15.2: invariant torus cohomology, twisted sectors, fixed loci, and blow-up modes. Section 14.5 supplies the Calabi–Yau cohomology interpretation.

7.1.1 Answer

7.1.1.1 Core point

The entries in Table 15.2 are obtained in two logically different steps.

First, one takes the ordinary Dolbeault cohomology of the covering torus and keeps only the forms invariant under the orbifold generator:

$$H^{p,q}_{\text{untw}}(T^6/\mathbb{Z}_N) = H^{p,q}(T^6)^{\mathbb{Z}_N}.$$

These dimensions are the numbers $h_0^{p,q}$ written in parentheses in the table.

Second, for every nonidentity element θ^k, one studies its fixed locus. A twisted string is closed only up to θ^k,

$$Z^i(\sigma + 2\pi) = e^{2\pi i k v_i} Z^i(\sigma),$$

so its center of mass is confined to $\text{Fix}(\theta^k)$. Massless twisted states are therefore organized by the cohomology of these fixed loci. Their Hodge bidegrees are shifted by the age of the twist, and one must then impose the orbifold projection by the full group.

The compact mathematical statement is

$$\boxed{H^{p,q}_{\text{orb}}([T^6/G]) = H^{p,q}(T^6)^G \oplus \bigoplus_{k=1}^{N-1} H^{p-a_k,q-a_k}(\text{Fix}(\theta^k))^G}, \qquad G = \mathbb{Z}_N.$$

When the fixed locus is disconnected, the last expression means the cohomology of the disjoint union of all components, followed by the G-projection. For the supersymmetric toroidal orbifolds in Table 15.2, these orbifold Hodge numbers agree with the Hodge numbers of a crepant resolution $\widetilde{X}$.

This formula will be derived and then applied explicitly to:

1. case 1, $T^6/\mathbb{Z}_3$ on $A_2 \times A_2 \times A_2$, where the twisted fixed loci are isolated points;
2. case 2, $T^6/\mathbb{Z}_4$ on $A_1 \times A_1 \times B_2 \times B_2$, where the θ^2 sector contains fixed tori and produces both twisted $(1,1)$ and twisted $(2,1)$ classes.

The second example explains why the total Hodge numbers depend on the lattice realization as well as on the twist vector.

7.1.1.2 Untwisted cohomology: invariant constant forms

On a complex torus, every Dolbeault class has a constant representative. If

$$V^{1,0\,*} = \mathrm{span}_{\mathbb{C}}\{dz^1, dz^2, dz^3\},$$

then

$$H^{p,q}(T^6) \cong \bigwedge^p V^{1,0\,*} \otimes \bigwedge^q \overline{V^{1,0\,*}}.$$

The generator acts on the one-forms by

$$\theta^* dz^i = e^{2\pi i v_i} dz^i, \qquad \theta^* d\bar{z}^{\bar{i}} = e^{-2\pi i v_i} d\bar{z}^{\bar{i}}.$$

Consequently,

$$\theta^* \left(dz^{i_1} \wedge \cdots \wedge dz^{i_p} \wedge d\bar{z}^{\bar{j}_1} \wedge \cdots \wedge d\bar{z}^{\bar{j}_q}\right) = e^{2\pi i \left(\sum_{a=1}^p v_{i_a} - \sum_{b=1}^q v_{j_b}\right)} (\cdots) \,.$$

The form survives precisely when

$$\sum_{a=1}^p v_{i_a} - \sum_{b=1}^q v_{j_b} \in \mathbb{Z}.$$

In particular,

$$h_0^{1,1} = \#\left\{(i,j) : v_i - v_j \in \mathbb{Z}\right\},$$

because a basis of $H^{1,1}(T^6)$ is $dz^i \wedge d\bar{z}^{\bar{j}}$. Similarly,

$$h_0^{2,1} = \#\left\{(i,j;k) : i < j,\ v_i + v_j - v_k \in \mathbb{Z}\right\}.$$

The supersymmetry condition

$$v_1 + v_2 + v_3 = 0 \pmod 1$$

implies

$$\theta^* \left(dz^1 \wedge dz^2 \wedge dz^3\right) = dz^1 \wedge dz^2 \wedge dz^3.$$

Thus the holomorphic volume form always survives, as stated around equation (15.3).

7.1.1.3 Twisted cohomology: fixed loci and the age shift

Consider the θ^k-twisted sector. Define

$$\alpha_i^{(k)} = \{k v_i\}, \qquad 0 \le \alpha_i^{(k)} < 1.$$

If $\alpha_i^{(k)} = 0$, the z^i direction is tangent to the fixed locus. If $\alpha_i^{(k)} \neq 0$, it is a rotated normal direction. For a fixed component F, the age is

$$a_k(F) = \sum_{\alpha_i^{(k)} \neq 0} \alpha_i^{(k)}.$$

In the present linear examples it is constant on all components of a given fixed locus. A class in $H^{r,s}(F)$ contributes after the age shift to

$$H_{\mathrm{orb}}^{r+a_k,\, s+a_k}.$$

The two cases needed below are especially simple.

For an isolated fixed point P,

$$H^{0,0}(P) = \mathbb{C}.$$

It therefore contributes one class of bidegree

$$(a_k, a_k).$$

Thus an age-one isolated fixed point contributes to $h^{1,1}$, whereas an age-two fixed point contributes to $h^{2,2}$.

For a fixed complex one-torus $F \cong T^2$ and age $a_k = 1$,

$$\begin{array}{ccc} H^{0,0}(F) & \longrightarrow & H^{1,1}_{\text{orb}}, \\ H^{1,0}(F) & \longrightarrow & H^{2,1}_{\text{orb}}, \\ H^{0,1}(F) & \longrightarrow & H^{1,2}_{\text{orb}}, \\ H^{1,1}(F) & \longrightarrow & H^{2,2}_{\text{orb}}. \end{array}$$

However, these classes survive only after projection by G. If a fixed torus is preserved by θ, one keeps only the θ-invariant forms on it. If several fixed tori are permuted, one must form invariant linear combinations across the whole orbit.

The fixed components themselves are found by solving

$$\theta^k z = z + \lambda, \qquad \lambda \in \Lambda.$$

If there are no invariant directions, the number of solutions modulo Λ is

$$\#\,\mathrm{Fix}(\theta^k) = \left|\det_{\mathbb{R}}(1-\theta^k)\right| = \prod_{i=1}^{3} \left|1 - e^{2\pi i k v_i}\right|^2 = \prod_{i=1}^{3} 4\sin^2(\pi k v_i).$$

If θ^k has invariant directions, this determinant vanishes because the fixed locus is positive-dimensional. One then takes the determinant only on the normal lattice to count components and separately analyzes how G permutes those components.

7.1.1.4 Example 1: case 1, $T^6/\mathbb{Z}_3$ on A_2^3

The twist vector is

$$\mathbf{v} = \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right).$$

Since $-\frac{2}{3} \equiv \frac{1}{3} \pmod 1$, the generator acts on all three complex coordinates by the same phase:

$$\theta : (z^1, z^2, z^3) \longmapsto (\omega z^1, \omega z^2, \omega z^3), \qquad \omega = e^{2\pi i/3}.$$

7.1.1.4.1 Untwisted $(1,1)$ classes Every basis element of $H^{1,1}(T^6)$ is

$$\omega_{i\bar{j}} = dz^i \wedge d\bar{z}^{\bar{j}}, \qquad i, j = 1, 2, 3.$$

Its phase is

$$\omega\bar{\omega} = 1.$$

Equivalently, $v_i - v_j = 0 \pmod 1$ for every pair (i,j). Therefore all nine torus $(1,1)$ classes survive:

$$\boxed{h_0^{1,1} = 3 \times 3 = 9}.$$

7.1.1.4.2 Untwisted $(2,1)$ classes A basis element is

$$dz^i \wedge dz^j \wedge d\bar{z}^{\bar{k}}, \qquad i < j.$$

Its phase is

$$\omega^2\bar{\omega} = \omega^2\omega^{-1} = \omega \neq 1.$$

Hence none of the nine $(2,1)$ classes survives:

$$\boxed{h_0^{2,1} = 0}.$$

This reproduces the parenthesized entries (9) and (0) in the first row of Table 15.2.

7.1.1.4.3 Fixed points in the θ sector Each A_2 root lattice is a hexagonal lattice preserved by a 120° rotation. On one complex torus, the number of fixed points is

$$|1-\omega|^2 = (1-\omega)(1-\bar{\omega}) = 3.$$

Equivalently, if the Coxeter action on an integral A_2 basis is represented by

$$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

then

$$|\det(1-R)| = \left|\det\begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}\right| = 3.$$

Because the six-torus is the product of three such factors,

$$\#\,\mathrm{Fix}(\theta) = 3^3 = 27.$$

For $k = 1$ the fractional twists are

$$\left(\alpha_1^{(1)}, \alpha_2^{(1)}, \alpha_3^{(1)}\right) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right),$$

so

$$a_1 = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1.$$

Every fixed point therefore supplies one twisted $(1,1)$ class. The 27 fixed points are individually fixed by the full $\mathbb{Z}_3$ action, so the orbifold projection does not identify them:

$$\boxed{\Delta_\theta h^{1,1} = 27, \qquad \Delta_\theta h^{2,1} = 0}.$$

Locally, each singularity is of type

$$\mathbb{C}^3/\mathbb{Z}_3, \qquad \frac{1}{3}(1,1,1).$$

Its crepant resolution introduces an exceptional divisor. The corresponding divisor class is the geometric realization of the age-one twisted $(1,1)$ state, and its Kähler parameter controls the blow-up of that singularity.

7.1.1.4.4 The θ^2 sector and why it does not give another 27 to $h^{1,1}$ The element θ^2 has the same 27 fixed points, but its fractional twists are

$$\left(\alpha_1^{(2)}, \alpha_2^{(2)}, \alpha_3^{(2)}\right) = \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right).$$

Thus

$$a_2 = \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = 2.$$

The point class in $H^{0,0}$ is now shifted to bidegree $(2,2)$, not $(1,1)$:

$$\Delta_{\theta^2} h^{2,2} = 27, \qquad \Delta_{\theta^2} h^{1,1} = 0.$$

This is the Hodge-dual partner of the θ contribution. The two CFT twisted sectors therefore do not give 54 independent Kähler moduli.

Combining the untwisted and age-one twisted contributions gives

$$\boxed{h^{1,1} = 9 + 27 = 36, \qquad h^{2,1} = 0}.$$

This is precisely case 1 of Table 15.2.

7.1.1.5 Example 2: case 2, $T^6/\mathbb{Z}_4$ on $A_1^2 \times B_2^2$

Now take

$$\mathbf{v} = \left(\frac{1}{4}, \frac{1}{4}, -\frac{1}{2}\right).$$

Choose the first two complex planes to be the two square B_2 tori and the third to be the $A_1 \times A_1$ torus. Then

$$\theta : (z^1, z^2, z^3) \longmapsto (iz^1, iz^2, -z^3).$$

7.1.1.5.1 Untwisted $(1,1)$ classes For $dz^i \wedge d\bar{z}^{\bar{j}}$ to be invariant, z^i and z^j must have the same eigenvalue. The first two coordinates both have eigenvalue i, while the third has eigenvalue -1. Hence the invariant forms are

$$dz^1 \wedge d\bar{z}^{\bar{1}}, \qquad dz^1 \wedge d\bar{z}^{\bar{2}}, \qquad dz^2 \wedge d\bar{z}^{\bar{1}}, \qquad dz^2 \wedge d\bar{z}^{\bar{2}},$$
$$dz^3 \wedge d\bar{z}^{\bar{3}}.$$

Thus

$$\boxed{h_0^{1,1} = 4 + 1 = 5}.$$

7.1.1.5.2 Untwisted $(2,1)$ classes Consider

$$\chi = dz^1 \wedge dz^2 \wedge d\bar{z}^{\bar{3}}.$$

Its phase is

$$i \cdot i \cdot (-1) = 1,$$

so it is invariant. It is the only invariant $(2,1)$ basis form. For example,

$$dz^1 \wedge dz^3 \wedge d\bar{z}^{\bar{1}}$$

has phase

$$i \cdot (-1) \cdot (-i) = -1,$$

and similarly for the remaining basis elements. Therefore

$$\boxed{h_0^{2,1} = 1}.$$

This reproduces the parenthesized (5) and (1) in the second row of Table 15.2.

7.1.1.5.3 The θ and θ^3 sectors: isolated fixed points In the θ sector, a quarter-turn on a square two-torus has

$$|1 - i|^2 = 2$$

fixed points, while inversion on a two-torus has

$$|1 - (-1)|^2 = 4$$

fixed points. Hence

$$\#\,\mathrm{Fix}(\theta) = 2 \cdot 2 \cdot 4 = 16.$$

The fractional twists are

$$\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right),$$

and therefore

$$a_1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1.$$

The θ sector consequently contributes

$$\boxed{\Delta_\theta h^{1,1} = 16, \qquad \Delta_\theta h^{2,1} = 0}.$$

The inverse element θ^3 has fractional twists

$$\left(\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right),$$

so

$$a_3 = \frac{3}{4} + \frac{3}{4} + \frac{1}{2} = 2.$$

It has the same 16 fixed points but contributes to $h^{2,2}$ rather than $h^{1,1}$:

$$\Delta_{\theta^3} h^{2,2} = 16.$$

7.1.1.5.4 The θ^2 sector: 16 fixed tori before the orbifold projection The square of the generator acts as

$$\theta^2 : (z^1, z^2, z^3) \longmapsto (-z^1, -z^2, z^3).$$

The first two coordinates must be two-torsion points, while z^3 is arbitrary. Each of the first two tori has four two-torsion points, so

$$\mathrm{Fix}(\theta^2) = \coprod_{a=1}^{16} F_a, \qquad F_a \cong T^2_{(3)}.$$

Thus there are 16 fixed two-tori before the remaining $\mathbb{Z}_4$ projection.

The fractional twists normal to a fixed torus are

$$\left(\frac{1}{2}, \frac{1}{2}, 0\right),$$

and hence

$$a_2 = \frac{1}{2} + \frac{1}{2} = 1.$$

Before projection, $H^{0,0}(F_a)$ could therefore give a $(1,1)$ class and $H^{1,0}(F_a)$ could give a $(2,1)$ class. The crucial next step is to determine the action of θ on the 16 components and on their one-forms.

7.1.1.5.5 Orbifold projection on the fixed tori Represent either square B_2 torus as

$$\mathbb{C}/(\mathbb{Z} + i\mathbb{Z}).$$

Its four points fixed by inversion are

$$0, \qquad \frac{1}{2}, \qquad \frac{i}{2}, \qquad \frac{1+i}{2}.$$

Multiplication by i acts on them as

$$0 \longmapsto 0, \qquad \frac{1+i}{2} \longmapsto \frac{1+i}{2}, \qquad \frac{1}{2} \longleftrightarrow \frac{i}{2},$$

where equality is understood modulo the lattice. Thus, on the four two-torsion points of each of the first two tori, θ has two one-element orbits and one two-element orbit.

On the Cartesian product of the two four-point sets:

- $2 \times 2 = 4$ fixed-torus components are preserved individually by θ;
- the remaining $16 - 4 = 12$ components form $12/2 = 6$ exchanged pairs.

Therefore the 16 components form

$$4 + 6 = 10$$

orbits under θ.

For $H^{0,0}$, each orbit supplies one invariant constant combination. After the age-one shift, this gives

$$\boxed{\Delta_{\theta^2} h^{1,1} = 10}.$$

The $H^{1,0}$ calculation is subtler. Along every fixed torus, the holomorphic one-form is dz^3, and

$$\theta^* dz^3 = -dz^3.$$

On each of the four individually preserved components, dz^3 is odd, so it is removed by the projection.

For an exchanged pair (F, F'), let η_F and $\eta_{F'}$ denote the copies of dz^3 supported on the two components. Then

$$\theta^* \eta_F = -\eta_{F'}, \qquad \theta^* \eta_{F'} = -\eta_F.$$

The antisymmetric combination

$$\eta_F - \eta_{F'}$$

is invariant:

$$\theta^*(\eta_F - \eta_{F'}) = -\eta_{F'} + \eta_F = \eta_F - \eta_{F'}.$$

Each of the six exchanged pairs therefore contributes one invariant $H^{1,0}$ class. After the age-one shift,

$$H^{1,0}(F) \longrightarrow H^{2,1}_{\text{orb}},$$

so

$$\boxed{\Delta_{\theta^2} h^{2,1} = 6}.$$

Complex conjugation similarly gives six $H^{1,2}$ classes.

7.1.1.5.6 Total for case 2 Collecting the contributions by sector,

$$h^{1,1} = \underbrace{5}_{\text{untwisted}} + \underbrace{16}_{\theta} + \underbrace{10}_{\theta^2} = 31,$$
$$h^{2,1} = \underbrace{1}_{\text{untwisted}} + \underbrace{0}_{\theta} + \underbrace{6}_{\theta^2} = 7.$$

Thus

$$\boxed{(h^{1,1}, h^{2,1}) = (31, 7)},$$

exactly as in case 2 of Table 15.2.

7.1.1.6 Why the twist vector alone does not determine the twisted answer

The untwisted count depends only on the eigenvalues $e^{2\pi i v_i}$ acting on the complex one-forms. This is why cases 2, 3, and 4, which all use the $\mathbb{Z}_4$ twist

$$\mathbf{v} = \left(\frac{1}{4}, \frac{1}{4}, -\frac{1}{2}\right),$$

all have

$$(h_0^{1,1}, h_0^{2,1}) = (5, 1).$$

Their total Hodge numbers differ:

$$(31, 7), \qquad (27, 3), \qquad (25, 1).$$

The reason is that the three cases use different six-dimensional lattices. The number of fixed components, the way θ permutes the fixed tori of θ^2, and the action on the cohomology of those tori all depend on the integral lattice automorphism. Thus the twist vector determines the local rotation angles and age shifts, but the lattice is also required to determine the global multiplicities.

7.1.1.7 Relation to the book's $T^6/\mathbb{Z}_3$ massless spectrum

The geometric count agrees directly with equations (15.21)–(15.22).

For type IIA on a Calabi-Yau threefold, the number of vector multiplets is $h^{1,1}$ and the number of hypermultiplets is $h^{2,1}+1$, where the extra one is the universal hypermultiplet. In the $\mathbb{Z}_3$ example,

$$h^{1,1} = 9_{\mathrm{untw}} + 27_{\mathrm{tw}}, \qquad h^{2,1} = 0.$$

Therefore type IIA has

$$9 \text{ untwisted vector multiplets} + 27 \text{ twisted vector multiplets},$$

together with one universal hypermultiplet, exactly as in equation (15.22).

For type IIB, the Kähler moduli instead occur in hypermultiplets. Thus the same cohomology gives ten untwisted hypermultiplets—nine from $h_0^{1,1} = 9$ plus the universal hypermultiplet—and 27 twisted hypermultiplets, as in equation (15.21).

This also makes precise the sentence following Table 15.2: the 27 age-one twisted classes are the Kähler moduli controlling the repair of the 27 orbifold singularities.

7.1.1.8 Checks

The age of a twist and that of its inverse obey

$$a_k + a_{N-k} = \operatorname{codim}_{\mathbb{C}} \operatorname{Fix}(\theta^k).$$

For the isolated $\mathbb{Z}_3$ fixed points,

$$1+2=3,$$

so the θ and θ^2 sectors contribute to the Hodge-dual groups $H^{1,1}$ and $H^{2,2}$. For the isolated $\mathbb{Z}_4$ fixed points, the same check is

$$1+2=3.$$

For the θ^2 fixed tori, the complex codimension is two and θ^2 is self-inverse:

$$1+1=2.$$

These relations ensure compatibility with Poincaré duality.

The resulting Euler characteristics are

$$\chi(\widetilde{X}_{\mathbb{Z}_3}) = 2(h^{1,1} - h^{2,1}) = 2(36-0) = 72,$$

and

$$\chi(\widetilde{X}_{\mathbb{Z}_4}) = 2(31-7) = 48.$$

7.1.1.9 Result

For $T^6/\langle\theta\rangle$ with

$$\theta : z^i \mapsto e^{2\pi i v_i} z^i,$$

the untwisted classes are selected by

$$\boxed{\sum_{a=1}^{p} v_{i_a} - \sum_{b=1}^{q} v_{j_b} \in \mathbb{Z}}.$$

The twisted contribution of θ^k is obtained from

$$\boxed{\alpha_i^{(k)} = \{k v_i\}, \qquad a_k = \sum_{\alpha_i^{(k)} \neq 0} \alpha_i^{(k)}, \qquad H^{r,s}(F) \mapsto H_{\mathrm{orb}}^{r+a_k,s+a_k}},$$

followed by the full G-projection on the cohomology of the union of fixed components.

For the two explicit examples,

orbifold	$(h_0^{1,1}, h_0^{2,1})$	θ contribution	θ^2 contribution	$(h^{1,1}, h^{2,1})$
$T^6/\mathbb{Z}_3$ on A_2^3	$(9,0)$	$(27,0)$	$(0,0)$ in these bidegrees	$(36,0)$
$T^6/\mathbb{Z}_4$ on $A_1^2 \times B_2^2$	$(5,1)$	$(16,0)$	$(10,6)$	$(31,7)$

In the $\mathbb{Z}_3$ example, the θ^2 sector contributes 27 classes to $h^{2,2}$; in the $\mathbb{Z}_4$ example, the θ^3 sector contributes 16 classes to $h^{2,2}$. They are the Hodge-dual partners of the age-one isolated-point contributions and are not additional $h^{1,1}$ moduli.

7.2 Q2: Origin of the enhanced orbifold gauge symmetry

In page 527: >While for smooth CY compactifications this leads to the gauge group $E_6 \times E_8$, in the orbifold limit it is enhanced to at least $E_6 \times G \times E_8$ where G can be $SU(3), SU(2) \times U(1)$ or $U(1) \times U(1)$, depending on the shift vector. The blow-up modes, which are the elements of the cohomology in the twisted sectors, are charged under G.

I want to know where the gauge bosons of G come from i.e. why the symmetry in this case can be enhanced.

> **Textbook location and related material.** Chapter 15, Section 15.2 around p. 527: standard embedding, orbifold gauge shift, enhanced group G, and charged blow-up modes. Sections 10.4–10.5 and 11.2–11.5 give the heterotic current-algebra and lattice machinery.

7.2.1 Answer

7.2.1.1 Core point

The gauge bosons of the extra factor G do **not** come from the twisted sectors and do not arise from wrapped branes on the collapsing cycles. They are ordinary perturbative heterotic gauge bosons already present in the ten-dimensional first E_8. More precisely, they are untwisted left-moving current-algebra states that survive the orbifold projection.

The difference between the smooth and orbifold compactifications is the group with which a gauge generator must commute:

$$\begin{array}{rcl} \text{smooth standard embedding} & : & \mathrm{Hol}(A^{\mathrm{bg}}) = SU(3), \\ \text{orbifold standard embedding} & : & \mathrm{Hol}_{\mathrm{orb}} = \langle \gamma_{\mathbf{V}} \rangle \subset SU(3), \end{array}$$

where $\gamma_{\mathbf{V}}$ is a finite-order element determined by the gauge shift $\mathbf{V}$.

A generator that commutes with the full continuous group $SU(3)$ automatically commutes with $\gamma_{\mathbf{V}}$, but the converse need not hold. Therefore

$$C_{E_8}(SU(3)) \subseteq C_{E_8}(\gamma_{\mathbf{V}}),$$

and the inclusion is generally strict. The two centralizers are

$$C_{E_8}(SU(3)) = E_6,$$

and, for the standard orbifold shifts considered here,

$$C_{E_8}(\gamma_{\mathbf{V}}) \supseteq E_6 \times C_{SU(3)}(\gamma_{\mathbf{V}}).$$

The book's extra factor is

$$\boxed{G_{\mathrm{enh}} = C_{SU(3)}(\gamma_{\mathbf{V}})}.$$

Thus the enhancement is a **larger-centralizer effect**: the singular orbifold background imposes only a discrete gauge holonomy, which projects out fewer of the original E_8 currents than a smooth connection with full $SU(3)$ holonomy.

7.2.1.2 What the book explicitly establishes

The discussion on pages 525–526 represents the heterotic gauge degrees of freedom by a level-one current algebra. In the bosonic realization, the generator of the orbifold point group acts on the sixteen chiral gauge bosons as the shift

$$X_L^I \longmapsto X_L^I + V^I,$$

which is equation (15.7). The order-N condition is

$$N\mathbf{V} \in \Lambda_{E_8\times E_8}.$$

The important source statements are:

1. only invariant currents give gauge bosons in the orbifold theory;
2. the unbroken gauge algebra is the subalgebra commuting with the gauge embedding of the orbifold group;
3. page 528 explicitly states that gauge bosons cannot arise from twisted sectors.

The last point is essential. The twisted fields mentioned in the question are charged scalar multiplets. They break the enhanced symmetry when they acquire expectation values, but they are not the source of its vector multiplets.

7.2.1.3 Gauge bosons as invariant current-algebra states

At level one, the first E_8 current algebra contains eight Cartan currents

$$H^I(z_L) = i\partial X_L^I(z_L),$$

and, for every root $\mathbf{P}$ of E_8, a root current of the schematic form

$$J_{\mathbf{P}}(z_L) = :e^{i\mathbf{P}\cdot\mathbf{X}_L(z_L)}: \ .$$

The normalization in this schematic exponential is fixed by the gauge-lattice periodicities. In the shift-vector convention of equations (15.8)–(15.11), the gauge shift acts as

$$\begin{aligned} H^I &\longmapsto H^I, \\ J_{\mathbf{P}} &\longmapsto e^{2\pi i\,\mathbf{P}\cdot\mathbf{V}} J_{\mathbf{P}}. \end{aligned}$$

Hence:

- every Cartan current survives an Abelian shift;
- a root current survives precisely if

$$\boxed{\mathbf{P}\cdot\mathbf{V}\in\mathbb{Z}}.$$

The surviving currents close under operator products and generate the unbroken current algebra.

A four-dimensional gauge-boson state has the schematic heterotic form

$$\widetilde{\psi}^{\mu}_{-1/2}|0;k\rangle_R \otimes J^T_{-1}|0\rangle_L,$$

where T is a gauge generator. Equivalently, its vertex operator contains

$$\mathcal{V}_A \sim \epsilon_\mu\,\widetilde{\psi}^{\mu} e^{-\widetilde{\phi}} e^{ik\cdot X} J^T.$$

The right-moving factor supplies the spacetime vector polarization, while the left-moving dimension-one current supplies the gauge generator. The orbifold projection retains this vector if

$$\gamma_{\mathbf{V}} T \gamma_{\mathbf{V}}^{-1} = T.$$

Therefore the vector multiplets are labeled precisely by

$$\operatorname{Lie} C_{E_8}(\gamma_{\mathbf{V}}).$$

Nothing new has to become massless in a twisted sector. At the orbifold point, currents already present in the ten-dimensional E_8 theory are simply projected less strongly than they are by a smooth $SU(3)$ bundle background.

7.2.1.4 Smooth standard embedding: why only E_6 remains

For the smooth standard embedding one identifies the gauge connection in an $SU(3) \subset E_8$ subgroup with the Calabi-Yau spin connection:

$$A_m^{\mathrm{bg}} = \omega_m.$$

Assuming the Calabi-Yau has holonomy exactly $SU(3)$, the gauge-bundle holonomy is also the full group $SU(3)$.

Expand a possible four-dimensional vector as

$$A_\mu(x, y) = \sum_a A_\mu^a(x)\, s_a(y),$$

where s_a is a section of the adjoint bundle. A massless gauge boson must have a covariantly constant internal wavefunction:

$$D_m^{\mathrm{bg}} s_a = 0.$$

Parallel transport around every closed loop then requires

$$h\, s_a\, h^{-1} = s_a \qquad \text{for every } h \in \mathrm{Hol}(A^{\mathrm{bg}}).$$

Thus the unbroken gauge algebra is

$$\mathfrak{g}_{4\mathrm{d}} = \mathrm{Lie}\, C_{E_8}(\mathrm{Hol}(A^{\mathrm{bg}}))\,.$$

For full $SU(3)$ holonomy,

$$\boxed{\mathfrak{g}_{\mathrm{smooth}} = \mathrm{Lie}\, C_{E_8}(SU(3)) = \mathfrak{e}_6}.$$

The same conclusion follows from the familiar branching

$$\mathbf{248} = (\mathbf{8}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{78}) \oplus (\mathbf{3}, \mathbf{27}) \oplus (\bar{\mathbf{3}}, \overline{\mathbf{27}})$$

under $SU(3) \times E_6$.

Only $(\mathbf{1}, \mathbf{78})$ is invariant under the full $SU(3)$ structure group. The $\mathbf{8}$ contains no singlet under the adjoint action of the entire $SU(3)$: equivalently, the Lie algebra $\mathfrak{su}(3)$ has no nonzero element commuting with all of $\mathfrak{su}(3)$. Therefore the smooth background has E_6, not $E_6 \times SU(3)$, as its four-dimensional gauge group.

This conclusion can also be seen from the ten-dimensional Yang–Mills action. For a mode $A_\mu(x)T$ independent of the internal coordinates,

$$F_{\mu m} = -\left[A_m^{\mathrm{bg}}, A_\mu T\right].$$

The term

$$\int_X \mathrm{tr}(F_{\mu m} F^{\mu m})$$

becomes a four-dimensional mass term proportional to

$$\int_X \mathrm{tr}\left([A_m^{\mathrm{bg}}, T][A^{\mathrm{bg}\, m}, T]\right).$$

It vanishes exactly for generators commuting with the background holonomy. A generic smooth $SU(3)$ connection therefore removes the would-be $SU(3)$ gauge vectors.

7.2.1.5 Orbifold standard embedding: only a discrete element must commute

In the orbifold CFT, the standard embedding is

$$\mathbf{V} = (v_1, v_2, v_3, 0, \dots, 0), \qquad \mathbf{V}' = 0.$$

The geometric point-group generator θ is embedded as the finite gauge transformation

$$\gamma_{\mathbf{V}} = \exp(2\pi i\, \mathbf{V} \cdot \mathbf{H}) \in SU(3) \subset E_8.$$

Away from the orbifold singularities, the corresponding idealized connection is locally flat. Its nontrivial information is the finite monodromy $\gamma_{\mathbf{V}}$ around the singular locus. Thus a gauge generator need commute only with this one finite-order element, not with every element of $SU(3)$:

$$\gamma_{\mathbf{V}} T \gamma_{\mathbf{V}}^{-1} = T.$$

Using the $SU(3) \times E_6$ branching:

- all $(\mathbf{1}, \mathbf{78})$ currents survive, giving E_6;
- the invariant part of $(\mathbf{8}, \mathbf{1})$ is
$$\operatorname{Lie} C_{SU(3)}(\gamma_{\mathbf{V}}),$$
giving G_{enh};
- the hidden second E_8 is unaffected because $\mathbf{V}' = 0$;
- for the four-dimensional supersymmetric twists in Table 15.1, all v_i are nonzero modulo one, so the mixed $(\mathbf{3}, \mathbf{27})$ and $(\bar{\mathbf{3}}, \overline{\mathbf{27}})$ currents are projected out as gauge bosons.

This yields

$$\boxed{E_6 \times G_{\text{enh}} \times E_8 \subseteq C_{E_8 \times E_8}(\gamma_{\mathbf{V}})}.$$

The book says "at least" because in more general embeddings or for special shifts there can be further invariant roots. For the minimal four-dimensional standard-embedding examples with no Wilson lines, the displayed group is the relevant result.

7.2.1.6 Computing G_{enh} from the three eigenvalues

Inside the $SU(3)$ used in the standard embedding, the shift element may be written in the defining representation as

$$\gamma_{\mathbf{V}} = \operatorname{diag}\left(e^{2\pi i v_1}, e^{2\pi i v_2}, e^{2\pi i v_3}\right),$$

with

$$v_1 + v_2 + v_3 = 0 \pmod 1.$$

The off-diagonal $SU(3)$ generator E_{ij} transforms as

$$\gamma_{\mathbf{V}} E_{ij} \gamma_{\mathbf{V}}^{-1} = e^{2\pi i (v_i - v_j)} E_{ij}.$$

Therefore

$$E_{ij} \text{ survives} \quad \Longleftrightarrow \quad v_i - v_j \in \mathbb{Z}.$$

The two Cartan generators always survive. The answer is then determined by the degeneracy pattern of the three eigenvalues:

eigenvalues of $\gamma_{\mathbf{V}}$	G_{enh}
$\lambda_1 = \lambda_2 = \lambda_3$	$SU(3)$
exactly two λ_i are equal	$SU(2) \times U(1)$
$\lambda_1, \lambda_2, \lambda_3$ all distinct	$U(1) \times U(1)$

This is the origin of the three possibilities listed on page 527.

7.2.1.7 Explicit $\mathbb{Z}_3$ example: the eight $SU(3)$ gauge bosons

For the $T^6/\mathbb{Z}_3$ model discussed later in the chapter,

$$\mathbf{v} = \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right), \qquad \mathbf{V} = \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}, 0, \dots, 0\right).$$

Modulo integers, all three phases are equal:

$$e^{2\pi i v_1} = e^{2\pi i v_2} = e^{2\pi i v_3} = \omega.$$

Thus

$$\gamma_{\mathbf{V}} = \omega\, \mathbf{1}_3.$$

This is a central element of $SU(3)$, so it commutes with the entire $SU(3)$:

$$C_{SU(3)}(\gamma_{\mathbf{V}}) = SU(3).$$

In current-algebra language, take the six $SU(3)$ roots

$$\mathbf{P}_{ij} = \mathbf{e}_i - \mathbf{e}_j, \qquad i \neq j, \qquad i, j \in \{1, 2, 3\}.$$

Their shift phases are controlled by

$$\mathbf{P}_{ij} \cdot \mathbf{V} = v_i - v_j.$$

Explicitly,

$$v_1 - v_2 = 0, \qquad v_1 - v_3 = 1, \qquad v_2 - v_3 = 1,$$

and the negatives are also integers. Hence all six root currents survive. Together with the two Cartan currents, they give

$$6 + 2 = 8$$

massless vector multiplets forming the adjoint $\mathbf{8}$ of $SU(3)$.

The first E_8 gauge group at this orbifold point is therefore

$$\boxed{E_6 \times SU(3)},$$

and including the unaffected hidden factor gives

$$\boxed{E_6 \times SU(3) \times E_8}.$$

These are precisely the untwisted gauge multiplets listed in equation (15.24).

Notice that the central element $\omega \mathbf{1}_3$ acts trivially on the adjoint $\mathbf{8}$ but nontrivially on the fundamental $\mathbf{3}$:

$$\mathbf{8} \longmapsto \mathbf{8}, \qquad \mathbf{3} \longmapsto \omega\, \mathbf{3}.$$

This is why the $SU(3)$ gauge currents survive while the mixed $(\mathbf{3}, \mathbf{27})$ currents do not become additional gauge bosons.

7.2.1.8 Explicit $\mathbb{Z}_4$ example: $SU(2) \times U(1)$

For

$$\mathbf{v} = \left(\frac{1}{4}, \frac{1}{4}, -\frac{1}{2}\right),$$

the three eigenvalues are

$$(i, i, -1).$$

Only the first two coincide. Equivalently,

$$v_1 - v_2 = 0, \qquad v_1 - v_3 = \frac{3}{4}, \qquad v_2 - v_3 = \frac{3}{4}.$$

Therefore the root currents $J_{\mathbf{e}_1 - \mathbf{e}_2}$ and $J_{\mathbf{e}_2 - \mathbf{e}_1}$ survive, while those connecting the third direction to the first two are projected out. The two Cartan currents remain.

The surviving algebra consists of:

- one $SU(2)$ Cartan generator and its two root generators;
- one additional commuting $U(1)$ Cartan generator.

Thus

$$\boxed{G_{\text{enh}} = SU(2) \times U(1)}$$

up to a finite global quotient.

If all three phases are distinct, none of the six off-diagonal $SU(3)$ currents survives, but the two Cartan currents do. This gives

$$\boxed{G_{\text{enh}} = U(1)^2}.$$

7.2.1.9 Why resolving the orbifold Higgses the extra group

The twisted sectors contain massless scalar fields localized at the orbifold fixed loci. Under the gauge shift, these states generally carry nonzero weights of G_{enh}. The book therefore identifies the blow-up modes with charged chiral multiplets.

Let b^A denote such fields. Their kinetic terms contain

$$\sum_A \left|D_\mu b^A\right|^2, \qquad D_\mu b^A = \partial_\mu b^A - igA^a_\mu (T_a b)^A.$$

If the resolution requires

$$\langle b^A \rangle \neq 0,$$

then

$$\sum_A |D_\mu b^A|^2 \supset g^2 A^a_\mu A^{\mu b} \sum_A (T_a\langle b\rangle)^A (T_b\langle b\rangle)^{A*}.$$

The gauge bosons for generators satisfying

$$T_a\langle b\rangle \neq 0$$

become massive. The unbroken group is the stabilizer of all blow-up expectation values.

In the $T^6/\mathbb{Z}_3$ example, equation (15.24) exhibits twisted fields charged as **3** under the enhanced $SU(3)$. Moving along suitable D-flat blow-up directions breaks this $SU(3)$ completely. As the book explains after equation (15.24), eight chiral-multiplet directions combine with the eight $SU(3)$ vector multiplets to form massive vector multiplets.

Geometrically, the singular orbifold with finite gauge monodromy is replaced by a smooth Calabi-Yau together with a smooth gauge connection whose holonomy fills out $SU(3)$. In the worldsheet description this is a deformation by twisted operators; in the four-dimensional description it is a Higgs effect. These are two descriptions of the same transition:

$$\boxed{\begin{array}{c}\text{orbifold point: discrete } \langle\gamma_{\mathbf{V}}\rangle \\ E_6\times G_{\text{enh}}\times E_8\end{array}} \\ \Big\Downarrow_{\langle b_{\text{blow-up}}\rangle\neq 0} \\ \boxed{\begin{array}{c}\text{smooth standard embedding: full } SU(3) \text{ holonomy} \\ E_6\times E_8\end{array}}\,.$$

The resolved Kähler moduli are neutral under the final $E_6 \times E_8$. There is no contradiction with their being charged under G_{enh} at the orbifold point: the smooth moduli are gauge-invariant coordinates on the D-flat Higgs quotient, not the individual charged fields used as coordinates at the enhanced-symmetry point.

7.2.1.10 Checks

The rank count is consistent. An Abelian shift lies in a Cartan torus and leaves all Cartan currents invariant. Hence the first E_8 retains rank eight:

$$\text{rank}(E_6) + \text{rank}(G_{\text{enh}}) = 6 + 2 = 8.$$

The smooth standard embedding has only rank-six E_6 because the continuous $SU(3)$ background does not leave its own Cartan generators covariantly constant.

For the $\mathbb{Z}_3$ example, the dimensions also agree:

$$\dim(E_6 \times SU(3)) = 78 + 8 = 86.$$

At the level of roots, this consists of 72 E_6 roots, six $SU(3)$ roots, and eight Cartan generators:

$$72 + 6 + 8 = 86.$$

Finally, the location of the states agrees with page 528:

$$\boxed{\text{enhanced gauge bosons: untwisted sector,} \qquad \text{charged blow-up fields: twisted sectors}}\,.$$

7.2.1.11 Result

For a heterotic orbifold gauge shift $\mathbf{V}$, the root current $J_{\mathbf{P}}$ survives when

$$\boxed{\mathbf{P}\cdot\mathbf{V}\in\mathbb{Z}},$$

so the orbifold gauge algebra is

$$\boxed{\mathfrak{g}_{\mathrm{orb}}=\mathrm{Lie}\,C_{E_8\times E_8}(\gamma_{\mathbf{V}})}.$$

For the standard embedding in the first E_8,

$$\mathbf{V}=(v_1,v_2,v_3,0,\dots,0),$$

this contains

$$\boxed{\mathfrak{e}_6\oplus\mathrm{Lie}\,C_{SU(3)}(\gamma_{\mathbf{V}})\oplus\mathfrak{e}_8}.$$

The extra factor is determined by

$$\boxed{G_{\mathrm{enh}}=\begin{cases}SU(3), & v_1=v_2=v_3 \pmod 1,\\ SU(2)\times U(1), & \text{exactly two } v_i \text{ are equal mod 1},\\ U(1)^2, & \text{all three } v_i \text{ are distinct mod 1.}\end{cases}}$$

The smooth standard embedding instead requires commutation with the full gauge-bundle holonomy:

$$\boxed{\mathfrak{g}_{\mathrm{smooth}}=\mathrm{Lie}\,C_{E_8}(SU(3))=\mathfrak{e}_6}.$$

Therefore the symmetry is enhanced at the orbifold point because

$$\boxed{\langle\gamma_{\mathbf{V}}\rangle\subset SU(3)\quad\Longrightarrow\quad C_{E_8}(SU(3))\subseteq C_{E_8}(\gamma_{\mathbf{V}})}.$$

The extra vectors are untwisted E_8 current-algebra states. Charged twisted blow-up fields acquire expectation values upon resolution and Higgs G_{enh}, returning the gauge group to $E_6\times E_8$.

7.3 Q3: T-duality orbit of the toroidal Type-II orientifolds

In page 536, it is said that one can check that the orientifolds in Table 10.3 on page 317 are related by T-dualities. For instance one can verifies that $T_2\Omega T_2^{-1}=\Omega I_2(-1)^{F_L}$.

Verify the T-duality relations in Table 10.3.

> **Textbook location and related material.** Chapter 15, Section 15.3 around p. 536, together with Table 10.3 on p. 317 and Section 10.6: the T-duality orbit of type-II orientifolds and the actions I_n and $(-1)^{F_L}$.

7.3.1 Answer

7.3.1.1 Core point

Start with the type-I orientifold

$$\frac{\text{type IIB}}{\Omega}$$

on T^6. Let $\mathcal{T}_A$ denote T-duality along a set A of n torus directions, and let I_A reflect those same target-space coordinates. With the spin-field phases used by the book, the complete operator relation on the physical closed-string Hilbert space is

$$\boxed{\mathcal{T}_A\Omega\mathcal{T}_A^{-1}=\Omega I_A\left[(-1)^{F_L}\right]^{\frac{n(n-1)}{2}}},\qquad n=|A|.$$

Only the parity of $\frac{n(n-1)}{2}$ matters. Equivalently,

$$\mathcal{T}_n \Omega \mathcal{T}_n^{-1} = \begin{cases} \Omega I_n, & n = 0, 1 \pmod 4, \\ \Omega I_n (-1)^{F_L}, & n = 2, 3 \pmod 4. \end{cases}$$

For $n = 0, 1, 2, 3, 4$, this gives exactly the five orientifold actions in Table 10.3:

n	type-II theory	$\mathcal{T}_n \Omega \mathcal{T}_n^{-1}$	O-plane
0	IIB	Ω	$O9$
1	IIA	ΩI_1	$O8$
2	IIB	$\Omega I_2 (-1)^{F_L}$	$O7$
3	IIA	$\Omega I_3 (-1)^{F_L}$	$O6$
4	IIB	ΩI_4	$O5$

There are three ingredients to verify:

1. the bosonic worldsheet fields give ΩI_n;
2. the lift of I_n to Ramond spinors produces the additional $(-1)^{F_L}$ for $n = 2, 3 \pmod 4$;
3. the fixed loci and the T-dual D-brane distribution reproduce the O-plane and Chan–Paton columns.

7.3.1.2 Bosonic conjugation: why a target-space reflection appears

For one dualized coordinate X^a, decompose

$$X^a(\tau, \sigma) = X_L^a(\tau + \sigma) + X_R^a(\tau - \sigma).$$

In the convention used around equation (10.211), T-duality is the asymmetric reflection

$$\mathcal{T}_a : \qquad X_L^a \longmapsto X_L^a, \qquad X_R^a \longmapsto -X_R^a.$$

Worldsheet parity exchanges the chiral halves:

$$\Omega : \qquad X_L^a \longleftrightarrow X_R^a.$$

Now conjugate Ω by $\mathcal{T}_a$. Acting on X_L^a gives

$$X_L^a \xrightarrow{\mathcal{T}_a^{-1}} X_L^a \xrightarrow{\Omega} X_R^a \xrightarrow{\mathcal{T}_a} -X_R^a,$$

while acting on X_R^a gives

$$X_R^a \xrightarrow{\mathcal{T}_a^{-1}} -X_R^a \xrightarrow{\Omega} -X_L^a \xrightarrow{\mathcal{T}_a} -X_L^a.$$

Therefore

$$\mathcal{T}_a \Omega \mathcal{T}_a^{-1} : \qquad (X_L^a, X_R^a) \longmapsto (-X_R^a, -X_L^a).$$

This is precisely worldsheet parity followed by the spacetime reflection

$$I_a : \qquad (X_L^a, X_R^a) \longmapsto (-X_L^a, -X_R^a).$$

Hence, on bosons,

$$\boxed{\mathcal{T}_a \Omega \mathcal{T}_a^{-1} = \Omega I_a},$$

which is equation (10.211).

The calculation factorizes over different coordinates. For a set A of n dualized coordinates,

$$\boxed{\mathcal{T}_A \Omega \mathcal{T}_A^{-1} = \Omega I_A \qquad \text{on the bosonic and NS sectors}}.$$

The worldsheet fermions in an NS sector transform in the vector representation and obey the same calculation. The subtlety is entirely in the Ramond ground states, because reflections act projectively on spinors.

7.3.1.3 Spinor lift of a reflection

Page 535 derives the action of a target-space reflection on spin fields. If the directions in A are reflected, then

$$I_A : \qquad S_\alpha \longmapsto (P_A)_\alpha{}^\beta S_\beta,$$

where equation (15.26) writes the lift in terms of gamma matrices. Its overall phase is fixed by the Majorana condition. It is useful to express the same lift as

$$P_A = \rho_{a_1}\rho_{a_2}\cdots\rho_{a_n},$$

where each ρ_a is the properly phased lift of a single-coordinate reflection.

The phases can be chosen, as in the book's light-cone discussion, so that

$$\rho_a^2 = 1.$$

Distinct reflection matrices anticommute:

$$\rho_a\rho_b = -\rho_b\rho_a, \qquad a \neq b.$$

The anticommutation is the spinorial statement that orthogonal reflections are represented by Clifford generators.

Squaring P_A requires moving the second copy of every ρ_a through the matrices to its left. The number of pairwise interchanges is

$$(n-1) + (n-2) + \cdots + 1 = \frac{n(n-1)}{2}.$$

Therefore

$$\boxed{P_A^2 = (-1)^{\frac{n(n-1)}{2}}}.$$

In particular,

n	0	1	2	3	4
P_A^2	$+1$	$+1$	-1	-1	$+1$

This reproduces the book's statement that reflection in two directions is a rotation by π and squares to -1 on a spinor.

Since $P_A^2 = \pm 1$,

$$P_A^{-1} = (-1)^{\frac{n(n-1)}{2}} P_A.$$

This inverse is the origin of the extra fermion-number operator in the conjugation of Ω.

7.3.1.4 Conjugating Ω on Ramond states

Equation (15.27) says that T-duality acts by P_A on the Ramond spin field of one chiral sector and trivially on the other. Worldsheet parity exchanges the two chiral sectors. Consequently, in

$$\mathcal{T}_A \Omega \mathcal{T}_A^{-1},$$

one of the two spin-field transformations appears as P_A^{-1}, whereas the geometric operation ΩI_A contains P_A.

Using

$$P_A^{-1} = (-1)^{\frac{n(n-1)}{2}} P_A,$$

the two actions differ by

$$(-1)^{\frac{n(n-1)}{2}}$$

on the affected Ramond factor and agree on its NS factor. Tracking the exchange of the two chiral sectors with the book's convention for Ω, and writing the final operator in the ordering used in Table 10.3, assigns this sign to the left-moving Ramond number. It is therefore precisely

$$\left[(-1)^{F_L}\right]^{\frac{n(n-1)}{2}}.$$

The label can be checked independently from

$$\Omega(-1)^{F_L} = (-1)^{F_R}\Omega,$$

which is the identity used by the book on page 318. Thus moving a fermion-number operator through Ω exchanges F_L and F_R; its placement and label must be kept together.

Thus the full operator relation is

$$\boxed{\mathcal{T}_A \Omega \mathcal{T}_A^{-1} = \Omega I_A \left[(-1)^{F_L}\right]^{\frac{n(n-1)}{2}}}.$$

The apparently innocuous bosonic identity $\mathcal{T}_A \Omega \mathcal{T}_A^{-1} = \Omega I_A$ therefore has to be corrected whenever the spin lift satisfies $P_A^2 = -1$. This occurs for $n = 2, 3 \pmod 4$.

7.3.1.5 Independent involution check

The same factor can be derived without tracking the detailed order of the spin matrices. On the complete closed-string Hilbert space,

$$I_A^2 = \left[(-1)^{F_L+F_R}\right]^{\frac{n(n-1)}{2}}.$$

Indeed, I_A^2 is P_A^2 on each Ramond factor and is $+1$ on an NS factor.

Worldsheet parity exchanges left and right fermion numbers:

$$\Omega(-1)^{F_L}\Omega^{-1} = (-1)^{F_R}.$$

Let

$$\mathcal{O}_n = \Omega I_n \left[(-1)^{F_L}\right]^{a_n}, \qquad a_n = \frac{n(n-1)}{2} \pmod 2.$$

Because Ω commutes with the symmetric target reflection I_n,

$$\begin{aligned}
\mathcal{O}_n^2 &= (\Omega I_n)^2 \left[(-1)^{F_L+F_R}\right]^{a_n} \\
&= I_n^2 \left[(-1)^{F_L+F_R}\right]^{a_n} \\
&= \left[(-1)^{F_L+F_R}\right]^{2a_n} \\
&= 1.
\end{aligned}$$

Thus every orientifold action in the table is a genuine involution on physical states.

For $n = 2$, this reduces exactly to the argument on page 318:

$$I_2^2 = (-1)^{F_L+F_R},$$

so

$$(\Omega I_2)^2 = (-1)^{F_L+F_R} \neq 1$$

on spacetime fermions. Inserting $(-1)^{F_L}$ gives

$$\boxed{\left(\Omega I_2 (-1)^{F_L}\right)^2 = 1}.$$

The $n = 3$ case has the same sign because $P_3^2 = -1$. For $n = 1$ and $n = 4$, $P_n^2 = +1$, so no fermion-number insertion is required.

7.3.1.6 Type IIA versus type IIB

A single T-duality reverses the chirality of the Ramond ground state in the chiral sector on which it acts. Therefore:

$$\begin{array}{r|l}
n \text{ even} & \text{type IIB} \longleftrightarrow \text{type IIB}, \\
n \text{ odd} & \text{type IIB} \longleftrightarrow \text{type IIA}.
\end{array}$$

This is equation (10.25) and the observation following equation (15.27). It immediately explains the distribution of the rows between the type-IIB and type-IIA blocks of Table 10.3.

7.3.1.7 Verification row by row

Choose, for definiteness, consecutive internal coordinates X^9, X^8, X^7, X^6. The cumulative T-duality chain is

$$\begin{aligned}\frac{IIB}{\Omega} &\overset{\mathcal{T}_9}{\longleftrightarrow} \frac{IIA}{\Omega I_9}\\ &\overset{\mathcal{T}_8}{\longleftrightarrow} \frac{IIB}{\Omega I_{89}(-1)^{F_L}}\\ &\overset{\mathcal{T}_7}{\longleftrightarrow} \frac{IIA}{\Omega I_{789}(-1)^{F_L}}\\ &\overset{\mathcal{T}_6}{\longleftrightarrow} \frac{IIB}{\Omega I_{6789}}.\end{aligned}$$

Here each displayed quotient is understood as the theory obtained from the original type-IIB orientifold after the cumulative set of T-dualities. Relabeling $I_9, I_{89}, I_{789}, I_{6789}$ as I_1, I_2, I_3, I_4 gives the notation in Table 10.3.

7.3.1.7.1 Row 1: type IIB with Ω For $n = 0$,

$$\frac{n(n-1)}{2} = 0.$$

There is no geometric reflection and no fermion-number factor:

$$\mathcal{O}_0 = \Omega.$$

The fixed locus fills all ten spacetime dimensions, so it is an O9-plane. This is the type-I theory.

7.3.1.7.2 Row 2: type IIA with ΩI_1 For $n = 1$,

$$\frac{n(n-1)}{2} = 0, \qquad P_1^2 = +1.$$

Thus

$$\boxed{\mathcal{T}_1 \Omega \mathcal{T}_1^{-1} = \Omega I_1}.$$

The odd number of T-dualities converts IIB to IIA. The reflection of one circle has two fixed points, so the O9-plane becomes two O8-planes.

In light-cone gauge, the book chooses the spin lift of the single reflection with the Majorana-compatible phase, for example

$$I_1 = i\Gamma^9\Gamma$$

on spinors. With this phase,

$$(\Omega I_1)^2 = 1,$$

as stated on page 536.

7.3.1.7.3 Row 3: type IIB with $\Omega I_2(-1)^{F_L}$ For $n = 2$,

$$\frac{n(n-1)}{2} = 1, \qquad P_2^2 = -1.$$

Therefore

$$\boxed{\mathcal{T}_2 \Omega \mathcal{T}_2^{-1} = \Omega I_2(-1)^{F_L}}.$$

For example, with the directions $8, 9$,

$$\boxed{\mathcal{T}_{89} \Omega \mathcal{T}_{89}^{-1} = \Omega I_{89}(-1)^{F_L}},$$

which is the relation quoted on pages 318 and 536.

The even number of dualities leaves the theory type IIB. Reflection of two circles has

$$2^2 = 4$$

fixed loci, so there are four O7-planes.

7.3.1.7.4 Row 4: type IIA with $\Omega I_3(-1)^{F_L}$ For $n = 3$,

$$\frac{n(n-1)}{2} = 3, \qquad P_3^2 = -1.$$

Hence

$$\boxed{\mathcal{T}_3 \Omega \mathcal{T}_3^{-1} = \Omega I_3 (-1)^{F_L}}.$$

Because n is odd, the dual theory is type IIA. The reflection has

$$2^3 = 8$$

fixed loci, giving eight O6-planes.

7.3.1.7.5 Row 5: type IIB with ΩI_4 For $n = 4$,

$$\frac{n(n-1)}{2} = 6, \qquad P_4^2 = +1.$$

The fermion-number factor occurs to an even power and drops out:

$$\boxed{\mathcal{T}_4 \Omega \mathcal{T}_4^{-1} = \Omega I_4}.$$

Four T-dualities return to type IIB. The reflection has

$$2^4 = 16$$

fixed loci, so there are sixteen O5-planes.

7.3.1.8 O-plane dimensions and multiplicities

T-duality along a direction parallel to a Dp-brane or Op-plane lowers its dimension:

$$D_p \longmapsto D_{p-1}, \qquad O_p \longmapsto O_{p-1}.$$

Starting from the type-I O9-plane and dualizing n internal directions therefore gives

$$p = 9 - n.$$

On each reflected circle, the equation

$$x^a = -x^a \pmod{2\pi R_a}$$

has the two solutions

$$x^a = 0, \qquad x^a = \pi R_a.$$

Thus I_n has

$$\boxed{2^n}$$

fixed components, each supporting an O_{9-n}-plane. This gives

n	0	1	2	3	4
number of O-planes	1	2	4	8	16
type	$O9$	$O8$	$O7$	$O6$	$O5$

and reproduces the third column of Table 10.3.

7.3.1.9 Chan–Paton groups and local tadpole cancellation

T-duality preserves the total number of D-branes in the upstairs count: the 32 type-I D9-branes become 32 D$(9-n)$-branes. Equation (10.212) gives the charge of a standard orientifold plane:

$$\mu_{O_p} = -2^{p-4}\mu_{D_p}.$$

For $p = 9-n$,

$$\mu_{O_{9-n}} = -2^{5-n}\mu_{D_{9-n}}.$$

Therefore local RR-charge cancellation at each fixed plane requires

$$\boxed{N_{\text{local}} = 2^{5-n} = \frac{32}{2^n}}$$

coincident D$(9-n)$-branes on each O$(9-n)$-plane.

The T-duals of the type-I $O9^-$ plane impose the orthogonal Chan–Paton projection. A stack of N_{local} branes on each such plane gives $SO(N_{\text{local}})$. Since there are 2^n fixed planes,

$$\boxed{G_{\text{CP}} = SO(2^{5-n})^{2^n}}.$$

Evaluating this formula gives

n	O-planes	D-branes per O-plane	G_{CP}
0	$1 \times O9$	32	$SO(32)$
1	$2 \times O8$	16	$SO(16)^2$
2	$4 \times O7$	8	$SO(8)^4$
3	$8 \times O6$	4	$SO(4)^8$
4	$16 \times O5$	2	$SO(2)^{16}$

This is the final column of Table 10.3.

As a charge check,

$$2^n\mu_{O_{9-n}} = 2^n\left(-2^{5-n}\mu_{D_{9-n}}\right) = -32\mu_{D_{9-n}},$$

which is canceled by the 32 T-dual D-branes for every n.

7.3.1.10 Check on the massless closed-string sectors

The conjugation can also be seen directly on massless states.

An NS–NS state has the schematic form

$$\psi^M_{-1/2}\widetilde{\psi}^N_{-1/2}|0;k\rangle_{\text{NS-NS}}.$$

Worldsheet parity exchanges M and N. Conjugation by $\mathcal{T}_A$ supplies a minus sign whenever an index lies in a dualized direction, so the polarization transforms as

$$E_{MN} \longmapsto (R_A)_M{}^P(R_A)_N{}^Q E_{QP},$$

which is exactly ΩI_A. Since $(-1)^{F_L} = +1$ in the left NS sector, the extra factor is invisible here.

For R–R and mixed R–NS/NS–R states, $\mathcal{T}_A$ acts on one spin field by P_A. Exchanging the two chiral sectors with Ω reverses the order of the spin lift and produces P_A^{-1}. The relation

$$P_A^{-1} = (-1)^{\frac{n(n-1)}{2}}P_A$$

then gives precisely the factor recorded by $(-1)^{F_L}$. This ensures that the transformed orientifold action has invariant fermions and preserves the same sixteen supercharges as type I.

7.3.1.11 Checks

The alternation between IIA and IIB is correct:

$$n = 0, 2, 4 \quad \Longrightarrow \quad IIB, \qquad n = 1, 3 \quad \Longrightarrow \quad IIA.$$

The fermion-number insertion has the independent pattern

$$\frac{n(n-1)}{2} = 0, 0, 1, 3, 6 \quad \text{for } n = 0, 1, 2, 3, 4.$$

Thus it occurs for $n = 2, 3$, not simply for odd or even n. These are exactly the two rows in Table 10.3 containing $(-1)^{F_L}$.

The O-plane dimension and number obey

$$p = 9 - n, \qquad N_{O_p} = 2^n.$$

The total O-plane charge is independent of n:

$$N_{O_p}\mu_{O_p} = -32\mu_{D_p},$$

as required by T-duality.

7.3.1.12 Result

For T-duality along a set A of n compact coordinates,

$$\boxed{\mathcal{T}_A : \quad X_L^a \mapsto X_L^a, \qquad X_R^a \mapsto -X_R^a},$$

and therefore the bosonic conjugation is

$$\boxed{\mathcal{T}_A \Omega \mathcal{T}_A^{-1} = \Omega I_A \quad \text{on NS states.}}$$

The spin lift satisfies

$$\boxed{P_A^2 = (-1)^{\frac{n(n-1)}{2}}},$$

so the complete orientifold action is

$$\boxed{\mathcal{T}_A \Omega \mathcal{T}_A^{-1} = \Omega I_A \left[(-1)^{F_L}\right]^{\frac{n(n-1)}{2}}}.$$

Consequently,

n	theory	orientifold	O-planes	local CP group
0	IIB	Ω	$O9$	$SO(32)$
1	IIA	ΩI_1	$2\,O8$	$SO(16)^2$
2	IIB	$\Omega I_2(-1)^{F_L}$	$4\,O7$	$SO(8)^4$
3	IIA	$\Omega I_3(-1)^{F_L}$	$8\,O6$	$SO(4)^8$
4	IIB	ΩI_4	$16\,O5$	$SO(2)^{16}$

.

This verifies every T-duality relation and every associated entry in Table 10.3.

7.4 Q4: Closed-string spectrum of the $T^4/\mathbb{Z}_2$ orientifold

In page 536-537, they obtained the massless closed string states for this $T^4/\mathbb{Z}_2$ orientifold construction. The detailed computation is omitted. Following their method, compute the resulting $SU(2) \times SU(2)$ representations.

> **Textbook location and related material.** Chapter 15, Section 15.3 (pp. 536–537) and Table 15.3: the $T^4/\mathbb{Z}_2$ chiral building blocks, orbifold/orientifold projections, and assembly into six-dimensional multiplets.

7.4.1 Answer

7.4.1.1 Core point

The computation consists of two successive projections.

First construct the type-IIB $T^4/\mathbb{Z}_2$ orbifold spectrum by tensoring a left-moving state and a right-moving state from Table 15.3 and retaining only equal I_4 parities:

$$q_{I_4}^{(L)} q_{I_4}^{(R)} = +1.$$

Then project the surviving closed-string states with

$$\mathcal{O} = \Omega R(-1)^{F_L},$$

where

$$R: \qquad y^1 \mapsto -y^1, \qquad y^2 \mapsto -y^2.$$

In the twisted sectors, the model additionally assigns the twist field the sign

$$\epsilon_{\text{tw}} = -1.$$

Carrying out these steps gives

	sector	$SU(2)_L \times SU(2)_R$ content
untwisted	NS–NS	$(\mathbf{3},\mathbf{3}) + 11(\mathbf{1},\mathbf{1})$
	R–R	$(\mathbf{3},\mathbf{1}) + (\mathbf{1},\mathbf{3}) + 6(\mathbf{1},\mathbf{1})$
	NS–$R + R$–NS	$2(\mathbf{2},\mathbf{3}) + 10(\mathbf{2},\mathbf{1})$
twisted	NS–NS	$48(\mathbf{1},\mathbf{1})$
	R–R	$16(\mathbf{1},\mathbf{1})$
	NS–$R + R$–NS	$32(\mathbf{2},\mathbf{1})$

.

The nontrivial multiplicities follow from how R acts on the internal degeneracy spaces and from the fact that Ω exchanges the two chiral factors. We now derive them one by one.

7.4.1.2 Chiral building blocks from Table 15.3

For one chiral half of the untwisted NS sector, Table 15.3 gives

$$V = (\mathbf{2},\mathbf{2})_+,$$
$$Y = 4(\mathbf{1},\mathbf{1})_-,$$

where the subscript is the I_4 eigenvalue.

The block V comes from an oscillator in one of the four transverse noncompact directions. The block Y comes from the four internal oscillators, one for each real coordinate

$$x^1,\ y^1,\ x^2,\ y^2.$$

Under R,

internal oscillator	x^1	y^1	x^2	y^2
R eigenvalue	$+1$	-1	$+1$	-1

so Y has two R-even and two R-odd basis states.

For one chiral half of the untwisted Ramond sector, define

$$A = 2(\mathbf{2},\mathbf{1})_-,$$
$$B = 2(\mathbf{1},\mathbf{2})_+.$$

The factor 2 in each block is an internal two-state degeneracy.

Equation (15.29) gives

$$R: \quad |s_1, s_2; s_3, s_4\rangle \longmapsto (-1)^{\frac{1}{2}+s_4}|s_1, s_2; -s_3, -s_4\rangle.$$

For either A or B, R exchanges the two internally degenerate states with a relative minus sign. In a suitable basis,

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J^2 = -\mathbf{1}_2.$$

The precise sign of J can be changed by rephasing the basis; $J^2 = -1$ is invariant and is what enters the projection.

At each of the 16 I_4 fixed points, the twisted chiral blocks are

$$C = 2(\mathbf{1},\mathbf{1})_+ \qquad \text{in the NS sector,}$$
$$D = (\mathbf{2},\mathbf{1})_+ \qquad \text{in the Ramond sector.}$$

The reflection acts on the two-dimensional degeneracy of C by another matrix J with $J^2 = -1$. The internal part of D is nondegenerate. Its geometric R phase and the phase assigned to the twist field enter the orientifold action only through their product; this model fixes that product by choosing the additional twisted-sector sign $\epsilon_{\mathrm{tw}} = -1$ on page 536.

7.4.1.3 Useful $SU(2)$ tensor products

The only nontrivial little-group identity needed is

$$\mathbf{2} \otimes \mathbf{2} = \mathrm{Sym}^2\, \mathbf{2} \oplus \bigwedge^2 \mathbf{2} = \mathbf{3} \oplus \mathbf{1}.$$

Consequently,

$$(\mathbf{2},\mathbf{2}) \otimes (\mathbf{2},\mathbf{2}) = (\mathbf{3},\mathbf{3}) \oplus (\mathbf{3},\mathbf{1}) \oplus (\mathbf{1},\mathbf{3}) \oplus (\mathbf{1},\mathbf{1}).$$

The symmetric and antisymmetric squares are

$$\mathrm{Sym}^2(\mathbf{2},\mathbf{2}) = (\mathbf{3},\mathbf{3}) \oplus (\mathbf{1},\mathbf{1}),$$
$$\bigwedge^2(\mathbf{2},\mathbf{2}) = (\mathbf{3},\mathbf{1}) \oplus (\mathbf{1},\mathbf{3}).$$

We will also repeatedly use the exchange operator

$$\tau(u \otimes v) = v \otimes u.$$

For a finite-dimensional space W and an operator M on W,

$$\mathrm{Tr}_{W\otimes W}\left[\tau(M \otimes M)\right] = \mathrm{Tr}_W(M^2).$$

If $\tau(M \otimes M)$ squares to one, the dimensions of its two eigenspaces are therefore

$$d_\pm = \frac{1}{2}\left[(\dim W)^2 \pm \mathrm{Tr}(M^2)\right].$$

7.4.1.4 Signs in the orientifold projection

The book uses the standard action of Ω:

- in NS–NS, Ω symmetrizes the two chiral factors;
- in R–R, Ω antisymmetrizes them.

Meanwhile,

$$(-1)^{F_L} = \begin{cases} +1, & \text{left NS}, \\ -1, & \text{left R}. \end{cases}$$

Therefore, in the untwisted diagonal sectors, the two minus signs in R–R cancel:

sector	$\mathcal{O}$ on identical chiral blocks
$NS\text{–}NS$	$+\tau(R\otimes R)$
$R\text{–}R$	$(-\tau)(R\otimes R)(-1) = +\tau(R\otimes R)$

Thus both untwisted diagonal sectors require the $+1$ eigenspace of

$$K_R = \tau(R\otimes R).$$

In a twisted sector the chosen extra sign reverses the projection:

$$\mathcal{O}_{\text{tw}} = -K_R$$

in both NS–NS and R–R. Hence an $\mathcal{O}_{\text{tw}}$-even state is a K_R-odd state.

In the mixed NS–R and R–NS sectors, Ω exchanges two different sectors. It pairs every state in NS–R with one state in R–NS, and precisely one linear combination survives. Therefore the orientifold projection divides the doubled mixed-sector count by two but does not otherwise change its little-group representation content.

7.4.1.5 Untwisted NS–NS sector

The tensor product of the two chiral NS spectra is

$$(V\oplus Y)_L \otimes (V\oplus Y)_R.$$

The I_4 projection retains equal parities:

$$\boxed{\mathcal{H}^{NS\text{–}NS}_{\text{untw}} = (V\otimes V)\oplus(Y\otimes Y)}.$$

The cross terms $V\otimes Y$ and $Y\otimes V$ have total I_4 eigenvalue -1 and are removed.

7.4.1.5.1 The $V\otimes V$ contribution The reflection R acts trivially on V, because V consists of noncompact oscillators. Hence the orientifold projection is ordinary symmetrization:

$$(V\otimes V)_{\mathcal{O}=+1} = \operatorname{Sym}^2 V.$$

Using the tensor-product identity above,

$$\boxed{\operatorname{Sym}^2(\mathbf{2},\mathbf{2}) = (\mathbf{3},\mathbf{3})\oplus(\mathbf{1},\mathbf{1})}.$$

7.4.1.5.2 The $Y\otimes Y$ contribution The space Y has dimension four and $R_Y^2 = 1$. The orientifold operator on $Y\otimes Y$ is

$$K_Y = \tau(R_Y\otimes R_Y).$$

Its trace is

$$\operatorname{Tr} K_Y = \operatorname{Tr}(R_Y^2) = 4.$$

Therefore its invariant subspace has dimension

$$\dim(Y \otimes Y)_{K_Y=+1} = \frac{1}{2}(4^2 + 4) = 10.$$

All these states are little-group singlets:

$$(Y \otimes Y)_{\mathcal{O}=+1} = 10(\mathbf{1}, \mathbf{1}).$$

One can display the ten states explicitly. If e_i is an R eigenbasis with eigenvalues $r_i = \pm 1$, the invariant tensors are

$$e_i \otimes e_i, \qquad e_i \otimes e_j + r_i r_j \, e_j \otimes e_i, \qquad i < j.$$

There are four diagonal tensors and six off-diagonal combinations.

Combining $V \otimes V$ and $Y \otimes Y$ gives

$$\boxed{\mathcal{H}^{NS\text{–}NS}_{\text{untw}} = (\mathbf{3}, \mathbf{3}) + 11(\mathbf{1}, \mathbf{1})}.$$

7.4.1.6 Untwisted R–R sector

The chiral R spectrum is $A \oplus B$, with opposite I_4 parities. Thus

$$\boxed{\mathcal{H}^{R\text{–}R}_{\text{untw}} = (A \otimes A) \oplus (B \otimes B)}.$$

The cross terms $A \otimes B$ and $B \otimes A$ are orbifold odd.

Consider $A \otimes A$. Write

$$A = (\mathbf{2}, \mathbf{1}) \otimes U_A, \qquad \dim U_A = 2,$$

where R acts as J on U_A. On $U_A \otimes U_A$, define

$$K_U = \tau_U (J \otimes J).$$

Since $J^2 = -1$,

$$\operatorname{Tr} K_U = \operatorname{Tr}(J^2) = -2.$$

The K_U eigenspaces therefore have dimensions

$$\dim U_+ = \frac{1}{2}(4 - 2) = 1, \qquad \dim U_- = \frac{1}{2}(4 + 2) = 3.$$

The exchange of the complete A factors is

$$\tau_A = \tau_{\mathbf{2}} \otimes \tau_U.$$

Hence the orientifold operator factorizes as

$$\tau_A (R \otimes R) = \tau_{\mathbf{2}} \otimes K_U.$$

An invariant state can arise in two ways:

1. the external doublet product is symmetric and the internal factor is K_U-even;
2. the external doublet product is antisymmetric and the internal factor is K_U-odd.

Since

$$\operatorname{Sym}^2 \mathbf{2} = \mathbf{3}, \qquad \bigwedge^2 \mathbf{2} = \mathbf{1},$$

the first possibility gives one triplet, while the second gives three singlets:

$$\boxed{(A \otimes A)_{\mathcal{O}=+1} = (\mathbf{3}, \mathbf{1}) + 3(\mathbf{1}, \mathbf{1})}.$$

Exactly the same calculation for

$$B = (\mathbf{1}, \mathbf{2}) \otimes U_B$$

gives

$$\boxed{(B \otimes B)_{\mathcal{O}=+1} = (\mathbf{1}, \mathbf{3}) + 3(\mathbf{1}, \mathbf{1})}.$$

Therefore

$$\boxed{\mathcal{H}^{R\text{–}R}_{\text{untw}} = (\mathbf{3}, \mathbf{1}) + (\mathbf{1}, \mathbf{3}) + 6(\mathbf{1}, \mathbf{1})}.$$

7.4.1.7 Untwisted mixed sectors

Consider first NS–R. Equal I_4 parity allows

$$V \otimes B \qquad \text{and} \qquad Y \otimes A.$$

For the first term,

$$\begin{aligned} V \otimes B &= (\mathbf{2}, \mathbf{2}) \otimes 2(\mathbf{1}, \mathbf{2}) \\ &= 2(\mathbf{2}, \mathbf{2} \otimes \mathbf{2}) \\ &= 2(\mathbf{2}, \mathbf{3}) + 2(\mathbf{2}, \mathbf{1}). \end{aligned}$$

For the second,

$$\begin{aligned} Y \otimes A &= 4(\mathbf{1}, \mathbf{1}) \otimes 2(\mathbf{2}, \mathbf{1}) \\ &= 8(\mathbf{2}, \mathbf{1}). \end{aligned}$$

Thus one orientation has

$$\mathcal{H}^{NS\text{–}R}_{\text{untw}} = 2(\mathbf{2}, \mathbf{3}) + 10(\mathbf{2}, \mathbf{1}).$$

The R–NS sector has an isomorphic set of states. The orientifold exchanges NS–R with R–NS and retains one linear combination from each pair. It therefore removes the overall doubling:

$$\boxed{\mathcal{H}^{NS\text{–}R+R\text{–}NS}_{\text{untw}} = 2(\mathbf{2}, \mathbf{3}) + 10(\mathbf{2}, \mathbf{1})}.$$

7.4.1.8 Twisted sectors at one fixed point

There are 16 I_4 fixed points. For the factorized rectangular torus used by the book, R maps each of them to itself, because each fixed coordinate is either 0 or a half-period. We may therefore compute the spectrum at one fixed point and multiply by 16.

The chiral twisted blocks are

$$C = 2(\mathbf{1}, \mathbf{1}), \qquad D = (\mathbf{2}, \mathbf{1}).$$

They are both I_4 even, so all diagonal and mixed tensor products relevant below survive the orbifold projection.

7.4.1.9 Twisted NS–NS sector

Before the orientifold projection,

$$C \otimes C = 4(\mathbf{1}, \mathbf{1}).$$

On the two-dimensional degeneracy space of C, R acts by J with $J^2 = -1$. Thus

$$K_C = \tau(J \otimes J)$$

has

$$\operatorname{Tr} K_C = \operatorname{Tr} J^2 = -2.$$

Its $+1$ and -1 eigenspaces have dimensions one and three:

$$\dim(K_C = +1) = 1, \qquad \dim(K_C = -1) = 3.$$

Because the chosen twisted-sector sign is $\epsilon_{\text{tw}} = -1$, the full orientifold operator is

$$\mathcal{O}_{\text{tw}} = -K_C.$$

An orientifold-even state must therefore lie in the $K_C = -1$ eigenspace. Hence one fixed point contributes

$$\boxed{3(\mathbf{1}, \mathbf{1})}$$

in NS–NS. Multiplying by 16 gives

$$\boxed{\mathcal{H}^{NS\text{–}NS}_{\text{tw}} = 48(\mathbf{1}, \mathbf{1})}.$$

7.4.1.10 Twisted R–R sector

At one fixed point,

$$D \otimes D = (\mathbf{2},\mathbf{1}) \otimes (\mathbf{2},\mathbf{1}) = (\mathbf{3},\mathbf{1}) \oplus (\mathbf{1},\mathbf{1}).$$

The symmetric square is $(\mathbf{3},\mathbf{1})$ and the antisymmetric square is $(\mathbf{1},\mathbf{1})$.

After including the standard R–R sign of Ω, the $(-1)^{F_L}$ sign, and the additional twisted-sector sign, the effective projection is

$$\mathcal{O}_{\rm tw} = -\tau$$

on $D \otimes D$. Thus the symmetric triplet is odd and the antisymmetric singlet is even:

$$\boxed{(D \otimes D)_{\mathcal{O}=+1} = (\mathbf{1},\mathbf{1})}.$$

With 16 fixed points,

$$\boxed{\mathcal{H}_{\rm tw}^{R\text{–}R} = 16(\mathbf{1},\mathbf{1})}.$$

7.4.1.11 Twisted mixed sectors

For one orientation and one fixed point,

$$C \otimes D = 2(\mathbf{1},\mathbf{1}) \otimes (\mathbf{2},\mathbf{1}) = 2(\mathbf{2},\mathbf{1}).$$

There is an isomorphic $D \otimes C$ contribution in the opposite R–NS sector. The orientifold exchanges the two orientations and leaves one linear combination, so one fixed point contributes

$$2(\mathbf{2},\mathbf{1}).$$

Multiplying by 16,

$$\boxed{\mathcal{H}_{\rm tw}^{NS\text{–}R+R\text{–}NS} = 32(\mathbf{2},\mathbf{1})}.$$

7.4.1.12 Assembly into six-dimensional $\mathcal{N} = (1,0)$ multiplets

The total untwisted bosonic content is

$$\begin{aligned} &[(\mathbf{3},\mathbf{3}) + 11(\mathbf{1},\mathbf{1})]_{NS\text{–}NS} \\ &\qquad \oplus [(\mathbf{3},\mathbf{1}) + (\mathbf{1},\mathbf{3}) + 6(\mathbf{1},\mathbf{1})]_{R\text{–}R} \\ &= (\mathbf{3},\mathbf{3}) + (\mathbf{3},\mathbf{1}) + (\mathbf{1},\mathbf{3}) + 17(\mathbf{1},\mathbf{1}). \end{aligned}$$

The untwisted fermions are

$$2(\mathbf{2},\mathbf{3}) + 10(\mathbf{2},\mathbf{1}).$$

Using Table 15.9:

- one gravity multiplet contains
$$(\mathbf{3},\mathbf{3}) + 2(\mathbf{2},\mathbf{3}) + (\mathbf{1},\mathbf{3});$$
- one tensor multiplet contains
$$(\mathbf{3},\mathbf{1}) + 2(\mathbf{2},\mathbf{1}) + (\mathbf{1},\mathbf{1}).$$

After subtracting these, the remaining untwisted states are

$$16(\mathbf{1},\mathbf{1}) + 8(\mathbf{2},\mathbf{1}).$$

One hypermultiplet contains four real singlet bosonic degrees of freedom and two $(\mathbf{2},\mathbf{1})$ fermionic degrees of freedom. The remainder is therefore four hypermultiplets. Thus

$$\boxed{\text{untwisted sector} = \text{gravity} + \text{tensor} + 4 \text{ hypermultiplets}}.$$

At each fixed point the twisted states are

$$\underbrace{3(\mathbf{1},\mathbf{1})+(\mathbf{1},\mathbf{1})}_{\text{4 bosonic singlets}}+\underbrace{2(\mathbf{2},\mathbf{1})}_{\text{fermions}},$$

which is exactly one hypermultiplet. Therefore

$$\boxed{\text{twisted sector} = 16 \text{ hypermultiplets}}.$$

This reproduces equation (15.30).

7.4.1.13 Checks

The untwisted bosonic degree count is

$$\begin{aligned} N_B^{\text{untw}} &= (9+11)+(3+3+6) \\ &= 20+12 \\ &= 32. \end{aligned}$$

The fermionic count is

$$N_F^{\text{untw}} = 2(2\cdot 3)+10(2\cdot 1) = 12+20 = 32.$$

In the twisted sector,

$$N_B^{\text{tw}} = 48+16 = 64,$$

and

$$N_F^{\text{tw}} = 32(2\cdot 1) = 64.$$

Thus bosonic and fermionic physical degrees of freedom match separately in the untwisted and twisted sectors.

The role of the twisted-sector sign can also be checked. If one chose

$$\epsilon_{\text{tw}} = +1$$

instead, the twisted NS–NS sector would retain the one-dimensional $K_C = +1$ space, while twisted R–R would retain the symmetric $(\mathbf{3},\mathbf{1})$. At each fixed point the result would be

$$(\mathbf{3},\mathbf{1})+2(\mathbf{2},\mathbf{1})+(\mathbf{1},\mathbf{1}),$$

which is a tensor multiplet rather than a hypermultiplet, precisely as stated by the book.

7.4.1.14 Result

The chiral blocks and orbifold parities are

	block	I_4
untwisted NS	$V=(\mathbf{2},\mathbf{2}),\quad Y=4(\mathbf{1},\mathbf{1})$	$+,\ -$
untwisted R	$A=2(\mathbf{2},\mathbf{1}),\quad B=2(\mathbf{1},\mathbf{2})$	$-,\ +$
twisted NS	$C=2(\mathbf{1},\mathbf{1})$	$+$
twisted R	$D=(\mathbf{2},\mathbf{1})$	$+$

Equal I_4 parities are tensored, after which one imposes

$$\mathcal{O} = \Omega R(-1)^{F_L}, \qquad \epsilon_{\text{tw}} = -1.$$

The result is

$$\boxed{\begin{aligned} \text{untwisted } NS\text{–}NS &: (\mathbf{3},\mathbf{3})+11(\mathbf{1},\mathbf{1}), \\ \text{untwisted } R\text{–}R &: (\mathbf{3},\mathbf{1})+(\mathbf{1},\mathbf{3})+6(\mathbf{1},\mathbf{1}), \\ \text{untwisted mixed} &: 2(\mathbf{2},\mathbf{3})+10(\mathbf{2},\mathbf{1}), \\ \text{twisted } NS\text{–}NS &: 48(\mathbf{1},\mathbf{1}), \\ \text{twisted } R\text{–}R &: 16(\mathbf{1},\mathbf{1}), \\ \text{twisted mixed} &: 32(\mathbf{2},\mathbf{1}). \end{aligned}}$$

These representations assemble as

$$\boxed{\text{untwisted: gravity} + \text{tensor} + 4 \text{ hypers}, \qquad \text{twisted: } 16 \text{ hypers.}}$$

7.5 Q5: Why the orientifold contains both x- and y-oriented O7-planes

In page 538-539, the prefactor in front of the massless tadpole indicates that it corresponds to an O7-plane, one occupies the six noncompact dimensions and two x-axes, the other occupies the six noncompact dimensions and two y-axes. I agree with this conclusion, but I also wander why the position of the O7-brane is like this. We have the orientifold action $\Omega R(-1)^{F_L}$ with $R : y_i \to -y_i, i = 1, 2$; so naively the orientifold plane may locate at the fixed locus of R i.e. it occupies the six noncompact dimensions and two x-axes? What about the other O7-plane? Or how to formally define the position of the O-plane?

> **Textbook location and related material.** Chapter 15, Section 15.3 (pp. 538–539): Klein-bottle lattice sums and the two O7-plane families. Sections 6.7 and 10.6 provide the general crosscap and orientifold fixed-locus interpretation.

7.5.1 Answer

7.5.1.1 Core point

Your fixed-locus argument is correct for the single group element

$$\mathcal{O} = \Omega R(-1)^{F_L}.$$

Its geometric part is R, and $\mathrm{Fix}(R)$ consists of loci extended along x^1, x^2 and localized in y^1, y^2. These are the $O7_x$-planes.

The missing point is that the background has already been divided by the orbifold generator I_4. The orientifold projection must therefore be imposed on the $T^4/\mathbb{Z}_2$ theory, not on the covering torus by itself. Closure of the full orientifold group gives two orientation-reversing elements:

$$\mathcal{O} = \Omega R(-1)^{F_L}, \qquad \mathcal{O}I_4 = \Omega R I_4(-1)^{F_L}.$$

The geometric part of the second element is RI_4. Since

$$R : (x^i, y^i) \mapsto (x^i, -y^i), \qquad I_4 : (x^i, y^i) \mapsto (-x^i, -y^i),$$

their product acts as

$$RI_4 : \qquad (x^i, y^i) \mapsto (-x^i, y^i).$$

Its fixed locus is localized in x^1, x^2 and extended along y^1, y^2. These are the $O7_y$-planes.

Thus the second family is not an extra object inserted by hand. It is required by the I_4 term in the orbifold projector in equation (15.31).

7.5.1.2 The full orientifold group and the two crosscaps

The full group is

$$\Gamma_{\text{or}} = \left\{1,\, I_4,\, \Omega R(-1)^{F_L},\, \Omega R I_4(-1)^{F_L}\right\}.$$

The first two elements preserve worldsheet orientation. The last two reverse it and therefore produce crosscap sectors.

Indeed, equation (15.31) contains

$$\begin{aligned}\mathcal{K} &= \int_0^\infty \frac{dt}{2t}\,\mathrm{Tr}\left[\Omega R(-1)^{F_L}\frac{1+I_4}{2}P_{\text{GSO}}e^{-4\pi t(L_0-c/24)}\right]\\ &= \frac{1}{2}\mathcal{K}_{\mathcal{O}} + \frac{1}{2}\mathcal{K}_{\mathcal{O}I_4}.\end{aligned}$$

The first term is a trace with insertion $\mathcal{O}$, whereas the second is a trace with insertion $\mathcal{O}I_4$. After transforming the Klein bottle to the tree channel, these become closed-string propagation between the corresponding crosscap states:

$$\widetilde{\mathcal{K}}_{\mathcal{O}} \sim \langle C_R|e^{-2\pi l H_{\rm cl}}|C_R\rangle, \qquad \widetilde{\mathcal{K}}_{\mathcal{O}I_4} \sim \langle C_{RI_4}|e^{-2\pi l H_{\rm cl}}|C_{RI_4}\rangle.$$

The target-space loci of these crosscaps are what are called the orientifold planes.

The factors Ω and $(-1)^{F_L}$ do not provide additional target-space coordinate transformations. The former creates the crosscap identification, while the latter affects supersymmetry and the relative NS–NS/R–R crosscap signs. The geometric support is determined by R for the first crosscap and by RI_4 for the second.

7.5.1.3 Formal definition of the O-plane locus

For an orientation-reversing element Ωg whose target-space part is the involution g, the crosscap condition has the schematic form

$$X(\tau=0,\sigma_{\rm ws}+\pi) = g(X(\tau=0,\sigma_{\rm ws}))\,.$$

In a coordinate reflected by g, the zero mode must obey

$$q=-q \pmod{2\pi R}.$$

It is therefore localized at

$$q=0 \qquad \text{or} \qquad q=\pi R.$$

In a coordinate left invariant by g, the crosscap is not localized. Consequently, the associated O-plane worldvolume is the noncompact spacetime times $\mathrm{Fix}(g)$ in the compact target.

This is also the book's definition in Chapter 9: after transforming the Klein bottle to the tree channel, the closed string couples to a crosscap source, and the target-space locus of that source is called an orientifold plane. Strictly speaking, it is an O7-plane rather than an "O7-brane": it is not a dynamical D-brane, has no transverse position modulus, and does not support open-string endpoints.

For an orbifold target, one must include every orientation-reversing group element:

$$\boxed{\mathcal{L}_O = \bigcup_{\substack{\gamma\in\Gamma_{\rm or}\\ \gamma \text{ reverses worldsheet orientation}}} \mathrm{Fix}(g_\gamma)\Big/\langle I_4\rangle}\,.$$

Here g_γ means the geometric part of γ. In the present model this gives

$$\mathcal{L}_O = [\mathrm{Fix}(R)\cup\mathrm{Fix}(RI_4)]\Big/\langle I_4\rangle.$$

This also gives a useful formulation directly on the quotient

$$X=T^4/\langle I_4\rangle.$$

The reflection induced by R acts on an orbifold point as

$$\widehat{R}:[p]\mapsto[Rp].$$

Such a point is fixed when

$$[Rp]=[p].$$

But the orbit represented by $[p]$ contains both p and I_4p. Hence there are two ways for this condition to hold:

$$Rp=p \pmod{\Lambda}, \qquad \text{or} \qquad Rp=I_4p \pmod{\Lambda}.$$

The first condition is the R branch. The second can be rewritten as

$$RI_4p=p \pmod{\Lambda},$$

and is the RI_4 branch. Therefore

$$\boxed{\mathrm{Fix}_X(\widehat{R}) = [\mathrm{Fix}_{T^4}(R)\cup\mathrm{Fix}_{T^4}(RI_4)]\Big/\langle I_4\rangle}\,.$$

This is the precise quotient-space version of the answer: even the fixed locus of the induced reflection on $T^4/\mathbb{Z}_2$ has both the x-oriented and the y-oriented branches.

7.5.1.4 Explicit fixed loci and their multiplicities

For the rectangular torus,

$$x^i \sim x^i + 2\pi R_x, \qquad y^i \sim y^i + 2\pi R_y.$$

For R, the fixed-point equations are

$$y^i = -y^i \pmod{2\pi R_y},$$

or

$$y^i \in \{0, \pi R_y\}, \qquad i = 1, 2,$$

while x^1, x^2 are arbitrary. Thus, in the covering geometry,

$$\boxed{\mathrm{Fix}(R) = \bigcup_{\epsilon_1,\epsilon_2\in\{0,1\}} \left\{y^1 = \epsilon_1 \pi R_y,\, y^2 = \epsilon_2 \pi R_y\right\} \cong 4T_x^2}.$$

Each component has worldvolume

$$\mathbb{R}^{1,5} \times T_x^2,$$

so it is an O7-plane.

For RI_4, the fixed-point equations are instead

$$x^i = -x^i \pmod{2\pi R_x},$$

or

$$x^i \in \{0, \pi R_x\}, \qquad i = 1, 2,$$

while y^1, y^2 are arbitrary. Hence

$$\boxed{\mathrm{Fix}(RI_4) = \bigcup_{\epsilon_1,\epsilon_2\in\{0,1\}} \left\{x^1 = \epsilon_1 \pi R_x,\, x^2 = \epsilon_2 \pi R_x\right\} \cong 4T_y^2}.$$

Each component has worldvolume

$$\mathbb{R}^{1,5} \times T_y^2,$$

and is the second kind of O7-plane.

Thus the upstairs covering geometry contains four $O7_x$ components and four $O7_y$ components. The book's singular phrase "an O7-plane along the two x-axes" or "an O7-plane along the two y-axes" refers to the corresponding crosscap source or family, not to only one connected component.

The two families intersect when both x^i and y^i are half-periods. These are precisely the 16 fixed points of I_4.

7.5.1.5 The two loci from the loop-channel lattice sums

The same result is encoded directly in equation (15.33). For one compact circle,

$$\Omega|m, n\rangle = |m, -n\rangle, \qquad I|m, n\rangle = |-m, -n\rangle,$$

where m and n are momentum and winding numbers.

Consider first the insertion $\mathcal{O} = \Omega R(-1)^{F_L}$.

- Along each x direction, R acts trivially. The insertion therefore acts as Ω, and a state contributes to the trace only when $n_x = 0$. The surviving sum is a momentum sum P_x.
- Along each y direction, R is a reflection. The combination ΩR maps

  $$|m_y, n_y\rangle \longmapsto |-m_y, n_y\rangle,$$

  so the trace requires $m_y = 0$. The surviving sum is a winding sum W_y.

There are two coordinates of each kind, so this insertion produces

$$\mathcal{K}_{\mathcal{O}}^{(0)} \propto P_x(t)^2 W_y(t)^2.$$

Momentum in x and winding in y are the closed-string zero-mode signature of Neumann-like crosscap conditions along x and Dirichlet-like localization along y. Thus this is the $O7_x$ contribution.

Now consider $\mathcal{O}I_4$. Its geometric action RI_4 reflects x and leaves y unchanged.

- Along x, the trace requires $m_x = 0$ and leaves a winding sum W_x.
- Along y, the trace requires $n_y = 0$ and leaves a momentum sum P_y.

Therefore

$$\mathcal{K}_{\mathcal{O}I_4}^{(0)} \propto P_y(t)^2 W_x(t)^2,$$

which is the $O7_y$ contribution. Equation (15.33) is exactly

$$\boxed{\mathcal{K}^{(0)} \propto P_x^2 W_y^2 + P_y^2 W_x^2}.$$

The notation "$(x \leftrightarrow y)$" in equation (15.35) is the tree-channel remnant of these two distinct insertions.

7.5.1.6 What can be read directly from the Klein-bottle trace

Your interpretation of the two terms is correct, with a small refinement in terminology. Expanding the orbifold projector gives

$$\mathcal{K} \supset \frac{1}{2}\,\mathrm{Tr}\left(\mathcal{O}P_{\mathrm{GSO}}e^{-4\pi t(L_0-c/24)}\right) + \frac{1}{2}\,\mathrm{Tr}\left(\mathcal{O}I_4 P_{\mathrm{GSO}}e^{-4\pi t(L_0-c/24)}\right).$$

Writing a general orientation-reversing element as

$$\gamma = \Omega g\,\Xi,$$

where g is its geometric target-space action and Ξ is any additional internal operator such as $(-1)^{F_L}$, the rule is

$$\boxed{\mathrm{Tr}_{\mathcal{H}}\left(\gamma\, q^{L_0-c/24}\bar{q}^{\bar{L}_0-\bar{c}/24}\right) \quad\longleftrightarrow\quad \text{crosscap associated with } \gamma, \qquad \mathcal{L}_{O,\gamma} = \mathrm{Fix}(g).}$$

Thus, in this model,

trace insertion	geometric part	O-plane locus
$\mathcal{O} = \Omega R(-1)^{F_L}$	R	$\mathrm{Fix}(R) = 4T_x^2$
$\mathcal{O}I_4 = \Omega RI_4(-1)^{F_L}$	RI_4	$\mathrm{Fix}(RI_4) = 4T_y^2$

It is therefore common, but slightly imprecise, to say that the planes are the "fixed loci of $\mathcal{O}$ and $\mathcal{O}I_4$." Since Ω reverses the worldsheet and $(-1)^{F_L}$ acts on the string Hilbert space, neither is an ordinary map of target-space points. The precise statement is that the planes are fixed loci of the geometric parts R and RI_4 of those two elements.

There are two complementary levels at which the trace supplies information:

1. **The insertion label identifies the geometric involution.** From $\mathcal{O}$ one extracts R, and from $\mathcal{O}I_4$ one extracts RI_4. Solving their fixed-point equations gives the actual loci.
2. **The zero-mode part verifies the wrapped and transverse directions.** A momentum sum in a coordinate signals that the crosscap is extended along that direction, whereas a winding sum signals localization in that direction. Hence

$$P_x^2 W_y^2 \quad\Longrightarrow\quad O7_x, \qquad P_y^2 W_x^2 \quad\Longrightarrow\quad O7_y.$$

After Poisson resummation, the corresponding volume factors provide the same information.

So in this unshifted geometric orientifold, one can indeed read off the O-plane locus from each term in the Klein-bottle trace by identifying the geometric part of its insertion and then examining its zero modes.

There is, however, a useful limitation to this statement. A volume prefactor by itself generally determines only how many directions are wrapped and which radii are parallel or transverse; it need not determine every absolute position or every local O-plane charge. Translated reflections, freely acting shifts, discrete torsion, and signs in the crosscap state can insert phases into the lattice sums or alter the distribution and type of O-planes. In such cases one must retain the complete insertion, including its lattice phases and crosscap signs, rather than infer the locus from the massless volume scaling alone. No such ambiguity occurs here because R and RI_4 are explicit unshifted reflections on the rectangular torus.

7.5.1.7 Why the Klein bottle does not describe open-string endpoints

The proposed endpoint interpretation is not correct for the Klein bottle. Although one often speaks broadly of "open–closed channel duality," the modular transformation of a Klein bottle does not turn it into an open-string diagram. A Klein bottle has no worldsheet boundary in either channel.

The relevant one-loop surfaces are distinguished as follows:

Worldsheet	Loop-channel description	Transverse-channel description
Klein bottle	Closed-string trace with an orientation-reversing insertion	Closed string propagating between two crosscaps
Annulus	Open-string trace	Closed string propagating between two D-brane boundaries
Möbius strip	Open-string trace with an orientation-reversing insertion	Closed string propagating between a D-brane boundary and a crosscap

A modular transformation changes the parametrization and propagation direction on the same worldsheet; it does not change its topology. In particular, it cannot turn either crosscap of a Klein bottle into a boundary. Although all three transverse-channel diagrams can be drawn as tubes, the states placed at the tube ends are different.

Schematically,

$$\begin{aligned}
\mathcal{K}_g &= \mathrm{Tr}_{\mathcal{H}_{\rm cl}}\left(\Omega g\,\Xi\, q^{H_{\rm cl}}\right) \quad \overset{\text{modular}}{\longleftrightarrow} \quad \langle C_g|e^{-2\pi l H_{\rm cl}}|C_g\rangle,\\
\mathcal{A}_{ab} &= \mathrm{Tr}_{\mathcal{H}^{\rm op}_{ab}}\left(q^{H_{\rm op}}\right) \quad \overset{\text{modular}}{\longleftrightarrow} \quad \langle B_a|e^{-2\pi l H_{\rm cl}}|B_b\rangle,\\
\mathcal{M}_{a,g} &= \mathrm{Tr}_{\mathcal{H}^{\rm op}_{a,g(a)}}\left(\Omega g\,\Xi\, q^{H_{\rm op}}\right) \quad \overset{\text{modular}}{\longleftrightarrow} \quad \langle B_a|e^{-2\pi l H_{\rm cl}}|C_g\rangle.
\end{aligned}$$

Numerical factors in the modular parameters have been suppressed here because the important point is the nature of the states and the ends of the transverse tube.

7.5.1.7.1 What the operator means in the Klein-bottle trace In

$$\mathrm{Tr}_{\mathcal{H}_{\rm cl}}\left(\Omega g\,\Xi\, e^{-4\pi t(L_0-c/24)}\right),$$

the trace is over the closed-string Hilbert space. The insertion means that, after propagating once around the loop-channel Euclidean time direction, the closed-string configuration is identified with its $\Omega g\,\Xi$ image. Because Ω reverses worldsheet orientation, this identification produces a nonorientable closed worldsheet.

After the modular transformation, the same surface is a tube with a crosscap at each end. There is still no open string and no open-string endpoint. The target-space fixed locus appears because the crosscap state satisfies

$$X(\sigma_{\rm ws}+\pi) = g(X(\sigma_{\rm ws}))\,.$$

For a reflected zero mode this reduces to

$$q = g(q) = -q \pmod{2\pi R},$$

which localizes the crosscap source at $\mathrm{Fix}(g)$. Thus the trace detects the O-plane through a crosscap condition, not through an endpoint condition.

For the two terms in equation (15.31),

$$\begin{aligned} \mathrm{Tr}_{\mathrm{cl}}\left(\mathcal{O}\, P_{\mathrm{GSO}} e^{-4\pi t(L_0 - c/24)}\right) &\longleftrightarrow |C_R\rangle \qquad \longrightarrow \mathrm{Fix}(R), \\ \mathrm{Tr}_{\mathrm{cl}}\left(\mathcal{O} I_4\, P_{\mathrm{GSO}} e^{-4\pi t(L_0 - c/24)}\right) &\longleftrightarrow |C_{RI_4}\rangle \qquad \longrightarrow \mathrm{Fix}(RI_4). \end{aligned}$$

Neither line contains a D-brane boundary state.

7.5.1.7.2 Where open strings actually enter Open strings enter only after D7-branes are introduced to cancel the R–R tadpoles. Their endpoints satisfy boundary conditions defined by the D7-brane boundary states $|B_a\rangle$. The annulus is a loop of such open strings, and its transverse channel is closed-string exchange between two D-branes.

The Möbius strip is the surface on which the orientifold operator acts inside an open-string trace. If a denotes a D-brane, then Ωg reverses the open-string orientation and maps the brane data to its image under g. An open-string sector can contribute to the Möbius trace only when it is mapped back to an equivalent sector. For an invariant brane this may require

$$g(a) = a.$$

For example, a D7-brane parallel to the x directions and localized at (y^1, y^2) is invariant under R only when

$$(y^1, y^2) \in \left\{(0,0),\, (\pi R_y, 0),\, (0, \pi R_y),\, (\pi R_y, \pi R_y)\right\}.$$

Such a D7-brane lies on an $O7_x$ component. Nevertheless, an open-string endpoint there is attached to the D7-brane, not to the O7-plane. The coincidence of their loci does not turn the O-plane into an object supporting open-string endpoints.

A D-brane need not always be individually fixed. A brane away from the fixed locus can occur together with its image $g(a)$; the orientifold then identifies the two brane sectors. This is why O-planes determine image rules and Chan–Paton projections without themselves carrying Chan–Paton gauge fields.

The logical chain is therefore

$$\boxed{\begin{aligned} \text{Klein-bottle insertion } \Omega g\, \Xi &\Longrightarrow \text{crosscap } |C_g\rangle \Longrightarrow \text{O-plane on } \mathrm{Fix}(g), \\ \text{D-brane boundary } |B_a\rangle &\Longrightarrow \text{open-string endpoints on } a, \\ \text{Möbius insertion } \Omega g\, \Xi &\Longrightarrow \text{orientifold identification of } a \text{ with } g(a). \end{aligned}}$$

7.5.1.8 Why the volume prefactors identify the wrapped directions

Poisson resummation gives

$$\sum_{m\in\mathbb{Z}} e^{-\pi t (m/\rho)^2} = \frac{\rho}{\sqrt{t}} \sum_{\widetilde{n}\in\mathbb{Z}} e^{-\pi \rho^2 \widetilde{n}^2/t},$$

and

$$\sum_{n\in\mathbb{Z}} e^{-\pi t (\rho n)^2} = \frac{1}{\rho\sqrt{t}} \sum_{\widetilde{m}\in\mathbb{Z}} e^{-\pi \widetilde{m}^2/(\rho^2 t)}.$$

After the Klein-bottle modular transformation $t = 1/(4l)$, the first lattice product therefore carries

$$P_x^2 W_y^2 \longrightarrow \left(\frac{\rho_x}{\rho_y}\right)^2 \times \text{tree-channel lattice sums},$$

while the second carries

$$P_y^2 W_x^2 \longrightarrow \left(\frac{\rho_y}{\rho_x}\right)^2 \times \text{tree-channel lattice sums}.$$

In the massless limit $l \to \infty$, all nonzero dual momentum and winding modes are exponentially suppressed. The remaining coefficients are

$$v_6 \left(\frac{\rho_x}{\rho_y}\right)^2, \qquad v_6 \left(\frac{\rho_y}{\rho_x}\right)^2.$$

The first coefficient scales as internal wrapped volume divided by transverse volume,

$$\frac{V_x}{V_y} = \frac{R_x^2}{R_y^2},$$

so it describes a source extended along x^1, x^2 and localized in y^1, y^2. The second has the reciprocal scaling and describes a source extended along y^1, y^2 and localized in x^1, x^2. This is why the tadpole calculation independently recovers the two fixed-locus geometries.

7.5.1.9 Checks

If the I_4 orbifold projector were absent, equation (15.31) would contain only $\mathcal{O}$. Then only $\mathrm{Fix}(R)$ and the $O7_x$ family would occur. The $O7_y$ family is therefore entirely tied to the fact that the orientifold is performed after the I_4 quotient.

The two fixed sets have real dimension two inside T^4. Adding the six noncompact dimensions gives an eight-dimensional worldvolume, which is precisely the worldvolume dimension of an O7-plane.

Finally, $(-1)^{F_L}$ does not change either fixed locus. It changes the action on spacetime fermions and the R–R part of the crosscap state. Likewise, the extra twisted-sector sign chosen on page 536 changes twisted-sector charges and the resulting multiplet projection, but not the geometric equations defining the two O7 loci.

7.5.1.10 Result

The orientifold group contains

$$\boxed{\Gamma_{\mathrm{or}} = \{1,\, I_4,\, \Omega R(-1)^{F_L},\, \Omega R I_4(-1)^{F_L}\}\,.}$$

Its two orientation-reversing elements have geometric parts

$$\begin{aligned} R &: (x^i, y^i) \mapsto (x^i, -y^i), \\ RI_4 &: (x^i, y^i) \mapsto (-x^i, y^i). \end{aligned}$$

Consequently,

$$\boxed{\begin{aligned} \mathrm{Fix}(R) = 4T_x^2 &\quad\Longrightarrow\quad 4\ O7_x \text{ components}, \\ \mathrm{Fix}(RI_4) = 4T_y^2 &\quad\Longrightarrow\quad 4\ O7_y \text{ components}. \end{aligned}}$$

Equivalently, on the quotient,

$$\boxed{[Rp] = [p] \quad\Longleftrightarrow\quad Rp = p \text{ or } Rp = I_4 p,}$$

which produces the same two branches.

The loop-channel zero modes and tree-channel tadpole prefactors are

$$\boxed{\begin{aligned} \Omega R(-1)^{F_L} : P_x^2 W_y^2 &\longrightarrow v_6 \left(\frac{\rho_x}{\rho_y}\right)^2 \quad \Longrightarrow O7_x, \\ \Omega R I_4(-1)^{F_L} : P_y^2 W_x^2 &\longrightarrow v_6 \left(\frac{\rho_y}{\rho_x}\right)^2 \quad \Longrightarrow O7_y. \end{aligned}}$$

More generally, the trace-to-locus rule is

$$\boxed{\gamma = \Omega g\, \Xi \quad\Longrightarrow\quad \mathrm{Tr}(\gamma \cdots) \text{ identifies the } \gamma\text{-crosscap}, \qquad \mathcal{L}_{O,\gamma} = \mathrm{Fix}(g),}$$

with the zero-mode lattice sums determining the parallel and transverse directions. This is a crosscap statement, not an open-string endpoint statement:

$$\boxed{\mathcal{K}:\ C_g\text{–}C_g, \qquad \mathcal{A}:\ B_a\text{–}B_b, \qquad \mathcal{M}:\ B_a\text{–}C_g.}$$

Thus your naive fixed-locus argument correctly finds the first family; the second comes from the equally mandatory orientation-reversing element obtained by multiplying the orientifold generator by the orbifold element I_4. Open-string endpoints appear only after D7-brane boundary states are introduced.

7.6 Q6: Derivation and massless interpretation of the fractional-D7 annulus

In page 539, they compute the amplitude for open string starting and ending on the same D7-brane in (15.36) and the result is (15.37). >Since ϑ_4 and ϑ_3 differ by the sign for the first excited level, this term implies that in the massless sector only the excitations $\psi^\mu_{-\frac{1}{2}}$ in the six non-compact directions survive the I_4 projection, whereas the excitations $\psi^i_{-\frac{1}{2}}$ in the internal directions $i = 5, \cdots, 8$ are projected out. The D7-brane therefore supports a six-dimensional gauge field and cannot be moved in the internal directions, i.e. it is frozen at $(y_1, y_2) = (0, 0)$ and has no continuous Wilson-line in the x-directions.

I am not such familiar with the computation of this partition function and its interpretation. Do the explicit computation for me and explain the above results from the computation.

> **Textbook location and related material.** Chapter 15, Section 15.3 around Eqs. (15.36)–(15.41), pp. 539–544: the fractional-D7 annulus, I_4 projection, tree-channel twisted exchange, and orientifold images.

7.6.1 Answer

7.6.1.1 Core point

The annulus in equation (15.36) is an open-string one-loop trace with two independent projections:

$$\mathcal{A} = \int_0^\infty \frac{dt}{2t}\, \mathrm{Tr}_{(D7,D7)}\left[\frac{1+I_4}{2}\, P_{\mathrm{GSO}}\, e^{-2\pi t(L_0 - c/24)}\right].$$

For a D7-brane parallel to the two x-axes, the trace has the following structure:

1. the six noncompact zero modes give the measure $v_6\, dt/t^4$;
2. the two Neumann x directions give Kaluza-Klein momentum sums;
3. the two Dirichlet y directions give winding sums;
4. the identity part of $(1 + I_4)/2$ gives the ordinary D7–D7 oscillator character;
5. the I_4-inserted part has no nonzero compact zero modes and changes the signs of all internal oscillators.

Combining these pieces gives exactly

$$\boxed{\begin{aligned}\mathcal{A}_{xx} = v_6 \int_0^\infty \frac{dt}{t^4}\, \frac{1}{2^6}(1_{\mathrm{NS}} - 1_{\mathrm{R}}) \\ \times\left[\frac{\vartheta_4^4}{\eta^{12}}(2it)\left(\sum_{m\in\mathbb{Z}} e^{-2\pi t(m/\rho_x)^2}\right)^2\left(\sum_{n\in\mathbb{Z}} e^{-2\pi t(\rho_y n)^2}\right)^2 + 4\,\frac{\vartheta_4^2\vartheta_3^2}{\eta^6\vartheta_2^2}(2it)\right].\end{aligned}}$$

The first term in brackets is the trace without I_4; the second is the trace with I_4.

At the massless NS level, before the orbifold projection, the physical states are

$$\psi^M_{-1/2}|0; k\rangle_{\mathrm{NS}}, \qquad M = \mu \text{ or } i.$$

Their I_4 eigenvalues are

$$I_4\psi^\mu_{-1/2}|0\rangle_{\rm NS} = +\psi^\mu_{-1/2}|0\rangle_{\rm NS}, \qquad I_4\psi^i_{-1/2}|0\rangle_{\rm NS} = -\psi^i_{-1/2}|0\rangle_{\rm NS}.$$

Therefore

$$\frac{1+I_4}{2}\psi^\mu_{-1/2}|0\rangle_{\rm NS} = \psi^\mu_{-1/2}|0\rangle_{\rm NS}, \qquad \frac{1+I_4}{2}\psi^i_{-1/2}|0\rangle_{\rm NS} = 0.$$

The surviving state is the six-dimensional gauge vector. The four removed internal states are precisely two transverse-position scalars and two Wilson-line scalars. We now derive every ingredient of the amplitude and this interpretation.

7.6.1.2 Geometry and open-string boundary conditions

Take

$$D7_x : \qquad \mathbb{R}^{1,5} \times T^2_x,$$

localized at a point (y^1, y^2) of the transverse T^2_y. Its boundary conditions are

directions	boundary condition	open-string zero modes
$\mathbb{R}^{1,5}$	NN	k_μ
x^1, x^2	NN	m_a/R_x
y^1, y^2	DD	$n_a R_y/\alpha'$

with $m_a, n_a \in \mathbb{Z}$.

The Dirichlet winding deserves emphasis. Both endpoints lie on the same point of the compact y-circle, but the string can stretch from that point to one of its images in the covering space:

$$Y(\pi) - Y(0) = 2\pi n R_y.$$

Its classical stretching energy is therefore proportional to

$$\left(\frac{nR_y}{\alpha'}\right)^2.$$

This produces the two types of lattice factor

$$P_x(t) = \sum_{m\in\mathbb{Z}} e^{-2\pi t(m/\rho_x)^2}, \qquad W_y(t) = \sum_{n\in\mathbb{Z}} e^{-2\pi t(\rho_y n)^2}.$$

Because there are two x and two y directions, the complete compact zero-mode factor is

$$P_x(t)^2 W_y(t)^2.$$

7.6.1.3 Why the D7-brane must be at an I_4 fixed position

The orbifold inversion maps

$$D7_x(y^1, y^2) \longmapsto D7_x(-y^1, -y^2).$$

The operator I_4 therefore maps the Hilbert space

$$\mathcal{H}_{(D7(y),D7(y))} \longrightarrow \mathcal{H}_{(D7(-y),D7(-y))}.$$

A trace with an I_4 insertion is defined within the same open-string Hilbert space only if the brane is mapped to itself:

$$(y^1, y^2) = (-y^1, -y^2) \pmod{2\pi R_y}.$$

Hence

$$\boxed{(y^1, y^2) \in \left\{(0,0),\ (\pi R_y, 0),\ (0, \pi R_y),\ (\pi R_y, \pi R_y)\right\}.}$$

The book chooses $(0,0)$ as a representative. Thus "frozen at $(0,0)$" means frozen at the particular fixed point selected in the example; the same calculation applies at any of the four fixed positions.

If the brane is away from these points, a consistent orbifold configuration must also contain its image at $(-y^1, -y^2)$. The individual $(D7(y), D7(y))$ trace with I_4 then vanishes because I_4 maps it to a distinct sector.

7.6.1.4 Zero modes in the identity and I_4 traces

Expanding the orbifold projector,

$$\mathcal{A} = \frac{1}{2}\mathcal{A}[1] + \frac{1}{2}\mathcal{A}[I_4].$$

For the identity trace, every momentum and winding state contributes:

$$\mathcal{A}[1]_{\text{zero}} = P_x(t)^2 W_y(t)^2.$$

For the inserted trace, I_4 reverses all four compact coordinates. It therefore acts on the zero-mode labels as

$$I_4 : \qquad |m_1, m_2; n_1, n_2\rangle \longmapsto |-m_1, -m_2; -n_1, -n_2\rangle.$$

The diagonal matrix element entering a trace is

$$\begin{aligned}&\langle m_1, m_2; n_1, n_2 | I_4 | m_1, m_2; n_1, n_2\rangle \\ &\quad = \delta_{m_1,-m_1}\delta_{m_2,-m_2}\delta_{n_1,-n_1}\delta_{n_2,-n_2}.\end{aligned}$$

For integer labels, all four conditions require

$$m_1 = m_2 = n_1 = n_2 = 0.$$

Consequently,

$$\boxed{\mathcal{A}[I_4]_{\text{zero}} = 1.}$$

This explains why the last term of equation (15.37) has no momentum or winding sums.

7.6.1.5 The six-dimensional measure and the factor 2^{-6}

The Gaussian trace over the six noncompact momenta is

$$\begin{aligned}V_6 \int \frac{d^6k}{(2\pi)^6} e^{-2\pi t\alpha' k^2} &= \frac{V_6}{(8\pi^2\alpha' t)^3} \\ &= \frac{v_6}{2^3 t^3},\end{aligned}$$

where

$$v_6 = \frac{V_6}{(4\pi^2\alpha')^3}.$$

Combining this with the annulus measure $dt/(2t)$ gives

$$v_6 \frac{dt}{t^4}\frac{1}{2^4}.$$

The orbifold projector contributes another factor $1/2$, and the GSO projector another factor $1/2$. Hence

$$\frac{1}{2^4} \times \frac{1}{2} \times \frac{1}{2} = \boxed{\frac{1}{2^6}},$$

which is the normalization displayed in equation (15.37) for a single Chan–Paton label.

7.6.1.6 Oscillator trace without I_4

In light-cone gauge there are eight physical bosonic and eight physical fermionic oscillators. The eight unreflected bosons give

$$Z_B[1] = \frac{1}{\eta^8}.$$

After the open-string GSO sum, the NS and R characters are equal by the Jacobi identity and occur with opposite spacetime-statistics signs. In the book's annulus convention this is written

$$Z_F[1] = (1_{\text{NS}} - 1_{\text{R}})\frac{\vartheta_4^4}{\eta^4}.$$

Therefore

$$\boxed{Z_{\mathrm{osc}}[1] = (1_{\mathrm{NS}} - 1_{\mathrm{R}})\frac{\vartheta_4^4}{\eta^{12}}.}$$

Multiplication by $P_x^2 W_y^2$ produces the first term in the square brackets of equation (15.37).

The symbol $(1_{\mathrm{NS}} - 1_{\mathrm{R}})$ records the cancellation of the complete vacuum amplitude between spacetime bosons and fermions. It should not be used to discard the amplitude before reading the spectrum: one first examines the NS and R characters separately.

7.6.1.7 Oscillator trace with I_4

The inversion I_4 acts trivially on the four physical noncompact transverse oscillators and as -1 on the four internal oscillators. The inserted trace therefore splits into four even and four odd real fields.

For an I_4-even bosonic oscillator,

$$\mathrm{Tr}\left(q^{nN}\right) = \sum_{N=0}^{\infty} q^{nN} = \frac{1}{1-q^n}.$$

For an I_4-odd bosonic oscillator,

$$\mathrm{Tr}\left(I_4 q^{nN}\right) = \sum_{N=0}^{\infty} (-1)^N q^{nN} = \frac{1}{1+q^n}.$$

The four internal real bosons form two complex bosons. Using the product formula for ϑ_2,

$$Z_{B,\mathrm{internal}}[I_4] = \left(\frac{2\eta}{\vartheta_2}\right)^2.$$

The four noncompact real bosons give η^{-4}, so

$$\boxed{Z_B[I_4] = \frac{1}{\eta^4}\left(\frac{2\eta}{\vartheta_2}\right)^2 = \frac{4}{\eta^2\vartheta_2^2}.}$$

This is the origin of the explicit factor 4 in equation (15.37). Equivalently, ϑ_2 has a leading factor 2, so the explicit 4 removes the factor contained in ϑ_2^2.

For an NS fermionic oscillator, the occupation number is $N = 0, 1$. An even oscillator contributes

$$\sum_{N=0}^{1} q^{rN} = 1 + q^r,$$

whereas an odd oscillator contributes

$$\sum_{N=0}^{1} (-1)^N q^{rN} = 1 - q^r.$$

The product formulas identify these two signs with ϑ_3 and ϑ_4. Since two complex fermions are even and two are odd, the inserted fermion character is

$$Z_F[I_4] = (1_{\mathrm{NS}} - 1_{\mathrm{R}})\frac{\vartheta_4^2\vartheta_3^2}{\eta^4}.$$

Therefore

$$\boxed{Z_{\mathrm{osc}}[I_4] = (1_{\mathrm{NS}} - 1_{\mathrm{R}})4\frac{\vartheta_4^2\vartheta_3^2}{\eta^6\vartheta_2^2}.}$$

Because the inserted trace has no compact zero-mode sum, this is exactly the second term in equation (15.37).

7.6.1.8 Reading the first excited level from ϑ_3 and ϑ_4

For vanishing compact momentum and winding, the NS mass formula is

$$\alpha' M^2 = N_{\mathrm{NS}} - \frac{1}{2}.$$

The GSO projection removes the $N_{\mathrm{NS}} = 0$ tachyon. The first allowed states have

$$N_{\mathrm{NS}} = \frac{1}{2},$$

and are created by one fermionic oscillator:

$$\psi^M_{-1/2}|0\rangle_{\mathrm{NS}}.$$

These are the massless states whose I_4 eigenvalues must be extracted from the first excited terms of the fermion characters.

Let $Q = e^{\pi i \tau}$. The relevant leading terms are

$$\vartheta_3(\tau) = 1 + 2Q + \mathcal{O}(Q^4), \qquad \vartheta_4(\tau) = 1 - 2Q + \mathcal{O}(Q^4).$$

The opposite signs of the first excited contribution are precisely the insertion eigenvalues. Four real fermions, or two complex fermions, contribute

$$\vartheta_3^2 = 1 + 4Q + \cdots, \qquad \vartheta_4^2 = 1 - 4Q + \cdots.$$

Thus the I_4-inserted character weights the first excited states in the two four-dimensional oscillator blocks with opposite signs. Up to the conventional overall phase assigned to the NS vacuum, this is

$$\mathrm{Tr}_{\mathrm{massless,NS}}\, I_4 = 4 - 4 = 0.$$

The four positive contributions are the noncompact excitations and the four negative contributions are the internal ones.

The trace without insertion counts all eight massless NS states:

$$\mathrm{Tr}_{\mathrm{massless,NS}}\, 1 = 4 + 4 = 8.$$

Applying the projector gives the total number

$$\mathrm{Tr}_{\mathrm{massless,NS}} \frac{1 + I_4}{2} = \frac{1}{2}(8 + 0) = 4.$$

More importantly, applying it state by state gives

state	I_4	$\frac{1}{2}(1+I_4)$
$\psi^\mu_{-1/2}\|0\rangle_{\mathrm{NS}}$	$+1$	kept
$\psi^i_{-1/2}\|0\rangle_{\mathrm{NS}}$	-1	removed

which is the statement made below equation (15.37).

This projection assumes that both endpoints lie on the same fractional-brane type. The Chan–Paton action is then

$$\lambda \longmapsto \gamma_{I_4} \lambda \gamma_{I_4}^{-1} = \lambda,$$

so the total I_4 parity is the oscillator parity displayed above. For an open string stretched between fractional branes with opposite twisted charges, the Chan–Paton factor contributes an additional minus sign. The projection is then reversed: the vector is removed while internal scalar states survive, in agreement with the discussion following equation (15.40).

7.6.1.9 From the surviving states to six-dimensional fields

Before the orbifold projection, a D7-brane supports an eight-dimensional gauge field. Reducing it on the wrapped T_x^2 and viewing the result in six dimensions gives

$$A_M \longrightarrow \left(A_\mu,\, A_{x^1},\, A_{x^2}\right),$$

together with the two transverse scalars

$$\Phi_{y^1},\, \Phi_{y^2}.$$

The correspondence with NS oscillators is

oscillator	six-dimensional field	geometric meaning
$\psi^\mu_{-1/2}$	A_μ	gauge vector
$\psi^{x^1}_{-1/2}, \psi^{x^2}_{-1/2}$	A_{x^1}, A_{x^2}	Wilson-line scalars
$\psi^{y^1}_{-1/2}, \psi^{y^2}_{-1/2}$	Φ_{y^1}, Φ_{y^2}	transverse-position scalars

Since I_4 leaves the noncompact coordinates invariant,

$$A_\mu \mapsto +A_\mu.$$

Since it reverses all four internal coordinates, a constant internal vector component and a transverse displacement are both odd:

$$A_{x^a} \mapsto -A_{x^a}, \qquad \Phi_{y^a} \mapsto -\Phi_{y^a}.$$

Therefore only A_μ survives.

The four states $\psi^\mu_{-1/2}|0\rangle$ are the four physical polarizations of a massless vector in six dimensions:

$$d-2 = 6-2 = 4.$$

The R-sector projection retains the corresponding gaugino states, so the result is a six-dimensional supersymmetric vector multiplet.

The absence of Φ_{y^a} means that an individual invariant brane has no infinitesimal transverse displacement. Moving it by δy would produce a brane at δy and an image at $-\delta y$, which is not a deformation of a single invariant fractional brane.

The absence of A_{x^a} means there is no continuous Wilson-line modulus. This does not rule out isolated discrete Wilson-line choices fixed by $A_x \sim -A_x$ modulo large gauge transformations; it says that there is no massless scalar generating a continuous Wilson-line deformation.

7.6.1.10 Tree-channel check and the meaning of "fractional"

Using

$$t = \frac{1}{2l},$$

Poisson resummation of the lattice sums, and the modular transformations of η and ϑ_a, equation (15.37) becomes

$$\boxed{\begin{aligned}\widetilde{\mathcal{A}}_{xx} = v_6 \int_0^\infty dl\, \frac{1}{2^7}\left(1_{\rm NS} - 1_{\rm R}\right) \\ \times \left[\left(\frac{\rho_x}{\rho_y}\right)^2 \frac{\vartheta_4^4}{\eta^{12}}(2il)\left(\sum_{n\in\mathbb{Z}} e^{-\pi l(\rho_x n)^2}\right)^2\left(\sum_{m\in\mathbb{Z}} e^{-\pi l(m/\rho_y)^2}\right)^2 \right. \\ \left. +4^2\,\frac{\vartheta_2^2\vartheta_3^2}{\eta^6\vartheta_4^2}(2il)\right].\end{aligned}}$$

This is equation (15.38).

The first term is closed-string exchange from the I_4-untwisted sector between two copies of the D7 boundary state. Its factor

$$\left(\frac{\rho_x}{\rho_y}\right)^2$$

has the correct scaling for a D7-brane extended in x^1, x^2 and localized in y^1, y^2.

The second term has no momentum or winding sums and transforms into a twisted closed-string character. It is exchange of I_4-twisted closed-string states. Hence the boundary state has both untwisted and twisted components. With the book's normalization, equation (15.39) is

$$|D7_x^{\pm}\rangle_{\text{frac}} = \frac{1}{2\mathcal{N}_{D7}}\left[\left(\frac{\rho_x}{\rho_y}\right)|D7_x\rangle_{\text{untw}} \pm 2\sum_{a=1}^{4}|D7_x, a\rangle_{\text{tw}}\right].$$

The four terms correspond to the four I_4 fixed points through which the x-wrapping D7-brane passes.

A twisted R–R field is localized at the orbifold fixed set. A brane carrying its charge cannot move continuously away from that set without violating charge conservation. This is the closed-string channel explanation of the same freezing already seen in the open-string spectrum.

Two fractional branes with opposite twisted charges have

$$(|D7_x\rangle_{\text{untw}} + |D7_x\rangle_{\text{tw}}) + (|D7_x\rangle_{\text{untw}} - |D7_x\rangle_{\text{tw}}) = 2|D7_x\rangle_{\text{untw}}.$$

Their twisted charges cancel, and together they can form an ordinary bulk D7-brane with movable brane-image degrees of freedom. Thus the statement that the brane is frozen applies to an individual fractional D7-brane, not to every possible D7-brane combination in the orbifold.

7.6.1.11 Why the annulus proves that the D7-brane carries twisted R–R charge

There are three steps:

1. the I_4 insertion in the open loop channel becomes an I_4 twist in the closed transverse channel;
2. the R contribution to that closed-string exchange is therefore an I_4-twisted R–R state;
3. a nonzero boundary-state overlap with an R–R state is an R–R source coupling and hence an R–R charge.

7.6.1.11.1 From an inserted open trace to a twisted closed string The I_4 contribution to equation (15.36) is

$$\mathcal{A}[I_4] = \mathrm{Tr}_{\mathcal{H}_{\text{op}}}\left[I_4 P_{\text{GSO}} e^{-2\pi t(L_0 - c/24)}\right].$$

In the open loop channel, the insertion means that fields are identified after one trip around Euclidean time by

$$X(\sigma, \tau + 2\pi t) = I_4 X(\sigma, \tau).$$

The annulus modular transformation exchanges its spatial and Euclidean-time cycles. In the transverse channel, the propagating closed string consequently obeys

$$X(\sigma + 2\pi, \tau) = I_4 X(\sigma, \tau).$$

This is the defining boundary condition of the I_4-twisted closed-string Hilbert space:

$$\boxed{\mathrm{Tr}_{\text{open}}\left(I_4 q^{H_{\text{op}}}\right) \quad \overset{S}{\leftrightarrow} \quad \text{closed-string propagation in } \mathcal{H}_{\text{cl}}^{(I_4\text{-tw})}.}$$

The theta functions in equations (15.37) and (15.38) implement this exchange:

$$4\,\frac{\vartheta_4^2\vartheta_3^2}{\eta^6\vartheta_2^2}(2it) \quad \xrightarrow{t=1/(2l)} \quad 4^2\,\frac{\vartheta_2^2\vartheta_3^2}{\eta^6\vartheta_4^2}(2il).$$

The expression on the right is the twisted closed-string character. Its lack of compact momentum and winding sums is another diagnostic: an I_4-twisted closed string is localized at an orbifold fixed point and has no translational zero mode in the four inverted directions.

7.6.1.11.2 Why the R term means R–R charge The shorthand

$$(1_{\rm NS} - 1_{\rm R})$$

in equation (15.38) contains a nonzero NS–NS exchange and a nonzero R–R exchange. They cancel only after the supersymmetric vacuum amplitude is summed; to determine charges, the two sectors must be kept separate.

In the transverse channel the annulus factorizes as

$$\widetilde{\mathcal{A}} = \langle B|e^{-2\pi l H_{\rm cl}}|B\rangle.$$

Inserting a complete set of closed-string states gives

$$\widetilde{\mathcal{A}} = \sum_{\alpha} \langle B|\alpha\rangle e^{-2\pi l E_\alpha} \langle \alpha|B\rangle.$$

For a long tube, $l \to \infty$, only the lightest closed-string states remain. If α is a massless twisted R–R state, its contribution is proportional to

$$\left|\langle \alpha_{\rm tw,R}|B\rangle\right|^2.$$

The overlap

$$\langle \alpha_{\rm tw,R}|B\rangle$$

is the disk one-point function of that R–R field in the presence of the D-brane. A nonzero R–R one-point function is the string-theory definition of a source coupling. Therefore

$$\boxed{\widetilde{\mathcal{A}}[I_4]_{\rm R} \neq 0 \quad \Longrightarrow \quad \langle \alpha_{\rm tw,R}|D7_x\rangle \neq 0 \quad \Longrightarrow \quad D7_x \text{ carries twisted R–R charge.}}$$

This is visible directly in the boundary state. Suppressing normalization,

$$|D7_x^{\pm}\rangle_{\rm frac} = |B\rangle_{\rm untw} \pm |B\rangle_{\rm tw}.$$

In a self-overlap, the twisted part occurs quadratically:

$$\langle D7_x^{\pm}|e^{-2\pi l H}|D7_x^{\pm}\rangle \supset \langle B_{\rm tw}|e^{-2\pi l H}|B_{\rm tw}\rangle,$$

so the sign of the twisted charge is invisible. To measure its sign, compare opposite fractional branes:

$$\langle D7_x^{+}|e^{-2\pi l H}|D7_x^{-}\rangle \supset -\langle B_{\rm tw}|e^{-2\pi l H}|B_{\rm tw}\rangle.$$

This is why the book obtains equation (15.37) with the sign of the I_4 term reversed for branes of opposite twisted charge.

7.6.1.11.3 Why the twisted field is $\int_{E_i} C_8$ The partition function establishes that the exchanged state is localized, twisted, and R–R. Identifying the spacetime differential form requires the geometric dictionary for the resolution of

$$T^4/\mathbb{Z}_2.$$

Near each fixed point, the singularity is locally

$$\mathbb{C}^2/\mathbb{Z}_2.$$

Resolving it introduces an exceptional cycle

$$E_i \simeq S^2$$

whose area tends to zero at the orbifold point. Associated with E_i is a localized harmonic two-form ω_i satisfying

$$\int_{E_j} \omega_i = \delta_{ij}.$$

In the democratic formulation of type-IIB supergravity, expand the R–R eight-form as

$$C_8(x,y) = \sum_{i=1}^{16} C_6^{(i)}(x) \wedge \omega_i(y) + \cdots .$$

Integrating over an exceptional cycle gives

$$\boxed{C_6^{(i)}(x) = \int_{E_i} C_8(x,y).}$$

Because ω_i is localized near the ith exceptional cycle, $C_6^{(i)}$ is the spacetime field associated with the ith twisted R–R sector. Although E_i has zero area at the orbifold point, it remains a vanishing homology cycle; its twisted state and R–R flux integral do not disappear.

The D7 Wess–Zumino coupling is

$$S_{\mathrm{WZ}} = \mu_7 \int_{W_8} C_8 + \cdots .$$

For a fractional D7 whose wrapped cycle contains an exceptional component at E_i, the corresponding part of its worldvolume is

$$W_8 \supset \mathbb{R}^{1,5} \times E_i.$$

Substituting the expansion of C_8 gives

$$\begin{aligned} \mu_7 \int_{\mathbb{R}^{1,5}\times E_i} C_8 &= \mu_7 \int_{\mathbb{R}^{1,5}} \left(\int_{E_i} C_8 \right) \\ &= \mu_7 \int_{\mathbb{R}^{1,5}} C_6^{(i)}. \end{aligned}$$

Thus the exceptional component of the fractional brane is electrically charged under

$$\boxed{C_6^{(i)} = \int_{E_i} C_8.}$$

The $D7_x$ wraps the (x^1, x^2) torus and passes through four I_4 fixed points. This is why equation (15.39) contains

$$\sum_{i=1}^{4} |D7_x, i\rangle_{\mathrm{tw}}$$

and why the brane couples to the four corresponding fields

$$C_6^{(i)} = \int_{E_i} C_8.$$

The two choices $D7_x^{\pm}$ have opposite coefficients for all these twisted components. Their sum has no exceptional charge and can become an ordinary movable bulk D7-brane.

7.6.1.11.4 What is and is not determined by the partition function From equations (15.37)–(15.38) alone one can read:

- the presence of an I_4-twisted closed-string exchange;
- its NS–NS and R–R parts;
- the absence of internal zero modes, hence localization at fixed points;
- a nonzero twisted R–R boundary-state coupling;
- the relative sign of the twisted charge by comparing $D7_x^+$ and $D7_x^-$ amplitudes.

The further identification

$$\text{twisted R–R state} \quad \longleftrightarrow \quad \int_{E_i} C_8$$

uses the geometric dictionary between orbifold twisted sectors and exceptional cycles, together with the D7 Wess–Zumino coupling. Thus the partition function proves that the charge is twisted R–R charge, while the resolved-orbifold geometry identifies the R–R potential that represents it.

7.6.1.12 Derivation of the orientifold image in equation (15.41)

Equation (15.41) states

$$\Omega R: \qquad |D7^{\pm}\rangle_{\text{frac}} \longmapsto |D7^{\mp}\rangle_{\text{frac}}.$$

Strictly, this is the action induced by the full orientifold generator

$$\mathcal{O} = \Omega R(-1)^{F_L}$$

on the GSO-complete boundary state; the book suppresses $(-1)^{F_L}$ in the label on the left-hand side of equation (15.41).

The result is not determined by the geometry of R alone. It follows from combining:

1. geometric invariance of the D7-brane under R;
2. the decomposition of a fractional boundary state into untwisted and twisted pieces;
3. the extra sign chosen for the orientifold action on I_4-twisted states on page 536.

7.6.1.12.1 Action on the untwisted component For a $D7_x$ at one of the fixed positions

$$(y^1, y^2) = (-y^1, -y^2) \pmod{2\pi R_y},$$

the reflection R maps the geometric brane to itself. Worldsheet parity exchanges the two ends of the open string, but a D7–D7 boundary condition is unchanged. The factor $(-1)^{F_L}$ supplies the action required for this supersymmetric type-IIB orientifold and does not convert this D7 into an anti-D7. With the common overall boundary-state phase fixed conventionally,

$$\boxed{\mathcal{O}|D7_x\rangle_{\text{untw}} = +|D7_x\rangle_{\text{untw}}.}$$

If a brane were not at an R-fixed position, the same equation would instead contain the boundary state of its geometrically reflected image. Equation (15.41) concerns the invariant fractional branes used in the construction.

7.6.1.12.2 Action on the twisted component On page 536 the model makes the discrete choice that $\mathcal{O}$ acts with an additional minus sign on every I_4-twisted-sector state. In the notation used earlier,

$$\epsilon_{\text{tw}} = -1.$$

The twisted boundary state $|D7_x, i\rangle_{\text{tw}}$ in equation (15.40) is a coherent state built on the localized twisted ground state $|t_i\rangle$. The oscillator part is carried into the corresponding boundary-state oscillator part by ΩR. The additional twist-field sign acts on its ground state and gives

$$\mathcal{O}|D7_x, i\rangle_{\text{tw}} = \epsilon_{\text{tw}}\,|D7_x, R(i)\rangle_{\text{tw}}.$$

For the four fixed points lying on an invariant $D7_x$, $R(i) = i$ modulo the torus lattice, so

$$\boxed{\mathcal{O}|D7_x, i\rangle_{\text{tw}} = -|D7_x, i\rangle_{\text{tw}}.}$$

More generally, if R permuted fixed points, the orientifold would apply the same minus sign while permuting the corresponding twisted boundary states. Here the relevant four fixed points are individually fixed.

7.6.1.12.3 Acting on the full fractional boundary state Write equation (15.39) as

$$|D7_x^{\pm}\rangle_{\text{frac}} = A\,|D7_x\rangle_{\text{untw}} \pm B\sum_{i=1}^{4}|D7_x, i\rangle_{\text{tw}},$$

where

$$A = \frac{1}{2\mathcal{N}_{D7}}\frac{\rho_x}{\rho_y}, \qquad B = \frac{1}{\mathcal{N}_{D7}}.$$

Using the two transformation laws above,

$$\begin{aligned}\mathcal{O}|D7_x^{\pm}\rangle_{\text{frac}} &= A\,\mathcal{O}|D7_x\rangle_{\text{untw}} \pm B\sum_{i=1}^{4}\mathcal{O}|D7_x,i\rangle_{\text{tw}}\\ &= A\,|D7_x\rangle_{\text{untw}} \mp B\sum_{i=1}^{4}|D7_x,i\rangle_{\text{tw}}\\ &= |D7_x^{\mp}\rangle_{\text{frac}}.\end{aligned}$$

Therefore

$$\boxed{\mathcal{O}: \qquad |D7_x^{+}\rangle_{\text{frac}} \longleftrightarrow |D7_x^{-}\rangle_{\text{frac}}.}$$

The identical reasoning applies to the $D7_y^{\pm}$ pair.

7.6.1.12.4 The same result from the twisted R–R coupling The relevant Wess–Zumino couplings have the schematic form

$$\begin{aligned}S_{\text{WZ}}^{\pm} &= \mu_{\text{ut}}\int_{\mathbb{R}^{1,5}\times\Pi_x} C_8^{\text{untw}}\\ &\pm \mu_{\text{tw}}\sum_{i=1}^{4}\int_{\mathbb{R}^{1,5}} C_6^{(i)}, \qquad C_6^{(i)} = \int_{E_i} C_8.\end{aligned}$$

The chosen orientifold parity leaves the relevant untwisted source component invariant but reverses the twisted one:

$$\mathcal{O}: \qquad C_8^{\text{untw}} \mapsto +C_8^{\text{untw}}, \qquad C_6^{(i)} \mapsto -C_6^{(i)}.$$

A source with $q_{\text{tw}} = +1$ must therefore be mapped to a source with $q_{\text{tw}} = -1$, and conversely. This reproduces the boundary-state derivation.

This also clarifies the terminology: $D7^{+}$ and $D7^{-}$ are parallel and have the same untwisted D7 charge and tension. They differ only in the sign of their localized twisted charge.

7.6.1.12.5 Why the pair is required and why its twisted tadpole cancels An orientifold-invariant D-brane configuration must contain complete orbits of $\mathcal{O}$. Since neither fractional brane is invariant by itself, the minimal invariant combination is

$$|D7_x^{+}\rangle_{\text{frac}} + |D7_x^{-}\rangle_{\text{frac}}.$$

Its twisted component cancels:

$$\begin{aligned}|D7_x^{+}\rangle_{\text{frac}} + |D7_x^{-}\rangle_{\text{frac}} &= 2A\,|D7_x\rangle_{\text{untw}}\\ &\quad + B\sum_i |D7_x,i\rangle_{\text{tw}} - B\sum_i |D7_x,i\rangle_{\text{tw}}\\ &= 2A\,|D7_x\rangle_{\text{untw}}.\end{aligned}$$

Consequently,

$$q_{\text{tw}}^{(+)} + q_{\text{tw}}^{(-)} = +1 - 1 = 0.$$

This is why the book says that introducing the fractional branes in orientifold-image pairs automatically cancels their twisted-sector tadpoles.

7.6.1.12.6 Dependence on the discrete orientifold choice The exchange in equation (15.41) is not universal for every orientifold of $T^4/\mathbb{Z}_2$. If one chooses the opposite twisted-sector action,

$$\epsilon_{\text{tw}} = +1,$$

then

$$\mathcal{O}|D7_x,i\rangle_{\text{tw}} = +|D7_x,i\rangle_{\text{tw}},$$

and the same calculation gives

$$\mathcal{O}|D7_x^{\pm}\rangle_{\text{frac}} = |D7_x^{\pm}\rangle_{\text{frac}}.$$

Each fractional brane is then separately invariant. This is precisely the alternative model mentioned below equation (15.41), whose closed twisted sectors contain tensor multiplets instead of hypermultiplets.

Thus equation (15.41) cannot be inferred from the annulus amplitude alone. The annulus establishes the existence of the two signs of twisted charge. Determining how $\Omega R(-1)^{F_L}$ maps those signs additionally requires the orientifold action on the twisted closed-string ground states, and the model fixed that action by the extra minus sign on page 536.

7.6.1.13 Checks

At the massless NS level,

$$N_{\text{before}} = 4_{\text{vector}} + 2_{\text{Wilson}} + 2_{\text{position}} = 8,$$

while after the I_4 projection,

$$N_{\text{after}} = 4_{\text{vector}}.$$

This is the correct bosonic content of a six-dimensional massless vector multiplet once the R-sector gauginos are included.

The trace with I_4 has no compact lattice sums because only

$$m_a = n_a = 0$$

is fixed by sign reversal. Under the annulus modular transformation, precisely this inserted trace becomes the twisted closed-string exchange. This matches the general orbifold rule that an insertion in the open loop channel becomes propagation in the corresponding twisted closed sector in the transverse channel.

Finally, changing the fixed position from $(0,0)$ to any of

$$(\pi R_y, 0), \qquad (0, \pi R_y), \qquad (\pi R_y, \pi R_y)$$

does not change the local oscillator projection. It only changes which fixed locus carries the fractional-brane boundary state.

7.6.1.14 Result

The trace decomposes as

$$\boxed{\mathcal{A} = \frac{1}{2}\mathcal{A}[1] + \frac{1}{2}\mathcal{A}[I_4],}$$

with

$$\boxed{\begin{aligned} \mathcal{A}[1]: &\quad P_x^2 W_y^2\,(1_{\text{NS}} - 1_{\text{R}})\frac{\vartheta_4^4}{\eta^{12}}, \\ \mathcal{A}[I_4]: &\quad (1_{\text{NS}} - 1_{\text{R}})4\frac{\vartheta_4^2\vartheta_3^2}{\eta^6\vartheta_2^2}. \end{aligned}}$$

The I_4 trace is possible only for

$$(y^1, y^2) = (-y^1, -y^2) \pmod{2\pi R_y},$$

and its compact zero modes obey

$$m_1 = m_2 = n_1 = n_2 = 0.$$

At the massless NS level,

$$\boxed{\begin{aligned} &\frac{1+I_4}{2}\psi^{\mu}_{-1/2}|0\rangle_{\text{NS}} = \psi^{\mu}_{-1/2}|0\rangle_{\text{NS}} &&\Longrightarrow A_\mu, \\ &\frac{1+I_4}{2}\psi^{x^a}_{-1/2}|0\rangle_{\text{NS}} = 0 &&\Longrightarrow \text{no continuous Wilson line,} \\ &\frac{1+I_4}{2}\psi^{y^a}_{-1/2}|0\rangle_{\text{NS}} = 0 &&\Longrightarrow \text{no transverse-position modulus.} \end{aligned}}$$

The transverse-channel interpretation is

$$\boxed{\mathcal{A}[1] \longleftrightarrow \text{untwisted closed-string exchange}, \qquad \mathcal{A}[I_4] \longleftrightarrow \text{twisted closed-string exchange}.}$$

Separating the R contribution and factorizing the long tube gives

$$\boxed{\widetilde{\mathcal{A}}[I_4]_{\mathrm{R}} \sim \sum_i \left|\langle C_6^{(i)}|D7_x\rangle\right|^2, \qquad C_6^{(i)} = \int_{E_i} C_8.}$$

Finally, the model's twisted-sector sign gives

$$\boxed{\begin{aligned} \mathcal{O}|B\rangle_{\mathrm{untw}} &= +|B\rangle_{\mathrm{untw}}, \\ \mathcal{O}|B\rangle_{\mathrm{tw}} &= -|B\rangle_{\mathrm{tw}}, \end{aligned} \qquad \Longrightarrow \qquad \mathcal{O}|D7^{\pm}\rangle_{\mathrm{frac}} = |D7^{\mp}\rangle_{\mathrm{frac}}.}$$

Hence an invariant pair has

$$\boxed{q_{\mathrm{tw}}^{(+)} + q_{\mathrm{tw}}^{(-)} = 0,}$$

which is equation (15.41) and its twisted-tadpole consequence.

Therefore the surviving object supports a six-dimensional gauge multiplet and is an I_4-charged fractional D7-brane pinned at the chosen orbifold fixed position.

7.7 Q7: Deriving the vertex-operator mass formula from conformal weights

How to derive the mass formula for a vertex operator? For example, in the chapter15.4, how to derive eq (15.52) for the vertex operator given in (15.51)?

> **Textbook location and related material.** Chapter 15, Section 15.4 around Eqs. (15.51)–(15.52): conformal weights of the general compactified vertex. Section 13.2 supplies the BRST and picture conventions.

7.7.1 Answer

7.7.1.1 Core point

The mass formula is the conformal-weight condition for a physical string vertex operator. The vertex written in equation (15.51) contains the matter and superghost factors but does not display the reparametrization ghosts $c\bar{c}$. Therefore it must have weights

$$(h, \bar{h}) = (1, 1).$$

Equivalently, the unintegrated BRST vertex $c\bar{c}V$ has total weights $(0, 0)$, because c and $\bar{c}$ each have weight -1.

In the convention used on pages 545–546, the z sector is called **right-moving** and the $\bar{z}$ sector **left-moving**. This is why h_{int} appears in the m_R^2 equation, while $\bar{h}_{\mathrm{int}}$ appears in the m_L^2 equation.

Suppressing normal-ordering symbols, equation (15.51) is

$$\begin{aligned} V(z,\bar{z}) = {} & \left(\bar{\partial}^{n_L} X^{\mu}(\bar{z})\right)^{m_L} \left(\partial^{n_{1R}} X^{\nu}(z)\right)^{m_{1R}} \left(\partial^{n_{2R}} \phi^i(z)\right)^{m_{2R}} \\ & \times e^{i\lambda_R \cdot \phi(z)} e^{q\phi(z)} V_{\mathrm{int}}(z,\bar{z}) e^{ik\cdot X(z,\bar{z})}. \end{aligned}$$

Here the indexed fields ϕ^i bosonize the four external right-moving fermions, whereas the unindexed ϕ bosonizes the $\beta\gamma$ superghost. The derivation consists of adding the weights of all these factors and setting the result equal to $(1, 1)$.

7.7.1.2 The plane-wave contribution

With the free-boson OPE

$$X^\mu(z,\bar z)X^\nu(w,\bar w) \sim -\frac{\alpha'}{2}\eta^{\mu\nu}[\log(z-w)+\log(\bar z-\bar w)],$$

the stress tensor gives

$$T_X(z)e^{ik\cdot X(w,\bar w)} \sim \frac{\alpha' k^2/4}{(z-w)^2}e^{ik\cdot X(w,\bar w)} + \frac{1}{z-w}\partial e^{ik\cdot X(w,\bar w)}.$$

Thus

$$h(e^{ik\cdot X}) = \bar h(e^{ik\cdot X}) = \frac{\alpha' k^2}{4}.$$

For a state of spacetime mass m, the mostly-plus convention gives

$$k^2 = -m^2,$$

and hence the plane wave contributes

$$\boxed{h_k = \bar h_k = -\frac{\alpha' m^2}{4}.}$$

This negative contribution is not pathological: the time component of the Lorentzian free boson has negative norm in target-space signature, and the physical-state condition balances it against positive oscillator and internal-CFT weights.

7.7.1.3 Oscillator and internal-CFT contributions

A chiral free boson has $h(\partial X) = 1$. More generally,

$$h(\partial^n X) = n, \qquad \bar h(\bar\partial^n X) = n.$$

Therefore the three oscillator blocks in (15.51) contribute

$$N_L = m_L n_L, \qquad N_{1R} = m_{1R} n_{1R}, \qquad N_{2R} = m_{2R} n_{2R}$$

for the displayed monomials. For a general product of oscillators, these become sums, for example

$$N_L = \sum_a m_{L,a} n_{L,a}.$$

The internal operator is defined to have weights

$$V_{\text{int}}(z,\bar z): \qquad (h,\bar h) = (h_{\text{int}}, \bar h_{\text{int}}).$$

It therefore contributes h_{int} to the right-moving z equation and $\bar h_{\text{int}}$ to the left-moving $\bar z$ equation.

7.7.1.4 The bosonized-fermion exponential

For the canonically normalized bosons representing the external right-moving fermions,

$$\phi^i(z)\phi^j(w) \sim -\delta^{ij}\log(z-w).$$

Their stress tensor is

$$T_\phi(z) = -\frac{1}{2}:\partial\phi\cdot\partial\phi:(z).$$

The standard exponential OPE then gives

$$T_\phi(z)e^{i\lambda_R\cdot\phi(w)} \sim \frac{\lambda_R^2/2}{(z-w)^2}e^{i\lambda_R\cdot\phi(w)} + \frac{1}{z-w}\partial e^{i\lambda_R\cdot\phi(w)}.$$

Consequently,

$$\boxed{h\left(e^{i\lambda_R\cdot\phi}\right) = \frac{1}{2}\lambda_R^2.}$$

This factor encodes whether the external right-moving state lies in an NS conjugacy class or a Ramond spinor conjugacy class; together with the picture charge q, λ_R forms the covariant-lattice vector $(\lambda_R, q) \in (D_{2,1})_R$.

7.7.1.5 The superghost-picture contribution

The bosonized superghost is not an ordinary free boson: it has a background charge. With

$$\beta = e^{-\phi}\partial\xi, \qquad \gamma = \eta e^{\phi},$$

its scalar stress tensor is

$$T_\phi(z) = -\frac{1}{2} :(\partial\phi)^2: (z) - \partial^2\phi(z).$$

Since

$$\phi(z)\phi(w) \sim -\log(z-w),$$

the double contraction with the first term contributes $-q^2/2$, while the background-charge term contributes $-q$. Hence

$$T_\phi(z)e^{q\phi(w)} \sim \frac{-\frac{1}{2}q^2 - q}{(z-w)^2}e^{q\phi(w)} + \frac{1}{z-w}\partial e^{q\phi(w)},$$

so that

$$\boxed{h(e^{q\phi}) = -\frac{1}{2}q(q+2) = -\frac{1}{2}q^2 - q.}$$

For example,

$$h(e^{-\phi}) = \frac{1}{2}, \qquad h(e^{-\phi/2}) = \frac{3}{8},$$

as required in the conventional NS picture $q=-1$ and Ramond picture $q=-\frac{1}{2}$.

7.7.1.6 The left-moving weight condition

Only three pieces of (15.51) contribute nontrivially to the antiholomorphic weight:

- the left-moving bosonic oscillators, with weight N_L;
- the internal operator, with weight $\bar{h}_{\text{int}}$;
- the plane wave, with weight $-\alpha' m^2/4$.

Therefore

$$\bar{h}(V) = N_L + \bar{h}_{\text{int}} - \frac{\alpha' m^2}{4}.$$

The physical condition $\bar{h}(V) = 1$ gives

$$\frac{\alpha' m^2}{4} = N_L + \bar{h}_{\text{int}} - 1.$$

The book defines the left-moving contribution m_L^2 by

$$\frac{\alpha'}{2}m_L^2 \equiv N_L + \bar{h}_{\text{int}} - 1.$$

Thus the first line of (15.52) is

$$\boxed{\frac{\alpha'}{2}m_L^2 = N_L + \bar{h}_{\text{int}} - 1.}$$

The constant -1 is simply the result of moving the required vertex weight 1 to the other side. In oscillator language it is the left-moving bosonic intercept.

7.7.1.7 The right-moving weight condition

Adding all holomorphic weights gives

$$h(V) = N_{1R} + N_{2R} + \frac{1}{2}\lambda_R^2 - \frac{1}{2}q^2 - q + h_{\text{int}} - \frac{\alpha' m^2}{4}.$$

Imposing $h(V) = 1$ yields

$$\frac{\alpha' m^2}{4} = \frac{1}{2}\lambda_R^2 - \frac{1}{2}q^2 - q + N_{1R} + N_{2R} + h_{\text{int}} - 1.$$

Defining the right-moving contribution m_R^2 by

$$\frac{\alpha'}{2}m_R^2 \equiv \frac{1}{2}\lambda_R^2 - \frac{1}{2}q^2 - q + N_{1R} + N_{2R} + h_{\text{int}} - 1,$$

we obtain the second line of (15.52):

$$\boxed{\frac{\alpha'}{2}m_R^2 = \frac{1}{2}\lambda_R^2 - \frac{1}{2}q^2 - q + N_{1R} + N_{2R} + h_{\text{int}} - 1.}$$

The familiar NS and R zero-point shifts are encoded here in a picture-covariant way: part of the right-moving conformal weight resides in the lattice exponential and part in $e^{q\phi}$.

7.7.1.8 Why the book writes $m^2 = m_L^2 + m_R^2$

There is only one spacetime momentum k^μ and one physical mass m. The symbols m_L^2 and m_R^2 are therefore not two independent masses. They are a bookkeeping convention that assigns half of the total mass formula to each chiral sector.

Indeed, either physical weight condition gives

$$N_L + \bar{h}_{\text{int}} - 1 = \frac{\alpha' m^2}{4},$$

and

$$\frac{1}{2}\lambda_R^2 - \frac{1}{2}q^2 - q + N_{1R} + N_{2R} + h_{\text{int}} - 1 = \frac{\alpha' m^2}{4}.$$

Comparing these equations with the definitions of m_L^2 and m_R^2 gives

$$m_L^2 = m_R^2 = \frac{m^2}{2}.$$

Consequently,

$$\boxed{m^2 = m_L^2 + m_R^2,}$$

which is the third line of (15.52).

The equality of the two chiral expressions is also the level-matching condition:

$$\boxed{N_L + \bar{h}_{\text{int}} = \frac{1}{2}\lambda_R^2 - \frac{1}{2}q^2 - q + N_{1R} + N_{2R} + h_{\text{int}}.}$$

Thus one should not first compute two unrelated masses and then add them. One computes the two chiral conformal weights, imposes level matching, and uses the book's definitions to package the common result as $m^2 = m_L^2 + m_R^2$.

7.7.1.9 Checks

For a massless NS state in the canonical picture $q = -1$, a right-moving fermion ψ^μ is represented by a lattice exponential with

$$\frac{1}{2}\lambda_R^2 = \frac{1}{2}.$$

The superghost supplies

$$-\frac{1}{2}q^2 - q = -\frac{1}{2} + 1 = \frac{1}{2}.$$

With $N_{1R} = N_{2R} = h_{\text{int}} = 0$, the right-moving non-plane-wave weight is therefore 1, and

$$m_R^2 = 0.$$

On the left, a factor $\bar{\partial}X^\mu$ has $N_L = 1$ and $\bar{h}_{\text{int}} = 0$, giving $m_L^2 = 0$. This is the standard massless heterotic vertex.

For a four-dimensional Ramond state in picture $q=-\frac{1}{2}$, the external spin field has

$$\lambda_R=\left(\pm\frac{1}{2},\pm\frac{1}{2}\right),\qquad \frac{1}{2}\lambda_R^2=\frac{1}{4}.$$

The superghost contributes 3/8. If the internal operator is an internal Ramond ground state of the $c_{\rm int}=9$ SCFT, then

$$h_{\rm int}=\frac{c_{\rm int}}{24}=\frac{3}{8}.$$

Hence

$$\frac{1}{4}+\frac{3}{8}+\frac{3}{8}=1,$$

again giving $m_R^2=0$.

Picture changing alters q and simultaneously alters the other right-moving factors, but it preserves their total conformal weight. The mass formula is therefore picture independent even though the separate terms in the second line of (15.52) are picture dependent.

7.7.1.10 Result

The conformal weights of the factors in (15.51) are

factor	h	$\bar h$
$e^{ik\cdot X}$	$-\alpha' m^2/4$	$-\alpha' m^2/4$
$(\bar\partial^{n_L}X)^{m_L}$	0	N_L
$(\partial^{n_{1R}}X)^{m_{1R}}$	N_{1R}	0
$(\partial^{n_{2R}}\phi^i)^{m_{2R}}$	N_{2R}	0
$e^{i\lambda_R\cdot\phi}$	$\lambda_R^2/2$	0
$e^{q\phi}$	$-q^2/2-q$	0
$V_{\rm int}$	$h_{\rm int}$	$\bar h_{\rm int}$

and the physical-state condition is

$$h(V)=\bar h(V)=1.$$

It follows that

$$\boxed{\begin{aligned}\frac{\alpha'}{2}m_L^2&=N_L+\bar h_{\rm int}-1,\\ \frac{\alpha'}{2}m_R^2&=\frac{1}{2}\lambda_R^2-\frac{1}{2}q^2-q+N_{1R}+N_{2R}+h_{\rm int}-1,\\ m^2&=m_L^2+m_R^2.\end{aligned}}$$

For a physical level-matched closed-string vertex,

$$\boxed{m_L^2=m_R^2=\frac{m^2}{2}.}$$

7.8 Q8: Momentum dependence of R–R vertices and derivative couplings

In page553-554 the discussion of type IIB (R, R) vertex operators. It is said that >The explicit k-dependence of the (R, R) vertex operators indicates that the corresponding massless fields have only derivative couplings in the lowenergy effective action. In other words they only appear through their field strength which implies that there are e.g. no fields charged with respect to the graviphoton.

Why the k-dependence of these vertex operators indicates this?

> **Textbook location and related material.** Chapter 15, Section 15.4 around pp. 553–554 and Eq. (15.84): type-IIB R–R vertices and their explicit momentum

factors. Section 13.2 gives the ten-dimensional R–R field-strength bispinor construction.

7.8.1 Answer

7.8.1.1 Core point

The relevant k-dependence is **not** the plane wave $e^{ik\cdot X}$, since every momentum-eigenstate vertex operator contains that factor. The important feature of equation (15.84) is the additional momentum multiplying the polarization:

$$V^A \propto \epsilon_\mu k_\nu (\text{spin fields})\, e^{ik\cdot X}, \qquad V^C \propto k_\mu (\text{spin fields})\, e^{ik\cdot X}, \qquad V^{\bar{C}} \propto k_\mu (\text{spin fields})\, e^{ik\cdot X}.$$

In Fourier space, multiplication by ik_μ is differentiation in spacetime:

$$\partial_\mu C(x) = \int \frac{d^4k}{(2\pi)^4}\, ik_\mu C(k) e^{ik\cdot x}.$$

Consequently, $k_\mu C(k)$ represents the one-form field strength dC, while the antisymmetric combination of $\epsilon_\mu k_\nu$ represents the graviphoton field strength $F_{\mu\nu} = 2\partial_{[\mu} A_{\nu]}$.

This is not an accidental feature of the chosen vertices. In the RNS formulation, a BRST-invariant R–R vertex is naturally labeled by an R–R **field-strength bispinor**, not directly by an R–R gauge potential. Equation (15.84) is the four-dimensional decomposition of that general statement.

7.8.1.2 The extra momentum in equation (15.84)

Suppressing the common superghost and plane-wave factors, the type-IIB vertices in (15.84) have the structure

$$\begin{aligned}
V^A &= \epsilon_\mu k_\nu \left[\bar{S}^\alpha \bar{\Sigma}\, \sigma^{\mu\nu}_\alpha{}^\beta S_\beta \Sigma + \bar{S}_{\dot\alpha} \bar{\Sigma}^\dagger\, \bar{\sigma}^{\mu\nu\dot\alpha}{}_{\dot\beta} S^{\dot\beta} \Sigma^\dagger\right] e^{-\frac{1}{2}(\phi+\bar\phi)} e^{ik\cdot X},\\
V^C &= k_\mu\, \bar{S}_{\dot\alpha} \bar{\Sigma}^\dagger \bar{\sigma}^{\mu\dot\alpha\beta} S_\beta \Sigma\, e^{-\frac{1}{2}(\phi+\bar\phi)} e^{ik\cdot X},\\
V^{\bar{C}} &= k_\mu\, \bar{S}^\alpha \bar{\Sigma} \sigma^\mu_{\alpha\dot\beta} S^{\dot\beta} \Sigma^\dagger\, e^{-\frac{1}{2}(\phi+\bar\phi)} e^{ik\cdot X}.
\end{aligned}$$

The spin fields and the internal spectral-flow operators determine the Lorentz representation and the internal R–R state. The factors $\epsilon_\mu k_\nu$ and k_μ determine which gauge-invariant spacetime tensor serves as the physical polarization.

For comparison, writing only

$$\epsilon_\mu (\text{spin fields})\, e^{ik\cdot X}$$

would describe the potential polarization itself. The book emphasizes that such a potential-level expression is not what appears in the BRST-invariant R–R vertex.

7.8.1.3 The graviphoton vertex is a field-strength vertex

The Lorentz generators appearing in the spin-field bilinears are antisymmetric:

$$\sigma^{\mu\nu} = -\sigma^{\nu\mu}, \qquad \bar\sigma^{\mu\nu} = -\bar\sigma^{\nu\mu}.$$

Therefore only the antisymmetric part of $\epsilon_\mu k_\nu$ contributes:

$$\epsilon_\mu k_\nu \sigma^{\mu\nu} = \frac{1}{2}\left(\epsilon_\mu k_\nu - \epsilon_\nu k_\mu\right)\sigma^{\mu\nu}.$$

With

$$F_{\mu\nu}(k) = i\left(k_\mu \epsilon_\nu - k_\nu \epsilon_\mu\right),$$

we obtain, up to the conventional Fourier factor,

$$\boxed{\epsilon_\mu k_\nu \sigma^{\mu\nu} \;\propto\; F_{\mu\nu}(k)\sigma^{\mu\nu}, \qquad \epsilon_\mu k_\nu \bar\sigma^{\mu\nu} \;\propto\; F_{\mu\nu}(k)\bar\sigma^{\mu\nu}.}$$

Thus V^A is really a vertex for the two chiral spinor components of $F_{\mu\nu}$, not for a bare A_μ.

In four-dimensional Lorentzian signature, $\sigma^{\mu\nu}$ and $\bar\sigma^{\mu\nu}$ select the two duality components

$$F^{\pm}_{\mu\nu} = \frac{1}{2}\left(F_{\mu\nu} \pm i\,{*F}_{\mu\nu}\right).$$

This is the reason for the two terms in V^A: as the book states, they encode the self-dual and anti-self-dual parts of the graviphoton field strength.

The vertex is also manifestly invariant under the linearized gauge transformation

$$\epsilon_\mu \longrightarrow \epsilon_\mu + a\,k_\mu,$$

because

$$k_\mu k_\nu \sigma^{\mu\nu} = k_\mu k_\nu \bar\sigma^{\mu\nu} = 0.$$

This is exactly the momentum-space version of

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda, \qquad F_{\mu\nu} \longrightarrow F_{\mu\nu}.$$

7.8.1.4 The scalar vertices contain dC and $d\bar C$

Equation (15.84) suppresses the scalar wavefunction, effectively setting its polarization to one. Restoring the Fourier coefficient $C(k)$, the prefactor of the vertex is $k_\mu C(k)$.

For a scalar mode,

$$C(x) = \int \frac{d^4k}{(2\pi)^4}\, C(k) e^{ik\cdot x},$$

so

$$k_\mu C(k) e^{ik\cdot x} = \frac{1}{i}\,\partial_\mu \left(C(k) e^{ik\cdot x}\right).$$

The tensor polarizations in the two scalar vertices are therefore

$$\mathcal{F}^C_\mu(k) \propto k_\mu C(k), \qquad \mathcal{F}^{\bar C}_\mu(k) \propto k_\mu \bar C(k).$$

In position space,

$$\boxed{\mathcal{F}^C_1 = dC, \qquad \mathcal{F}^{\bar C}_1 = d\bar C.}$$

A naive spin-field operator with a coefficient $C(k)$ but no k_μ would represent the R–R potential rather than its field strength. As the book says, that naive expression is not BRST invariant.

There is a useful direct check. The scalar bispinor polarization has the form

$$\mathcal{F}_{\dot\alpha\beta} \propto k_\mu \bar\sigma^\mu_{\dot\alpha\beta} C(k).$$

Acting once more with the momentum slash gives

$$k_\nu \sigma^\nu \mathcal{F} \propto k_\nu \sigma^\nu k_\mu \bar\sigma^\mu C(k) \propto k^2 C(k).$$

It vanishes for the massless state, $k^2 = 0$, which is precisely the form of the R–R BRST constraint.

7.8.1.5 Why BRST invariance gives a field-strength bispinor

The general ten-dimensional R–R vertex in the symmetric $(-\frac{1}{2}, -\frac{1}{2})$ picture has the schematic form

$$V_{\text{R–R}} = c\bar c\, e^{-\phi/2} S^A\, e^{-\bar\phi/2} \bar S^B\, \mathcal{F}_{AB}(k)\, e^{ik\cdot X}.$$

Here $\mathcal{F}_{AB}$ is a bispinor. To see the origin of its BRST constraint, recall that the BRST charge contains the worldsheet-supercurrent term

$$Q_{\text{BRST}} \supset \oint \frac{dz}{2\pi i}\, \gamma(z) T_F(z), \qquad T_F(z) \supset \psi_\mu \partial X^\mu(z).$$

The two OPEs responsible for the momentum constraint are

$$\partial X^\mu(z) e^{ik\cdot X(w)} \sim -\frac{i\alpha'}{2}\frac{k^\mu}{z-w} e^{ik\cdot X(w)},$$

and

$$\psi^\mu(z) S^A(w) \sim \frac{1}{(z-w)^{1/2}} (\Gamma^\mu)^A{}_C S^C(w).$$

Together with the superghost OPE, these produce an unwanted BRST singularity proportional to

$$k_\mu (\Gamma^\mu)^A{}_C \mathcal{F}_{AB}.$$

Its vanishing gives the first Dirac equation. Repeating the same calculation in the other chiral sector gives the second:

$$\boxed{k_\mu \Gamma^\mu \mathcal{F} = 0, \qquad \mathcal{F}\, k_\mu \Gamma^\mu = 0.}$$

Expand the bispinor in antisymmetrized gamma matrices:

$$\mathcal{F} = \sum_n \frac{1}{n!} F^{(n)}_{\mu_1\cdots\mu_n} \Gamma^{\mu_1\cdots\mu_n}.$$

For type IIB the sum contains the odd-degree R–R field strengths $F_1, F_3, F_5, ...$, with the appropriate duality relations. Gamma-matrix identities turn the two bispinor Dirac equations into

$$k \wedge F^{(n)} = 0, \qquad k \lrcorner F^{(n)} = 0.$$

In position space these are

$$dF^{(n)} = 0, \qquad d{*}F^{(n)} = 0,$$

namely the linearized Bianchi identity and equation of motion. Locally, the first equation implies

$$F^{(n)} = dC^{(n-1)}.$$

Equivalently, in momentum space,

$$\boxed{F^{(n)}(k) = i\, k \wedge C^{(n-1)}(k).}$$

The explicit k in (15.84) is therefore the compactified remnant of the universal relation $F = dC$.

7.8.1.6 From a momentum factor to a derivative coupling

Consider a local term in the low-energy effective action that is linear in an R–R scalar:

$$S_{\text{int}} = \int d^4x\, \partial_\mu C(x)\, \mathcal{O}^\mu(x).$$

After Fourier transformation, its vertex contains

$$ik_\mu C(k)\, \mathcal{O}^\mu.$$

Conversely, if every local amplitude with one external C state contains an explicit $k_\mu C(k)$, then the corresponding local interaction contains a derivative on C. A constant mode,

$$C(x) = C_0,$$

has $dC = 0$ and drops out of such perturbative local couplings. This is the perturbative shift symmetry

$$C \longrightarrow C + C_0.$$

For the vector, the analogous local interactions are built from

$$F_{\mu\nu} = 2\partial_{[\mu} A_{\nu]},$$

for example the kinetic term

$$S_A = -\frac{1}{4g_A^2} \int d^4x \, F_{\mu\nu} F^{\mu\nu}.$$

An external A_μ leg extracted from an interaction expressed through $F_{\mu\nu}$ necessarily brings down a factor

$$i(k_\mu \epsilon_\nu - k_\nu \epsilon_\mu).$$

This is exactly the structure exhibited by V^A.

There is a small S-matrix qualification. The argument applies directly to **local contact terms**, or equivalently to the local one-particle-irreducible low-energy action. A complete amplitude can contain a massless exchange pole proportional to $1/k$ that compensates a soft momentum. Such a nonlocal pole describes propagation through intermediate states, not a local nonderivative R–R coupling.

7.8.1.7 Why this excludes perturbative matter charged under the graviphoton

If a field Ψ had charge Q under the graviphoton, its kinetic term would contain the covariant derivative

$$D_\mu \Psi = \left(\partial_\mu - iQA_\mu\right) \Psi.$$

Expanding $|D_\mu \Psi|^2$ gives a minimal coupling

$$S_{\text{min}} \supset \int d^4x \, A_\mu J^\mu,$$

where, for a charged complex scalar,

$$J^\mu = iQ \left(\Psi^* \partial^\mu \Psi - \Psi \partial^\mu \Psi^*\right).$$

The corresponding three-point vertex contains the graviphoton polarization ϵ_μ itself:

$$\mathcal{A}_3 \propto \epsilon_\mu J^\mu.$$

It is not forced to carry the graviphoton momentum in the antisymmetric combination $k_{[\mu} \epsilon_{\nu]}$. Therefore a minimal coupling requires the potential A_μ, rather than only $F_{\mu\nu}$.

The perturbative closed-string R–R vertex, however, supplies only $F_{\mu\nu}$. Accordingly, the perturbative bulk spectrum has no matter field with a minimal electric coupling to this R–R graviphoton. This is the precise content of the book's statement that there are no fields charged with respect to the graviphoton.

This conclusion is stronger than the ordinary Ward identity. Gauge invariance alone permits

$$\int A_\mu J^\mu$$

when $\partial_\mu J^\mu = 0$. What excludes the coupling here is the string-theoretic fact that the perturbative R–R state is represented by a field-strength vertex and that perturbative fundamental closed strings carry no R–R charge.

Neutral fields may still interact with the graviphoton through field-strength couplings, for example Pauli-type terms schematically of the form

$$F_{\mu\nu} \bar{\psi} \Gamma^{\mu\nu} \psi.$$

Such an interaction does not make ψ electrically charged: the distinction is between an $F_{\mu\nu}$ coupling and a minimal $A_\mu J^\mu$ coupling.

7.8.1.8 Important scope of the statement

The statement is about the perturbative closed-string bulk theory being discussed in this section. It does not mean that type-II string theory has no R–R-charged objects.

D-branes carry R–R charge and have Wess–Zumino couplings

$$S_{\mathrm{WZ}} = \mu_p \int_{D_p} \mathbf{C}_{\mathrm{RR}} \wedge e^{2\pi\alpha' F_{\mathrm{wv}} + B}.$$

After compactification, a wrapped D-brane can become a particle charged under a four-dimensional R–R vector, including a graviphoton in an appropriate compactification. Such states are nonperturbative from the viewpoint of the closed-string genus expansion. Their potential-level coupling is therefore not contradicted by the perturbative sphere-vertex argument. In the RNS description of D-brane couplings, one must also treat superghost pictures and boundaries carefully; asymmetric-picture R–R vertices can display the potential rather than only the symmetric-picture field strength.

Likewise, supergravity may contain gauge-invariant Chern–Simons terms or modified field strengths, schematically

$$\widetilde{F} = dC + \text{terms involving other form potentials.}$$

An action in a particular duality frame can then contain an undifferentiated potential in a topological term while remaining gauge invariant up to a total derivative. Nonperturbative effects can also break a continuous R–R axion shift symmetry to a discrete subgroup. Thus "only derivative couplings" should be read as the linearized perturbative bulk statement encoded by the physical R–R vertex, not as a prohibition of every potential-level or nonperturbative coupling.

7.8.1.9 Checks

Under a graviphoton gauge transformation,

$$\epsilon_\mu \mapsto \epsilon_\mu + a k_\mu,$$

the vertex changes by

$$\delta V^A \propto a\, k_\mu k_\nu \sigma^{\mu\nu} + a\, k_\mu k_\nu \bar{\sigma}^{\mu\nu} = 0.$$

Thus the explicit vertex depends only on the gauge-equivalence class of A_μ.

In the soft limit,

$$k_\mu \longrightarrow 0,$$

the local vertices $V^A, V^C, V^{\bar{C}}$ vanish at least linearly in k. A nonzero constant graviphoton potential is locally pure gauge, and a constant R–R scalar has zero perturbative field strength. Both conclusions agree with the spacetime interpretation.

Finally, the two terms in V^A contain $\sigma^{\mu\nu}$ and $\bar{\sigma}^{\mu\nu}$. They therefore reconstruct the two duality components F^+ and F^- of one real graviphoton field strength, rather than describing two independent vector potentials.

7.8.1.10 Result

The extra momentum dependence in (15.84) has the direct spacetime meaning

$$\boxed{\begin{aligned} &\epsilon_\mu k_\nu \sigma^{\mu\nu} \propto F^+_{\mu\nu}\sigma^{\mu\nu}, \\ &\epsilon_\mu k_\nu \bar{\sigma}^{\mu\nu} \propto F^-_{\mu\nu}\bar{\sigma}^{\mu\nu}, \\ &k_\mu C(k) \longleftrightarrow \frac{1}{i}\,\partial_\mu C(x), \\ &k_\mu \bar{C}(k) \longleftrightarrow \frac{1}{i}\,\partial_\mu \bar{C}(x). \end{aligned}}$$

The general R–R spin-field polarization obeys

$$\boxed{\not{k}\,\mathcal{F} = 0, \qquad \mathcal{F}\,\not{k} = 0, \qquad \mathcal{F} = \sum_n \frac{1}{n!} F^{(n)}\Gamma_{(n)}, \qquad F^{(n)} = dC^{(n-1)}.}$$

Therefore the physical R–R vertices describe

$$\boxed{\text{R–R field strengths, not bare R–R potentials.}}$$

For the perturbative local bulk action this means

$$\boxed{C, \bar{C} \text{ enter through } dC, d\bar{C}, \qquad A_\mu \text{ enters through } F_{\mu\nu},}$$

so there is no perturbative minimal coupling $A_\mu J^\mu$ and hence no perturbative matter charged under the R–R graviphoton. R–R-charged D-branes and gauge-invariant topological terms lie outside this restricted statement.

7.9 Q9: BRST invariance, current superfields, and picture changing

In page 556: >In the latter case, BRST invariance demands that $J^a(z)$ is the upper component of a two-dimensional superfield, i.e. it can be obtained via picture changing from a dimension $\frac{1}{2}$ (fermionic) field $\hat{J}^a(z)$.

Then they derive the vertex operator (15.91) and (15.92).

I have no idea how to derive this from the BRST invariance. Show the explicit computation.

> **Textbook location and related material.** Chapter 15, Section 15.4 around p. 556 and Eqs. (15.91)–(15.92): heterotic gauge-current vertices, superfield components, BRST invariance, and picture changing. See also Sections 12.1 and 13.2.

7.9.1 Answer

7.9.1.1 Core point

The statement follows from the γG term in the RNS BRST charge. For an integrated picture-zero operator $W(z)$ of weight one, BRST invariance does not require $G(z)W(w)$ to vanish. It requires the BRST variation to be a total derivative. If

$$G(z)W(w) \sim \frac{A(w)}{(z-w)^2} + \frac{B(w)}{z-w} + \text{finite},$$

then the γG term in Q_{B} produces

$$(\partial\gamma)A + \gamma B.$$

This is a total derivative precisely when

$$B = \partial A,$$

because then

$$(\partial\gamma)A + \gamma\partial A = \partial(\gamma A).$$

The $N = 1$ superconformal algebra supplies exactly such a pair:

$$A = \hat{J}^a, \qquad W = J^a = G_{-1/2}\hat{J}^a,$$

where $\hat{J}^a$ is a weight-1/2 superprimary. Indeed,

$$G(z)J^a(w) \sim \frac{\hat{J}^a(w)}{(z-w)^2} + \frac{\partial\hat{J}^a(w)}{z-w}.$$

Therefore the γG contribution to the BRST variation is

$$\partial\left(\gamma\hat{J}^a\right),$$

which integrates to zero. This is the explicit reason that the weight-one current must be the upper component of a weight-1/2 superfield.

There is also an important finite-momentum subtlety. Acting with the **full** picture-changing operator on (15.91) produces not only J^a, but also a term proportional to

$$(k \cdot \psi)\hat{J}^a.$$

Thus the complete finite-momentum picture-zero matter vertex is

$$\epsilon_\mu \bar{\partial} X^\mu \left[J^a + c_X (k \cdot \psi)\hat{J}^a\right] e^{ik\cdot X},$$

where c_X is convention dependent. Equation (15.92) displays only the first term. It is exact at $k = 0$, and it can be read as shorthand for the internal-current part, but at nonzero momentum the omitted term is required by a literal BRST/picture-changing computation.

7.9.1.2 The relevant part of the heterotic BRST charge

The right-moving, holomorphic half of the heterotic string is an RNS superstring. Bosonizing

$$\beta = e^{-\phi}\partial\xi, \qquad \gamma = \eta e^{\phi},$$

the BRST charge contains

$$Q_{\mathrm{B}} = \oint \frac{dz}{2\pi i}\left[c\left(T_{\mathrm{m}} + T_{\mathrm{gh}}\right) + \gamma G_{\mathrm{m}} - \frac{1}{4} b\gamma^2 + \cdots\right].$$

Only two parts are needed for the present argument:

- cT imposes the conformal-weight and Virasoro-primary conditions;
- γG_{m} imposes the worldsheet-supersymmetry constraint.

The matter supercurrent splits into external and internal pieces,

$$G_{\mathrm{m}} = G_{\mathrm{st}} + G_{\mathrm{int}},$$

with

$$G_{\mathrm{st}} \propto \psi_\mu \partial X^\mu.$$

The antiholomorphic half of the heterotic string is bosonic. Its BRST conditions give the usual spacetime mass-shell and transversality constraints

$$k^2 = 0, \qquad k^\mu \epsilon_\mu = 0.$$

The new condition discussed around (15.91) comes from the holomorphic γG_{m} term.

7.9.1.3 The weight-$1/2$ lower component

Let $\hat{J}^a$ be an internal NS field satisfying

$$h(\hat{J}^a) = \frac{1}{2}.$$

For it to be an $N = 1$ superconformal primary, its state must obey

$$L_n|\hat{J}^a\rangle = 0 \quad (n > 0), \qquad G_r|\hat{J}^a\rangle = 0 \quad (r > 0).$$

Its upper component is defined by

$$\boxed{|J^a\rangle = G^{\mathrm{int}}_{-1/2}|\hat{J}^a\rangle.}$$

The commutator

$$[L_0, G_{-1/2}] = \frac{1}{2} G_{-1/2}$$

immediately gives

$$h(J^a) = h(\hat{J}^a) + \frac{1}{2} = 1.$$

Thus

$$\mathcal{J}^a(z,\theta) = \hat{J}^a(z) + \theta J^a(z)$$

is a fermionic $N = 1$ current superfield with lower component of weight $1/2$ and upper component of weight one.

7.9.1.4 Deriving the two supercurrent OPEs

The NS $N = 1$ superconformal algebra is

$$\{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}.$$

The definition $J^a = G_{-1/2}\hat{J}^a$ gives the first component OPE

$$\boxed{G(z)\hat{J}^a(w) \sim \frac{J^a(w)}{z-w}.}$$

To obtain the OPE with the upper component, act with $G_{1/2}$:

$$\begin{aligned} G_{1/2}|J^a\rangle &= G_{1/2}G_{-1/2}|\hat{J}^a\rangle \\ &= \{G_{1/2}, G_{-1/2}\}|\hat{J}^a\rangle - G_{-1/2}G_{1/2}|\hat{J}^a\rangle \\ &= 2L_0|\hat{J}^a\rangle \\ &= |\hat{J}^a\rangle. \end{aligned}$$

Here $G_{1/2}|\hat{J}^a\rangle = 0$ and $2h(\hat{J}^a) = 1$. Similarly,

$$G_{-1/2}|J^a\rangle = G_{-1/2}^2|\hat{J}^a\rangle = L_{-1}|\hat{J}^a\rangle.$$

Through the state-operator correspondence, these two relations give

$$\boxed{G(z)J^a(w) \sim \frac{\hat{J}^a(w)}{(z-w)^2} + \frac{\partial\hat{J}^a(w)}{z-w}.}$$

This is the component transformation law of an $N = 1$ superfield whose lower component has weight 1/2.

7.9.1.5 Explicit BRST variation of a picture-zero current

Consider first the internal operator $J^a(w)$ at zero spacetime momentum. The supercurrent part of its BRST variation is

$$\left[\oint_w \frac{dz}{2\pi i}\,\gamma(z)G(z), J^a(w)\right].$$

Expand

$$\gamma(z) = \gamma(w) + (z-w)\partial\gamma(w) + \cdots$$

and insert the OPE just derived:

$$\gamma(z)G(z)J^a(w) \sim \gamma(z)\left[\frac{\hat{J}^a(w)}{(z-w)^2} + \frac{\partial\hat{J}^a(w)}{z-w}\right].$$

The contour residue is

$$\begin{aligned} \delta_{\gamma G}J^a &= (\partial\gamma)\hat{J}^a + \gamma\partial\hat{J}^a \\ &= \boxed{\partial\big(\gamma\hat{J}^a\big)\,.} \end{aligned}$$

The cT term likewise gives a total derivative for a weight-one primary:

$$\delta_{cT}J^a = c\,\partial J^a + (\partial c)J^a = \partial(cJ^a).$$

Consequently,

$$[Q_{\mathrm{B}}, J^a] = \partial\big(cJ^a + \gamma\hat{J}^a\big)$$

up to terms that vanish in the small Hilbert space or belong to the standard ghost completion. Therefore

$$\boxed{\left[Q_{\mathrm{B}}, \int dz\, J^a(z)\right] = 0.}$$

This is the promised explicit BRST computation.

7.9.1.6 Why an arbitrary weight-one current fails

Now suppose that K^a is merely a bosonic weight-one field, without assuming it belongs to a superfield. Write its most general relevant supercurrent OPE as

$$G(z)K^a(w) \sim \frac{A^a(w)}{(z-w)^2} + \frac{B^a(w)}{z-w} + \cdots.$$

The same contour calculation gives

$$\delta_{\gamma G} K^a = (\partial\gamma)A^a + \gamma B^a.$$

This is a total derivative only if

$$B^a = \partial A^a.$$

Dimensional analysis gives

$$h(A^a) = \frac{1}{2}.$$

Calling $A^a = \hat{J}^a$, the required OPE becomes

$$G(z)K^a(w) \sim \frac{\hat{J}^a(w)}{(z-w)^2} + \frac{\partial\hat{J}^a(w)}{z-w}.$$

By the $N=1$ algebra, this is precisely the OPE of the upper component

$$K^a = G_{-1/2}\hat{J}^a,$$

up to possible null or BRST-trivial operators. Thus a generic Kac–Moody current of weight one is not sufficient: it must extend to a super-Kac–Moody multiplet.

7.9.1.7 Deriving the canonical vertex (15.91)

The complete integrated canonical-picture matter vertex is

$$\boxed{V^a_{-1} = \epsilon_\mu\, \bar{\partial}X^\mu(\bar{z})\, \hat{J}^a(z)\, e^{-\phi(z)} e^{ik\cdot X(z,\bar{z})}.}$$

Its conformal weights are easily checked. In the right-moving sector,

$$h(\hat{J}^a) = \frac{1}{2}, \qquad h(e^{-\phi}) = \frac{1}{2}, \qquad h(e^{ik\cdot X}) = 0 \quad (k^2=0),$$

so

$$h(V^a_{-1}) = 1.$$

In the left-moving sector,

$$\bar{h}(\bar{\partial}X^\mu) = 1, \qquad \bar{h}(e^{ik\cdot X}) = 0,$$

so

$$\bar{h}(V^a_{-1}) = 1.$$

The corresponding unintegrated representative is

$$U^a_{-1} = c\bar{c}\, V^a_{-1}.$$

The cT part of BRST closure gives $k^2 = 0$ and the left-moving vector condition $k \cdot \epsilon = 0$. On the right, the factor $e^{-\phi}$ supplies half a unit of weight, while the superprimary $\hat{J}^a$ supplies the other half. Because $G(z)\hat{J}^a(w)$ has at most the allowed simple pole, the γG term produces no uncancelled BRST residue in picture -1.

This derives equation (15.91).

7.9.1.8 Picture changing to picture zero

The picture-changing operator is

$$\mathcal{X}(z) = \{Q_{\mathrm{B}}, \xi(z)\} = e^{\phi} G_{\mathrm{m}} + \text{ghost terms}.$$

For the integrated matter vertex, the displayed term supplies the matter part of the picture-zero representative; the omitted ghost terms contribute BRST-exact pieces or the ghost completion of an unintegrated operator.

The superghost OPE is

$$e^{\phi(z)} e^{-\phi(w)} \sim (z-w).$$

The internal supercurrent gives

$$G_{\mathrm{int}}(z) \hat{J}^a(w) \sim \frac{J^a(w)}{z-w}.$$

Their product has a finite coincident limit:

$$e^{\phi} G_{\mathrm{int}}\, \hat{J}^a e^{-\phi} \longrightarrow J^a.$$

This is the internal term shown in equation (15.92).

However, the full supercurrent also acts on the plane wave. Using

$$\partial X^\mu(z) e^{ik\cdot X(w,\bar{w})} \sim -\frac{i\alpha'}{2} \frac{k^\mu}{z-w} e^{ik\cdot X(w,\bar{w})},$$

one finds

$$G_{\mathrm{st}}(z) e^{ik\cdot X(w,\bar{w})} \sim \frac{c_X\, k\cdot\psi(w)}{z-w} e^{ik\cdot X(w,\bar{w})},$$

where c_X contains the convention-dependent factor of $i\alpha'$ and the normalization of G_{st}. Hence

$$e^{\phi} G_{\mathrm{st}}\, \hat{J}^a e^{-\phi} e^{ik\cdot X} \longrightarrow c_X (k\cdot\psi) \hat{J}^a e^{ik\cdot X}.$$

The complete matter result of picture changing is therefore

$$\boxed{V_0^a = \epsilon_\mu \bar{\partial} X^\mu \left[J^a + c_X (k\cdot\psi) \hat{J}^a \right] e^{ik\cdot X}.}$$

Equivalently, if

$$\mathcal{U}^a = \hat{J}^a e^{ik\cdot X},$$

then the bracket is the full upper component

$$\boxed{\mathcal{W}^a = G_{-1/2}^{\mathrm{tot}} \mathcal{U}^a = J^a e^{ik\cdot X} + c_X (k\cdot\psi) \hat{J}^a e^{ik\cdot X}.}$$

The superconformal algebra then guarantees

$$G(z) \mathcal{W}^a(w) \sim \frac{\mathcal{U}^a(w)}{(z-w)^2} + \frac{\partial \mathcal{U}^a(w)}{z-w},$$

because $h(\mathcal{U}^a) = 1/2$ on shell. Its BRST variation is therefore

$$\delta_{\gamma G} \mathcal{W}^a = \partial(\gamma \mathcal{U}^a).$$

7.9.1.9 How equation (15.92) should be read

Equation (15.92) prints

$$V^a_{\mathrm{book},0} = \epsilon_\mu \bar{\partial} X^\mu J^a e^{ik\cdot X}.$$

At zero momentum this agrees exactly with the full result:

$$k^\mu = 0 \quad \Longrightarrow \quad V_0^a = \epsilon_\mu \bar{\partial} X^\mu J^a.$$

This is the limit in which J^a itself is the conserved worldsheet current generating the spacetime gauge symmetry.

At nonzero momentum, a literal application of the full picture-changing operator produces the additional term

$$c_X\, \epsilon_\mu \bar{\partial} X^\mu (k\cdot\psi)\hat{J}^a e^{ik\cdot X}.$$

It is not removed merely by $k\cdot\epsilon = 0$, because k contracts with ψ^ν whereas ϵ_μ contracts with the independent left-moving oscillator $\bar{\partial}X^\mu$.

This is also visible by comparing with equation (15.87). There the book picture-changes a canonical vertex with lower component Λ^i and obtains

$$\left[\Lambda^i k\cdot\psi + 2\hat{\Lambda}^i\right] e^{ik\cdot X}.$$

The same calculation with

$$\Lambda^i \longrightarrow \hat{J}^a, \qquad 2\hat{\Lambda}^i \longrightarrow J^a$$

necessarily gives

$$\left[(k\cdot\psi)\hat{J}^a + J^a\right] e^{ik\cdot X},$$

up to normalization.

Therefore the precise conclusion is:

1. equation (15.91) is the complete canonical -1-picture matter vertex;
2. the J^a term in (15.92) is derived from the internal part of picture changing;
3. as printed, (15.92) is exact at $k=0$ or is shorthand for the internal-current contribution;
4. at finite k, BRST invariance requires the full upper component of $\hat{J}^a e^{ik\cdot X}$, including $(k\cdot\psi)\hat{J}^a$.

7.9.1.10 Checks

The lower component has the correct weight for picture -1:

$$h(\hat{J}^a) + h(e^{-\phi}) = \frac{1}{2} + \frac{1}{2} = 1.$$

The two terms in the full picture-zero bracket both have weight one:

$$h(J^a) = 1, \qquad h((k\cdot\psi)\hat{J}^a) = \frac{1}{2} + \frac{1}{2} = 1.$$

At $k=0$, the momentum-dependent completion vanishes and the BRST computation reduces to

$$[Q_{\mathrm{B}}, J^a] = \partial(cJ^a + \gamma\hat{J}^a),$$

so the integrated current is BRST invariant.

If J^a were not an upper component, its OPE with G would give

$$G(z)J^a(w) \sim \frac{A^a}{(z-w)^2} + \frac{B^a}{z-w}, \qquad B^a \neq \partial A^a,$$

and the uncancelled variation

$$(\partial\gamma)A^a + \gamma B^a$$

would not be a total derivative. The corresponding integrated operator would therefore fail BRST invariance, as the book states.

7.9.1.11 Result

For a weight-1/2 $N = 1$ superprimary $\hat{J}^a$,

$$\boxed{\begin{aligned} J^a &= G^{\text{int}}_{-1/2}\hat{J}^a, \\ G(z)\hat{J}^a(w) &\sim \frac{J^a(w)}{z-w}, \\ G(z)J^a(w) &\sim \frac{\hat{J}^a(w)}{(z-w)^2} + \frac{\partial\hat{J}^a(w)}{z-w}. \end{aligned}}$$

The last OPE implies

$$\boxed{\delta_{\gamma G} J^a = (\partial\gamma)\hat{J}^a + \gamma\partial\hat{J}^a = \partial(\gamma\hat{J}^a),}$$

which is the explicit BRST derivation of the superfield condition.

The canonical vertex is

$$\boxed{V^a_{-1} = \epsilon_\mu \bar{\partial} X^\mu \hat{J}^a e^{-\phi} e^{ik\cdot X}, \qquad k^2 = k\cdot\epsilon = 0.}$$

Full picture changing gives

$$\boxed{V^a_0 = \epsilon_\mu \bar{\partial} X^\mu \left[J^a + c_X (k\cdot\psi)\hat{J}^a\right] e^{ik\cdot X}.}$$

Thus the book's displayed expression

$$V^a_{\text{book},0} = \epsilon_\mu \bar{\partial} X^\mu J^a e^{ik\cdot X}$$

is the $k = 0$ form, or the internal-current part, of the complete picture-zero representative. The requirement that J^a be the upper component of $\hat{J}^a$ is exactly what makes the γG BRST variation a total derivative.

Chapter 8

String Scattering Amplitudes and Low-Energy Effective Field Theory

8.1 Q1: On-shell string amplitudes and off-shell effective vertices

When computing string scattering amplitudes by computing the correlation functions of vertex operators, we can only compute on-shell scattering amplitudes since BRST-invariance forces the vertex operators to be on-shell. In the resulting low-energy effective action, what is an off-shell amplitude? Why these off-shell amplitudes are also important that we cannot neglect them?

> **Textbook location and related material.** Chapter 16, Sections 16.1 and 16.3 (pp. 585–588, 615–622): on-shell string amplitudes, low-energy expansions, field redefinitions, and construction of an effective action. BRST on-shell conditions come from Section 13.2.

8.1.1 Answer

8.1.1.1 Core point

The phrase **off-shell amplitude** is somewhat loose. An amplitude in the strict physical sense is an S-matrix element, and its external states are on shell. What the low-energy field theory possesses away from the mass shell are instead **off-shell Green functions** and **off-shell vertices**. They are obtained from the action without imposing the external equations of motion. Only after one computes connected Green functions, amputates their external propagators, takes the external momenta to their poles, and contracts with physical polarizations does one obtain the on-shell S-matrix.

Thus there is no contradiction:

$$\boxed{\text{first-quantized string theory computes the on-shell } S\text{-matrix}}$$

whereas

$$\boxed{S_{\text{eff}}[\Phi] \text{ is an off-shell functional whose on-shell } S\text{-matrix agrees with string theory.}}$$

The second box does **not** mean that the string S-matrix uniquely determines every term in S_{eff}. It determines the effective action only up to transformations that leave the on-shell S-matrix invariant, most importantly local field redefinitions, gauge-fixing choices, total derivatives, and operators proportional to the lower-order equations of motion.

This is precisely the ambiguity mentioned by the book in §16.1. The book explicitly states that BRST invariance restricts the scattered string states to be on shell and that this produces ambiguities in reconstructing the low-energy action. Section 16.3 then gives the practical matching prescription:

construct $\mathcal{L}_{2\mathrm{pt}}$, match the three-point amplitudes with $\mathcal{L}_{3\mathrm{pt}}$, and at four points subtract the massless exchange graphs already generated by $\mathcal{L}_{3\mathrm{pt}}$; the remaining analytic terms determine local interactions in $\mathcal{L}_{4\mathrm{pt}}$. The action so obtained is one convenient off-shell representative of an entire on-shell equivalence class.

Off-shell information is nevertheless indispensable for using a field theory. It supplies the equations of motion, propagators, interaction vertices attached to internal lines, response to sources and backgrounds, and the starting point for loop calculations. The precise statement is therefore not that every off-shell vertex is an independent physical datum that must be retained. Rather:

> a conventional local effective-action description requires off-shell data,
> but one must not assign invariant meaning to a particular off-shell continuation.

8.1.1.2 From an action to off-shell vertices

Let $\Phi^I(x)$ be the massless spacetime fields. Around a chosen background, write the effective action schematically as

$$S_{\mathrm{eff}}[\Phi] = S_2[\Phi] + S_3[\Phi] + S_4[\Phi] + \cdots,$$

where S_n contains terms of order Φ^n. Its quadratic part has the momentum-space form

$$S_2 = \frac{1}{2}\int \frac{d^dk}{(2\pi)^d}\, \Phi^I(-k) K_{IJ}(k) \Phi^J(k).$$

After gauge fixing, the inverse of K_{IJ} is the propagator. For a scalar,

$$K(k) \sim k^2 + m^2, \qquad G(k) \sim \frac{i}{k^2 + m^2 - i\epsilon}.$$

The physical mass shell is the zero locus of the inverse propagator,

$$K(k) = 0.$$

For tensor or gauge fields this equation must be supplemented by the physical-polarization conditions. For example, a massless vector satisfies

$$k^2 = 0, \qquad k \cdot \epsilon = 0, \qquad \epsilon_\mu \sim \epsilon_\mu + k_\mu \lambda.$$

The n-point 1PI vertex is obtained by differentiating the effective action:

$$\Gamma^{(n)}_{I_1\cdots I_n}(x_1, \dots, x_n) = \left.\frac{\delta^n \Gamma}{\delta\Phi^{I_1}(x_1)\cdots\delta\Phi^{I_n}(x_n)}\right|_{\Phi=\Phi_0}.$$

After Fourier transformation, translation invariance gives

$$\Gamma^{(n)}(k_1, \dots, k_n) = (2\pi)^d \delta^{(d)}\left(\sum_{i=1}^{n} k_i\right) \hat{\Gamma}^{(n)}(k_1, \dots, k_n).$$

Momentum conservation is already imposed here, but the individual momenta need not satisfy

$$k_i^2 = -m_i^2.$$

The function $\hat{\Gamma}^{(n)}$ at such general momenta is what is often called an off-shell amplitude. More accurately, it is an amputated off-shell 1PI vertex. It depends on the chosen fields and, in gauge theories, on gauge fixing.

The physical scattering amplitude is obtained only after restricting the external momenta to the poles of the exact propagators and contracting the external indices with physical wave functions. Schematically, the LSZ relation has the form

$$\mathcal{A}_n \sim \lim_{K_i(k_i)\to 0} \prod_{i=1}^{n} \left[Z_i^{-1/2} K_i(k_i) \right] G^{(n)}_{\mathrm{conn}}(k_1, \dots, k_n),$$

where $G^{(n)}_{\mathrm{conn}}$ is the connected Green function before amputation. The factors K_i remove the external propagators, and the final restriction puts the external particles on shell. For gauge fields, only the BRST cohomology classes of external polarizations survive.

This field-theory distinction mirrors the worldsheet statement. A BRST-invariant string vertex operator satisfies

$$Q_{\mathrm{B}} V_i = 0,$$

which includes the target-space mass-shell and polarization constraints. A BRST-exact change,

$$V_i \longrightarrow V_i + Q_{\mathrm{B}} \Lambda_i,$$

decouples from the physical amplitude. Consequently, the ordinary worldsheet correlator directly produces the analogue of the final, LSZ-reduced quantity, not a unique function $\hat{\Gamma}^{(n)}$ at arbitrary k_i^2.

8.1.1.3 Why internal lines are off shell

Even when all external particles are on shell, the momenta carried by internal lines are generally off shell. Consider a four-point process with external momenta k_1, k_2, k_3, k_4 and an s-channel internal momentum

$$q = k_1 + k_2, \qquad s = -q^2$$

in the mostly-plus convention.

The external conditions $k_i^2 = -m_i^2$ do not imply

$$q^2 = -m_{\mathrm{int}}^2.$$

The field-theory exchange contribution therefore uses cubic vertices evaluated with one off-shell leg:

$$\mathcal{A}_4^{(s)\text{-exchange}} = V_3(k_1, k_2, -q) \frac{i}{K(q)} V_3(q, k_3, k_4).$$

There are analogous t- and u-channel terms. The full tree-level field-theory amplitude is

$$\mathcal{A}_4^{\mathrm{EFT}} = \mathcal{A}_4^{(s)\text{-exchange}} + \mathcal{A}_4^{(t)\text{-exchange}} + \mathcal{A}_4^{(u)\text{-exchange}} + \mathcal{A}_4^{\text{contact}}.$$

Only this complete sum, with its external legs put on shell, is required to agree with the string amplitude.

This is the content of the matching procedure described in §16.3. The poles of the string four-point amplitude due to massless states must be reproduced by exchange diagrams constructed from the lower-point effective interactions. Once those poles are subtracted, the part analytic in the external momenta is expanded as

$$\mathcal{A}_{4,\text{analytic}} = c_0 + c_2\alpha' k^2 + c_4(\alpha' k^2)^2 + \cdots,$$

with the kinematic tensors understood. These terms are represented by local contact operators in $\mathcal{L}_{4\text{pt}}$ of the schematic form

$$D^n F^m, \qquad m \le 4,$$

with the appropriate powers of α' fixed by dimensions. As the book notes below equation (16.130), terms with $m > 4$ require amplitudes with more than four external gauge fields.

Equivalently, the full string amplitude contains massive-string poles such as

$$\frac{1}{q^2 + M_N^2}, \qquad M_N^2 \sim \frac{N}{\alpha'}.$$

At $|q^2| \ll M_N^2$, a massive propagator admits the expansion

$$\frac{1}{q^2 + M_N^2} = \frac{1}{M_N^2}\left(1 - \frac{q^2}{M_N^2} + \frac{q^4}{M_N^4} - \cdots\right).$$

Each term becomes a local higher-derivative interaction among the light fields. This is how an on-shell string amplitude determines the derivative expansion of an off-shell low-energy action.

8.1.1.4 Field redefinitions and the non-uniqueness of off-shell vertices

The central subtlety can be seen with a scalar field. Let

$$S_2[\phi] = \frac{1}{2}\int \frac{d^dk}{(2\pi)^d}\,\phi(-k)K(k)\phi(k), \qquad K(k) = k^2 + m^2,$$

and add the local cubic operator

$$\Delta S = \frac{a}{\Lambda^p}\int d^dx\,\phi^2 K\phi.$$

Here $K\phi = 0$ is the free equation of motion. Upon symmetrizing the three identical external legs, this operator changes the cubic off-shell vertex by a term proportional to

$$\Delta V_3(k_1, k_2, k_3) \propto \frac{a}{\Lambda^p}\left[K(k_1) + K(k_2) + K(k_3)\right].$$

If all three external legs are on shell, then $K(k_i) = 0$, and hence

$$\Delta V_3|_{\text{on shell}} = 0.$$

The string three-point S-matrix therefore cannot determine the coefficient a. Nevertheless, if the third leg is an internal line carrying momentum q, then

$$\Delta V_3(k_1, k_2, q) \propto K(q)$$

for on-shell k_1 and k_2. In an exchange graph this factor can cancel the propagator:

$$\Delta V_3(k_1, k_2, -q)\frac{i}{K(q)}V_3(q, k_3, k_4) \propto iV_3(q, k_3, k_4).$$

The resulting contribution is local and has the form of a contact interaction. Thus changing an off-shell cubic vertex changes the division of a four-point amplitude into

$$\text{exchange part} + \text{contact part}.$$

But it need not change their on-shell sum. Indeed, perform the local field redefinition

$$\phi \longrightarrow \phi - \frac{a}{\Lambda^p}\phi^2.$$

To first order in a, its change of the quadratic action is

$$\delta S_2 = \int d^dx\, \frac{\delta S_2}{\delta\phi}\,\delta\phi = -\frac{a}{\Lambda^p}\int d^dx\,(K\phi)\phi^2,$$

which removes ΔS. At the same time, this redefinition changes the quartic and higher interactions already present in the action. Those induced changes precisely compensate the altered exchange graphs. The equivalence theorem therefore gives

$$S_{\text{eff}}[\phi] \quad \text{and} \quad S'_{\text{eff}}[\phi'] \qquad \Longrightarrow \qquad \mathcal{S}[S_{\text{eff}}] = \mathcal{S}[S'_{\text{eff}}]$$

for local, perturbatively invertible field redefinitions, with the usual consistent treatment of wave-function normalization and quantum Jacobians.

More generally, if

$$E_I(\Phi) = \frac{\delta S_{\text{eff}}}{\delta\Phi^I},$$

then an infinitesimal field redefinition

$$\Phi^I \longrightarrow \Phi^I - F^I(\Phi, \partial\Phi, \ldots)$$

changes the action by

$$\Delta S_{\text{eff}} = -\int d^dx\, F^I E_I + O(F^2).$$

Therefore operators proportional to the equations of motion are redundant in the on-shell operator basis. They can change off-shell vertices while leaving the physical S-matrix unchanged.

This also explains the book's discussion around equation (16.116). A Weyl rescaling of the metric together with field redefinitions converts the Einstein-frame action into the string-frame action. The two actions have different off-shell couplings—for example, the dilaton is distributed differently among the kinetic terms—but describe the same perturbative on-shell physics when fields and external states are translated consistently.

8.1.1.5 What on-shell string amplitudes do and do not determine

Suppose two local low-energy actions satisfy

$$S'_{\text{eff}} = S_{\text{eff}} + \int d^dx\, F^I E_I + \int d^dx\, \partial_\mu J^\mu + O(F^2).$$

They may have different off-shell vertices, but they give the same on-shell scattering amplitudes. Consequently, matching string amplitudes determines

$$\boxed{S_{\text{eff}} \quad \text{modulo field redefinitions, total derivatives, and gauge-fixing choices.}}$$

At a fixed order in α', this is usually implemented by choosing a basis of local gauge- and diffeomorphism-invariant operators and eliminating redundant operators with integrations by parts, algebraic identities, and the lower-order equations of motion. On-shell string amplitudes then determine the coefficients of the remaining independent operators.

For example, a curvature-squared correction may be written in several bases involving

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \qquad R_{\mu\nu}R^{\mu\nu}, \qquad R^2.$$

A local metric redefinition of the schematic form

$$g_{\mu\nu} \longrightarrow g_{\mu\nu} + \alpha' \left(aR_{\mu\nu} + bg_{\mu\nu}R\right)$$

shifts the coefficients of terms involving the leading Einstein equation. Hence the on-shell graviton S-matrix cannot determine all three coefficients separately in an arbitrary off-shell basis. It determines the field-redefinition-invariant combinations. This is why the book can say that higher-derivative terms are inferred from amplitudes while also warning that the resulting action is ambiguous.

The same qualification applies to gauge dependence. An off-shell gauge-field Green function contains longitudinal components and generally depends on the gauge-fixing parameter. A physical external state is transverse and belongs to BRST cohomology, so these gauge-dependent components cancel or decouple in the on-shell S-matrix.

8.1.1.6 Why an off-shell action is still necessary

Although its detailed form is not unique, an off-shell action is needed for several distinct reasons.

8.1.1.6.1 It generates internal propagation and complete higher-point amplitudes An n-point S-matrix element is not generally one n-field contact vertex. It is a sum of contact and exchange graphs. Internal momenta are off shell, so one needs the quadratic operator, its inverse propagator, and vertices defined away from the external mass shell. The book's four-point matching prescription is an explicit use of precisely this off-shell field-theory machinery.

One must retain the **complete action in a chosen field basis**. It would be inconsistent to delete a term merely because its contribution vanishes when every leg attached to that vertex is put on shell, while leaving all other vertices unchanged. Such a term may contribute when one of those legs is internal. It may be removed only through a consistent field redefinition, including all induced changes in the remaining interactions.

8.1.1.6.2 The equations of motion come from variations away from a solution A classical background Φ_0 is determined by

$$\left.\frac{\delta S_{\text{eff}}}{\delta \Phi^I(x)}\right|_{\Phi_0} = 0.$$

This derivative compares the action at configurations infinitesimally displaced from Φ_0. If the action existed only on solutions, its first variation—and therefore its equations of motion—could not be defined.

The same observation applies to tadpoles. A one-point function is the first derivative of the effective action. A nonzero tadpole means that the chosen background is not a stationary point. This is not an ordinary scattering amplitude between asymptotic particles, but it is essential for deciding whether perturbation theory has been set up around a valid vacuum.

8.1.1.6.3 Potentials and vacuum structure are intrinsically off shell For scalar fields that are constant in spacetime,

$$\Gamma[\phi_{\text{const}}] = -\,\text{Vol}_d\; V_{\text{eff}}(\phi).$$

A generic value of ϕ is not a solution because

$$\frac{dV_{\text{eff}}}{d\phi} \neq 0.$$

Nevertheless, the entire function V_{eff} is needed to locate vacua, test stability, calculate masses from its Hessian, and study rolling or tunnelling solutions. On-shell scattering about one stationary point determines only local information accessible around that vacuum and may not determine a global potential without additional input.

8.1.1.6.4 Quantum calculations use off-shell loop momenta In a loop integral, an internal momentum is integrated over all values:

$$\int \frac{d^d q}{(2\pi)^d} \frac{N(q,k_i)}{K_1(q)K_2(q+k_1)\cdots}.$$

The internal lines are not restricted to the zeros of their inverse propagators. Off-shell Feynman rules are therefore necessary for ordinary perturbative calculations in the effective theory. The final on-shell observable remains invariant under consistent local field redefinitions and gauge choices, even though intermediate integrands and off-shell Green functions change.

8.1.1.6.5 Symmetries are most usefully encoded off shell Gauge invariance and diffeomorphism invariance relate vertices of different orders. As §16.3 emphasizes, a covariant kinetic term already contains interactions with gauge bosons and arbitrarily many gravitons when expanded in fluctuations. Writing a gauge- and diffeomorphism-invariant action makes these relations manifest and guarantees the Ward or Slavnov–Taylor identities needed for cancellations among diagrams.

Supersymmetry may close only modulo equations of motion if no auxiliary-field formulation is available. This is another sense in which an on-shell symmetry algebra can be sufficient for the physical spectrum while an off-shell action remains the practical object used to organize interactions.

8.1.1.7 How string theory can discuss genuinely off-shell configurations

The limitation belongs to the ordinary first-quantized vertex-operator prescription, not to every possible formulation of string theory.

In covariant string field theory, the string field Ψ is not restricted to BRST cohomology. For example, the schematic cubic open-string-field-theory action begins as

$$S_{\text{SFT}}[\Psi] = \frac{1}{2}\langle \Psi, Q_{\text{B}}\Psi\rangle + \frac{g_s}{3}\langle \Psi, \Psi * \Psi\rangle + \cdots.$$

The equation $Q_{\text{B}}\Psi = 0$ emerges from the linearized equation of motion instead of being imposed before the field is inserted. String field theory therefore supplies off-shell vertices and propagators, although these depend on choices such as gauge fixing, local coordinates on punctured surfaces, and the definition of the string field. Its on-shell S-matrix is independent of those choices.

There is also the worldsheet sigma-model route discussed earlier in the book. One allows general spacetime-dependent background couplings and computes their beta functions. The conditions

$$\beta^I = 0$$

are spacetime equations of motion in a particular field or renormalization scheme. Away from $\beta^I = 0$, the worldsheet theory is not an exact conformal background, so this is not an ordinary correlator of physical BRST-invariant scattering vertices. Scheme changes of the beta functions correspond closely to spacetime field redefinitions, again reflecting the non-uniqueness of off-shell data.

8.1.1.8 Checks

1. **A free external leg.** For a scalar, LSZ multiplies the connected Green function by $K(k) = k^2 + m^2$ and then takes $K(k) \to 0$. Terms in an amputated vertex proportional to $K(k)$ vanish for that external leg. This is the elementary origin of the equation-of-motion ambiguity.

2. **An internal leg.** For $q = k_1 + k_2$, there is no reason for $K(q)$ to vanish. A factor $K(q)$ in a vertex can cancel $1/K(q)$ from the propagator, converting what looks like exchange into a local contact contribution. This confirms that contact and exchange pieces separately depend on the off-shell convention.

3. **A field redefinition.** The identity

$$S[\Phi + \delta\Phi] = S[\Phi] + \int d^d x \, \frac{\delta S}{\delta \Phi^I} \delta\Phi^I + O(\delta\Phi^2)$$

shows directly that equation-of-motion operators are generated or removed by changing field coordinates. Their coefficients cannot be on-shell observables.

4. **Einstein frame versus string frame.** The two frames have visibly different dilaton and metric couplings off shell, yet they are related by invertible field redefinitions. Physical amplitudes agree after the external states and normalizations are translated. This is a concrete example already present in §16.3.

8.1.1.9 Result

An off-shell n-point vertex is

$$\Gamma^{(n)}_{I_1 \cdots I_n} = \left. \frac{\delta^n \Gamma}{\delta\Phi^{I_1} \ldots \delta\Phi^{I_n}} \right|_{\Phi_0}, \qquad \sum_i k_i = 0,$$

with no requirement that

$$K_i(k_i) = 0.$$

The physical amplitude is the pole residue with external legs on shell:

$$\mathcal{A}_n \sim \lim_{K_i(k_i) \to 0} \prod_i Z_i^{-1/2} K_i(k_i) \, G^{(n)}_{\text{conn}} \Big|_{\text{physical polarizations}}.$$

At four points,

$$\mathcal{A}_4^{\text{EFT}} = \sum_{s,t,u} V_3 \frac{i}{K} V_3 + V_4,$$

so internal off-shell vertices and propagators are required even though the external states are on shell.

Under a local field redefinition,

$$\Phi^I \to \Phi^I - F^I, \qquad S_{\text{eff}} \to S_{\text{eff}} - \int d^d x \, F^I \frac{\delta S_{\text{eff}}}{\delta \Phi^I} + O(F^2),$$

the off-shell vertices change but the on-shell S-matrix does not. Therefore the string amplitudes determine

$$\boxed{S_{\text{eff}} \text{ only up to field redefinitions and other on-shell-trivial terms.}}$$

The correct conclusion is:

Off-shell Green functions are necessary ingredients of the conventional action formalism, but they are not separately physical or uniquely fixed by first-quantized string amplitudes.

8.2 Q2: Explicit three-tachyon sphere amplitude

I want to see some explicit computation for the correlation function of vertex operators. Compute the simplest three-point amplitude of the closed bosonic string three-tachyon eq(16.24).

> **Textbook location and related material.** Chapter 16, Section 16.2 around Eq. (16.24): the closed-bosonic three-tachyon sphere correlator, conformal Killing group fixing, and ghost insertions.

8.2.1 Answer

8.2.1.1 Core point

Equation (16.24) is the simplest closed-string tree amplitude because the worldsheet is the sphere and the conformal Killing group $SL(2,\mathbb{C})$ fixes all three vertex positions. There is therefore no modulus and no remaining position integral.

The complete gauge-fixed correlator is

$$\mathcal{A}_3 = C_{S^2} g_c^3 \left\langle c\bar{c}\, V_T(k_1; z_1, \bar{z}_1) c\bar{c}\, V_T(k_2; z_2, \bar{z}_2) c\bar{c}\, V_T(k_3; z_3, \bar{z}_3) \right\rangle_{S^2},$$

where

$$V_T(k_i; z_i, \bar{z}_i) =: e^{ik_i\cdot X(z_i,\bar{z}_i)} :, \qquad k_i^2 = \frac{4}{\alpha'}.$$

The answer factorizes into a ghost correlator, a nonzero-mode matter correlator, and the integral over the spacetime zero mode of X^μ:

$$\mathcal{A}_3 = C_{S^2} g_c^3 \underbrace{\langle c_1 c_2 c_3\rangle \langle \bar{c}_1 \bar{c}_2 \bar{c}_3\rangle}_{\text{ghosts}} \underbrace{\left\langle \prod_{i=1}^{3} : e^{ik_i\cdot X_i} : \right\rangle'}_{\text{matter nonzero modes}} \underbrace{\int d^d x_0\, e^{ix_0\cdot \sum_i k_i}}_{\text{matter zero mode}}.$$

The three factors are

$$\langle c_1 c_2 c_3\rangle \langle \bar{c}_1 \bar{c}_2 \bar{c}_3\rangle = |z_{12} z_{13} z_{23}|^2,$$

$$\left\langle \prod_{i=1}^{3} : e^{ik_i\cdot X_i} : \right\rangle' = \prod_{i<j} |z_{ij}|^{\alpha' k_i\cdot k_j},$$

and

$$\int d^d x_0\, e^{ix_0\cdot\sum_i k_i} = (2\pi)^d \delta^{(d)}\left(\sum_i k_i\right).$$

For three on-shell tachyons, momentum conservation implies

$$\alpha' k_i \cdot k_j = -2, \qquad i \neq j.$$

There is a factor-of-two typo in the sentence immediately below equation (16.24): the book prints $\frac{\alpha'}{2} k_i \cdot k_j = -2$. The derivation below shows that the correct condition is

$$\boxed{\frac{\alpha'}{2} k_i \cdot k_j = -1, \qquad i \neq j.}$$

The corrected value is also required for the matter and ghost position dependences to cancel. Equation (16.24) itself is correct.

Consequently,

$$\prod_{i<j} |z_{ij}|^{\alpha' k_i\cdot k_j} = |z_{12} z_{13} z_{23}|^{-2},$$

which exactly cancels the ghost factor. Restoring the delta function that the book suppresses,

$$\boxed{\mathcal{A}_3 = (2\pi)^d \delta^{(d)}(k_1 + k_2 + k_3) C_{S^2} g_c^3.}$$

After adopting the book's convention of omitting the momentum-conserving delta function, this is precisely

$$\boxed{\mathcal{A}_3 = C_{S^2} g_c^3,}$$

which is equation (16.24).

8.2.1.2 Why three $c\bar{c}$ insertions appear

Before gauge fixing, the formal closed-string sphere amplitude is

$$\mathcal{A}_3 \sim g_c^3 C_{S^2} \int \frac{d^2 z_1\, d^2 z_2\, d^2 z_3}{\mathrm{Vol}\, SL(2,\mathbb{C})} \left\langle V_T(k_1; z_1, \bar{z}_1) V_T(k_2; z_2, \bar{z}_2) V_T(k_3; z_3, \bar{z}_3) \right\rangle.$$

The Möbius group acts by

$$z \longmapsto \frac{az+b}{cz+d}, \qquad ad - bc = 1,$$

and has three complex parameters. On a sphere with three punctures, these parameters can move any three distinct points to any other three distinct points. Thus all three z_i are gauge, rather than moduli.

The Faddeev-Popov determinant for fixing z_1, z_2, z_3 is represented by three holomorphic c ghosts and three antiholomorphic $\bar{c}$ ghosts. Equivalently, the sphere has three holomorphic c zero modes, corresponding to the conformal Killing vectors

$$1, \qquad z, \qquad z^2,$$

and the correlator vanishes unless all three are saturated. Therefore every one of the three fixed matter vertices becomes an unintegrated vertex,

$$U_T(k_i; z_i, \bar{z}_i) = c(z_i)\bar{c}(\bar{z}_i)V_T(k_i; z_i, \bar{z}_i).$$

The tachyon matter vertex has weights

$$h_T = \bar{h}_T = \frac{\alpha' k_i^2}{4}.$$

BRST invariance requires $h_T = \bar{h}_T = 1$, so

$$k_i^2 = \frac{4}{\alpha'}.$$

Since c and $\bar{c}$ have weights -1 and -1, respectively, the unintegrated vertex $c\bar{c}V_T$ has total weights $(0,0)$. This is why its correlator can be independent of the three fixed positions.

8.2.1.3 Matter correlator from Wick's theorem

We now calculate the matter part explicitly. The book's free-field convention is

$$\langle X^\mu(z) X^\nu(w)\rangle = -\frac{\alpha'}{2}\eta^{\mu\nu}\log(z-w),$$

together with

$$\langle \bar{X}^\mu(\bar{z}) \bar{X}^\nu(\bar{w})\rangle = -\frac{\alpha'}{2}\eta^{\mu\nu}\log(\bar{z}-\bar{w}).$$

Combining the two chiralities gives

$$\langle X^\mu(z,\bar{z}) X^\nu(w,\bar{w})\rangle = -\frac{\alpha'}{2}\eta^{\mu\nu}\log|z-w|^2.$$

For normal-ordered exponentials, self-contractions within a single vertex are omitted. Wick's theorem retains only contractions between distinct vertices. If

$$A_i = ik_i \cdot X(z_i, \bar{z}_i),$$

then

$$\left\langle \prod_{i=1}^{3} :e^{A_i}: \right\rangle' = \exp\left(\sum_{i<j}\langle A_i A_j\rangle\right).$$

For $i \neq j$,

$$\begin{aligned}
\langle A_i A_j\rangle &= i^2 k_{i\mu}k_{j\nu}\langle X^\mu(z_i,\bar{z}_i)X^\nu(z_j,\bar{z}_j)\rangle \\
&= (-1)k_{i\mu}k_{j\nu}\left[-\frac{\alpha'}{2}\eta^{\mu\nu}\log|z_{ij}|^2\right] \\
&= \frac{\alpha'}{2}k_i\cdot k_j \log|z_{ij}|^2.
\end{aligned}$$

Exponentiating this result gives

$$\left\langle \prod_{i=1}^{3} :e^{ik_i \cdot X(z_i,\bar{z}_i)}: \right\rangle' = \exp\left[\frac{\alpha'}{2}\sum_{i<j} k_i \cdot k_j \log|z_{ij}|^2\right]$$
$$= \prod_{i<j} |z_{ij}|^{\alpha' k_i \cdot k_j}$$
$$= |z_{12}|^{\alpha' k_1 \cdot k_2}|z_{13}|^{\alpha' k_1 \cdot k_3}|z_{23}|^{\alpha' k_2 \cdot k_3}.$$

This is the three-point specialization of the Koba-Nielsen factor.

8.2.1.4 The zero mode and momentum conservation

The preceding Wick contraction concerns the oscillating part of X^μ. To see the remaining factor, split

$$X^\mu(z,\bar{z}) = x_0^\mu + X'^\mu(z,\bar{z}),$$

where x_0^μ is constant on the sphere. The product of the three exponentials contains

$$\prod_{i=1}^{3} e^{ik_i \cdot x_0} = e^{ix_0 \cdot (k_1+k_2+k_3)}.$$

The path integral over x_0 is an ordinary Fourier integral:

$$\int d^d x_0 \, e^{ix_0 \cdot (k_1+k_2+k_3)} = (2\pi)^d \delta^{(d)}(k_1+k_2+k_3).$$

Thus the full matter correlator is

$$\left\langle \prod_{i=1}^{3} :e^{ik_i \cdot X_i}: \right\rangle = (2\pi)^d \delta^{(d)}\left(\sum_i k_i\right) \prod_{i<j} |z_{ij}|^{\alpha' k_i \cdot k_j}.$$

The paragraph immediately preceding equation (16.24) states that the book assumes momentum conservation and drops this delta function. It is therefore absent from the printed final answer, not from the underlying path integral.

8.2.1.5 Deriving the pairwise on-shell products

For the closed bosonic tachyon,

$$k_i^2 = -m_T^2 = \frac{4}{\alpha'}.$$

Momentum conservation is

$$k_1 + k_2 + k_3 = 0.$$

For example,

$$k_1 + k_2 = -k_3.$$

Squaring both sides yields

$$k_1^2 + k_2^2 + 2k_1 \cdot k_2 = k_3^2.$$

Substituting the three mass-shell conditions,

$$\frac{4}{\alpha'} + \frac{4}{\alpha'} + 2k_1 \cdot k_2 = \frac{4}{\alpha'},$$

so

$$k_1 \cdot k_2 = -\frac{2}{\alpha'}.$$

Cyclically,

$$\boxed{k_1 \cdot k_2 = k_1 \cdot k_3 = k_2 \cdot k_3 = -\frac{2}{\alpha'}.}$$

This gives the matter factor

$$\left\langle \prod_{i=1}^{3} : e^{ik_i \cdot X_i} : \right\rangle' = |z_{12}z_{13}z_{23}|^{-2}.$$

Thus the correct condition to use in the step below equation (16.24) is

$$\frac{\alpha'}{2} k_i \cdot k_j = -1, \qquad i \neq j.$$

The printed right-hand side -2 would instead give $\alpha' k_i \cdot k_j = -4$ and hence the matter factor $|z_{12}z_{13}z_{23}|^{-4}$. Multiplication by the ghost factor would then leave $|z_{12}z_{13}z_{23}|^{-2}$, contradicting the required $SL(2, \mathbb{C})$ independence. This provides an independent check that the printed value is a typo.

8.2.1.6 Ghost correlator

The bc system has the basic OPE

$$b(z)c(w) \sim \frac{1}{z - w}.$$

On the sphere, the normalized correlator saturating the three holomorphic c zero modes is

$$\langle c(z_1)c(z_2)c(z_3) \rangle = z_{12}z_{13}z_{23}.$$

This formula can be seen directly from the mode expansion. Only the three zero modes can contribute:

$$c(z) = c_1 + zc_0 + z^2 c_{-1} + \text{nonzero-mode terms}.$$

With the standard sphere normalization

$$\langle 0|c_{-1}c_0c_1|0\rangle = 1,$$

the coefficient of $c_{-1}c_0c_1$ in the product of the three $c(z_i)$ is the Vandermonde determinant

$$\begin{aligned}\langle c(z_1)c(z_2)c(z_3) \rangle &= \det \begin{pmatrix} z_1^2 & z_1 & 1 \\ z_2^2 & z_2 & 1 \\ z_3^2 & z_3 & 1 \end{pmatrix} \\ &= (z_1 - z_2)(z_1 - z_3)(z_2 - z_3) \\ &= z_{12}z_{13}z_{23}.\end{aligned}$$

The antiholomorphic sector similarly gives

$$\langle \bar{c}(\bar{z}_1)\bar{c}(\bar{z}_2)\bar{c}(\bar{z}_3)\rangle = \bar{z}_{12}\bar{z}_{13}\bar{z}_{23}.$$

Therefore

$$\begin{aligned}\langle c_1c_2c_3\rangle\langle \bar{c}_1\bar{c}_2\bar{c}_3\rangle &= (z_{12}z_{13}z_{23})(\bar{z}_{12}\bar{z}_{13}\bar{z}_{23})\\ &= |z_{12}z_{13}z_{23}|^2.\end{aligned}$$

This is exactly the Faddeev-Popov factor displayed in equations (16.5)-(16.6).

8.2.1.7 Cancellation of the insertion-point dependence

Multiplying the ghost and matter nonzero-mode factors gives

$$\begin{aligned}&\langle c_1c_2c_3\rangle\langle \bar{c}_1\bar{c}_2\bar{c}_3\rangle \left\langle \prod_{i=1}^{3} :e^{ik_i\cdot X_i}: \right\rangle'\\ &= |z_{12}z_{13}z_{23}|^2|z_{12}z_{13}z_{23}|^{-2}\\ &= 1.\end{aligned}$$

This cancellation is not accidental. Each matter tachyon vertex is a $(1,1)$ primary, while $c\bar{c}$ has weights $(-1,-1)$. Hence every unintegrated operator $c\bar{c}V_T$ has weights $(0,0)$, and the gauge-fixed three-point correlator cannot depend on which three distinct points were chosen.

It follows that

$$\begin{aligned}\mathcal{A}_3 &= C_{S^2}g_c^3(2\pi)^d\delta^{(d)}(k_1+k_2+k_3)\\ &\quad\times |z_{12}z_{13}z_{23}|^2|z_{12}z_{13}z_{23}|^{-2}\\ &= (2\pi)^d\delta^{(d)}(k_1+k_2+k_3)C_{S^2}g_c^3.\end{aligned}$$

Dropping the delta function gives equation (16.24).

8.2.1.8 Explicit check with $z_1 = 0$, $z_2 = 1$, $z_3 = \infty$

The usual choice is

$$z_1 = 0, \qquad z_2 = 1, \qquad z_3 = \infty.$$

To treat the point at infinity carefully, first set $z_3 = R$ and take $|R| \to \infty$. Then

$$z_{12} = -1, \qquad z_{13} = -R, \qquad z_{23} = 1-R.$$

The ghost factor behaves as

$$|z_{12}z_{13}z_{23}|^2 = |R|^2|1-R|^2 \sim |R|^4.$$

The matter factor behaves as

$$|z_{12}z_{13}z_{23}|^{-2} = |R|^{-2}|1-R|^{-2} \sim |R|^{-4}.$$

Their product is one before and after the limit:

$$\lim_{|R|\to\infty} \left[|R|^2|1-R|^2\right]\left[|R|^{-2}|1-R|^{-2}\right] = 1.$$

Equivalently, since $U_T = c\bar{c}V_T$ has weights $(0,0)$, its insertion at infinity is simply defined by the conformal limit of the complete operator. Separating the matter and ghost pieces makes each piece singular, but their product is finite.

8.2.1.9 Checks

1. **Closed-string coupling counting.** Each of the three vertices contributes g_c, while the sphere normalization satisfies $C_{S^2} \propto g_c^{-2}$. Therefore

$$\mathcal{A}_3 \propto g_c^{3-2} = g_c,$$

in agreement with the general closed-string tree-level rule $\mathcal{A}_N \propto g_s^{N-2}$.

2. **No modulus integral.** The complex dimension of the moduli space of a sphere with N marked points is $N-3$. For $N=3$ this is zero. The three-point answer is consequently a constant rather than a beta or gamma function of Mandelstam invariants.

3. **Worldsheet scaling.** The matter vertex has weight $+1$ at each holomorphic and antiholomorphic insertion; the attached $c\bar{c}$ factor has weight -1 in each chirality. The total unintegrated vertex has weight zero, explaining the exact position independence.

4. **Spacetime interpretation.** After removing the universal delta function, the amplitude is momentum independent. At this order it therefore matches a local cubic tachyon interaction in spacetime. Its precise coefficient and phase depend on the normalization conventions collected in g_c and C_{S^2}; the book postpones fixing C_{S^2} until factorization and unitarity are imposed.

5. **Formal nature of the tachyon amplitude.** The bosonic-string tachyon signals that the perturbative vacuum is unstable. The correlator is nevertheless a perfectly well-defined on-shell CFT amplitude, understood with the usual analytic continuation of momenta when necessary, and is the cleanest example for learning the worldsheet calculation.

8.2.1.10 Result

The unintegrated tachyon vertex is

$$U_T(k_i; z_i, \bar{z}_i) = c(z_i)\bar{c}(\bar{z}_i) : e^{ik_i \cdot X(z_i,\bar{z}_i)} :, \qquad k_i^2 = \frac{4}{\alpha'}.$$

The matter correlator is

$$\left\langle \prod_{i=1}^{3} : e^{ik_i \cdot X_i} : \right\rangle = (2\pi)^d \delta^{(d)}\left(\sum_i k_i\right) \prod_{i<j} |z_{ij}|^{\alpha' k_i \cdot k_j}.$$

Using

$$k_1 + k_2 + k_3 = 0, \qquad k_i^2 = \frac{4}{\alpha'},$$

one obtains

$$\alpha' k_i \cdot k_j = -2, \qquad \left\langle \prod_i : e^{ik_i \cdot X_i} : \right\rangle' = |z_{12}z_{13}z_{23}|^{-2}.$$

The ghost correlator is

$$\langle c_1 c_2 c_3 \rangle \langle \bar{c}_1 \bar{c}_2 \bar{c}_3 \rangle = |z_{12}z_{13}z_{23}|^2.$$

Therefore

$$\boxed{\mathcal{A}_3 = (2\pi)^d \delta^{(d)}(k_1 + k_2 + k_3) C_{S^2} g_c^3,}$$

or, suppressing the delta function as in the book,

$$\boxed{\mathcal{A}_3 = C_{S^2} g_c^3.}$$

8.3 Q3: Mixed left-right propagators on the upper half-plane

In page603, due to the presence of the boundary along the real axis, there are non-trivial two-point functions between left- and right-moving fields. Compute the $\langle X^\mu(z)\overline{X}^\nu(\overline{w})\rangle$ and $\langle \psi^\mu(z)\overline{\psi}(\overline{w})\rangle$ correlation functions in eq (16.56) in details.

> **Textbook location and related material.** Chapter 16, Section 16.2 around p. 603 and Eqs. (16.56)–(16.58): the upper-half-plane doubling trick and mixed left–right propagators in the presence of a D-brane.

8.3.1 Answer

8.3.1.1 Core point

On the full complex plane, the holomorphic and antiholomorphic fields are independent, so mixed contractions such as

$$\langle X^\mu(z)\bar{X}^\nu(\bar{w})\rangle_{\mathbb{C}}$$

vanish. On the upper half-plane, however, the real boundary glues left and right movers. For a Dp-brane this gluing is encoded by

$$D^\mu{}_\nu = \mathrm{diag}(+1_{\mathrm{N}}, -1_{\mathrm{D}}),$$

so a right mover can be replaced by a holomorphic image field at the reflected point $\bar{w}$ in the lower half-plane. The ordinary plane propagators then immediately give

$$\boxed{\left\langle X^\mu(z)\bar{X}^\nu(\bar{w})\right\rangle_{\mathbb{H}_+} = -\frac{\alpha'}{2} D^{\mu\nu}\log(z-\bar{w})}$$

and

$$\boxed{\left\langle \psi^\mu(z)\bar{\psi}^\nu(\bar{w})\right\rangle_{\mathbb{H}_+} = \frac{D^{\mu\nu}}{z-\bar{w}}.}$$

These are the first two mixed correlators in equation (16.56). The sign difference between Neumann and Dirichlet directions is entirely contained in $D^{\mu\nu}$.

I interpret the second correlator in the question as $\langle \psi^\mu(z)\bar{\psi}^\nu(\bar{w})\rangle$, with the missing target-space index ν restored as in equation (16.56).

The derivation has three steps:

1. derive the left-right gluing conditions from Neumann and Dirichlet boundary conditions;
2. extend, or double, the fields from $\mathbb{H}_+$ to the full plane;
3. use the standard holomorphic plane two-point functions between a point and its image.

8.3.1.2 From Neumann and Dirichlet conditions to the gluing matrix

Write the Euclidean worldsheet coordinate as

$$z = x + iy, \qquad \bar{z} = x - iy, \qquad y \geq 0.$$

The boundary is $y = 0$. With

$$\partial = \frac{1}{2}(\partial_x - i\partial_y), \qquad \bar{\partial} = \frac{1}{2}(\partial_x + i\partial_y),$$

the tangential and normal derivatives are, up to the orientation convention for the outward normal,

$$\partial_t = \partial_x = \partial + \bar{\partial}, \qquad \partial_n = \partial_y = i(\partial - \bar{\partial}).$$

Split the closed-string coordinate into chiral pieces:

$$X^\mu(z, \bar{z}) = X^\mu(z) + \bar{X}^\mu(\bar{z}).$$

For a world-volume direction X^α, the Neumann condition is

$$\partial_n X^\alpha\big|_{y=0} = 0.$$

Because the left-moving piece depends only on z and the right-moving piece only on $\bar{z}$, this gives

$$\partial X^\alpha(z)\big|_{z=x} = \bar{\partial}\bar{X}^\alpha(\bar{z})\big|_{\bar{z}=x}.$$

After integration along the boundary,

$$X^\alpha(x) = +\bar{X}^\alpha(x) + \text{constant}.$$

The additive constant is irrelevant for contractions and can be suppressed.

For a transverse direction X^i, the D-brane fixes the endpoint position:

$$X^i(z, \bar{z})\big|_{y=0} = y_0^i.$$

The boundary value is constant along the real axis, so

$$\partial_t X^i\big|_{y=0} = 0.$$

This implies

$$\partial X^i(z)\big|_{z=x} = -\,\bar{\partial}\bar{X}^i(\bar{z})\big|_{\bar{z}=x},$$

and hence

$$X^i(x) = -\bar{X}^i(x) + y_0^i.$$

Suppressing the position constant, the two cases combine into

$$\boxed{X^\mu(x) = D^\mu{}_\nu \bar{X}^\nu(x),}$$

where

$$D^\alpha{}_\beta = \delta^\alpha{}_\beta, \qquad D^i{}_j = -\delta^i{}_j, \qquad D^\alpha{}_i = D^i{}_\alpha = 0.$$

The fermionic boundary condition follows either from the boundary term in the variation of the NSR fermion action or by requiring that worldsheet supersymmetry preserve the bosonic boundary conditions. In the sign convention chosen in equation (16.54), it is

$$\boxed{\psi^\mu(x) = D^\mu{}_\nu \bar{\psi}^\nu(x).}$$

Thus

$$\psi^\alpha(x) = +\bar{\psi}^\alpha(x) \qquad (\mathrm{N}),$$

whereas

$$\psi^i(x) = -\bar{\psi}^i(x) \qquad (\mathrm{D}).$$

There can be a common sign multiplying all fermionic gluing conditions, associated with the choice of spin structure or boundary-state sign. Equation (16.54) has chosen this common sign to be $+1$, so it will not be displayed below.

The matrix D is a target-space reflection. In particular,

$$D^\mu{}_\rho D^\rho{}_\nu = \delta^\mu{}_\nu, \qquad D^T \eta D = \eta.$$

It is therefore legitimate to use $D^{-1} = D$ when undoing the doubling transformation.

8.3.1.3 The doubling trick

Define a single holomorphic bosonic field on the full complex plane by

$$\mathbb{X}^\mu(\zeta) = \begin{cases} X^\mu(\zeta), & \operatorname{Im}\zeta > 0, \\ D^\mu{}_\rho \bar{X}^\rho(\zeta), & \operatorname{Im}\zeta < 0. \end{cases}$$

Here $\bar{X}^\rho(\zeta)$ in the second line means the analytic right-moving field continued in its own argument into the lower half-plane. At the boundary $\zeta = x$, the two definitions agree because of the gluing condition.

Similarly define the doubled fermion

$$\Psi^\mu(\zeta) = \begin{cases} \psi^\mu(\zeta), & \operatorname{Im}\zeta > 0, \\ D^\mu{}_\rho \bar{\psi}^\rho(\zeta), & \operatorname{Im}\zeta < 0. \end{cases}$$

The doubled fields are ordinary holomorphic free fields on the full plane. Their two-point functions are therefore the plane correlators from equation (16.2):

$$\langle \mathbb{X}^\mu(\zeta)\mathbb{X}^\nu(\xi)\rangle_{\mathbb{C}} = -\frac{\alpha'}{2}\eta^{\mu\nu}\log(\zeta-\xi),$$

$$\langle \Psi^\mu(\zeta)\Psi^\nu(\xi)\rangle_{\mathbb{C}} = \frac{\eta^{\mu\nu}}{\zeta-\xi}.$$

The entire effect of the boundary has now been moved into the definition of the fields below the real axis.

8.3.1.4 Bosonic mixed correlator

Let $z, w \in \mathbb{H}_+$. The reflected point $\bar{w}$ lies in $\mathbb{H}_-$. From the lower-half-plane definition,

$$\mathbb{X}^\lambda(\bar{w}) = D^\lambda{}_\rho \bar{X}^\rho(\bar{w}).$$

Multiplying by D and using $D^2 = 1$ gives

$$\bar{X}^\nu(\bar{w}) = D^\nu{}_\lambda \mathbb{X}^\lambda(\bar{w}).$$

The desired mixed correlator is therefore

$$\begin{aligned}
\left\langle X^\mu(z)\bar{X}^\nu(\bar{w})\right\rangle_{\mathbb{H}_+} &= D^\nu{}_\lambda \left\langle \mathbb{X}^\mu(z)\mathbb{X}^\lambda(\bar{w})\right\rangle_{\mathbb{C}} \\
&= D^\nu{}_\lambda \left[-\frac{\alpha'}{2}\eta^{\mu\lambda}\log(z-\bar{w})\right] \\
&= -\frac{\alpha'}{2}\eta^{\mu\lambda}D^\nu{}_\lambda \log(z-\bar{w}).
\end{aligned}$$

Define

$$D^{\mu\nu} = \eta^{\mu\lambda}D^\nu{}_\lambda.$$

For the diagonal reflection matrix used here, $D^{\mu\nu} = D^{\nu\mu}$. Hence

$$\boxed{\left\langle X^\mu(z)\bar{X}^\nu(\bar{w})\right\rangle_{\mathbb{H}_+} = -\frac{\alpha'}{2}D^{\mu\nu}\log(z-\bar{w}).}$$

This correlator is holomorphic in z and antiholomorphic in w, as it must be. The logarithm has its singularity at $z = \bar{w}$, the image of w across the boundary.

8.3.1.5 Fermionic mixed correlator

The same argument applies to the worldsheet fermion. For the reflected point,

$$\bar{\psi}^\nu(\bar{w}) = D^\nu{}_\lambda \Psi^\lambda(\bar{w}).$$

Therefore

$$\begin{aligned}
\left\langle \psi^\mu(z)\bar{\psi}^\nu(\bar{w})\right\rangle_{\mathbb{H}_+} &= D^\nu{}_\lambda \left\langle \Psi^\mu(z)\Psi^\lambda(\bar{w})\right\rangle_{\mathbb{C}} \\
&= D^\nu{}_\lambda \frac{\eta^{\mu\lambda}}{z-\bar{w}} \\
&= \frac{D^{\mu\nu}}{z-\bar{w}}.
\end{aligned}$$

Thus

$$\boxed{\left\langle \psi^\mu(z)\bar{\psi}^\nu(\bar{w})\right\rangle_{\mathbb{H}_+} = \frac{D^{\mu\nu}}{z-\bar{w}}.}$$

No additional minus sign arises from the doubling calculation itself. The relative Dirichlet sign mentioned below equation (16.56) is precisely the -1 already present in $D^i{}_j$.

8.3.1.6 Neumann and Dirichlet components

It is useful to write the formulas separately. Along Neumann directions,

$$D^{\alpha\beta} = \eta^{\alpha\beta},$$

so

$$\boxed{\left\langle X^{\alpha}(z)\bar{X}^{\beta}(\bar{w})\right\rangle = -\frac{\alpha'}{2}\eta^{\alpha\beta}\log(z-\bar{w}),}$$

$$\boxed{\left\langle \psi^{\alpha}(z)\bar{\psi}^{\beta}(\bar{w})\right\rangle = \frac{\eta^{\alpha\beta}}{z-\bar{w}}.}$$

Along spatial Dirichlet directions,

$$D^{ij} = -\delta^{ij},$$

and hence

$$\boxed{\left\langle X^{i}(z)\bar{X}^{j}(\bar{w})\right\rangle = +\frac{\alpha'}{2}\delta^{ij}\log(z-\bar{w}),}$$

$$\boxed{\left\langle \psi^{i}(z)\bar{\psi}^{j}(\bar{w})\right\rangle = -\frac{\delta^{ij}}{z-\bar{w}}.}$$

Mixed Neumann-Dirichlet components vanish because

$$D^{\alpha i} = D^{i\alpha} = 0.$$

8.3.1.7 Equivalent method-of-images Green function

The doubling result can be repackaged as the Green function of the full nonchiral coordinate

$$X^{\mu}(z,\bar{z}) = X^{\mu}(z) + \bar{X}^{\mu}(\bar{z}).$$

Expanding its two-point function gives four chiral contractions:

$$\begin{aligned}
&\langle X^{\mu}(z,\bar{z})X^{\nu}(w,\bar{w})\rangle_{\mathbb{H}_+} \\
&\quad = \langle X^{\mu}(z)X^{\nu}(w)\rangle + \langle \bar{X}^{\mu}(\bar{z})\bar{X}^{\nu}(\bar{w})\rangle \\
&\qquad + \langle X^{\mu}(z)\bar{X}^{\nu}(\bar{w})\rangle + \langle \bar{X}^{\mu}(\bar{z})X^{\nu}(w)\rangle.
\end{aligned}$$

Using the same-chirality plane propagators and the mixed propagators derived above,

$$\boxed{\begin{aligned}
&\langle X^{\mu}(z,\bar{z})X^{\nu}(w,\bar{w})\rangle_{\mathbb{H}_+} \\
&\qquad = -\frac{\alpha'}{2}\eta^{\mu\nu}\log|z-w|^2 - \frac{\alpha'}{2}D^{\mu\nu}\log|z-\bar{w}|^2.
\end{aligned}}$$

The first term is the direct source at w; the second is its image at $\bar{w}$. For a Neumann direction the image has the same sign, whereas for a Dirichlet direction it has the opposite sign.

This immediately verifies the boundary conditions. For Neumann directions,

$$G_{\mathrm{N}}(z,w) = -\frac{\alpha'}{2}\left[\log|z-w|^2 + \log|z-\bar{w}|^2\right],$$

and the normal derivative vanishes at $z = x \in \mathbb{R}$:

$$\partial_n G_{\mathrm{N}}(z,w)\big|_{z=x} = 0.$$

For Dirichlet directions,

$$G_{\mathrm{D}}(z,w) = -\frac{\alpha'}{2}\left[\log|z-w|^2 - \log|z-\bar{w}|^2\right].$$

At $z = x \in \mathbb{R}$, the two distances are equal,

$$|x-w| = |x-\bar{w}|,$$

so

$$G_{\mathrm{D}}(z,w)\big|_{z=x} = 0,$$

as appropriate for fluctuations of a coordinate fixed on the boundary.

8.3.1.8 Boundary-limit checks and equation (16.58)

For a Neumann coordinate, the gluing condition gives

$$X^\alpha(x,\bar{x}) = X^\alpha(x) + \bar{X}^\alpha(x) = 2X^\alpha(x).$$

Consequently,

$$\begin{aligned}\langle X^\alpha(x)X^\beta(x')\rangle_{\text{boundary}} &= 4\left[-\frac{\alpha'}{2}\eta^{\alpha\beta}\log|x-x'|\right]\\ &= -2\alpha'\eta^{\alpha\beta}\log|x-x'|,\end{aligned}$$

which is the first line of equation (16.58).

For a Dirichlet coordinate the boundary value itself does not fluctuate, but its normal derivative does. At the boundary,

$$\partial_n X^i = i(\partial - \bar{\partial})X^i = 2i\partial X^i,$$

where the last equality uses $\partial X^i = -\bar{\partial}\bar{X}^i$. From

$$\langle \partial X^i(x)\partial X^j(x')\rangle = -\frac{\alpha'}{2}\frac{\delta^{ij}}{(x-x')^2},$$

we obtain

$$\begin{aligned}\langle \partial_n X^i(x)\partial_n X^j(x')\rangle &= (2i)^2\left[-\frac{\alpha'}{2}\frac{\delta^{ij}}{(x-x')^2}\right]\\ &= 2\alpha'\frac{\delta^{ij}}{(x-x')^2},\end{aligned}$$

which is the second line of equation (16.58), up to the immaterial choice of orientation for each normal derivative.

For fermions, moving both points to the boundary and using the chiral field gives

$$\langle\psi^\mu(x)\psi^\nu(x')\rangle = \frac{\eta^{\mu\nu}}{x-x'},$$

for both Neumann and Dirichlet directions. The differing left-right sign has already been absorbed into the gluing relation $\bar\psi = D\psi$. This reproduces the third line of (16.58).

8.3.1.9 Why there was no mixed correlator on the sphere

On the sphere or full plane, the path integral separates into independent holomorphic and antiholomorphic sectors:

$$X^\mu(z,\bar z) = X^\mu(z) + \bar X^\mu(\bar z),$$

with no boundary condition relating the two. One therefore has

$$\left\langle X^\mu(z)\bar X^\nu(\bar w)\right\rangle_{\mathbb{C}} = 0, \qquad \left\langle \psi^\mu(z)\bar\psi^\nu(\bar w)\right\rangle_{\mathbb{C}} = 0.$$

The disk boundary reduces the independent left and right conformal symmetries to a diagonal subgroup and imposes the gluing conditions. A right mover is then not an independent field: it is the reflected image of a left mover. This is the conceptual origin of equation (16.56).

8.3.1.10 Checks

1. **Image-point singularity.** The denominators depend on $z-\bar w$, not $z-w$. A mixed contraction becomes singular when the left-moving insertion approaches the mirror of the right-moving insertion. For two points strictly inside $\mathbb{H}_+$, $z=\bar w$ cannot occur; the singularity is reached when the points approach the boundary together.

2. **Neumann sign.** For $D=+1$, the image source has the same sign as the original source. This makes the normal derivative of the scalar Green function vanish at the real axis.

3. **Dirichlet sign.** For $D=-1$, the image source has the opposite sign. This makes the boundary value of the fluctuating scalar Green function vanish.

4. **Fermionic sign.** The standard plane OPE has a positive numerator, $\eta^{\mu\nu}/(z-w)$. Replacing the right mover by its image multiplies it by D, giving a plus sign for Neumann components and a minus sign for Dirichlet components.

5. **Upper half-plane versus unit disk.** On $\mathbb{H}_+$, reflection is simply $w\mapsto\bar w$, so the doubled fermion acquires no conformal Jacobian. On the unit disk, the image map is $w\mapsto 1/\bar w$; because ψ has weight $1/2$, its transformation includes the square-root Jacobian mentioned in footnote 10 on page 603.

8.3.1.11 Result

The boundary gluing conditions are

$$X^\mu(x) = D^\mu{}_\nu \bar X^\nu(x), \qquad \psi^\mu(x) = D^\mu{}_\nu\bar\psi^\nu(x),$$

with

$$D^\mu{}_\nu = \mathrm{diag}(+1_{\mathrm{N}}, -1_{\mathrm{D}}), \qquad D^2 = 1.$$

The doubling trick replaces a right mover at $\bar w\in\mathbb{H}_-$ by a holomorphic image multiplied by D. Using

$$\langle X^\mu(z) X^\nu(w)\rangle_{\mathbb{C}} = -\frac{\alpha'}{2}\eta^{\mu\nu}\log(z-w),$$

$$\langle \psi^\mu(z)\psi^\nu(w)\rangle_{\mathbb{C}} = \frac{\eta^{\mu\nu}}{z-w},$$

one obtains

$$\boxed{\left\langle X^\mu(z)\bar{X}^\nu(\bar{w})\right\rangle_{\mathbb{H}_+} = -\frac{\alpha'}{2}D^{\mu\nu}\log(z-\bar{w}),}$$

$$\boxed{\left\langle \psi^\mu(z)\bar{\psi}^\nu(\bar{w})\right\rangle_{\mathbb{H}_+} = \frac{D^{\mu\nu}}{z-\bar{w}}.}$$

Equivalently, the full bosonic image-charge Green function is

$$\boxed{\langle X^\mu(z,\bar{z})X^\nu(w,\bar{w})\rangle_{\mathbb{H}_+} = -\frac{\alpha'}{2}\eta^{\mu\nu}\log|z-w|^2 - \frac{\alpha'}{2}D^{\mu\nu}\log|z-\bar{w}|^2.}$$

8.4 Q4: Why vanishing sigma-model beta functions are spacetime equations of motion

Why the vanishing of the sigma-model β function gives string equation of motion i.e. the equation of motion derived by string effective field theory action?

In my understanding, sigma-model (Polyakov) action shows how string couples to background fields which are string excitations (and we only consider the massless part). The vanishing of β function is a somehow consistency condition required by the definition of the string theory and this constraint the massless fields. But why these constraints are exactly equivalent to the equation of motion from the effective action of these massless fields?

> **Textbook location and related material.** Chapter 14, Section 14.1 (pp. 427–432): sigma-model beta functions and background equations; Chapter 16, Sections 16.3–16.4 (pp. 615–630): the matching spacetime effective actions and supergravities.

8.4.1 Answer

8.4.1.1 Core point

Your understanding is essentially correct. The missing link is that the worldsheet sigma model and the spacetime effective action are not two unrelated theories that happen to impose the same constraints. They are two descriptions of the dynamics of the same string degrees of freedom:

- in the worldsheet description, $G_{\mu\nu}(X)$, $B_{\mu\nu}(X)$, and $\Phi(X)$ are position-dependent couplings multiplying worldsheet operators;
- in the spacetime description, those same coupling functions are dynamical fields;
- failure of the worldsheet theory to be Weyl invariant is a tadpole, or failure of stationarity, for the corresponding spacetime string field.

The precise relation is not generally

$$\beta^I = \frac{\delta S_{\text{eff}}}{\delta\lambda^I}.$$

Instead it has the schematic form

$$\boxed{\frac{\delta S_{\text{eff}}}{\delta \lambda^I(x)} = \mathcal{K}_{IJ}(\lambda, \partial\lambda, \ldots)\, \beta^J(x),}$$

up to target-space gauge transformations and higher-order corrections. The kernel $\mathcal{K}_{IJ}$ is a field-space metric or, more generally, an invertible differential operator. It also permits mixing among the metric and dilaton equations. Therefore the invariant statement is

$$\boxed{\beta^I = 0 \quad \Longleftrightarrow \quad \frac{\delta S_{\text{eff}}}{\delta \lambda^I} = 0,}$$

provided $\mathcal{K}$ is nondegenerate after gauge redundancies have been removed and both sides are compared in the same renormalization scheme and at the same order in α' and g_s.

Equations (14.5) and (14.8) give an explicit leading-order demonstration. Varying the string-frame action (14.8) gives equations that are invertible linear combinations of the three Weyl-anomaly coefficients in (14.5). Thus their common zero locus is exactly the same, even though an individual raw Euler-Lagrange derivative is not identical to an individual beta function.

8.4.1.2 Background fields are worldsheet couplings

The bosonic part of the NS-NS sigma model in equation (14.3) is

$$S_\sigma = -\frac{1}{4\pi\alpha'} \int_\Sigma d^2\sigma \sqrt{-h} \left[h^{ab} \partial_a X^\mu \partial_b X^\nu G_{\mu\nu}(X) + \epsilon^{ab} \partial_a X^\mu \partial_b X^\nu B_{\mu\nu}(X) + \alpha' \Phi(X) R^{(2)} \right].$$

From the two-dimensional viewpoint, this is a quantum field theory with infinitely many couplings: every value of $G_{\mu\nu}(x)$, $B_{\mu\nu}(x)$, and $\Phi(x)$ is part of the coupling data. Schematically,

$$S_\sigma = S_* + \sum_I \int_\Sigma d^2z\, \lambda^I(X) \mathcal{O}_I(z, \bar{z}).$$

For example,

$$\mathcal{O}^G_{\mu\nu} \sim \partial X^\mu \bar{\partial} X^\nu, \qquad \mathcal{O}^B_{\mu\nu} \sim \partial X^{[\mu} \bar{\partial} X^{\nu]}, \qquad \mathcal{O}^\Phi \sim R^{(2)}.$$

Expanding a background around flat spacetime makes the relation to string states explicit:

$$G_{\mu\nu}(X) = \eta_{\mu\nu} + h_{\mu\nu}(X).$$

Fourier expanding

$$h_{\mu\nu}(X) = \int \frac{d^dk}{(2\pi)^d}\, h_{\mu\nu}(k) e^{ik\cdot X}$$

turns the perturbation of the sigma-model action into a coherent sum of integrated graviton vertex operators,

$$\delta S_\sigma \sim \int \frac{d^dk}{(2\pi)^d}\, h_{\mu\nu}(k) \int_\Sigma d^2z\, \partial X^\mu \bar{\partial} X^\nu e^{ik\cdot X}.$$

Thus a spacetime field profile is literally a collection of worldsheet couplings to string vertex operators. This is why a condition on sigma-model couplings can become a differential equation for spacetime fields.

8.4.1.3 Why Weyl invariance requires the beta functions to vanish

After fixing the worldsheet metric to conformal gauge,

$$h_{ab} = e^{2\omega}\hat{h}_{ab},$$

the Weyl factor ω is a gauge degree of freedom. It must not become a physical field merely because of a quantum anomaly. Equivalently, the trace of the renormalized worldsheet stress tensor must vanish, apart from the central-charge term whose matter and ghost contributions must cancel.

For spacetime-dependent couplings, the local Weyl anomaly has the form displayed schematically in equation (14.4):

$$T^a{}_a = \beta^G_{\mu\nu}\, h^{ab}\partial_a X^\mu \partial_b X^\nu + \beta^B_{\mu\nu}\, \epsilon^{ab}\partial_a X^\mu \partial_b X^\nu + \beta^\Phi R^{(2)} + \text{total derivatives}.$$

The operators multiplying the independent coefficients are independent local worldsheet operators. Consequently, quantum Weyl invariance requires the corresponding Weyl-anomaly coefficients to vanish, modulo redundant operators generated by target-space gauge transformations:

$$\beta^G_{\mu\nu} = 0, \qquad \beta^B_{\mu\nu} = 0, \qquad \beta^\Phi = 0.$$

At linear order around flat space these are just the BRST physical-state conditions on the graviton, antisymmetric tensor, and dilaton. At nonlinear order they include the interactions among those fields. The sigma-model beta functions are therefore nonlinear completions of the free mass-shell and polarization constraints.

8.4.1.4 The leading beta functions in the book

For the type-II NS-NS fields, equation (14.5) gives

$$\beta^G_{\mu\nu} = \alpha' \left(R_{\mu\nu} - \frac{1}{4} H_{\mu\rho\sigma} H_\nu{}^{\rho\sigma} + 2\nabla_\mu \nabla_\nu \Phi \right) + O(\alpha'^2),$$

$$\beta^B_{\mu\nu} = \alpha' \left(-\frac{1}{2} \nabla^\rho H_{\rho\mu\nu} + H_{\rho\mu\nu} \nabla^\rho \Phi \right) + O(\alpha'^2),$$

and

$$\beta^\Phi = \frac{1}{4}\Delta + \alpha' \left[(\nabla\Phi)^2 - \frac{1}{2}\nabla^2\Phi - \frac{1}{24}H^2 \right] + O(\alpha'^2), \qquad \Delta = d - d_{\text{crit}}.$$

The first two equations say that the metric and B-field couplings cease to run only when the spacetime fields satisfy particular differential equations. The constant term in β^Φ is the central-charge deficit, including the ghost contribution.

8.4.1.5 The string-frame action

The spacetime action displayed in equation (14.8) is, up to an overall normalization,

$$\boxed{S_{\text{eff}} = \int d^d x\, \sqrt{-G}\, e^{-2\Phi} \left[R + 4(\nabla\Phi)^2 - \frac{1}{12}H^2 - \frac{\Delta}{\alpha'} \right].}$$

This is the tree-level NS-NS action in string frame to the derivative order shown in (14.5). The factor $e^{-2\Phi}$ reflects the sphere Euler characteristic and hence the closed-string tree-level factor g_s^{-2}.

We now vary this action and compare the resulting equations directly with the beta functions.

8.4.1.6 Variation with respect to the B-field

Since

$$H_{\mu\nu\rho} = 3\partial_{[\mu} B_{\nu\rho]},$$

its variation is

$$\delta H_{\mu\nu\rho} = 3\nabla_{[\mu} \delta B_{\nu\rho]}.$$

Only the H^2 term varies:

$$\delta_B S_{\text{eff}} = -\frac{1}{6} \int d^d x \sqrt{-G}\, e^{-2\Phi} H^{\mu\nu\rho} \delta H_{\mu\nu\rho}.$$

Using antisymmetry,

$$H^{\mu\nu\rho} \delta H_{\mu\nu\rho} = 3 H^{\mu\nu\rho} \nabla_\mu \delta B_{\nu\rho},$$

and integrating by parts gives

$$\delta_B S_{\text{eff}} = \frac{1}{2} \int d^d x\, \delta B_{\nu\rho}\, \nabla_\mu \left(\sqrt{-G}\, e^{-2\Phi} H^{\mu\nu\rho} \right).$$

Hence the B-field equation is

$$\nabla_\mu \left(e^{-2\Phi} H^{\mu\nu\rho} \right) = 0,$$

or

$$\nabla_\mu H^{\mu\nu\rho} - 2(\nabla_\mu \Phi) H^{\mu\nu\rho} = 0.$$

Multiplying by $-\alpha'/2$ and lowering indices gives exactly

$$\boxed{\beta^B_{\nu\rho} = 0.}$$

Thus the B-field beta function and Euler-Lagrange equation agree up to a nonzero normalization.

8.4.1.7 Variation with respect to the dilaton

Define

$$\mathcal{L} = R + 4(\nabla\Phi)^2 - \frac{1}{12} H^2 - \frac{\Delta}{\alpha'}.$$

Both the prefactor $e^{-2\Phi}$ and the explicit kinetic term depend on Φ. The variation is

$$\delta_\Phi S_{\text{eff}} = \int d^d x \sqrt{-G}\, e^{-2\Phi} \left[-2\mathcal{L}\, \delta\Phi + 8 \nabla_\mu \Phi\, \nabla^\mu \delta\Phi \right].$$

Integrating the second term by parts with the weighted measure gives

$$\int \sqrt{-G}\, e^{-2\Phi} 8 \nabla_\mu \Phi\, \nabla^\mu \delta\Phi = -8 \int \sqrt{-G}\, e^{-2\Phi} \left[\nabla^2 \Phi - 2(\nabla\Phi)^2 \right] \delta\Phi.$$

Therefore the dilaton equation is

$$\boxed{\mathcal{E}_\Phi \equiv R + 4\nabla^2\Phi - 4(\nabla\Phi)^2 - \frac{1}{12}H^2 - \frac{\Delta}{\alpha'} = 0.}$$

This is not equal to $\beta^\Phi = 0$ by itself. Instead, taking the trace of $\beta^G_{\mu\nu}$ gives

$$\frac{1}{\alpha'}G^{\mu\nu}\beta^G_{\mu\nu} = R - \frac{1}{4}H^2 + 2\nabla^2\Phi + O(\alpha').$$

Combining this with β^Φ yields

$$\boxed{\mathcal{E}_\Phi = \frac{1}{\alpha'}\left(G^{\mu\nu}\beta^G_{\mu\nu} - 4\beta^\Phi\right) + O(\alpha').}$$

This is the first explicit instance of field-space mixing: the dilaton Euler-Lagrange equation is a linear combination of the metric and dilaton Weyl-anomaly coefficients.

8.4.1.8 Variation with respect to the metric

It is useful to do the metric variation without first imposing the dilaton equation. Varying with respect to the inverse metric, the relevant identity is

$$\delta\big(\sqrt{-G}\, fR\big) = \sqrt{-G}\left[f\left(R_{\mu\nu} - \frac{1}{2}G_{\mu\nu}R\right) + \left(G_{\mu\nu}\nabla^2 - \nabla_\mu\nabla_\nu\right) f\right]\delta G^{\mu\nu}$$

up to a boundary term. Here $f = e^{-2\Phi}$, so

$$\frac{\nabla_\mu\nabla_\nu f}{f} = -2\nabla_\mu\nabla_\nu\Phi + 4\nabla_\mu\Phi\nabla_\nu\Phi,$$

$$\frac{\nabla^2 f}{f} = -2\nabla^2\Phi + 4(\nabla\Phi)^2.$$

The other metric variations are

$$\delta\sqrt{-G} = -\frac{1}{2}\sqrt{-G}\, G_{\mu\nu}\delta G^{\mu\nu},$$

$$\delta\big((\nabla\Phi)^2\big) = \nabla_\mu\Phi\nabla_\nu\Phi\,\delta G^{\mu\nu},$$

and, because H^2 contains three inverse metrics,

$$\delta H^2 = 3H_{\mu\rho\sigma}H_\nu{}^{\rho\sigma}\delta G^{\mu\nu}.$$

After collecting these terms, the coefficient of $\delta G^{\mu\nu}$ can be written particularly transparently in terms of the dilaton equation already found:

$$\delta_G S_{\text{eff}} = \int d^dx\,\sqrt{-G}\,e^{-2\Phi}\left[R_{\mu\nu} + 2\nabla_\mu\nabla_\nu\Phi - \frac{1}{4}H_{\mu\rho\sigma}H_\nu{}^{\rho\sigma} - \frac{1}{2}G_{\mu\nu}\mathcal{E}_\Phi\right]\delta G^{\mu\nu}.$$

Thus a convenient representative of the metric Euler-Lagrange equation is

$$\boxed{\mathcal{E}^G_{\mu\nu} \equiv R_{\mu\nu} + 2\nabla_\mu\nabla_\nu\Phi - \frac{1}{4}H_{\mu\rho\sigma}H_\nu{}^{\rho\sigma} - \frac{1}{2}G_{\mu\nu}\mathcal{E}_\Phi = 0.}$$

Comparison with $\beta^G_{\mu\nu}$ gives

$$\boxed{\mathcal{E}^G_{\mu\nu} = \frac{1}{\alpha'}\beta^G_{\mu\nu} - \frac{1}{2}G_{\mu\nu}\mathcal{E}_\Phi + O(\alpha').}$$

The term proportional to $G_{\mu\nu}\mathcal{E}_\Phi$ is why the raw metric variation is not literally $\beta^G_{\mu\nu}/\alpha'$. Once the dilaton equation is imposed, however, the metric equation reduces precisely to $\beta^G_{\mu\nu} = 0$ at this order.

8.4.1.9 Equivalence of the complete sets of equations

The three variations can now be compared with the three beta functions without any hand-waving. To the order displayed in the book,

$$\mathcal{E}^B_{\mu\nu} = 0 \quad \Longleftrightarrow \quad \beta^B_{\mu\nu} = 0,$$

$$\mathcal{E}_\Phi = \frac{1}{\alpha'}\left(G^{\mu\nu}\beta^G_{\mu\nu} - 4\beta^\Phi\right) + O(\alpha'),$$

$$\mathcal{E}^G_{\mu\nu} = \frac{1}{\alpha'}\beta^G_{\mu\nu} - \frac{1}{2}G_{\mu\nu}\mathcal{E}_\Phi + O(\alpha').$$

Therefore:

1. If all beta functions vanish, then $\mathcal{E}_\Phi = 0$, $\mathcal{E}^G_{\mu\nu} = 0$, and $\mathcal{E}^B_{\mu\nu} = 0$.
2. Conversely, if all Euler-Lagrange equations vanish, the metric relation gives $\beta^G_{\mu\nu} = 0$, the dilaton relation then gives $\beta^\Phi = 0$, and the B-field equation gives $\beta^B_{\mu\nu} = 0$.

Hence

$$\boxed{\left\{\frac{\delta S_{\text{eff}}}{\delta G} = 0, \frac{\delta S_{\text{eff}}}{\delta B} = 0, \frac{\delta S_{\text{eff}}}{\delta \Phi} = 0\right\} \quad \Longleftrightarrow \quad \{\beta^G = 0, \beta^B = 0, \beta^\Phi = 0\}}$$

to the common order in α' and g_s. The assertion is equality of the two solution sets, not component-by-component identity between every variational derivative and the beta function carrying the same label.

More generally, if all couplings are denoted by λ^I, the relation takes the schematic form

$$\frac{1}{\sqrt{-G}}\frac{\delta S_{\text{eff}}}{\delta \lambda^I(x)} = e^{-2\Phi}\mathcal{K}_{IJ}(\lambda)\beta^J(x),$$

where $\mathcal{K}_{IJ}$ is a local field-space kernel. After quotienting by redundant directions associated with diffeomorphisms and gauge transformations, $\mathcal{K}_{IJ}$ is nondegenerate perturbatively. It may mix fields, exactly as the explicit G-Φ relations above do, but it does not change the common zero locus.

8.4.1.10 Why the worldsheet and spacetime computations encode the same dynamics

This agreement is structural rather than accidental. A variation of a background field changes the worldsheet action by an integrated vertex operator:

$$\delta S_\sigma = \delta\lambda^I \int d^2z\, \mathcal{O}_I(z,\bar{z}).$$

Thus the background coupling λ^I is the spacetime field whose fluctuation is created by $\mathcal{O}_I$. At linear order, requiring $\mathcal{O}_I$ to be a marginal, BRST-invariant vertex operator gives the free spacetime equation of motion. At nonlinear order, operator products generate the beta functions; their vanishing is the condition that the deformed worldsheet theory remain conformal.

The spacetime effective action packages the same string amplitudes in target-space language. Its first variation is the one-point function, or tadpole, of the corresponding string state in the chosen background. A nonzero beta function is likewise a failure of the background to solve the string consistency conditions. Both computations are therefore measuring the same obstruction: the background is off shell.

String field theory makes this relation especially direct. The string field contains the coefficients of vertex operators, including $G_{\mu\nu}$, $B_{\mu\nu}$, and Φ among its massless components. Its classical equation of motion is the nonlinear BRST consistency condition. Expanding that equation in components gives the spacetime field equations, while interpreting the same deformation as a two-dimensional QFT gives the conformal-invariance conditions. The low-energy effective action is the massless, derivative-expanded form of this string-field-theoretic dynamics.

This also explains why reconstructing the action from on-shell scattering amplitudes is not a third, unrelated mechanism. The amplitudes determine interaction vertices modulo field redefinitions and terms proportional to lower-order equations of motion. The sigma-model beta functions determine the same equivalence class by demanding quantum Weyl invariance.

8.4.1.11 What “exactly” means and what it does not mean

Several qualifications are essential.

First, one should use the Weyl-anomaly coefficients. Ordinary renormalization-group beta functions can differ from them by target-space diffeomorphisms or B-field gauge transformations. Those differences correspond to redundant worldsheet operators and do not alter the physical conformal background.

Second, the comparison must be made in a common renormalization scheme. A local field redefinition,

$$\lambda'^I = \lambda'^I(\lambda),$$

changes the beta functions as a vector field on the space of couplings,

$$\beta'^I = \frac{\partial \lambda'^I}{\partial \lambda^J}\beta^J,$$

and rewrites the spacetime action in the new variables. Their component formulas can change, but the condition $\beta^I = 0$ and the physical content of the Euler-Lagrange equations do not.

Third, orders must be matched. A one-loop sigma-model calculation produces the leading two-derivative spacetime equations. Higher sigma-model loops produce higher powers of α' and must be compared with higher-derivative terms such as curvature-squared or curvature-quartic corrections in the effective action. Separately, worldsheet genus counts powers of g_s: equation (14.8) is a sphere-level, or closed-string tree-level, action, so quantum spacetime corrections require higher-genus contributions. The sigma-model loop expansion in α' and the genus expansion in g_s are distinct expansions.

Fourth, equation (14.8) is a truncation to the NS-NS massless fields. If additional massless fields, Ramond-Ramond backgrounds, open-string fields, or massive string modes are sourced, their equations and beta functions must be included. Setting omitted fields to zero is valid only when that is a consistent truncation.

Finally, the effective action is not unique as a formula. Total derivatives, local field redefinitions, and terms proportional to lower-order equations of motion can change its appearance without changing on-shell physics. The invariant statement is the equivalence class of equations and their solution space.

8.4.1.12 Checks

Critical, constant-dilaton, fluxless limit. Take $\Delta = 0$, $\nabla\Phi = 0$, and $H = 0$. The beta functions give

$$\beta^G_{\mu\nu} = \alpha' R_{\mu\nu}, \qquad \beta^\Phi = 0,$$

so conformal invariance requires $R_{\mu\nu} = 0$. The action reduces to the Einstein-Hilbert action. Its raw metric equation is $R_{\mu\nu} - \frac{1}{2}G_{\mu\nu}R = 0$; tracing it for $d \neq 2$ gives $R = 0$, and hence $R_{\mu\nu} = 0$. This simple example already shows why identical solution sets do not require identical component expressions.

Derivative counting. Each leading beta function contains two target-space derivatives: $R_{\mu\nu}$ has two derivatives of G, ∇H has two derivatives of B, and $\nabla^2\Phi$ has two derivatives of Φ. This matches the two-derivative action (14.8).

Gauge invariance. The action depends on B only through $H = dB$, so it is invariant under $B \mapsto B + d\Lambda$. Correspondingly, the B equation is a divergence equation, and its beta function is defined modulo the same redundant gauge direction.

Linearized spectrum. Expanding around $G_{\mu\nu} = \eta_{\mu\nu}$, constant Φ, and $B = 0$, the linearized beta-function equations reproduce the massless wave equations and transversality conditions after gauge fixing. These are the same conditions obtained from BRST invariance of the massless vertex operators.

8.4.1.13 Result

The sigma-model background fields are worldsheet couplings,

$$S_\sigma = S_\sigma[\lambda^I], \qquad \lambda^I = \{G_{\mu\nu}, B_{\mu\nu}, \Phi, ...\},$$

and quantum Weyl invariance requires

$$\boxed{\beta^I = 0.}$$

At leading order, the same equations follow from

$$\boxed{S_{\text{eff}} = \int d^dx \sqrt{-G}\, e^{-2\Phi} \left[R + 4(\nabla\Phi)^2 - \frac{1}{12}H^2 - \frac{d - d_{\text{crit}}}{\alpha'}\right].}$$

Explicitly,

$$\mathcal{E}^B = 0 \Longleftrightarrow \beta^B = 0,$$

$$\mathcal{E}_\Phi = \frac{1}{\alpha'}\left(G^{\mu\nu}\beta^G_{\mu\nu} - 4\beta^\Phi\right) + O(\alpha'),$$

$$\mathcal{E}^G_{\mu\nu} = \frac{1}{\alpha'}\beta^G_{\mu\nu} - \frac{1}{2}G_{\mu\nu}\mathcal{E}_\Phi + O(\alpha').$$

Consequently,

$$\boxed{\frac{\delta S_{\text{eff}}}{\delta\lambda^I} = 0 \quad \Longleftrightarrow \quad \beta^I = 0,}$$

provided the same fields, α' order, string-loop order, renormalization scheme, and gauge-equivalence class are used. The variational derivatives and beta functions need not be literally identical components; they are related by an invertible field-space transformation and have the same physical zero locus.

Chapter 9

Type-II Compactifications with D-Branes and Fluxes

9.1 Q1: Special-Lagrangian deformations and Wilson-line cohomology

In page650, it is said that: >Additional non-chiral matter transforming in the adjoint representation of $U(N_a)$ arises from open strings stretched between branes in the same stack, i.e. branes lying on top of each other. Geometrically, these correspond to deformations of the three-cycle inside the Calabi-Yau or to continuous Wilson lines along non-trivial one-cycles inside the three-cycles. They are counted by $H^0(\Sigma, N_\Sigma)$ and $H^1(\Sigma)$ respectively (Σ is a special Lagrangian inside the Calabi-Yau).

Why the number of these states are counted by these cohomologies?

> **Textbook location and related material.** Chapter 17, Section 17.2 around p. 650: same-stack open strings, geometric brane deformations, Wilson lines, and adjoint matter. Sections 14.4–14.5 give the normal-bundle and special-Lagrangian geometry.

9.1.1 Answer

9.1.1.1 Core point

There are three related but logically distinct statements in the quoted passage:

1. an arbitrary infinitesimal deformation of an embedded submanifold is a section of its normal bundle;
2. a deformation of a **special Lagrangian** submanifold is much more constrained, and its massless deformation space is isomorphic to $H^1(\Sigma, \mathbb{R})$;
3. an infinitesimal continuous Wilson line is a closed one-form modulo an exact one-form, so it is also represented by $H^1(\Sigma, \mathbb{R})$.

The first statement explains the appearance of $H^0(\Sigma, N_\Sigma)$. The second and third explain why the actual finite multiplicity of massless adjoint matter is $b_1(\Sigma)$.

More precisely, the book's $H^0(\Sigma, N_\Sigma)$ should not be read as saying that the multiplicity is literally

$$\dim H^0(\Sigma, N_\Sigma)$$

with H^0 the space of all smooth normal sections. That space is infinite-dimensional. It is the kinematic space in which the normal deformation field takes values. The massless supersymmetric modes are the sections satisfying the linearized special-Lagrangian equations. McLean's theorem gives

$$\boxed{T_{[\Sigma]}\mathcal{M}_{\text{sLag}} \cong \mathcal{H}^1(\Sigma) \cong H^1_{\text{dR}}(\Sigma, \mathbb{R})},$$

and therefore

$$\dim_{\mathbb{R}} \mathcal{M}_{\mathrm{sLag}} = b_1(\Sigma).$$

The Wilson-line moduli form another real space of dimension $b_1(\Sigma)$. Four-dimensional $\mathcal{N} = 1$ supersymmetry pairs each real geometric modulus with one real Wilson-line modulus. Thus a stack with gauge group $U(N_a)$ has

$$\boxed{b_1(\Sigma) \text{ complex scalars in } \operatorname{Adj} U(N_a)}.$$

Here "non-chiral matter" means that the adjoint representation is real and therefore produces no net four-dimensional gauge chirality. The fields can still be packaged into $\mathcal{N} = 1$ chiral supermultiplets.

9.1.1.2 Why same-stack strings give adjoint-valued fields

Consider N_a coincident D6-branes wrapping

$$\mathbb{R}^{1,3} \times \Sigma.$$

An open string whose two endpoints lie on this stack carries Chan–Paton labels (r, s), with $r, s = 1, \dots, N_a$. Its state is therefore accompanied by an $N_a \times N_a$ matrix λ. Under a gauge transformation $U \in U(N_a)$,

$$\lambda \longmapsto U\lambda U^{-1}.$$

Consequently the corresponding worldvolume fields take values in

$$\operatorname{End}(\mathbb{C}^{N_a}) \cong \mathfrak{u}(N_a),$$

which is the adjoint representation.

Before compactification, the massless bosonic fields on a D6-brane are a seven-dimensional gauge field together with three transverse scalar fields. After wrapping the D6-brane on Σ:

- the components A_μ, $\mu = 0, \dots, 3$, give the four-dimensional gauge field;
- the internal components A_m, $m = 1, 2, 3$ along Σ, give possible Wilson-line scalars;
- the three transverse scalar fields combine geometrically into a section of the rank-three real normal bundle N_Σ.

The question is which internal profiles of these seven-dimensional fields have zero Kaluza–Klein mass. Those are precisely the zero modes of the appropriate internal differential operators.

9.1.1.3 Arbitrary embedded deformations and $H^0(\Sigma, N_\Sigma)$

Let $i : \Sigma \hookrightarrow X$ be the embedding. An infinitesimal variation of the embedding is a vector field along Σ,

$$\delta i \in \Gamma(i^* TX).$$

Using the induced metric, it decomposes into tangent and normal parts,

$$\delta i = (\delta i)^{\parallel} + (\delta i)^{\perp}.$$

The tangent part changes only the parametrization of Σ; it is generated by an infinitesimal diffeomorphism of the worldvolume and does not change the embedded submanifold. The physical infinitesimal displacement is therefore the normal part,

$$v = (\delta i)^{\perp} \in \Gamma(N_\Sigma).$$

Equivalently,

$$T_{[i]} \frac{\operatorname{Emb}(\Sigma, X)}{\operatorname{Diff}(\Sigma)} \cong \Gamma(N_\Sigma) = H^0(\Sigma, N_\Sigma).$$

This is the precise kinematic meaning of the first cohomology group in the quoted sentence. A zeroth cohomology group records global sections: one must choose the normal displacement consistently over all of Σ, not independently in unrelated coordinate patches.

There is an important qualification. Since Σ is a real three-manifold and N_Σ is a real smooth vector bundle, $\Gamma(N_\Sigma)$ is generically infinite-dimensional. Most of its sections do not preserve the Lagrangian or special condition. They change the D6-brane energy and are not massless moduli. The BPS and zero-mode equations reduce this kinematic space to a finite-dimensional subspace.

This differs from the familiar statement that deformations of a **complex** submanifold are counted by holomorphic sections of a holomorphic normal bundle. In that B-brane setting, a Dolbeault group such as $H^0(\Sigma, N_\Sigma^{1,0})$ can already be finite-dimensional. A special Lagrangian D6-brane is an A-brane, and the relevant finite-dimensional theorem is instead McLean's theorem.

9.1.1.4 The Lagrangian condition identifies normal vectors with one-forms

Because Σ is Lagrangian,

$$i^*\omega = 0.$$

This implies that J exchanges tangent and normal directions:

$$J(T\Sigma) = N_\Sigma, \qquad J(N_\Sigma) = T\Sigma.$$

Indeed, if $u, w \in T\Sigma$, then

$$g(Ju, w) = \omega(u, w) = 0,$$

so Ju is orthogonal to $T\Sigma$ and hence normal. Since both bundles have real rank three, this is an isomorphism.

The metric then identifies a normal vector field v with a one-form on Σ. A convenient sign convention is

$$\alpha_v = -i^*(\iota_v\omega), \qquad \alpha_v(w) = -\omega(v, w), \quad w \in T\Sigma.$$

The map

$$N_\Sigma \longrightarrow T^*\Sigma, \qquad v \longmapsto \alpha_v,$$

is a pointwise bundle isomorphism. Thus the three normal components of the D6-brane deformation may equivalently be treated as the components of a one-form on Σ.

To see this explicitly, choose at a point of Σ an oriented orthonormal tangent basis e_1, e_2, e_3. Then Je_1, Je_2, Je_3 is a normal basis. If

$$v = v^I Je_I,$$

then, in the corresponding dual tangent coframe e^I,

$$\alpha_v = v^I e^I.$$

No information about v is lost in passing to α_v.

9.1.1.5 Linearizing the Lagrangian condition gives $d\alpha_v = 0$

Let $i_t : \Sigma \hookrightarrow X$ be a one-parameter family of embeddings with $i_0 = i$ and normal velocity v. For every differential form ρ on X,

$$\left.\frac{d}{dt}\right|_{t=0} i_t^*\rho = i^*(\mathcal{L}_v\rho).$$

Using Cartan's formula,

$$\mathcal{L}_v\rho = d(\iota_v\rho) + \iota_v(d\rho).$$

The Calabi–Yau Kähler form is closed, $d\omega = 0$. Therefore

$$\left.\frac{d}{dt}\right|_{t=0} i_t^*\omega = i^*d(\iota_v\omega) = -d\alpha_v.$$

For the deformed cycle to remain Lagrangian to first order, this variation must vanish. Hence

$$\boxed{d\alpha_v = 0}.$$

Thus infinitesimal Lagrangian deformations correspond locally to closed one-forms. If one imposed only the Lagrangian condition, exact one-forms would also occur; the special condition supplies the second equation needed for harmonicity.

9.1.1.6 Linearizing the special condition gives $d^\dagger\alpha_v = 0$

For the phase θ selected by the orientifold and the preserved $\mathcal{N} = 1$ supersymmetry, the second special-Lagrangian condition is

$$i^* \operatorname{Im}(e^{-i\theta}\Omega) = 0.$$

Since Ω is closed,

$$\frac{d}{dt}\bigg|_{t=0} i_t^* \operatorname{Im}(e^{-i\theta}\Omega) = d\, i^*\big(\iota_v \operatorname{Im}(e^{-i\theta}\Omega)\big)\,.$$

On a special Lagrangian three-cycle there is a pointwise identity

$$i^*(\iota_v \operatorname{Im}(e^{-i\theta}\Omega)) = *_\Sigma \alpha_v.$$

With a different sign convention for α_v the right-hand side acquires an overall minus sign, which does not affect the resulting equation.

The identity can be checked in an adapted orthonormal frame. Let f^I be the coframe dual to the normal vectors Je_I. At the chosen point one may write

$$\omega = \sum_{I=1}^{3} e^I \wedge f^I, \qquad e^{-i\theta}\Omega = (e^1 + if^1) \wedge (e^2 + if^2) \wedge (e^3 + if^3).$$

For $v = v^I Je_I$, contracting the terms in $\operatorname{Im}(e^{-i\theta}\Omega)$ containing one normal coframe gives

$$i^*(\iota_v \operatorname{Im}(e^{-i\theta}\Omega)) = v^1 e^2 \wedge e^3 - v^2 e^1 \wedge e^3 + v^3 e^1 \wedge e^2 = *_\Sigma(v^I e^I) = *_\Sigma \alpha_v.$$

The linearized special condition is consequently

$$d(*_\Sigma \alpha_v) = 0.$$

On a three-manifold, for a one-form α,

$$d^\dagger \alpha = -*_\Sigma\, d *_\Sigma \alpha.$$

Therefore

$$\boxed{d^\dagger \alpha_v = 0}.$$

Combining the two linearized equations gives

$$d\alpha_v = 0, \qquad d^\dagger \alpha_v = 0,$$

or equivalently

$$\Delta_1 \alpha_v = 0.$$

Hence α_v is a harmonic one-form.

9.1.1.7 McLean's theorem and the finite deformation count

The preceding calculation determines the tangent space to the special-Lagrangian deformation problem:

$$\big\{v \in \Gamma(N_\Sigma) : \delta_v(i^*\omega) = 0, \quad \delta_v i^* \operatorname{Im}(e^{-i\theta}\Omega) = 0\big\} \cong \mathcal{H}^1(\Sigma).$$

Hodge theory gives a unique harmonic representative of every de Rham cohomology class, so

$$\mathcal{H}^1(\Sigma) \cong H^1_{\mathrm{dR}}(\Sigma, \mathbb{R}).$$

McLean's theorem says more than this linear identification: for a compact special Lagrangian, these infinitesimal deformations are unobstructed. Locally, the moduli space of special Lagrangian deformations is a smooth manifold whose tangent space is $H^1(\Sigma, \mathbb{R})$. Therefore

$$\boxed{\#\{\text{real geometric moduli}\} = b_1(\Sigma)}.$$

This is the precise content behind the book's statement on the following page that "for special Lagrangian cycles the number of these two kinds of moduli are equal." The normal-bundle notation describes where the deformation field lives; the special-Lagrangian equations and McLean's theorem determine its finite set of massless zero modes.

The same result also has a direct Kaluza–Klein interpretation. Under the bundle isomorphism $N_\Sigma \cong T^*\Sigma$, the quadratic operator governing supersymmetric normal fluctuations reduces to the one-form Hodge Laplacian. If

$$\Delta_1 \eta_n = \lambda_n \eta_n,$$

then the corresponding four-dimensional field has a Kaluza–Klein mass proportional to λ_n. The massless modes have $\lambda_n = 0$, so they are exactly the harmonic one-forms.

9.1.1.8 Why continuous Wilson lines are counted by $H^1(\Sigma)$

Now consider the internal part a of the D6-brane gauge field. First take a single brane, or the Abelian part of a coincident stack. Infinitesimally,

$$a \in \Omega^1(\Sigma).$$

A Wilson-line modulus is a deformation of a flat connection, so its linearized field strength must vanish:

$$da = 0.$$

An infinitesimal gauge transformation with parameter $\lambda \in \Omega^0(\Sigma)$ acts by

$$a \longmapsto a + d\lambda.$$

The space of inequivalent infinitesimal flat connections is therefore

$$\frac{\ker(d : \Omega^1 \to \Omega^2)}{\operatorname{im}(d : \Omega^0 \to \Omega^1)} = H^1_{\mathrm{dR}}(\Sigma, \mathbb{R}).$$

Equivalently, in a gauge such as $d^\dagger a = 0$, the zero-mode equation is

$$\Delta_1 a = 0.$$

Thus one may expand

$$a(x, y) = \sum_{I=1}^{b_1(\Sigma)} \xi^I(x) \eta_I(y) + \text{massive modes},$$

where the η_I form a basis of harmonic one-forms. Each coefficient $\xi^I(x)$ is a real massless scalar in four dimensions.

The terminology "Wilson line" comes from its holonomy around a closed one-cycle $\gamma \subset \Sigma$,

$$W_\gamma = \exp\left(i \oint_\gamma A \right).$$

If $a = d\lambda$ is pure gauge, its integral around every closed cycle vanishes. A non-exact closed one-form can have nonzero periods and changes these holonomies. Globally, for a $U(1)$ connection the continuous moduli space is the torus

$$H^1(\Sigma, \mathbb{R}) \big/ 2\pi H^1(\Sigma, \mathbb{Z}) \cong \operatorname{Hom}(H_1(\Sigma, \mathbb{Z}), U(1))_{\text{continuous}},$$

whose real dimension is $b_1(\Sigma)$. Torsion in $H_1(\Sigma, \mathbb{Z})$ can produce discrete Wilson lines, but discrete choices do not give continuous four-dimensional scalar fields and are not counted by b_1.

For a $U(N_a)$ stack with a background flat bundle E, the correct infinitesimal space is

$$H^1(\Sigma, \operatorname{ad} E).$$

For the trivial flat connection used implicitly in the quoted counting,

$$H^1(\Sigma, \operatorname{ad} E) \cong H^1(\Sigma, \mathbb{R}) \otimes \mathfrak{u}(N_a).$$

This makes the adjoint Chan–Paton representation explicit.

9.1.1.9 Pairing into complex adjoint scalars

Choose a harmonic basis η_I, $I = 1, \dots, b_1(\Sigma)$. Through the special-Lagrangian isomorphism, a normal deformation can be written as

$$\alpha_v(x,y) = \sum_{I=1}^{b_1(\Sigma)} s^I(x)\eta_I(y),$$

while the Wilson-line fluctuation is

$$a(x,y) = \sum_{I=1}^{b_1(\Sigma)} \xi^I(x)\eta_I(y).$$

Thus for each I there are two real four-dimensional scalar fields:

$$(s^I, \xi^I).$$

They are paired by the supersymmetry preserved by the special Lagrangian D6-brane. After a convention-dependent normalization, one can form complex coordinates

$$\Phi^I = \xi^I + i\, c_I s^I,$$

where the factors c_I depend on the normalization of the harmonic forms, the D6-brane action, and the chosen four-dimensional fields. The invariant statement is the counting:

$$\underbrace{b_1(\Sigma)}_{\text{real sLag deformations}} + \underbrace{b_1(\Sigma)}_{\text{real Wilson lines}} = \underbrace{b_1(\Sigma)}_{\text{complex scalars}} .$$

Because both types of open-string zero modes carry the same endpoint Chan–Paton matrices, every Φ^I transforms in $\mathrm{Adj}\, U(N_a)$.

9.1.1.10 Checks and qualifications

For a factorized toroidal cycle $\Sigma = T^3$,

$$b_1(T^3) = 3.$$

There are three independent harmonic one-forms dx^1, dx^2, dx^3, hence three real Wilson-line moduli and three real special-Lagrangian position/deformation moduli. They combine into three complex adjoint scalars, in agreement with the familiar toroidal D6-brane spectrum.

For $\Sigma = S^3$,

$$b_1(S^3) = 0.$$

The special Lagrangian is locally rigid and there are no continuous Wilson lines, so there are no adjoint chiral multiplets of this type. Notice that $\Gamma(N_{S^3})$ is nevertheless infinite-dimensional. This is a simple demonstration that the book's $H^0(\Sigma, N_\Sigma)$ cannot, by itself, be the literal finite multiplicity of massless fields.

The counting above assumes a compact smooth special Lagrangian without boundary, a fixed ambient Calabi–Yau structure, no worldvolume flux, and the trivial flat Chan–Paton bundle. A nontrivial flat bundle replaces ordinary cohomology by $H^1(\Sigma, \mathrm{ad}\, E)$. Orientifold projections can retain only particular parity eigenspaces, and fluxes or other effects can lift some modes. For a generic non-invariant brane and its orientifold image giving $U(N_a)$, the book is using the unprojected $b_1(\Sigma)$ count described above.

9.1.1.11 Result

The hierarchy of spaces is

$$\boxed{\begin{aligned} \text{all infinitesimal embedded deformations:} &\quad v \in H^0(\Sigma, N_\Sigma) = \Gamma(N_\Sigma), \\ \alpha_v : &\quad v \longmapsto -i^*(\iota_v \omega) \in \Omega^1(\Sigma), \\ \text{linearized sLag conditions:} &\quad d\alpha_v = 0, \qquad d^\dagger \alpha_v = 0, \\ \text{massless geometric moduli:} &\quad T_{[\Sigma]}\mathcal{M}_{\text{sLag}} \cong \mathcal{H}^1(\Sigma) \cong H^1(\Sigma, \mathbb{R}), \\ \text{continuous Wilson lines:} &\quad \frac{\{a \in \Omega^1 : da = 0\}}{\{a \sim a + d\lambda\}} \cong H^1(\Sigma, \mathbb{R}). \end{aligned}}$$

Therefore,

$$\boxed{b_1(\Sigma) \text{ real geometric modes} + b_1(\Sigma) \text{ real Wilson-line modes} = b_1(\Sigma) \text{ complex adjoint scalars}}.$$

9.2 Q2: No-scale structure and the tree-level vacuum energy

In page661, they derived the holomorphic superpotential for type IIB on Calabi-Yau threefold with non-trivial three-form flux

$$W = \frac{1}{\tilde{\kappa}_{10}^2} \int_X \Omega \wedge G_3$$

The holomorphic three-form depends on the complex structure moduli, G_3 on the complex scalar but W is independent of the Kähler moduli. The Kähler potential, which is also needed to compute the scalar potential, depends on the complex-structure moduli, on the Kähler-moduli and on τ. The dependence on the Kähler moduli features the so-called no-scale-structure, which means that supersymmetric minima have vanishing vacuum energy.

I am interested with this no-scale-structure. What is its concrete statement, and to what extend it leads to the vanishing of vacuum energy?

> **Textbook location and related material.** Chapter 17, Section 17.3 around pp. 657–663, especially p. 661: the Gukov–Vafa–Witten superpotential, Kähler potential, F-term potential, and no-scale structure.

9.2.1 Answer

9.2.1.1 Core point

The concrete no-scale statement is an identity satisfied by the Kähler potential of the Kähler moduli:

$$\boxed{K^{\alpha\bar{\beta}} K_\alpha K_{\bar{\beta}} = 3}$$

in four-dimensional Planck units. Together with

$$\partial_{T_\alpha} W = 0,$$

this implies

$$D_\alpha W = K_\alpha W$$

and hence

$$K^{\alpha\bar{\beta}} D_\alpha W D_{\bar{\beta}} \overline{W} = 3|W|^2.$$

This term cancels the universal negative term $-3|W|^2$ in the $\mathcal{N} = 1$ F-term potential.

The cancellation does **not** imply that the scalar potential vanishes identically. It removes only the Kähler-modulus part. At tree level the result is

$$\boxed{V_F = e^K K^{A\bar{B}} D_A W D_{\bar{B}} \overline{W} \geq 0,}$$

where A, B run over the complex-structure moduli and the axiodilaton. Therefore:

- V_F is zero when $D_A W = 0$ for all complex-structure and axiodilaton directions;
- this zero-energy point need not be supersymmetric, because $D_\alpha W = K_\alpha W$ is nonzero when $W \neq 0$;
- a fully supersymmetric vacuum must also obey $D_\alpha W = 0$, which in a no-scale model forces $W = 0$ and therefore gives a supersymmetric Minkowski vacuum;
- the Kähler moduli are not stabilized at this order;
- corrections that change the Kähler potential or introduce T_α dependence in W generally break the cancellation.

Thus there are two distinct conclusions:

$$\boxed{\begin{array}{llllll} D_A W = 0, & W \neq 0 & \Longrightarrow & V_F = 0, & D_\alpha W \neq 0, & \text{nonsupersymmetric no-scale Minkowski,} \\ D_A W = 0, & W = 0 & \Longrightarrow & V_F = 0, & D_I W = 0, & \text{supersymmetric Minkowski.} \end{array}}$$

The sentence in the book that "supersymmetric minima have vanishing vacuum energy" refers to the second line. The same no-scale cancellation actually allows the larger first class of zero-energy but nonsupersymmetric minima, as the book explains immediately below equation (17.58).

9.2.1.2 The generic $\mathcal{N} = 1$ F-term potential

For four-dimensional $\mathcal{N} = 1$ supergravity, the F-term scalar potential is

$$V_F = e^K \left(K^{I\bar{J}} D_I W D_{\bar{J}} \overline{W} - 3|W|^2 \right),$$

where

$$D_I W = \partial_I W + K_I W.$$

The index I runs over every chiral multiplet. We use $M_{\mathrm{P}} = 1$ here. The book retains κ_4 and writes

$$V_F = e^{\kappa_4^2 K} \left(K^{I\bar{J}} D_I W D_{\bar{J}} \overline{W} - 3\kappa_4^2 |W|^2 \right), \qquad D_I W = \partial_I W + \kappa_4^2 K_I W.$$

The first term is nonnegative because the Kähler metric is positive definite. The second term is negative. Consequently, in a generic $\mathcal{N} = 1$ theory, a supersymmetric point

$$D_I W = 0 \quad \text{for every } I$$

has

$$V_{\text{susy}} = -3 e^K |W|^2.$$

It is therefore:

$$\begin{array}{r|c} W = 0 & \text{supersymmetric Minkowski,} \\ W \neq 0 & \text{supersymmetric anti-de Sitter.} \end{array}$$

The special feature of the flux compactification is that the Kähler moduli occur in a sector whose positive F-term contribution is forced to reproduce exactly $3|W|^2$. That is the no-scale structure.

9.2.1.3 Tree-level Kähler potential in type IIB

For the Calabi–Yau orientifold compactification considered around equation (17.58), the tree-level Kähler potential has the schematic separated form

$$K = K_{\text{cs}}(z, \bar{z}) + K_\tau(\tau, \bar{\tau}) + K_{\mathrm{K}}(T, \bar{T}),$$

with

$$K_{\mathrm{cs}} = -\log\left(i\int_X \Omega(z)\wedge\overline{\Omega(z)}\right),$$

$$K_\tau = -\log[-i(\tau-\bar{\tau})],$$

and

$$\boxed{K_{\mathrm{K}} = -2\log\mathcal{V}(T+\bar{T}).}$$

Additive constants and convention-dependent powers of $\tilde{\kappa}_{10}$ do not affect the no-scale identity.

The flux superpotential is

$$W(z,\tau) = \frac{1}{\tilde{\kappa}_{10}^2}\int_X \Omega(z)\wedge(F_3-\tau H_3).$$

It depends on z^i through Ω and on τ through G_3, but at string tree level

$$\boxed{\partial_{T_\alpha}W = 0.}$$

The absence of T_α from W is only one half of the no-scale mechanism. The other half is the precise homogeneity of $\mathcal{V}$.

9.2.1.4 Deriving the no-scale identity from volume homogeneity

Let

$$x^\alpha = T_\alpha + \overline{T}_\alpha.$$

The real parts of the T_α are four-cycle volumes:

$$\tau_\alpha = \frac{\partial\mathcal{V}}{\partial v^\alpha} = \frac{1}{2}\kappa_{\alpha\beta\gamma}v^\beta v^\gamma,$$

while

$$\mathcal{V} = \frac{1}{6}\kappa_{\alpha\beta\gamma}v^\alpha v^\beta v^\gamma.$$

Under a rescaling $v^\alpha\mapsto\lambda v^\alpha$,

$$\mathcal{V}\mapsto\lambda^3\mathcal{V},\qquad \tau_\alpha\mapsto\lambda^2\tau_\alpha.$$

Therefore, when regarded as a function of the four-cycle volumes, $\mathcal{V}$ is homogeneous of degree 3/2:

$$\mathcal{V}(\lambda x) = \lambda^{3/2}\mathcal{V}(x).$$

Euler's theorem for homogeneous functions gives

$$x^\alpha\partial_{x^\alpha}\mathcal{V} = \frac{3}{2}\mathcal{V}.$$

Because

$$K_{\mathrm{K}} = -2\log\mathcal{V},$$

we obtain

$$x^\alpha K_\alpha = -2\,\frac{x^\alpha\partial_{x^\alpha}\mathcal{V}}{\mathcal{V}} = -3.$$

Here $\partial/\partial T_\alpha$ acts as $\partial/\partial x^\alpha$ on a function that depends only on $T+\bar{T}$.

Differentiate the last identity with respect to x^β:

$$K_{\bar{\beta}} + x^\alpha K_{\alpha\bar{\beta}} = 0.$$

Multiplying by the inverse Kähler metric yields

$$K^{\gamma\bar{\beta}}K_{\bar{\beta}} = -x^\gamma.$$

Contracting once more with K_γ gives

$$K^{\gamma\bar{\beta}} K_\gamma K_{\bar{\beta}} = -x^\gamma K_\gamma = 3.$$

Therefore

$$\boxed{K^{\alpha\bar{\beta}} K_\alpha K_{\bar{\beta}} = 3.}$$

This is not an accidental one-modulus relation. It follows from the degree-3/2 homogeneity of the Calabi–Yau volume as a function of divisor volumes and therefore holds for any number of Kähler moduli at the classical level.

In the dimensionful convention of equation (17.56), the corresponding formulas are

$$K^{\alpha\bar{\beta}} K_\alpha K_{\bar{\beta}} = \frac{3}{\kappa_4^2}, \qquad D_\alpha W = \kappa_4^2 K_\alpha W.$$

Consequently,

$$K^{\alpha\bar{\beta}} D_\alpha W D_{\bar{\beta}} \overline{W} = 3\kappa_4^2 |W|^2,$$

which cancels precisely the $-3\kappa_4^2|W|^2$ term printed in equation (17.56).

9.2.1.5 Explicit one-modulus calculation

The example in footnote 21 has one Kähler modulus and

$$K = -3\log(T+\overline{T}) + \widetilde{K}(\phi_A, \overline{\phi}_{\bar{A}}), \qquad W = W(\phi_A).$$

Then

$$K_T = -\frac{3}{T+\overline{T}}, \qquad K_{T\overline{T}} = \frac{3}{(T+\overline{T})^2}, \qquad K^{T\overline{T}} = \frac{(T+\overline{T})^2}{3}.$$

Hence

$$K^{T\overline{T}} K_T K_{\overline{T}} = \frac{(T+\overline{T})^2}{3} \frac{9}{(T+\overline{T})^2} = 3.$$

Since $\partial_T W = 0$,

$$D_T W = K_T W.$$

It follows that

$$K^{T\overline{T}} |D_T W|^2 = K^{T\overline{T}} K_T K_{\overline{T}} |W|^2 = 3|W|^2.$$

Substitution into the full potential gives

$$\begin{aligned} V_F &= e^K \left[K^{A\bar{B}} D_A W D_{\bar{B}} \overline{W} + K^{T\overline{T}} |D_T W|^2 - 3|W|^2 \right] \\ &= e^K K^{A\bar{B}} D_A W D_{\bar{B}} \overline{W}. \end{aligned}$$

This is the formula displayed in compressed form in footnote 21.

9.2.1.6 What remains after the cancellation

For several Kähler moduli, the same calculation gives

$$\begin{aligned} V_F &= e^K [K^{A\bar{B}} D_A W D_{\bar{B}} \overline{W} + K^{\alpha\bar{\beta}} D_\alpha W D_{\bar{\beta}} \overline{W} - 3|W|^2] \\ &= e^K K^{A\bar{B}} D_A W D_{\bar{B}} \overline{W}. \end{aligned}$$

Thus V_F is positive semidefinite, not identically zero:

$$V_F \geq 0.$$

It vanishes precisely when

$$D_A W = 0$$

for all complex-structure and axiodilaton fields, assuming the corresponding Kähler metric is nonsingular and positive definite.

At such a point the fluxes have fixed z^i and τ, but the potential has no dependence that fixes T_α. After inserting the solution for z^i and τ, one obtains a constant

$$W_0 = W(z_\star, \tau_\star),$$

while

$$V_F(T, \bar{T}) = 0$$

along the entire Kähler-moduli space at this order. This degenerate family of flat directions is the origin of the name "no-scale."

9.2.1.7 Relation to ISD flux and the Hodge decomposition of G_3

The four-dimensional result is the effective-supergravity version of the ten-dimensional positive-semidefinite potential in equations (17.53)–(17.55). For harmonic three-form flux on a Calabi–Yau threefold, the imaginary self-dual and imaginary anti-self-dual components are

$$*_6 G_3^{\mathrm{ISD}} = i G_3^{\mathrm{ISD}}, \qquad *_6 G_3^{\mathrm{IASD}} = -i G_3^{\mathrm{IASD}}.$$

The flux potential, after the tadpole and source contribution has been included, is proportional to the norm of the IASD component:

$$V_{\mathrm{flux}} \sim \frac{1}{\operatorname{Im}\tau\, \mathcal{V}^2} \int_X G_3^{\mathrm{IASD}} \wedge *_6 \overline{G_3^{\mathrm{IASD}}} \geq 0.$$

Therefore its zero-energy minima obey

$$G_3^{\mathrm{IASD}} = 0,$$

so G_3 is ISD.

In the Calabi–Yau Hodge decomposition, and under the harmonic/primitivity assumptions used in this discussion,

$$\begin{array}{c|l} \mathrm{ISD} & G_3^{(2,1)} + G_3^{(0,3)}, \\ \mathrm{IASD} & G_3^{(1,2)} + G_3^{(3,0)}. \end{array}$$

The complex-structure equations $D_i W = 0$ remove the $(1,2)$ part, while the axiodilaton equation $D_\tau W = 0$ removes the $(3,0)$ part. These equations therefore impose the ISD condition but can leave a $(0,3)$ component.

The value of the superpotential detects precisely that remaining component:

$$W = \frac{1}{\tilde{\kappa}_{10}^2} \int_X \Omega^{(3,0)} \wedge G_3^{(0,3)}.$$

Thus:

- primitive $(2,1)$ flux has $W = 0$ and preserves $\mathcal{N} = 1$ supersymmetry;
- a $(0,3)$ component gives $W \neq 0$, is still ISD, and can have zero no-scale potential, but it breaks supersymmetry.

This is why the condition "ISD" is sufficient for a tree-level zero-energy minimum but is not, by itself, sufficient for a supersymmetric minimum.

9.2.1.8 Why a fully supersymmetric no-scale vacuum forces $W = 0$

Unbroken four-dimensional $\mathcal{N} = 1$ supersymmetry requires

$$D_I W = 0$$

for **all** chiral multiplets, including the Kähler moduli. Since W is independent of them,

$$D_\alpha W = K_\alpha W.$$

The no-scale Euler identity gives

$$x^\alpha K_\alpha = -3.$$

If every $D_\alpha W$ vanishes, then

$$0 = x^\alpha D_\alpha W = x^\alpha K_\alpha W = -3W.$$

Therefore

$$\boxed{\text{full supersymmetry} \quad \Longrightarrow \quad W = 0.}$$

At such a point,

$$D_I W = 0, \qquad W = 0,$$

and the generic supergravity expression gives

$$\boxed{V_{\text{susy}} = 0.}$$

In flux language, $W = 0$ removes $G_3^{(0,3)}$, and the supersymmetric flux is primitive of type $(2,1)$, exactly as stated below equation (17.58).

This explains the book's statement more precisely. A generic supersymmetric $\mathcal{N} = 1$ vacuum may be AdS when $W \neq 0$, but in this tree-level no-scale compactification the Kähler-modulus F-term equations themselves force $W = 0$. Hence the supersymmetric flux vacua in this approximation are Minkowski.

9.2.1.9 Zero vacuum energy does not imply supersymmetry

Suppose instead that

$$D_A W = 0, \qquad W_0 \neq 0.$$

The no-scale potential still vanishes:

$$V_F = 0.$$

However,

$$D_\alpha W = K_\alpha W_0 \neq 0.$$

Equivalently, the Kähler-modulus auxiliary fields are nonzero:

$$F^\alpha = -e^{K/2} K^{\alpha\bar{\beta}} D_{\bar{\beta}} \overline{W} \neq 0.$$

The gravitino mass is also nonzero,

$$m_{3/2} = e^{K/2} |W_0|.$$

Supersymmetry is therefore spontaneously broken even though the classical vacuum energy vanishes.

The equality $V_F = 0$ is a cancellation between the positive supersymmetry-breaking contribution

$$K^{\alpha\bar{\beta}} D_\alpha W D_{\bar{\beta}} \overline{W} = 3|W_0|^2$$

and the negative supergravity term $-3|W_0|^2$. It is not a statement that every auxiliary field vanishes.

This distinction is particularly important:

$$\boxed{V = 0 \not\Longrightarrow \text{unbroken supersymmetry.}}$$

The ISD flux $G_3^{(0,3)}$ provides the standard string-theory realization of this nonsupersymmetric no-scale Minkowski situation.

9.2.1.10 Scope and limitations of the vanishing-energy result

The cancellation is powerful but limited. The tree-level conclusion requires all of the following.

9.2.1.10.1 The remaining F-terms must vanish No-scale structure cancels only the Kähler-modulus block and the universal negative term. If

$$D_iW \neq 0 \quad \text{or} \quad D_\tau W \neq 0,$$

then

$$V_F = e^K K^{A\bar{B}} D_A W D_{\bar{B}} \overline{W} > 0.$$

Thus no-scale structure does not make an arbitrary flux configuration a Minkowski vacuum.

9.2.1.10.2 D-terms must vanish as well The full scalar potential is

$$V = V_F + V_D.$$

Gauge fluxes, charged fields, or D7-branes can generate a positive-semidefinite D-term potential. A zero-energy vacuum also requires

$$V_D = 0.$$

The F-term no-scale identity does not cancel a nonzero D-term.

9.2.1.10.3 The Kähler moduli are not stabilized At the no-scale minimum,

$$V_F = 0$$

for a continuous family of T_α. The tree-level flux potential fixes the complex structure and axiodilaton but leaves the overall volume and other Kähler moduli massless. A vacuum with exactly flat moduli is not yet a fully stabilized phenomenological vacuum.

9.2.1.10.4 Corrections break the classical identity or the T-independence of W The relation

$$K^{\alpha\bar{\beta}} K_\alpha K_{\bar{\beta}} = 3$$

uses the classical homogeneous Kähler potential $K_{\mathrm{K}} = -2\log \mathcal{V}$. Higher-derivative α' corrections and string-loop corrections change K and generally change the left-hand side away from 3.

Nonperturbative effects such as Euclidean D3-brane instantons or gaugino condensation can generate

$$W = W_0 + \sum_\alpha A_\alpha e^{-a_\alpha T_\alpha}.$$

Then

$$\partial_{T_\alpha} W \neq 0,$$

so $D_\alpha W$ is no longer simply $K_\alpha W$ and the cancellation no longer follows. These effects are precisely what are used in constructions such as KKLT or the Large Volume Scenario to stabilize Kähler moduli. The resulting vacuum energy can be negative, zero, or positive depending on the complete stabilization and uplift mechanism; no-scale structure alone no longer fixes it.

9.2.1.10.5 Warping and backreaction require the appropriate effective theory The book states equation (17.58) while initially neglecting backreaction and then explains that the consistent solution is a warped Calabi–Yau geometry. The leading Giddings–Kachru–Polchinski effective theory retains the relevant no-scale structure, but strong warping, localized sources, and corrections must be incorporated consistently. The simple unwarped formulas should therefore be understood as the leading two-derivative approximation.

Consequently, the no-scale cancellation is not a solution of the cosmological-constant problem. It is a classical structural cancellation in a controlled approximation. Once the effects required to stabilize the Kähler moduli are included, the vacuum energy must be computed again.

9.2.1.11 Result

At type-IIB tree level,

$$K = K_{\mathrm{cs}}(z,\bar{z}) + K_\tau(\tau,\bar{\tau}) - 2\log\mathcal{V}(T+\bar{T}),$$
$$W = \frac{1}{\tilde{\kappa}_{10}^2}\int_X \Omega(z)\wedge(F_3-\tau H_3), \qquad \partial_{T_\alpha}W = 0.$$

Homogeneity of $\mathcal{V}$ gives

$$\boxed{K^{\alpha\bar{\beta}}K_\alpha K_{\bar{\beta}} = 3, \qquad D_\alpha W = K_\alpha W.}$$

Therefore

$$\boxed{\begin{aligned} V_F &= e^K\left[K^{A\bar{B}}D_AWD_{\bar{B}}\overline{W} + K^{\alpha\bar{\beta}}D_\alpha WD_{\bar{\beta}}\overline{W} - 3|W|^2\right]\\ &= e^KK^{A\bar{B}}D_AWD_{\bar{B}}\overline{W} \geq 0. \end{aligned}}$$

The vacuum classes are

conditions	V_F	supersymmetry	Kähler moduli
$D_AW=0,\ W=0$	0	unbroken	unfixed
$D_AW=0,\ W\neq 0$	0	broken by $D_\alpha W$	unfixed
$D_AW\neq 0$	>0	broken	generically still unstabilized at tree level

Thus no-scale structure ensures that fully supersymmetric flux vacua are Minkowski and also permits nonsupersymmetric tree-level Minkowski vacua. It does not make all vacua have zero energy, does not stabilize the Kähler moduli, and does not protect the zero energy once the no-scale structure is broken by corrections or additional effects.

Chapter 10

String Dualities and M-Theory

10.1 Q1: String, Einstein, and modified Einstein frames

In page690 eq(18.26), they used the so-called modified Einstein frame: >The frame we have used in (18.26) is called modified Einstein frame. It is related to the Einstein frame through a rescaling of the metric by the constant (asymptotic) value of the dilaton.

There are at least three frames used in this book: string frame, Einstein frame and this modified Einstein frame. List the low-energy effective action in each frame and show how they are related, and the special properties of each frame, and in which discussion which frame is used.

> **Textbook location and related material.** Chapter 18, Sections 18.3 and 18.5 (pp. 680–684, 688–705), especially Eq. (18.26): string, Einstein, and asymptotically normalized modified Einstein frames for brane solutions.

10.1.1 Answer

10.1.1.1 Core point

The three frames do not describe three different theories. They are three choices of field variables, distinguished by which metric is called the spacetime metric. Their exact relation in ten dimensions is

$$\boxed{G^{(E)}_{MN} = e^{-\Phi/2} G^{(S)}_{MN}, \qquad g^{(mE)}_{MN} = e^{-\phi/2} G^{(S)}_{MN} = \sqrt{g_s}\, G^{(E)}_{MN}},$$

where

$$\Phi = \Phi_0 + \phi, \qquad g_s = e^{\Phi_0}, \qquad \phi \xrightarrow{r\to\infty} 0.$$

Equivalently,

$$G^{(S)}_{MN} = e^{\Phi/2} G^{(E)}_{MN} = e^{\phi/2} g^{(mE)}_{MN}, \qquad G^{(E)}_{MN} = \frac{1}{\sqrt{g_s}} g^{(mE)}_{MN}.$$

Thus the ordinary and modified Einstein metrics differ only by a constant. They have the same local advantages: in both frames the Ricci scalar has no dilaton prefactor and the graviton and dilaton kinetic terms are diagonal. The reason for introducing the modified frame is instead the boundary condition. If the string-frame metric approaches the coordinate Minkowski metric, then

$$G^{(S)}_{MN} \to \eta_{MN}, \qquad g^{(mE)}_{MN} \to \eta_{MN}, \qquad G^{(E)}_{MN} \to \frac{\eta_{MN}}{\sqrt{g_s}}.$$

Consequently, the modified frame is convenient for computing the ADM mass or tension of the brane solutions without performing a final constant conversion.

The rest of the answer first writes the action in all three frames, then derives the Weyl transformation explicitly, and finally explains where each frame is used in the book.

10.1.1.2 String-frame action

For the bosonic type-II fields, omitting the Chern–Simons terms for the moment, the string-frame action can be written schematically as

$$S_S = \frac{1}{2\tilde{\kappa}_{10}^2}\int d^{10}x\,\sqrt{-G^{(S)}}\left[e^{-2\Phi}\left(R_S + 4(\partial\Phi)_S^2 - \frac{1}{2\cdot 3!}H_{MNP}H_S^{MNP}\right) - \sum_{n\in\mathrm{RR}}\frac{1}{2n!}F_{n\,S}^2\right] + S_{\mathrm{CS}}.$$

For type IIA the non-democratic R–R degrees are F_2, F_4, while for type IIB they are F_1, F_3, F_5. In the IIB pseudo-action the F_5 term has one extra factor of $1/2$,

$$-\frac{1}{4}|F_5|^2,$$

and the self-duality condition $F_5 = *F_5$ is imposed after varying the pseudo-action.

This is the structure displayed explicitly in equation (16.134). The decisive feature is that every tree-level NS–NS term has the common factor $e^{-2\Phi}$, whereas the R–R kinetic terms do not. This difference is not arbitrary. A closed-string sphere has Euler characteristic $\chi = 2$ and hence contributes g_s^{-2}; the NS–NS bulk action is therefore naturally organized with $e^{-2\Phi}$. The R–R fields used in (16.134) have been normalized so that their kinetic terms have no such exponential.

It is useful to rewrite the same action in the Chapter 18 normalization. Split off the asymptotic dilaton and define

$$\kappa_{10} = g_s\tilde{\kappa}_{10}, \qquad \widetilde{F}_{\mathrm{RR}} = g_s F_{\mathrm{RR}}, \qquad \widetilde{H}_3 = H_3.$$

Then

$$S_S = \frac{1}{2\kappa_{10}^2}\int d^{10}x\,\sqrt{-G^{(S)}}\left[e^{-2\phi}\left(R_S + 4(\partial\phi)_S^2 - \frac{1}{2\cdot 3!}H_{3\,S}^2\right) - \sum_{n\in\mathrm{RR}}\frac{1}{2n!}\widetilde{F}_{n\,S}^2\right] + S_{\mathrm{CS}}.$$

Indeed,

$$\frac{e^{-2\Phi}}{\tilde{\kappa}_{10}^2} = \frac{e^{-2\phi}}{\kappa_{10}^2}, \qquad \frac{F_n^2}{\tilde{\kappa}_{10}^2} = \frac{\widetilde{F}_n^2}{\kappa_{10}^2}.$$

This second form makes the passage to equation (18.26) transparent.

The special properties of string frame are:

- $G_{MN}^{(S)}$ is the metric that appears directly in the Polyakov sigma-model action,

$$S_\sigma = -\frac{1}{4\pi\alpha'}\int d^2\sigma\sqrt{-h}\,h^{\alpha\beta}G_{MN}^{(S)}(X)\partial_\alpha X^M\partial_\beta X^N + \cdots.$$

- The world-sheet beta functions directly give the string-frame equations of motion. This is why equation (14.8), derived from the vanishing of the sigma-model beta functions, is in string frame.
- The factors $e^{-2\Phi}$ for a sphere and $e^{-\Phi}$ for a disk make string-loop counting manifest. In particular, the string-frame fundamental-string tension is independent of g_s, a D-brane tension scales as g_s^{-1}, and an NS5-brane tension scales as g_s^{-2}.
- The Buscher T-duality rules, equations (14.15)–(14.16), take their standard form for $G^{(S)}$, B_2, and Φ.
- The disadvantage is that $e^{-2\Phi}R_S$ produces kinetic mixing between the trace of the metric fluctuation and the dilaton. Thus the graviton and dilaton are not diagonal variables at quadratic order.

10.1.1.3 Ordinary Einstein-frame action

Define the ordinary Einstein metric by equation (16.141),

$$\boxed{G_{MN}^{(E)} = e^{-\Phi/2}G_{MN}^{(S)}}.$$

In this frame the type-II bulk action is

$$S_E = \frac{1}{2\tilde{\kappa}_{10}^2} \int d^{10}x \sqrt{-G^{(E)}} \left[R_E - \frac{1}{2}(\partial\Phi)_E^2 - \frac{e^{-\Phi}}{2\cdot 3!} H_{3\,E}^2 - \sum_{n\in\mathrm{RR}} \frac{e^{(5-n)\Phi/2}}{2n!} F_{n\,E}^2 \right] + S_{\mathrm{CS}}.$$

For $n = p+2$, the R–R exponent is

$$\frac{5-n}{2} = \frac{3-p}{2} = a_p.$$

Thus, for example,

$$F_2 : \ e^{3\Phi/2}, \qquad F_3 : \ e^{\Phi}, \qquad F_4 : \ e^{\Phi/2}, \qquad F_5 : \ e^{0}.$$

The NS–NS three-form instead has $e^{-\Phi}$, corresponding to $a_1 = -1$. The coincidence $a_3 = 0$ for the R–R five-form explains why the D3-brane can have a constant dilaton.

The defining properties of Einstein frame are now visible directly:

- The gravitational term is the canonical Einstein–Hilbert term $\sqrt{-G^{(E)}}R_E$, with no dilaton-dependent multiplier.
- The scalar kinetic term is canonically normalized as $-\frac{1}{2}(\partial\Phi)^2$.
- At quadratic order around a constant-dilaton background, the graviton and dilaton propagators are diagonal. This is the point made after equation (16.115): the vertex operators used there create orthogonal graviton and dilaton states, so their low-energy action naturally first appears in Einstein frame.
- The ordinary Einstein metric is the natural metric for spacetime duality symmetries. In type IIB, the action can be organized using
$$\tau = C_0 + ie^{-\Phi},$$
and the classical $SL(2,\mathbb{R})$ transformation leaves $G_{MN}^{(E)}$ invariant. This is the form used in equations (16.141)–(16.144) and in the S-duality discussion of Section 18.6.

The cost is that the direct relation to the world-sheet expansion is less visible: the various form-field terms now carry different dilaton exponentials. Also, if one keeps the same coordinates in which $G^{(S)} \to \eta$, the standard Einstein metric does not approach η but rather $\eta/\sqrt{g_s}$.

10.1.1.4 Modified Einstein-frame action and equation (18.26)

The modified Einstein frame removes only the fluctuating part of the dilaton from the string metric:

$$\boxed{g_{MN}^{(mE)} = e^{-\phi/2} G_{MN}^{(S)}}.$$

Since $\Phi = \Phi_0 + \phi$, comparison with the ordinary Einstein metric gives

$$g_{MN}^{(mE)} = e^{\Phi_0/2} G_{MN}^{(E)} = \sqrt{g_s}\, G_{MN}^{(E)}.$$

This is exactly the constant rescaling mentioned in the quoted paragraph.

With the physical gravitational coupling and rescaled form fields defined above, the action becomes

$$S_{mE} = \frac{1}{2\kappa_{10}^2} \int d^{10}x \sqrt{-g^{(mE)}} \left[R_{mE} - \frac{1}{2}(\partial\phi)_{mE}^2 - \frac{e^{-\phi}}{2\cdot 3!} H_{3\,mE}^2 - \sum_{n\in\mathrm{RR}} \frac{e^{(5-n)\phi/2}}{2n!} \widetilde{F}_{n\,mE}^2 \right] + S_{\mathrm{CS}}.$$

If only one $(p+2)$-form is retained, this reduces precisely to equation (18.26):

$$\boxed{S = \frac{1}{2\kappa_d^2} \int d^dx \sqrt{-g} \left(R - \frac{1}{2} g^{MN} \partial_M\phi \partial_N\phi - \frac{1}{2(p+2)!} e^{a_p\phi} \widetilde{F}_{p+2}^2 \right)},$$

with

$$a_1 = -1 \quad \text{for } H_3, \qquad a_p = \frac{3-p}{2} \quad \text{for an R–R } F_{p+2}.$$

The special properties of the modified Einstein frame are:

- The local kinetic terms are just as canonical as in ordinary Einstein frame, because the two metrics differ only by a constant.
- The fluctuating field obeys $\phi \to 0$, and the brane metrics are chosen to obey $g^{(mE)}_{MN} \to \eta_{MN}$. ADM masses and tensions obtained from the asymptotic falloff can therefore be read in standard Minkowski units.
- All constant g_s dependence has been moved into κ_{10}, the physical brane tensions, and the rescaled R–R fields. The remaining exponentials depend only on the local fluctuation ϕ.
- It is especially convenient when comparing supergravity brane solutions with the physical tensions previously extracted from string amplitudes.

10.1.1.5 Explicit Weyl-rescaling derivation

The numerical exponents are worth deriving, because this is where the distinction between Φ and ϕ matters.

For a Weyl rescaling in D dimensions,

$$G_{MN} = e^{2\omega} g_{MN},$$

one has

$$\sqrt{-G} = e^{D\omega}\sqrt{-g},$$
$$R[G] = e^{-2\omega}\left[R[g] - 2(D-1)\nabla^2\omega - (D-1)(D-2)(\partial\omega)^2\right],$$

and for an n-form,

$$F^2_{n\,G} = e^{-2n\omega} F^2_{n\,g},$$

because raising each of its n indices contributes one inverse metric.

For the string-to-modified-Einstein transformation in ten dimensions,

$$G^{(S)}_{MN} = e^{\phi/2} g^{(mE)}_{MN}, \qquad \omega = \frac{\phi}{4}.$$

The overall exponential multiplying the Ricci scalar is

$$e^{10\omega} e^{-2\phi} e^{-2\omega} = e^{8\omega - 2\phi} = e^{2\phi-2\phi} = 1.$$

This proves that the Ricci scalar has no dilaton prefactor. More explicitly,

$$\sqrt{-G^{(S)}}e^{-2\phi}R_S = \sqrt{-g^{(mE)}}\left[R_{mE} - \frac{9}{2}\nabla^2\phi - \frac{9}{2}(\partial\phi)^2\right].$$

The $\nabla^2\phi$ term is a total derivative and is discarded when the appropriate boundary term is understood. The original string-frame dilaton term becomes

$$\sqrt{-G^{(S)}}e^{-2\phi}\,4(\partial\phi)^2_S = \sqrt{-g^{(mE)}}\,4(\partial\phi)^2_{mE}.$$

Combining it with the $-\frac{9}{2}(\partial\phi)^2$ generated by the curvature gives

$$4 - \frac{9}{2} = -\frac{1}{2},$$

which is the canonical dilaton kinetic term in (18.26).

For the NS–NS three-form, the three factors are

$$\sqrt{-G^{(S)}}:\ e^{5\phi/2}, \qquad e^{-2\phi}, \qquad H^2_{3\,S}:\ e^{-3\phi/2}H^2_{3\,mE}.$$

Their product is

$$e^{5\phi/2}e^{-2\phi}e^{-3\phi/2} = e^{-\phi},$$

so $a_1 = -1$.

For an R–R n-form there is no initial $e^{-2\phi}$ factor. Therefore

$$\sqrt{-G^{(S)}}F_{nS}^2 = \sqrt{-g^{(mE)}}e^{5\phi/2}e^{-n\phi/2}F_{nmE}^2 = \sqrt{-g^{(mE)}}e^{(5-n)\phi/2}F_{nmE}^2.$$

Putting $n = p + 2$ gives

$$a_p = \frac{5-(p+2)}{2} = \frac{3-p}{2},$$

exactly as stated below equation (18.26).

Finally, the constant rescaling from ordinary to modified Einstein frame also reproduces every constant in (18.26). Starting with

$$G_{MN}^{(E)} = g_s^{-1/2}g_{MN}^{(mE)}, \qquad \Phi = \Phi_0 + \phi,$$

the gravitational term supplies g_s^{-2}:

$$\sqrt{-G^{(E)}}R_E = g_s^{-5/2}g_s^{1/2}\sqrt{-g^{(mE)}}R_{mE} = g_s^{-2}\sqrt{-g^{(mE)}}R_{mE}.$$

Since $\kappa_{10}^2 = g_s^2\tilde{\kappa}_{10}^2$,

$$\frac{g_s^{-2}}{\tilde{\kappa}_{10}^2} = \frac{1}{\kappa_{10}^2}.$$

For an R–R form the remaining factor is canceled by $\widetilde{F}_n = g_s F_n$. This explains why equation (18.26) contains κ_{10} and $\widetilde{F}$, rather than the Chapter 16 quantities $\tilde{\kappa}_{10}$ and F.

10.1.1.6 Brane source actions in the three frames

The same transformation is visible in the Dp-brane action. In string frame,

$$S_{Dp}^{(S)} = -\mu_p\int_{\Sigma_{p+1}} e^{-\Phi}\sqrt{-\det P[G^{(S)}]} + \mu_p\int_{\Sigma_{p+1}} P[C_{p+1}] + \cdots.$$

Using $G^{(S)} = e^{\Phi/2}G^{(E)}$, the induced $(p+1)$-dimensional volume element transforms as

$$\sqrt{-\det P[G^{(S)}]} = e^{(p+1)\Phi/4}\sqrt{-\det P[G^{(E)}]}.$$

Hence

$$S_{Dp}^{(E)} = -\mu_p\int e^{(p-3)\Phi/4}\sqrt{-\det P[G^{(E)}]} + \mu_p\int P[C_{p+1}] + \cdots.$$

Since $a_p = (3-p)/2$, this exponential is $e^{-a_p\Phi/2}$.

In modified Einstein frame, define

$$\tau_p = \frac{\mu_p}{g_s}, \qquad \widetilde{C}_{p+1} = g_s C_{p+1}.$$

Then

$$S_{Dp}^{(mE)} = -\tau_p\int e^{-a_p\phi/2}\sqrt{-\det P[g^{(mE)}]} + \tau_p\int P[\widetilde{C}_{p+1}] + \cdots,$$

which is exactly the source structure written in equation (18.48). The common coefficient τ_p of the DBI and Wess–Zumino terms makes the BPS equality between tension and charge manifest in the normalization of Section 18.5.

10.1.1.7 Masses and tensions under the constant rescaling

The string and modified Einstein metrics have the same asymptotic normalization, so an asymptotic observer finds

$$m_S^2 = m_{mE}^2.$$

Because

$$G_{MN}^{(E)} = g_s^{-1/2}g_{MN}^{(mE)},$$

their inverse metrics obey

$$(G^{(E)})^{MN} = g_s^{1/2}(g^{(mE)})^{MN}.$$

Applying the mass-shell condition in the two metrics gives

$$\boxed{m_S^2 = m_{mE}^2 = \frac{m_E^2}{\sqrt{g_s}}},$$

which is the relation stated on page 690.

For a p-brane tension, which is mass per spatial p-volume, the corresponding constant Weyl factor gives

$$\tau_p^{(E)} = g_s^{(p+1)/4}\tau_p^{(mE)}.$$

For example,

$$\tau_{F1}^{(E)} = \sqrt{g_s}\,\tau_{F1}^{(S)}, \qquad \tau_{D1}^{(E)} = \sqrt{g_s}\,\tau_{D1}^{(S)} = \frac{1}{\sqrt{g_s}}\frac{1}{2\pi\alpha'},$$

so the fundamental string and D-string tensions are exchanged by $g_s \leftrightarrow 1/g_s$ in ordinary Einstein frame, as in equation (18.81).

10.1.1.8 Which frame is used where in the book

The book changes frame according to the question being studied:

1. **World-sheet sigma models, beta functions, and T-duality: string frame.** The nonlinear sigma-model metric in equation (14.1) is $G^{(S)}$. The beta-function action (14.8) and the Buscher rules (14.15)–(14.16) are string-frame formulas.

2. **Ten-dimensional supergravity actions as obtained from string perturbation theory: mostly string frame.** Equations (16.118), (16.134), and (16.146) display the heterotic/type-II/type-I actions in string frame because their dilaton dependence keeps track of sphere and disk topology.

3. **Diagonal graviton–dilaton variables and type-IIB $SL(2)$ symmetry: ordinary Einstein frame.** Equation (16.103) is identified as Einstein frame because the graviton and dilaton decouple at quadratic order. Equations (16.141)–(16.144) use ordinary Einstein frame to make the type-IIB $SL(2,\mathbb{R})$ symmetry manifest; this metric is invariant under the classical duality.

4. **Flux compactifications: frame stated case by case.** For example, the type-IIB flux action in equation (17.48) is in Einstein frame, whereas the type-II supersymmetry variations in equation (17.64) are explicitly in string frame. One must not combine their formulas without converting the metric, gamma matrices, and spinors consistently.

5. **The brane solutions of Section 18.5: modified Einstein frame unless the book explicitly labels a string-frame expression.** Equation (18.26), the equations of motion (18.27), the ansatz (18.28), the general solution (18.34), the ADM tension calculation, and the source action (18.48) all use $g^{(mE)}$. The sentence "from here on, in this section, Einstein frame always means modified Einstein frame" applies specifically to Section 18.5. The supersymmetry variations used in (18.52) are nevertheless imported in string frame, so the book explicitly converts the background when applying them. Table 18.3 then lists each ten-dimensional solution in both the modified Einstein frame and string frame.

6. **The S-duality tension comparison of Section 18.6: ordinary Einstein frame.** Equation (18.81) uses the standard Einstein-frame scaling $\tau_{F1}^{(E)} \propto \sqrt{g_s}$ and $\tau_{D1}^{(E)} \propto 1/\sqrt{g_s}$. These would not be the scalings in the modified frame, whose asymptotic masses agree with string frame. Thus the special convention declared inside Section 18.5 should not be silently extended to all later occurrences of the words "Einstein frame."

7. **DBI actions and perturbative brane tensions: naturally string frame, often converted to modified Einstein frame for supergravity matching.** The disk factor $e^{-\Phi}$ and the scaling $\tau_{Dp} \sim g_s^{-1}$ are most transparent in string frame. Equation (18.48) is their modified-Einstein-frame form, chosen to match the normalization of the supergravity charge and ADM tension.

10.1.1.9 Checks

For a D3-brane, $p = 3$, so

$$a_3 = 0.$$

Its R–R five-form kinetic term and its modified-Einstein-frame DBI volume term have no local dilaton factor. This is consistent with the constant-dilaton D3 solution (18.59) and with the self-duality of the D3-brane under type-IIB S-duality.

For the fundamental string, the relevant NS–NS field strength is H_3, so $a_1 = -1$. Its modified-Einstein-frame source factor is

$$e^{-a_1\phi/2} = e^{\phi/2},$$

which is exactly what follows by applying $G^{(S)} = e^{\phi/2} g^{(mE)}$ to a two-dimensional Nambu–Goto area element.

For an R–R D1-brane, $a_1 = (3-1)/2 = 1$, so its source factor is $e^{-\phi/2}$. The opposite signs for F1 and D1 are the local dilaton version of their exchange under type-IIB S-duality.

10.1.1.10 Result

The complete chain of metric redefinitions is

$$\boxed{\Phi = \Phi_0 + \phi, \quad g_s = e^{\Phi_0}, \quad G^{(E)} = e^{-\Phi/2} G^{(S)}, \quad g^{(mE)} = e^{-\phi/2} G^{(S)} = \sqrt{g_s}\, G^{(E)}}.$$

The characteristic bulk actions are

$$\boxed{S_S \sim \int \sqrt{-G^{(S)}} \left[e^{-2\Phi} \left(R_S + 4(\partial\Phi)^2 - \frac{1}{2}|H_3|^2 \right) - \frac{1}{2} \sum_{\mathrm{RR}} |F_n|^2 \right]},$$

$$\boxed{S_E \sim \int \sqrt{-G^{(E)}} \left[R_E - \frac{1}{2}(\partial\Phi)^2 - \frac{1}{2} e^{-\Phi}|H_3|^2 - \frac{1}{2} \sum_{\mathrm{RR}} e^{(5-n)\Phi/2} |F_n|^2 \right]},$$

and

$$\boxed{S_{mE} \sim \int \sqrt{-g^{(mE)}} \left[R_{mE} - \frac{1}{2}(\partial\phi)^2 - \frac{1}{2} e^{-\phi}|H_3|^2 - \frac{1}{2} \sum_{\mathrm{RR}} e^{(5-n)\phi/2} |\widetilde{F}_n|^2 \right]}.$$

Here the omitted common prefactors, factorials, Chern–Simons terms, and the type-IIB five-form qualification are those stated above. For a single field strength F_{p+2}, the last action is equation (18.26), with

$$\boxed{a_p = -1 \text{ for NS–NS } H_3, \qquad a_p = \frac{3-p}{2} \text{ for R–R } F_{p+2}}.$$

Conceptually: string frame is adapted to the world-sheet and perturbative g_s counting; ordinary Einstein frame is adapted to canonical gravity and S-duality; modified Einstein frame is adapted to asymptotically Minkowski brane solutions and direct ADM mass/tension comparisons.

10.2 Q2: Deriving the BPS warp-factor condition

In page692 eq(18.33) is said to be derived from the requirement that the p-brane solution preserves half of the supercharges. This equation is very simple and clear, how to derived it?

> **Textbook location and related material.** Chapter 18, Section 18.5 (pp. 688–705): the supergravity p-brane ansatz, Killing-spinor/BPS conditions, harmonic functions, and the relation among warp factors.

10.2.1 Answer

10.2.1.1 Core point

The statement to derive is

$$\boxed{nA + \tilde{n}B = \text{const}}, \qquad n = p+1, \qquad \tilde{n} = d - n - 2.$$

For an asymptotically Minkowski solution, $A, B \to 0$ as $r \to \infty$, so the constant is zero.

This condition is a consequence of the first-order Killing-spinor equations. A charged brane imposes one algebraic projector on the supersymmetry parameter. That projector leaves one half of the spinors. Once it is imposed, the longitudinal and transverse components of the gravitino variation contain the same radial flux function. Their relative gamma-matrix coefficients force

$$A' = -\frac{2\tilde{n}}{\Delta(d-2)}u(r), \qquad B' = \frac{2n}{\Delta(d-2)}u(r).$$

Therefore

$$nA' + \tilde{n}B' = -\frac{2n\tilde{n}}{\Delta(d-2)}u + \frac{2n\tilde{n}}{\Delta(d-2)}u = 0.$$

Integration gives equation (18.33).

The book does not perform this calculation at page 692. It first imposes (18.33) while solving the bosonic equations, then later gives the relevant Killing-spinor equations in (18.52)–(18.55). Footnote 9 on page 697 says that one could instead insert the unsolved ansatz (18.28)–(18.29) directly into those supersymmetry variations. The derivation below supplies that omitted calculation.

10.2.1.2 The metric ansatz and the meaning of the coefficients

The metric in equation (18.28) is

$$ds^2 = e^{2A(r)}\eta_{\mu\nu}dx^\mu dx^\nu + e^{2B(r)}\delta_{mn}dy^m dy^n,$$

with

$$\mu = 0, \dots, n-1, \qquad m = 1, \dots, q, \qquad q = d - n = \tilde{n} + 2.$$

Thus $\tilde{n}$ is not the number of transverse dimensions; it is two less:

$$\tilde{n} = q - 2.$$

This explains why (18.33) is not simply a statement about the determinant of the metric.

Choose the orthonormal coframe

$$e^{\hat{\mu}} = e^A dx^\mu, \qquad e^{\hat{m}} = e^B dy^m.$$

The nonzero spin-connection components are precisely those displayed in equation (18.53). Schematically, the parts relevant for a radial Killing spinor are

$$\nabla_{\hat{\mu}}\epsilon \;\propto\; e^{-B}A'\Gamma_{\hat{\mu}\hat{r}}\epsilon, \qquad \nabla_{\hat{i}}\epsilon \;\propto\; e^{-B}B'\Gamma_{\hat{i}\hat{r}}\epsilon,$$

where $\hat{i}$ is tangent to the transverse sphere and

$$\Gamma_{\hat{r}} = \frac{y^m}{r}\Gamma_{\hat{m}}.$$

The first term measures the radial variation of the longitudinal vielbein, while the second measures the radial variation of the transverse vielbein.

For an electric p-brane, the field strength has an orthonormal component

$$F_{\hat{r}\hat{0}\cdots\hat{p}}(r).$$

Its gamma-matrix contraction is proportional to

$$F_{\hat{r}\hat{0}\cdots\hat{p}}\Gamma^{\hat{r}\hat{0}\cdots\hat{p}} = F_{\hat{r}\hat{0}\cdots\hat{p}}\Gamma^{\hat{r}}\Gamma_{\parallel}, \qquad \Gamma_{\parallel} = \Gamma^{\hat{0}\cdots\hat{p}}.$$

Consequently the form-field term in the supersymmetry variation can cancel the spin connection only on spinors obeying a brane projector.

10.2.1.3 The brane projector and the loss of half the supercharges

Equation (18.54) has the form

$$\left(1 - s\,\mathcal{K}_p\right)\epsilon = 0, \qquad \mathcal{K}_p = (-1)^p \Gamma^{\hat{0}\cdots\hat{p}}\mathcal{P}_{p+2}, \qquad s = \pm 1,$$

up to the book's choice of which sign is called a brane. The gamma-matrix and type-II chirality factors obey

$$\mathcal{K}_p^2 = 1.$$

Hence

$$\Pi_s = \frac{1}{2}(1 + s\mathcal{K}_p)$$

is a projector. Its two eigenspaces have equal dimension, so

$$\Pi_s \epsilon = \epsilon$$

retains 16 of the original 32 type-II supercharges. Changing s reverses the R–R charge and gives the anti-brane projector.

The projector is important for two separate reasons:

1. it is the algebraic statement that the configuration is at most 1/2 BPS;
2. it replaces $\Gamma_{\parallel}\mathcal{P}_{p+2}$ by the number s inside the differential Killing-spinor equations.

After this replacement, all flux terms have the same gamma structures as the spin-connection terms. The Killing-spinor equations therefore reduce to ordinary first-order equations for the radial functions.

10.2.1.4 General first-order BPS equations

For those Einstein-dilaton-form systems in (18.26) that arise as supersymmetric type-II or eleven-dimensional supergravity truncations, the longitudinal gravitino equation, the angular transverse gravitino equation, and the dilatino equation reduce to

$$A' = -\frac{2\tilde{n}}{\Delta(d-2)}u(r),$$

$$B' = \frac{2n}{\Delta(d-2)}u(r),$$

$$\phi' = \frac{2\zeta a_p}{\Delta}u(r), \qquad \Delta = a_p^2 + \frac{2n\tilde{n}}{d-2}.$$

Here $\zeta = +1$ for the electric representative and $\zeta = -1$ for its magnetic dual, while the definition of u can absorb the orientation sign s. At this stage $u(r)$ is simply the one common function built from the orthonormal form-field component and the dilaton exponential. The radial gravitino equation separately determines the radial dependence of ϵ.

The crucial fact is the relative coefficient of the first two equations. It comes from the gamma-matrix trace subtraction in the form-field contribution to the gravitino variation: the flux acts with weight $\tilde{n}$ along a world-volume direction and with weight n along a transverse direction, with opposite signs. The overall normalization of u, the value of a_p, and the electric/magnetic sign do not affect their ratio.

Taking the weighted sum immediately gives

$$\begin{aligned}\frac{d}{dr}(nA + \tilde{n}B) &= nA' + \tilde{n}B' \\ &= -\frac{2n\tilde{n}}{\Delta(d-2)}u + \frac{2\tilde{n}n}{\Delta(d-2)}u \\ &= 0.\end{aligned}$$

Therefore

$$\boxed{nA + \tilde{n}B = C}.$$

The boundary conditions used below equation (18.28) are

$$A(r) \to 0, \qquad B(r) \to 0, \qquad \phi(r) \to 0 \qquad (r \to \infty),$$

so

$$\boxed{C = 0}.$$

Solving the remaining Maxwell and dilaton equations identifies

$$u(r) = (\log H(r))'$$

in a convenient normalization. Integration then gives

$$A = -\frac{2\tilde{n}}{\Delta(d-2)} \log H, \qquad B = \frac{2n}{\Delta(d-2)} \log H, \qquad \phi = \frac{2\zeta a_p}{\Delta} \log H,$$

which is equation (18.34). Notice that (18.33) follows before one knows the explicit harmonic function $H = 1 + N\alpha/r^{\tilde{n}}$.

10.2.1.5 Explicit derivation for a type-II Dp-brane

It is useful to see the cancellation without hiding it inside the general coefficients. Equations (18.52) are written in string frame, whereas A and B in (18.33) are modified-Einstein-frame warp factors. Write

$$ds_S^2 = e^{2A_S(r)} \eta_{\mu\nu} dx^\mu dx^\nu + e^{2B_S(r)} \delta_{mn} dy^m dy^n.$$

Because

$$G_{MN}^{(S)} = e^{\phi/2} g_{MN}^{(mE)},$$

the warp factors are related by

$$\boxed{A_S = A + \frac{\phi}{4}, \qquad B_S = B + \frac{\phi}{4}}.$$

Define the oriented orthonormal flux function

$$\mathcal{Q}(r) = s\, e^{\Phi + B_S} F_{\hat{r}\hat{0}\cdots\hat{p}}(r),$$

where numerical signs associated with $\mathcal{P}_{p+2}$ are absorbed into s. Insert the spin connection (18.53) and the projector (18.54) into the Dp-brane equations (18.52). The four independent components become

$$\delta\Psi_\mu = 0: \qquad A_S' - \frac{1}{4}\mathcal{Q} = 0,$$

$$\delta\Psi_i = 0: \qquad B_S' + \frac{1}{4}\mathcal{Q} = 0,$$

$$\delta\lambda = 0: \qquad \phi' + \frac{3-p}{4}\mathcal{Q} = 0,$$

and

$$\delta\Psi_r = 0: \qquad \epsilon' - \frac{1}{8}\mathcal{Q}\,\epsilon = 0.$$

The overall sign of every $\mathcal{Q}$ term flips if the opposite orientation is chosen, but the relations obtained by eliminating $\mathcal{Q}$ are unchanged.

The first three equations imply

$$\boxed{B_S' = -A_S', \qquad \phi' = (p-3)A_S'}.$$

With the asymptotic conditions $A_S, B_S, \phi \to 0$, they integrate to

$$B_S = -A_S, \qquad \phi = (p-3)A_S.$$

The radial gravitino equation gives

$$\epsilon(r) = e^{A_S(r)/2}\epsilon_0, \qquad \Pi_s\epsilon_0 = \epsilon_0.$$

Thus the solution has a nonzero Killing spinor satisfying one rank-half projector.

Now convert back to the modified Einstein frame:

$$\begin{aligned} A &= A_S - \frac{\phi}{4} = \left(1 - \frac{p-3}{4}\right) A_S = \frac{7-p}{4} A_S, \\ B &= B_S - \frac{\phi}{4} = -\left(1 + \frac{p-3}{4}\right) A_S = -\frac{p+1}{4} A_S. \end{aligned}$$

In ten dimensions,

$$n = p+1, \qquad \tilde{n} = 10 - (p+1) - 2 = 7 - p.$$

Therefore

$$\begin{aligned} nA + \tilde{n}B &= (p+1)\frac{7-p}{4} A_S + (7-p)\left(-\frac{p+1}{4} A_S\right) \\ &= 0. \end{aligned}$$

This is equation (18.33) derived directly from the type-II Killing-spinor equations quoted later in the same section.

For $p = 3$, the coefficient $3 - p$ in the dilatino equation vanishes. Then $\phi' = 0$, as expected for a D3-brane. The brane projector is obtained from the gravitino/five-form equation rather than from the dilatino equation alone, and the same warp-factor result still follows.

10.2.1.6 Why the condition makes the second-order equations simple

There is also a useful geometric explanation of why the authors impose precisely this combination. For a radial scalar $f(r)$,

$$\sqrt{-g} \;\propto\; e^{nA+qB} r^{q-1}, \qquad g^{rr} = e^{-2B}.$$

Hence

$$\begin{aligned} \nabla^2 f &= \frac{1}{\sqrt{-g}} \partial_r \left(\sqrt{-g}\, g^{rr} f'\right) \\ &= e^{-2B} \left[f'' + \left(\frac{q-1}{r} + nA' + (q-2)B' \right) f' \right]. \end{aligned}$$

Since $q - 2 = \tilde{n}$,

$$\nabla^2 f = e^{-2B} \left[f'' + \left(\frac{q-1}{r} + nA' + \tilde{n}B' \right) f' \right].$$

When the BPS condition holds,

$$nA' + \tilde{n}B' = 0,$$

the extra warp-factor derivative disappears. Apart from the overall e^{-2B}, the radial operator becomes the flat transverse Laplacian,

$$f'' + \frac{q-1}{r} f'.$$

This is why the remaining equations collapse to the statement that H is harmonic in the transverse space:

$$\frac{1}{r^{q-1}} \frac{d}{dr} \left(r^{q-1} \frac{dH}{dr} \right) = 0 \qquad (r \neq 0).$$

Thus (18.33) is often called the harmonic or extremal gauge condition. In this presentation it is not imposed merely as a convenient coordinate gauge: supersymmetry selects it for the isotropic BPS ansatz.

10.2.1.7 Checks

For a ten-dimensional Dp-brane, $\Delta = 4$, and equation (18.34) gives

$$A = -\frac{7-p}{16}\log H, \qquad B = \frac{p+1}{16}\log H.$$

Therefore

$$(p+1)A + (7-p)B = 0.$$

For the M2-brane, $d = 11$, $n = 3$, and $\tilde{n} = 6$. Its metric is

$$ds^2 = H^{-2/3}ds_{\parallel}^2 + H^{1/3}ds_{\perp}^2,$$

so

$$A = -\frac{1}{3}\log H, \qquad B = \frac{1}{6}\log H,$$

and

$$3A + 6B = 0.$$

For the M5-brane, $n = 6$ and $\tilde{n} = 3$:

$$ds^2 = H^{-1/3}ds_{\parallel}^2 + H^{2/3}ds_{\perp}^2,$$

so

$$A = -\frac{1}{6}\log H, \qquad B = \frac{1}{3}\log H,$$

and again

$$6A + 3B = 0.$$

These checks show that the relation is independent of the presence of a dilaton and survives electric/-magnetic duality.

10.2.1.8 Result

The brane projector

$$\Pi_s\epsilon = \epsilon, \qquad \Pi_s = \frac{1}{2}(1 + s\mathcal{K}_p), \qquad \mathcal{K}_p^2 = 1,$$

leaves one half of the supercharges. After imposing it, the Killing-spinor equations reduce to

$$\boxed{A' = -\frac{2\tilde{n}}{\Delta(d-2)}u, \qquad B' = \frac{2n}{\Delta(d-2)}u, \qquad \phi' = \frac{2\zeta a_p}{\Delta}u}.$$

The first two equations imply

$$\boxed{\frac{d}{dr}(nA + \tilde{n}B) = 0},$$

and hence

$$\boxed{nA + \tilde{n}B = \text{const} = 0}$$

for asymptotically Minkowski boundary conditions.

For a Dp-brane, the same result follows explicitly from

$$B_S' = -A_S', \qquad \phi' = (p-3)A_S', \qquad A = A_S - \frac{\phi}{4}, \qquad B = B_S - \frac{\phi}{4}.$$

The simplicity of (18.33) is therefore not accidental: it is the flux-independent combination of the longitudinal and transverse first-order BPS equations.

10.3 Q3: Supersymmetry and the slope rule for (p, q) five-brane webs

Consider the type IIB (p, q) 5-brane web. The 5-branes extends in the 01234 directions and one direction in the $x^5 - x^6$ plane. Suppose we have D5-brane along x^5 and NS5-brane along x^6, we can find the susy condition

$$\text{D5} : \epsilon_L = \Gamma^0\Gamma^1\Gamma^2\Gamma^3\Gamma^4\Gamma^5\epsilon_R$$
$$\text{NS5} : \epsilon_L = \Gamma^0\Gamma^1\Gamma^2\Gamma^3\Gamma^4\Gamma^6\epsilon_L \ , \ \epsilon_R = -\Gamma^0\Gamma^1\Gamma^2\Gamma^3\Gamma^4\Gamma^6\epsilon_R$$

Show that 8 of the 32 susy are preserved. In the construction of (p, q) 5-brane web, one crucial fact is that a (p, q) 5-brane along the direction with slope $\delta x^5 + i\delta x^6 \propto p + \tau q$ in the $x^5 - x^6$ plane, where $\tau = C_0 + ie^{-\Phi}$ can be set to $\tau = i$. Derive this fact.

> **Textbook location and related material.** Chapter 18, Section 18.6 (pp. 705–716): type-IIB $SL(2, \mathbb{Z})$, (p, q) objects, and five-brane webs. Section 18.4 supplies the projector and central-charge logic.

10.3.1 Answer

10.3.1.1 Core point

There are two related statements.

First, the D5-brane along (012345) and the NS5-brane along (012346) impose two independent but commuting half-BPS projectors. Therefore their common eigenspace has

$$\frac{1}{2} \cdot \frac{1}{2} \cdot 32 = 8$$

real supercharges. This is the minimal $\mathcal{N} = 1$ supersymmetry of the common five-dimensional spacetime (01234).

Second, a general $(p, q)_5$ charge has the complex BPS phase

$$e^{i\alpha_{p,q}} = \frac{p + \tau q}{|p + \tau q|}.$$

Its charge-space projector preserves the same eight spinors as the D5–NS5 intersection only if its spatial angle θ in the x^5-x^6 plane equals this phase:

$$\boxed{\theta = \alpha_{p,q} = \arg(p + \tau q)}.$$

Equivalently,

$$\boxed{\delta x^5 + i\delta x^6 \ \propto \ p + \tau q}.$$

The slope rule and the normalized-coordinate convention $\tau = i$ are equations (11) and (16)–(18) of the original five-brane-web construction by Aharony, Hanany, and Kol. The supersymmetry projectors used below follow the book's page 698 conventions.

10.3.1.2 Writing the two conditions as projectors

The type-IIB supersymmetry parameter is

$$\epsilon = \begin{pmatrix} \epsilon_L \\ \epsilon_R \end{pmatrix},$$

where ϵ_L and ϵ_R are Majorana–Weyl spinors of the same chirality, each with 16 real components.

Define

$$\Gamma_D = \Gamma^{012345}, \qquad \Gamma_N = \Gamma^{012346}.$$

For a product of six gamma matrices containing one timelike direction,

$$\Gamma_D^2 = \Gamma_N^2 = 1.$$

The D5 condition in the question,

$$\epsilon_L = \Gamma_D \epsilon_R,$$

also implies

$$\epsilon_R = \Gamma_D \epsilon_L.$$

Therefore it is equivalent to

$$K_D \epsilon = \epsilon, \qquad K_D = \Gamma_D \sigma_1,$$

because

$$\Gamma_D \sigma_1 \begin{pmatrix} \epsilon_L \\ \epsilon_R \end{pmatrix} = \begin{pmatrix} \Gamma_D \epsilon_R \\ \Gamma_D \epsilon_L \end{pmatrix}.$$

Similarly, the two NS5 conditions are equivalent to

$$K_N \epsilon = \epsilon, \qquad K_N = \Gamma_N \sigma_3,$$

since

$$\Gamma_N \sigma_3 \begin{pmatrix} \epsilon_L \\ \epsilon_R \end{pmatrix} = \begin{pmatrix} \Gamma_N \epsilon_L \\ -\Gamma_N \epsilon_R \end{pmatrix}.$$

Both operators are involutions:

$$K_D^2 = K_N^2 = 1.$$

Thus

$$\Pi_D = \frac{1}{2}(1 + K_D), \qquad \Pi_N = \frac{1}{2}(1 + K_N)$$

are rank-half projectors.

10.3.1.3 Why the two projectors commute

Let

$$\Gamma_{(5)} = \Gamma^{01234},$$

so

$$\Gamma_D = \Gamma_{(5)} \Gamma^5, \qquad \Gamma_N = \Gamma_{(5)} \Gamma^6.$$

Because $\Gamma_{(5)}$ contains five gamma matrices, both Γ^5 and Γ^6 anticommute with it. It follows that

$$\Gamma_D \Gamma_N = -\Gamma_N \Gamma_D.$$

The Pauli matrices also anticommute:

$$\sigma_1 \sigma_3 = -\sigma_3 \sigma_1.$$

The two minus signs cancel in the full operators:

$$\begin{aligned} K_D K_N &= (\Gamma_D \Gamma_N)(\sigma_1 \sigma_3) \\ &= (-\Gamma_N \Gamma_D)(-\sigma_3 \sigma_1) \\ &= K_N K_D. \end{aligned}$$

Hence

$$[K_D, K_N] = 0.$$

The two projectors are independent: their product contains $\Gamma^{56}\sigma_2$ and is not ± 1. They can therefore be diagonalized simultaneously, and their common $+1$ eigenspace has dimension

$$\dim \ker(1 - K_D) \cap \ker(1 - K_N) = \frac{32}{2 \cdot 2} = 8.$$

10.3.1.4 The same count directly in terms of ϵ_L, ϵ_R

The NS5 condition on ϵ_R is

$$\Gamma_N \epsilon_R = -\epsilon_R.$$

Since $\Gamma_N^2 = 1$, its -1 eigenspace has dimension eight. Choose any ϵ_R in this eigenspace and define

$$\epsilon_L = \Gamma_D \epsilon_R.$$

Then, using $\{\Gamma_D, \Gamma_N\} = 0$,

$$\Gamma_N \epsilon_L = \Gamma_N \Gamma_D \epsilon_R = -\Gamma_D \Gamma_N \epsilon_R = \Gamma_D \epsilon_R = \epsilon_L.$$

Thus the NS5 condition on ϵ_L is automatic. No further constraint is produced, so the solution space is parametrized by the eight allowed components of ϵ_R.

This also shows why it is essential that the D5 and NS5 occupy different directions in the x^5-x^6 plane. The gamma-matrix anticommutation is exactly what converts the -1 NS5 eigenvalue of ϵ_R into the $+1$ eigenvalue required for ϵ_L.

10.3.1.5 The projector for a general angled (p, q) five-brane

Let the oriented spatial direction of a five-brane in the plane be

$$\hat{t}_\theta = \cos\theta\, \partial_5 + \sin\theta\, \partial_6,$$

and define

$$\Gamma_\theta = \cos\theta\, \Gamma^5 + \sin\theta\, \Gamma^6.$$

In the convention of the question,

$$(1,0)_5 = \text{D5}, \qquad (0,1)_5 = \text{NS5}.$$

10.3.1.5.1 The charge-space matrix from the IIB BPS algebra The two Pauli matrices in the elementary projectors tell us how the two elementary five-brane charges enter the type-IIB supersymmetry algebra:

$$\text{D5 charge}: \ \sigma_1, \qquad \text{NS5 charge}: \ \sigma_3$$

at the canonical point $\tau = i$. More invariantly, the bare integer charge vector (p, q) is an $SL(2)$ doublet, whereas the supersymmetry spinors transform under the local $SO(2)$ associated with the coset

$$\frac{SL(2,\mathbb{R})}{SO(2)}.$$

The axio-dilaton supplies the coset vielbein that converts the integer $SL(2)$ charge vector into a local orthonormal $SO(2)$ vector.

A convenient real coset representative is

$$V(\tau) = \frac{1}{\sqrt{\tau_2}} \begin{pmatrix} 1 & \tau_1 \\ 0 & \tau_2 \end{pmatrix}.$$

Thus the local orthonormal charge vector is

$$\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = V(\tau) \begin{pmatrix} p \\ q \end{pmatrix} = \frac{1}{\sqrt{\tau_2}} \begin{pmatrix} p + \tau_1 q \\ \tau_2 q \end{pmatrix},$$

or, in complex notation,

$$z_1 + i z_2 = \frac{p + \tau q}{\sqrt{\tau_2}}.$$

The common positive factor $1/\sqrt{\tau_2}$ affects the Einstein-frame magnitude but not the direction of this vector. In the string-frame normalization convention used for five-brane webs, the corresponding five-brane central-charge matrix acting on the IIB spinor doublet is

$$\boxed{\mathcal{Z}_5(p, q; \tau) = T_{\text{D5}} \left[(p + \tau_1 q)\sigma_1 + \tau_2 q\, \sigma_3\right]}$$

or equivalently

$$\boxed{\mathcal{Z}_5 = T_{\mathrm{D5}}\left[\mathrm{Re}(p+\tau q)\sigma_1 + \mathrm{Im}(p+\tau q)\sigma_3\right].}$$

This is the precise step that was missing from the previous answer. It is the central-charge, or zero-world-volume-flux limit, of the $SL(2,\mathbb{Z})$-covariant kappa-symmetry construction of a type-IIB (p,q) five-brane; see, for example, the covariant super-five-brane construction.

Since

$$\sigma_1^2 = \sigma_3^2 = 1, \qquad \{\sigma_1, \sigma_3\} = 0,$$

the square of this matrix is

$$\begin{aligned}\mathcal{Z}_5^2 &= T_{\mathrm{D5}}^2\left[(p+\tau_1 q)^2 + (\tau_2 q)^2\right]\mathbf{1}\\ &= T_{\mathrm{D5}}^2 |p+\tau q|^2 \mathbf{1}.\end{aligned}$$

The largest eigenvalue of the central-charge matrix gives the BPS bound. Thus, in the string-frame convention used for the web,

$$T_{(p,q)_5} = T_{\mathrm{D5}}\,|p+\tau q|.$$

For a saturated brane, divide the central-charge matrix by its tension:

$$\begin{aligned}\widehat{\mathcal{Z}}_5 &= \frac{\mathcal{Z}_5}{T_{(p,q)_5}}\\ &= \frac{\mathrm{Re}(p+\tau q)}{|p+\tau q|}\sigma_1 + \frac{\mathrm{Im}(p+\tau q)}{|p+\tau q|}\sigma_3.\end{aligned}$$

Defining

$$e^{i\alpha} = \frac{p+\tau q}{|p+\tau q|}$$

gives

$$\boxed{\widehat{\mathcal{Z}}_5 = \cos\alpha\,\sigma_1 + \sin\alpha\,\sigma_3}.$$

It is an involution,

$$\widehat{\mathcal{Z}}_5^2 = 1,$$

as required for the internal part of a kappa-symmetry or supersymmetry projector.

At $\tau = i$ this statement is especially transparent:

$$\mathcal{Z}_5 = T_{\mathrm{D5}}(p\sigma_1 + q\sigma_3),$$

so the Pauli factor is simply the unit vector in the two-dimensional D5–NS5 charge plane. At a general τ, the integer charges remain (p,q), but the scalar background dresses them into local orthonormal components proportional to $(p+\tau_1 q, \tau_2 q)$, with the common factor $1/\sqrt{\tau_2}$ shown above.

There is an equivalent rotation description. Let

$$U_\alpha = \exp\left(\frac{i\alpha}{2}\sigma_2\right).$$

Using the Pauli algebra,

$$U_\alpha \sigma_1 U_\alpha^{-1} = \cos\alpha\,\sigma_1 + \sin\alpha\,\sigma_3 = \widehat{\mathcal{Z}}_5.$$

The half-angle in U_α is present because it acts on the spinor doublet; conjugation rotates the Pauli-vector by the full angle α.

10.3.1.5.2 Why the normalized central charge enters a supersymmetry projector The remaining step follows directly from the BPS algebra. For a static planar five-brane, the energy-density term and the five-brane central-charge term in the type-IIB supercharge anticommutator can be organized, after suppressing the positive charge-conjugation matrix, as

$$\{Q,Q\}_{(p,q)_5} \sim T_{(p,q)_5}\mathbf{1} - \Gamma_{(5)}\Gamma_\theta\,\mathcal{Z}_5.$$

The overall sign here is fixed so that the elementary D5 condition is $K_D\epsilon = \epsilon$. Positivity of $\{Q, Q\}$ gives the BPS inequality

$$T_{(p,q)_5} \geq \|\mathcal{Z}_5\|.$$

A supersymmetry generated by ϵ is preserved precisely when the corresponding eigenvalue of this anticommutator vanishes. For a saturated brane, $T_{(p,q)_5} = \|\mathcal{Z}_5\|$, so

$$\left[T_{(p,q)_5}\mathbf{1} - \Gamma_{(5)}\Gamma_\theta\mathcal{Z}_5\right]\epsilon = 0.$$

Dividing by the tension gives

$$\Gamma_{(5)}\Gamma_\theta\frac{\mathcal{Z}_5}{T_{(p,q)_5}}\epsilon = \epsilon.$$

Therefore the operator that must appear in the preserved-supersymmetry condition is not the unnormalized charge matrix, but

$$\boxed{K_{(p,q)}(\theta) = \Gamma_{(5)}\Gamma_\theta\widehat{\mathcal{Z}}_5}, \qquad \widehat{\mathcal{Z}}_5 = \frac{\mathcal{Z}_5}{T_{(p,q)_5}}.$$

This also explains why normalization by the BPS tension is compulsory: it makes $K_{(p,q)}^2 = 1$, so $(1 + K_{(p,q)})/2$ is a genuine rank-half projector.

10.3.1.5.3 The spacetime gamma matrix from the geometric rotation The reference D5-brane lies along x^5 and has spacetime factor

$$\Gamma_{(5)}\Gamma^5 = \Gamma^{012345}.$$

Rotate its final tangent direction through angle θ in the x^5-x^6 plane. The spin lift of this rotation is

$$S_\theta = \exp\left(-\frac{\theta}{2}\Gamma^{56}\right).$$

The Clifford algebra gives

$$S_\theta\Gamma^5 S_\theta^{-1} = \cos\theta\,\Gamma^5 + \sin\theta\,\Gamma^6 = \Gamma_\theta.$$

Since $\Gamma_{(5)} = \Gamma^{01234}$ is invariant under rotations in the 56 plane,

$$S_\theta\left(\Gamma_{(5)}\Gamma^5\right)S_\theta^{-1} = \Gamma_{(5)}\Gamma_\theta.$$

10.3.1.5.4 Combining the independent rotations Start from the reference D5 involution

$$K_D = \Gamma_{(5)}\Gamma^5\sigma_1.$$

The spatial rotation S_θ acts only on spacetime spinor indices, while U_α acts only on the IIB (L, R) doublet. Hence they commute. The involution for the rotated, charge-dressed brane is obtained by conjugation:

$$\begin{aligned} K_{(p,q)}(\theta) &= S_\theta U_\alpha K_D U_\alpha^{-1} S_\theta^{-1} \\ &= \left[S_\theta(\Gamma_{(5)}\Gamma^5)S_\theta^{-1}\right]\left[U_\alpha\sigma_1U_\alpha^{-1}\right] \\ &= \Gamma_{(5)}\Gamma_\theta\left(\cos\alpha\,\sigma_1 + \sin\alpha\,\sigma_3\right). \end{aligned}$$

Therefore

$$\boxed{K_{(p,q)}(\theta) = \Gamma_{(5)}\Gamma_\theta\left(\cos\alpha\,\sigma_1 + \sin\alpha\,\sigma_3\right)}.$$

This formula is thus the product of two independently normalized factors:

$$\boxed{K_{(p,q)} = \underbrace{\Gamma_{(5)}\Gamma_\theta}_{\text{oriented world-volume}}\ \underbrace{\widehat{\mathcal{Z}}_5}_{\text{normalized IIB charge matrix}}}.$$

It is the standard BPS form “world-volume gamma matrix times normalized central charge.”

It obeys

$$K_{(p,q)}(\theta)^2 = 1,$$

so an isolated angled (p, q) five-brane is half-BPS.

At $\tau = i$ this reproduces the two elementary cases:

(p, q)	α	θ	$K_{(p,q)}$
$(1, 0)$	0	0	$\Gamma^{012345}\sigma_1 = K_D$
$(0, 1)$	$\pi/2$	$\pi/2$	$\Gamma^{012346}\sigma_3 = K_N.$

10.3.1.6 Alignment of the charge phase and spatial angle

Now restrict to a spinor ϵ satisfying

$$K_D\epsilon = \epsilon, \qquad K_N\epsilon = \epsilon.$$

The two mixed gamma–Pauli structures act as

$$\Gamma_{(5)}\Gamma^5\sigma_3\epsilon = \Gamma^{56}\epsilon,$$

and

$$\Gamma_{(5)}\Gamma^6\sigma_1\epsilon = -\Gamma^{56}\epsilon.$$

To see the first identity, for example, use $K_N\epsilon = \epsilon$ to replace

$$\sigma_3\epsilon = \Gamma_{(5)}\Gamma^6\epsilon.$$

Expanding the general projector gives

$$\begin{aligned} K_{(p,q)}(\theta)\epsilon = \; & \cos\theta\cos\alpha\, K_D\epsilon + \sin\theta\sin\alpha\, K_N\epsilon \\ & + \cos\theta\sin\alpha\,\Gamma_{(5)}\Gamma^5\sigma_3\epsilon \\ & + \sin\theta\cos\alpha\,\Gamma_{(5)}\Gamma^6\sigma_1\epsilon. \end{aligned}$$

Using the common D5–NS5 conditions,

$$K_{(p,q)}(\theta)\epsilon = \left[\cos(\theta-\alpha) - \sin(\theta-\alpha)\Gamma^{56}\right]\epsilon.$$

Since

$$(\Gamma^{56})^2 = -1,$$

the operator

$$[\cos(\theta-\alpha) - 1] - \sin(\theta-\alpha)\Gamma^{56}$$

has a nontrivial kernel only if

$$\theta - \alpha = 0 \pmod{2\pi}$$

for an oriented leg. Reversing both the geometric orientation and the charge vector gives the same unoriented line.

Therefore

$$\boxed{\theta = \alpha = \arg(p + \tau q)},$$

which is equivalent to

$$\boxed{\delta x^5 + i\delta x^6 = \ell\,(p + \tau q), \qquad \ell > 0}.$$

This proves that adding any number of correctly oriented (p, q) legs does not impose a third independent projector: every such leg preserves the same eight-dimensional common eigenspace.

10.3.1.7 Force balance gives the same slope rule

The supersymmetry result has a simple mechanical consequence. Treat the plane as the complex plane. The tension vector of an oriented leg is

$$\mathcal{T}_{p,q} = T_{(p,q)_5} e^{i\theta}.$$

Using the BPS tension and the angle rule,

$$\mathcal{T}_{p,q} = T_{\mathrm{D5}}|p+\tau q| \frac{p+\tau q}{|p+\tau q|} = T_{\mathrm{D5}}(p+\tau q).$$

At a junction, charge conservation is

$$\sum_i p_i = 0, \qquad \sum_i q_i = 0.$$

It immediately implies

$$\sum_i \mathcal{T}_{p_i,q_i} = T_{\mathrm{D5}}\left(\sum_i p_i + \tau \sum_i q_i\right) = 0.$$

Thus the same relation that aligns the supersymmetry projector also guarantees zero net force at every vertex.

10.3.1.8 How τ is normalized to i

Write

$$\tau = \tau_1 + i\tau_2, \qquad \tau_2 > 0.$$

The physical slope rule says

$$\delta x^5 + i\delta x^6 = \ell\,(p+\tau q)\,.$$

Separating real and imaginary parts,

$$\delta x^5 = \ell(p+\tau_1 q), \qquad \delta x^6 = \ell\tau_2 q.$$

Now introduce coordinates in the real basis $(1,\tau)$:

$$\boxed{x^5 = \tilde{x}^5 + \tau_1\tilde{x}^6, \qquad x^6 = \tau_2\tilde{x}^6}.$$

Equivalently,

$$\boxed{\tilde{x}^5 = x^5 - \frac{\tau_1}{\tau_2}x^6, \qquad \tilde{x}^6 = \frac{x^6}{\tau_2}}.$$

Then every (p,q) leg obeys

$$\delta\tilde{x}^5 = \ell p, \qquad \delta\tilde{x}^6 = \ell q,$$

or

$$\boxed{\delta\tilde{x}^5 + i\delta\tilde{x}^6 = \ell(p+iq)}.$$

Thus, in the normalized web diagram,

$$\boxed{\tau_{\mathrm{web}} = i},$$

and the slope is simply

$$\delta\tilde{x}^5 : \delta\tilde{x}^6 = p : q.$$

For $\tau_1 \neq 0$, this transformation shears the plane; for $\tau_2 \neq 1$, it rescales one direction. Indeed, the physical Euclidean metric becomes

$$(dx^5)^2 + (dx^6)^2 = \left(d\tilde{x}^5 + \tau_1 d\tilde{x}^6\right)^2 + \tau_2^2(d\tilde{x}^6)^2.$$

Therefore the normalized diagram is not claiming that the physical axio-dilaton has dynamically changed to i while every other datum remains fixed. It is a convenient affine drawing of the web in which the (p,q) charge lattice itself supplies the coordinate grid. Physical lengths and tensions must still be interpreted with the transformed metric and tension scale.

For the decoupled low-energy five-dimensional theory, one can simultaneously rescale and shear the web while holding its BPS mass parameters fixed. In that restricted sense τ is redundant, which is the sense intended in the five-brane-web construction.

10.3.1.9 Distinction from an $SL(2,\mathbb{Z})$ duality

At the level of classical type-IIB supergravity, the continuous group $SL(2,\mathbb{R})$ acts transitively on the upper half-plane. The real matrix

$$M_\tau = \begin{pmatrix} 1/\sqrt{\tau_2} & -\tau_1/\sqrt{\tau_2} \\ 0 & \sqrt{\tau_2} \end{pmatrix} \in SL(2,\mathbb{R})$$

sends

$$\tau \longmapsto \frac{a\tau + b}{c\tau + d} = i.$$

However, as the book stresses around equation (18.76), the exact string-theory duality group is $SL(2,\mathbb{Z})$, not all of $SL(2,\mathbb{R})$. A generic M_τ is not integral and does not preserve the integer (p,q) charge lattice. Consequently,

$$\text{generic } \tau \quad \nsim_{SL(2,\mathbb{Z})} \quad i.$$

The safe statement is therefore:

- one may set $\tau = i$ in the abstract normalized web coordinates without changing the web's low-energy combinatorial data;
- one may also map τ to i at the level of classical supergravity by an $SL(2,\mathbb{R})$ transformation;
- one may not generally claim that an arbitrary full type-IIB string vacuum is $SL(2,\mathbb{Z})$-equivalent to the vacuum at $\tau = i$.

10.3.1.10 Checks

For a $(1,1)_5$ leg at $\tau = i$,

$$\alpha_{1,1} = \arg(1+i) = \frac{\pi}{4}.$$

Thus it must run at 45° and has projector

$$K_{(1,1)} = \Gamma_{(5)} \frac{\Gamma^5 + \Gamma^6}{\sqrt{2}} \frac{\sigma_1 + \sigma_3}{\sqrt{2}}.$$

On the common D5–NS5 eigenspace this acts as the identity, so it introduces no additional supersymmetry breaking.

At the elementary three-leg vertex with incoming charges

$$(1,0) + (0,1) + (-1,-1) = 0,$$

the three tension vectors at $\tau = i$ are proportional to

$$1, \qquad i, \qquad -1-i,$$

which sum to zero.

If $C_0 = 0$ but $g_s \neq 1$, then

$$\tau = \frac{i}{g_s}.$$

A (p,q) leg has physical slope

$$\tan\theta = \frac{q}{g_s p}.$$

The normalized rescaling $\tilde{x}^6 = g_s x^6$ turns this into

$$\tan\tilde{\theta} = \frac{q}{p},$$

which is precisely the $\tau = i$ drawing convention.

10.3.1.11 Result

The elementary projectors are

$$\boxed{K_D = \Gamma^{012345}\sigma_1, \qquad K_N = \Gamma^{012346}\sigma_3}.$$

They satisfy

$$K_D^2 = K_N^2 = 1, \qquad [K_D, K_N] = 0,$$

and are independent, so

$$\boxed{32 \longrightarrow 16 \longrightarrow 8}.$$

The scalar-dressed charge matrix, its BPS norm, and its normalized form are

$$\boxed{\begin{aligned} \mathcal{Z}_5 &= T_{\mathrm{D5}}\left[\mathrm{Re}(p+\tau q)\sigma_1 + \mathrm{Im}(p+\tau q)\sigma_3\right], \\ T_{(p,q)_5} &= \|\mathcal{Z}_5\| = T_{\mathrm{D5}}|p+\tau q|, \\ \widehat{\mathcal{Z}}_5 &= \frac{\mathcal{Z}_5}{T_{(p,q)_5}} = \cos\alpha\,\sigma_1 + \sin\alpha\,\sigma_3, \qquad e^{i\alpha} = \frac{p+\tau q}{|p+\tau q|}. \end{aligned}}$$

BPS saturation gives

$$\left[T_{(p,q)_5}\mathbf{1} - \Gamma^{01234}\Gamma_\theta \mathcal{Z}_5\right]\epsilon = 0,$$

so the preserved-supersymmetry involution must be the world-volume gamma matrix multiplied by the normalized charge matrix.

For a general leg,

$$\boxed{K_{(p,q)}(\theta) = \Gamma^{01234}\left(\cos\theta\,\Gamma^5 + \sin\theta\,\Gamma^6\right)\left(\cos\alpha\,\sigma_1 + \sin\alpha\,\sigma_3\right), \qquad e^{i\alpha} = \frac{p+\tau q}{|p+\tau q|}}.$$

It preserves the common eight supercharges exactly when

$$\boxed{\theta = \alpha \quad \Longleftrightarrow \quad \delta x^5 + i\delta x^6 \propto p + \tau q}.$$

Finally, for $\tau = \tau_1 + i\tau_2$,

$$\boxed{x^5 = \tilde{x}^5 + \tau_1\tilde{x}^6, \qquad x^6 = \tau_2\tilde{x}^6}$$

turns the slope rule into

$$\boxed{\delta\tilde{x}^5 + i\delta\tilde{x}^6 \propto p + iq}.$$

This is the precise meaning of setting $\tau = i$ in a normalized (p,q) five-brane web.

10.4 Q4: Type-I D-string and heterotic-string spectrum match

In page710-711 they showed the massless excitations of the D-string in type I are precisely the fields appearing on the world-sheet of the heterotic string. What are the massless excitations of a D-brane? Derive this equivalence in details.

> **Textbook location and related material.** Chapter 18, Section 18.6: type-I/heterotic duality and the D-string realization of the heterotic string. Sections 9.4 and 13.3 provide the type-I projections and D-brane spectrum.

10.4.1 Answer

10.4.1.1 Core point

For an ordinary oriented type-II Dp-brane, the massless open strings with both endpoints on the brane form the dimensional reduction of ten-dimensional $\mathcal{N}=1$ super-Yang–Mills to $p+1$ dimensions:

$$\boxed{A_M \longrightarrow (A_\alpha, \Phi^i)\,, \qquad \lambda_{10} \longrightarrow \lambda_{p+1}}.$$

Here A_α is a gauge field tangent to the Dp-brane, the $9-p$ fields Φ^i describe transverse fluctuations of the brane, and λ_{p+1} is the dimensionally reduced ten-dimensional Majorana–Weyl gaugino. For N coincident oriented branes, all these fields are in the adjoint of $U(N)$.

The type-I D-string is more special because type I is the orientifold

$$\text{type I} = \frac{\text{type IIB}}{\Omega} + 32 \text{ D9-branes.}$$

Consequently, the massless fields on a single D1 receive contributions from two kinds of open strings:

1. D1–D1 strings give eight transverse bosons and eight right-moving fermions;
2. D1–D9 and D9–D1 strings together give 32 real left-moving fermions.

Thus the final physical world-sheet fields are

$$\boxed{X^i(\sigma,\tau), \qquad \psi^A(\sigma^-), \qquad \lambda^a(\sigma^+)},$$

with

$$i = 2, \dots, 9, \qquad A = 1, \dots, 8, \qquad a = 1, \dots, 32.$$

These are exactly the Green–Schwarz world-sheet fields of the ten-dimensional $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic string: eight transverse non-chiral bosons, eight right-moving Green–Schwarz fermions, and 32 left-moving real fermions generating the $\mathfrak{so}(32)_1$ current algebra.

The derivation has three essential ingredients: the ordinary D-brane open-string spectrum, the type-I Ω projection in the D1–D1 sector, and the change of oscillator moding in the eight mixed D1–D9 directions.

10.4.1.2 Massless open strings on a general Dp-brane

Take a flat Dp-brane along $x^0, \dots, x^p$. An open string whose two endpoints lie on it obeys

$$\partial_\sigma X^\alpha\big|_{\partial\Sigma} = 0, \qquad \alpha = 0, \dots, p,$$

and

$$\delta X^i\big|_{\partial\Sigma} = 0, \qquad i = p+1, \dots, 9.$$

The first are Neumann–Neumann boundary conditions and the second are Dirichlet–Dirichlet boundary conditions. The open-string center-of-mass momentum therefore has only world-volume components:

$$k^M = (k^\alpha, 0), \qquad k^2 = k_\alpha k^\alpha.$$

For NN or DD boundary conditions, the world-sheet fermions are half-integer moded in the NS sector and integer moded in the R sector. The mass formulas are

$$\alpha' M^2_{\mathrm{NS}} = N_{\mathrm{osc}} - \frac{1}{2}, \qquad \alpha' M^2_{\mathrm{R}} = N_{\mathrm{osc}}.$$

10.4.1.2.1 NS sector: gauge field and transverse scalars The NS ground state has

$$\alpha' M^2 = -\frac{1}{2}$$

and is removed by the open-string GSO projection. The first GSO-even states have one fermionic oscillator:

$$\epsilon_M b^M_{-1/2}|0;k\rangle_{\mathrm{NS}}.$$

Their oscillator number is $N_{\mathrm{osc}} = \frac{1}{2}$, so they are massless. Splitting the polarization into NN and DD components gives

$$\epsilon_\alpha b^\alpha_{-1/2}|0;k\rangle_{\mathrm{NS}} \quad \longleftrightarrow \quad A_\alpha,$$

and

$$\phi_i b^i_{-1/2}|0;k\rangle_{\mathrm{NS}} \quad \longleftrightarrow \quad \Phi^i.$$

The super-Virasoro constraint $G_{1/2}|\mathrm{phys}\rangle = 0$ gives

$$k^\alpha \epsilon_\alpha = 0.$$

Null states identify

$$\epsilon_\alpha \sim \epsilon_\alpha + k_\alpha \Lambda,$$

which is the linearized gauge invariance

$$A_\alpha \sim A_\alpha + \partial_\alpha \Lambda.$$

There is no analogous condition on ϕ_i, because the open string has no momentum k^i in a Dirichlet direction. The Φ^i are therefore world-volume scalars.

Their geometrical meaning follows either directly from the Dirichlet boundary value or by T-duality:

$$\boxed{X^i_{\mathrm{brane}} = 2\pi\alpha' \Phi^i}$$

for a single brane, up to the conventional normalization of the scalar. A constant expectation value of Φ^i moves the D-brane in the x^i direction.

10.4.1.2.2 R sector: the world-volume gaugino In the R sector the zero-point energy vanishes. The ground states are already massless and the zero modes obey

$$\{b_0^M, b_0^N\} = \eta^{MN}.$$

Thus they generate the ten-dimensional Clifford algebra and the R ground states form a ten-dimensional spinor. The GSO projection selects one Majorana–Weyl chirality, leaving 16 real components before the massless Dirac equation is imposed:

$$\Gamma_{11} u = +u.$$

The zero-mode super-Virasoro constraint is

$$G_0|u;k\rangle_{\mathrm{R}} = 0 \quad \Longleftrightarrow \quad k_\alpha \Gamma^\alpha u = 0.$$

For a nonzero null world-volume momentum this leaves eight on-shell fermionic polarizations. Together with A_α and Φ^i, these are the fields obtained by reducing ten-dimensional $\mathcal{N} = 1$ super-Yang–Mills to the Dp-brane world-volume.

For a Chan–Paton stack of N identical oriented branes, the states carry endpoint labels $|r, s\rangle$, $r, s = 1, \ldots, N$. Hence

$$A_\alpha, \Phi^i, \lambda \in \mathrm{End}(\mathbb{C}^N),$$

and the low-energy gauge group is $U(N)$. The abelian scalars give the center-of-mass position, while eigenvalue differences measure relative brane positions.

10.4.1.3 A useful zero-point-energy formula for mixed boundary conditions

The D1–D9 sector differs from the D1–D1 sector because it has mixed boundary conditions. It is useful to derive the relevant zero-point energy once.

For one real bosonic or fermionic oscillator, zeta-function regularization gives

$$\sum_{n=1}^{\infty} n = -\frac{1}{12}, \qquad \sum_{n=0}^{\infty}\left(n+\frac{1}{2}\right) = \frac{1}{24}.$$

Consequently,

	integer modes	half-integer modes
boson	$-\frac{1}{24}$	$+\frac{1}{48}$
fermion	$+\frac{1}{24}$	$-\frac{1}{48}$

where the opposite sign in the fermionic row comes from fermionic normal ordering.

For an NN or DD direction, the NS fermion is half-integer moded, so one transverse boson–fermion pair contributes

$$-\frac{1}{24} - \frac{1}{48} = -\frac{1}{16}.$$

For an ND or DN direction, the moding is reversed: the boson is half-integer moded and the NS fermion is integer moded. Such a pair contributes

$$\frac{1}{48} + \frac{1}{24} = +\frac{1}{16}.$$

If ν of the eight light-cone transverse directions are mixed, then

$$\boxed{a_{\mathrm{NS}}(\nu) = (8-\nu)\left(-\frac{1}{16}\right) + \nu\left(\frac{1}{16}\right) = -\frac{1}{2} + \frac{\nu}{8}}.$$

In the R sector, bosons and fermions have the same moding for every type of boundary condition and cancel pairwise:

$$\boxed{a_{\mathrm{R}} = 0}.$$

For a string with both endpoints on the same D-brane, $\nu = 0$ and $a_{\mathrm{NS}} = -\frac{1}{2}$, reproducing the spectrum above. For a D1–D9 string, $\nu = 8$ and therefore

$$\boxed{a_{\mathrm{NS}} = +\frac{1}{2}, \qquad a_{\mathrm{R}} = 0}.$$

This single calculation explains why the D1–D9 sector has massless fermions but no massless bosons.

10.4.1.4 The D1–D1 NS sector after the type-I projection

The D1 extends along x^0, x^1. Thus a D1–D1 string has NN boundary conditions for $M = \alpha = 0, 1$ and DD boundary conditions for $M = i = 2, \ldots, 9$.

Before the orientifold projection, its massless NS states are

$$b^{\alpha}_{-1/2}|0;k\rangle_{\mathrm{NS}} \quad \longleftrightarrow \quad A_{\alpha},$$

and

$$b^{i}_{-1/2}|0;k\rangle_{\mathrm{NS}} \quad \longleftrightarrow \quad X^{i}.$$

The notation X^i is natural here because these scalars are precisely the transverse embedding coordinates of the D-string.

Equation (9.75) gives the action of world-sheet parity on the fermionic oscillators:

$$\Omega b^M_r \Omega^{-1} = \eta_M e^{i\pi r} b^M_r, \qquad \eta_\alpha = +1, \qquad \eta_i = -1.$$

The sign difference between η_α and η_i is the difference between NN and DD boundary conditions. With the type-I convention (9.76),

$$\Omega|0\rangle_{\rm NS} = -i|0\rangle_{\rm NS}.$$

For $r = -\frac{1}{2}$,

$$e^{i\pi r} = e^{-i\pi/2} = -i.$$

Therefore

$$\begin{aligned}\Omega\left(b^M_{-1/2}|0\rangle_{\rm NS}\right) &= \eta_M(-i)(-i)b^M_{-1/2}|0\rangle_{\rm NS}\\ &= -\eta_M b^M_{-1/2}|0\rangle_{\rm NS}.\end{aligned}$$

The result is

state	Ω eigenvalue	type-I projection
$b^{0,1}_{-1/2}\|0\rangle_{\rm NS}$	-1	removed,
$b^{i}_{-1/2}\|0\rangle_{\rm NS}$	$+1$	kept.

Thus

$$\boxed{\text{D1–D1 NS sector:}\qquad X^i(\sigma,\tau),\quad i=2,\dots,9}.$$

For a single type-I D1 the Chan–Paton group is $O(1) \cong \mathbb{Z}_2$, so there is no continuous world-volume gauge field. Notice also that A_0, A_1 would carry no local propagating polarization in two dimensions even before the projection. The eight surviving scalars are the Goldstone modes for the eight translations broken by placing the D1 at a definite position in $\mathbb{R}^8$.

10.4.1.5 The D1–D1 R sector and its chirality

All ten D1–D1 directions are either NN or DD, so all R-sector fermions are integer moded. The ten zero modes generate Cl(1, 9):

$$b_0^M = \frac{1}{\sqrt{2}}\Gamma^M.$$

After the open-string GSO projection, choose the positive-chirality spinor

$$\Gamma_{11} = +1.$$

Under

$$SO(1,9) \supset SO(1,1) \times SO(8),$$

it decomposes in the book's conventions as

$$\boxed{\mathbf{16} \longrightarrow \mathbf{8}_s^+ \oplus \mathbf{8}_c^-}.$$

The superscript is the two-dimensional chirality. To determine which summand survives, use the weight basis

$$|s\rangle = |s_0; s_1, s_2, s_3, s_4\rangle, \qquad s_I = \pm\frac{1}{2}, \quad I = 0,\dots,4.$$

For a D1, equation (9.75) at $r = 0$ says that Ω commutes with the two NN zero modes and anticommutes with the eight DD zero modes:

$$\Omega b_0^\alpha \Omega^{-1} = +b_0^\alpha, \qquad \Omega b_0^i \Omega^{-1} = -b_0^i.$$

Thus its action on the spinor module is, up to the ground-state phase fixed by (9.76), the transverse chirality operator. The book obtains

$$\Omega|s\rangle = -e^{i\pi(s_1+s_2+s_3+s_4)}|s\rangle.$$

Let n_- be the number of minus signs among $s_1, \dots, s_4$. Then

$$s_1 + s_2 + s_3 + s_4 = 2 - n_-,$$

and hence

$$\Omega|s\rangle = -(-1)^{n_-}|s\rangle.$$

The invariant states have odd n_-. Those are precisely the $\mathbf{8}_c$ weights. Positive ten-dimensional chirality then correlates them with negative $SO(1,1)$ chirality, so

$$\boxed{\mathbf{16} \xrightarrow{\Omega} \mathbf{8}_c^-}.$$

The remaining physical-state condition is the two-dimensional massless Dirac equation

$$k_\alpha \Gamma^\alpha u_- = 0.$$

In position space, multiply by Γ^0 and use $\Gamma^{01}u_- = -u_-$. With mostly-plus signature,

$$\left(-\partial_\tau + \Gamma^{01}\partial_\sigma\right) u_- = 0 \quad \Longrightarrow \quad \left(\partial_\tau + \partial_\sigma\right) u_- = 0.$$

Therefore

$$u_- = u_-(\tau - \sigma) = u_-(\sigma^-),$$

which is right-moving in the convention of the book. We obtain

$$\boxed{\text{D1–D1 R sector:} \qquad \psi^A(\sigma^-), \quad A = 1, \ldots, 8}.$$

These are the eight right-moving Green–Schwarz fermions paired by supersymmetry with the eight transverse bosons X^i.

10.4.1.6 The D1–D9 and D9–D1 sectors

A D9 fills all ten dimensions. A D1–D9 string therefore has

$$\begin{array}{r|l} x^{0,1} & \text{NN} \\ x^2, \ldots, x^9 & \text{DN} \end{array}$$

and hence eight mixed directions:

$$\nu = 8.$$

10.4.1.6.1 Absence of massless bosons The zero-point formula gives

$$\alpha' M_{\text{NS}}^2 = N_{\text{osc}} + \frac{1}{2}.$$

Since $N_{\text{osc}} \geq 0$, the entire NS sector is massive at its lowest level:

$$M_{\text{NS}}^2 \geq \frac{1}{2\alpha'}.$$

Thus the D1–D9 strings supply no additional massless bosonic fields.

10.4.1.6.2 The 32 massless chiral fermions The R-sector zero-point energy is zero, so its ground state is massless. In an ND or DN direction, however, the R fermions are half-integer moded. Therefore the eight fields in directions $2, \ldots, 9$ have no R zero modes. Only the common NN directions $0, 1$ have zero modes:

$$\{b_0^\alpha, b_0^\beta\} = \eta^{\alpha\beta}, \qquad \alpha, \beta = 0, 1.$$

The ground state is consequently only a two-dimensional spinor, rather than a ten-dimensional spinor:

$$|s_0, a\rangle_{\text{R}}, \qquad s_0 = \pm\frac{1}{2}.$$

The label $a = 1, \ldots, 32$ is carried by the D9 endpoint because tadpole cancellation requires 32 D9-branes.

The GSO projection keeps

$$s_0 = +\frac{1}{2}.$$

Equivalently, the mutual-locality condition between D1–D1 and D1–D9 vertex operators selects this chirality. The physical state equation is again

$$k_\alpha \Gamma^\alpha u_+ = 0.$$

Now $\Gamma^{01} u_+ = +u_+$, so

$$(-\partial_\tau + \partial_\sigma)\, u_+ = 0 \quad \Longrightarrow \quad (\partial_\tau - \partial_\sigma)\, u_+ = 0.$$

It follows that

$$u_+ = u_+(\tau + \sigma) = u_+(\sigma^+),$$

which is left-moving in the book's convention.

An oppositely oriented string belongs to the D9–D1 sector. World-sheet parity reverses the string orientation and exchanges the two sectors:

$$\Omega : \quad \mathcal{H}_{\mathrm{D1-D9}} \longleftrightarrow \mathcal{H}_{\mathrm{D9-D1}}.$$

The orientifold projection therefore identifies their states rather than adding a second independent copy. The result is exactly

$$\boxed{\text{D1–D9 plus D9–D1:} \qquad \lambda^a(\sigma^+), \quad a = 1, \ldots, 32},$$

where the λ^a are real left-moving Majorana–Weyl fermions transforming in the vector representation of $SO(32)$.

10.4.1.7 Matching to the heterotic Green–Schwarz world-sheet

Collecting the two sectors gives

D1 sector	field	$SO(8) \times SO(32)$	two-dimensional propagation
$\mathrm{D1-D1}$, NS	X^i	$(\mathbf{8}_v, \mathbf{1})$	left and right
$\mathrm{D1-D1}$, R	ψ^A	$(\mathbf{8}_c, \mathbf{1})$	right-moving
$\mathrm{D1-D9/D9-D1}$, R	λ^a	$(\mathbf{1}, \mathbf{32})$	left-moving.

This is exactly the physical Green–Schwarz description of the heterotic string in static or light-cone gauge:

- X^i are the eight transverse embedding coordinates;
- ψ^A are their eight right-moving supersymmetric partners;
- λ^a are the 32 left-moving fermions of the gauge sector.

The gauge currents are the bilinears

$$\boxed{J^{ab} = i\lambda^a \lambda^b, \qquad J^{ab} = -J^{ba}}.$$

There are

$$\frac{32 \cdot 31}{2} = 496$$

such currents, and their operator products generate the level-one affine algebra

$$\widehat{\mathfrak{so}}(32)_1.$$

This explains how open strings ending on the 32 D9-branes become the left-moving gauge degrees of freedom of the dual heterotic closed string.

The matching of chiral central charges is also exact. In light-cone gauge,

$$c_L = 8 + \frac{32}{2} = 24,$$

while

$$c_R = 8 + \frac{8}{2} = 12.$$

These are the heterotic light-cone matter central charges: the left-moving side is bosonic and the right-moving side is supersymmetric.

10.4.1.8 How the heterotic GSO projection arises on the D-string

The equality of local massless fields is not yet the complete global statement. Thirty-two free left-moving fermions have four $\mathfrak{so}(32)_1$ conjugacy classes, conventionally denoted

$$O, \qquad V, \qquad S, \qquad C.$$

The $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic string keeps the root and one spinor class and removes the vector and conjugate-spinor classes:

$$\boxed{O \oplus S \quad \text{kept}, \qquad V \oplus C \quad \text{projected out}}$$

up to the convention that exchanges S and C. In the fermionic formulation, this is implemented by summing over spin structures for all 32 fermions.

On one type-I D1 the continuous gauge field was projected out, but its Chan–Paton group is not trivial:

$$O(1) \cong \mathbb{Z}_2.$$

Therefore a D1 world-sheet with nontrivial cycles can carry flat $\mathbb{Z}_2$ gauge bundles. On a torus,

$$H^1(T^2, \mathbb{Z}_2) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_2,$$

so there are four choices of holonomy:

$$(w_A, w_B) \in \{(+,+), (+,-), (-,+), (-,-)\}.$$

Because the D1–D9 fermions are charged under this $O(1)$, a negative holonomy reverses the sign of λ^a around the corresponding cycle. Summing over the four flat bundles is therefore precisely the common spin-structure sum for the 32 left-moving fermions. It produces the heterotic GSO projection and refines the perturbative Lie algebra $\mathfrak{so}(32)$ to the global gauge group

$$\boxed{\mathrm{Spin}(32)/\mathbb{Z}_2}.$$

Thus both the local fields and their global projection agree.

10.4.1.9 Checks

For the D1–D1 strings, $\nu = 0$, so

$$a_{\mathrm{NS}} = -\frac{1}{2}.$$

One $b_{-1/2}$ oscillator raises the level by $\frac{1}{2}$ and gives a massless state, as required.

For the D1–D9 strings, $\nu = 8$, so

$$a_{\mathrm{NS}} = +\frac{1}{2}.$$

There is no possible nonnegative oscillator level that makes the NS state massless. The R ground state remains massless because $a_{\mathrm{R}} = 0$.

The Ω projection retains eight D1–D1 bosons and eight D1–D1 fermions, so the supersymmetric transverse sector is balanced:

$$8_B = 8_F.$$

The 32 additional fermions are all of the opposite world-sheet chirality and form the heterotic gauge sector rather than additional supersymmetric partners.

Finally, Ω identifies the D1–D9 and D9–D1 orientations. Without this identification one would incorrectly obtain 64 rather than 32 real chiral fermions.

10.4.1.10 Result

For an oriented type-II Dp-brane,

$$\boxed{\begin{array}{ll} \mathrm{NS}: & b^{\alpha}_{-1/2}|0;k\rangle \leftrightarrow A_{\alpha}, \qquad b^{i}_{-1/2}|0;k\rangle \leftrightarrow \Phi^{i}, \\ \mathrm{R}: & |u;k\rangle, \qquad \Gamma_{11}u = u, \qquad k_{\alpha}\Gamma^{\alpha}u = 0. \end{array}}$$

This is ten-dimensional $\mathcal{N} = 1$ super-Yang–Mills reduced to $p + 1$ dimensions.

For ν mixed ND or DN directions,

$$\boxed{a_{\mathrm{NS}} = -\frac{1}{2} + \frac{\nu}{8}, \qquad a_{\mathrm{R}} = 0}.$$

For the type-I D1,

$$\boxed{\begin{array}{c|c} \mathrm{D1-D1,\ NS} & 8\,X^{i}(\sigma,\tau) \\ \mathrm{D1-D1,\ R} & 8\,\psi^{A}(\sigma^{-}) \\ \mathrm{D1-D9/D9-D1,\ NS} & \text{no massless states} \\ \mathrm{D1-D9/D9-D1,\ R} & 32\,\lambda^{a}(\sigma^{+}). \end{array}}$$

Therefore

$$\boxed{\mathcal{H}^{\text{type I D1}}_{\text{massless}} = \{8X^{i},\, 8\psi^{A}_{R},\, 32\lambda^{a}_{L}\} = \mathcal{H}^{\text{Spin}(32)/\mathbb{Z}_2\text{ heterotic}}_{\text{world-sheet}}}.$$

The four flat $O(1) \cong \mathbb{Z}_2$ bundles on a toroidal D1 world-sheet implement the heterotic spin-structure sum and hence the heterotic GSO projection.

10.5 Q5: Deriving the dilaton exponents in the M-theory/type-IIA metric ansatz

In equation (18.105), the ansatz is

$$ds^2_{11} = e^{-2\Phi/3} g_{\mu\nu} dx^{\mu} dx^{\nu} + e^{4\Phi/3}(dx^{11} + C_{\mu} dx^{\mu})^2 \ , \ \mu, \nu = 0, \cdots, 9$$

I want to know why the factors $e^{-2\Phi/3}$ and $e^{4\Phi/3}$ are chosen like these? It seems that from the 11d action (18.104) to 10d IIA (16.134a), we have the relation

$$\frac{1}{\kappa^2_{10}}\sqrt{-g}R_{d=10} \sim \frac{2\pi g_s\sqrt{\alpha'}}{\kappa^2_{11}}\sqrt{-G}R_{d=11} \implies \sqrt{-G}R_{d=11} = \sqrt{-g}R_{d=10}$$

When we take the determinant of the 11d metric we find

$$\sqrt{-G} = e^{2\Phi/3} \cdot e^{-10\Phi/3}\sqrt{-g} = e^{-8\Phi/3}\sqrt{-g}$$

What about the Ricci scalar part?

> **Textbook location and related material.** Chapter 18, Section 18.7 (pp. 716–724): reduction of eleven-dimensional supergravity on a circle and the M-theory/type-IIA metric ansatz. Section 16.4 lists the ten- and eleven-dimensional actions being matched.

10.5.1 Answer

10.5.1.1 Core point

Your determinant is correct:

$$\boxed{\sqrt{-G_{11}} = e^{-8\Phi/3}\sqrt{-g_{10}}}.$$

The missing point is that the eleven-dimensional Ricci scalar is not equal to the ten-dimensional one. Its term proportional to R_{10} carries the inverse Weyl factor

$$R_{11} \supset e^{2\Phi/3} R_{10}.$$

Therefore

$$\sqrt{-G_{11}}\, R_{11} \supset \sqrt{-g_{10}}\, e^{-8\Phi/3} e^{2\Phi/3} R_{10} = \sqrt{-g_{10}}\, e^{-2\Phi} R_{10}.$$

This is exactly the string-frame gravitational coupling. Equation (16.134a) does not contain a bare $\sqrt{-g}\, R$ term; its NS–NS part is

$$\boxed{\sqrt{-g}\, e^{-2\Phi} \left[R + 4(\partial\Phi)^2 - \frac{1}{2}|H_3|^2 \right]}.$$

The full metric reduction gives, before integrating by parts,

$$\boxed{R_{11} = e^{2\Phi/3} \left[R_{10} + \frac{14}{3}\Box\Phi - \frac{16}{3}(\partial\Phi)^2 - \frac{1}{4} e^{2\Phi} F_2^2 \right]},$$

where $F_2 = dC_1$. Consequently,

$$\boxed{\sqrt{-G_{11}} R_{11} = \sqrt{-g} \left\{ e^{-2\Phi} \left[R + \frac{14}{3}\Box\Phi - \frac{16}{3}(\partial\Phi)^2 \right] - \frac{1}{4} F_2^2 \right\}}.$$

After integrating the $\Box\Phi$ term by parts, its contribution combines with $-\frac{16}{3}(\partial\Phi)^2$ to give $+4(\partial\Phi)^2$. The rest of the answer derives these statements and explains why the two exponents in (18.105) are uniquely selected.

10.5.1.2 The target is the type-IIA string-frame action

Ignoring the Chern–Simons term, the relevant bosonic type-IIA action is

$$S_{\mathrm{IIA}} = \frac{1}{2\tilde{\kappa}_{10}^2} \int d^{10}x \sqrt{-g} \left\{ e^{-2\Phi} \left[R + 4(\partial\Phi)^2 - \frac{1}{2}|H_3|^2 \right] - \frac{1}{2}|F_2|^2 - \frac{1}{2}|F_4|^2 \right\}.$$

Thus the two classes of fields have different dilaton behavior:

$$\begin{aligned} \text{NS–NS fields } (g_{\mu\nu}, \Phi, B_2): &\qquad e^{-2\Phi}, \\ \text{R–R fields } (C_1, C_3): &\qquad \text{no string-frame dilaton prefactor.} \end{aligned}$$

The ten-dimensional metric $g_{\mu\nu}$ in (18.105) is specifically the string-frame metric. Choosing another ten-dimensional frame would change the first exponent in the eleven-dimensional ansatz.

If the asymptotic coupling has already been extracted from the gravitational constant, write

$$\Phi = \Phi_0 + \phi, \qquad g_s = e^{\Phi_0}.$$

Then the same local statement becomes

$$\frac{e^{-2\Phi}}{\tilde{\kappa}_{10}^2} = \frac{e^{-2\phi}}{\kappa_{10}^2}, \qquad \kappa_{10} = g_s \tilde{\kappa}_{10}.$$

This constant convention does not change any of the following local reduction formulas.

10.5.1.3 Why the exponents are fixed

Begin with the more general Kaluza–Klein ansatz

$$ds_{11}^2 = e^{2a\Phi} g_{\mu\nu} dx^\mu dx^\nu + e^{2b\Phi} \left(dx^{11} + C_\mu dx^\mu \right)^2.$$

There are two unknown constants, a and b.

The determinant scales as

$$\sqrt{-G_{11}} = e^{(10a+b)\Phi}\sqrt{-g}.$$

The part of the eleven-dimensional Ricci scalar containing the ten-dimensional Ricci scalar scales as

$$R_{11} \supset e^{-2a\Phi} R_{10}.$$

Hence the reduced Einstein–Hilbert density contains

$$\sqrt{-G_{11}}R_{11} \supset \sqrt{-g}\, e^{(8a+b)\Phi} R_{10}.$$

To reproduce the string-frame factor $e^{-2\Phi}$, one must impose

$$\boxed{8a + b = -2}.$$

The off-diagonal metric component is the Kaluza–Klein gauge field:

$$C_\mu \equiv \frac{G_{\mu\,11}}{G_{11\,11}}.$$

In type IIA it is the R–R one-form, so its kinetic term must not carry the NS–NS factor $e^{-2\Phi}$. The higher-dimensional curvature contains

$$R_{11} \supset -\frac{1}{4} e^{(2b-4a)\Phi} F_{\mu\nu}F^{\mu\nu},$$

where the indices on F_2 are raised with $g_{\mu\nu}$. After multiplying by the determinant, its dilaton factor is

$$e^{(10a+b)\Phi} e^{(2b-4a)\Phi} = e^{(6a+3b)\Phi}.$$

Requiring no dilaton prefactor gives

$$6a + 3b = 0 \qquad \Longleftrightarrow \qquad \boxed{2a + b = 0}.$$

Solving the two equations,

$$8a + b = -2, \qquad 2a + b = 0,$$

gives

$$\boxed{a = -\frac{1}{3}, \qquad b = \frac{2}{3}}.$$

Therefore

$$e^{2a\Phi} = e^{-2\Phi/3}, \qquad e^{2b\Phi} = e^{4\Phi/3},$$

which proves the exponents in equation (18.105).

This argument also shows that the exponents are not chosen merely to make the determinant simple. They simultaneously arrange the string-frame NS–NS prefactor and the R–R normalization of the Kaluza–Klein vector.

10.5.1.4 Determinant of the full Kaluza–Klein metric

For the actual values $a = -1/3$, $b = 2/3$, the metric components are

$$\begin{aligned} G_{\mu\nu} &= e^{-2\Phi/3} g_{\mu\nu} + e^{4\Phi/3} C_\mu C_\nu, \\ G_{\mu\,11} &= e^{4\Phi/3} C_\mu, \\ G_{11\,11} &= e^{4\Phi/3}. \end{aligned}$$

Using the Schur complement,

$$\begin{aligned} \det G_{11} &= G_{11\,11} \det\left(G_{\mu\nu} - \frac{G_{\mu\,11}G_{\nu\,11}}{G_{11\,11}}\right) \\ &= e^{4\Phi/3} \det\left(e^{-2\Phi/3} g_{\mu\nu}\right) \\ &= e^{4\Phi/3} e^{-20\Phi/3} \det g \\ &= e^{-16\Phi/3} \det g. \end{aligned}$$

Thus

$$\boxed{\sqrt{-G_{11}} = e^{-8\Phi/3}\sqrt{-g}}.$$

The cancellation of the $C_\mu C_\nu$ terms is why the Kaluza–Klein vector does not occur in the determinant.

10.5.1.5 Explicit reduction of the Ricci scalar

The curvature calculation is clearest in two steps. Define

$$h_{\mu\nu} = e^{-2\Phi/3} g_{\mu\nu}, \qquad \sigma = \frac{2\Phi}{3}, \qquad \eta = dx^{11} + C_1.$$

Then

$$ds_{11}^2 = h_{\mu\nu} dx^\mu dx^\nu + e^{2\sigma}\eta^2.$$

10.5.1.5.1 Step 1: ordinary circle reduction For a one-dimensional fiber with all fields independent of x^{11}, direct evaluation of the Christoffel symbols, or equivalently the Cartan structure equations, gives

$$\boxed{R_{11} = R[h] - \frac{1}{4}e^{2\sigma} F_{\mu\nu} F^{\mu\nu}_{(h)} - 2\Box_h \sigma - 2(\partial\sigma)_h^2}.$$

The four terms have a transparent origin:

- $R[h]$ is the curvature of the ten-dimensional base;
- $F_2 = dC_1$ measures the twisting of the circle over the base;
- the last two terms arise because the local circle radius e^σ varies over the base.

For $C_1 = 0$, this reduces to the familiar scalar-warped-product formula

$$R_{11} = R[h] - 2e^{-\sigma}\Box_h e^\sigma.$$

10.5.1.5.2 Step 2: Weyl-transform from $h_{\mu\nu}$ to the string metric Write

$$h_{\mu\nu} = e^{2\omega} g_{\mu\nu}, \qquad \omega = -\frac{\Phi}{3}.$$

In $d = 10$, the Weyl transformation of the Ricci scalar is

$$R[h] = e^{-2\omega}\left[R[g] - 18\Box\omega - 72(\partial\omega)^2\right].$$

Substituting $\omega = -\Phi/3$ gives

$$\boxed{R[h] = e^{2\Phi/3}\left[R[g] + 6\Box\Phi - 8(\partial\Phi)^2\right]}.$$

For a scalar f,

$$\Box_h f = e^{-2\omega}\left[\Box f + 8\,\partial_\mu\omega\,\partial^\mu f\right].$$

Therefore, with $\sigma = 2\Phi/3$,

$$\boxed{\Box_h\sigma = e^{2\Phi/3}\left[\frac{2}{3}\Box\Phi - \frac{16}{9}(\partial\Phi)^2\right]},$$

and

$$\boxed{(\partial\sigma)_h^2 = e^{2\Phi/3}\frac{4}{9}(\partial\Phi)^2}.$$

Finally, because a two-form contains two inverse metrics,

$$F_{\mu\nu}F^{\mu\nu}_{(h)} = e^{4\Phi/3} F_{\mu\nu} F^{\mu\nu}_{(g)}.$$

10.5.1.5.3 Step 3: combine all terms Substitution into the circle-reduction formula gives

$$R_{11} = e^{2\Phi/3}\Bigg[R + 6\Box\Phi - 8(\partial\Phi)^2 \\ - \frac{4}{3}\Box\Phi + \frac{32}{9}(\partial\Phi)^2 - \frac{8}{9}(\partial\Phi)^2 \\ - \frac{1}{4}e^{2\Phi}F_2^2\Bigg].$$

Combining coefficients,

$$6 - \frac{4}{3} = \frac{14}{3},$$

and

$$-8 + \frac{32}{9} - \frac{8}{9} = -\frac{16}{3}.$$

Hence

$$\boxed{R_{11} = e^{2\Phi/3}\left[R + \frac{14}{3}\Box\Phi - \frac{16}{3}(\partial\Phi)^2 - \frac{1}{4}e^{2\Phi}F_2^2\right]}.$$

This is the Ricci-scalar factor missing from the determinant-only argument.

10.5.1.6 Combining the determinant and Ricci scalar

Multiplying by

$$\sqrt{-G_{11}} = e^{-8\Phi/3}\sqrt{-g}$$

gives

$$\sqrt{-G_{11}}R_{11} = \sqrt{-g}\Bigg\{e^{-2\Phi}\left[R + \frac{14}{3}\Box\Phi - \frac{16}{3}(\partial\Phi)^2\right] \\ - \frac{1}{4}F_2^2\Bigg\}.$$

The F_2 term has no dilaton factor, exactly as required for an R–R field in string frame.

Now discard a boundary term and integrate by parts:

$$\int d^{10}x\,\sqrt{-g}\,e^{-2\Phi}\Box\Phi = -\int d^{10}x\,\sqrt{-g}\,\partial_\mu(e^{-2\Phi})\partial^\mu\Phi \\ = 2\int d^{10}x\,\sqrt{-g}\,e^{-2\Phi}(\partial\Phi)^2.$$

It follows that

$$\frac{14}{3}\Box\Phi - \frac{16}{3}(\partial\Phi)^2 \quad \xrightarrow{\text{inside the action}} \quad \left(\frac{28}{3} - \frac{16}{3}\right)(\partial\Phi)^2 = 4(\partial\Phi)^2.$$

Thus, suppressing the constant circle integral for the moment, the eleven-dimensional Einstein–Hilbert term reduces to

$$\boxed{\int d^{11}x\,\sqrt{-G}\,R_{11} \longrightarrow \int dx^{11}\int d^{10}x\,\sqrt{-g}\left[e^{-2\Phi}\left(R + 4(\partial\Phi)^2\right) - \frac{1}{4}F_2^2\right]}$$

This formula isolates the local field dependence. The constant normalization of x^{11}, its asymptotic proper radius, and the gravitational couplings are treated separately below.

10.5.1.7 Reduction of the eleven-dimensional three-form

The same exponents also give the correct NS–NS/R–R split of the three-form. Equation (18.105) writes

$$A_3 = C_3 + B_2 \wedge dx^{11},$$

or, in explicitly Kaluza–Klein-invariant form,

$$dA_3 = F_4 + H_3 \wedge \eta, \qquad \eta = dx^{11} + C_1,$$

with

$$H_3 = dB_2, \qquad F_4 = dC_3 - H_3 \wedge C_1.$$

For a four-form entirely along the ten-dimensional base, four inverse base metrics give

$$|F_4|^2_{11} = e^{8\Phi/3}|F_4|^2_{10}.$$

The determinant cancels this factor:

$$\sqrt{-G}\,|F_4|^2_{11} = \sqrt{-g}\,|F_4|^2_{10}.$$

Thus F_4 is R–R and has no string-frame dilaton prefactor.

For $H_3 \wedge \eta$, the three inverse base metrics contribute $e^{2\Phi}$, while the inverse fiber metric contributes $e^{-4\Phi/3}$. Hence

$$|H_3 \wedge \eta|^2_{11} = e^{2\Phi/3}|H_3|^2_{10},$$

and therefore

$$\sqrt{-G}\,|H_3 \wedge \eta|^2_{11} = \sqrt{-g}\,e^{-2\Phi}|H_3|^2_{10}.$$

Thus H_3 is NS–NS and acquires precisely the same $e^{-2\Phi}$ factor as R and the dilaton kinetic term.

Combining the metric and form reductions gives

$$\boxed{\begin{aligned} S_{10} = \frac{1}{2\tilde{\kappa}^2_{10}} \int d^{10}x\sqrt{-g}\Big\{ & e^{-2\Phi}\left[R + 4(\partial\Phi)^2 - \frac{1}{2}|H_3|^2\right] \\ & - \frac{1}{2}|F_2|^2 - \frac{1}{2}|F_4|^2\Big\} + S_{\mathrm{CS}}, \end{aligned}}$$

which is the bosonic type-IIA string-frame action.

10.5.1.8 What the gravitational-coupling relation does and does not imply

Equation (16.134a) uses the g_s-independent coupling $\tilde{\kappa}_{10}$ together with the full dilaton Φ. Equation (18.106), by contrast, uses the physical coupling

$$\kappa_{10} = g_s\tilde{\kappa}_{10}$$

and the asymptotically normalized fluctuation $\phi = \Phi - \Phi_0$. Indeed,

$$\frac{e^{-2\Phi}}{\tilde{\kappa}^2_{10}} = \frac{e^{-2\phi}}{\kappa^2_{10}}.$$

After this constant normalization has been made, integrating a field configuration independent of x^{11} over the circle produces the overall factor

$$\frac{2\pi R_{11}}{2\kappa^2_{11}}.$$

Equation (18.106),

$$\boxed{\kappa^2_{11} = 2\pi R_{11}\kappa^2_{10}},$$

then gives

$$\frac{2\pi R_{11}}{2\kappa_{11}^2} = \frac{1}{2\kappa_{10}^2}.$$

This matches the overall normalization of the integrated ten-dimensional action.

It does not imply

$$\sqrt{-G}\, R_{11} = \sqrt{-g}\, R_{10}.$$

That would incorrectly omit the local dilaton, its derivatives, and the Kaluza–Klein field strength. Written in the full-dilaton convention of equation (16.134a), the actual local relation, modulo a total derivative, is

$$\boxed{\sqrt{-G}\, R_{11} \doteq \sqrt{-g}\left[e^{-2\Phi}\left(R + 4(\partial\Phi)^2\right) - \frac{1}{4}F_2^2\right]},$$

where $\doteq$ denotes equality inside the action after discarding the boundary term. In the physical-coupling convention used in (18.106), replace Φ in the NS–NS factor by the normalized fluctuation $\phi = \Phi - \Phi_0$.

The factor

$$2\pi g_s\sqrt{\alpha'} = 2\pi R_{11}$$

in your calculation is therefore responsible only for the relation between κ_{11} and κ_{10}. It cannot be used to equate the two curvature densities pointwise.

10.5.1.9 The same reduction in ten-dimensional Einstein frame

The relation between string and ordinary Einstein metrics is

$$g_{\mu\nu}^{(E)} = e^{-\Phi/2} g_{\mu\nu}^{(S)}.$$

Substituting

$$g_{\mu\nu}^{(S)} = e^{\Phi/2} g_{\mu\nu}^{(E)}$$

into (18.105) gives

$$\boxed{ds_{11}^2 = e^{-\Phi/6} ds_{10,E}^2 + e^{4\Phi/3}(dx^{11} + C_1)^2}.$$

In this form, the condition that the ten-dimensional Einstein–Hilbert term have no dilaton prefactor is manifest:

$$8\left(-\frac{1}{12}\right) + \frac{2}{3} = 0.$$

The reduced metric sector becomes

$$\boxed{\sqrt{-G} R_{11} \doteq \sqrt{-g_E}\left[R_E - \frac{1}{2}(\partial\Phi)^2 - \frac{1}{4}e^{3\Phi/2}F_2^2\right]}.$$

Thus equation (18.105) can equivalently be understood as the standard Einstein-frame Kaluza–Klein ansatz, rewritten using $g_S = e^{\Phi/2} g_E$.

10.5.1.10 Geometrical meaning of the fiber exponent

For a constant asymptotic dilaton $\Phi_0 = \log g_s$ and a dimensionless coordinate x^{11} of period 2π, the physical eleven-dimensional line element is

$$ds_{\text{phys}}^2 = \ell_{11}^2\left[g_s^{-2/3} g_{\mu\nu} dx^\mu dx^\nu + g_s^{4/3}(dx^{11})^2\right].$$

Consequently,

$$\alpha' = \ell_{11}^2 g_s^{-2/3}, \qquad R_{11} = \ell_{11} g_s^{2/3}.$$

These imply

$$\boxed{\ell_{11} = g_s^{1/3}\sqrt{\alpha'}, \qquad R_{11} = g_s\sqrt{\alpha'}}.$$

Thus the factor $e^{4\Phi/3}$ also has an immediate geometrical interpretation: its square root, $e^{2\Phi/3}$, measures the M-theory circle radius in eleven-dimensional Planck units,

$$\boxed{\frac{R_{11}}{\ell_{11}} = g_s^{2/3}}.$$

10.5.1.11 Checks

If Φ is constant and $C_1 = 0$, then

$$R_{11} = e^{2\Phi/3} R_{10},$$

so

$$\sqrt{-G}R_{11} = \sqrt{-g}\, e^{-2\Phi} R_{10}.$$

This already checks the principal Weyl factor without any derivative terms.

If Φ varies but $C_1 = 0$, the integration-by-parts identity gives

$$\frac{14}{3}\Box\Phi - \frac{16}{3}(\partial\Phi)^2 \;\longrightarrow\; 4(\partial\Phi)^2,$$

which is the characteristic string-frame dilaton kinetic coefficient.

If Φ is constant but $F_2 \neq 0$, then

$$\sqrt{-G}R_{11} \supset -\frac{1}{4}\sqrt{-g}\, F_2^2,$$

with no dilaton prefactor. This confirms that the Kaluza–Klein vector is the type-IIA R–R one-form rather than an NS–NS gauge field.

10.5.1.12 Result

For

$$ds_{11}^2 = e^{2a\Phi} ds_{10,S}^2 + e^{2b\Phi}(dx^{11} + C_1)^2,$$

string-frame reduction requires

$$\boxed{8a + b = -2, \qquad 2a + b = 0},$$

and hence

$$\boxed{a = -\frac{1}{3}, \qquad b = \frac{2}{3}}.$$

Therefore

$$\boxed{ds_{11}^2 = e^{-2\Phi/3} ds_{10,S}^2 + e^{4\Phi/3}(dx^{11} + C_1)^2}.$$

The determinant and curvature are

$$\boxed{\sqrt{-G} = e^{-8\Phi/3}\sqrt{-g}},$$

and

$$\boxed{R_{11} = e^{2\Phi/3}\left[R + \frac{14}{3}\Box\Phi - \frac{16}{3}(\partial\Phi)^2 - \frac{1}{4}e^{2\Phi}F_2^2\right]}.$$

Thus, modulo a total derivative,

$$\boxed{\sqrt{-G}R_{11} \doteq \sqrt{-g}\left[e^{-2\Phi}\left(R + 4(\partial\Phi)^2\right) - \frac{1}{4}F_2^2\right]}.$$

Finally,

$$\boxed{\kappa_{11}^2 = 2\pi R_{11}\kappa_{10}^2}$$

matches the overall coefficients after integration over the circle; it does not assert equality of the eleven- and ten-dimensional local curvature densities.

10.6 Q6: Deriving the M2 two-form charge from the Green–Schwarz action

In page 720, the M-theory susy algebra is given by

$$\{Q_\alpha, Q_\beta\} = -(\Gamma^\mu C^{-1})_{\alpha\beta}P_\mu - \frac{1}{2}(\Gamma^{\mu\nu}C^{-1})_{\alpha\beta}Z_{\mu\nu}$$

It is said that this can be derived from the world-volume action of the extended objects in the Green-Schwarz formulation. How to derive this? Show the derivations.

Textbook location and related material. Chapter 18, Sections 18.4 and 18.7: the M-theory supersymmetry algebra, M2/M5 central charges, and Green–Schwarz branes. The explicit Wess–Zumino descent derivation supplements the book.

10.6.1 Answer

10.6.1.1 Core point

The statement can be made precise as follows. Start with the covariant Green–Schwarz action of a single M2-brane in flat eleven-dimensional superspace. Its Nambu–Goto term is **strictly** invariant under rigid spacetime supersymmetry. Its Wess–Zumino term is invariant only **up to a total derivative**. Noether's theorem therefore gives a supercharge containing an improvement term inherited from that total derivative. When the canonical Dirac bracket of two improved supercharges is evaluated,

- the ordinary part gives the total momentum P_M;
- the Wess–Zumino improvement gives

$$Z^{MN} = T_{\mathrm{M2}} \int_{\Sigma_2} dX^M \wedge dX^N.$$

This is the two-form charge in equation (18.117), and the last expression is equation (18.118) with $p = 2$. Thus the two-form term is not inserted by hand: it is the Noether two-cocycle caused by the quasi-invariance of the Green–Schwarz Wess–Zumino term.

There is one point of scope to make explicit. The algebra on page 720 actually also permits the five-form term

$$-\frac{1}{5!}(\Gamma^{M_1\cdots M_5}C^{-1})_{\alpha\beta} Z_{M_1\cdots M_5}.$$

The formula in the question is the momentum-plus-M2 truncation. I will derive the M2 term explicitly. The M5 term follows from the same Noether mechanism applied to the Green–Schwarz/PST M5 action, with an additional contribution involving its chiral world-volume two-form. This distinction agrees with the direct M2 discussion in Townsend's superalgebra lectures and the direct M5 Noether calculation of Sorokin and Townsend.

10.6.1.2 The flat-superspace Green–Schwarz M2 action

Let

$$\xi^i = (\tau, \sigma^1, \sigma^2)$$

be world-volume coordinates. The embedding fields are

$$X^M(\xi), \qquad \Theta^\alpha(\xi), \qquad M = 0, \ldots, 10, \qquad \alpha = 1, \ldots, 32.$$

The fundamental supersymmetry-invariant one-form is

$$\boxed{\Pi^M = dX^M - i\bar{\Theta}\Gamma^M d\Theta}, \qquad \Pi_i^M = \partial_i X^M - i\bar{\Theta}\Gamma^M \partial_i \Theta.$$

The induced metric is

$$g_{ij} = \Pi_i^M \Pi_j^N \eta_{MN}.$$

The Green–Schwarz action is

$$\boxed{S_{\mathrm{M2}} = -T_{\mathrm{M2}} \int_{\mathcal{M}_3} d^3\xi \sqrt{-\det g} + T_{\mathrm{M2}} \int_{\mathcal{M}_3} B_3}.$$

Here B_3 is a super-three-form potential. Its exterior derivative is the invariant closed super-four-form

$$H_4 = dB_3 = -i\, d\bar{\Theta}\Gamma_{MN} d\Theta \wedge \Pi^M \wedge \Pi^N,$$

up to an overall sign fixed by the orientation chosen for the M2-brane. Closure,

$$dH_4 = 0,$$

uses the eleven-dimensional Fierz identity which symmetrically contracts one factor of $C\Gamma^M$ with one factor of $C\Gamma_{MN}$. This closed four-form is the superspace cocycle underlying the membrane Wess–Zumino coupling.

The action also has local κ-symmetry. That gauge symmetry removes half of the components of Θ, but it is not the origin of the extra term in the spacetime algebra. The extra term comes from rigid supersymmetry and the Wess–Zumino coupling. The role of κ-symmetry will reappear in the BPS check.

10.6.1.3 Rigid supersymmetry and strict versus quasi-invariance

Rigid eleven-dimensional supersymmetry acts as

$$\delta_\varepsilon \Theta = \varepsilon, \qquad \delta_\varepsilon X^M = i\bar{\varepsilon}\Gamma^M\Theta, \qquad d\varepsilon = 0.$$

It follows directly that

$$\begin{aligned}\delta_\varepsilon \Pi^M &= d(\delta_\varepsilon X^M) - i\,\delta_\varepsilon\bar{\Theta}\Gamma^M d\Theta - i\bar{\Theta}\Gamma^M d(\delta_\varepsilon\Theta)\\ &= i\bar{\varepsilon}\Gamma^M d\Theta - i\bar{\varepsilon}\Gamma^M d\Theta - 0\\ &= 0.\end{aligned}$$

Therefore

$$\delta_\varepsilon g_{ij} = 0, \qquad \delta_\varepsilon\left(\sqrt{-\det g}\right) = 0.$$

The Nambu–Goto part of the action is exactly invariant.

Because H_4 is constructed only from the invariant forms Π^M and $d\Theta$, it is also strictly invariant:

$$\delta_\varepsilon H_4 = 0.$$

But this does **not** imply that a chosen potential B_3 is strictly invariant. Since $H_4 = dB_3$,

$$d(\delta_\varepsilon B_3) = \delta_\varepsilon H_4 = 0.$$

Locally, the Poincare lemma gives

$$\boxed{\delta_\varepsilon B_3 = d\Lambda_2(\varepsilon)}.$$

Consequently,

$$\delta_\varepsilon S_{\mathrm{M2}} = T_{\mathrm{M2}}\int_{\mathcal{M}_3} d\Lambda_2(\varepsilon) = T_{\mathrm{M2}}\int_{\partial\mathcal{M}_3}\Lambda_2(\varepsilon).$$

For a closed membrane, or for boundary conditions that kill the last integral, the action is supersymmetric. Nevertheless, the Lagrangian density has changed by a divergence, and that divergence must be retained in the Noether current.

At the lowest nontrivial order in Θ, the descent two-form has the structure

$$\Lambda_2(\varepsilon) = -\frac{i}{2}\bar{\varepsilon}\Gamma_{MN}\Theta\, dX^M \wedge dX^N + O(\Theta^3).$$

The higher-order terms replace the dX^M by the appropriate supersymmetric combinations and are needed for the exact current. They do not change the bosonic topological charge that survives in the final algebra.

10.6.1.4 Noether's theorem for a quasi-invariant Lagrangian

For fields ϕ^A with a variation

$$\delta L = \partial_i K^i,$$

the Noether current is not merely the canonical momentum times $\delta\phi^A$. It is

$$\boxed{j^i = \frac{\partial L}{\partial(\partial_i \phi^A)}\delta\phi^A - K^i}.$$

Indeed, using the Euler–Lagrange equations,

$$\begin{aligned}\delta L &= \frac{\partial L}{\partial \phi^A}\delta\phi^A + \frac{\partial L}{\partial(\partial_i\phi^A)}\partial_i\delta\phi^A \\ &= \left[\frac{\partial L}{\partial\phi^A} - \partial_i \frac{\partial L}{\partial(\partial_i\phi^A)}\right]\delta\phi^A + \partial_i\left[\frac{\partial L}{\partial(\partial_i\phi^A)}\delta\phi^A\right],\end{aligned}$$

and hence on shell

$$\partial_i j^i = 0.$$

For the membrane supersymmetry, write

$$j^i(\varepsilon) = \bar{\varepsilon} j^i, \qquad Q(\varepsilon) = \int_{\Sigma_2} d^2\sigma\, j^0(\varepsilon) = \bar{\varepsilon}Q.$$

If $P_M(\sigma)$ and $\mathcal{P}_\alpha(\sigma)$ denote the canonical momenta conjugate to X^M and Θ^α, the supercharge has the following relevant terms:

$$\boxed{Q_\alpha = \int_{\Sigma_2} d^2\sigma \left[\mathcal{P}_\alpha - iP_M(C\Gamma^M\Theta)_\alpha - \frac{iT_{\mathrm{M2}}}{2}\epsilon^{rs}\partial_r X^M \partial_s X^N (C\Gamma_{MN}\Theta)_\alpha + O(\Theta^3)\right]}.$$

The first two terms are the expected generator of

$$\delta\Theta = \varepsilon, \qquad \delta X^M = i\bar{\varepsilon}\Gamma^M\Theta.$$

The third term is the crucial Wess–Zumino improvement $-K^0$. If the total derivative in δB_3 were discarded, this term would be missed and one would incorrectly recover only the unextended super-Poincare algebra.

The displayed formula keeps precisely the terms needed to expose the bosonic momentum and membrane charge. The omitted $O(\Theta^3)$ terms are not optional in the exact supercharge: together with the fermionic constraints they supersymmetrically complete the result. Their brackets either cancel fermion-dependent pieces, combine into world-volume diffeomorphism and κ-symmetry constraints, or vanish on the physical constraint surface.

10.6.1.5 Canonical bracket of two supercharges

The fermionic canonical relation is schematically

$$\{\Theta^\alpha(\sigma), \mathcal{P}_\beta(\sigma')\}_D = \delta^\alpha{}_\beta\, \delta^{(2)}(\sigma - \sigma').$$

Strictly speaking, the Green–Schwarz fermionic constraints contain both first- and second-class parts, so the physical calculation uses a Dirac bracket. For the two field-independent bosonic terms we seek, it is enough to display the cross-brackets of $\mathcal{P}_\alpha$ with the terms linear in Θ.

Split the charge density as

$$Q_\alpha = Q_\alpha^{(0)} + Q_\alpha^{(P)} + Q_\alpha^{(Z)} + \cdots,$$

where

$$\begin{aligned}Q_\alpha^{(0)} &= \int d^2\sigma\, \mathcal{P}_\alpha, \\ Q_\alpha^{(P)} &= -i\int d^2\sigma\, P_M(C\Gamma^M\Theta)_\alpha, \\ Q_\alpha^{(Z)} &= -\frac{iT_{\mathrm{M2}}}{2}\int d^2\sigma\, \epsilon^{rs}\partial_r X^M \partial_s X^N (C\Gamma_{MN}\Theta)_\alpha.\end{aligned}$$

First consider the momentum term. Since $C\Gamma^M$ is symmetric in eleven dimensions,

$$\begin{aligned}&\{Q_\alpha^{(0)}, Q_\beta^{(P)}\}_D + \{Q_\alpha^{(P)}, Q_\beta^{(0)}\}_D \\ &= -i(C\Gamma^M)_{\beta\alpha} \int d^2\sigma\, P_M(\sigma) - i(C\Gamma^M)_{\alpha\beta} \int d^2\sigma\, P_M(\sigma) \\ &= -2i(C\Gamma^M)_{\alpha\beta} P_M,\end{aligned}$$

where

$$P_M = \int_{\Sigma_2} d^2\sigma\, P_M(\sigma).$$

Now consider the Wess–Zumino improvement. In eleven dimensions $C\Gamma^{MN}$ is also symmetric on its spinor indices. Therefore

$$\begin{aligned}&\{Q_\alpha^{(0)}, Q_\beta^{(Z)}\}_D + \{Q_\alpha^{(Z)}, Q_\beta^{(0)}\}_D \\ &= -iT_{\mathrm{M2}}(C\Gamma_{MN})_{\alpha\beta} \int_{\Sigma_2} d^2\sigma\, \epsilon^{rs} \partial_r X^M \partial_s X^N.\end{aligned}$$

Define

$$\boxed{Z^{MN} = T_{\mathrm{M2}} \int_{\Sigma_2} d^2\sigma\, \epsilon^{rs} \partial_r X^M \partial_s X^N}.$$

Since

$$dX^M \wedge dX^N = \epsilon^{rs} \partial_r X^M \partial_s X^N\, d\sigma^1 \wedge d\sigma^2,$$

this is equivalently

$$\boxed{Z^{MN} = T_{\mathrm{M2}} \int_{\Sigma_2} dX^M \wedge dX^N}.$$

Combining the two cross-brackets gives the physical part of the classical charge algebra,

$$\boxed{\{Q_\alpha, Q_\beta\}_D = -2i \left[(C\Gamma^M)_{\alpha\beta} P_M + \frac{1}{2}(C\Gamma^{MN})_{\alpha\beta} Z_{MN}\right] + \text{constraints}}.$$

The factor $1/2$ has the familiar tensor meaning: with unrestricted sums over M, N, every antisymmetric pair occurs twice,

$$\frac{1}{2}\Gamma^{MN} Z_{MN} = \sum_{M<N} \Gamma^{MN} Z_{MN}.$$

On the physical constraint surface, the constraint terms vanish. Passing from the classical Grassmann Dirac bracket to the quantum anticommutator, normalizing Q as in the book, and moving C according to the book's placement of spinor indices gives

$$\boxed{\{Q_\alpha, Q_\beta\} = -(\Gamma^M C^{-1})_{\alpha\beta} P_M - \frac{1}{2}(\Gamma^{MN} C^{-1})_{\alpha\beta} Z_{MN}}.$$

Thus the overall minus signs in the question are convention dependent, but the relative tensor structure, the factor $1/2$, and the topological charge are fixed.

10.6.1.6 The same result from the Wess–Zumino descent cocycle

The canonical computation has a useful conceptual reformulation. Because

$$\delta_\varepsilon B_3 = d\Lambda_2(\varepsilon),$$

two supersymmetry transformations produce the two-form descent cocycle

$$\Omega_2(\varepsilon_1, \varepsilon_2) = \delta_{\varepsilon_1} \Lambda_2(\varepsilon_2) - \delta_{\varepsilon_2} \Lambda_2(\varepsilon_1) - \Lambda_2([\varepsilon_1, \varepsilon_2]).$$

The commutator of two rigid supersymmetries is an ordinary translation, and evaluating the field-independent part of this expression gives

$$\Omega_2(\varepsilon_1, \varepsilon_2) \propto \bar{\varepsilon}_2 \Gamma_{MN} \varepsilon_1 \, dX^M \wedge dX^N.$$

Therefore the algebra of the integrated Noether generators contains

$$T_{\mathrm{M2}} \bar{\varepsilon}_2 \Gamma_{MN} \varepsilon_1 \int_{\Sigma_2} dX^M \wedge dX^N.$$

This is exactly the term found in the canonical bracket. It also explains why the extension can be present even though the supersymmetry transformations close locally on X^M and Θ: the extension is carried by the boundary/improvement data of the action, not by an additional local transformation of those fields.

10.6.1.7 Why Z^{MN} is topological and conserved

The integrand is closed identically:

$$d(dX^M \wedge dX^N) = 0.$$

Suppose Σ_2 and Σ_2' are spatial membrane slices bounding a segment $\mathcal{W}_3$ of the world-volume. Then

$$\begin{aligned} Z^{MN}(\Sigma_2') - Z^{MN}(\Sigma_2) &= T_{\mathrm{M2}} \int_{\partial \mathcal{W}_3} dX^M \wedge dX^N \\ &= T_{\mathrm{M2}} \int_{\mathcal{W}_3} d(dX^M \wedge dX^N) \\ &= 0. \end{aligned}$$

This is conservation without using the membrane equations of motion. Equivalently, the world-volume current

$$J^{iMN} = T_{\mathrm{M2}} \epsilon^{ijk} \partial_j X^M \partial_k X^N$$

obeys

$$\partial_i J^{iMN} = 0$$

identically, because second derivatives are symmetric whereas ϵ^{ijk} is antisymmetric.

There is an important global qualification. If Σ_2 is closed and the functions X^M are globally single-valued in noncompact target-space directions, then

$$dX^M \wedge dX^N = d(X^M dX^N)$$

is exact and its integral vanishes. A nonzero finite charge requires a nontrivial wrapped two-cycle, winding in compact directions, suitable asymptotic conditions for an infinite membrane, or an open membrane with boundary data.

For example, take a rectangular target two-torus and

$$\begin{aligned} X^1 &= R_1(m_1 \sigma^1 + n_1 \sigma^2), \\ X^2 &= R_2(m_2 \sigma^1 + n_2 \sigma^2), \qquad 0 \le \sigma^r < 2\pi. \end{aligned}$$

Then

$$\int_{\Sigma_2} dX^1 \wedge dX^2 = (2\pi)^2 R_1 R_2 (m_1 n_2 - m_2 n_1),$$

and hence

$$Z^{12} = T_{\mathrm{M2}} (2\pi)^2 R_1 R_2 (m_1 n_2 - m_2 n_1).$$

The integer determinant is the oriented wrapping number. Continuous deformations of the embedding cannot change it.

In the conventions of the book,

$$T_{\mathrm{M2}} = \frac{2\pi}{\ell_{11}^3},$$

so the world-volume result is precisely

$$\boxed{Z^{MN} = \frac{2\pi}{\ell_{11}^3} \int_{\Sigma_2} dX^M \wedge dX^N},$$

the $p = 2$ case of equation (18.118).

10.6.1.8 Short check: a static planar M2-brane and its projector

Put a static M2-brane along the x^1, x^2 directions and choose static gauge,

$$X^0 = \tau, \qquad X^1 = \sigma^1, \qquad X^2 = \sigma^2.$$

Per unit spatial area,

$$P_0 = T_{\mathrm{M2}}, \qquad Z_{12} = \pm T_{\mathrm{M2}},$$

where the sign is the membrane orientation. With $C = \Gamma^0$ in a Majorana basis, the algebra becomes, up to the common positive normalization,

$$\{Q, Q\} = T_{\mathrm{M2}} \left(1 \mp \Gamma^{012}\right).$$

Since

$$(\Gamma^{012})^2 = 1, \qquad \operatorname{tr} \Gamma^{012} = 0,$$

the two eigenspaces of Γ^{012} each have dimension 16. The unbroken supersymmetries satisfy

$$\boxed{\Gamma^{012}\varepsilon = \pm\varepsilon}.$$

This is exactly the κ-symmetry projector of a static M2-brane. Positivity of the supercharge anticommutator gives

$$P_0 \geq |Z_{12}|,$$

and the fundamental M2 saturates the bound because its charge density equals its tension. This reproduces equation (18.119) and checks both the interpretation and the normalization of the two-form charge.

10.6.1.9 Extension to the M5-brane

For the M5-brane one repeats the same logical steps with its six-dimensional Green–Schwarz/PST action. The Wess–Zumino sector contains the pullbacks of the six-form potential dual to the eleven-dimensional three-form and terms involving the chiral world-volume three-form field strength. Its supersymmetry variation is again an exact form. The corresponding Noether improvement produces

$$Z^{M_1 \cdots M_5} = T_{\mathrm{M5}} \int_{\Sigma_5} dX^{M_1} \wedge \cdots \wedge dX^{M_5},$$

which enters as

$$-\frac{1}{5!} (\Gamma^{M_1 \cdots M_5} C^{-1})_{\alpha\beta} Z_{M_1 \cdots M_5}.$$

The M5 world-volume chiral two-form also allows dissolved M2 charge, so the complete M5 Noether algebra can contain both the two-form and five-form charges. The simple statement “M2 gives Z_2 and M5 gives Z_5” is correct for pure branes, but the full M5 world-volume algebra is richer.

10.6.1.10 Result

The Green–Schwarz M2 action and its rigid supersymmetry are

$$S_{\mathrm{M2}} = -T_{\mathrm{M2}} \int \sqrt{-\det g} + T_{\mathrm{M2}} \int B_3, \qquad \delta\Theta = \varepsilon, \qquad \delta X^M = i\bar{\varepsilon}\Gamma^M \Theta.$$

The invariant pullback satisfies

$$\Pi^M = dX^M - i\bar{\Theta}\Gamma^M d\Theta, \qquad \delta\Pi^M = 0,$$

while the Wess–Zumino potential is only quasi-invariant:

$$\delta B_3 = d\Lambda_2(\varepsilon).$$

The resulting Noether supercharge contains the improvement

$$Q_\alpha \supset -\frac{iT_{\mathrm{M2}}}{2}\int_{\Sigma_2} d^2\sigma\,\epsilon^{rs}\partial_r X^M \partial_s X^N (C\Gamma_{MN}\Theta)_\alpha.$$

Its canonical Dirac bracket gives

$$\boxed{Z^{MN} = T_{\mathrm{M2}}\int_{\Sigma_2} dX^M \wedge dX^N = \frac{2\pi}{\ell_{11}^3}\int_{\Sigma_2} dX^M \wedge dX^N}.$$

After imposing the world-volume constraints and translating to the book's convention,

$$\boxed{\{Q_\alpha, Q_\beta\} = -(\Gamma^M C^{-1})_{\alpha\beta}P_M - \frac{1}{2}(\Gamma^{MN}C^{-1})_{\alpha\beta}Z_{MN}}.$$

Thus the momentum term comes from the strictly invariant kinetic part of the Green–Schwarz action, whereas the two-form extension comes from the exact-form supersymmetry variation of the Wess–Zumino term.

10.7 Q7: Total seven-brane monodromy, tadpole cancellation, and the number 18

In page 727 it is said that: >At most 18 of the 24 7-branes can be D7-branes, while the remaining ones are more general (p, q) 7-branes. An easy way to see that not all 24 branes can be D7's is that the total monodromy must be trivial, as a cycle enclosing all 24 branes can be contracted to a point. This is nothing but the tadpole condition for the seven-branes with a compact transverse space.

The statements are too brief. Why the requirement of trivial total monodromy is the tadpole cancellation condition? Why this implies there are at most 18 D7-branes?

> **Textbook location and related material.** Chapter 18, Section 18.8 (pp. 724–733): seven-brane monodromies, elliptic K3, F-theory tadpoles, and the moduli/rank count associated with the number 18.

10.7.1 Answer

10.7.1.1 Core point

There are two logically distinct statements in the quoted paragraph:

1. **Global monodromy/tadpole condition.** Because the transverse space is the compact sphere $B = \mathbb{P}^1$, the ordered product of the monodromies around all seven-branes must be the identity:

   $$\boxed{K_{\mathrm{tot}} = K_{24}K_{23}\cdots K_1 = \mathbf{1}}.$$

   This is the exact, non-Abelian $SL(2,\mathbb{Z})$ version of the statement that the integrated seven-brane charge on a compact space must vanish.

2. **The number 18.** The product condition immediately proves that the 24 branes cannot all be D7-branes, because their product would be $T^{24} \neq \mathbf{1}$. It does **not**, by itself, prove the sharper inequality "number of D7-branes $\leq$ 18." The number 18 comes from the dimension of the elliptic-K3 complex-structure moduli space and equivalently from the rank of the eight-dimensional vector-multiplet sector.

Accordingly, the phrase "at most 18 D7-branes" should be read as "at most 18 independent mutually local brane gauge fields/relative-position multiplets can be put in one D7 duality frame." If it is read as a literal bound on the number of D7 constituents counted with multiplicity, it is too strong. I will derive each statement and return to this qualification explicitly.

10.7.1.2 Seven-branes are monodromy defects

The type-IIB axio-dilaton is

$$\tau = C_0 + ie^{-\Phi}.$$

A D7-brane is magnetically charged under the axion C_0. Near a D7-brane at $z = z_i$, the local solution has the logarithmic form

$$\tau(z) = \tau_{\text{reg}}(z) + \frac{1}{2\pi i}\log(z - z_i).$$

On taking $z - z_i$ once counterclockwise around the origin,

$$\log(z - z_i) \longmapsto \log(z - z_i) + 2\pi i,$$

and therefore

$$\tau \longmapsto \tau + 1.$$

This is the $SL(2, \mathbb{Z})$ transformation

$$\boxed{T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}}, \qquad K_{\text{D7}} = T.$$

A general (p, q) seven-brane is obtained by conjugating a D7-brane by a duality transformation. In the convention of equation (18.89), its monodromy is

$$\boxed{K_{[p,q]} = M(p, q) = \begin{pmatrix} 1 + pq & p^2 \\ -q^2 & 1 - pq \end{pmatrix}}.$$

Although every individual $K_{[p,q]}$ is conjugate to T, different such matrices generally do not commute. Consequently, "total seven-brane charge" is not globally an ordinary sum of integers. It is encoded by an **ordered product** of monodromies.

10.7.1.3 Why the total monodromy on $\mathbb{P}^1$ is trivial

Remove the 24 seven-brane positions from the base:

$$B^{\circ} = \mathbb{P}^1 \setminus \{z_1, \dots, z_{24}\}.$$

Choose a common base point z_* and loops γ_i based at z_*, with γ_i going once around z_i. The fundamental group of a sphere with 24 punctures has the presentation

$$\pi_1(B^{\circ}, z_*) = \langle \gamma_1, \dots, \gamma_{24} \mid \gamma_1\gamma_2 \cdots \gamma_{24} = 1 \rangle.$$

Geometrically, a loop enclosing every puncture can be moved to the opposite side of the sphere, where it encloses no puncture, and then contracted.

The one-cycles (a, b) of the elliptic fiber form an integral rank-two local system over B°. Parallel transport defines a monodromy representation

$$\rho : \pi_1(B^{\circ}, z_*) \longrightarrow SL(2, \mathbb{Z}), \qquad \rho(\gamma_i) = K_i.$$

Applying ρ to the fundamental-group relation gives

$$\rho(\gamma_1 \cdots \gamma_{24}) = \rho(1) = \mathbf{1}.$$

Depending on whether monodromies act on row or column vectors and on the convention for composing paths, this is written as either

$$K_1 K_2 \cdots K_{24} = \mathbf{1}$$

or

$$K_{24} \cdots K_2 K_1 = \mathbf{1}.$$

The invariant content is

$$\boxed{K_{\text{tot}} = \mathbf{1}}.$$

This condition says that after transporting an arbitrary fiber cycle around all singular fibers, it must return to itself. Otherwise the elliptic fibration, and equivalently the type-IIB fields patched by $SL(2, \mathbb{Z})$, would not be globally well defined.

10.7.1.4 Why this is the seven-brane tadpole condition

First consider a perturbative region in which all sources are mutually local and a single axion field C_0 can be used globally away from them. Locally,

$$F_1 = dC_0.$$

A D7-brane is a magnetic source, so the Bianchi identity takes the schematic form

$$dF_1 = 2\pi \sum_i Q_i \, \delta^{(2)}(z - z_i),$$

where $Q_i = +1$ for a D7-brane and negative sources such as orientifold planes must be included when present.

Integrating over a compact transverse surface B gives

$$\int_B dF_1 = 2\pi \sum_i Q_i.$$

Because B has no boundary,

$$\int_B dF_1 = \int_{\partial B} F_1 = 0,$$

and hence

$$\boxed{\sum_i Q_i = 0}.$$

This is precisely tadpole cancellation: the zero-momentum source for the field dual to C_0 must vanish. Equivalently, the sourced field equation has a solution on a compact space only if its integrated source is zero. It is the same elementary obstruction as the impossibility of solving

$$\nabla^2 \phi = \sum_i Q_i \delta^{(2)}(z - z_i)$$

on a compact manifold unless $\sum_i Q_i = 0$, because integrating the left-hand side gives zero.

For mutually local D7-type sources, a source of charge Q_i has monodromy

$$K_i = T^{Q_i}.$$

Since these matrices commute,

$$K_{\text{tot}} = \prod_i T^{Q_i} = T^{\sum_i Q_i}.$$

Thus

$$\sum_i Q_i = 0 \qquad \Longleftrightarrow \qquad K_{\text{tot}} = \mathbf{1}.$$

This explicitly proves the equivalence between the integrated Bianchi identity and trivial total monodromy in the perturbative, Abelian case.

In a genuine F-theory background, different (p, q) branes are mutually nonlocal. There is no global duality frame in which all their charges are integers sourcing the same C_0, and the matrices K_i do not commute. The additive equation $\sum_i Q_i = 0$ must then be replaced by its exact group-valued form,

$$\boxed{K_{24} \cdots K_1 = \mathbf{1}}.$$

This is why the book calls trivial total monodromy "the tadpole condition": it is the nonperturbative $SL(2, \mathbb{Z})$ generalization of the integrated seven-brane Bianchi identity.

One can also see the global condition geometrically. In an elliptic Calabi–Yau fibration,

$$f \in H^0(B, K_B^{-4}), \qquad g \in H^0(B, K_B^{-6}),$$

so

$$\Delta = 4f^3 + 27g^2 \in H^0(B, K_B^{-12}).$$

For $B = \mathbb{P}^1$,

$$K_B^{-12} \cong \mathcal{O}(24),$$

and therefore

$$\deg \Delta = 24.$$

This explains why the compact solution has 24 elementary discriminant zeros. However, "24 discriminant zeros" does not mean "net D7 charge +24": the zeros can carry different, noncommuting (p, q) monodromies whose ordered product is trivial.

10.7.1.5 Why 24 ordinary D7-branes are impossible

If all 24 branes were D7-branes in one fixed duality frame, every monodromy would equal T. Their total monodromy would be

$$K_{\text{tot}} = T^{24} = \begin{pmatrix} 1 & 24 \\ 0 & 1 \end{pmatrix},$$

which is not the identity. Therefore

$$\boxed{\text{the 24 elementary seven-branes cannot all be D7-branes in one global duality frame}}.$$

Some of the branes must have noncommuting monodromies conjugate to T so that the full ordered product can become $\mathbf{1}$.

Notice exactly what has been established:

$$K_{\text{tot}} = \mathbf{1} \quad \Longrightarrow \quad N_{\text{D7}} \neq 24.$$

No inequality $N_{\text{D7}} \leq 18$ follows from this one calculation. The integer 18 requires additional global information.

10.7.1.6 Where the number 18 actually comes from

The elliptic K3 is written in Weierstrass form as

$$y^2 = x^3 + f(u)x + g(u),$$

with

$$f \in H^0(\mathbb{P}^1, \mathcal{O}(8)), \qquad g \in H^0(\mathbb{P}^1, \mathcal{O}(12)).$$

A degree-eight homogeneous polynomial has nine coefficients, and a degree-twelve homogeneous polynomial has thirteen. Thus (f, g) initially contain

$$9 + 13 = 22$$

complex parameters.

They are not all physical:

- $PGL(2, \mathbb{C})$ reparametrizations of the base $\mathbb{P}^1$ remove three complex parameters;
- the weighted rescaling of the Weierstrass coordinates, equivalently the common rescaling $f \mapsto w^2 f$, $g \mapsto w^3 g$ used by the book, removes one more.

Therefore the number of complex-structure moduli is

$$\boxed{22 - 3 - 1 = 18}.$$

The 24 roots of Δ are consequently not 24 freely specifiable points. Their positions obey the relations required for them to arise from the same pair (f, g). Only 18 complex combinations are independent.

In eight dimensions a vector multiplet with sixteen supercharges contains one complex scalar. In a local D7 description this scalar is the transverse position of the brane. Hence the 18 independent complex deformations are paired with 18 independent vector multiplets:

$$\boxed{n_V = 18, \qquad G_{\text{generic, brane}} = U(1)^{18}}.$$

The eight-dimensional supergravity multiplet contains two additional graviphotons, so the total generic Abelian rank is

$$18 + 2 = 20.$$

This matches the dual heterotic string on T^2, whose vector-multiplet moduli are organized by the Narain lattice $\Gamma^{2,18}$.

The naive 24 world-volume $U(1)$ fields are therefore not 24 independent eight-dimensional gauge fields. Global $SL(2,\mathbb{Z})$ identifications, couplings to bulk form fields, and Stückelberg relations remove or identify six combinations. This is the physical content behind the book's number 18.

This moduli-and-spectrum argument is the one given in the original construction of F-theory on K3; see Vafa, *Evidence for F-Theory*.

10.7.1.7 The precise meaning of "at most 18 D7-branes"

The safest precise statement is

$$\boxed{\text{the independent seven-brane vector-multiplet rank is at most 18}}.$$

Equivalently, at most 18 independent mutually local brane degrees of freedom can be described simultaneously as ordinary D7 degrees of freedom in one perturbative duality frame. At enhanced points these $U(1)$ fields become the Cartan generators of a non-Abelian algebra, whose rank is still at most 18.

This is not the same as a literal bound on the number of mutually local D7 constituents. For example, an elliptic K3 can have a Kodaira I_{19} fiber. Its monodromy is

$$K_{I_{19}} = T^{19},$$

and in the seven-brane description it is a stack of 19 mutually local D7-type constituents. The associated algebra is

$$A_{18} \cong \mathfrak{su}(19),$$

whose rank is 18. Such elliptic K3 surfaces are known explicitly; see the classification result on maximal singular fibers of elliptic K3 surfaces.

Thus:

- the monodromy condition rules out **24** D7-branes in one frame;
- the moduli/spectrum calculation gives at most **18 independent vector multiplets or gauge rank**;
- a coincident stack may contain 19 D7 constituents while still contributing only rank 18.

The book's wording is therefore a useful shorthand in the discussion of the generic Abelian spectrum, but it should not be interpreted as a universal constituent-counting theorem.

10.7.1.8 Short check: the perturbative orientifold limit

The weak-coupling orientifold limit gives a concrete Abelian version of the same cancellation. F-theory on K3 becomes type IIB on

$$T^2/\mathbb{Z}_2$$

with four $O7^-$-planes and sixteen D7-branes. In D7 charge units,

$$Q_{\mathrm{D7}} = +1, \qquad Q_{O7^-} = -4,$$

so

$$16Q_{\mathrm{D7}} + 4Q_{O7^-} = 16 - 16 = 0.$$

The corresponding monodromies may be written

$$K_{\mathrm{D7}} = T, \qquad K_{O7^-} = -T^{-4}.$$

Since four factors of $-\mathbf{1}$ multiply to $\mathbf{1}$,

$$T^{16}(-T^{-4})^4 = T^{16}T^{-16} = \mathbf{1}.$$

Nonperturbatively, each $O7^-$ splits into a pair of mutually nonlocal seven-branes. Thus the F-theory discriminant count is

$$16 \text{ D7 constituents} + 4 \times 2 \text{ nonlocal constituents} = 24,$$

while the total monodromy and the D7 tadpole both cancel.

This limit also shows why "24 seven-branes" and "24 D7-branes" are different statements. The first counts elementary discriminant zeros; the second assumes that every zero can be placed in the same perturbative duality frame, which global monodromy forbids.

10.7.1.9 Result

On the punctured sphere,

$$\pi_1(\mathbb{P}^1 \setminus \{z_i\}) = \langle \gamma_1, \dots, \gamma_{24} \mid \gamma_1 \cdots \gamma_{24} = 1 \rangle \,.$$

Therefore the monodromy representation obeys

$$\boxed{K_{\text{tot}} = K_{24} \cdots K_1 = \mathbf{1}}.$$

For mutually local sources this reduces to the integrated axion Bianchi identity,

$$dF_1 = 2\pi \sum_i Q_i \delta_i, \qquad 0 = \int_{\mathbb{P}^1} dF_1 = 2\pi \sum_i Q_i,$$

so trivial total monodromy is precisely the non-Abelian form of seven-brane tadpole cancellation.

For 24 D7-branes,

$$K_{\text{tot}} = T^{24} = \begin{pmatrix} 1 & 24 \\ 0 & 1 \end{pmatrix} \neq \mathbf{1},$$

so the 24 branes cannot all be D7-branes in one duality frame.

The number of independent elliptic-K3 deformations is instead

$$\boxed{(9+13) - 3 - 1 = 18},$$

giving 18 vector multiplets and maximal independent brane gauge rank 18. Hence the precise conclusion is

$$\boxed{\begin{array}{c}\text{at most 18 independent D7-type gauge/position multiplets,} \\ \text{not necessarily at most 18 D7 constituents}\end{array}}.$$

10.8 Q8: Four orientifold bulk vectors and heterotic $U(1)^{20}$

In page 731-732, they considered compactifying the type IIB string on a T^2 and taking the orientifold quotient $\Omega I_2(-1)^{F_L}$. This leads to the gauge group $SO(8)^4$ since locally there are four O7-planes and on each O7-plane there are four dynamical D7-branes. While they said that: >In the orientifold description, the remaining four Abelian gauge bosons known to be present in F-theory……

Why there are four remaining Abelian gauge bosons in the orientifold description?

In the next page they also discussed heterotic string on T^2 and they also mentioned >the gauge group for generic Wilson-lines of the heterotic string on T^2 is also $U(1)^{20}$

How to derive this fact?

Make these two points clear. If they were studied somewhere in this book, search and find the discussions.

Textbook location and related material. Chapter 18, Section 18.8: the $SO(8)^4$ orientifold point and heterotic dual on T^2. Sections 10.2–10.4 provide the Narain lattice and toroidal vector counting.

10.8.1 Answer

10.8.1.1 Core point

At the orientifold point, the complete eight-dimensional gauge group is more precisely

$$\boxed{SO(8)^4 \times U(1)^4}$$

up to possible global discrete quotients. The factors have different origins:

- $SO(8)^4$ comes from open strings on the four D7/O7 stacks;
- $U(1)^4$ comes from the closed-string fields

$$B_{\mu a}, \qquad C_{\mu a}, \qquad a = 1, 2.$$

There are two vectors from the NS–NS two-form and two from the R–R two-form. The combined orientifold projection keeps these components even, although neither set would survive the geometric reflection in isolation.

On the heterotic side, the generic gauge group follows from the equally direct count

$$\boxed{16 \text{ Cartan vectors} + 2 \text{ metric vectors} + 2\ B\text{-field vectors} = 20 \text{ vectors}}.$$

Generic Wilson lines make all non-Cartan root gauge bosons massive, leaving

$$\boxed{G_{\text{het,generic}} = U(1)^{16} \times U(1)^2_{\text{KK}} \times U(1)^2_{\text{wind}} = U(1)^{20}}.$$

In world-sheet language this is

$$U(1)^{18}_L \times U(1)^2_R.$$

The book develops the ingredients earlier in Chapter 10:

- Section 10.2, especially equation (10.44), derives the $2D$ vectors $U(1)^D_L \times U(1)^D_R$ of a T^D compactification.
- Section 10.4, equations (10.115)–(10.127), introduces heterotic Wilson lines and proves that the generic rank is $16 + 2D$.
- Section 10.6 and Table 10.3 identify the supersymmetric orientifold $\Omega I_2(-1)^{F_L}$ and its $SO(8)^4$ Chan–Paton group.

I will now derive both counts explicitly.

10.8.1.2 Rank supplied by the four D7/O7 stacks

The involution

$$I_2 : (y^1, y^2) \longmapsto (-y^1, -y^2)$$

has four fixed points on T^2. After quotienting, these are the four $O7^-$-planes on

$$T^2/\mathbb{Z}_2 \simeq \mathbb{P}^1.$$

Local tadpole cancellation places four dynamical D7-branes on each orientifold plane. The orientifold projection of the Chan–Paton factors gives

$$SO(8)$$

at each fixed point, and therefore

$$G_{\text{open}} = SO(8)^4.$$

Since

$$\operatorname{rank} SO(8) = 4,$$

the open-string sector supplies

$$\operatorname{rank} G_{\text{open}} = 4 \times 4 = 16.$$

This is not yet the entire rank-20 gauge sector of F-theory on K3. The missing four vectors are closed-string zero modes.

10.8.1.3 Reduction of the type-IIB two-forms

Split the ten-dimensional coordinates as

$$x^M = (x^\mu, y^a), \qquad \mu = 0, \dots, 7, \qquad a = 1, 2.$$

Before imposing the orientifold projection, decompose the NS–NS two-form:

$$B_2 = \frac{1}{2} B_{\mu\nu}\, dx^\mu \wedge dx^\nu + B_{\mu a}\, dx^\mu \wedge dy^a + \frac{1}{2} B_{ab}\, dy^a \wedge dy^b.$$

For each internal index a, the component

$$B_{\mu a}(x)$$

has one noncompact index and is therefore an eight-dimensional vector potential. Since $a = 1, 2$, B_2 would produce two vectors.

The R–R two-form decomposes in the same way:

$$C_2 = \frac{1}{2} C_{\mu\nu}\, dx^\mu \wedge dx^\nu + C_{\mu a}\, dx^\mu \wedge dy^a + \frac{1}{2} C_{ab}\, dy^a \wedge dy^b,$$

and its two components

$$C_{\mu 1}, \qquad C_{\mu 2}$$

give two more eight-dimensional vector potentials.

Thus dimensional reduction gives the candidate set

$$\boxed{B_{\mu 1},\ B_{\mu 2},\ C_{\mu 1},\ C_{\mu 2}}.$$

It remains to verify that the orientifold projection retains them.

10.8.1.4 Explicit orientifold parity calculation

The orientifold generator is

$$\mathcal{O} = \Omega I_2 (-1)^{F_L}.$$

The three factors act as follows.

First, world-sheet orientation reversal gives the standard type-IIB parities

$$\Omega: \qquad B_2 \longmapsto -B_2, \qquad C_2 \longmapsto +C_2.$$

The first sign follows because the world-sheet coupling

$$\int_{\Sigma_2} B_2$$

changes sign when the world-sheet orientation is reversed. The second sign is also familiar from the type-I quotient: the R–R two-form C_2 is retained by Ω.

Second,

$$(-1)^{F_L}$$

acts trivially on NS–NS fields and by a minus sign on R–R fields:

$$(-1)^{F_L}: \qquad B_2 \longmapsto +B_2, \qquad C_2 \longmapsto -C_2.$$

Third, under the reflection

$$I_2 : y^a \longmapsto -y^a, \qquad dy^a \longmapsto -dy^a.$$

A tensor component with exactly one internal index therefore acquires a minus sign:

$$I_2 : \qquad B_{\mu a} \longmapsto -B_{\mu a}, \qquad C_{\mu a} \longmapsto -C_{\mu a}.$$

Multiplying the three signs gives

component	Ω	$(-1)^{F_L}$	I_2	$\mathcal{O}$
$G_{\mu a}$	$+$	$+$	$-$	$-$
$B_{\mu a}$	$-$	$+$	$-$	$+$
$C_{\mu a}$	$+$	$-$	$-$	$+$

Therefore

$$\boxed{\mathcal{O}B_{\mu a} = +B_{\mu a}, \qquad \mathcal{O}C_{\mu a} = +C_{\mu a}},$$

while the metric Kaluza–Klein vectors $G_{\mu a}$ are projected out.

Hence exactly four bulk Abelian vectors survive:

$$\boxed{2\ B_{\mu a} + 2\ C_{\mu a} = 4\ U(1) \text{ gauge bosons}}.$$

There is a small geometric subtlety here. The quotient is $\mathbb{P}^1$, and

$$b_1(\mathbb{P}^1) = 0,$$

so these vectors should not be interpreted as ordinary reduction on harmonic one-forms of $\mathbb{P}^1$. They are most transparently obtained on the covering torus. The one-form dy^a is odd under I_2, but this geometric minus sign is canceled by the intrinsic Ω or $\Omega(-1)^{F_L}$ parity of the corresponding two-form. On the quotient, the fields are therefore modes valued in the orientifold sign local system rather than ordinary one-form zero modes.

Combining open and closed sectors gives

$$\boxed{G_{\text{8d}} = SO(8)^4 \times U(1)^4, \qquad \operatorname{rank} G_{\text{8d}} = 16 + 4 = 20}.$$

Of these four Abelian fields, two belong to the eight-dimensional supergravity multiplet and are the graviphotons; the other two belong to vector multiplets. Which linear combinations of $B_{\mu a}$ and $C_{\mu a}$ are assigned to these multiplets depends on the duality frame and the scalar background, but the $2+2$ multiplet split and the total number four are invariant.

10.8.1.5 Ten-dimensional heterotic gauge fields and generic Wilson lines

The ten-dimensional heterotic gauge group is either

$$E_8 \times E_8$$

or

$$\text{Spin}(32)/\mathbb{Z}_2.$$

Both have rank 16. Let H_I, $I = 1, \dots, 16$, denote Cartan generators. Compactification on T^2 allows two commuting Wilson lines

$$W_a = P\exp\left(i \oint_{\gamma_a} A\right), \qquad a = 1, 2.$$

Because

$$\pi_1(T^2) = \mathbb{Z} \oplus \mathbb{Z}$$

is Abelian, W_1 and W_2 must commute. They can therefore be conjugated into the maximal torus:

$$W_a = \exp\left(2\pi i\, A_a^I H_I\right).$$

Decompose the ten-dimensional adjoint algebra into Cartan and root generators,

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Delta_{\mathfrak{g}}} \mathfrak{g}_\alpha.$$

For a root generator E_α,

$$W_a E_\alpha W_a^{-1} = e^{2\pi i \alpha\cdot A_a} E_\alpha.$$

The corresponding gauge boson is unbroken only if it is invariant around both cycles:

$$\alpha \cdot A_a \in \mathbb{Z}, \qquad a = 1,2.$$

For generic values of the Wilson lines, no nonzero root satisfies both conditions. All root gauge bosons become massive, while the Cartan generators always commute with W_a. Therefore

$$\boxed{E_8 \times E_8 \ \text{or}\ \ \mathrm{Spin}(32)/\mathbb{Z}_2 \xrightarrow{\text{generic } W_1, W_2} U(1)^{16}}.$$

In Kaluza–Klein language the mass shift can be seen schematically as

$$p_a = \frac{1}{R_a}\left(m_a + \alpha\cdot A_a\right), \qquad m_a \in \mathbb{Z}.$$

A ten-dimensional root gauge boson can remain massless only if integers m_a can cancel both Wilson-line phases. At a generic point this is impossible for every $\alpha \neq 0$; at special Wilson lines some roots satisfy the condition and a non-Abelian group reappears.

10.8.1.6 Four additional heterotic vectors from the torus

The ten-dimensional heterotic NS–NS sector contains the metric G_{MN} and the two-form B_{MN}. Reducing on T^2 gives

$$G_{\mu a}, \qquad B_{\mu a}, \qquad a = 1,2.$$

The two $G_{\mu a}$ are ordinary Kaluza–Klein gauge fields. An internal diffeomorphism with an x-dependent translation parameter,

$$y^a \longmapsto y^a + \xi^a(x),$$

acts on them as

$$\delta G_{\mu a} \sim \partial_\mu \xi_a,$$

which is an Abelian gauge transformation. They therefore generate

$$U(1)^2_{\mathrm{KK}}.$$

States carrying integer momentum m_a around T^2 are electrically charged under these vectors.

The two $B_{\mu a}$ arise from the gauge symmetry

$$\delta B_2 = d\Lambda_1.$$

Taking an internal component $\Lambda_a(x)$ gives

$$\delta B_{\mu a} = \partial_\mu \Lambda_a,$$

so they generate

$$U(1)^2_{\mathrm{wind}}.$$

String states with winding numbers n^a are charged under these vectors.

Consequently the complete generic heterotic group is

$$\begin{aligned} G_{\mathrm{het,generic}} &= U(1)^{16}_{\mathrm{Cartan}} \times U(1)^2_{\mathrm{KK}} \times U(1)^2_{\mathrm{wind}} \\ &= \boxed{U(1)^{20}}. \end{aligned}$$

This field-theory reduction already proves the statement in the book.

10.8.1.7 World-sheet and Narain-lattice derivation

The string derivation organizes the same fields more naturally into left- and right-moving combinations. For a T^D compactification, the metric and two-form vectors combine schematically as

$$A^a_{\mu,L} \sim G_{\mu a} + B_{\mu a}, \qquad A^a_{\mu,R} \sim G_{\mu a} - B_{\mu a}.$$

Thus a closed string compactified on T^D has

$$U(1)^D_L \times U(1)^D_R.$$

This is the result derived from the massless vector states in equation (10.44).

The heterotic left-moving sector has, in addition, sixteen chiral internal bosons whose currents generate the rank-16 gauge lattice. Therefore

$$\boxed{G_{\text{het,generic}}(T^D) = U(1)^{16+D}_L \times U(1)^D_R}.$$

For $D = 2$,

$$\boxed{G_{\text{het,generic}}(T^2) = U(1)^{18}_L \times U(1)^2_R = U(1)^{20}}.$$

The charges lie in the even self-dual Narain lattice

$$\Gamma^{16+D,D}.$$

For $D = 2$ this is

$$\Gamma^{18,2}.$$

At a generic point in the Narain moduli space, the only massless vectors are the 20 Cartan currents just counted. Extra left-moving gauge bosons appear when the lattice contains vectors satisfying

$$p_R = 0, \qquad p_L^2 = 2.$$

Such vectors form a simply laced root system. Hence at special values of the metric, B-field, and Wilson lines, part of

$$U(1)^{18}_L$$

is enhanced to an ADE group. The two right-moving $U(1)^2_R$ are in the supergravity multiplet and do not undergo this non-Abelian enhancement.

The Narain moduli space quoted in equation (10.127) becomes, for $D = 2$,

$$\mathcal{M}_{\text{Narain}} = \frac{O(2,18)}{O(2) \times O(18)} \Big/ O(2,18;\mathbb{Z}).$$

This is the same $O(2,18)$ structure that appears in the eight-dimensional F-theory/heterotic duality.

10.8.1.8 Matching the $SO(8)^4$ orientifold point to the heterotic theory

At the special orientifold point, the rank-20 gauge algebra is

$$\boxed{\mathfrak{so}(8)^{\oplus 4} \oplus \mathfrak{u}(1)^{\oplus 4}}.$$

The four $\mathfrak{so}(8)$ factors have total Cartan rank

$$4 \times 4 = 16.$$

On the heterotic side, this means that special Wilson lines have made the roots of

$$D_4^{\oplus 4}$$

massless inside the left-moving Narain lattice. Sixteen left-moving Cartan generators are thereby enhanced from

$$U(1)^{16} \quad \text{to} \quad SO(8)^4,$$

without changing the rank.

Four Abelian fields remain:

- two unenhanced left-moving $U(1)$ fields in vector multiplets;
- two right-moving $U(1)$ graviphotons in the supergravity multiplet.

Thus

$$\operatorname{rank}\left[SO(8)^4 \times U(1)^4\right] = 16 + 4 = 20,$$

exactly matching both the orientifold reduction and the heterotic Narain count.

The duality chain quoted by the book,

$$\frac{\text{type IIB on } T^2}{\Omega I_2(-1)^{F_L}} \overset{T}{\longleftrightarrow} \text{type I on } T^2 \overset{S}{\longleftrightarrow} \text{Spin}(32)/\mathbb{Z}_2 \text{ heterotic on } T^2 \overset{T}{\longleftrightarrow} E_8 \times E_8 \text{ heterotic on } T^2,$$

requires these rank-20 spectra to agree. The four orientifold bulk vectors $B_{\mu a}, C_{\mu a}$ are mapped by this sequence of T- and S-dualities to the four toroidal heterotic vectors constructed from $G_{\mu a}$ and $B_{\mu a}$.

10.8.1.9 Checks

For a circle, $D = 1$, the same heterotic formula gives

$$U(1)_L^{17} \times U(1)_R,$$

of total rank 18. This agrees with the general result $16 + 2D$ in equation (10.127).

For T^4, $D = 4$, it gives

$$U(1)_L^{20} \times U(1)_R^4,$$

of total rank 24. This is exactly the group displayed in equations (18.97) and (18.101) in the book's discussion of heterotic/type-IIA duality in six dimensions.

Finally, at the $SO(8)^4$ point,

$$\operatorname{rank} SO(8)^4 = 16$$

and the four bulk vectors restore the universal total rank

$$16 + 4 = 20.$$

10.8.1.10 Result

The orientifold projection retains

$$\begin{aligned} B_{\mu a}: &\quad (-1)_\Omega (+1)_{(-1)^{F_L}} (-1)_{I_2} = +1, \\ C_{\mu a}: &\quad (+1)_\Omega (-1)_{(-1)^{F_L}} (-1)_{I_2} = +1, \end{aligned}$$

for $a = 1, 2$. Hence

$$\boxed{G_{\text{orientifold}} = SO(8)^4 \times U(1)^4, \qquad U(1)^4 \leftrightarrow \{B_{\mu 1}, B_{\mu 2}, C_{\mu 1}, C_{\mu 2}\}}.$$

For the heterotic string,

$$\boxed{G_{10} \xrightarrow{\text{generic Wilson lines}} U(1)^{16}}$$

and the torus supplies

$$\boxed{G_{\mu a} : U(1)^2_{\text{KK}}, \qquad B_{\mu a} : U(1)^2_{\text{wind}}}.$$

Therefore

$$\boxed{G_{\text{het,generic}}(T^2) = U(1)^{16+2+2} = U(1)^{20} = U(1)_L^{18} \times U(1)_R^2}.$$

At special Narain moduli, lattice vectors with

$$p_R = 0, \qquad p_L^2 = 2$$

become additional massless gauge bosons and enhance a subgroup of $U(1)_L^{18}$ to an ADE group, while the total rank remains 20.

10.9 Q9: Localized matter at intersecting D-branes and F-theory seven-branes

In page 732, they briefly mentioned lower dimensional compactifications of F-theory: >The singularity enhances further where these co-dimension one submanifolds, i.e. the 7-branes, intersect. Similar to intersecting D-branes, this is where additional matter fields are localized.

Why intersecting D-branes lead to additional localized matter fields? Is this discussed somewhere in this book?

> **Textbook location and related material.** Chapter 18, Section 18.8 for F-theory localization, read with Chapter 17, Section 17.2 for matter at intersecting D-branes and its multiplicity/chirality.

10.9.1 Answer

10.9.1.1 Core point

The reason is very concrete in perturbative string theory. Suppose two D-brane stacks a and b intersect. Besides the aa and bb open strings whose two endpoints lie on the same stack, string theory necessarily contains an ab sector whose two endpoints obey different D-brane boundary conditions:

$$X(\tau,0) \in \Pi_a, \qquad X(\tau,\pi) \in \Pi_b.$$

The center of mass of such a string can move freely only in directions common to both branes. In the relative internal directions, the boundary conditions force its zero mode to sit at

$$\Pi_a \cap \Pi_b.$$

Consequently, the ab state is a field on the intersection world-volume, rather than on either full brane world-volume.

At the intersection, the shortest string connecting the two branes has zero length. Its stretching contribution to the mass therefore vanishes. The Ramond sector has a massless fermionic ground state; when the intersection preserves supersymmetry, an appropriate Neveu–Schwarz state is also massless and combines with it into a matter multiplet. If the stacks contain N_a and N_b branes, the two endpoints give Chan–Paton indices and the localized state transforms as

$$\boxed{(\mathbf{N}_a, \overline{\mathbf{N}}_b) \quad \text{under} \quad U(N_a) \times U(N_b)}.$$

Thus these are matter fields charged under two gauge factors, not additional gauge bosons in either adjoint representation.

This is derived explicitly earlier in the book. The main discussion is Section 10.6, especially equations (10.188)–(10.197), Figures 10.3–10.4, and Table 10.2. The book first proves that the center-of-mass position is fixed at an intersection, then derives the fractionally moded open string, its massless Ramond state, and its bifundamental Chan–Paton representation. Section 15.3 gives an explicit orientifold example in Table 15.10, and Section 17.2 and Table 17.2 generalize the multiplicity of charged chiral matter to topological intersection numbers of D6-brane cycles.

10.9.1.2 Same-brane strings give gauge fields; mixed strings give matter

Consider stacks a and b with world-volume gauge groups

$$U(N_a) \times U(N_b).$$

There are four oriented open-string sectors:

$$aa, \qquad bb, \qquad ab, \qquad ba.$$

An aa string carries a Chan–Paton matrix $(\lambda_{aa})^i{}_j$ with both indices belonging to stack a. It transforms as

$$\lambda_{aa} \longmapsto U_a \lambda_{aa} U_a^{-1},$$

so its massless vector state is in the adjoint of $U(N_a)$. Because both endpoints may move together anywhere along stack a, this gauge field propagates on the entire world-volume of that stack. The same statement applies to the bb sector.

An ab string instead carries one index $i = 1, \dots, N_a$ at its first endpoint and one index $\bar{j} = 1, \dots, N_b$ at its second endpoint:

$$(\lambda_{ab})^i{}_{\bar{j}}.$$

It transforms as

$$\lambda_{ab} \longmapsto U_a \lambda_{ab} U_b^{-1},$$

and therefore lies in

$$(\mathbf{N}_a, \overline{\mathbf{N}}_b).$$

Reversing the orientation gives the ba sector in

$$(\overline{\mathbf{N}}_a, \mathbf{N}_b).$$

These two orientations are related by CPT in the lower-dimensional theory. They do not in general constitute two independent chiral multiplets.

The representation is already the first indication that this state is matter. A gauge boson of $U(N_a)$ would have to lie in $(\mathrm{adj}_a, \mathbf{1})$, whereas the ab state is charged under both groups.

10.9.1.3 Boundary conditions pin the string to the intersection

Now reproduce the local calculation in equations (10.188)–(10.191). Take a two-dimensional plane with coordinates (X, Y). Put brane a along the X-axis and let brane b meet it at a relative angle $\Delta\phi$.

At the endpoint $\sigma = 0$ on brane a, X is Neumann and Y is Dirichlet:

$$\sigma = 0: \qquad \partial_\sigma X = 0, \qquad \partial_\tau Y = 0.$$

At the endpoint $\sigma = \pi$ on brane b, the tangent coordinate is Neumann and the normal coordinate is Dirichlet:

$$\begin{aligned} \sigma = \pi: \qquad \cos(\Delta\phi)\, \partial_\sigma X + \sin(\Delta\phi)\, \partial_\sigma Y = 0, \\ -\sin(\Delta\phi)\, \partial_\tau X + \cos(\Delta\phi)\, \partial_\tau Y = 0. \end{aligned}$$

To isolate the center-of-mass zero mode, take the part constant in both σ and τ:

$$X(\tau, \sigma) = x_0, \qquad Y(\tau, \sigma) = y_0.$$

The Dirichlet condition at the first endpoint says that the point lies on brane a:

$$y_0 = 0.$$

The Dirichlet condition at the second endpoint says that it lies on brane b:

$$-\sin(\Delta\phi) x_0 + \cos(\Delta\phi) y_0 = 0.$$

For a transverse intersection, $\sin(\Delta\phi) \neq 0$, and hence

$$y_0 = 0, \qquad x_0 = 0.$$

The only allowed constant position is therefore the intersection point. If the intersection is located at a point P rather than the origin, the same calculation gives

$$(x_0, y_0) = P.$$

This is the precise meaning of localization. It does not mean that the state is localized in the noncompact spacetime. If the two D-branes share d noncompact directions, their common Neumann boundary conditions still give momentum zero modes k_μ in those directions. The state propagates as a d-dimensional field, but its internal wavefunction is supported at the brane intersection.

10.9.1.4 Fractional moding and the missing translational zero mode

Introduce the complex coordinate

$$Z = \frac{1}{\sqrt{2}}(X + iY)$$

and the fractional angle

$$\nu = \frac{\Delta\phi}{\pi}.$$

Solving the two different endpoint conditions gives the shifted oscillator modes displayed in equation (10.191):

$$Z(\tau, \sigma) = i\sqrt{\frac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \left[\frac{\alpha_{n+\nu}}{n+\nu} e^{-i(n+\nu)(\tau-\sigma)} + \frac{\tilde{\alpha}_{n-\nu}}{n-\nu} e^{-i(n-\nu)(\tau+\sigma)} \right].$$

The important structural fact is that the modes are shifted by $\pm\nu$. For $\nu \neq 0$, there is no ordinary zero-frequency oscillator and no term

$$z_0 + 2\alpha' p\tau$$

describing free translation in this plane. This is the oscillator-language version of the elementary zero-mode calculation above.

When $\nu \to 0$, the branes become parallel. The ordinary translational zero mode reappears, and the string is no longer pinned to an isolated intersection. Thus localization is not an extra assumption: it is encoded directly in the open-string boundary conditions.

The world-sheet fermions acquire the corresponding shifted modes

$$r + \nu, \qquad r - \nu,$$

where

$$r \in \mathbb{Z} \quad \text{in the R sector}, \qquad r \in \mathbb{Z} + \frac{1}{2} \quad \text{in the NS sector}.$$

This is why the book describes the *ab* string as a “twisted open string’ ’: the relative brane angle plays the same role as a twist acting on a closed-string coordinate.

10.9.1.5 Why some localized states become massless

For two nonintersecting branes separated by a shortest distance L, the classical stretching energy is

$$E_{\text{stretch}} = T_{\text{F}} L = \frac{L}{2\pi\alpha'}.$$

It contributes

$$M^2_{\text{stretch}} = \left(\frac{L}{2\pi\alpha'}\right)^2$$

to the open-string mass. The schematic mass formula is therefore

$$M^2 = \left(\frac{L}{2\pi\alpha'}\right)^2 + \frac{1}{\alpha'}(N + a).$$

At an intersection,

$$L = 0,$$

so the geometrical mass obstruction disappears.

For supersymmetric open strings at angles, the bosonic and fermionic zero-point energies cancel in the Ramond sector:

$$a_{\text{R}} = 0.$$

The GSO-projected Ramond ground state is therefore massless. For D6-branes wrapping three-cycles and sharing four-dimensional Minkowski space, it gives a four-dimensional Weyl fermion. Reversing the orientation supplies its CPT conjugate, and the orientation of the internal intersection fixes the four-dimensional chirality.

The NS sector is angle dependent. For relative angles $0 \leq \nu_I < 1$ in three complex internal planes, its vacuum energy is

$$a_{\mathrm{NS}} = -\frac{1}{2} + \frac{1}{2}\sum_{I=1}^{3} \nu_I .$$

The four light scalar masses obtained from the lowest NS states can be organized as

$$\begin{aligned}
\alpha' M_1^2 &= \frac{1}{2}(-\nu_1 + \nu_2 + \nu_3), \\
\alpha' M_2^2 &= \frac{1}{2}(\nu_1 - \nu_2 + \nu_3), \\
\alpha' M_3^2 &= \frac{1}{2}(\nu_1 + \nu_2 - \nu_3), \\
\alpha' M_4^2 &= 1 - \frac{1}{2}(\nu_1 + \nu_2 + \nu_3).
\end{aligned}$$

Different choices of oriented representatives for the angles permute these expressions. The supersymmetry condition in the book,

$$\sum_{I=1}^{3} \Delta\phi^I = 0 \pmod{2\pi},$$

is precisely the condition that one of these scalars be massless. It then combines with the Ramond fermion into a four-dimensional $\mathcal{N} = 1$ chiral multiplet. With more special angle relations, Table 10.2 shows how the fields instead assemble into $\mathcal{N} = 2$ hypermultiplets or $\mathcal{N} = 4$ vector multiplets.

Thus an intersection always creates a distinct localized open-string sector. It does not, by itself, guarantee a healthy supersymmetric matter multiplet: for nonsupersymmetric angles some NS states can be massive or tachyonic. The statement on page 732 concerns supersymmetric F-theory compactifications, where the relevant localized states organize into supersymmetric matter multiplets.

10.9.1.6 Intersection number gives the multiplicity and chirality

On a compact space, two wrapped branes may intersect several times. For one T^2, let their one-cycles have wrapping numbers

$$\Pi_a = n_a[a] + m_a[b], \qquad \Pi_b = n_b[a] + m_b[b].$$

Their oriented intersection number is equation (10.193):

$$\boxed{I_{ab} = \Pi_a \circ \Pi_b = n_a m_b - m_a n_b}.$$

For transverse intersections,

$$|I_{ab}|$$

counts the distinct intersection points, so there is one copy of the localized open-string ground state at each point. The sign of I_{ab} records the local orientation and therefore the chirality convention.

For factorized D6-branes on

$$T^6 = T_1^2 \times T_2^2 \times T_3^2,$$

equation (10.195) gives

$$\boxed{I_{ab} = \prod_{I=1}^{3} \left(n_a^I m_b^I - m_a^I n_b^I\right)}.$$

Hence the net chiral spectrum contains

$$I_{ab}$$

copies, with sign interpreted as chirality, in the bifundamental representation

$$(\mathbf{N}_a, \overline{\mathbf{N}}_b).$$

Section 17.2 replaces this toroidal formula by the general homological pairing

$$I_{ab} = \Pi_a \circ \Pi_b$$

for special Lagrangian three-cycles in a Calabi–Yau orientifold. This is why charged chiral multiplicities are topological and remain invariant under continuous deformations that do not cross a wall where intersections are created or annihilated in oppositely oriented pairs.

10.9.1.7 Translation into F-theory geometry

Now return to the statement on page 732. In an F-theory compactification, a seven-brane wraps a component

$$D_a \subset B$$

of the discriminant locus

$$\Delta = 4f^3 + 27g^2 = 0.$$

Since D_a has complex codimension one in the base, gauge fields associated with the singular elliptic fiber over D_a propagate on the external spacetime together with the internal divisor D_a.

At a generic point of D_a, a collection of fibral curves collapses according to the root lattice of a gauge algebra $\mathfrak{g}_a$. If another discriminant component D_b meets it, then over

$$\Sigma_{ab} = D_a \cap D_b$$

the fiber degenerates further. Additional fibral curves become vanishing cycles only over this complex-codimension-two locus. This is the geometric meaning of "the singularity enhances further.' '

Locally, the enhanced algebra often organizes the matter representation through a decomposition

$$\mathrm{adj}(G_{\mathrm{enh}}) \longrightarrow (\mathrm{adj}\, G_a, \mathbf{1}) \oplus (\mathbf{1}, \mathrm{adj}\, G_b) \oplus \bigoplus_{\mathbf{R}} (\mathbf{R}_a, \mathbf{R}_b)\,.$$

The first two terms are the gauge fields already present along D_a and D_b. The off-diagonal terms are charged under both groups and are interpreted as matter localized on Σ_{ab}.

For example, a local enhancement

$$SU(N_a) \times SU(N_b) \subset SU(N_a + N_b)$$

gives schematically

$$\begin{aligned} \mathrm{adj}\, SU(N_a + N_b) \longrightarrow\ & (\mathrm{adj}_a, \mathbf{1}) \oplus (\mathbf{1}, \mathrm{adj}_b) \oplus (\mathbf{1}, \mathbf{1}) \\ & \oplus (\mathbf{N}_a, \overline{\mathbf{N}}_b) \oplus (\overline{\mathbf{N}}_a, \mathbf{N}_b). \end{aligned}$$

The last two terms are precisely the representations of the perturbative ab and ba open strings.

10.9.1.8 M-theory explanation of F-theory localization

The nonperturbative F-theory statement is especially precise in the dual M-theory description. Resolve the singular elliptic fiber. Gauge bosons arise from M2-branes wrapping fibral curves corresponding to roots. If $C_{\mathbf{w}}$ is an additional fibral curve associated with a matter weight $\mathbf{w}$, then the wrapped M2-brane has mass

$$M_{\mathbf{w}} = T_{\mathrm{M2}}\, \mathrm{Vol}(C_{\mathbf{w}}).$$

At a generic point of D_a, this particular curve is absent as an independent vanishing curve or has nonzero volume, so the corresponding state is massive. Over the intersection locus Σ_{ab}, the fiber splits or enhances and

$$\mathrm{Vol}(C_{\mathbf{w}}) \longrightarrow 0.$$

The wrapped M2-brane therefore becomes massless only there. Since the vanishing cycle exists only above Σ_{ab}, its wavefunction can propagate along the external spacetime and along Σ_{ab}, but not over the whole of D_a or D_b. This is the F-theory version of the missing open-string translational zero mode.

In a weakly coupled type-IIB limit, these wrapped-M2 states reduce to ordinary strings stretched between D7-brane stacks. For genuinely mutually nonlocal (p, q) seven-branes, there need not be a globally valid perturbative open-string description; the states are more properly described by (p, q) string junctions or by the vanishing fibral curves. The localization principle remains the same.

10.9.1.9 What “localized matter’ ’ means in different F-theory dimensions

For F-theory on an elliptically fibered Calabi–Yau threefold with base surface B_2, the seven-branes wrap curves in B_2. Two such curves meet at points. The matter field is localized at those points internally and propagates in the six noncompact dimensions.

For F-theory on an elliptically fibered Calabi–Yau fourfold with base threefold B_3, the seven-branes wrap surfaces. Two surfaces intersect along a complex curve:

$$\Sigma_{ab} = D_a \cap D_b.$$

This is the familiar matter curve of four-dimensional F-theory compactifications. The geometry determines which charged representations can occur. A nontrivial G_4 flux determines the net number of four-dimensional chiral minus antichiral zero modes. This is why the paragraph immediately following the quoted sentence adds that chiral matter in the fourfold case requires a nontrivial F_4 or G_4 background.

It is therefore useful to distinguish three statements:

1. The codimension-two fiber enhancement produces matter representations localized on Σ_{ab}.
2. The internal Dirac equation on Σ_{ab} determines the actual lower-dimensional zero modes.
3. Gauge flux, represented by G_4 in F-theory, determines their net chirality in four dimensions.

10.9.1.10 Checks

If the two branes are separated and do not intersect, then $L > 0$ and the ab string has

$$M^2 \supset \left(\frac{L}{2\pi\alpha'}\right)^2 > 0.$$

There is no massless localized matter.

If the two branes are parallel and coincident, then $\nu = 0$ and the translational zero mode returns. The mixed strings can propagate along the full common brane world-volume and usually enlarge the adjoint gauge algebra; this is gauge enhancement, rather than matter confined to an isolated internal intersection.

If the branes intersect transversely at $|I_{ab}|$ points, the local boundary-condition problem is identical at each point. Hence the same bifundamental state occurs $|I_{ab}|$ times, exactly as equations (10.193) and (10.195) require.

10.9.1.11 Result

For an ab open string,

$$X(\tau, 0) \in \Pi_a, \qquad X(\tau, \pi) \in \Pi_b$$

forces its internal zero mode to obey

$$X_0 \in \Pi_a \cap \Pi_b.$$

At an intersection, the stretching length is zero:

$$M^2 = \left(\frac{L}{2\pi\alpha'}\right)^2 + \frac{N+a}{\alpha'}, \qquad L = 0.$$

The massless Ramond state, together with its NS partner when supersymmetry is preserved, gives localized matter transforming as

$$\boxed{(\mathbf{N}_a, \overline{\mathbf{N}}_b) \quad \text{with net multiplicity} \quad I_{ab} = \Pi_a \circ \Pi_b}.$$

For factorized cycles on T^6,

$$\boxed{I_{ab} = \prod_{I=1}^{3} \left(n_a^I m_b^I - m_a^I n_b^I\right)}.$$

In F-theory,

$$\boxed{D_a \cap D_b = \Sigma_{ab} \quad \Longrightarrow \quad \text{codimension-two fiber enhancement} \quad \Longrightarrow \quad \text{matter localized on } \Sigma_{ab}}.$$

Equivalently, an additional fibral curve collapses only over Σ_{ab},

$$M_{\mathbf{w}} = T_{\mathrm{M2}}\,\mathrm{Vol}(C_{\mathbf{w}}) \longrightarrow 0,$$

which is the nonperturbative geometric counterpart of a zero-length open string at a D-brane intersection.

10.10 Q10: The regimes $g_s \ll 1$, $g_sN \ll 1$, and $g_sN \gg 1$

In page 734: >The D-brane picture is valid at weak string coupling, $g_s \gg 1$, where the string is perturbative and open strings end on D-branes which are heavy and themselves not in the perturbative spectrum of the theory. This picture also assumes that the back-reaction of N coincident branes on Minkowski space is small, which is the case if $g_sN \gg 1$. Otherwise their distortion of space-time cannot be neglected. In that situation we have a description as solutions of the SUGRA equations of motion with fluxes. This also requires g_s small but g_sN large, because otherwise the SUGRA description (18.26) which ignores string loop effects and α' corrections due to massive string modes is not valid and the equations of motions (18.27) and therefore also their brane solutions have to be corrected.

I am confused with these statements: 1. The limit $g_s \gg 1$ is easy to understand: we can consider the supergravity as an tree-level effective field theory description without higher string loop corrections. 2. How the second limit $g_sN \gg 1$ relates to the back-reaction of N coincident branes on the Minkowski space? 3. Why we need g_sN large compared to g_s i.e. $N \gg 1$? To ignore α' corrections we need to make sure that the curvature of the SUGRA solution is small compared to $1/\alpha'$, how to obtain the requirement for g_sN?

> **Textbook location and related material.** Chapter 18, Section 18.9 around p. 734: open-string, supergravity, and large-N regimes for coincident branes; the D3 near-horizon scale and suppression of α' corrections.

10.10.1 Answer

10.10.1.1 Core point

The apparent contradiction comes from two reversed inequalities in the question. The printed text on page 734 says

$$\boxed{g_s \ll 1, \qquad g_sN \ll 1 \quad \text{for the perturbative D-brane picture in nearly flat space}}$$

and

$$\boxed{g_s \ll 1, \qquad g_sN \gg 1 \quad \text{for the classical supergravity brane geometry}}.$$

It does not say $g_s \gg 1$, and it does not say that back-reaction is small for $g_sN \gg 1$.

The two parameters control different approximations:

- g_s controls closed-string loops. Tree-level string theory requires $g_s \ll 1$.
- g_sN controls the collective strength of N D-branes and, for D3-branes, the curvature radius in string units.

For N D3-branes the exact relation quoted in equation (18.149) is

$$L^4 = 4\pi g_s N \alpha'^2.$$

The curvature scale is $|\mathrm{Riemann}|^{1/2} \sim L^{-2}$, so

$$\alpha'|\mathrm{Riemann}|^{1/2} \sim \frac{\alpha'}{L^2} = \frac{1}{\sqrt{4\pi g_s N}}.$$

Therefore

$$\boxed{\alpha' \text{ corrections small} \quad \Longleftrightarrow \quad L \gg \ell_s \quad \Longleftrightarrow \quad g_s N \gg 1}.$$

Combining this with $g_s \ll 1$ is possible only if N is large. This is the essential large-N window:

$$\boxed{\frac{1}{N} \ll g_s \ll 1}$$

up to the numerical factors suppressed in the parametric inequalities.

The book derives these facts in two places. Section 18.5, especially pages 695–696, obtains the D-brane gravitational strength $g_s N$ directly from Newton's constant and the Dp-brane tension. Section 18.9 then specializes to D3-branes in equations (18.149), (18.187), and (18.188), relating $g_s N$ to the radius L and to the 't Hooft coupling.

10.10.1.2 Why tree level requires $g_s \ll 1$, not $g_s \gg 1$

A connected closed-string world-sheet of genus h carries the dilaton weight

$$g_s^{-\chi} = g_s^{2h-2}, \qquad \chi = 2 - 2h.$$

Relative to the sphere contribution, each additional handle contributes a factor

$$g_s^2.$$

Consequently, higher closed-string loops are suppressed when

$$\boxed{g_s \ll 1}.$$

If $g_s \gg 1$, the perturbative genus expansion is not controlled. Type-IIB string theory may then admit an S-dual weakly coupled description, but it is not the same weakly coupled D-brane expansion being discussed on page 734.

It is also important that "tree-level string theory" and "supergravity" are not identical approximations. Classical supergravity requires two independent suppressions:

$$\boxed{\text{string loops:} \quad e^{\Phi} \ll 1, \qquad \text{massive-string or derivative corrections:} \quad \alpha' |\text{Riemann}| \ll 1}.$$

For the D3-brane solution the dilaton is constant,

$$e^{\Phi} = g_s,$$

so the first condition is simply $g_s \ll 1$. The second condition will lead to $g_s N \gg 1$.

10.10.1.3 The source strength of one Dp-brane

The ten-dimensional bulk action begins schematically as

$$S_{\text{bulk}} = \frac{1}{2\kappa_{10}^2} \int d^{10}x \sqrt{-G}\, R + \cdots,$$

while N coincident Dp-branes contribute the source action

$$S_{\text{source}} = -N\tau_p \int d^{p+1}\xi \sqrt{-\det G_{\text{ind}}} + \cdots.$$

In the conventions used in the book,

$$2\kappa_{10}^2 = (2\pi)^7 g_s^2 \alpha'^4,$$

and the physical Dp-brane tension is

$$\tau_p = \frac{1}{(2\pi)^p g_s \alpha'^{(p+1)/2}}.$$

The tension is large at weak coupling,

$$\tau_p \sim \frac{1}{g_s},$$

which is why a D-brane is not an ordinary perturbative closed-string state. Nevertheless, its gravitational field is governed not by τ_p alone but by Newton's constant times the tension:

$$\begin{aligned}\kappa_{10}^2 N\tau_p &= \frac{(2\pi)^7}{2} g_s^2 \alpha'^4 \frac{N}{(2\pi)^p g_s \alpha'^{(p+1)/2}} \\ &= \frac{(2\pi)^{7-p}}{2} g_s N\, \alpha'^{(7-p)/2}.\end{aligned}$$

Thus the factor $1/g_s$ in the D-brane tension cancels one of the two powers of g_s in κ_{10}^2, leaving

$$\boxed{\kappa_{10}^2 N\tau_p \sim g_s N \ell_s^{7-p}}.$$

This is the elementary origin of the combination $g_s N$.

10.10.1.4 Linearized back-reaction on Minkowski space

Let

$$G_{MN} = \eta_{MN} + h_{MN}$$

and linearize the Einstein equation around flat space. Suppressing tensor indices and numerical coefficients, a stack localized in the $9-p$ transverse directions obeys

$$\nabla_\perp^2 h \sim \kappa_{10}^2 N\tau_p\, \delta^{(9-p)}(y).$$

For $p < 7$, the Green function in $9-p$ transverse dimensions behaves as

$$G_\perp(r) \sim \frac{1}{r^{7-p}}.$$

Therefore

$$h(r) \sim \frac{\kappa_{10}^2 N\tau_p}{r^{7-p}} \sim g_s N \left(\frac{\ell_s}{r}\right)^{7-p}.$$

At distances of order the string length,

$$r \sim \ell_s,$$

the dimensionless metric perturbation is parametrically

$$h(\ell_s) \sim g_s N.$$

Hence

$$\boxed{g_s N \ll 1 \quad\Longrightarrow\quad \text{weak back-reaction outside the string-scale core}}.$$

In this regime it is consistent to treat the D-branes as hypersurfaces embedded in nearly flat Minkowski space and quantize open strings ending on them.

Conversely,

$$\boxed{g_s N \gtrsim 1 \quad\Longrightarrow\quad \text{the flat-background approximation fails}}.$$

The branes must then be replaced by their back-reacted geometry.

There is a small qualification. For every nonzero source, the linearized field becomes large sufficiently close to the idealized brane position. The claim $g_s N \ll 1$ means that the breakdown occurs inside the string-scale region, where the geometric supergravity description was not trustworthy anyway and the microscopic open-string description takes over.

10.10.1.5 The same result from the nonlinear Dp-brane solution

The full BPS solution for N coincident Dp-branes, with $p < 7$, contains a harmonic function of the form

$$H_p(r) = 1 + \frac{R_p^{7-p}}{r^{7-p}},$$

where

$$R_p^{7-p} = c_p\, g_s N \alpha'^{(7-p)/2}.$$

The constant c_p depends only on p and on normalization conventions. Thus

$$\left(\frac{R_p}{\ell_s}\right)^{7-p} = c_p\, g_s N.$$

If

$$g_s N \ll 1,$$

then $R_p \ll \ell_s$: the would-be strongly warped region is smaller than the string length and is not a macroscopic supergravity throat. Outside the string-scale core, $H_p \simeq 1$.

If instead

$$g_s N \gg 1,$$

then $R_p \gg \ell_s$: the region in which H_p differs substantially from one is macroscopic in string units. The geometry is strongly deformed away from Minkowski space and back-reaction cannot be ignored.

This explains an important distinction:

$$\boxed{\text{large back-reaction does not mean large curvature}}.$$

A geometry can differ greatly from flat space but vary over a very large radius R_p. In that case its curvature is small even though its warping is strong. This is precisely the regime in which classical supergravity is useful.

10.10.1.6 Explicit D3-brane derivation of the curvature condition

For D3-branes the book gives

$$ds^2 = H_3^{-1/2}\eta_{\mu\nu}dx^\mu dx^\nu + H_3^{1/2}\left(dr^2 + r^2 d\Omega_5^2\right),$$

with

$$H_3(r) = 1 + \frac{L^4}{r^4}, \qquad L^4 = 4\pi g_s N\alpha'^2.$$

The same relation immediately measures the back-reaction. At $r \sim \ell_s$,

$$H_3 - 1 \sim \frac{L^4}{\ell_s^4} = 4\pi g_s N.$$

Thus the flat-space description requires $g_s N \ll 1$, up to the inessential factor 4π.

In the near-horizon region $r \ll L$, the metric becomes

$$ds^2 = \frac{r^2}{L^2}\eta_{\mu\nu}dx^\mu dx^\nu + \frac{L^2}{r^2}dr^2 + L^2 d\Omega_5^2,$$

which is

$$AdS_5(L) \times S^5(L).$$

Both factors have curvature radius L. Their Riemann tensors scale as

$$R_{MNPQ} \sim \frac{1}{L^2}.$$

Although the ten-dimensional scalar curvature cancels between the AdS_5 and S^5 factors,

$$R_{AdS_5} + R_{S^5} = 0,$$

the spacetime is not flat. For example,

$$R_{MNPQ}R^{MNPQ} = \frac{80}{L^4}.$$

The appropriate dimensionless derivative-expansion parameter is therefore

$$\alpha' \sqrt{R_{MNPQ}R^{MNPQ}} \sim \frac{\alpha'}{L^2} = \frac{1}{\sqrt{4\pi g_s N}}.$$

Thus

$$\boxed{L \gg \ell_s \quad \Longleftrightarrow \quad 4\pi g_s N \gg 1}.$$

The leading higher-derivative term in type-IIB string theory is schematically

$$\alpha'^3 R^4.$$

Relative to the two-derivative Einstein term, its size on a background of radius L is controlled by

$$\left(\frac{\alpha'}{L^2}\right)^3 \sim (4\pi g_s N)^{-3/2}.$$

This makes the suppression of α' corrections at large $g_s N$ explicit.

10.10.1.7 Why large N is necessary

Classical D3-brane supergravity requires both

$$g_s \ll 1$$

and

$$g_s N \gg 1.$$

Dividing the second inequality by the first parameter shows that the number of branes must be large:

$$N \gg \frac{1}{g_s} \gg 1.$$

Equivalently, there must be a parametric window

$$\boxed{\frac{1}{N} \ll g_s \ll 1}.$$

It is slightly misleading to say that "$g_s N$ must be large compared with g_s." The two quantities control different effects:

- each individual D-brane is a weak gravitational source of order g_s;
- N coincident branes source the same fields coherently, giving a total strength of order $g_s N$;
- closed-string loops remain controlled by g_s itself.

Large N therefore permits a useful separation:

$$\boxed{\text{each brane is weakly coupled} \quad \text{but} \quad \text{the stack has a large classical field}}.$$

This is analogous to a classical electromagnetic field produced by many individually weak sources.

10.10.1.8 Gauge-theory interpretation and the 't Hooft coupling

For N coincident D3-branes, the low-energy open-string theory is four-dimensional $\mathcal{N} = 4$ $U(N)$ super-Yang–Mills. The book uses

$$g_{\mathrm{YM}}^2 = 2\pi g_s.$$

Its 't Hooft coupling is therefore

$$\lambda = g_{\mathrm{YM}}^2 N = 2\pi g_s N.$$

Equation (18.188) can be written as

$$L^4 = 4\pi g_s N \alpha'^2 = 2\lambda \alpha'^2.$$

Hence

$$\frac{\alpha'}{L^2} = \frac{1}{\sqrt{2\lambda}}.$$

The two descriptions have complementary perturbative regimes:

$$\boxed{\lambda \ll 1 \quad \Longleftrightarrow \quad \text{weakly coupled open-string gauge theory}}$$

whereas

$$\boxed{\lambda \gg 1 \quad \Longleftrightarrow \quad \text{small-curvature gravitational description}}.$$

This is the strong/weak character of the AdS/CFT correspondence.

At fixed λ,

$$g_s = \frac{\lambda}{2\pi N}.$$

Thus taking $N \to \infty$ suppresses string loops. To have both large λ and small g_s, the more precise parametric window is

$$\boxed{1 \ll \lambda \ll 2\pi N}.$$

One can also see bulk quantum-loop suppression directly. Compactification on the S^5 of radius L gives parametrically

$$G_5 \sim \frac{G_{10}}{L^5}.$$

The dimensionless five-dimensional loop parameter is

$$\frac{G_5}{L^3} \sim \frac{g_s^2 \alpha'^4}{L^8} \sim \frac{1}{N^2}.$$

Therefore the classical bulk limit is simultaneously the planar large-N limit of the gauge theory.

10.10.1.9 The three regimes should not be conflated

For D3-branes, the approximations can be summarized as follows:

regime	coupling conditions	appropriate description
Weak flat-space D-brane regime	$g_s \ll 1$, $g_s N \ll 1$	Perturbative open strings; weakly coupled world-volume gauge theory; back-reaction neglected
Classical but string-scale curved regime	$g_s \ll 1$, $g_s N \sim 1$	String loops are small, but the full α'-corrected classical string background is needed
Classical supergravity regime	$g_s \ll 1$, $g_s N \gg 1$	Back-reacted D3 geometry; both string loops and α' corrections suppressed

regime	coupling conditions	appropriate description
Quantum-string regime	$g_s \not\ll 1$	String loops cannot be neglected; a dual formulation or full quantum string theory is required

The two perturbatively useful descriptions do not overlap:

$$\text{weak open-string/gauge theory:} \quad g_s N \ll 1,$$

$$\text{weakly curved supergravity:} \quad g_s N \gg 1.$$

AdS/CFT asserts that they are two descriptions of the same system and, in its strongest form, extends beyond either perturbative limit.

10.10.1.10 General Dp-brane caveat

The particularly clean condition $g_s N \gg 1$ for small curvature applies to the D3-brane throat because the D3 solution has a constant dilaton and one constant curvature radius L.

For a general Dp-brane,

$$e^{\Phi} = g_s H_p^{(3-p)/4}$$

varies with position, and so do the curvature and the effective gauge coupling. In the decoupling variables one commonly organizes the local conditions using

$$g_{\text{eff}}^2(U) \sim g_{\text{YM}}^2 N\, U^{p-3}.$$

Small curvature requires $g_{\text{eff}}^2(U) \gg 1$, while loop suppression imposes a separate condition on the local dilaton. Thus for $p \neq 3$ the trustworthy supergravity region can be only a range of radial positions. The page-734 discussion immediately specializes to D3-branes, for which

$$g_{\text{eff}}^2 = \lambda$$

is independent of scale and the two conditions reduce to the simple statements above.

10.10.1.11 Checks

For one D3-brane at weak coupling,

$$N = 1, \qquad g_s \ll 1,$$

one necessarily has $g_s N \ll 1$. The perturbative D-brane description is valid, but

$$L \lesssim \ell_s,$$

so the corresponding classical supergravity throat is string-scale and cannot be trusted.

For fixed small g_s and sufficiently large N,

$$g_s \ll 1, \qquad g_s N \gg 1,$$

the throat satisfies $L \gg \ell_s$. The metric is far from Minkowski space near the stack, but its curvature is small. There is no contradiction between strong back-reaction and a valid derivative expansion.

Finally, in the 't Hooft limit

$$N \to \infty, \qquad \lambda \text{ fixed},$$

one has

$$g_s = \frac{\lambda}{2\pi N} \to 0.$$

This suppresses string loops, but supergravity additionally requires

$$\lambda \gg 1.$$

10.10.1.12 Result

The two transcription corrections are

$$\boxed{\begin{array}{lcl} \text{flat-space D-brane picture} & : & g_s \ll 1, \quad g_s N \ll 1, \\ \text{classical SUGRA brane picture} & : & g_s \ll 1, \quad g_s N \gg 1. \end{array}}$$

The collective source strength is

$$\boxed{\kappa_{10}^2 N \tau_p \sim g_s N \alpha'^{(7-p)/2}}$$

and therefore, for $p < 7$,

$$h(r) \sim g_s N \left(\frac{\ell_s}{r}\right)^{7-p}.$$

For D3-branes,

$$\boxed{H_3 = 1 + \frac{L^4}{r^4}, \qquad L^4 = 4\pi g_s N \alpha'^2}$$

so the α' expansion parameter is

$$\boxed{\frac{\alpha'}{L^2} = \frac{1}{\sqrt{4\pi g_s N}} = \frac{1}{\sqrt{2\lambda}}}.$$

Hence

$$\boxed{g_s \ll 1 \text{ suppresses string loops}, \qquad g_s N \gg 1 \text{ suppresses curvature corrections}}.$$

Both can hold simultaneously because

$$\boxed{N \gg 1, \qquad \frac{1}{N} \ll g_s \ll 1}.$$

10.11 Q11: Minkowski space as a section of the projective null cone

In page 738-739 they discussed the conformally compactified Minkowski space. They embed $\mathbb{R}^{d-1,1}$ with coordinates $(y^0, \cdots, y^{d-1})$ into $\mathbb{R}^{d,2}$ with coordinates $(y^0, \cdots, y^{d-1}, y^d, y^{d+1})$. They said that the Minkowski space is the intersection of the light-cone with the hyperplane $v = y^{d+1} - y^d = 1$. How to see this? Then they intersect the light-cone with the sphere $(y^0)^2 + (y^{d+1})^2 + \sum_{i=1}^{d}(y^i)^2 = 2$, the resulting space is topologically $S^1 \times S^{d-1}$. Is this the conformally compactified Minkowski space? How it is relate to the common Minkowski space?

> **Textbook location and related material.** Chapter 18, Section 18.9 around pp. 738–739: the projective null cone in $\mathbb{R}^{d,2}$, conformal compactification, and the Einstein cylinder.

10.11.1 Answer

10.11.1.1 Core point

Yes. The construction is the standard projective-null-cone description of conformally compactified Minkowski space. The logical steps are:

1. The ambient null cone in $\mathbb{R}^{d,2}$ has dimension $d+1$, one dimension too many.
2. Points differing by a nonzero rescaling represent the same conformal point:

$$Y^M \sim \lambda Y^M.$$

3. On the open patch $v \neq 0$, one may use this rescaling to impose $v = 1$. The cone equation then determines $u = x^2$, leaving exactly the d arbitrary Minkowski coordinates x^μ.

4. The patch $v = 1$ misses the rays with $v = 0$. These are precisely the points at conformal infinity.
5. Intersecting the cone with the Euclidean sphere fixes the scale without discarding the $v = 0$ rays. The result is the compact space $S^1 \times S^{d-1}$, or more precisely its antipodal quotient if one identifies Y and $-Y$.

Thus the precise global statement is

$$\boxed{\overline{\mathbb{R}^{d-1,1}} = \frac{\{Y \in \mathbb{R}^{d,2} \setminus \{0\} \mid Y^2 = 0\}}{Y \sim \lambda Y} \simeq \frac{S^1 \times S^{d-1}}{\mathbb{Z}_2}}.$$

The book writes $S^1 \times S^{d-1}$ because it is using the double cover, or equivalently quotienting only by positive rescalings. Ordinary Minkowski space is an open dense patch of this compact space.

10.11.1.2 Ambient metric and the null cone

To avoid confusing ambient and physical coordinates, denote coordinates on $\mathbb{R}^{d,2}$ by

$$Y^M = (Y^\mu, Y^d, Y^{d+1}), \qquad \mu = 0, \dots, d-1.$$

The ambient metric is

$$\eta_{MN} = \mathrm{diag}(-1, +1, \dots, +1, -1).$$

Therefore

$$Y^2 = \eta_{\mu\nu} Y^\mu Y^\nu + (Y^d)^2 - (Y^{d+1})^2.$$

Define

$$u = Y^{d+1} + Y^d, \qquad v = Y^{d+1} - Y^d.$$

Since

$$(Y^d)^2 - (Y^{d+1})^2 = -uv,$$

the null-cone equation $Y^2 = 0$ becomes equation (18.163b):

$$\boxed{\eta_{\mu\nu} Y^\mu Y^\nu = uv}.$$

The differential ambient metric is correspondingly

$$dS^2 = \eta_{\mu\nu} dY^\mu dY^\nu - du\, dv.$$

The cone has dimension

$$(d+2) - 1 = d + 1.$$

The extra dimension is the radial scaling direction

$$Y^M \longmapsto \lambda Y^M.$$

Because the cone equation is homogeneous, a whole ray lies on the cone whenever one nonzero point of that ray does. Conformal compactification uses the ray, not a particular representative on it.

10.11.1.3 Why the section $v = 1$ is exactly Minkowski space

Restrict first to the open part of the cone on which

$$v \neq 0.$$

Using the scaling $Y \sim \lambda Y$, choose the representative with

$$v = 1.$$

Let

$$x^\mu = Y^\mu$$

on this section and define

$$x^2 = \eta_{\mu\nu} x^\mu x^\nu.$$

The cone equation now gives

$$x^2 = u.$$

Consequently every point on the $v = 1$ section is uniquely of the form

$$(Y^\mu, u, v) = (x^\mu, x^2, 1),$$

with arbitrary

$$x^\mu \in \mathbb{R}^{d-1,1}.$$

Conversely, every $x^\mu \in \mathbb{R}^{d-1,1}$ produces a null ambient vector of this form. This is already a one-to-one identification between the section and Minkowski space.

In terms of the original final two coordinates,

$$Y^{d+1} = \frac{u+v}{2} = \frac{1+x^2}{2},$$

$$Y^d = \frac{u-v}{2} = \frac{x^2-1}{2}.$$

The embedding is therefore

$$\boxed{Y^M(x) = \left(x^\mu, \frac{x^2-1}{2}, \frac{x^2+1}{2}\right)}.$$

It is manifestly null:

$$\begin{aligned} Y^2 &= x^2 + \frac{(x^2-1)^2}{4} - \frac{(x^2+1)^2}{4} \\ &= 0. \end{aligned}$$

The identification is not merely topological. It also gives the Minkowski metric. Since $dv = 0$ on the $v = 1$ section,

$$dS^2\big|_{v=1} = \eta_{\mu\nu} dx^\mu dx^\nu - du\, dv = \eta_{\mu\nu} dx^\mu dx^\nu.$$

Equivalently, $dY^d = dY^{d+1} = x_\mu dx^\mu$, so the last two contributions to the ambient line element cancel. Thus

$$\boxed{\mathcal{N} \cap \{v = 1\} \cong \mathbb{R}^{d-1,1}}$$

as a metric space.

10.11.1.4 Why projectivization is needed for the conformal group

The group $SO(d, 2)$ acts linearly on Y^M and preserves the null cone. A generic $SO(d, 2)$ transformation does not preserve the particular section $v = 1$. If

$$Y^M(x) \longmapsto Y'^M,$$

one must rescale the transformed vector back to the section:

$$Y'^M \longmapsto \frac{Y'^M}{v'}.$$

The induced transformation of the physical coordinates is

$$x'^\mu = \frac{Y'^\mu}{v'}.$$

Because the denominator depends on x, a linear ambient transformation becomes a nonlinear conformal transformation of Minkowski space.

For some x, one may obtain

$$v' = 0.$$

In the ordinary Minkowski chart, this means that x'^{μ} has been sent to infinity. This is exactly what happens for a special conformal transformation at the point where its denominator vanishes. The projective cone includes these $v = 0$ rays, allowing $SO(d, 2)$ to act globally.

This explains why ordinary Minkowski space alone is insufficient: it is not closed under finite conformal transformations. Its conformal compactification is.

10.11.1.5 Why the sphere section is $S^1 \times S^{d-1}$

Now fix the scale globally by intersecting the null cone with the Euclidean sphere used in the book:

$$(Y^0)^2 + (Y^{d+1})^2 + \sum_{i=1}^{d} (Y^i)^2 = 2.$$

Introduce

$$A = (Y^0)^2 + (Y^{d+1})^2, \qquad B = \sum_{i=1}^{d} (Y^i)^2.$$

The null-cone equation (18.163a) says

$$A = B.$$

The sphere equation says

$$A + B = 2.$$

Therefore

$$A = B = 1.$$

That is,

$$(Y^0)^2 + (Y^{d+1})^2 = 1$$

and

$$\sum_{i=1}^{d} (Y^i)^2 = 1.$$

The first equation defines S^1, while the second defines S^{d-1}. Hence

$$\boxed{\mathcal{N} \cap \{|Y|_{\mathrm{E}}^2 = 2\} \simeq S^1 \times S^{d-1}}.$$

This sphere is not being used as a spacetime metric condition. Its role is only to choose a finite-norm representative of every null ray. Every positive ray intersects it exactly once, including rays with $v = 0$, so unlike the affine section $v = 1$, it gives a compact global section.

10.11.1.6 The antipodal $\mathbb{Z}_2$ subtlety

If projective equivalence allows every nonzero real scaling,

$$Y \sim \lambda Y, \qquad \lambda \in \mathbb{R}^{\times},$$

then

$$Y \sim -Y.$$

Both Y and $-Y$ lie on the sphere section. Thus the sphere intersects each unoriented projective ray twice, and the true projective cone is

$$\boxed{\mathbb{P}\mathcal{N} \simeq \frac{S^1 \times S^{d-1}}{\mathbb{Z}_2}},$$

where the $\mathbb{Z}_2$ acts antipodally on both factors.

If instead one quotients only by positive scalings,

$$\lambda > 0,$$

then the rays are oriented and the result is the double cover

$$S^1 \times S^{d-1}.$$

The book states the topology of this double cover and suppresses the antipodal quotient. Both conventions are common; locally they give the same conformal geometry.

For Lorentzian physics one often goes one step further and unwraps the timelike circle:

$$S^1 \times S^{d-1} \longrightarrow \mathbb{R} \times S^{d-1}.$$

This universal cover is the Einstein static cylinder and avoids the closed timelike curves associated with periodic cylinder time.

10.11.1.7 Explicit conformal map to the Einstein cylinder

Parametrize the sphere section by

$$Y^0 = \sin T, \qquad Y^{d+1} = \cos T,$$

and write the unit vector on S^{d-1} as

$$Y^d = -\cos\chi, \qquad Y^a = \sin\chi\,\hat{n}^a, \qquad a = 1, \dots, d-1,$$

where

$$\hat{n} \in S^{d-2}, \qquad 0 \le \chi \le \pi.$$

The induced ambient metric on this section is

$$\boxed{ds^2_{\text{cyl}} = -dT^2 + d\chi^2 + \sin^2\chi\, d\Omega^2_{d-2}}.$$

This is the metric on the Lorentzian Einstein cylinder $S^1 \times S^{d-1}$.

For this parametrization,

$$v = Y^{d+1} - Y^d = \cos T + \cos\chi.$$

On the patch $v \neq 0$, rescale to $v = 1$ and define Minkowski coordinates:

$$x^\mu = \frac{Y^\mu}{v}.$$

Writing $t = x^0$ and

$$r^2 = \sum_{a=1}^{d-1} (x^a)^2,$$

one obtains

$$\boxed{t = \frac{\sin T}{\cos T + \cos\chi}, \qquad r = \frac{\sin\chi}{\cos T + \cos\chi}}.$$

Equivalently,

$$\boxed{t + r = \tan\left(\frac{T+\chi}{2}\right), \qquad t - r = \tan\left(\frac{T-\chi}{2}\right)}.$$

Substitution gives the conformal relation

$$\begin{aligned} ds^2_{\text{M}} &= -dt^2 + dr^2 + r^2 d\Omega^2_{d-2} \\ &= \frac{1}{(\cos T + \cos\chi)^2}\left(-dT^2 + d\chi^2 + \sin^2\chi\, d\Omega^2_{d-2}\right). \end{aligned}$$

Thus

$$\boxed{ds^2_{\text{cyl}} = \Omega^2 ds^2_{\text{M}}, \qquad \Omega = \cos T + \cos\chi}.$$

The cylinder metric is not equal to the Minkowski metric; it is conformally related to it. That is why the result is called a conformal compactification.

10.11.1.8 Minkowski space is a diamond inside the cylinder

Choose the representative with

$$\Omega = \cos T + \cos \chi > 0.$$

The tangent formulas imply

$$-\pi < T + \chi < \pi, \qquad -\pi < T - \chi < \pi.$$

Equivalently,

$$\boxed{|T| + \chi < \pi}.$$

This is the familiar diamond-shaped Minkowski region in the Einstein cylinder.

The ordinary Minkowski metric becomes singular when

$$\Omega = 0,$$

but the rescaled cylinder metric remains regular. The new finite loci are the points at infinity:

- future null infinity $\mathscr{I}^+$ lies on $T + \chi = \pi$;
- past null infinity $\mathscr{I}^-$ lies on $T - \chi = -\pi$;
- future and past timelike infinity are

$$i^+ : (T, \chi) = (\pi, 0), \qquad i^- : (T, \chi) = (-\pi, 0);$$

- spatial infinity is

$$i^0 : (T, \chi) = (0, \pi).$$

Accordingly, conformal compactification does not make the physical Minkowski metric regular at infinity. It replaces it by a conformally equivalent metric that extends smoothly to a boundary at finite coordinate distance.

10.11.1.9 Why equation (18.165) uses the patch $u \neq 0$

There is a small change of chart between equations (18.163) and (18.165) that the book does not emphasize.

The sentence before equation (18.164) uses the $v \neq 0$ chart and fixes

$$v = 1.$$

Equation (18.165), however, uses the other affine chart,

$$u \neq 0,$$

and writes

$$(Y^\mu, u, v) = \left(u x^\mu, u, u x^2\right).$$

Projectively setting $u = 1$ gives

$$(Y^\mu, u, v) = \left(x^\mu, 1, x^2\right),$$

which is another copy of Minkowski space.

Let $x_v^\mu = Y^\mu / v$ be the coordinates in the v-chart and $x_u^\mu = Y^\mu / u$ those in the u-chart. On their overlap, the cone equation gives

$$\frac{u}{v} = x_v^2,$$

and hence

$$\boxed{x_u^\mu = \frac{x_v^\mu}{x_v^2}}.$$

The transition function is conformal inversion. Thus the two affine sections are not contradictory; they are two conformally related Minkowski charts on the same projective null cone.

10.11.1.10 Checks

The dimension count is

$$\dim \mathbb{R}^{d,2} - 1_{\text{ cone equation}} - 1_{\text{ projective scaling}} = d,$$

as required for compactified d-dimensional Minkowski space.

On the $v = 1$ section,

$$Y^d = \frac{x^2-1}{2}, \qquad Y^{d+1} = \frac{x^2+1}{2},$$

so

$$Y^{d+1} - Y^d = 1$$

and $Y^2 = 0$ identically. This directly verifies the book's claim.

At the center of the Minkowski diamond,

$$T = 0, \qquad \chi = 0,$$

the conformal map gives

$$t = 0, \qquad r = 0.$$

As $T + \chi \to \pi$, one has

$$t + r \to +\infty,$$

showing explicitly how null infinity is moved to a finite cylinder coordinate.

10.11.1.11 Result

The ambient cone is

$$\mathcal{N} = \left\{Y \neq 0 : \eta_{\mu\nu} Y^\mu Y^\nu = uv\right\}, \qquad u = Y^{d+1} + Y^d, \quad v = Y^{d+1} - Y^d.$$

On the $v = 1$ section,

$$\boxed{u = x^2, \qquad Y^M(x) = \left(x^\mu, \frac{x^2-1}{2}, \frac{x^2+1}{2}\right), \qquad ds^2_{\text{ind}} = \eta_{\mu\nu} dx^\mu dx^\nu}.$$

The global conformal compactification is the projective null cone:

$$\boxed{\overline{\mathbb{R}^{d-1,1}} = \mathbb{P}\mathcal{N} \simeq \frac{S^1 \times S^{d-1}}{\mathbb{Z}_2}}.$$

The book's $S^1 \times S^{d-1}$ is its double cover.

With

$$\Omega = \cos T + \cos \chi,$$

ordinary Minkowski space is the open diamond

$$\boxed{|T| + \chi < \pi}$$

and its metric is related to the Einstein-cylinder metric by

$$\boxed{ds^2_{\text{M}} = \Omega^{-2} ds^2_{\text{cyl}}, \qquad ds^2_{\text{cyl}} = -dT^2 + d\chi^2 + \sin^2 \chi \, d\Omega^2_{d-2}}.$$

The added locus $\Omega = 0$ is conformal infinity. Equation (18.165) uses the complementary $u \neq 0$ chart, whose coordinates are related to those of the $v \neq 0$ chart by inversion:

$$\boxed{x^\mu_u = \frac{x^\mu_v}{x^2_v}}.$$

10.12 Q12: Finite-temperature $\mathcal{N} = 4$ SYM and its thermal stress tensor

In page759, they interpret the result with $\mathcal{N} = 4$ SYM at finite temperature on the boundary. They said the energy-momentum tensor is the one of a relativistic plasma at rest in a thermal state

$$\langle T_{\mu\nu} \rangle = \frac{\pi^2}{8} T^4 N^2 \mathrm{diag}(3, 1, 1, 1)$$

Give a brief introduction on $\mathcal{N} = 4$ SYM at finite temperature and this energy-momentum tensor.

> **Textbook location and related material.** Chapter 18, Section 18.9 around p. 759: finite-temperature $\mathcal{N} = 4$ SYM, the AdS black brane, holographic thermodynamics, and the thermal stress tensor.

10.12.1 Answer

10.12.1.1 Core point

The formula on page 759 is the equilibrium stress tensor of strongly coupled, large-N $\mathcal{N} = 4$ $SU(N)$ super-Yang–Mills theory:

$$\boxed{\left\langle T_{\mu\nu} \right\rangle = \frac{\pi^2}{8} N^2 T^4 \,\mathrm{diag}(3, 1, 1, 1)}.$$

It contains two logically distinct ingredients.

First, equilibrium symmetries and conformal invariance fix the tensor structure:

$$\left\langle T_{\mu\nu} \right\rangle = \mathrm{diag}(\epsilon, p, p, p), \qquad \epsilon = 3p.$$

Second, AdS/CFT fixes the strong-coupling pressure:

$$p = \frac{\pi^2}{8} N^2 T^4.$$

Therefore

$$\epsilon = 3p = \frac{3\pi^2}{8} N^2 T^4.$$

The coefficient is not determined by conformal symmetry alone. It is the classical-supergravity prediction in the limit

$$N \gg 1, \qquad \lambda = g_{\mathrm{YM}}^2 N \gg 1, \qquad g_s \ll 1.$$

At weak coupling the tensor has the same perfect-fluid and traceless form but a different coefficient.

10.12.1.2 Zero-temperature $\mathcal{N} = 4$ SYM

Four-dimensional $\mathcal{N} = 4$ SYM with gauge group $SU(N)$ contains only adjoint fields:

- one gauge field A_μ;
- six real scalar fields Φ^I, $I = 1, \ldots, 6$;
- four Weyl fermions λ^A, $A = 1, \ldots, 4$.

All fields transform in the adjoint representation, whose dimension is

$$\dim SU(N) = N^2 - 1.$$

The theory has sixteen Poincare supercharges and an $SU(4)_R \simeq SO(6)_R$ R-symmetry. Its beta function vanishes:

$$\beta(g_{\mathrm{YM}}) = 0.$$

Consequently, on flat space and at zero temperature it is an exactly conformal quantum field theory for every value of the coupling.

For N coincident D3-branes, the microscopic open-string theory initially has gauge group $U(N)$. Its diagonal $U(1)$ describes the free center-of-mass motion of the stack and decouples from the interacting sector. The nontrivial CFT is therefore the $SU(N)$ theory. Classical gravity retains only the leading large-N scaling

$$N^2 - 1 = N^2\left(1 - \frac{1}{N^2}\right) \longrightarrow N^2.$$

10.12.1.3 What finite temperature means

The thermal theory is defined by the canonical density matrix

$$\rho_{\text{th}} = \frac{e^{-\beta H}}{Z(\beta)}, \qquad Z(\beta) = \operatorname{Tr} e^{-\beta H}, \qquad \beta = \frac{1}{T}.$$

Thermal expectation values are

$$\langle \mathcal{O} \rangle_T = \operatorname{Tr}\left(\rho_{\text{th}} \mathcal{O}\right).$$

In the Euclidean path integral, this is implemented by compactifying Euclidean time:

$$\tau_E \sim \tau_E + \beta.$$

Bosons are periodic and fermions are antiperiodic:

$$\Phi(\tau_E + \beta) = \Phi(\tau_E), \qquad \lambda(\tau_E + \beta) = -\lambda(\tau_E).$$

The unequal boundary conditions break supersymmetry at every nonzero temperature. Thus "finite-temperature $\mathcal{N} = 4$ SYM" means the thermal state of a theory whose zero-temperature Lagrangian has $\mathcal{N} = 4$ supersymmetry; the thermal state itself is not supersymmetric.

The temperature introduces a scale, so the thermal state is not invariant under dilatations. Nevertheless, the operator identity expressing conformal invariance still constrains its equation of state. On flat spacetime there is no curvature contribution to the Weyl anomaly, and hence

$$\left\langle T^{\mu}{}_{\mu} \right\rangle_T = 0.$$

10.12.1.4 Perfect-fluid form from equilibrium symmetries

The thermal equilibrium state at rest is:

- invariant under time translations;
- invariant under spatial translations;
- invariant under spatial rotations;
- homogeneous and isotropic;
- at zero chemical potential in the discussion of the book.

These symmetries imply

$$\langle T_{0i} \rangle = 0,$$

and rotational invariance implies

$$\langle T_{ij} \rangle = p\, \delta_{ij}.$$

Defining

$$\epsilon = \langle T_{00} \rangle,$$

one obtains

$$\boxed{\langle T_{\mu\nu} \rangle = \operatorname{diag}(\epsilon, p, p, p)}.$$

Covariantly, the equilibrium stress tensor of a relativistic perfect fluid is

$$\boxed{T_{\mu\nu} = (\epsilon + p) u_\mu u_\nu + p\, \eta_{\mu\nu}},$$

where

$$u^\mu u_\mu = -1.$$

In the rest frame and in the mostly-plus convention,

$$u^\mu = (1,0,0,0), \qquad u_\mu = (-1,0,0,0),$$

so this formula gives

$$T_{00} = \epsilon, \qquad T_{ij} = p\,\delta_{ij}.$$

This is also the origin of the boosted expression in equation (18.231). Once $\epsilon = 3p$,

$$T_{\mu\nu} = p\left(\eta_{\mu\nu} + 4u_\mu u_\nu\right).$$

10.12.1.5 Conformal invariance fixes $\epsilon = 3p$

Using

$$\eta_{\mu\nu} = \mathrm{diag}(-,+,+,+),$$

the trace is

$$\begin{aligned}\langle T^\mu{}_\mu\rangle &= \eta^{\mu\nu}\langle T_{\mu\nu}\rangle \\ &= -\epsilon + 3p.\end{aligned}$$

Flat-space conformal invariance requires the trace to vanish:

$$-\epsilon + 3p = 0.$$

Therefore

$$\boxed{\epsilon = 3p}.$$

Dimensional analysis supplies the temperature dependence. In four dimensions, $T_{\mu\nu}$ has mass dimension four, and a homogeneous thermal state has no scale other than T. Hence

$$p = N^2 T^4 f(\lambda) + O(N^0),$$

where $f(\lambda)$ is a dimensionless function of the exactly marginal 't Hooft coupling. Conformal symmetry fixes the power T^4 and the relation $\epsilon = 3p$, but it does not fix $f(\lambda)$.

10.12.1.6 The dual AdS black brane

At large N and large λ, the thermal state has a classical gravitational dual. It is not empty $AdS_5 \times S^5$, which represents the vacuum, but an asymptotically AdS black three-brane.

Restoring the AdS radius L, the metric in the coordinates of equation (18.227) is

$$ds^2 = L^2\left[\frac{dU^2}{U^2 f(U)} - U^2 f(U)\,dt^2 + U^2 d\mathbf{x}^2 + d\Omega_5^2\right],$$

where

$$f(U) = 1 - \frac{U_0^4}{U^4}.$$

The event horizon is at

$$U = U_0.$$

After Wick rotation $t = -i\tau_E$, regularity of the Euclidean metric at the horizon requires a definite period for τ_E. This period gives the Hawking temperature

$$\boxed{T = T_{\mathrm{BH}} = \frac{U_0}{\pi}}.$$

AdS/CFT identifies the Hawking temperature of the bulk black brane with the temperature of the boundary SYM state.

The translationally invariant planar horizon explains why the boundary state is homogeneous. Its rotational symmetry in the three $\mathbf{x}$ directions explains why all three pressures are equal.

10.12.1.7 Derivation from the holographic stress tensor

The book first derives the answer through holographic renormalization. In its Fefferman–Graham expansion, the normalizable boundary datum is chosen as

$$g^{(2)}_{\mu\nu} = c\,\mathrm{diag}(3,1,1,1), \qquad c = \frac{1}{\rho_0^2}.$$

For a flat four-dimensional boundary, equation (18.220) gives

$$\langle T_{\mu\nu}\rangle = \frac{4L^3}{16\pi G_5} g^{(2)}_{\mu\nu} = \frac{L^3}{4\pi G_5} g^{(2)}_{\mu\nu}.$$

Equation (18.225) relates ρ_0 to the Hawking temperature:

$$T = \frac{1}{\pi}\sqrt{\frac{2}{\rho_0}}.$$

Therefore

$$\frac{1}{\rho_0^2} = \frac{\pi^4 T^4}{4}.$$

Dimensional reduction of type-IIB supergravity on S^5 gives

$$\boxed{\frac{L^3}{G_5} = \frac{2N^2}{\pi}}$$

at leading order in large N. Substituting these relations yields

$$\begin{aligned}\langle T_{\mu\nu}\rangle &= \frac{L^3}{4\pi G_5}\frac{\pi^4 T^4}{4}\,\mathrm{diag}(3,1,1,1)\\ &= \frac{\pi^2}{8} N^2 T^4\,\mathrm{diag}(3,1,1,1).\end{aligned}$$

This is equation (18.228).

10.12.1.8 Independent thermodynamic derivation from the horizon

The same coefficient follows from the Bekenstein–Hawking entropy. The spatial horizon area per unit boundary volume is

$$\frac{A_{\mathrm{h}}}{V_3} = L^3 U_0^3.$$

Therefore the entropy density is

$$s = \frac{A_{\mathrm{h}}}{4G_5 V_3} = \frac{L^3 U_0^3}{4G_5}.$$

Using

$$U_0 = \pi T, \qquad \frac{L^3}{G_5} = \frac{2N^2}{\pi},$$

one obtains

$$\boxed{s = \frac{\pi^2}{2} N^2 T^3}.$$

At zero chemical potential, the thermodynamic relations are

$$s = \frac{\partial p}{\partial T}, \qquad \epsilon = Ts - p.$$

Taking $p(0) = 0$ and integrating gives

$$p(T) = \int_0^T s(T')\,dT' = \frac{\pi^2}{8} N^2 T^4.$$

Then

$$\epsilon = Ts - p = \frac{3\pi^2}{8}N^2T^4.$$

Hence

$$\boxed{\epsilon = 3p, \qquad p = \frac{\pi^2}{8}N^2T^4, \qquad s = \frac{\pi^2}{2}N^2T^3}.$$

10.12.1.9 Comparison with the free SYM plasma

At zero coupling, the energy density can be computed by counting massless adjoint degrees of freedom. For each adjoint generator, the bosonic degrees of freedom are

$$n_b = 2_{\text{ gauge polarizations}} + 6_{\text{ scalars}} = 8.$$

The four Weyl fermions contribute

$$n_f = 4 \times 2 = 8$$

fermionic degrees of freedom. The energy density of a free relativistic gas is

$$\epsilon_{\text{free}} = \frac{\pi^2}{30}\left(n_b + \frac{7}{8}n_f\right)(N^2-1)T^4.$$

Since

$$n_b + \frac{7}{8}n_f = 8 + 7 = 15,$$

this gives

$$\boxed{\epsilon_{\text{free}} = \frac{\pi^2}{2}(N^2-1)T^4 \simeq \frac{\pi^2}{2}N^2T^4}.$$

The free pressure is

$$p_{\text{free}} = \frac{1}{3}\epsilon_{\text{free}} \simeq \frac{\pi^2}{6}N^2T^4.$$

The strong-coupling holographic result is therefore

$$\boxed{\frac{\epsilon_{\text{strong}}}{\epsilon_{\text{free}}} = \frac{p_{\text{strong}}}{p_{\text{free}}} = \frac{3}{4}}$$

at leading order in large N. This is equation (18.229).

Unlike the Weyl-anomaly coefficients of $\mathcal{N}=4$ SYM, the thermal free energy is not protected against coupling dependence. The factor $3/4$ does not mean that one quarter of the fields disappear. It means that interactions change the thermodynamic free energy. Finite-λ corrections arise from α' corrections in the bulk, while finite-N corrections arise from string loops.

10.12.1.10 What "relativistic plasma at rest" does and does not mean

The diagonal form

$$\mathrm{diag}(\epsilon, p, p, p)$$

means:

- there is no momentum or energy flux in the chosen rest frame, so $T_{0i} = 0$;
- there is no shear stress in equilibrium;
- the pressure is isotropic;
- the state is homogeneous and time independent.

It does not mean that the strongly coupled plasma is a free ideal gas. The displayed tensor is only the zeroth-order equilibrium form. If temperature and velocity vary slowly in spacetime, hydrodynamics adds derivative corrections:

$$T_{\mu\nu} = (\epsilon + p)u_\mu u_\nu + p\eta_{\mu\nu} - 2\eta_{\text{shear}}\sigma_{\mu\nu} + \cdots.$$

Those corrections vanish for the static homogeneous state in equation (18.228). This is why the next part of the book proceeds from the equilibrium black brane to holographic hydrodynamics.

10.12.1.11 Checks

The tensor is traceless:

$$\eta^{\mu\nu}\left\langle T_{\mu\nu}\right\rangle = \frac{\pi^2}{8}N^2T^4(-3+1+1+1) = 0.$$

It is conserved because it is constant:

$$\partial^\mu\left\langle T_{\mu\nu}\right\rangle = 0.$$

The thermodynamic identity also checks:

$$Ts = T\left(\frac{\pi^2}{2}N^2T^3\right) = \frac{\pi^2}{2}N^2T^4 = \epsilon + p.$$

Finally, as $T \to 0$,

$$\epsilon, p, s \longrightarrow 0,$$

and the planar black-brane horizon disappears, returning to the vacuum Poincare patch of AdS_5.

10.12.1.12 Result

Finite-temperature $\mathcal{N} = 4$ SYM is defined by

$$\rho_{\text{th}} = Z^{-1}e^{-H/T},$$

with periodic bosons and antiperiodic fermions on the Euclidean thermal circle. The thermal boundary conditions break supersymmetry.

Homogeneity, isotropy, and conformal invariance imply

$$\boxed{\left\langle T_{\mu\nu}\right\rangle = (\epsilon+p)u_\mu u_\nu + p\eta_{\mu\nu}, \qquad \epsilon = 3p}.$$

In the large-N, large-λ holographic regime,

$$\boxed{T = \frac{U_0}{\pi}, \qquad s = \frac{\pi^2}{2}N^2T^3, \qquad p = \frac{\pi^2}{8}N^2T^4, \qquad \epsilon = \frac{3\pi^2}{8}N^2T^4}.$$

Therefore, in the rest frame,

$$\boxed{\left\langle T_{\mu\nu}\right\rangle = \frac{\pi^2}{8}N^2T^4\,\text{diag}(3,1,1,1)}.$$

For the free theory,

$$\epsilon_{\text{free}} \simeq \frac{\pi^2}{2}N^2T^4,$$

so

$$\boxed{\epsilon_{\text{strong}} = \frac{3}{4}\epsilon_{\text{free}}}$$

at leading order in the classical-supergravity limit.

Chapter A

Essential Notation and Detailed Roadmap of the Textbook

This appendix is designed as a re-entry map. The notation list retains only symbols used repeatedly across the book and the conversations. The roadmap then follows the logic of *Basic Concepts of String Theory* chapter by chapter and section by section, with printed page numbers from the book.

A.1 Essential notation and conventions

A.1.1 World-sheet geometry and conformal field theory

- $\sigma^a = (\tau, \sigma)$ are world-sheet coordinates; $z, \bar{z}$ are Euclidean complex coordinates. The world-sheet metric is h_{ab}, while G_{MN} denotes the target-space metric.
- $T(z)$ and $\bar{T}(\bar{z})$ are the holomorphic and antiholomorphic stress tensors. Their modes are L_n and $\bar{L}_n$, with central charges c and $\bar{c}$.
- A local operator has conformal weights $(h, \bar{h})$. A physical integrated closed-string matter insertion has $(h, \bar{h}) = (1, 1)$; an unintegrated insertion includes $c\bar{c}$ ghosts and has total weight $(0, 0)$.
- On the torus, $\tau = \tau_1 + i\tau_2$ is the complex-structure modulus and $q = e^{2\pi i\tau}$. The modular generators act by $S : \tau \mapsto -1/\tau$ and $T : \tau \mapsto \tau + 1$.
- $\eta(\tau)$ is the Dedekind eta function and ϑ_i are Jacobi theta functions. A characteristic is written $\vartheta\left[\begin{smallmatrix}\alpha\\ \beta\end{smallmatrix}\right]$.
- In an RCFT, $\chi_i(\tau)$ is the character of the representation i; S and T are its modular matrices. Charge conjugation is $C_{ij} = \delta_{i,j^*}$, and $S^2 = C$.

A.1.2 Quantization, ghosts, and superstrings

- α' is the inverse string tension scale, with $T_{\mathrm{F1}} = 1/(2\pi\alpha')$; conventions for l_s differ by harmless factors of 2π.
- N, $\bar{N}$ denote oscillator levels. The normal-ordering intercept is $a = 1$ for the bosonic string, $a_{\mathrm{NS}} = 1/2$, and $a_{\mathrm{R}} = 0$ in the RNS string.
- (b, c) are reparametrization ghosts. (β, γ) are superconformal ghosts, often bosonized using ϕ; q_{pic} denotes picture number when confusion with the torus nome is possible.
- Q_{BRST} is the BRST charge. Physical states are BRST cohomology classes: $Q_{\mathrm{BRST}}|\Psi\rangle = 0$ modulo $|\Psi\rangle \sim |\Psi\rangle + Q_{\mathrm{BRST}}|\Lambda\rangle$.
- NS and R denote antiperiodic and periodic cylinder fermions, respectively. $(-1)^F$ is world-sheet fermion parity; $(-1)^{F_L}$ is left-moving spacetime fermion number.
- Ω denotes world-sheet orientation reversal. I_n is reflection of n target coordinates. $D^{\mu}{}_{\nu}$ is the D-brane gluing matrix, $+1$ along Neumann and -1 along Dirichlet directions.

A.1.3 Compactification and geometry

- X is a compactification manifold, often a Calabi–Yau threefold. ω is its Kähler form and Ω its nowhere-vanishing holomorphic volume form.

- $g_{i\bar{j}}$ is a Kähler metric; barred indices are kept explicit. $H^{p,q}(X)$ and $h^{p,q}$ denote Dolbeault cohomology and Hodge numbers.
- For a cycle $\Sigma \subset X$, N_Σ is its normal bundle. A special-Lagrangian deformation is represented by a harmonic one-form, so its infinitesimal moduli are counted by $b_1(\Sigma)$.
- T, U, and S commonly denote Kähler, complex-structure, and axiodilaton moduli, with the precise assignment stated locally. K and W are the four-dimensional Kähler potential and superpotential.
- In heterotic compactifications, P and Q denote weight and root lattices when discussing Lie algebras. For an affine algebra, k is the level, ψ the highest root, and λ an integrable highest weight satisfying $(\psi, \lambda) \leq k$.

A.1.4 Couplings, branes, and dualities

- Φ is the ten-dimensional dilaton and $g_s = e^{\Phi_0}$ its asymptotic exponential. The string and Einstein metrics obey $G^{(E)}_{MN} = e^{-\Phi/2} G^{(S)}_{MN}$ in ten dimensions.
- κ_d is the d-dimensional gravitational coupling. τ_p or T_p is a Dp-brane tension; μ_p is its R–R charge in conventions that distinguish the two.
- B_2 is the NS–NS two-form, $H_3 = dB_2$, and C_p denotes R–R potentials. The gauge-invariant D-brane combination is $P[B_2] + 2\pi\alpha' F$.
- Γ^M are spacetime gamma matrices. BPS projectors act on supersymmetry parameters and reduce the number of preserved supercharges.
- $g_s N$ measures the collective backreaction of N coincident D-branes. Perturbative branes require $g_s N \ll 1$, whereas a weakly curved classical near-horizon geometry requires $g_s N \gg 1$ together with $g_s \ll 1$.

A.2 The book's overall argument

The book follows a deliberate chain:

$$
\begin{aligned}
\text{classical string} &\longrightarrow \text{quantization and CFT} \longrightarrow \text{BRST and perturbative amplitudes}\\
&\longrightarrow \text{superstrings and compactification}\\
&\longrightarrow \text{D-branes, fluxes, dualities, M/F-theory, and AdS/CFT.}
\end{aligned}
$$

The first six chapters construct the bosonic string and its perturbation theory. Chapters 7–9 add world-sheet supersymmetry, spacetime fermions, GSO projection, D-branes, and orientifolds. Chapters 10–13 build the algebraic technology needed for compactified superstrings: Narain lattices, affine Lie algebras, superconformal theories, BRST vertices, and covariant lattices. Chapters 14–17 translate world-sheet consistency into spacetime geometry and effective field theory, then add branes and fluxes. Chapter 18 reorganizes the perturbative theories into a nonperturbative network and ends with holography.

A.3 Chapter-by-chapter and section-by-section roadmap

A.3.1 Chapter 1: Introduction (p. 1)

The introduction motivates strings as quantum gravity rather than hadronic models. It emphasizes the unavoidable massless closed-string spin-two state, the role of supersymmetry and anomaly cancellation, the five ten-dimensional superstring theories, D-branes, dualities, M-theory, compactification, and the gauge/gravity correspondence. Read it as a list of destinations whose mechanisms are supplied later.

A.3.2 Chapter 2: The Classical Bosonic String (pp. 7–34)

A.3.2.1 2.1 The relativistic particle (p. 7)

The point-particle action introduces reparametrization invariance, the einbein formulation, constraints, and gauge fixing. This miniature model previews the Polyakov formulation: an auxiliary world-volume metric makes symmetries transparent, and its equation of motion returns the geometric action.

A.3.2.2 2.2 The Nambu–Goto action (p. 10)

The world-line is replaced by a world-sheet, and proper length by area. The induced metric gives the Nambu–Goto action and the string tension sets the fundamental scale. The square-root form is geometrically direct but inconvenient for quantization.

A.3.2.3 2.3 The Polyakov action and its symmetries (p. 12)

An independent world-sheet metric produces a quadratic action classically equivalent to Nambu–Goto. The section develops target-space Poincaré symmetry, world-sheet diffeomorphisms, Weyl symmetry, equations of motion, stress-tensor constraints, conformal gauge, residual conformal symmetry, and open-string Neumann/Dirichlet boundary conditions. These constraints later become Virasoro conditions.

A.3.2.4 2.4 Oscillator expansions (p. 23)

Classical solutions are decomposed into left- and right-moving modes. Open and closed boundary conditions determine the allowed oscillators, zero modes encode center-of-mass position and momentum, and the constraints become relations among mode coefficients. This is the classical input for canonical quantization.

A.3.2.5 2.5 Examples of classical string solutions (p. 31)

Rotating and stretched strings illustrate how energy, angular momentum, and tension produce Regge behavior. The examples connect the geometric world-sheet description with the particle spectrum anticipated after quantization.

A.3.3 Chapter 3: The Quantized Bosonic String (pp. 35–62)

A.3.3.1 3.1 Canonical quantization of the bosonic string (p. 35)

Oscillator Poisson brackets become commutators, and the Fock space initially contains timelike negative-norm excitations. Virasoro constraints, normal ordering, and the intercept define physical states; null states encode gauge redundancy.

A.3.3.2 3.2 Light-cone quantization of the bosonic string (p. 42)

Light-cone gauge solves the constraints explicitly and leaves only transverse oscillators. Lorentz-algebra closure fixes the critical dimension and intercept, giving a manifestly positive physical Hilbert space at the cost of manifest covariance.

A.3.3.3 3.3 Spectrum of the bosonic string (p. 46)

The open and closed mass formulae, level matching, tachyon, massless vector, graviton, antisymmetric tensor, and dilaton are identified. The little-group decomposition explains the spacetime spins carried by oscillator states.

A.3.3.4 3.4 Covariant path-integral quantization (p. 53)

The Polyakov functional integral is gauge fixed, producing determinants and a sum/integral over inequivalent metrics. This viewpoint prepares the ghost system, moduli integration, and higher-genus perturbation theory.

A.3.3.5 3.5 Appendix: the Virasoro algebra (p. 59)

The central extension of the conformal algebra is derived from oscillator normal ordering. The anomaly connects representation theory, the critical dimension, and the consistency of the gauge-fixed quantum theory.

A.3.4 Chapter 4: Introduction to Conformal Field Theory (pp. 63–106)

A.3.4.1 4.1 General introduction (p. 63)

The section develops conformal transformations, complex coordinates, operator-product expansions, primary fields, Ward identities, radial quantization, the state–operator correspondence, the Virasoro algebra, and basic representation theory. It supplies the language in which a string background is a two-dimensional CFT.

A.3.4.2 4.2 Application to closed string theory (p. 85)

Free-boson CFT is matched to the closed-string oscillator construction. Vertex operators encode string states, conformal weights impose mass-shell conditions, and correlation functions become the local ingredients of amplitudes.

A.3.4.3 4.3 Boundary conformal field theory (p. 93)

A boundary identifies left and right conformal transformations, leaving a diagonal algebra. Open-string spectra, boundary conditions, the doubling trick, and open–closed channel duality are formulated in CFT language.

A.3.4.4 4.4 Free-boson boundary states (p. 101)

Neumann and Dirichlet conditions are imposed as closed-string gluing equations. Their coherent-state solutions represent D-branes in the closed channel and reproduce cylinder amplitudes through overlaps.

A.3.4.5 4.5 Crosscap states for the free boson (p. 103)

Orientation-reversing identifications produce crosscap gluing conditions. Crosscap states are the closed-channel representation of unoriented world-sheets and become the basic description of orientifold planes.

A.3.5 Chapter 5: Parametrization Ghosts and BRST Quantization (pp. 107–120)

A.3.5.1 5.1 The ghost system as a conformal field theory (p. 107)

Faddeev–Popov gauge fixing of world-sheet diffeomorphisms introduces the (b, c) system. Its OPEs, conformal weights, stress tensor, central charge, zero modes, and ghost-number selection rules compensate the unphysical metric degrees of freedom.

A.3.5.2 5.2 BRST quantization (p. 110)

Matter and ghost constraints are combined into a nilpotent BRST charge. Nilpotency encodes criticality; physical states are cohomology classes, BRST-exact states decouple, and vertex operators acquire their precise ghost dressing. This is the covariant definition used in amplitude calculations.

A.3.6 Chapter 6: String Perturbation Theory and One-Loop Amplitudes (pp. 121–174)

A.3.6.1 6.1 String perturbation expansion (p. 121)

Splitting and joining strings generates world-sheets classified by topology. The dilaton coupling weights a surface by its Euler characteristic, turning g_s into the genus expansion parameter and organizing closed, open, and unoriented diagrams.

A.3.6.2 6.2 Polyakov path integral for the closed bosonic string (p. 126)

Gauge fixing separates Weyl/diffeomorphism directions from moduli and conformal Killing transformations. Ghost zero modes, vertex positions, and the integration over moduli space are derived rather than inserted ad hoc.

A.3.6.3 6.3 The torus partition function (p. 146)

The genus-one vacuum amplitude is evaluated both as a path integral and as a Hamiltonian trace. Modular invariance identifies equivalent tori and restricts integration to a fundamental domain with invariant measure; level matching is enforced by the real part of τ.

A.3.6.4 6.4 Torus partition functions for rational CFTs (p. 152)

The torus amplitude is expanded in finitely many characters. Modular matrices S, T, charge conjugation, and integer multiplicity matrices classify modular invariants and connect geometry of the torus with fusion and representation data.

A.3.6.5 6.5 The cylinder partition function (p. 157)

The annulus is described as an open-string one-loop trace or a closed string propagating between boundaries. The modular transformation between channels exposes tadpoles and long-range forces, and the measure $dt/(2t)$ reflects its single real modulus and residual automorphism.

A.3.6.6 6.6 Boundary states and cylinder amplitudes for RCFTs (p. 163)

Ishibashi states solve chiral gluing conditions, while Cardy states impose open–closed consistency. The Cardy condition relates boundary coefficients to the modular S matrix and ensures nonnegative integer open-string spectra.

A.3.6.7 6.7 Crosscap states, Klein bottle, and Möbius strip amplitudes (p. 166)

Unoriented one-loop surfaces are written in direct and transverse channels. Crosscap coefficients, parity projections, Möbius consistency, and tadpole factorization provide the general RCFT framework for orientifolds.

A.3.6.8 6.8 Appendix: D-brane tension (p. 171)

The massless part of the cylinder is compared with field-theory graviton/dilaton exchange. This fixes the normalization and tension of a D-brane and illustrates how world-sheet factorization determines spacetime source couplings.

A.3.7 Chapter 7: The Classical Fermionic String (pp. 175–194)

A.3.7.1 7.1 Motivation for the fermionic string (p. 175)

World-sheet fermions are introduced to remove central defects of the bosonic theory and to build spacetime supersymmetric spectra. The section motivates the RNS formulation.

A.3.7.2 7.2 Superstring action and its symmetries (p. 176)

The Polyakov action is extended to a locally supersymmetric world-sheet theory. Global and local supersymmetry, zweibein, gravitino, and their gauge transformations identify the supergravity multiplet whose redundancies must be fixed.

A.3.7.3 7.3 Superconformal gauge (p. 180)

Gauge fixing leaves free bosons and Majorana fermions plus superconformal constraints. The supercurrent joins the stress tensor, and residual transformations form the super-Virasoro symmetry used in quantization.

A.3.7.4 7.4 Oscillator expansions (p. 189)

Boundary conditions yield Ramond and Neveu–Schwarz moding. Mode expansions, canonical brackets, and classical super-Virasoro generators set up the different sectors of the quantum RNS string.

A.3.7.5 7.5 Appendix: spinor algebra in two dimensions (p. 192)

Two-dimensional gamma matrices, chirality, Majorana conditions, and spinor bilinears fix conventions underlying the action and supersymmetry transformations.

A.3.8 Chapter 8: The Quantized Fermionic String (pp. 195–222)

A.3.8.1 8.1 Canonical quantization (p. 195)

Bosonic and fermionic modes are quantized and the super-Virasoro algebra acquires central terms. Physical-state conditions and normal-ordering constants differ between NS and R sectors.

A.3.8.2 8.2 Light-cone quantization (p. 200)

Superconformal constraints eliminate longitudinal modes, leaving transverse bosons and fermions. Closure and zero-point energies select ten dimensions and provide a ghost-free description.

A.3.8.3 8.3 Spectrum and GSO projection (p. 202)

The unprojected NS tachyon and R spinor are reorganized by the GSO projection. It removes the tachyon, selects chirality, produces equal bosonic and fermionic degeneracies, and yields spacetime supersymmetry.

A.3.8.4 8.4 Path-integral quantization (p. 208)

Gauge fixing local world-sheet supersymmetry introduces the (β, γ) superghosts in addition to (b, c). Central-charge cancellation and spin-structure sums anticipate the covariant BRST formalism.

A.3.8.5 8.5 Appendix: Dirac matrices and spinors in d dimensions (p. 210)

Clifford algebras, Weyl and Majorana conditions, chirality, and dimensional dependence are collected for use in spectra, R-sector ground states, supersymmetry, and brane projectors.

A.3.9 Chapter 9: Superstrings (pp. 223–262)

A.3.9.1 9.1 Spin structures and the superstring partition function (p. 223)

Fermion boundary conditions on the torus are summed with GSO phases. Theta functions organize the four spin structures; modular invariance fixes their combination, and the Jacobi identity makes the supersymmetric vacuum amplitude vanish.

A.3.9.2 9.2 Boundary states for fermions (p. 232)

Fermionic gluing conditions are solved in NS–NS and R–R sectors. Spin labels, zero modes, channel duality, and GSO projection determine the consistent boundary-state combinations.

A.3.9.3 9.3 D-branes (p. 235)

Bosonic and fermionic boundary states combine into BPS D-branes. Their R–R charges, tensions, preserved supersymmetry, open-string spectra, and cancellation of NS–NS attraction against R–R repulsion are derived.

A.3.9.4 9.4 The type I string (p. 241)

The Ω orientifold of type IIB introduces an O9-plane and D9-branes. Klein-bottle, annulus, and Möbius tadpoles factorize, requiring 32 D9-branes and producing the anomaly-free $SO(32)$ theory.

A.3.9.5 9.5 Stable non-BPS branes (p. 251)

Orientifold and orbifold projections can remove tachyons from brane–antibrane constructions. K-theoretic charges and stability beyond the BPS sector are motivated through explicit spectra.

A.3.9.6 9.6 Appendix: theta functions and twisted fermion partition functions (p. 254)

Products, characteristics, modular transformations, and identities of theta functions are compiled and related to fermionic traces with twisted boundary conditions.

A.3.10 Chapter 10: Toroidal Compactifications and the Ten-Dimensional Heterotic String (pp. 263–320)

A.3.10.1 10.1 Motivation (p. 263)

Compactification is introduced as the bridge from critical ten dimensions to lower-dimensional physics, with moduli and new momentum/winding quantum numbers.

A.3.10.2 10.2 Toroidal compactification of the closed bosonic string (p. 264)

Momentum and winding form left/right lattices controlled by the metric and B-field. Mass formulae, level matching, T-duality, enhanced gauge symmetry, and the Narain moduli space emerge from lattice geometry.

A.3.10.3 10.3 Toroidal partition functions (p. 280)

The compact zero modes are summed into lattice theta functions. Modular invariance requires even self-dual Lorentzian lattices, explaining why consistency is encoded arithmetically.

A.3.10.4 10.4 The $E_8 \times E_8$ and $SO(32)$ heterotic string theories (p. 284)

Left-moving bosonic and right-moving supersymmetric sectors are combined. Even self-dual 16-dimensional Euclidean lattices select the two heterotic gauge groups, and the massless gauge/gravity spectrum is read off.

A.3.10.5 10.5 Toroidal orbifolds (p. 294)

Discrete quotients introduce untwisted projections and twisted sectors localized at fixed loci. Modular invariance, level matching, fixed-point degeneracy, and gauge embeddings determine consistent spectra and furnish solvable singular compactifications.

A.3.10.6 10.6 D-branes on toroidal compactifications (p. 308)

T-duality converts Wilson lines into positions and Neumann into Dirichlet directions. Wrapped branes, images, orientifold planes, Chan–Paton projections, and tadpoles are organized geometrically; Table 10.3 summarizes a key T-duality orbit.

A.3.11 Chapter 11: CFT II - Lattices and Kac–Moody Algebras (pp. 321–354)

A.3.11.1 11.1 Kac–Moody algebras (p. 321)

Currents generate affine Lie algebras through their OPEs. The level, central extension, Sugawara stress tensor, and conformal weights connect spacetime gauge symmetry to chiral CFT data.

A.3.11.2 11.2 Lattices and Lie algebras (p. 327)

Roots, coroots, weights, Weyl reflections, conjugacy classes, and ADE lattices are reviewed. This arithmetic controls both current-algebra representations and modular invariant compactification lattices.

A.3.11.3 11.3 Frenkel–Kac–Segal construction (p. 337)

Simply-laced level-one current algebras are realized by free bosons compactified on root lattices. Cartan currents arise from derivatives and step operators from lattice vertex operators with cocycles.

A.3.11.4 11.4 Fermionic construction and bosonization (p. 340)

Free fermion bilinears generate current algebras, while bosonization translates fermions into exponential bosonic vertices. The four D_n conjugacy classes encode NS/R sectors and spinor/vector representations.

A.3.11.5 11.5 Unitary representations and characters (p. 343)

Affine primaries, integrability bounds, finite representation sets at fixed level, characters, and modular transformations are derived. This makes Kac–Moody theories rational CFTs.

A.3.11.6 11.6 Highest-weight representations of $\widehat{\mathfrak{su}}(2)_k$ (p. 349)

The general theory is made explicit: allowed spins, null vectors, characters, modular S matrix, and fusion illustrate how level truncates representation theory.

A.3.12 Chapter 12: CFT III - Superconformal Field Theory (pp. 355–390)

A.3.12.1 12.1 $N = 1$ superconformal symmetry (p. 355)

The stress tensor and supercurrent form the $N = 1$ algebra. NS/R representations, superprimaries, spectral properties, and minimal models provide the internal-CFT language needed for spacetime supersymmetric strings.

A.3.12.2 12.2 $N = 2$ superconformal symmetry (p. 370)

Two supercurrents and a $U(1)$ current enlarge the algebra. Spectral flow relates NS and R sectors, charges BPS states, and turns internal $N = 2$ symmetry into spacetime supersymmetry after compactification.

A.3.12.3 12.3 Chiral ring and topological conformal field theory (p. 382)

Chiral primaries saturate a unitarity bound and close under nonsingular OPEs. Twisting converts a supercharge into a BRST operator, producing topological sectors whose ring data encode protected geometry.

A.3.13 Chapter 13: Covariant Vertex Operators, BRST, and Covariant Lattices (pp. 391–426)

A.3.13.1 13.1 Bosonization and first-order systems (p. 391)

General (b, c)-type systems and the superghosts are bosonized. Background charge, conformal weights, zero-mode anomalies, spin fields, and picture number prepare a covariant treatment of RNS vertices.

A.3.13.2 13.2 Covariant vertex operators, BRST, and picture changing (p. 403)

Physical NS and R vertices are constructed in different pictures. BRST closure imposes mass shell, transversality, and Dirac equations; picture-changing relates equivalent representatives and makes superghost charge balance possible in amplitudes.

A.3.13.3 13.3 D-branes and spacetime supersymmetry (p. 411)

Fermion gluing on the upper half-plane is lifted to spin fields and supercharges. A D-brane preserves a diagonal combination $Q_L + PQ_R$, and gamma-matrix chirality selects allowed BPS Dp-branes in type IIA/IIB.

A.3.13.4 13.4 The covariant lattice (p. 415)

Bosonized fermions and ghosts are assembled into a lattice description. GSO locality becomes a lattice condition, conjugacy classes organize sectors, and one-loop characters acquire a uniform covariant form.

A.3.13.5 13.5 Heterotic strings in covariant lattice description (p. 422)

Right-moving superstring and left-moving gauge lattices are combined. Modular invariance and lattice extensions classify ten-dimensional heterotic constructions and prepare four-dimensional covariant-lattice models.

A.3.14 Chapter 14: String Compactifications (pp. 427–520)

A.3.14.1 14.1 Conformal invariance and spacetime geometry (p. 427)

The nonlinear sigma model treats G, B, and Φ as world-sheet couplings. Vanishing Weyl-anomaly coefficients gives their spacetime field equations order by order in α', establishing the world-sheet origin of gravitational dynamics.

A.3.14.2 14.2 T-duality and Buscher rules (p. 432)

Gauging an isometry and integrating out different fields derives the Buscher transformation of G, B, and Φ. T-duality is extended from a circle spectrum to general sigma-model backgrounds.

A.3.14.3 14.3 Compactification (p. 439)

Kaluza–Klein reduction relates internal geometry to lower-dimensional fields, masses, gauge symmetries, and supersymmetry. Holonomy and covariantly constant spinors identify the geometry required by unbroken supercharges.

A.3.14.4 14.4 Mathematical preliminaries (p. 450)

Complex, Hermitian, Kähler, symplectic, and Riemannian geometry are reviewed together with differential forms, cohomology, characteristic classes, bundles, and index-theoretic tools. This is the reference section for the later Calabi–Yau analysis.

A.3.14.5 14.5 Calabi–Yau manifolds (p. 473)

Calabi–Yau holonomy, the Kähler form, holomorphic volume form, Hodge diamond, moduli, and special geometry are developed. Topological data are connected to harmonic representatives and deformations.

A.3.14.6 14.6 Type-II compactification on Calabi–Yau threefolds (p. 487)

Ten-dimensional fields are expanded in harmonic forms to obtain four-dimensional $\mathcal{N} = 2$ multiplets. Type IIA and IIB exchange Kähler and complex-structure sectors under mirror symmetry.

A.3.14.7 14.7 Heterotic compactification on Calabi–Yau threefolds (p. 497)

Supersymmetry and anomaly cancellation constrain holomorphic vector bundles through the Hermitian Yang–Mills equations and Bianchi identity. The standard embedding, surviving gauge group, chiral index, and bundle moduli connect geometry to particle spectra.

A.3.14.8 14.8 Appendix: Riemannian geometry and Hodge duals (p. 508)

Curvature, spin connection, form conventions, Hodge stars, and dimensional-reduction identities are collected for calculations in Chapters 14–18.

A.3.15 Chapter 15: CFTs for Type-II and Heterotic String Vacua (pp. 521–584)

A.3.15.1 15.1 Motivation (p. 521)

The chapter shifts from large-radius geometry to exact world-sheet constructions. Orbifolds, orientifolds, Gepner models, and covariant lattices provide complementary solvable points in the string-vacuum moduli space.

A.3.15.2 15.2 Supersymmetric orbifold compactifications (p. 522)

Twist vectors preserving supersymmetry act on tori and gauge lattices. Untwisted and twisted spectra, fixed points, Hodge numbers, standard embeddings, modular constraints, and blow-up modes connect singular CFT data with smooth Calabi–Yau geometry.

A.3.15.3 15.3 Orientifold compactifications (p. 534)

World-sheet parity is combined with geometric involutions and $(-1)^{F_L}$. Closed projections, O-plane loci, Klein-bottle tadpoles, fractional D-branes, annulus/Möbius amplitudes, and open spectra are worked out in concrete toroidal models.

A.3.15.4 15.4 World-sheet aspects of supersymmetric compactifications (p. 544)

Internal $N = 2$ SCFT, bosonization, spectral flow, GSO conditions, and covariant vertices are used to derive massless type-II and heterotic spectra. R–R fields appear through field-strength vertices, while internal current superfields generate gauge bosons.

A.3.15.5 15.5 Gepner models (p. 567)

Tensor products of $N = 2$ minimal models with the correct total central charge give exactly solvable Calabi–Yau compactifications. Simple-current projections, spectral flow, massless spectra, and the relation to geometry are explained.

A.3.15.6 15.6 Four-dimensional heterotic strings via covariant lattices (p. 578)

Covariant lattice methods construct modular invariant heterotic vacua directly in four dimensions. Lattice vectors encode spacetime supersymmetry, gauge symmetry, chirality, and matter representations.

A.3.16 Chapter 16: String Scattering Amplitudes and Low-Energy Effective Field Theory (pp. 585–640)

A.3.16.1 16.1 Generalities (p. 585)

String amplitudes are defined as moduli integrals of BRST-invariant vertex correlators, with coupling and topology factors. The section explains what data are on shell and how factorization constrains consistency.

A.3.16.2 16.2 String scattering amplitudes (p. 588)

Explicit sphere, disk, and related amplitudes produce Veneziano/Virasoro–Shapiro structures. Ghost fixing, Koba–Nielsen factors, upper-half-plane propagators, D-brane scattering, poles, residues, and soft limits are computed.

A.3.16.3 16.3 From amplitudes to the low-energy effective field theory (p. 615)

The α' expansion separates massless poles from contact terms. Matching on-shell amplitudes determines effective interactions up to field redefinitions and equation-of-motion terms, explaining both the power and limits of S-matrix reconstruction.

A.3.16.4 16.4 Effective supergravities in ten and eleven dimensions (p. 622)

The bosonic actions and field contents of type IIA, type IIB, heterotic/type I, and eleven-dimensional supergravity are summarized. Frame changes, dual field strengths, Chern–Simons terms, and self-duality conventions prepare the duality chapter.

A.3.16.5 16.5 Effective action for D-branes (p. 630)

The Dirac–Born–Infeld and Chern–Simons actions encode tension, gauge fields, transverse scalars, NS–NS couplings, and R–R charges. Pullbacks, the combination $B + 2\pi\alpha' F$, and curvature/derivative corrections translate disk amplitudes into brane dynamics.

A.3.16.6 16.6 Appendix: integrals in Veneziano and Virasoro–Shapiro amplitudes (p. 636)

Beta-function and complex-plane integrals used in tree amplitudes are derived and analytically continued, clarifying their pole expansions.

A.3.17 Chapter 17: Type-II Compactifications with D-Branes and Fluxes (pp. 641–674)

A.3.17.1 17.1 Brane worlds and fluxes (p. 641)

D-branes localize gauge sectors and fluxes generate potentials for moduli. The section sets the phenomenological and geometric questions: supersymmetry, chirality, tadpoles, and stabilization.

A.3.17.2 17.2 Supersymmetry, tadpoles, and massless spectra (p. 644)

Orientifolded Calabi–Yau compactifications with D-branes are constrained by calibration conditions and R–R charge cancellation. Intersection numbers determine chiral matter, same-stack strings give adjoints, and anomalies are tied to tadpoles and Green–Schwarz couplings.

A.3.17.3 17.3 Flux compactifications (p. 657)

NS–NS and R–R fluxes induce the Gukov–Vafa–Witten superpotential. Flux quantization, D3 tadpoles, F-term conditions, imaginary self-duality, and no-scale structure explain how complex-structure moduli and the axiodilaton are stabilized while Kähler moduli remain flat at tree level.

A.3.17.4 17.4 Fluxes and $SU(3)$ structure group (p. 663)

Allowing torsion generalizes Calabi–Yau holonomy to $SU(3)$ structure. Intrinsic torsion classes, non-closed J and Ω, flux supersymmetry equations, and warped backgrounds organize solutions beyond Ricci-flat compactifications.

A.3.17.5 17.5 Appendix (p. 670)

Technical conventions and geometric identities used in the flux and brane analysis are collected for reference.

A.3.18 Chapter 18: String Dualities and M-Theory (pp. 675–774)

A.3.18.1 18.1 General remarks (p. 675)

Duality is presented as an equivalence between descriptions with different expansion parameters or geometries. Perturbative theories become coordinate patches on a larger nonperturbative structure.

A.3.18.2 18.2 Simple examples: modular invariance and T-duality (p. 677)

Earlier exact symmetries are reinterpreted as prototypes of duality: modular transformations identify world-sheet descriptions, while T-duality exchanges momentum/winding, radius, brane dimension, and type IIA/IIB.

A.3.18.3 18.3 Extended objects: generalities (p. 680)

p-branes couple electrically and magnetically to form fields, with tensions and charges controlled by g_s and frame choice. Dirac quantization and world-volume actions set the vocabulary for nonperturbative states.

A.3.18.4 18.4 Central charges and the BPS bound (p. 684)

Extended charges appear in supersymmetry algebras. Positivity gives BPS inequalities; saturation implies shortened multiplets, preserved projector equations, and protected masses/tensions that can be followed across strong coupling.

A.3.18.5 18.5 Brane solutions in supergravity (p. 688)

Symmetry and supersymmetry determine warped metrics, dilaton profiles, form fluxes, and harmonic functions. Fundamental strings, NS5-branes, Dp-branes, and their near-horizon regions connect source actions with classical backreaction.

A.3.18.6 18.6 Nonperturbative dualities (p. 705)

Type-IIB self-duality organizes (p, q) strings and five-branes; type I and heterotic strings are S-dual; compactified theories exhibit U-duality. Brane spectra and BPS quantities supply quantitative tests beyond perturbation theory.

A.3.18.7 18.7 M-theory (p. 716)

Strongly coupled type IIA opens an eleventh dimension. Reduction of eleven-dimensional supergravity yields IIA fields, while M2/M5 branes reduce or wrap to strings and D-branes. The supersymmetry algebra unifies their charges.

A.3.18.8 18.8 F-theory (p. 724)

The varying type-IIB axiodilaton becomes the complex structure of an auxiliary elliptic fiber. Seven-brane monodromies, elliptic K3, orientifold limits, heterotic duality, singular fibers, gauge enhancement, and localized matter are encoded geometrically.

A.3.18.9 18.9 AdS/CFT correspondence (p. 733)

The D3-brane near-horizon limit produces $AdS_5 \times S^5$ and leads to the equivalence with four-dimensional $\mathcal{N} = 4$ SYM. The section develops the parameter map, conformal compactification, bulk/boundary dictionary, correlation functions, Wilson loops, black branes, and finite-temperature thermodynamics, ending with holography as the strong-coupling culmination of the book.

A.4 Rapid re-entry guide

When returning after a long break, use the following routes:

- For **world-sheet foundations**, revisit Sections 2.3, 3.2, 4.1–4.3, 5.2, and 6.2–6.3.
- For **superstring spectra and D-branes**, revisit Sections 8.3, 9.1–9.4, 13.2–13.3, and 16.5.
- For **modular invariance, lattices, and exact CFT**, revisit Sections 6.4–6.7, 10.2–10.5, 11.1–11.6, 12.1–12.3, and 15.2–15.6.
- For **Calabi–Yau compactification**, revisit Sections 14.1 and 14.4–14.7, followed by 17.2–17.4.
- For **effective actions and amplitudes**, revisit Sections 16.1–16.5 and the beta-function argument of Section 14.1.
- For **dualities and modern geometry**, read Chapter 18 in order; its logic depends especially on BPS bounds (18.4), brane solutions (18.5), and the compactification technology of Chapters 10, 14, and 17.

The shortest conceptual reconstruction is: world-sheet gauge symmetry forces CFT and BRST; modular consistency and supersymmetry restrict the perturbative theories; compactification converts CFT data into geometry and lower-dimensional fields; D-branes and fluxes make gauge sectors, chirality, and potentials; BPS protection lets those objects be followed to strong coupling; dualities then reveal M-theory, F-theory, and holography as different geometric organizations of the same underlying degrees of freedom.